

Passage 1	2011-01-08-1	Crown of Thorns Starfish and Coral Reefs
Passage 2	2011-01-08-2	The Rise of Moscow
Passage 3	2011-01-08-3	Forest Succession
Passage 4	2013-01-12-1	England's Economy in the Sixteenth Century
Passage 5	2013-01-12-2	Documenting the Incas
Passage 6	2013-01-12-3	What Controls Flowering
Passage 7	2014-03-02-1	Cell Theory
Passage 8	2014-03-02-2	The Great Depression
Passage 9	2014-03-02-3	Poikilotherms
Passage 10	2014-03-16-1	Costs and Benefits of Social Life
Passage 11	2014-03-16-2	Comets
Passage 12	2014-03-16-3	Agricultural Society in Eighteenth-Century British America
Passage 13	2014-03-22-1	The Beringia Landscape
Passage 14	2014-03-22-2	Wind pollination
Passage 15	2014-03-22-3	Feeding Strategies in the Ocean
Passage 16	2014-03-23-1	From Fish to Terrestrial Vertebrates
Passage 17	2014-03-23-2	Early Modern Industrialization
Passage 18	2014-03-23-3	Environmental Impact of the Anasazi
Passage 19	2014-04-12-1	Surface Fluids on Venus and Earth
Passage 20	2014-04-12-2	Water Management in Early Agriculture
Passage 21	2014-04-12-3	Plant and Animal Life of the Pacific Islands
Passage 22	2014-04-19-1	The Cosmological Principle

Passage	23	2014-04-19-2	Distribution of Seaweeds
Passage	24	2014-04-19-3	Bird Colonies
Passage	25	2014-04-27-1	Urban Development in the United States in the Nineteenth
Passage	26	2014-04-27-2	The Climate of Japan
Passage	27	2014-04-27-3	Economic Decline in Europe During the Fourteenth Centu



Passage 28	2014-05-11-1	Determining the Ages of the Planets and the Universe
Passage 29	2014-05-11-2	Costs and Benefits of Dispersal
Passage 30	2014-05-11-3	The Multiplier Effect
Passage 31	2014-05-17-1	Cognitive Maps in Animals
Passage 32	2014-05-17-2	The Use of the Camera Obscura
Passage 33	2014-05-17-3	Habitat Selection
Passage 34	2014-05-24-1	Sumerian Contributions
Passage 35	2014-05-24-2	Impacts and Mass Extinctions
Passage 36	2014-05-24-3	Milankovitch Cycles and Glaciation
Passage 37	2014-05-25-1	The Long History of Overexploitation
Passage 38	2014-05-25-2	How Animals in Rain Forests Make Themselves Heard
Passage 39	2014-05-25-3	Pest Control
Passage 40	2014-06-15-1	Early life-forms and Earth's Atmosphere
Passage 41	2014-06-15-2	Constraints on Natural Selection
Passage 42	2014-06-15-3	Newspapers in Western Europe
Passage 43	2014-06-21-1	Climate Change and the Natufian People
Passage 44	2014-06-21-2	Metamorphosis
Passage 45	2014-06-21-3	The Origin of Coral Reefs
Passage 46	2014-06-28-1	Southwest Agriculture
Passage 47	2014-06-28-2	The Development of Printing
Passage 48	2014-06-28-3	Earth's Energy Cycle
Passage 49	2014-07-06-1	Removing Dams

Passage 50	2014-07-06-2	Coral Reef Communities
Passage 51	2014-07-06-3	Sociality in Animals
Passage 52	2014-07-12-1	Understanding Ancient Mesoamerican Art
Passage 53	2014-07-12-2	How Soil is Formed
Passage 54	2014-07-12-3	The Emergence of Civilization



Passage 55	2014-08-16-1	Hunting and the Setting of Inner Eurasia
Passage 56	2014-08-16-2	Written Records
Passage 57	2014-08-16-3	Temperature Regulation in Marine Organisms
Passage 58	2014-08-30-1	Mesolithic Complexity in Scandinavia
Passage 59	2014-08-30-2	Evolution of the Flowering Plants
Passage 60	2014-08-30-3	How Herding Can Provide Safety
Passage 61	2014-10-18-1	The Origins of Writing
Passage 62	2014-10-18-2	Soil Fertilization
Passage 63	2014-10-18-3	The Collapse of the Mays
Passage 64	2014-12--1	Forms of Locomotion
Passage 65	2014-12--2	How Plants and Animals Arrived in the Hawaiian Islands
Passage 66	2014-12--3	Mexican Mural Art
Passage 67	2015-01-10-1	Artisans in Sixteenth-Century Europe
Passage 68	2015-01-10-2	Disease and History
Passage 69	2015-01-10-3	Earth's Atmosphere
Passage 70	2015-01-11-1	Birdsong
Passage 71	2015-01-11-2	The Role of Diapause
Passage 72	2015-01-11-3	The Plow and the Horse in Medieval Europe
Passage 73	2015-01-25-1	The Origins of Plant and Animal Domestication
Passage 74	2015-01-25-3	Preventing Overgrowth among Tree Branches
Passage 75	2015-01-31-1	Effects of the Commercial Revolution
Passage 76	2015-01-31-2	European Context of the Scientific Revolution

Passage	77	2015-02-01-1	Greek Sacred Groves and Parks
Passage	78	2015-02-01-2	The Chaco Phenomenon
Passage	79	2015-02-01-3	Dinosaurs and Parental Care
Passage	80	2015-03-07-1	Colonial America and the Navigation Acts
Passage	81	2015-03-07-2	Mass Production: Method and Impact



Passage 81	2015-03-07-2	Mass Production: Method and Impact
Passage 82	2015-03-07-3	Art and Culture of Pacific Northwest Communities
Passage 83	2015-04-12-1	Architectural Change in Eighth-Century Japan
Passage 84	2015-04-12-2	Origin of the Solar System
Passage 85	2015-tfyd-30-1	Attempts at Determining Earth's Age
Passage 86	2015-tfyd-30-2	Water Supply in Venice
Passage 87	2015-tfyd-30-3	The Early History of Motion Pictures
Passage 89	2015-tfyd-30-10	Early Theories of Continental Drift
Passage 90	2015-tfyd-30-11	Bioluminescence in Marine Creatures
Passage 91	2015-tfyd-30-22	Egypt Circa 3100 B.C
Passage 92	2015-tfyd-30-24	Population Growth in Nineteenth-Century Europe
Passage 93	2015-tfyd-30-25	Costs of Quitting a Job
Passage 94	2015-tfyd-30-26	Stone Tools and Pottery Fragments
Passage 95	2015-tfyd-30-27	The Role of the Ocean in Controlling Climate
Passage 96	2015-tfyd-30-28	Animal Behavior
Passage 97	2015-tfyd-30-29	Grain in Colonial North America
Passage 98	2015-tfyd-30-30	Sea Turtle Hatchling Strategies for Navigation
Passage 99	2015-kl5-12-1	Controversy about Causing Emotion
Passage 100	2015-kl5-12-2	Oxygen and nitrogen are major components
Passage 101	2015-kl5-12-3	The Distribution of Gliding Animals
Passage 102	2015-kl5-12-4	The Sentinel Behavior of Meerkats
Passage 103	2015-kl5-12-5	Portraits as Art

Passage 104	2015-ks-12-7	Agriculture in Medieval Japan
Passage 105	2015-ks-12-8	Grassland Environment
Passage 106	2015-ks-12-9	The British Economy Under the Roman Empire
Passage 107	2015-ks-12-10	Foundational and Keystone Species



Passage	108	2015-03-14-1	The First Eyes
Passage	109	2015-03-14-2	The Origin of Earth's Atmosphere
Passage	110	2015-04-18-1	Historical Trends in European Urban Design
Passage	111	2015-04-18-2	Origins of the Megaliths
Passage	112	2015-05-09-1	Reconstructing Ancient Environments
Passage	113	2015-05-09-3	Vocalization in Frogs
Passage	114	2015-05-24-1	The Geographical Distribution of Gliding Animals
Passage	115	2015-05-24-2	Saving Soil and Cropland
Passage	116	2015-05-24-3	Poetry Writing Course to be Canceled
Passage	117	2015-05-30-1	Thermal Stratification
Passage	118	2015-05-30-2	Live Performance
Passage	119	2015-05-30-3	Earthquake Prediction
Passage	120	2015-06-13-2	Gondwana
Passage	121	2015-06-13-3	Olmec Art
Passage	122	2015-06-14-2	Mesopotamian and Egyptian Settlement Patterns
Passage	123	2015-06-14-3	Extinctions at the End of the Cretaceous
Passage	124	2015-07-05-1	Agriculture in the Late Ottoman Empire
Passage	125	2015-07-05-2	Elements of Life
Passage	126	2015-07-05-3	Primitive and Advanced Termite Species
Passage	127	2015-07-11-1	Individual Performance and the Presence of Others
Passage	128	2015-07-11-2	The Identification of the Genetic Material
Passage	129	2015-07-11-3	How Birds Acquire Their Songs

Passage	130	2015-07-11-4	Cereals and Legumes: A Partnership
Passage	131	2015-07-11-5	Alaska and Bark Beetles
Passage	132	2015-07-11-6	Motor Development in Children
Passage	133	2015-07-11-7	Bison and Humans
Passage	134	2015-07-12-1	The Western Roman Empire in the Fifth Century
Passage	135	2015-07-12-2	Features of Tropical Mammals
Passage	136	2015-07-12-3	Honeybee Society
Passage	137	2015-08-22-2	The Upper Paleolithic Revolution
Passage	138	2015-08-22-3	Two Kinds of Lizards
Passage	139	2015-09-05-2	The Cambrian Explosion
Passage	140	2015-09-05-3	Weak Electric Systems in Fish
Passage	141	2015-09-12-1	Europe in the High Middle Ages
Passage	142	2015-09-12-2	Gliding and Soaring
Passage	143	2015-09-12-3	Pleistocene Extinctions
Passage	144	2015-09-19-2	Models of Egg Development
Passage	145	2015-09-19-3	Population Revolution in Eighteenth-Century Europe
Passage	146	2015-11-15-3	Tree Species Identification in Tropical Rain Forests
Passage	147	2015-11-15-4	The Extinction of the Dinosaurs

Crown of Thorns Starfish and Coral Reefs

The crown of thorns starfish, *Acanthaster Tlanci*, is large, twenty-five to thirty-five centimeters in diameter, and has seven to twenty-one arms that are covered in spines. It feeds primarily on coral and is found from the Indian Ocean to the west coast of Central America, usually at quite low population densities. Since the mid-1950s, population outbreaks at densities four to six times greater than normal have occurred at the same time in places such as Hawaii, Tahiti, Panama, and the Great Barrier Reef. The result has often been the loss of a fifty percent to nearly one hundred percent of the coral cover over large areas.

A single *Acanthaster* can consume five to six square meters of coral polyps per year, and dense populations can destroy up to six square kilometers per year and move on rapidly. *Acanthasters* show a preference for branching corals, especially *Acroporids*. After an outbreak in a particular area, it is common to find that *Acroporids* have been selectively removed, leaving a mosaic of living and dead corals. In places where *Acroporids* previously dominated the community devastation can be almost complete, and local areas of reefs have collapsed.

Areas of dead coral are usually colonized rapidly by algae and often are later colonized by sponges and soft corals. Increases in abundance of plant-eating fish and decreases in abundance of coral-feeding fish accompany these changes. Coral larvae settle among the algae and eventually establish flourishing coral colonies. In ten to fifteen years the reefs often return to about the same percentage of coral cover as before. Development of a four-species diversity takes about twenty years.

Two schools of thought exist concerning the cause of these outbreaks. One group holds that they are natural phenomena that have occurred many times in the past, citing old men's recollections of earlier outbreaks and evidence from traditional cultures. The other group maintains that recent human activities ranging from physical coral destruction through pollution to predator removal have triggered these events.

One theory, the adult aggregation hypothesis, maintains that most species is more abundant than we realize when a storm destroys coral and causes a food shortage. The adult *Acanthasters* converge on remaining portions of healthy coral and feed hungrily. Certainly there have been outbreaks of *Acanthaster* following large storms, but there is little evidence that the storms have caused the enough reef damage to create a food shortage for these starfish.

Two other hypotheses attempt to explain the increased abundance of

Acanthaster after episodes of high terrestrial runoff following storms. The first hypothesis is that low salinity and high temperatures favor the survival of the starfish larvae. The second hypothesis emphasizes the food web aspect, suggesting that strong fresh water runoff brings additional nutrients to the coastal waters, stimulating phytoplankton production and promoting more rapid development and better survival of the starfish larvae.

Those favoring anthropogenic (human influenced) causes have pointed to the large proportion of outbreaks that have been near centers of human populations. It has been suggested that coral polyps are the main predators of the starfish larvae. Destruction of coral by blasting and other bad land use practices would reduce predation on the starfish larvae and cause a feedback in which increases in Acanthaster populations cause still further coral destruction. Unfortunately, there are too few documented instances of physical destruction of coral being followed by outbreaks of Acanthaster for these hypotheses to be fully supported.

Another group of hypothesis focuses on removal of Acanthaster's predators. Some have suggested that the predators might have been killed off by pollution whereas others have suggested that the harvesting of vertebrate and invertebrate predators of Acanthaster could have reduced mortality and caused increased abundance of adults. The problem with this group of hypothesis is that it is difficult to understand how reduced predation would lead to sudden increases in Acanthaster numbers in several places at the same time in specific years. It seems probable that there is no single explanation but that there are elements of the truth in several of the hypotheses. That is there are natural processes that have led to outbreaks in the past, but human impact has increased the frequency and severity of the outbreaks.

Paragraph 1

The crown of thorns starfish, Acanthaster Tlanci, is large, twenty-five to thirty-five centimeters in diameter, and has seven to twenty-one arms that are covered in spines. It feeds primarily on coral and is found from the Indian Ocean to the west coast of Central America, usually at quite low population densities. Since the mid-1950s, population outbreaks at densities four to six times greater than normal have occurred at the same time in places such as Hawaii, Tahiti, Panama, and the Great Barrier Reef. The result has often been the loss of a fifty percent to nearly one hundred percent of the coral cover over large areas.

1. According to paragraph 1 all of the following statements about crown of thorns starfish are true **EXCEPT**

- Crown of thorns starfish usually have several arms covered in spines.
- Crown of thorns starfish's main food is coral.
- Crown of thorns starfish normally live in high population densities.
- Crown of thorns starfish have caused a significant reduction in coral cover over large areas.

2. The word "outbreak" in the passage is closest in meaning to

- brief appearance
- sudden increase
- exposure
- observation

Paragraph 2

A single Acanthaster can consume five to six square meters of coral polyps per year, and dense populations can destroy up to six square kilometers per year and move on rapidly. Acanthasters show a preference for branching corals, especially Acroporids. After an outbreak in a particular area, it is common to find that Acroporids have been selectively removed, leaving a mosaic of living and dead corals. In places where Acroporids previously dominated the community devastation can be almost complete, and local areas of reefs have collapsed.

3. What can be inferred from paragraph 2 about a coral reef that has a small number of Acroporids

- It will collapse completely after Acanthaster consumes the Acroporids. ○ It will lose the Acroporids before the other corals when Acanthaster attack it.
- It will have a mixture of living and dead Acroporids after an Acanthaster attack.
- It will have a dense population of Acanthaster for several years.

A 选项，珊瑚礁并不会崩塌，因为除了一小部分鹿角珊瑚被吃掉以外，其它珊瑚仍保留，故排除。B 选项，鹿角珊瑚会首先被吃掉，符合原文，故正确答案为 B 选项。C 选项，活和死的鹿角珊瑚会镶嵌在一起，但根据原文我们知道，棘冠海星一定会先把鹿角珊瑚全部吃完，留下其它珊瑚，不会留下活的鹿角珊瑚，故排除。D 选项，本段并没有提到和棘冠海星密度有关的信息，故排除。

Paragraph 3

Areas of dead coral are usually colonized rapidly by algae and often are later colonized by sponges and soft corals. Increases in abundance of plant-eating fish and decreases in abundance of coral-feeding fish accompany these changes. Coral larvae settle among the algae and eventually establish flourishing coral colonies. In ten to fifteen years the reefs often return to about the same percentage of coral cover as before. Development of a four-species diversity takes about twenty years.

4. The word "accompany" in the passage is closest in meaning to

- occur along with

- ☐ speed up
- ☐ precede
- ☐ end

5. Which of the following is **NOT** mentioned in paragraph 3 as a consequence of the destruction of an area of a coral by Acanthaster

- ☐ Algae colonize the dead area.
- ☐ Plant-eating fish increase in number.
- ☐ Coral larvae disappear.
- ☐ Species diversity recovers fully over time.

Paragraph 5

One theory, the adult aggregation hypothesis, maintains that most species is more abundant than we realize when a storm destroys coral and causes a food shortage. The adult Acanthasters **converge** on remaining portions of healthy coral and feed hungrily. Certainly there have been outbreaks of Acanthaster following large storms, but there is little evidence that the storms have caused the enough reef damage to create a food shortage for these starfish.

6. The word "converge" in the passage is closest in meaning to

- ☐ rest
- ☐ come together
- ☐ spread out
- ☐ depend

7. According to paragraph 5, what is an important weakness of the adult aggregation hypothesis

- ☐ It is based on studies of the deeper parts of coral reefs.
- ☐ It fails to explain the abundance of Acanthaster in the deeper parts of coral reefs.
- ☐ It fails to explain why Acanthaster feed hungrily on healthy coral after a storm.
- ☐ It is not supported by evidence that storms result in food shortages for Acanthaster.

Paragraph 6

Two other hypotheses attempt to explain the increased abundance of Acanthaster after episodes of high terrestrial runoff following storms. The first hypothesis is that low salinity and high temperatures favor the survival of the

starfish larvae. The second hypothesis emphasizes the food web aspect, suggesting that strong fresh water runoff brings additional nutrients to the coastal waters, stimulating phytoplankton production and promoting more rapid development and better survival of the starfish larvae.

8. What can be inferred from paragraph 6 about the diet of *Acanthaster* larvae
- ☐ Phytoplankton is an important food source for them.
 - ☐ Storm water runoff washes away important nutrients they need.
 - ☐ When water temperature rises they have difficulty finding enough food.
 - ☐ Storms add needed salt to their diet.

Paragraph 7

Those favoring anthropogenic (human influenced) causes have pointed to the large proportion of outbreaks that have been near centers of human populations. It has been suggested that coral polyps are the main predators of the starfish larvae. Destruction of coral by blasting and other bad land use practices would reduce predation on the starfish larvae and cause a feedback in which increases in *Acanthaster* populations cause still further coral destruction. Unfortunately, there are too few documented instances of physical destruction of coral being followed by outbreaks of *Acanthaster* for these hypotheses to be fully supported.

9. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Blasting and other bad land use practices would cause the destruction of coral and increase the number of predators that feed on *Acanthaster* larvae.
- ☐ Bad land use practices would reduce predation on *Acanthaster* larvae and increase the starfish population which would cause more coral destruction.
- ☐ A reduction of bad land use practices would reduce coral destruction by increasing the survival of *Acanthaster* larvae and their predators.
- ☐ The destruction of coral through land use activities would cause a decrease in the number of predators that feed on *Acanthaster* larvae.

Paragraph 8

Another group of hypothesis focuses on removal of *Acanthaster*'s predators. Some have suggested that the predators might have been killed off by pollution whereas others have suggested that the harvesting of vertebrate and invertebrate predators of *Acanthaster* could have reduced mortality and caused increased abundance of adults. The problem with this group of hypothesis is that it is difficult to understand how reduced predation would lead to sudden

increases in Acanthaster numbers in several places at the same time in specific years. It seems probable that there is no single explanation but that there are elements of the truth in several of the hypotheses. That is there are natural processes that have led to outbreaks in the past, but human impact has increased the frequency and severity of the outbreaks.

10. Why does the author state that "it is difficult to understand how reduced predation would lead to sudden increases in Acanthaster numbers in several places at the same time in specific years"?

- ☒ To indicate the difficulty in proving that increases in Acanthaster numbers occur at the same time in different areas of the world.
- ☐ To identify a problem with a hypothesis that associates the increase in the population of Acanthaster with a removal of its predators.
- ☐ To argue against the hypothesis that human activities cause sudden population outbreaks of Acanthaster in different places at the same time.
- ☐ To argue against the view that the predators of Acanthaster decline because of pollution and harvesting by humans.

A 选项说，阴影句是为了说明“很难证明棘冠海星在同一时间在世界的不同地方数量增长”，即又把阴影句的内容重复了一遍。阴影句的修辞目的显然不可能不是为了证明这句话本身，故 A 排除。C 选项说，阴影句是为了反驳“人类活动导致海星在各地同时数量增长”，但这里并不存在反驳关系，阴影句只是提出了该假说存在的问题，故 C 选项排除。D 选项说，阴影句是为了反驳“污染和捕捞造成了海星天敌数量的减少”，同理，这里不存在反驳关系，故 D 选项排除。

11. Why does the author conclude in paragraph 8 about the causes of sudden population increases of Acanthaster?

- ☐ A sudden population increase in the Acanthaster population could be attributed to natural process as well as human activities.
- ☐ There are no elements of truth in the hypothesis proposed to explain sudden population increases of Acanthaster.
- ☐ It is possible to offer a single explanation for sudden population increases of Acanthaster.
- ☐ Natural processes have been responsible for sudden population increases of Acanthaster in the past while present outbreaks are due to human activities

12. The word "severity" in the passage is closest in meaning to

- ☐ lands
- ☐ efficiency
- ☐ speed
- ☐ seriousness

Paragraph 2:

■ A single Acanthaster can consume five to six square meters of coral polyps per year, and dense populations can destroy up to six square kilometers per year and move on rapidly. ■ Acanthaster show a preference for branching corals, especially Acroporids. ■ After an outbreak in a particular area, it is common to find that Acroporids have been selectively removed, leaving a mosaic of living

and dead corals. ■ In places where Acroporids previously dominated the community devastation can be almost complete, and local areas of reefs have collapsed.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

The crown of thorns starfish is a coral reef predator that preys on coral polyps.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Attempts have been made to explain the sudden increases in Acanthaster populations since the mid-1950s and their impact on coral reefs.

Answer Choices

1. Larvae Acanthaster populations have preyed on coral reefs and caused the loss of coral cover over large areas and the collapse of some coral reef communities.

2. Acanthaster outbreaks have been attributed to natural phenomena such as stroms that have created favorable conditions for the rapid development of the starfish larvae.

3. Acanthaster outbreaks are the results of natural processes that have led to some increases in the past, as well as of human impact that has sped up the frequency and severity of the outbreaks.

4. There are several hypotheses trying to explain the Acanthaster outbreaks but none have elements of the truth because they contradict each other.

5. The crown of thorns starfish, Acanthaster Tlanci, is large, twenty-five to thirty-five centimeters in diameter, and has seven to twenty-one arms that are covered in spines. 选项对应文章第一句话，虽然 E 选项是原文的正确信息，但是是与本文主旨不相关的细节，故排除

6. The favorable anthropogenic causes, such as destruction of coral by blasting and other bad land use practices, have provided Acanthaster more food sources and therefore stimulate its population outbreaks.

The Rise of Moscow

The rise of Moscow during medieval times was a fundamental development in Russian history. Moscow began with very little and for a long time could not be compared to such flourishing principalities as Novgorod or Galicia. Even in its own area, the northeast, it was junior to old centers like Rostov and Suzdal. In accounting for Moscow's rise, historians have emphasized several factors or rather groups of factors.

First, attention may be given to the doctrine of geographic causation. It stresses the decisive importance of the location on Moscow for the later expansion of the Muscovite state (the medieval state centered in Moscow) and includes several lines of argument. Moscow lay as a crossing of three roads. The most important was the way from the historically crucial city of Kiev and the declining south to the growing northeast. In fact Moscow has been described as the first stopping and setting point in the northeast. But it also profited from moments in other directions, including the reverse. Thus it seems immigrants came to Moscow after the Mongol devastation of the lands further to the northeast. Moscow was also situated on a bend of the Moscow River that flows from the northwest to the southeast into the Oka, the largest western tributary of the Volga River. To speak more broadly of water communications which span and unite European Russia, Moscow has the rare fortune of being located near the headwaters of four major rivers: the Oka, the Volga, the Don, and the Dnieper. This offered marvelous opportunities for expansion across the flowing plain, especially as there were no mountains or other natural obstacles to hem in the young principality.

In another sense too, Moscow benefited from a central position. It stood in the midst of lands inhabited by the Russian people which, so the argument runs, provided a proper setting for a natural growth in all directions. In fact some specialists have tried to estimate precisely how close to the geographic center of the Russian people Moscow was situated, noting also such circumstances as proximity to the land dividing the two main dialects of the Great Russian language. Central location within Russia, to make an additional point, cushioned Moscow from outside invaders. Thus, for example, it was the city of Novgorod, not Moscow, that continuously had to meet enemies from the northwest, while in the southeast Riazan absorbed the first blows from the direction. All in all, the considerable significance of the location on Moscow cannot be denied although this geographic factor has generally been assigned less relative weight by recent scholars.

The economic argument is linked in part to the geographic. The Moscow River

served as an important trade artery, and as the Muscovite principality expanded around its waterways, it profited by and in turn helped to promote increasing economic intercourse. One school of thought has treated the expansion of Moscow largely in terms of the growth of a common market. Another economic approach emphasizes the success of the Muscovite princes in developing agriculture in their domains and supporting colonization. These princes clearly outdistance their rivals in obtaining peasants to settle on their lands. As a further advantage, they managed to maintain in their realm a relative peace and security highly beneficial to economic life.

The last view introduces another key factor in explaining the Muscovite rise: the role of the rulers of Moscow. Moscow has generally been considered fortunate in its princes. Sheer luck constituted an important part of the picture. For several generations, the princes of Moscow had the advantage of male succession without interruption or conflict. In particular, for a long time the sons of the princes of Moscow were lucky not to have uncles competing for the Muscovite seat. When the classic power struggle between royal uncles and nephews finally erupted under Basil II(reigned 1425-1462), direct succession from father to son possessed sufficient standing and support in the principality of Moscow to overcome the challenge. The principality has also been considered fortunate because its early rulers, descending from the youngest son of Alexander Nevskii (1220?-1263) and thus representing a junior princely branch, found it expedient to devote themselves to their small holdings instead of neglecting them for more ambitious undertakings elsewhere.

Paragraph 1

The rise of Moscow during medieval times was a fundamental development in Russian history. Moscow began with very little and for a long time could not be compared to such flourishing principalities as Novgorod or Galicia. Even in its own area, the northeast, it was junior to old centers like Rostov and Suzdal. In accounting for Moscow's rise, historians have emphasized several factors or rather groups of factors. 即使是在莫斯科内部，东北地区的发展也不如罗斯托夫、苏兹达尔等老城区中心成熟。

1. The word "flourishing" in the passage is closest in meaning to

- ☐ powerful
- ☐ fortunate
- ☐ neighboring
- ☐ prosperous

Paragraph 2

First, attention may be given to the doctrine of geographic causation. It stresses the decisive importance of the location on Moscow for the later expansion of the

Muscovite state (the medieval state centered in Moscow) and includes several lines of argument. Moscow lay as a crossing of three roads. The most important was the way from the historically crucial city of Kiev and the declining south to the growing northeast. In fact Moscow has been described as the first stopping and setting point in the northeast. But it also profited from moments in other directions, including the reverse. Thus it seems immigrants came to Moscow after the Mongol devastation of the lands further to the northeast. Moscow was also situated on a bend of the Moscow River that flows from the northwest to the southeast into the Oka, the largest western tributary of the Volga River. To speak more broadly of water communications which span and unite European Russia, Moscow has the rare fortune极其幸运 of being located near the headwaters of four major rivers: the Oka, the Volga, the Don, and the Dnieper. This offered marvelous opportunities for expansion across the flowing plain, especially as there were no mountains or other natural obstacles to hem in the young principality. Moscow lay as a crossing of three roads.莫斯科位于三条道路的交叉口

2.The word "doctrine" in the passage is closest in meaning to

- ☐ principle
- ☐ role
- ☐ problem
- ☐ power

3.Why does the author include the information that Moscow has been described as the first stopping and setting point in the northeast?

“In fact Moscow has been described as the first stopping and setting point in the northeast. But it also profited from moments in other directions.....”这句话中，“in fact”说明阴影句是对上文内容的补充说明，而“but”带有转折关系，说明下文会展开新的层次，所以答案一定不在后文中，而是在“in fact”之前。

- ☐ To explain why certain historically important areas near Moscow have fallen into a decline.
- ☐ To show why Moscow attracted populations of immigrants from lines located further to the northeast.
- ☐ To emphasize the importance of the location on Moscow in accounting for its growth.
- ☐ To challenge the idea that immigrants to Moscow came from only one direction.

4. Which of the following can be inferred from paragraph 2 about residents of Moscow during medieval times?

- ☐ Most retained strong cultural and economic ties to Kiev and the South.
- ☐ Some were Mongols who came from areas northeast of Moscow.
- ☐ Some had fled to Moscow to escape violent attacks from the northeast.
- ☐ Most eventually moved on to other locations in the northeast.

5. According to paragraph 2, all of the following were true of Moscow **EXCEPT**

- It was near several important waterways.
- It was surrounded by open areas that were easily crossed.
- It was located at a place where several travel routes came together.
- It was protected from attack by its position in the bend of a river.

Paragraph 3

In another sense too, Moscow benefited from a central position. It stood in the midst of lands inhabited by the Russian people which, so the argument runs提到, provided a proper setting for a natural growth in all directions. In fact some specialists have tried to estimate precisely how close to the geographic center of the Russian people Moscow was situated, noting also such circumstances as proximity to the land dividing the two main dialects of the Great Russian language. Central location within Russia, to make an additional point, cushioned Moscow from outside invaders. Thus, for example, it was the city of Novgorod, not Moscow, that continuously had to meet enemies from the northwest, while in the southeast Riazan absorbed the first blows from the direction. All in all, the considerable significance of the location on Moscow cannot be denied although this geographic factor has generally been assigned less relative weight by recent scholars.

6. The word "cushioned" in the passage is closest in meaning to
- excluded
 - divided
 - distracted
 - protected
7. According to paragraph 3 specialists have tried to determine which of the following
- Why Moscow seems to grow naturally in all directions
 - How close Moscow was to the geographic population center of the Russian people
 - Where the dividing line between the two major dialects of Great Russian was located
 - Which of the two major dialects of Great Russian was more commonly used
8. According to paragraph 3 which of the following best represents the view of recent scholars about the role of geography in Moscow's rise
- Geography cannot account for many aspects of Moscow's rise.
 - The role of geography in Moscow's rise is still not fully understood.
 - Geography was important in Moscow's rise but not as important as early scholars thought it was.

- Although geography played a role in Moscow's rise this role was fairly minor.
- D 选项说，地理因素的作用很小，但原文中说“the considerable significance of the location on Moscow cannot be denied”，所以其实地理因素的作用是很大的，与原文矛盾，故排除

Paragraph 4

The **economic argument** 经济学说是 linked in part to the geographic. The Moscow River served as an important trade artery, and as the Muscovite principality expanded around its waterways, it profited by and in turn helped to promote increasing economic intercourse. **One school of thought** 一种学派 has treated the expansion of Moscow largely in terms of the growth of a common market. Another economic approach emphasizes the success of the Muscovite princes in developing agriculture in their domains and supporting colonization. These princes clearly outdistance their rivals in obtaining peasants to settle on their lands. As a further advantage, they managed to maintain in their realm a relative peace and security highly beneficial to economic life.

9. The word "intercourse" in the passage is closest in meaning to

- **exchange**
- growth
- influence
- independence

“inter”这个词根的意思是“在.....之间；相互”，

10. Which of the following is mentioned in paragraph 4 as a factor in the economic success of the Muscovite principality?

- improvements made to waterways used in trade
- efforts of Muscovite princes to introduce new agricultural methods
- conditions of comparative peace and security on Muscovite lands
- groups of peasants settling land in other areas of Russia

Paragraph 5

The last view introduces another key factor in explaining the Muscovite rise: the role of the rulers of Moscow. Moscow has generally been considered fortunate in its princes. Sheer luck constituted an important part of the picture. For several generations, the princes of Moscow had the advantage of male succession without interruption or conflict. In particular, for a long time the sons of the princes of Moscow were lucky not to have uncles competing for the Muscovite seat. When the classic power struggle between royal uncles and nephews finally erupted under Basil II (reigned 1425-1462), direct succession from father to son possessed sufficient standing and support in the principality of Moscow to overcome the challenge. The principality has also been considered fortunate because its early rulers, descending from the youngest son of Alexander Nevskii (1220?-1263) and thus representing a junior princely branch, found it expedient to devote themselves to their small holdings instead of neglecting

them for more ambitious undertakings elsewhere.

11. According to paragraph 5 why has Moscow generally been considered fortunate in its princes

- ☐ For a long period of time royal succession proceeded peacefully and without interruption from one generation to the next.
- ☐ Through most of Moscow's history royal nephews were unable to take power away from each other.
- ☐ In general Moscow's princes were almost always willing to share their power with their uncles.
- ☐ Over the course of time the Muscovite rules of succession from father to son was relaxed considerably.

12. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

The principality has also been considered fortunate because its early rulers, descending from the youngest son of Alexander Nevskii (1220?-1263) and thus representing a junior princely branch, found it expedient to devote themselves to their small holdings instead of neglecting them for more ambitious undertakings elsewhere. 他们认为相比在别处实现他们的雄心壮志，致力于发展自己所拥有的小片土地才是权宜之计。expedient权宜的

- ☐ It is fortunate that Moscow's early rulers were descended from an unambitious princely branch with small holdings.
- ☐ Moscow benefited from its early rulers' desire to concentrate their energy on the care of their own small properties.
- ☐ Moscow was fortunate in that it traced its rulers back to the youngest son of Alexander Nevskii.
- ☐ Because most of Moscow's early rulers were junior princes their properties were fortunately relatively small and easy to care for.

Paragraph 4:

The economic argument is linked in part to the geographic. ■ The Moscow river served as an important trade artery, and as the Muscovite principality expanded around its waterways, it profited by and in turn helped to promote increasing economic intercourse. ■ One school of thought has treated the expansion of Moscow largely in terms of the growth of a common market. ■ Another economic approach emphasizes the success of the Muscovite princes in developing agriculture in their domains and supporting colonization. ■ These princes clearly outdistance their rivals in obtaining peasants to settle on their lands. As a further advantage, they managed to maintain in their realm a

relative peace and security highly beneficial to economic life.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Scholars have debated the relative importance of various economic factors in the principality's overall success.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Historians point to several factors to account for Moscow's rise during medieval times.

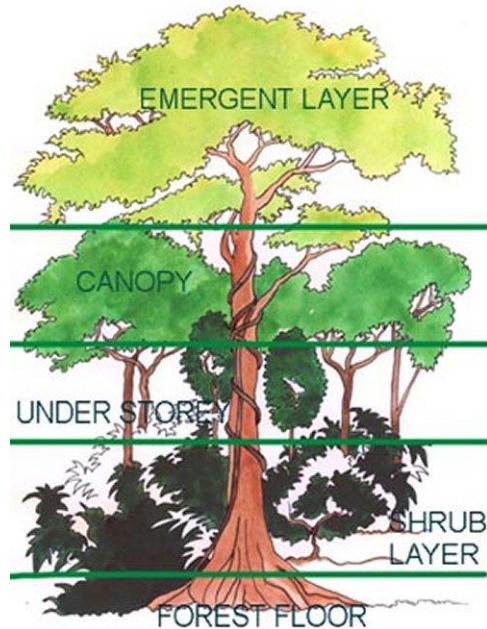
-
-
-

Answer Choices

1. Moscow was located near the geographic center of Russian-speaking peoples and was situated advantageously with respect to enemies, travel, and expansion.
2. Moscow expanded despite its early difficulty in its establishing command of its local waterways.
3. Moscow was politically stable because its early rulers generally succeeded each other steadily and without violence.
4. Moscow enjoyed numerous economic advantages due to expanding trade, successful farming and the relative absence of war.
5. Because some of Moscow's princes were descended from ancestors with relatively small holdings they were slow to improve life for the principality's residents.
6. Because its princes were deeply involved in economic activities such as agriculture and trade Moscow did not experience the power struggle that troubled other principalities.

Forest Succession

Layers of a Forest



Succession is a continuous change in the species composition, structure, and function of a forest through time following disturbance. Each stage of succession is referred to as a successional sere. The final stage of succession, which is generally self-replacing, is referred to as the climax sere. There are two major types of succession: primary and secondary. Primary succession is the establishment of vegetation on bare rocks or radically disturbed soil. Secondary succession is the reestablishment of vegetation following a disturbance that killed or removed the vegetation but did not greatly affect the soil. Volcanic eruptions, retreating glaciers, and bare sand dunes are examples of sites subject to primary succession, while clear-cutting of forests, wild fires, and hurricanes are examples of sites subject to secondary succession. Hundreds to thousands of years are required for primary succession to reach the climax sere, compared to decades to hundreds of years for it to occur in secondary succession. A longer time is needed to reach the climax sere for primary than secondary succession because soil development must first take place in primary succession. The rate of succession is dependent upon the extent of the disturbance and the availability of appropriate seeds for recolonization.

What morphological (structural) and ecophysiological characteristics determine the species composition and abundance in succession? In general, nitrogen fixing plants (plants that can make use of atmospheric nitrogen) are important early succession species in primary succession because nitrogen is not derived from the weathering of rock and little or no organic matter is present in the soil.

Weedy plants are common early successional species because of their rapid growth and high reproductive rates, while stress-tolerant species are common late successional species.

The structure of a forest changes as well in secondary succession. Depending on the type and the severity of the disturbance, a moderate to large amount of dead organic matter from the previous forest remains on the site immediately from the disturbance. The leaf area of the forest is at a minimum and slowly increases as new vegetation occupies the site. Following a disturbance, such as a fire, the new canopy (the uppermost spreading and branching layer of a forest) is largely composed of similar-aged, or even-aged, trees. Light, nutrient, and water availability are highest during the early successional sere because the vegetation has not completely occupied the site. Canopy closure, or maximum leaf area, can occur within several years after disturbance in some tropical forests, but may take three to fifty years in evergreen forests.

In the second stage of forest development there is tree mortality caused by competition for light, nutrients and water. The intense intraspecies (within a species) and interspecies (between species) competition for light, nutrients and water induces the mortality of plants that are shaded or have one or more life-history characteristics that are not well adapted to the changing environment. The third stage of forest development is characterized by openings in the overstory canopy, caused by tree mortality, and the renewed growth of understory in response to increased light reaching the forest floor. Consequently, the forest canopy becomes more complex, or multilayered. The final stage of forest development, the climax or old growth stage, is characterized by a species composition that in theory can continue to replace itself unless a catastrophic disturbance occurs. Unique characteristics of old growth forests include large accumulation of standing and fallen dead trees--referred to as coarse woody debris. Also, the annual input of forest litter is dominated by coarse woody debris compared to the earlier stages of forest development, when leaf and fine root debris were the dominant sources of nutrients and organic matter input into the soil.

Some ecosystems may never reach the latter stages of succession if natural disturbances (fire, flooding, hurricanes, etc) are frequent. A pyric climax refers to an ecosystem that never reaches the potential climax vegetation defined by climate because of frequent fires. The ecotone, a boundary, between grassland and forest is a pyric climax, and only with fire suppression have woodlands and forests began to advance into these regions.

Paragraph 1.

Succession is a continuous change in the species composition, structure, and function of a forest through time following disturbance. Each stage of succession is referred to as a successional sere. The final stage of succession, which is generally self-replacing, is referred to as the climax sere. There are two major types of succession: primary and secondary. Primary succession is the establishment of vegetation on bare rocks or radically disturbed soil. Secondary succession is the reestablishment of vegetation following a disturbance that killed or removed the vegetation but did not greatly affect the soil. Volcanic eruptions, retreating glaciers, and bare sand dunes are examples of sites subject to primary succession, while clear-cutting of forests, wild fires, and hurricanes are examples of sites subject to secondary succession. Hundreds to thousands of years are required for primary succession to reach the climax sere, compared to decades to hundreds of years for it to occur in secondary succession. A longer time is needed to reach the climax sere for primary than secondary succession because soil development must first take place in primary succession. The rate of succession is dependent upon the extent 程度 of the disturbance and the availability of appropriate seeds for recolonization.

1. According to paragraph 1 each of the following can create a site that is likely to be subject to secondary succession **EXCEPT**

- ☐ large scale forest clearances
- ☐ volcanic eruptions
- ☐ hurricanes
- ☐ major forest fires

2. Paragraph 1 supports which of the following statements about disturbances

- ☐ They occur in a series of stages.
- ☐ They can result from natural causes or human activities.
- ☐ They may contribute to increased volcanic eruptions.
- ☐ They have no impact on the speed with which succession occurs.

题干关键词“disturbance”在整段中重复出现，无法通过定位词缩小阅读范围，因此保险起见我们需要阅读整段文字，然后用排除法做题。A 选项说干扰分好几个阶段进行。而文章中只提到“Each stage of succession is referred to as a successional sere.”这句话是说演替的每一阶段都被称为一个演替系列。整段文字完全没有提到 disturbance 分阶段进行，故排除。B 选项说干扰可以由自然因素和人为因素造成。我们定位到这句话“Volcanic eruptions, retreating glaciers, and bare sand dunes are examples of sites subject to primary succession, while clear-cutting of forests, wild fires, and hurricanes are examples of sites subject to secondary succession.”这句话中例举了原生演替和次生演替发生的场所，其中火山爆发、冰川退缩、裸露沙丘、野火、飓风都属于自然因素，而森林砍伐很明显属于人为因素，因此 B 选项正确

3. According to paragraph 1 the main factor that explains why it takes so long for a primary succession to reach climax is the time it takes for

- ☐ glaciers to retreat
- ☐ dead vegetation to decay
- ☐ soil to form
- ☐ seeds to mature after recolonization

Paragraph 2

What **morphological (structural)** and ecophysiological characteristics determine the species composition and abundance in succession? In general,

nitrogen-fixing plants (plants that can make use of atmospheric nitrogen) are important early succession species in primary succession because nitrogen is not derived from the **weathering of the rock**岩石风化和 little or no organic matter is present in the soil. **Weedy**杂草丛生的 plants are common early successional species because of their rapid growth and high reproductive rates, while **stress-tolerant**抗压的 species are common late successional species.

4. Paragraph 2 supports the idea that nitrogen-fixing plant species are heavily represented in the early stages of a primary succession because they

- ☐ do not require soil that contains nitrogen.
- ☐ generally grow more rapidly than other plants.
- ☐ tend to be more stress tolerant than other plants.
- ☐ tend to be less weedy than other plants.

Paragraph 3

The structure of a forest changes as well in secondary succession. Depending on the type and the **severity** of the disturbance, a moderate to large amount of dead organic matter from the previous forest remains on the site immediately from the disturbance. The leaf area of the forest is at a minimum and slowly increases as new vegetation occupies the site. Following a disturbance, such as a fire, the new canopy (the uppermost spreading and branching layer of a forest) is largely composed of similar-aged, or even-aged, trees. Light, nutrient, and water availability are highest during the early successional sere because the vegetation has not completely occupied the site. Canopy closure, or maximum leaf area, can occur within several years after disturbance in some tropical forests, but may take three to fifty years in evergreen forests.

5. The word "**severity**" in the passage is closest in meaning to

- ☐ suddenness
- ☐ seriousness
- ☐ location
- ☐ timing

6. According to paragraph 3 the earliest stage of a secondary succession is characterized by each of the following **EXCEPT**

- ☐ an incomplete covering of vegetation
- ☐ a canopy made up of trees that vary widely in age
- ☐ relatively ample nutrients available in the soil
- ☐ relatively plentiful light reaching growing plants

Paragraph 4

In the second stage of forest development there is tree mortality caused by competition for light, nutrients and water. The intense intraspecies (within a species) and interspecies (between species) competition for light, nutrients and water induces the mortality of plants that are shaded or have one or more life-history characteristics that are not well adapted to the changing environment. The third stage of forest development is characterized by openings in the overstory canopy, caused by tree mortality, and the renewed growth of understory in response to increased light reaching the forest floor. Consequently, the forest canopy becomes more complex, or multilayered. The final stage of forest development, the climax or old growth stage, is characterized by a species composition that in theory can continue to replace itself unless a catastrophic disturbance occurs. Unique characteristics of old growth forests include large accumulation of standing and fallen dead trees--referred to as coarse woody debris. Also, the annual input of forest litter is dominated by coarse woody debris compared to the earlier stages of forest development, when leaf and fine root debris were the dominant sources of nutrients and organic matter input into the soil.

7. The word "induces" in the passage is closest in meaning to

- ☐ explains
- ☐ increases
- ☐ disappears with
- ☐ brings about

8. The word "consequently" in the passage is closest in meaning to

- ☐ in other words
- ☐ nevertheless
- ☐ as a result
- ☐ basically

9. The word "catastrophic" in the passage is closest in meaning to

- ☐ extremely harmful
- ☐ very complex
- ☐ different
- ☐ long-lasting

10. According to paragraph 4 which of the following is a characteristic of old growth forests

- ☐ improved ability to withstand major disturbances
- ☐ increasing amounts of woody debris
- ☐ a decline in the mortality rate of tree species

- increasing reliance on leaf and fine root detritus as a source of nutrients

11. According to paragraph 4 one difference between the second and the third stage in forest development is that in the third stage

- there is more intraspecific competition than interspecific competition.
- the growth of understory and suppressed substory trees slows down.
- sudden changes in the environment are less frequent.
- the canopy becomes more complex.

Paragraph 5

Some ecosystems may never reach the latter stages of succession if natural disturbances (fire, flooding, hurricanes, etc) are frequent. A pyric climax refers to an ecosystem that never reaches the potential climax vegetation defined by climate because of frequent fires. The ecotone, a boundary, between grassland and forest is a pyric climax, and only with fire suppression have woodlands and forests begun to advance into these regions.

12. Why does the author discuss pyric climax in the passage?

- To help explain the idea that a forest may not necessarily reach the old growth stage.
- To suggest that fire is a natural disturbance least likely to prevent an ecosystem from reaching its potential climax stage.
- To acknowledge that the pattern of succession following a disturbance is a very different in ecosystems such as grasslands than it is in forests.
- To help explain how the boundary areas of an ecosystem affect the way its structure changes during succession.

Paragraph 1:

Succession is a continuous change in the species composition, structure, and function of a forest through time following disturbance. Each stage of succession is referred to as a successional sere. The final stage of succession, which is generally self-replacing, is referred to as the climax sere. There are two major types of succession: primary and secondary. Primary succession is the establishment of vegetation on bare rocks or radically disturbed soil. Secondary succession is the reestablishment of vegetation following a disturbance that killed or removed the vegetation but did not greatly affect the soil. Volcanic eruptions, retreating glaciers, and bare sand dunes are examples of sites subject to primary succession, while clear-cutting of forests, wild fires, and hurricanes are examples of sites subject to secondary succession. ■ Hundreds to thousands of years are required for primary succession to reach the climax sere, compared to decades to hundreds of years for it to occur in secondary

succession. ■ A longer time is needed to reach the climax sere for primary than secondary succession because soil development must first take place in primary succession. ■ The rate of succession is dependent upon the extent of the disturbance and the availability of appropriate seeds for recolonization. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Tree species that have small, light seeds that are dispersed by wind or transported by animals recolonize a disturbed area more quickly than species with large seeds.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Succession is a continuous change in the species composition, structure, and function of a forest through time following disturbances.

-
-
-

Answer Choices

1. Primary succession occurs at sites where soil must be developed and thus takes a far longer to complete than secondary succession, which occurs where relatively undisturbed soil already exists.
2. The second stage of forest development is much shorter in boreal forests than it is in tropical forests.
3. Old growth forest differs from earlier succession stages in that most soil nutrients come from leaves and fine root debris rather than from dead trees and other coarse woody debris.
4. Early in secondary succession resources are relatively abundant and vegetation increases until canopy closure, after which competition for resources brings about increased plant mortality.
5. With a rising tree mortality, openings in the canopy develop, leading to layered plant growth beneath the overstory canopy, and eventually a climax stage is reached.
6. The effective suppression of grassland wild fires has provided opportunities for the ecotone to proceed into a forest.

2013-01-12

Passage 4 2013-01-12-1

England's Economy in the Sixteenth Century

In the last half of the sixteenth century England emerged as a commercial and manufacturing power in Europe due to a combination of demographic, agricultural and industrial factors. The population of England and Wales grew rapidly from about 2.5 million in the 1520s to more than 3.5 million in 1580, reaching about 4.5 million in 1610. Reduced mortality rates and increased fertility, the latter probably generated by expanding work opportunities in manufacturing and farming (leading to earlier marriage and more children), explained this rapid rise in population. While epidemics and plague occasionally took their toll, the people in England still suffered less than did those in continental Europe. Furthermore, the country had been pulled out of the war that occurred in France and central Europe during the same period.

England provides the prominent example of the expansion of agricultural production well before the general European agricultural revolution of the eighteenth and nineteenth centuries. A larger population stimulated the increased woollen through crop civilization. English agriculture became more efficient and market-oriented than almost anywhere else on the continent. Between 1450 and 1640 the yield of grain per acre increased by at least thirty percent. In sharp contrast with farming in Spain, English land owners brought more dense marshes and woodlands into cultivation.

The great land estates of the English society largely remained intact and many wealthy land owners aggressively increased the size of their holdings, a precondition for increased productivity. Marriages between the children of landowners also increased the size of land estates. Primogeniture (the full inheritance of land by the eldest son) helped prevent land from being subdivided. Younger sons of independent land owners left the family and went to find other respective locations. Larger farms contributed more to commercialized farming at the time when an expanding population pushed up demand and prices. Farmland owners turned part of their land into pasture land for sheep in order to adapt to developing woollen trade.

Some of the great land owners as well as Yeomen (farmers whose holdings and security of land tenure guaranteed their prosperity and status), organized their holdings in the interest efficiency. Many farmers selected crops for sales in growing London market. In their quest for greater profits, many land owners put their squeeze on their tenants. Between 1580 and 1620 land lords raised rents and altered conditions of land tenure in their favor, preferring shorter phases and forcing tenants to pay an entry fee before agreeing to rent them land. Landlords evicted those who could not afford annual, more

onerous terms. But they also pushed tenants toward more productive farming methods, including crop rotation.

England's exceptional economic development also drew the country's natural resources, including iron, timber, and coal, extracted in far greater quantity than elsewhere in the continent. New industrial development expanded the production of iron and pewter in and around the city of Birmingham.

But above all textile manufacturing transformed English economy. Woolens, which accounted for eighty percent of the exports, worsteds (sturdy yarn spun from combed wool fibers), and other cloth found eager buyers in England as well as in the continent. Moreover, late in the sixteenth century as English merchants began making forays across the Atlantic these textiles were also sold in the Americas. Cloth manufacturers undercut production by urban craftspeople by "putting out" work to the villages and farms of the countryside. In such domestic industry poor rural women could spin and make cading (combing fibers in preparation for spin) in their homes.

The English textile trade was closely tied to Antwerp, in the Spanish Netherlands, where workers dyed English cloth. The entrepreneur Sir Thomas Gresham became England's representative there. He so enhanced the reputation of English business in that region that English merchants could operate on credit---the most prominent achievement for sixteenth century. He also advised the government to explore the economic possibilities of Americas, which led to the first concerted efforts at colonization, undertaken with commercial profits in mind.

Paragraph 1

In the last half of the sixteenth century England emerged as a commercial and manufacturing power in Europe due to a combination of demographic, agricultural and industrial factors. The population of England and Wales grew rapidly from about 2.5 million in the 1520s to more than 3.5 million in 1580, reaching about 4.5 million in 1610. Reduced mortality rates and increased fertility, the latter probably **generated** by expanding work opportunities in manufacturing and farming (leading to earlier marriage and more children), explained this rapid rise in population. While epidemics and plague occasionally took their toll, the people in England still suffered less than did those in continental Europe. Furthermore, the country had been pulled out of the war that occurred in France and central Europe during the same period.

1. The word "generated" in the passage is closest in meaning to
Produced
strengthened
followed
dominated

2. The word "furthermore" in the passage is closest in meaning to

however
in addition
similarly
in general

3. All of the following are mentioned in Paragraph 1 as developments that led to England's emergence as a commercial and manufacturing power **EXCEPT**

- ☐ rapid population growth
- ☐ an increase in the number of jobs
- ☐ a trend toward earlier marriages
- ☐ an end to epidemics and a plague

Paragraph 2

England provides the prominent example of the expansion of agricultural production well before the general European agricultural revolution of the eighteenth and nineteenth centuries. A larger population stimulated the increased woollen through crop civilization. English agriculture became more efficient and market-oriented than almost anywhere else on the continent. Between 1450 and 1640 the yield of grain per acre increased by at least thirty percent. In sharp contrast with farming in Spain, English land owners brought more dense marshes and woodlands into cultivation.

4. Why does the author mention that “English land owners brought more dense marshes and woodlands into cultivation” ?

- ☐ To give an example of a way in which the English increased agricultural production
- ☐ To emphasize the influence of English agriculture on the agriculture of other European countries
- ☐ To emphasize that Spanish agriculture needed to improve their farming methods
- ☐ To discuss an intended consequence of crop specialization

Paragraph 3

The great land estates of the English society largely remained intact and many wealthy land owners aggressively increased the size of their holdings, a precondition for increased productivity. Marriages between the children of landowners also increased the size of land estates. Primogeniture (the full inheritance of land by the eldest son) helped prevent land from being subdivided. Younger sons of independent land owners left the family and went to find other respective locations. Larger farms were conducive more to commercialized farming at the time when an expanding population pushed up demand and prices. Farmland owners turned part of their land into pasture land for sheep in order to adapt to developing woollen trade.

5. The phrase "conducive to" in the passage is closest in meaning to
- ☐ a result of
 - ☐ favorable to
 - ☐ able to
 - ☐ expanded to
6. According to paragraph 3 why did wealthy land owners increase the size of the land they owned?
- ☐ To avoid the necessity of arranged marriages between their children and the children of other land owners
 - ☐ To ensure that there were enough land to divide between their older and younger sons
 - ☐ To increase the number of goods that their farms produced
 - ☐ To provide more jobs for members of the colony
 - ☐
7. Why does the author mention that farmland owners turned part of their land into pasture land for sheep ?
- ☐ To explain why the younger sons of land owners had to leave the family land and find other jobs
 - ☐ To provide an example of a way in which large farms were eager to improve their business
 - ☐ To explain how the practice of primogeniture worked
 - ☐ To describe a strategy land owners used to give their lands from being subdivided.

Paragraph 4

Some of the great land owners as well as Yeomen (farmers whose holdings and security of land tenure guaranteed their prosperity and status), organized their holdings in the interest efficiency. Many farmers selected crops for sales in growing London market. In their quest for greater profits, many land owners put their squeeze on their tenants. Between 1580 and 1620 land lords raised rents and altered conditions of land tenure in their favor, preferring shorter phases and forcing tenants to pay an entry fee before agreeing to rent them land. Landlords evicted those who could not afford annual, more onerous terms. But they also pushed tenants toward more productive farming methods, including crop rotation.

8. The word "quest" in the passage is closest in meaning to
- ☐ organization
 - ☐ exchange
 - ☐ search
 - ☐ argument
9. Which of the following is **NOT** mentioned in paragraph 4 as a way that

English landlords tried to increase the efficiency of their farms?

- ☐ encouraging the use of methods as crop rotation
- ☐ selecting crops on the basis of demand in the London market
- ☐ increasing rents and reducing the length of phases
- ☐ hiring Yeomen to work on their farms instead of relying on tenants

Paragraph 5

England's exceptional economic development also drew the country's natural resources, including iron, timber, and coal, extracted in far greater quantity than elsewhere in the continent. New industrial development expanded the production of iron and pewter in and around the city of Birmingham.

10. The word "exceptional " in the passage is closest in meaning to

- ☐ predictable
- ☐ initial
- ☐ extraordinary
- ☐ complex

Paragraph 6

But above all textile manufacturing transformed English economy. Woolens, which accounted for eighty percent of the exports, worsteds (sturdy yarn spun from combed wool fibers), and other cloth found eager buyers in England as well as in the continent. Moreover, late in the sixteenth century as English merchants began making forays across the Atlantic these textiles were also sold in the Americas. Cloth manufacturers undercut production by urban craftspeople by "putting out" work to the villages and farms of the countryside. In such domestic industry poor rural women could spin and make cading (combing fibers in preparation for spin) in their homes.

11. Select TWO answer choices that according to paragraph 6 indicate changes that occurred in the English economy.

To receive credit you must select two answer choices.

- ☐ The textile trade products became more profitable than manufacturing them.
- ☐ Local sales of textile products increased by eighty percent.
- ☐ English textile merchants expanded their markets inside and outside England.
- ☐ Part of English cloth manufacturing moved to rural areas.

Paragraph 7

The English textile trade was closely dyed to Antwerp, in the Spanish Netherlands, where workers died English cloth. ■ The entrepreneur Sir Thomas Gresham became England's representative there. ■ He so enhanced the reputation of English business in that region that English merchants could

operate on credit---the most prominent achievement for sixteenth century.
 ■ He also advised the government to explore the economic possibilities of Americas, which led to the first concerted efforts at colonization, undertaken with commercial profits in mind. ■

12. Which of the following statements about merchants during the sixteenth century can be inferred from the information in paragraph 7?

- ☐ Most merchant activity at this time was controlled by Spain.
- ☐ The textile market was less profitable for merchants than were other areas of trade.
- ☐ Merchants from different countries in Europe rarely operated in the same regions.
- ☐ During this period most European merchants did not operate on credit.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

In addition to this achievement in Antwerp, Gresham convinced the government at home in England to authorize actions that would make trading in the rest of Europe even more profitable for English merchants.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

In the last half of sixteenth century England emerged as a commercial and manufacturing power in Europe.

-
-
-

Answer Choices

1. It is somewhat surprising that England was able to expand its economy during the sixteenth century because its neighbors were involved in many wars during this period.
2. In order to make more money England nobles acquired more land, pushed for more efficient farming methods, and introduced new land tenure conditions.
3. Two England's economic strengths are its use of natural resources, and its

textile manufacturing, which found increased markets at home as well in Europe and in the Americas.

4. England experienced an agricultural revolution much earlier than the rest of Europe and agriculture became more efficient and market-oriented.
5. Many of the changes that land owners made to land renting rules not only made their lands more profitable but also made a life for their peasants.
6. Though he was a successful ambassador to the Spanish Netherlands, Thomas Gresham was unable to convince the English government to start colonies in America.



Documenting the Incas

The Incans ruled a vast empire in western South America when the Spaniards encountered them in the sixteenth century. Although the Incas had no writing system of their own, historical information about Incas is available to researchers because early Spaniards wrote documents about them. However, there are drawbacks to use the written record. First, the Spanish writers were describing activities and institutions that were very different from their own, but they often described Inca culture in terms of their own society. As an example, consider the list of kings given by the Incas. As presented in the historical chronology, Spanish sources indicate there were thirteen kings who ruled sequentially. The names were given to them by Inca informants. However, one school of thought in Inca studies suggests that the names were not actual people, but, rather, titles filled by different individuals. Thus, the number of actual kings may have been fewer, and several titles may have been filled at the same time. The early Spanish writers, being unfamiliar with such a system of titles, simply translated it into something they were familiar with (a succession of kings). Given that the Inca empire expanded only during the time of the last four kings, or as a result of the actions of the individuals in those four positions, this question is not deemed significant for an understanding of the Incas. But the example shows that biases and inaccuracies may have been introduced inadvertently from the very beginning of the written Spanish reports about the Incas. Moreover, early writers often copied information from each other---so misinformation was likely to be passed on and accepted as true by later scholars.

Second, both Spanish writers and Incan informants sometimes had motives for being deliberately deceitful. For example, in an effort to gain status in the Spaniards' eyes, Incas might say that they formerly had been more important in the Inca empire than they actually were. Spanish officials as well were occasionally untruthful when it served their purposes. For example, Spaniards might deliberately underreport the productivity of a region under their authority so they could sell the additional products and keep the money, rather than hand it over to the Spanish Crown.

Third, it should be noted that the Spaniards' main sources of information were the Incas themselves, often members of the Inca ruling class. Therefore, what was recorded was the Incas' point of view about their own history and empire. Some modern authorities question whether the history of Incas happened as they said it did. Although some of their history is certainly more myth than truth, many, if not most, scholars agree that the history of the last four Inca kings is probably accurate. The same is true of other things told to the Spanish

writers: the more recently an event is said to have occurred, the more likely it is to have actually happened.

A fourth problem relates to the nature of the Inca conquests of the other people in the Americas before the Spanish arrived and how accurate the accounts of those conquests are---whether related by the Spaniards or by the Incas on whom they relied. It was certainly in the Inca's interest to describe themselves as invincible and just. However, lacking accounts by conquered people about their interactions with the Incas, it is unknown how much of the information of the Inca conquest as related by the ruling class is factual.

Finally, there is a certain vagueness in the historical record regarding places and names. Many Spanish writers listed places they had visited within the empire, including both provinces and towns. However, other writers traveling along the same routes sometimes recounted different lists of places. In addition, it is difficult to identify the exact locations of towns and other geographic points of reference because of the widespread movements of people over the past five centuries.

For all these reasons, the historical record must be carefully evaluated to determine whether it is accurate and to verify the locations of past events. One approach is to cross-check information from a number of authors. Another approach is to conduct archaeological research. Regardless of the problems, historical documents review some important information about the Incas.

Paragraph 1

The Incans ruled a vast empire in western South America when the Spaniards encountered them in the sixteenth century. Although the Incas had no writing system of their own, historical information about Incas is available to researchers because early Spaniards wrote documents about them. However, there are drawbacks to use the written record. First, the Spanish writers were describing activities and institutions 【that were very different from their own】 , but they often described Inca culture in terms of their own society. As an example, consider the list of kings given by the Incas. As presented in the historical chronology, Spanish sources indicate there were thirteen kings who ruled sequentially. The names were given to them by Inca informants. However, one school of thought in Inca studies suggests that the names were not actual people, but, rather, titles filled by different individuals. Thus, the number of actual kings may have been fewer, and several titles may have been filled at the same time. The early Spanish writers, being unfamiliar with such a system of titles, simply translated it into something they were familiar with (a succession of kings). Given that the Inca empire expanded only during the time of the last four kings, or as a result of the actions of the individuals in

those four positions, this question is not deemed significant for an understanding of the Incas. But the example shows that biases and inaccuracies may have been introduced inadvertently from the very beginning of the written Spanish reports about the Incas. Moreover, early writers often copied information from each other---so misinformation was likely to be passed on and accepted as true by later scholars.

1. The word "vast" in the passage is closest in meaning to
ancient
wealthy
powerful
extensive

2. The word "drawbacks" in the passage is closest in meaning to
requirements
techniques
exceptions
disadvantages

3. The word "sequentially" in the passage is closest in meaning to
Secretly
one after another
formerly
in partnership with each other

4. According to paragraph 1, why does the written record about the Incan civilization depend on the reports of Spaniards?

- ☐ The Incas destroyed their written records to prevent the Spaniards from benefiting from them.
- ☐ The Incas did not have a writing system.
- ☐ The Spaniards destroyed all records written by the Incas.
- ☐ Incan records were written on materials that do not preserve well.

5. According to paragraph 1, sixteenth century's Spaniards may have been incorrect about which of the following aspects of the Incan empire

- ☐ its size during the reigns of the last four kings
- ☐ the total number of Incan kings
- ☐ the kinds of power that Incan kings were able to exercise
- ☐ the extent to which Incan kings were able to control activities within their empire

6. Which of the following can be inferred from paragraph 1 about the Incan system of rulers?

- ☐ It was first introduced when the expansion of the Incan empire began.

- It required that multi-rulers share a particular title at the same time.
- It was sometimes confusing to the Incan informants of the Spaniards.
- It was unlike the system used in sixteenth century Spain.

Paragraph 2

Second, both Spanish writers and Incan informants sometimes had motives for being deliberately deceitful. For example, in an effort to gain status in the Spaniards' eyes, Incas might say that they formerly had been more important in the Inca empire than they actually were. Spanish officials as well were occasionally untruthful when it served their purposes. For example, Spaniards might deliberately underreport the productivity of a region under their authority so they could sell the additional products and keep the money, rather than hand it over to the Spanish Crown.

7. Which of the following is mentioned in paragraph 2 as a possible motive for deliberate inaccuracy in official Spanish reports of the Incas?

- The desire of some Spanish officials to appear more important than they really were
- The need to please Spanish rulers by making productivity seem greater than it really was
- The desire of the Incas to make their empire seem more successful than it really was
- The desire of most Spanish officials to enrich themselves

Paragraph 3

Third, it should be noted that the Spaniards' main sources of information were the Incas themselves, often members of the Inca ruling class. Therefore, what was recorded was the Incas' point of view about their own history and empire. Some modern authorities question whether the history of Incas happened as they said it did. Although some of their history is certainly more myth than truth, many, if not most, scholars agree that the history of the last four Inca kings is probably accurate. The same is true of other things told to the Spanish writers: the more recently an event is said to have occurred, the more likely it is to have actually happened.

8. Why does the author indicate that the Spaniards' main sources of information were the Incas themselves

- To argue that the Spaniards made great efforts to obtain the most information
- To explain why some scholars think that the documentary history of the Incan empire may not be correct
- To question the idea that more recent events in the Incan empire are more likely to be accurate than are more ancient ones
- To explain how scholars are able to determine that the history of the last

four Incan kings is probably correct

Paragraph 4

A fourth problem relates to the nature of the Inca conquests of the other people in the Americas before the Spanish arrived and how accurate the accounts of those conquests are---whether related by the Spaniards or by the Incas on whom they relied. It was certainly in the Inca's interest to describe themselves as invincible and just. However, lacking accounts by conquered people about their interactions with the Incas, it is unknown how much of the information of the Inca conquest as related by the ruling class is factual.

9. The word "just" in the passage is closest in meaning to
- ☐ fair
 - ☐ strong
 - ☐ independent
 - ☐ wise
10. According to paragraph 4, why is there some doubt whether Spanish accounts of the Incan conquests of other people are accurate?
- ☐ The Spaniards included some information about which Incan informants disagreed.
 - ☐ The conquered people's tales of the Incan conquests sometimes differed from the Inca's tales of them
 - ☐ The Spanish accounts of the Incan conquest were based on information from Incan informants only.
 - ☐ Some archaeological evidence does not support the Spanish accounts.

Paragraph 5

Finally, there is a certain vagueness in the historical record regarding places and names. Many Spanish writers listed places they had visited within the empire, including both provinces and towns. However, other writers traveling along the same routes sometimes recounted different lists of places. In addition, it is difficult to identify the exact locations of towns and other geographic points of reference because of the widespread movements of people over the past five centuries.

11. According to paragraph 5, what is a reason that place names in the historical record are sometimes vague?
- ☐ When people moved they often gave new provinces and towns the same names as places they came from.
 - ☐ Different writers in historical record listed different names for places along the same routes.
 - ☐ Most writers of the historical record traveled little within the empire.
 - ☐ The names of provinces and towns often did not reflect geographic points

of reference.

Paragraph 6

For all these reasons, the historical record must be carefully evaluated to determine whether it is accurate and to verify the locations of past events. One approach is to cross-check information from a number of authors. Another approach is to conduct archaeological research. Regardless of the problems, historical documents review some important information about the Incas.

12. Which of the following is mentioned in paragraph 6 as a method for verifying accuracy of an account of a past event?

- ☐ Comparing accounts of the event given by different writers
- ☐ Investigating whether a particular author's accounts of other events have proved accurate
- ☐ Investigating whether the author of an account actually presented as the event described.
- ☐ Determining whether the author of an account was able to crosscheck information with multiple informants

Paragraph 1

The Incans ruled a vast empire in western South America when the Spaniards encountered them in the sixteenth century. ■ Although the Incas had no writing system of their own, historical information about Incas is available to researchers because early Spaniards wrote documents about them. ■ However, there are drawbacks to use the written record. ■ First, the Spanish writers were describing activities and institutions that were very different from their own, but they often described Inca culture in terms of their own society. ■ As an example, consider the list of kings given by the Incas. As presented in the historical chronology, Spanish sources indicate there were thirteen kings who ruled sequentially. The names were given to them by Inca informants. However, one school of thought in Inca studies suggests that the names were not actual people, but, rather, titles filled by different individuals. Thus, the number of actual kings may have been fewer, and several titles may have been filled at the same time. The early Spanish writers, being unfamiliar with such a system of titles, simply translated it into something they were familiar with (a succession of kings). Given that the Inca empire expanded only during the time of the last four kings, or as a result of the actions of the individuals in those four positions, this question is not deemed significant for an understanding of the Incas. But the example shows that biases and inaccuracies may have been introduced inadvertently from the very beginning of the written Spanish reports about the Incas. Moreover, early writers often copied information from each other---so misinformation was likely to be passed on and accepted as true by later scholars.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Official accounts, legal papers, and letters written during that period provide a wealth of information about the Incan empire and the Incas' way of life.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

The accuracy of the historical record concerning the Incas is problematic.

1. Numerous aspects of Incan life were recorded in the sixteenth century but many historical documents created at this time were lost or destroyed.
2. Many Spaniards were interested primarily in the expansion of the Incan empire and therefore ignored other periods of Incan history.
- Mostly, — principally,
3. The fact 【that the Incan informants remained sources of information about themselves】 as well as inconsistencies in place names call the sound aspects of the accounts into question.
4. Descriptions of Incan society may often be influenced by the cultural biases of people writing about Incas.
5. Incorrect information may sometimes have been knowingly included in historical records because it was advantageous to the person including such information.
6. The movement of people due to the Incan conquest and the settlement of the Spanish has caused much confusion about place names in the historical record.

What Controls Flowering

The timing of flowering and seed production is precisely tuned to a plant's physiology and the rigors of its environment. In temperate climate plants must flower early enough so that their seeds can mature before the deadly winds of autumn. Depending on how quickly the seed and food develop flowering may occur in spring as it does in oaks; in summer as in lettuces; or even in autumn as in asters.

What environmental cues do plants use to determine the seasons? Most cues such as temperature or water availability are quite variable: autumn can be warm; a late snow could fall in spring; also summer might be unusually cool and wet. So the only reliable cue is day length: longer days always mean that spring and summer are coming; shorter days foretell the onset of autumn and winter.

With respect to flowering botanists classify plants as day neutral, long day or short day. A day neutral plant flowers as soon as it has sufficiently grown and developed regardless of the length of day. The neutral plants include tomatoes, corn, snapdragons and roses. Although the naming is traditional, long day and short day plants are better described as short night and long night plants because their flowering actually depends on the duration of continuous darkness rather than on day length. Short night plants (which include lettuces, spinach, iris, clover and petunias) flower when the length of darkness is shorter than a species' specific critical period. Long night plants (including asters, potatoes, soy beans, goldenrod and cockleburs) flower when the length of uninterrupted darkness is longer than the species' specific critical period. Thus spinach is classified as a short night plant because it flowers only if the night is shorter than eleven hours (its critical period), and the cocklebur is a long night plant because it flowers only if an uninterrupted darkness lasts more than 8.5 hours. Both of these plants will flower with ten-hour nights.

Plant scientists can induce flowering

Plant scientists can induce flowering in the cocklebur by exposing leaves to long nights (longer than its 8.5 hour critical period) in a special chamber, while the rest of the plant continues to experience short nights. Clearly, a signal that induces flowering transmitted from the leaf to the flowering bud. Plant physiologists have been attempting for decades to isolate these elusive signaling molecule often called florigen (literally, flowering maker). Some researchers believe they are close to demonstrating a flower's stimulating substance for specific type of plant. Using genetic manipulation, it is likely, however, that interactions among multiple and yet unidentified plant hormones stimulate or inhibit flowering, and that these chemicals may differ among plant species. Researchers have had more success in determining how plants measure the length of uninterrupted darkness, which is a crucial

stimulus for producing whatever substance control flowering.

To measure continuous darkness, a plant needs two things: some sort of metabolic clock to measure time (the duration of darkness) and a light detecting system to set the clock. Virtually all organisms have an internal biological clock that measures the time even without environmental cues. In most organisms including plants, the biological clock is poorly understood, but we know that the environmental cues, particularly light, can reset the clock. How do plants detect light? The light detecting system of plants is a pigment in leaves called phytochrome (literally, plant color).

Plants seem to use the phytochrome system in combination with their internal biological clocks to detect the duration of continuous darkness. Cockleburs, for example, flower under the schedule of sixteen hours of darkness and eight hours of light. However, interrupting the middle of the dark period with just a minute or two of lights prevents flowering. Thus their flowering is controlled by the length of continuous darkness. It is evident that even brief exposure to sunlight or white light will reset their biological clocks. The color of the light used for the light exposure is also important. A nighttime flash of pure red light inhibits flowering, while flash of light at the far-red end of the spectrum has no effect on flowering, as if no light were detected.

Paragraph 1

The timing of flowering and seed production is precisely tuned to a plant's physiology and the rigors of its environment. In temperate climate plants must flower early enough so that their seeds can mature before the deadly winds of autumn. Depending on how quickly the seed and food develop, flowering may occur in spring as it does in oaks; in summer as in lettuces; or even in autumn as in asters.

1. The word "rigors" in the passage is closest in meaning to
 - ☐ characteristics
 - ☐ advantages
 - ☐ stability
 - ☐ difficulties
2. The word "precisely" in the passage is closest in meaning to
 - ☐ severely
 - ☐ quickly
 - ☐ exactly
 - ☐ efficiently
3. Paragraph 1 suggests which of the following are the seeds of asters in temperate climate?
 - ☐ They mature a short time after the plant blooms.
 - ☐ They begin to develop in autumn and mature in winter.
 - ☐ They were more plentiful than those of oaks or lettuces.

- They are not designed to survive temperate climate.

Paragraph 2

What environmental cues do plants use to determine the seasons? Most cues such as temperature or water availability are quite variable: autumn can be warm; a late snow could fall in spring; also summer might be unusually cool and wet. So the only reliable cue is day length: longer days always mean that spring and summer are coming; shorter days foretell the onset of autumn and winter.

Paragraph 3

With respect to flowering botanists classify plants as day neutral, long day or short day. A day neutral plant flowers as soon as it has sufficiently grown and developed regardless of the length of day. The neutral plants include tomatoes, corn, snapdragons and roses. Although the naming is traditional, long day and short day plants are better described as short night and long night plants because their flowering actually depends on the duration of continuous darkness rather than on day length. Short night plants (which include lettuces, spinach, iris, clover and petunias) flower when the length of darkness is shorter than a species' specific critical period. Long night plants (including asters, potatoes, soy beans, goldenrod and cockleburs) flower when the length of uninterrupted darkness is longer than the species' specific critical period. Thus spinach is classified as a short night plant because it flowers only if the night is shorter than eleven hours (its critical period), and the cocklebur is a long night plant because it flowers only if an uninterrupted darkness lasts more than 8.5 hours. Both of these plants will flower with ten-hour nights.

4. The author mentions "tomatoes, corn, snapdragons and roses" in order to
- Name some of the most well-developed and commonly grown domestic plants
 - Provide examples of plants whose flowering is not determined by length of day
 - Demonstrate that the neutral plants are more common than long-day or short-day plants
 - Support the claim that some plants flower at night as well as during the day

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

Although the naming is traditional, long day and short day plants are better described as short night and long night plants because their flowering actually depends on the duration of continuous darkness rather than on day length.

- Traditionally, long day and short day plants are better known as short night and long night plants because they actually flower during periods of continuous darkness.
- The naming of long day and short day plants is traditional and depends on whether they flower during long days or long nights.
- Whether plant is better described in terms of day length or night depends on whether it flowers during the day or during continuous darkness.
- Despite their traditional names, the flowering of what are called long day and short day plants actually depends on the length of continuous darkness rather than on day length.

6. According to paragraph 3, why is spinach considered a short night plant?

- It will not flower if it experiences too many hours of uninterrupted darkness.
- It flowers after certain number of short nights.
- It flowers as soon as the nights become shorter than the days.
- It flowers for only a short period of time usually at night.

Paragraph 4

Plant scientists can induce flowering in the cocklebur by exposing leaves to long nights (longer than its 8.5 hour critical period) in a special chamber, while the rest of the plant continues to experience short nights. Clearly, a signal 【that induces flowering】 transmitted from the leave to the flowering bud. Plant physiologists have been attempting for decades to isolate these elusive signaling molecule often called florigen (literally, flowering maker). Some researchers believe they are close to demonstrating a flower's stimulating substance for specific type of plant. Using genetic manipulation, it is likely, however, that interactions among multiple and yet unidentified plant hormones stimulate or inhibit flowering, and that these chemicals may differ among plant species. Researchers have had more success in determining how plants measure the length of uninterrupted darkness, which is a crucial stimulus for producing whatever substance control flowering.

7. The word "induce" in the passage is closest in meaning to

- bring about
- prevent
- settle down
- control

8. According to paragraph 4 what had experiments with cockleburs revealed about the flowering process?

- Flowering is triggered not by multiple chemicals but by a single hormone in the bud.
- The entire plant must experience the crucial period of darkness in order to

flower.

- The signal to begin flowering can originate in the leaves.
- Flowering is not always related to length of darkness.

9. According to paragraph 4, which of the following is true of the signal for the cocklebur to flower?

- Only certain leaves produced signal.
- The signal must arise from the flower bud.
- The signaling molecule has not been chemically identified.
- The length of uninterrupted darkness cockleburs need in order to signal varies a lot.

10. The word "inhibit" in the passage is closest in meaning to

- speed up
- restrain
- contribute to
- prolong

Paragraph 5

To measure continuous darkness, a plant needs two things: some sort of metabolic clock to measure time (the duration of darkness) and a light detecting system to set the clock. Virtually all organisms have an internal biological clock that measures the time even without environmental cues. In most organisms including plants, the biological clock is poorly understood, but we know that the environmental cues, particularly light, can reset the clock. How do plants detect light? The light detecting system of plants is a pigment in leaves called phytochrome (literally, plant color).

11. According to paragraph 5, what is the function of plants' phytochrome?

- measuring time durations in the absence of environmental cues
- helping reset the plant's biological clock through light detection
- stimulating the growth for plants' leaves
- determining the color of the plants' leaves

Paragraph 6

Plants seem to use the phytochrome system in combination with their internal biological clocks to detect the duration of continuous darkness. Cockleburs, for example, flower under the schedule of sixteen hours of darkness and eight hours of light. However, interrupting the middle of the dark period with just a minute or two of lights prevents flowering. Thus their flowering is controlled by the length of continuous darkness. It is evident that even brief exposure to sunlight or white light will reset their biological clocks. The color of the light used for the light exposure is also important. A nighttime flash of pure red light inhibits flowering, while flash of light at the far-red end of the spectrum

has no effect on flowering, as if no light were detected.

12. According to paragraph 6, which of the following types of light will not reset a cocklebur's biological clock?

- ☐ White light
- ☐ sunlight
- ☐ pure red light
- ☐ far-red light

Paragraph 6

Plants seem to use the phytochrome system in combination with their internal biological clocks to detect the duration of continuous darkness. Cockleburs, for example, flower under the schedule of sixteen hours of darkness and eight hours of light. However, interrupting the middle of the dark period with just a minute or two of lights prevents flowering. ■ Thus their flowering is controlled by the length of continuous darkness. ■ It is evident that even brief exposure to sunlight or white light will reset their biological clocks. ■ The color of the light used for the light exposure is also important. A nighttime flash of pure red light inhibits flowering, while flash of light at the far-red end of the spectrum has no effect on flowering, as if no light were detected.■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

But not all types of light will have this effect.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Attempts have been made to explain what controls the flowering.

-
-
-

Answer Choices

1. Day neutral plants flower as soon as they have sufficiently developed but the flowering of short night and long night plants is dependent on the duration of

continuous darkness.

2. Scientists do not yet know what chemical or combination of chemicals is responsible for the signal to flower, but they do know that the signal travels from leaf to bud.

3. Plants measure darkness through an internal biological clock which is reset when the plant's phytochrome detects certain types of light.

4. Short night plants flower if any part of the plant is exposed to continuous darkness, but long night plants will not flower if even one leaf experiences a flash of light.

5. Day neutral plants use the phytochrome system to detect environmental conditions other than light, which may affect the viability of their flowers and seeds.

6. Once a flower has started to bloom, the color and duration of light it is exposed to is important, because some kind of light will prevent flowering while others will have no effect.



2014-03-02

Passage 7 2014-03-02-1

Cell Theory

The study of cells--cell biology--began in 1660, when English physicist Robert Hooke melted strands of spun glass to create lenses that he focused on bee stingers, fish scales, fly legs, feathers, and any type of insect he could hold still. When he looked at cork, which is the bark from a type of oak tree, it appeared to be divided into little boxes, which were remnants of cells that were once alive. Hooke called these units "cells" because they looked like the cubicles (cellae) where monks studied and prayed. Although Hooke did not realize the significance of his observation, he was the first person to see the outlines of cells.

In 1673, Antony van Leeuwenhoek of Holland improved lenses further. He used only a single lens, but due to its quality, it was more effective at magnifying and produced a clearer image than most two-lens microscopes then available. One of his first objects of study was tartar scraped from his own teeth, and he observed that it contained many very small animalcules(microscopic organisms). Over the next few years, Leeuwenhoek built more than 500 microscopes that opened a vast new world to the human eye and mind. He viewed bacteria and other microorganisms--life that people had not known existed. However, he failed to see the single-celled "animalcules" reproduce, and therefore he perpetuated the popular idea at the time that life arises from the nonliving or from nothing. Nevertheless, he described with remarkable accuracy microorganisms and microscopic parts of larger organisms, including human red blood cells.

Despite the accumulation of microscopists' drawings of cells made during the seventeenth and

eighteenth centuries, the cell theory --the idea that the cell is the fundamental unit of all life--did not emerge until the nineteenth century. Historians attribute the delay to poor technology--for example, crude microscopes and a lack of procedures to preserve and study living cells without damaging them. Neither the evidence itself nor early interpretations of it suggested that all organisms were composed of cells. Hooke had not observed actual cells but rather what they had left behind: the cell walls. Leeuwenhoek made important observations, but he did not methodically describe or categorize the structures that cells had in common.

In the nineteenth century, more powerful microscopes, with better magnification and illumination, revealed details of life at the subcellular level. In the early 1830s, Scottish surgeon Robert Brown noted a roughly circular structure in cells from orchid plants. Finding the structure in every orchid cell, he then identified it in all cells from a variety of other organisms. He named it the “nucleus,” a term that had remained in use. Brown memorialized the importance of the structure he discovered, but today we know the nucleus houses DNA for complex cells.

The cell theory finally emerged in 1839 when German biologists Matthias J. Schleiden and Theodore Schwann made careful comparisons of plants and animals. Schleiden first noted that cells were the basic units of plants, and then Schwann compared animal cells to plant cells. After observing many different plant and animal cells, they concluded that cells were “elementary particles of organisms, the unit of structure and function.” Schleiden and Schwann described the components of the cell as a cell body and nucleus contained within a surrounding membrane. Schleiden called a cell a “peculiar little organism” and realized that a cell can be a living entity on its own; but the new theory also recognized that in large plants and animals, cells are part of a larger living organism.

Many cell biologists extended Schleiden and Schwann’s observations and ideas. German physiologist Rudolph Virchow added the important corollary in 1855 that all cells come from preexisting cells, contradicting the still-popular idea that life can arise from the nonliving or from nothingness. Virchow’s statement also challenged the popular concept that cells develop on their own from the inside out, the nucleus forming a cell body around itself, and then the cell body growing a cell membrane. Virchow’s observation set the stage for descriptions of cell division in the 1870s and 1880s. Virchow was ahead of his time because he hypothesized that abnormal cells cause diseases that affect the whole body.

词汇题:

remnants:

- A. walls
- B. remains
- C. reflections
- D. structures

According to paragraph 2, which of the following did Antony van Leeuwenhoek NOT do?

- A. observe bacteria
- B. watch single-celled organisms reproduce

- C. feel individual cells of larger organisms
- D. inspect materials scraped from his teeth under a lens

Which of the following statements about microscopes is supported by the information presented in paragraph 2?

- A. two-lens microscopes were not available in Holland in Leeuwenhoek's time
- B. Leeuwenhoek preferred two-lens microscopes to one-lens microscopes
- C. The quality of Leeuwenhoek's microscope was better than that of most two-lens microscopes available in 1673
- D. two-lens microscopes were invented by Leeuwenhoek

Why does the author state that "Leeuwenhoek failed to see the single-celled "animalcules" reproduce"?

- A. to argue that Leeuwenhoek intentionally ignored some of what he saw with his microscope
- B. to suggest that Leeuwenhoek was not as important a researcher as he has traditionally been considered
- C. to explain why Leeuwenhoek did not challenge the idea that life could emerge from nonliving material
- D. to contrast Leeuwenhoek's observations with those of earlier researchers



The Great Depression

The term the Great Depression refers to the severe economic crisis of the 1930s that affected the United States and eventually much of the world. It is considered the worst and longest period of high unemployment and low business activity in modern times. The 1920s was a period of great prosperity in the United States. During this period many Americans opposed government regulation of the economy in principle. They believed that in periods of economic setback the government should cut expenditures and wait for the economy to improve. This prevailing attitude was to change profoundly by the time the Great Depression was over.

By 1929, many weaknesses in the American economy became evident. The prospect for construction was adversely affected by the rising costs of construction and land. Farmers were facing difficulties as surpluses depressed prices for agricultural products. Consequently the potential market for consumer goods in farm areas dried up and banks and other financial institutions with loans outstanding to farmers were weakened. Many people lacked savings. An employment insurance, or other cushions were vulnerable to economic adversity. Especially if they had bought goods on credit as advertisements had urged them to do. And when they suffered, so did the manufacturers and retailers who served them.

The most dramatic and obvious cause of the Depression was the collapse of the stock market in 1929. During the prosperous years between 1925 and 1929 stock market prices rose about three hundred percent. By 1929 over a million stock buyers were forcing prices to rise much faster than corporate profits. It was a process that could not continue indefinitely. In one terrible month starting on October 24, 1929 (now known as black Thursday), prices on the New York Stock Exchange alone fell in value about 26 million dollars, a decline of nine percent, and prices proceeded downward for another two and half years until by 1932 the stock market had lost nearly ninety percent of its value. It seems safe to assert that the enormous loss of wealth by individuals and institutions played a major role in the subsequent failure of the United States economy.

The Depression passed through to distinct stages. The first, which lasted from 1929 to 1931, was a period of reduced economic activity. In 1930 for example, industrial production was only 26 percent below its peak level of 1929. The second stage was much more severe; it lasted from 1931 until 1936. This stage was related not only to domestic events but also to serious financial problems in Germany, Austria and Great Britain that were induced primarily by a sharp decline in American investment, purchasing, and lending abroad. International commodity prices fell to absurd depth. Repercussions were severe in the United States, especially on agricultural prices and on banks with extensive foreign holdings. Some 2,300 banks with deposits of nearly 1.7 billion dollar failed in the United States in 1931.

It was an economic catastrophe. From 1929 to 1933 national income declined from 87.8 billion dollar to 40.2 billion dollar. More than 100,000 businesses failed. The number of unemployment swelled to 14 million by early 1933. Farm income, salaries, and manufacturing wages plummeted.

The trauma of the Great Depression was devastating. Two to three million people moved from cities back to the country in the hope of being able to scratch a bare living from the soil. Family tensions multiplied, the number of marriages and birth declined, and poverty generally increased.

One of the most significant long-term impacts was the change that it brought in attitudes toward business and government. Loss of a popular confidence in the nation's business and financial leadership was almost complete by 1933. Lack of trust increased particularly after several investigations revealed the extent to which bankers had misused depositors' money and manipulated the stock market during the 1920s. Indeed, investment bankers, along with President Hubert Hoover, became the chief scapegoats of the debacle. Gradually, Americans began to look to the government for protection. By 1935 a majority of Americans finally concluded that the government should provide jobs for the unemployed whenever private businesses put unable to do so. The old system of relatively unregulated free enterprise had to give way to a new system of government intending to assure the successful functioning of the economy.



Poikilotherms

Physiologically, animals can be divided into three groups: homeotherms, which maintain a fairly constant internal temperature regardless of the external temperature; poikilotherms, which have a body temperature that varies according to the surrounding environmental temperature; and heterotherms, which sometimes maintain a fairly constant body temperature and sometimes do not.

Poikilotherms, such as amphibians and insects, have a high thermal or heat conductance between the body and the environment, and a low metabolic rate. For this reason body temperature__and thus tissue temperature changes with environmental temperature. Being ectothermic and maintaining body temperature by using sources of heat energy such as solar radiation rather than metabolism has advantages. Prisoners of environmental temperatures, poikilotherms of temperate regions, such as snakes, become highly active only when temperature is adequately warm. Because their metabolic activity declines with decreasing temperature, these animals become sluggish in the cool of morning and evening. Similarly, they have to restrict their activity to the late spring, summer and early fall. During periods of intense physical activity, when energy consumption is high, poikilotherms depend on the anaerobic (without oxygen) breakdown of glycogen (sugars stored for energy). These breakdown results in an accumulation of lactic acid in the tissues, and the lactic acid can be oxidised only after activity ceases. Anaerobic metabolism severely limits bursts of poikilothermic activity to a few minutes, because of physical exhaustion. This tendency to become exhausted is one reason that so many predatory terrestrial poikilotherms, such as snakes and alligators, secure prey by ambush rather than by chase.

Because they do not depend on internally generated body heat, poikilotherms can reduce metabolic activity during periods of temperature extremes and of food or water shortage. Low energy demands enable poikilotherms to colonize areas of limited food and water, such as deserts. Because they do not have the problem of metabolic heat loss, poikilotherms are not limited to any minimum size or definite shape. Many are single celled, such as paramecia and amoebas; others such as earthworms, millipedes, and snakes have cylindrical bodies. Such characteristics enable poikilotherms to exploit resources and habitats unavailable to homeotherms. On the other hand, the same metabolic restrictions imposed an upper size limit: poikilotherms would not be able to absorb enough heat to warm a very large body. For this reason, some paleontologists argue that large dinosaurs had to be endothermic (maintaining a constant body temperature by means of metabolism). A counter argument is that large ectotherms could develop and maintain body temperatures above air temperatures in a tropical environment because their low surface-to-volume ratio would limit cooling.

Although poikilotherms as a group do not have a lower limit to body size, size is nevertheless an important aspect of their life. The rate of heating and cooling in a poikilotherm decreases as the size increases. Because of its small size, a beetle heats and cools quickly, but rarely can it raise its

uniform temperature above that of the surrounding environmental temperature. However, it does have the ability to control its temperature by moving in and out of the sunlight.

For very large terrestrial poikilotherms, such as tortoises, life is considerably different. Because they heat slowly, they need a much higher environmental temperature than smaller poikilotherms need to reach the same temperature. Further, because large poikilotherms have difficulty locating shelter from environmental extremes, they are more or less restricted to environments with small seasonal fluctuations in temperatures. Large poikilotherms, however, have the advantage of moving freely about in space and time during the day because their large body mass buffers their body temperatures.

Many poikilothermic animals of temperate and arctic regions must endure long periods of below-freezing temperatures in winter. Some intertidal invertebrates of high latitudes and certain aquatic insects survive the cold by actually freezing and then thawing when the temperature moderates. Other animals, particularly Arctic and Antarctic fish and many insects, resist freezing because of their presence of glycerol in their body fluids. Glycerol protects against damage from freezing and lowers the animal's freezing point.

In addition, many insects exhibiting frost hardiness enter a resting stage diapause, characterized by a cessation of feeding, growth, mobility and reproduction. Among many insects, diapause is a genetically determined, obligatory resting stage before development can proceed—it prevents the appearance of a sensitive stage of development at a time when low temperatures would kill.

2014-03-16

Passage 10 2014-03-16-1

Costs and Benefits of Social Life

Many think that the reason so many animals live with others of their species is that social creatures are higher up the evolutionary scale and so are better adapted and leave more offspring than do animals that live solitary lives. However, in each and every species, generation after generation, relatively social and relatively solitary types compete unconsciously with one another in ways that determine who leaves more offspring on average. In some species, the more social individuals have won out, but in a large majority, it is the solitary types that have consistently left more surviving descendants on average.

But how can living alone ever be superior to living together? Under some conditions, a cost-benefit comparison favors solitary life over a more social existence. For example, among most social species, animals have to expend time and energy competing for social status. Those that do not occupy the top positions regularly have to signal their submissive state to their superiors if they are to be permitted to remain in the group. This can take up a major share of a social subordinate's life. In fact, even in small social groups there are both subtle competition and not-so-subtle competition.

Social groups also offer opportunities for reproductive interference. Breeding males that live in close association with more attractive rivals may lose their mates to these individuals. In addition, sociality has two other potential disadvantages. The first is heightened competition for food, which occurs in animals as different as colonial fieldfares (a kind of songbird) and groups of lions, whose females are often pushed from their food by hungry males. The second is increased vulnerability to parasites and disease, which plague social species of all sorts. While it is true that some social animals have evolved special responses designed to combat parasites and disease, those responses can only reduce, but cannot totally eliminate, the damage caused by those threats, and the responses may even carry their own costs. Thus, honeybees warm their hives in response to an infestation by a fungal pathogen, which apparently helps kill the heat-sensitive fungus, but at the price of time and energy expended by the heat-producing workers.

If social living carries a heightened risk of infection, then the larger the group, the greater the risk. This prediction holds for cliff swallows, which pack their nests side by side in colonies composed of anywhere from a handful of birds to several thousand pairs. The more swallows nesting together, the greater the chance that at least one bird will be infested with swallow bugs, which can then readily spread from one nest to another.

The parasites and fungi that make life miserable for swallows and other social creatures demonstrate that if sociality is to evolve, the assorted costs of living together must be outweighed by compensatory benefits. Cliff swallows may join others to take advantage of the improved foraging that comes from following companions to good feeding sites, while other animals, such

as male imperial penguins, save thermal energy by huddling shoulder to shoulder during the brutal Antarctica winter. Still others, such as lionesses, join forces to fend off enemies of their own species.

The most widespread fitness benefit for social animals, however, probably is improved protection against predators. Many studies have shown that animals in groups gain by reducing the individual risk of being captured, or by spotting danger sooner, or by attacking their enemies in groups. Males in nesting colonies of bluegill sunfish cooperate in driving egg-eating bullhead catfish away from their nests at the bottom of a freshwater lake. While bluegills have adopted social behavior to avoid predation, closely related species that nest alone have evolved means to protect themselves while nesting alone. Thus, the solitary pumpkinseed sunfish, a member of the same genus as the bluegill, has powerful biting jaws and so can repel egg-eating enemies on its own, whereas bluegills have small, delicate mouths good only for inhaling small, soft-bodied insect larvae. Pumpkinseed sunfish are in no way inferior to or less well adapted than bluegills because they are solitary; they simply gain less through social living, which makes solitary nesting the adaptive tactic for them.

Paragraph 2 But how can living alone ever be superior to living together? Under some conditions, a cost-benefit comparison favors solitary life over a more social existence. For example, among most social species, animals have to expend time and energy competing for social status. Those that do not occupy the top positions regularly have to signal their submissive state to their superiors if they are to be permitted to remain in the group. This can take up a major share of a social subordinate's life. In fact, even in small social groups there are both subtle competition and not-so-subtle competition.

1. Which of the following can be inferred from paragraph 2 as a reason for saying that under some conditions a solitary life is superior to a more social life?
 - ☐ Solitary animals can compete more successfully against animals of other species than social animals can.
 - ☐ Solitary animals can hide from predators more easily than groups of animals can.
 - ☐ Solitary animals do not need to spend time and energy competing for social status.
 - ☐ Solitary animals do not need large quantities of food to survive.

Paragraph 3 Social groups also offer opportunities for reproductive interference. Breeding males that live in close association with more attractive rivals may lose their mates to these individuals. In addition, sociality has two other **potential** disadvantages. The first is heightened competition for food, which occurs in animals as different as colonial fieldfares (a kind of songbird) and groups of lions, whose females are often pushed from their food by hungry males. The second is increased vulnerability to parasites and disease, which plague social species of all sorts. While it is true that some social animals have evolved special responses designed to combat parasites and disease, those responses can only reduce, but cannot totally eliminate, the damage caused by those threats, and the responses may even carry their own costs. Thus, **honeybees warm their**

hives in response to an infestation by a fungal pathogen, which apparently helps kill the heat-sensitive fungus, but at the price of time and energy expended by the heat-producing workers.

2. The word "potential" in the passage is closest in meaning to
 - ☐ possible
 - ☐ serious
 - ☐ common
 - ☐ related

3. Which of the following is associated in paragraph 3 with reproductive interference in social animals
 - ☐ Females being pushed away from their food by hungry males
 - ☐ Breeding males losing their mates to more attractive males
 - ☐ Parasites and disease infecting the eggs of breeding females
 - ☐ Females warming their habitats to care a fungal pathogen

4. According to paragraph 3 which of the following is likely to increase when animals live with other members of their species rather than alone
 - ☐ reproductive success
 - ☐ unpredictable behavior among males
 - ☐ resistance to disease
 - ☐ competition for food

5. In paragraph 3 why does author discuss "honeybees warming their hives in response to an infestation by a fungal pathogen"?
 - ☐ To give an example of how living in groups benefits animals
 - ☐ To illustrate how social species combat parasites and disease
 - ☐ To indicate that worker bees are the most active in fighting parasites and disease
 - ☐ To demonstrate that avoiding a disadvantage of social living can carry its own costs

Paragraph 4 If social living carries a heightened risk of infection, then the larger the group, the greater the risk. This prediction holds for cliff swallows, which pack their nests side by side in colonies composed of anywhere from a handful of birds to several thousand pairs. The more swallows nesting together, the greater the chance that at least one bird will be infested with swallow bugs, which can then readily spread from one nest to another.

Paragraph 5 The parasites and fungi that make life miserable for swallows and other social creatures demonstrate that if sociality is to evolve, the assorted costs of living together must be outweighed by compensatory benefits. Cliff swallows may join others to take advantage of the improved foraging that comes from following companions to good feeding sites, while other animals, such as male imperial penguins, save thermal energy by huddling shoulder to shoulder during the brutal Antarctica winter. Still others, such as lionesses, join forces to fend off enemies of their own species.

6. The word "heightened" in the passage is closest in meaning to
- ☐ long-term
 - ☐ increased
 - ☐ real
 - ☐ unnecessary
7. Paragraphs 4 and 5 support all of the following statements about cliff swallows **EXCEPT**
- ☐ They feed primarily on bugs found in their nests.
 - ☐ They sometimes live in groups of only a few birds.
 - ☐ They sometimes locate good feeding sites by following other cliff swallows.
 - ☐ They build their nests close to the nests of other cliff swallows.
8. The word "assorted" in the passage is closest in meaning to
- ☐ usual
 - ☐ obvious
 - ☐ various
 - ☐ severe
9. Which of the following is **NOT** identified in paragraph 5 as a need that some animals can better satisfy when they live with other members of their species?
- ☐ The need to find food
 - ☐ The need to locate breeding sites
 - ☐ The need to preserve body heat
 - ☐ The need to protect themselves from attackers

Paragraph 6 The most widespread fitness benefit for social animals, however, probably is improved protection against predators. Many studies have shown that animals in groups gain by reducing the individual risk of being captured, or by spotting danger sooner, or by attacking their enemies in groups. Males in nesting colonies of bluegill sunfish cooperate in driving egg-eating bullhead catfish away from their nests at the bottom of a freshwater lake. While bluegills have adopted social behavior to avoid predation, closely related species that nest alone have evolved means to protect themselves while nesting alone. Thus, the solitary pumpkinseed sunfish, a member of the same genus as the bluegill, has powerful biting jaws and so can repel egg-eating enemies on its own, whereas bluegills have small, delicate mouths good only for inhaling small, soft-bodied insect larvae. Pumpkinseed sunfish are in no way inferior to or less well adapted than bluegills because they are solitary; they simply gain less through social living, which makes solitary nesting the adaptive **tactic** for them.

10. In paragraph 6, why does the author compare the bluegill sunfish to the pumpkinseed sunfish?
- ☐ To show that nesting has different costs and benefits for solitary organisms than it has for organisms living with other members of the same species
 - ☐ To provide evidence that social nesting is usually superior to solitary nesting

- ☐ To argue that, in closely related species, the feeding organs of solitary organisms usually differ from those of social organisms
- ☐ To show that closely related species can be social or solitary depending on which behavior better enables them to fight off predators

Paragraph 6 is marked with an arrow [→]

11. The word “tactic” in the passage is closest in meaning to

- ☐ strategy
- ☐ development
- ☐ result
- ☐ characteristic

Paragraph 3

Social groups also offer opportunities for reproductive interference. Breeding males that live in close association with more attractive rivals may lose their mates to these individuals. In addition, sociality has two other potential disadvantages. The first is heightened competition for food, which occurs in animals as different as colonial fieldfares (a kind of songbird) and groups of lions, whose females are often pushed from their food by hungry males. ■ The second is increased vulnerability to parasites and disease, which plague social species of all sorts. ■ While it is true that some social animals have evolved special responses designed to combat parasites and disease, those responses can only reduce, but cannot totally eliminate, the damage caused by those threats, and the responses may even carry their own costs. ■ Thus, honeybees warm their hives in response to an infestation by a fungal pathogen, which apparently helps kill the heat-sensitive fungus, but at the price of time and energy expended by the heat-producing workers. ■

12. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Likewise, termites can reduce their risk of some infectious diseases by associating with other termites, but only at the expense of decreased physiological functioning.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

13. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

There are costs and benefits associated with living in groups.

•

-
-

Answer choices

Solitary animals are sometimes thought to be higher up the evolutionary scale because they leave more surviving descendants on average than social animals do.

Social animals have evolved fewer mechanisms for reducing the costs of competing for social status than they have for reducing the risks of exposure to disease and parasites.

The need for protection from predators makes animals adopt social living, but animals that have the ability to individually fight off predators benefit more from solitary living.

Social living has advantages such as making it possible for animals to become more successful at locating food, but it also has risks such as increased competition for mates.

The numerous disadvantages of living together lead animals to adopt social living only when the benefits are greater than the costs.

Animals that live alone such as pumpkinseed fish need to evolve a special mouth or some other special organ to help them combat predators.



Comets

Comets are among the most interesting and unpredictable bodies in the solar system. They are made of frozen gases (water vapor, ammonia, methane, carbon dioxide, and carbon monoxide) that hold together small pieces of rocky and metallic materials. Many comets travel in very elongated orbits that carry them far beyond Pluto. These long-period comets take hundreds of thousands of years to complete a single orbit around the Sun. However, a few short-period comets (those having an orbital period of less than 200 years), such as Halley's Comet, make a regular encounters with the inner solar system.

When a comet first becomes visible from Earth, it appears very small, but as it approaches the Sun, solar energy begins to vaporize the frozen gases, producing a glowing head called the coma. The size of the coma varies greatly from one comet to another. Extremely rare ones exceed the size of the Sun, but most approximate the size of Jupiter. Within the coma, a small glowing nucleus with a diameter of only a few kilometers can sometimes be detected. As comets approach the Sun, some develop a tail that extends for millions of kilometers. Despite the enormous size of their tails and comas, comets are relatively small members of the solar system.

The observation that the tail of a comet points away from the Sun in a slightly curved manner led early astronomers to propose that the Sun has a repulsive force that pushes the particles of the coma away, thereby forming the tail. Today, two solar forces are known to contribute to this formation. One, radiation pressure, pushes dust particles away from the coma. The second, known as solar wind, is responsible for moving the ionized gases, particularly carbon monoxide. Sometimes a single tail composed of both dust and ionized gases is produced, but often two tails—one of dust, the other, a blue streak of ionized gases—are observed.

As a comet moves away from the Sun, the gases forming the coma recondense, the tail disappears, and the comet returns to distant space. Material that was blown from the coma to form the tail is lost from the comet forever. Consequently, it is believed that most comets cannot survive more than a few hundred close orbits of the Sun. Once all the gases are expelled, the remaining material—a swarm of tiny metallic and stony particles—continues the orbit without a coma or a tail.

Comets apparently originate in two regions of the outer solar system. Most short-period comets are thought to orbit beyond Neptune in a region called the Kuiper belt, in honor of the astronomer Gerald Kuiper. During the past decade over a hundred of these icy bodies have been discovered. Most Kuiper belt comets move in nearly circular orbits that lie roughly in the same plane as the planets. A chance collision between two comets, or the gravitational influence of one of the Jovian planets—Jupiter, Saturn, Uranus, and Neptune—may occasionally alter the orbit of a comet in these regions enough to send it to the inner solar system and into our view.

Unlike short-period comets, long-period comets have elliptical orbits that are not confined to the plane of the solar system. These comets appear to be distributed in all directions from the Sun,

forming a spherical shell around the solar system, called the Oort cloud, after the Dutch astronomer Jan Oort. Millions of comets are believed to orbit the Sun at distances greater than 10,000 times the Earth-Sun distance. The gravitational effect of a distant passing star is thought to send an occasional Oort cloud comet into a highly eccentric orbit that carries it toward the Sun. However, only a tiny portion of the Oort cloud comets have orbits that bring them into the inner solar system.

The most famous short-period comet is Halley's Comet, named after English astronomer Edmond Halley. Its orbital period averages 76 years, and every one of its 30 appearances since 240 B.C. has been recorded by Chinese astronomers. When seen in 1910, Halley's Comet had developed a tail nearly 1.6 million kilometers (1 million miles) long and was visible during daylight hours. Its most recent approach occurred in 1986.

Paragraph 1

Comets are among the most interesting and unpredictable bodies in the solar system. They are made of frozen gases (water vapor, ammonia, methane, carbon dioxide, and carbon monoxide) that hold together small pieces of rocky and metallic materials. Many comets travel in very elongated orbits that carry them far beyond Pluto. These long-period comets take hundreds of thousands of years to complete a single orbit around the Sun. However, a few short-period comets (those having an orbital period of less than 200 years), such as Halley's Comet, make a regular encounters with the inner solar system.

1. All of the following questions can be answered based on the facts presented in paragraph 1 EXCEPT:

- ☐ Do any comets travel to regions of space beyond our solar system?
- ☐ Do most short-period comets enter the inner solar system?
- ☐ What are comets composed of?
- ☐ Why are comets regarded as among the most unpredictable bodies in the solar system?

Paragraph 1 is marked with an arrow [→]

Paragraph 2

When a comet first becomes visible from Earth, it appears very small, but as it approaches the Sun, solar energy begins to vaporize the frozen gases, producing a glowing head called the coma. The size of the coma varies greatly from one comet to another. Extremely rare ones **exceed** the size of the Sun, but most approximate the size of Jupiter. Within the coma, a small glowing nucleus with a diameter of only a few kilometers can sometimes be **detected**. As comets approach the Sun, some develop a tail that extends for millions of kilometers. Despite the enormous size of their tails and comas, comets are relatively small members of the solar system.

2. The word "**exceed**" in the passage is closest in meaning to

- ☐ become
- ☐ grow to
- ☐ approach

- ☐ go beyond
3. The word “detected” in the passage is closest in meaning to
- ☐ noticed
 - ☐ created
 - ☐ expanded
 - ☐ displayed
4. Paragraph 2 supports the idea that comets appear very small when they first become visible from Earth because
- ☐ the glow from the nucleus hides the rest of the comet from view
 - ☐ they are still too far away from the Sun to have developed a coma or a tail
 - ☐ their various gases have mostly become vaporized
 - ☐ they are relatively small when compared to the visible planets
- Paragraph 2 is marked with an arrow [→]
5. Paragraph 2 supports which of the following statements about comet size?
- ☐ The size of a comet is affected by the addition of gases absorbed as the comet passes the Sun.
 - ☐ The size of a comet’s tail is less variable than the size of its coma.
 - ☐ The coma of most comets is smaller than the Sun.
 - ☐ The size of a comet cannot be accurately determined until it nears Earth.
- Paragraph 2 is marked with an arrow [→]

Paragraph 3

The observation that the tail of a comet points away from the Sun in a slightly curved manner led early astronomers to propose that the Sun has a repulsive force that pushes the particles of the coma away, thereby forming the tail. Today, two solar forces are known to contribute to this formation. One, radiation pressure, pushes dust particles away from the coma. The second, known as solar wind, is responsible for moving the ionized gases, particularly carbon monoxide. Sometimes a single tail composed of both dust and ionized gases is produced, but often two tails—one of dust, the other, a blue streak of ionized gases—are observed.

6. The word “propose” in the passage is closest in meaning to
- ☐ offer the theory
 - ☐ prove
 - ☐ agree
 - ☐ discover
7. According to paragraph 3, what is true about comets’ tails?
- ☐ Their shape led early astronomers to draw false conclusions about the Sun.
 - ☐ They consist mostly of ionized gases that had been blown out of the Sun.
 - ☐ Radiation pressure and solar wind both play a role in their formation.
 - ☐ Their dust content decreases as a comet nears the Sun.

Paragraph 3 is marked with an arrow [→]

Paragraph 4

As a comet moves away from the Sun, the gases forming the coma recondense, the tail disappears, and the comet returns to distant space. Material that was blown from the coma to form the tail is lost from the comet forever. Consequently, it is believed that most comets cannot survive more than a few hundred close orbits of the Sun 和C并不对应. Once all the gases are expelled, the remaining material—a swarm of tiny metallic and stony particles—continues the orbit without a coma or a tail.

8. Select TWO answer choices that according to paragraph 4 describe how a comet changes as it travels away from the Sun. To receive credit, you must select TWO answers.

- ☐ It loses its metallic and stony particles.
- ☐ It loses all the material that was in its tail.
- ☐ Its orbital period becomes shorter.
- ☐ The gases that were in its coma recondense.

Paragraph 3 is marked with an arrow [→]

Paragraph 5

Comets apparently originate in two regions of the outer solar system. Most short-period comets are thought to orbit beyond Neptune in a region called the Kuiper belt, in honor of the astronomer Gerald Kuiper. During the past decade over a hundred of these icy bodies have been discovered. Most Kuiper belt comets move in nearly circular orbits that lie roughly in the same plane as the planets. A chance collision between two comets, or the gravitational influence of one of the Jovian planets—Jupiter, Saturn, Uranus, and Neptune—may occasionally alter the orbit of a comet in these regions enough to send it to the inner solar system and into our view.

9. The word “roughly” in the passage is closest in meaning to

- ☐ frequently
- ☐ approximately
- ☐ typically
- ☐ clearly

Paragraph 6

Unlike short-period comets, long-period comets have elliptical orbits that are not confined to the plane of the solar system. These comets appear to be distributed in all directions from the Sun, forming a spherical shell around the solar system, called the Oort cloud, after the Dutch astronomer Jan Oort. Millions of comets are believed to orbit the Sun at distances greater than 10,000 times the Earth-Sun distance. The gravitational effect of a distant passing star is thought to send an occasional Oort cloud comet into a highly eccentric orbit that carries it toward the Sun. However, only a tiny portion of the Oort cloud comets have orbits that bring them into the inner solar system.

10. According to paragraphs 5 and 6, compared to the orbits of short-period comets, the orbits

of long-period comets are

- ☐ more circular
- ☐ more likely to result in a comet's entering the inner solar system
- ☐ more likely to be affected by a passing star
- ☐ more likely to lie in the same plane as the planets

Paragraph 5 and 6 are marked with an arrow [→]

11. According to paragraph 6, which of the following is true about the Oort cloud?

- ☐ It is almost 10,000 times the size of the Sun.
- ☐ It is formed by the orbits of long-period comets orbiting far from the Sun.
- ☐ It is produced by the gravitational effects of distant stars.
- ☐ It can disrupt the orbit of a comet enough to send it on a path toward the Sun.

Paragraph 6 is marked with an arrow [→]

Paragraph 7

The most famous short-period comet is Halley's Comet, named after English astronomer Edmond Halley. ■ Its orbital period averages 76 years, and every one of its 30 appearances since 240 B.C. has been recorded by Chinese astronomers. ■ When seen in 1910, Halley's Comet had developed a tail nearly 1.6 million kilometers (1 million miles) long and was visible during daylight hours. ■ Its most recent approach occurred in 1986. ■

12. According to paragraph 7, all of the following are true of Halley's Comet EXCEPT:

- ☐ On at least one appearance, it had a tail of well over a million kilometers in length.
- ☐ Each of its last 30 appearances has been recorded by astronomers in China.
- ☐ It is the only comet that can be observed during daylight hours.
- ☐ It averages 76 years between appearances in the inner solar system.

Paragraph 7 is marked with an arrow [→]

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

It was not until 1705, however, that Halley determined that each of these appearances was actually a reappearance of the same object.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Short –period comets complete their orbits in less than 200 years while long-period comets may take hundreds of thousands of years to complete an orbit.

-
-
-

Answer Choices

The size of a comet's nucleus is determined by two solar forces, radiation pressure and solar wind.错

Comets accelerate as they move away from the Sun in part because they lose material from the coma each time they pass near the Sun.错

Comets are at times sent into the inner solar system as a result of a collision or a gravitational disturbance.

Although a comet's coma and tail may be

quite large, these features do not develop until a comet is close enough to the Sun for solar energy to vaporize its frozen gases.

While short-period comets have circular orbits that lie close to the plane of the planets, long-period comets have more elliptical orbits that form a spherical shell around the solar system.

Although studies of Halley's Comet and other short-period comets provide some clues, astronomers do not yet fully understand how and when comets formed.错

Agricultural Society in Eighteenth-Century British America

In the northern American colonies, especially New England, tight-knit farming families, organized in communities of several thousand people, dotted the landscape by the mid-eighteenth century. New Englanders staked their future on a mixed economy. They cleared forests for timber used in barrels, ships, houses, and barns. They plumbed the offshore waters for fish to feed local populations. And they cultivated and grazed as much of the thin-soiled, rocky hills and bottomlands as they could recover from the forest.

The farmers of the middle colonies-Pennsylvania, Delaware, New Jersey, and New York-set their wooden plows to much richer soils than New Englanders did. They enjoyed the additional advantage of setting an area already partly cleared by Native Americans who had relied more on agriculture than had New England tribes. Thus favored, mid-Atlantic farm families produced modest surpluses of corn, wheat, beef, and pork. By the mid-eighteenth century, ships from New York and Philadelphia were carrying these foodstuffs not only to the West Indies, always a primary market, but also to areas that could no longer feed themselves-England, Spain, Portugal, and even New England.

In the North, the broad ownership of land distinguished farming society from every other agricultural region of the Western world. Although differences in circumstances and ability led gradually toward greater social stratification, in most communities, the truly rich and terribly poor were few and the gap between them small compared with European society. Most men other than indentured servants (servants contracted to work for a specific number of years) lived to purchase or inherit a farm of at least 50 acres. With their family's labor, they earned a decent existence and provided a small inheritance for each of their children. Settlers valued land highly, for owning land ordinarily guaranteed both economic independence and political rights.

By the eighteenth century, amid widespread property ownership, a rising population pressed against a limited land supply, especially in New England. Family farms could not be divided and subdivided indefinitely, for it took at least fifty acres(of which only a quarter could usually be cropped) to support a single family. In Concord, Massachusetts, for example, the founders had worked farms averaging about 250 acres. A century later, in the 1730s, the average farm had shrunk by two thirds, as farm owners struggled to provide an inheritance for the three or four sons that the average marriage produced.

The decreasing fertility of the soil compounded the problem of dwindling farm size in New England. When land had been plentiful, farmers planted crops in the same field for three years and then let it lie fallow (unplanted) in pasture seven years or more until it regained its fertility. But on the smaller farms of the eighteenth century, farmers had reduced fallow time to only a year or two. Such intense use of the soil reduced crop yields, forcing farmers to plow marginal land or shift to livestock production.

The diminishing size and productivity of family farms forced many New Englanders to move to

the frontier or out of the area altogether in the eighteenth century. "Many of our old towns are too full of inhabitants for husbandry, many of them living on small shares of land, " complained one writer. In Concord, one of every four adult males migrated from town every decade from the 1740s on, and in many towns migration out was even greater. Some drifted south to New York and Pennsylvania. Others sought opportunities as artisans in the coastal towns or took to the sea. More headed for the colonies, western frontier or north into New Hampshire and the eastern frontier of Maine. Several thousand New England families migrated even farther north to the Annapolis Valley of Nova Scotia. Throughout New England after the early eighteenth century, most farmers' sons knew that their destiny lay elsewhere.

Wherever they took up farming, northern cultivators engaged in agricultural work routines that were far less intense than in the south. The growing season was much shorter, and the cultivation of cereal crops required incessant labor only during spring planting and autumn harvesting. This less burdensome work rhythm let many northern cultivators to fill out their calendars with intermittent work as clockmakers, shoemakers, carpenters, and weavers.

Paragraph 1 In the northern American colonies, especially New England, tight-knit farming families, organized in communities of several thousand people, dotted the landscape by the mid-eighteenth century. New Englanders staked their future on a mixed economy. They cleared forests for timber used in barrels, ships, houses, and barns. They plumbed the offshore waters for fish to feed local populations. And they cultivated and grazed as much of the thin-soiled, rocky hills and bottomlands洼地 as they could recover from the forest.

1. Paragraph 1 mentions all of the following as economic activities that eighteenth century New Englanders practiced EXCEPT

- ☐ raising crops
- ☐ catching fish
- ☐ trading goods
- ☐ selling timber

Paragraph 2 The farmers of the middle colonies-Pennsylvania, Delaware, New Jersey, and New York-set their wooden plows to much richer soils than New Englanders did. They enjoyed the additional advantage of setting an area already partly cleared by Native Americans who had relied more on agriculture than had New England tribes. Thus favored, mid-Atlantic farm families produced modest surpluses of corn, wheat, beef, and pork. By the mid-eighteenth century, ships from New York and Philadelphia were carrying these foodstuffs not only to the West Indies, always a primary market, but also to areas that could no longer feed themselves-England, Spain, Portugal, and even New England.

2. Paragraph 1 and 2 support all of the following statements about the economies of British colonies in America EXCEPT

- ☐ The middle colonies engaged in international trade.
- ☐ The middle colonies had agricultural advantages the northern colonies did not.
- ☐ The northern colonies participated in a variety of economic enterprises.

☐ The middle colonies were less prosperous than the northern colonies.

3. According to paragraph 2, one advantage that farmers in the middle colonies had over farmers in New England was better

- ☐ plows
- ☐ soil
- ☐ ways of shipping farm products to market
- ☐ relations with local native American tribes

4. The word “modest” in the passage is closest in meaning to

- ☐ mixed
- ☐ moderate
- ☐ growing
- ☐ constant

CD本质是类似的，而B和modest本身的意思也很接近

Paragraph 3 In the North, the broad ownership of land distinguished farming society from every other agricultural region of the Western world. Although differences in circumstances and ability led gradually toward greater social stratification, in most communities, the truly rich and terribly poor were few and the gap between them small compared with European society. Most men other than indentured受契约束缚的 servants (servants contracted to work for a specific number of years) lived to purchase or inherit a farm of at least 50 acres. With their family's labor, they earned a decent existence and provided a small inheritance for each of their children. Settlers valued land highly, for owning land ordinarily guaranteed both economic independence and political rights.

5. According to paragraph 3 in what way did farming society in the northern colonies differ from farming societies in the rest of the western world

- ☐ The differences between social classes were much greater.
- ☐ People lived much closer together.
- ☐ The proportion of land owners was much higher.
- ☐ Many more families had servants.

Paragraph 4 By the eighteenth century, amid widespread property ownership, a rising population pressed against a limited land supply, especially in New England. Family farms could not be divided and subdivided indefinitely, for it took at least fifty acres (of which only a quarter could usually be cropped) to support a single family. In Concord, Massachusetts, for example, the founders had worked farms averaging about 250 acres. A century later, in the 1730s, the average farm had shrunk by two thirds, as farm owners struggled to provide an inheritance for the three or four sons that the average marriage produced.

6. The word “indefinitely” in the passage is closest in meaning to

- ☐ fairly
- ☐ safely
- ☐ more than once
- ☐ without limit

7. Why does author include a discussion of "Concord, Massachusetts"

- ☐ To give an example of the type of inheritance farm owners generally provided for their sons
- ☐ To help explain why the farms started by the founders averaged at least 250 acres
- ☐ To indicate that New England farms were always inherited by the oldest sons from their fathers
- ☐ To help illustrate how limited the overall land supply was in New England

Paragraph 5 The decreasing fertility of the soil compounded the problem of **dwindling** **dwindle** 减少 farm size in New England. When land had been plentiful, farmers planted crops in the same field for three years and then let it lie fallow (unplanted) in pasture seven years or more until it regained its fertility. But on the smaller farms of the eighteenth century, farmers had reduced fallow time to only a year or two. Such intense use of the soil reduced crop yields, forcing farmers to plow marginal land or shift to livestock production.

8. The word "compounded" in the passage is closest in meaning to

- ☐ added to
- ☐ resulted from
- ☐ led to
- ☐ occurred before

9. According to paragraph 5, what causes the crop yields in New England to fail

- ☐ The shift to livestock production by many farmers
- ☐ The decreased amount of time that fields were left fallow
- ☐ The practice of planting crops in the same field for three years in a row
- ☐ The reduced size of the average field

Paragraph 6 The diminishing size and productivity of family farms forced many New Englanders to move to the frontier or out of the area altogether in the eighteenth century. "Many of our old towns are too full of inhabitants for **husbandry** 饲养、务农, many of them living on small shares of land," complained one writer. In Concord, one of every four adult males migrated from town every decade from the 1740s on, and in many towns migration out was even greater. Some drifted south to New York and Pennsylvania. Others sought opportunities as artisans in the coastal towns or took to the sea. More headed for the colonies, western frontier or north into New Hampshire and the eastern frontier of Maine. Several thousand New England families migrated even farther north to the Annapolis Valley of Nova Scotia. Throughout New England after the early eighteenth century, most farmers' sons knew that their destiny lay elsewhere.

10. According to paragraph 6 why did many New Englanders move out of the area in the eighteenth century

- ☐ They wanted to live in towns rather than on farms.
- ☐ Their farms no longer provided them with good living.
- ☐ There was unequal distribution of males and females in New England.
- ☐ They were being crowded out by migrants from outside New England.

Paragraph 7 Whereever they took up farming, northern cultivators engaged in agricultural work routines that were far less intense than in the south. The growing season was much shorter, and the cultivation of cereal crops required incessant labor only during spring planting and autumn harvesting. This less **burdensome** work rhythm let many northern cultivators to fill out their calendars with **intermittent work as clockmakers, shoemakers, carpenters, and weavers.**

11. The word “burdensome” in the passage is closet in meaning to

- ☐ frequent
- ☐ productive
- ☒ difficult
- ☐ well-paid

12. Why does the author include the information about the “intermittent work as clockmakers, shoemakers, carpenters, and weavers” that northern cultivators engaged in?

- ☐ To suggest that northern cultivators were not as skilled at agricultural work as southern cultivators were
- ☐ To indicate an economic effect of the shorter northern growing season on northern cultivators
- ☐ To challenge the claim that work routines in the north were less intense than they were in the south
- ☐ To emphasize that northern workers tried to change their agriculturally centered economy

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Paragraph 6 The diminishing size and productivity of family farms forced many New Englanders to move to the frontier or out of the area altogether in the eighteen century. "Many of our old towns are too full of inhabitants for husbandry, many of them living on small shares of land, " complained one writer. In Concurd, one of every four adult males migrated from town every decade from the 1740s on, and in many towns migration out was even greater. ■Some drifted south to New York and Pennsylvania. ■Others sought opportunities as artisans in the coastal towns or took to the sea. ■More headed for the colonies, western frontier or north into New Hampshire and the eastern frontier of Maine. ■Several thousand New England families migrated even farther north to the Annapolis Valley of Nova Scotia. Throughout New England after the early eighteenth century, most farmers' sons knew that their destiny lay elsewhere.

A third of Northampton's men over 21 years old left, joining the stream of departing New Englanders.

C

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below.

Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

In eighteenth century British America agriculture was more productive and profitable in the middle colonies than in New England.

-
-
-

Answer Choices

By the mid-eighteenth century shipping had become important to the economy of the middle colonies where farmers produced large (moderate) surpluses of foodstuffs for trade with Europe and elsewhere.

The labor provided by indentured servants allowed most New England farmers to raise enough food and livestock to earn a living and leave a comfortable inheritance for their children.

Declining farm size forced farmers to greatly reduce the time fields were left fallow, and this more intensive use of relatively poor soil resulted in seriously decreased fertility and lowered crop yields.

Land ownership was far more important to New Englanders than to people in the middle colonies because it was necessary for political rights and economic independence only in the North.

Land ownership was widespread in the North but a shortage of farmland and the practice of dividing family farms among the sons had left the average farm barely big enough to support a family.

The reduced size and productivity of northern farms forced many farmers to move to other regions or to take up other occupations at least during those periods when little work was required on a farm.

2014-03-22

The Beringia Landscape

During the peak of the last ice age, northeast Asia (Siberia) and Alaska were connected by a broad land mass called the Bering Land Bridge. This land bridge existed because so much of Earth's water was frozen in the great ice sheets that sea levels were over 100 meters lower than they are today. Between 25,000 and 10,000 years ago, Siberia, the Bering Land Bridge, and Alaska shared many environmental characteristics. These included a common mammalian fauna of large mammals, a common flora composed of broad grasslands as well as wind-swept dunes and tundra, and a common climate with cold, dry winters and somewhat warmer summers. The recognition that many aspects of the modern flora and fauna were present on both sides of the Bering Sea as remnants of the ice-age landscape led to this region being named Beringia.

It is through Beringia that small groups of large mammal hunters, slowly expanding their hunting territories, eventually colonized North and South America. On this archaeologists generally agree, but that is where the agreement stops. One broad area of disagreement in expanding the peopling of the Americas is the domain of paleoecologists, but it is critical to understanding human history: what was Beringia like?

The Beringian landscape was very different from what it is today. Broad, windswept valleys, glaciated mountains, sparse vegetation, and less moisture created a rather forbidding land mass. This land mass supported herds of now-extinct species of mammoth, bison, and horse and somewhat modern versions of caribou, musk ox, elk, and saiga antelope. These grazers supported in turn a number of impressive carnivores, including the giant short-faced bear, the saber-tooth cat, and a large species of lion.

The presence of mammal species that require grassland vegetation has led Arctic biologist Dale Guthrie to argue that while cold and dry, there must have been broad areas of dense vegetation to support herds of mammoth, horse, and bison. Further, nearly all of the ice-age fauna had teeth that indicate an adaptation to grasses and sedges; they could not have been supported by a modern flora of mosses and lichens. Guthrie has also demonstrated that the landscape must have been subject to intense and continuous winds, especially in winter. He makes this argument based on the anatomy of horse and bison, which do not have the ability to search for food through deep snow cover. They need landscapes with strong winds that remove the winter snows, exposing the dry grasses beneath. Guthrie applied the term "mammoth steppe" to characterize this landscape.

In contrast, Paul Colinvaux has offered a counterargument based on the analysis of pollen in lake sediments dating to the last ice age. He found that the amount of pollen recovered in these sediments is so low that the Beringian landscape during the peak of the last glaciation was more

likely to have been what he termed a “polar desert,” with little or only sparse vegetation. In no way was it possible that this region could have supported large herds of mammals and thus, human hunters. Guthrie has argued against this view by pointing out that radiocarbon analysis of mammoth, horse, and bison bones from Beringian deposits revealed that the bones date to the period of most intense glaciation.

The argument seemed to be at a standstill until a number of recent studies resulted in a spectacular suite of new finds. The first was the discovery of a 1,000-square-kilometer preserved patch of Beringian vegetation dating to just over 17,000 years ago—the peak of the last ice age. The plants were preserved under a thick ash fall from a volcanic eruption. Investigations of the plants found grasses, sedges, mosses, and many other varieties in a nearly continuous cover, as was predicted by Guthrie. But this vegetation had a thin root mat with no soil formation, demonstrating that there was little long-term stability in plant cover, a finding supporting some of the arguments of Colinvaux. A mixture of continuous but thin vegetation supporting herds of large mammals is one that seems plausible and realistic with the available data.

Paragraph 1

During the peak of the last ice age, northeast Asia (Siberia) and Alaska were connected by a broad land mass called the Bering Land Bridge. This land bridge existed because so much of Earth’s water was frozen in the great ice sheets that sea levels were over 100 meters lower than they are today. Between 25,000 and 10,000 years ago, Siberia, the Bering Land Bridge, and Alaska shared many environmental characteristics. These included a common mammalian fauna of large mammals, a common flora composed of broad grasslands as well as wind-swept dunes and tundra, and a common climate with cold, dry winters and somewhat warmer summers. The recognition that many aspects of the modern flora and fauna were present on both sides of the Bering Sea as remnants of the ice-age landscape led to this region being named Beringia.

1. The word “remnants” in the passage is closest in meaning to
 - ☐ remains
 - ☐ evidence
 - ☐ results
 - ☐ reminders

Paragraph 2

It is through Beringia that small groups of large mammal hunters, slowly expanding their hunting territories, eventually colonized North and South America. On this archaeologists generally agree, but that is where the agreement stops. One broad area of disagreement in expanding the peopling of the Americas is the domain of paleoecologists, but it is critical to understanding human history: what was Beringia like?

2. The word “domain” in the passage is closest in meaning to
 - ☐ field of expertise
 - ☐ challenge
 - ☐ interest

- responsibility

Paragraph 3

The Beringian landscape was very different from what it is today. Broad, windswept valleys, glaciated mountains, sparse vegetation, and less moisture created a rather forbidding land mass. This land mass supported herds of now-extinct species of mammoth, bison, and horse and somewhat modern versions of caribou, musk ox, elk, and saiga antelope 赛加羚羊. These grazers supported in turn a number of impressive carnivores, including the giant short-faced bear, the saber-tooth 上犬齿的 ; 有军刀形的 cat, and a large species of lion.

3. According to paragraph 3, all of the following are true of the Beringian landscape EXCEPT:

- There was little vegetation.
- The mammal species there all survived into modern versions.
- The climate was drier than it is today.
- There were mountains with glaciers.

Paragraph 3 is marked with an arrow [→]

4. The purpose of paragraph 3 is to

- contrast today's Beringian landscape with other landscapes in the American continent
- describe the Beringian landscape during the last ice age
- explain why so many Beringian species became extinct during the last ice age
- summarize the information about Beringia that historians agree on

Paragraph 3 is marked with an arrow [→]

Paragraph 4

The presence of mammal species that require grassland vegetation has led Arctic biologist Dale Guthrie to argue that while cold and dry, there must have been broad areas of dense vegetation to support herds of mammoth, horse, and bison. Further, nearly all of the ice-age fauna had teeth that indicate an adaptation to grasses and sedges; they could not have been supported by a modern flora of mosses and lichens. Guthrie has also demonstrated that the landscape must have been subject to intense and continuous winds, especially in winter. He makes this argument based on the anatomy of horse and bison, which do not have the ability to search for food through deep snow cover. They need landscapes with strong winds that remove the winter snows, exposing the dry grasses beneath. Guthrie applied the term "mammoth steppe" 大草原 to characterize this landscape.

5. The word "continuous" in the passage is closest in meaning to

- unpredictable
- very cold
- dangerous
- uninterrupted

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave

out essential information

- ☐ According to biologist Dale Guthrie, mammal species require broad areas of vegetation to survive.
 - ☐ Dale Guthrie is an Arctic biologist who argued that broad areas of dense vegetation were surely enough to attract mammals such as mammoth, horse, and bison to Beringia.
 - ☐ Dale Guthrie argued that Beringia, though cold and dry, must have had enough dense vegetation to support the herds of mammoth, horse, and bison that lived there.
 - ☐ As long as Beringia was cold and dry, argued Dale Guthrie, dense vegetation grew in order to support the herds of mammoth, horse, and bison—the mammal species present there.
7. According to paragraph 4, Guthrie believes that the teeth of ice-age fauna support which of the following conclusions?
- ☐ Large mammals would not have been able to survive in the Beringian landscape
 - ☐ Grasslands were part of the Beringian landscape.
 - ☐ Strong winds exposed dry grasses under the snow.
 - ☐ Horses and bison did not have the ability to search for food through deep snow cover.

Paragraph 4 is marked with an arrow [→]

8. According to paragraph 4, which of the following statements is true of the relationship between ice-age Beringian animals and their environment?
- ☐ When present in sufficient quantities, lichens and mosses provide enough nutrients to satisfy the needs of herds of large mammals.
 - ☐ The anatomy of certain animals present in that environment provides information about the intensity of winds there at that time.
 - ☐ The structure of the teeth of most ice-age fauna indicates that they preyed on animals such as the mammoth, horse, and bison.
 - ☐ Horses and bison are large enough that their feet can easily penetrate deep snow and uncover areas where they can feed on plant material.

Paragraph 5

In contrast, Paul Colinvaux has offered a counterargument based on the analysis of pollen in lake sediments dating to the last ice age. He found that the amount of pollen recovered in these sediments is so low that the Beringian landscape during the peak of the last glaciation was more likely to have been what he termed a “polar desert,” with little or only sparse vegetation. In no way was it possible that this region could have supported large herds of mammals and thus, human hunters. Guthrie has argued against this view by pointing out that radiocarbon analysis of mammoth, horse, and bison bones from Beringian deposits revealed that the bones date to the period of most intense glaciation.

9. In paragraph 5, the amount of pollen in Beringian lake sediments from the last ice age is used to explain
- ☐ how long the ice age lasted
 - ☐ how important pollen is as a source of food
 - ☐ how many different kinds of plants produce pollen

- how little vegetation must have been present at that time

Paragraph 5 is marked with an arrow [→]

10. According to paragraph 5, how did Dale Guthrie use the information about radiocarbon analysis of bones from Beringian deposits?

- To suggest that Colinvaux should have used different methods to measure the amount of pollen in ice-age lake sediments
- To argue that the large Beringian mammals must have eaten plants that produce little, if any, pollen
- To show that the conclusions that Colinvaux drew from the analysis of pollen in ice-age lake sediments cannot be correct
- To explain why so-called polar deserts are incapable of supporting such large animals as mammoth, horse, and bison

Paragraph 5 is marked with an arrow [→]

Paragraph 6

The argument seemed to be at a **standstill** 停顿 until a number of recent studies resulted in a spectacular suite of new finds. The first was the discovery of a 1,000-square-kilometer preserved patch of Beringian vegetation dating to just over 17,000 years ago—the peak of the last ice age. The plants were preserved under a thick ash fall from a volcanic eruption. Investigations of the plants found grasses, **sedges** 沙草, mosses, and many other varieties in a nearly continuous cover, as was predicted by Guthrie. But this vegetation had a thin root mat with no soil formation, demonstrating that there was little long-term stability in plant cover, a finding supporting some of the arguments of Colinvaux. A mixture of continuous but thin vegetation supporting herds of large mammals is one that seems plausible and realistic with the available data.

11. The word “plausible” in the passage is closest in meaning to

- understandable
- believable
- sustainable
- necessary

12. Which of the following best describes the organization of paragraph 6?

- Two contrasting views are presented, and a study that could decide between them is proposed.
- An argument is offered, and reasons both for and against the argument are presented.
- A claim is made, and a study supporting the claim is described.
- New information is presented, and the information is used to show that two competing explanations can each be seen as correct in some way

Paragraph 6 is marked with an arrow [→]

Paragraph 3

The Beringian landscape was very different from what it is today. ■ Broad, windswept valleys, glaciated mountains, sparse vegetation, and less moisture created a rather forbidding land mass.

■ This land mass supported herds of now-extinct species of mammoth, bison, and horse and somewhat modern versions of caribou, mush ox, elk, and saiga antelope. ■ These grazers supported in turn a number of impressive carnivores, including the giant short-faced bear, the saber-tooth cat, and a large species of lion. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Nevertheless, large animals managed to survive in Beringia.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

During the last ice age, human hunters pursued large animals across Beringia, a land whose climatic characteristics have been in dispute.

-
-
-

Answer choices

Strong evidence indicates that large mammals like mammoth and bison survived in the harsh ice-age Beringian landscape.

Carnivores such as the saber-tooth cat were primarily responsible for the disappearance of the largest of the grazing animals, but the harsh winters caused some grazers to die of starvation.

The discovery that grasses, sedges, and mosses survived under the thick ash from a large volcanic eruption proved that the ice-age Beringian plant cover was extremely resistant to climatic extremes.

Beringian mammals crossed easily from northeastern Asia to Alaska across the Bering Land Bridge, though there are indications that they usually went back to Asia for the brief, but warm, summers.

Analysis of ice-age sediments uncovered very small amounts of pollen, suggesting that Beringia lacked the quantity of vegetation needed to support large herds of mammals.

Recent discoveries suggested that shallow-rooted plants created a fairly continuous cover over ice-age Beringia, though the cover most likely was variable and uncertain in any one location.

Wind pollination

Pollen, a powdery substance, which is produced by flowering plants and contains male reproductive cells, is usually carried from plant to plant by insects or birds, but some plants rely on the wind to carry their pollen. Wind pollination is often seen as being primitive and wasteful in costly pollen and yet it is surprisingly common, especially in higher latitudes. Wind is very good at moving pollen a long way; pollen can be blown for hundreds of kilometers, and only birds can get pollen anywhere near as far. The drawback is that wind is obviously unspecific as to where it takes the pollen. It is like trying to get a letter to a friend at the other end of the village by climbing onto the roof and throwing an armful of letters into the air and hoping that one will end up in the friend's garden. For the relatively few dominant tree species that make up temperate forests, where there are many individuals of the same species within pollen range, this is quite a safe gamble. If a number of people in the village were throwing letters off roofs, your friend would be bound to get one. By contrast, in the tropics, where each tree species has few, widely scattered individuals, the chance of wind blowing pollen to another individual is sufficiently slim that animals are a safer bet as transporters of pollen. Even tall trees in the tropics are usually not wind pollinated despite being in windy conditions. In a similar way, trees in temperate forests that are insect pollinated tend to grow as solitary, widely spread individuals.

Since wind-pollinated flowers have no need to attract insects or other animals, they have dispensed with bright petals, nectar, and scent. These are at best a waste and at worst an impediment to the transfer of pollen in the air. The result is insignificant-looking flowers and catkins (dense cylindrical clusters of small, petalless flowers).

Wind pollination does, of course, require a lot of pollen. Birch and hazel trees can produce 5.5 and 4 million grains per catkin, respectively. There are various adaptations to help as much of the pollen go as far as possible. Most deciduous wind-pollinated trees (which shed their leaves every fall) produce their pollen in the spring while the branches are bare of leaves to reduce the surrounding surfaces that "compete" with the stigmas (the part of the flower that receives the pollen) for pollen. Evergreen conifers, which do not shed their leaves, have less to gain from spring flowering, and, indeed, some flower in the autumn or winter.

Pollen produced higher in the top branches is likely to go farther, it is windier (and gustier) and the pollen can be blown farther before hitting the ground. Moreover, dangling catkins like hazel hold the pollen in until the wind is strong enough to bend them, ensuring that pollen is only shed into the air when the wind is blowing hard. Weather is also important. Pollen is shed primarily when the air is dry to prevent too much sticking to wet surfaces or being knocked out of the air by rain. Despite these adaptations, much of the pollen fails to leave the top branches, and only between 0.5 percent and 40 percent gets more than 100 meters away from the parent. But once this far, significant quantities can go a kilometer or more. Indeed, pollen can travel many thousands of kilometers at high altitudes. Since all this pollen is floating around in the air, it is no wonder that wind-pollinated trees are a major source of allergies.

Once the pollen has been snatched by the wind, but not everything is left to chance. Windborne pollen is dry, rounded, smooth, and generally smaller than that of insect-pollinated plants. But size is a two-edged sword. Small grains may be blown farther but they are also more prone to be whisked past the waiting stigma because smaller particles tend to stay trapped in the fast-moving air that flows around the stigma. But stigmas create turbulence, which slows the air speed around them and may help pollen stick to them.

Paragraph 1

Pollen, a powdery substance, which is produced by flowering plants and contains male reproductive cells, is usually carried from plant to plant by insects or birds, but some plants rely on the wind to carry their pollen. Wind pollination is often seen as being primitive and wasteful in costly pollen and yet it is surprisingly common, especially in higher **latitudes**. Wind is very good at moving pollen a long way; pollen can be blown for hundreds of kilometers, and only birds can get pollen anywhere near as far. The **drawback** is that wind is obviously unspecific as to where it takes the pollen. It is like trying to get a letter to a friend at the other end of the village by climbing onto the roof and throwing an armful of letters into the air and hoping that one will end up in the friend's garden. For the relatively few dominant tree species that make up temperate forests, where there are many individuals of the same species within pollen range, this is quite a safe gamble. If a number of people in the village were throwing letters off roofs, your friend would be bound to get one. By contrast, in the tropics, where each tree species has few, widely scattered individuals, the chance of wind blowing pollen to another individual is sufficiently slim that animals are a safer bet as transporters of pollen. Even tall trees in the tropics are usually not wind pollinated despite being in windy conditions. In a similar way, trees in temperate forests that are insect pollinated tend to grow as solitary, widely spread individuals.

1. The word "drawback" in the passage is closest in meaning to
 - ☐ other side of the issue
 - ☐ objection
 - ☐ concern
 - ☐ problem
2. Which of the following can be inferred from paragraph 1 about pollen production?
 - ☐ Pollen production requires a significant investment of energy and resources on the part of the plant.
 - ☐ The capacity to produce pollen in large quantities is a recent development in the evolutionary history of plants.
 - ☐ Plants in the tropics generally produce more pollen than those in temperate zones.
 - ☐ The highest levels of pollen production are found in plants that depend on insects or birds to carry their pollen.

Paragraph 1 is marked with an arrow [→]

3. According to paragraph 1, wind-pollinated trees are most likely to be found
 - ☐ in temperate forests safe gamble安全的赌博

- ☐ at lower latitudes
- ☐ in the tropics
- ☐ surrounded by trees of many different species

Paragraph 1 is marked with an arrow [→]

4. Paragraph 1 supports which of the following as the reason animals are a safer bet than wind as pollinators when the individual trees of a species are widely separated?

- ☐ Animals tend to carry pollen from a given flower further than the wind does.
- ☐ Animals serve as pollinators even where there is little wind to disperse the pollen.
- ☐ An animal that visits a flower is likely to deliberately visit other flowers of the same species and pollinate them.
- ☐ Birds and insects fly in all directions, not just the direction the wind is blowing at a given moment.

Paragraph 1 is marked with an arrow [→]

5. In paragraph 1, the author compares pollen moved by wind with letters thrown off roofs in order to

- ☐ explain why there are relatively few species of trees that depend on wind pollination
- ☐ compare natural, biological processes with human social practices
- ☐ make a point about the probability of wind-blown pollen reaching a tree of the same species
- ☐ argue against the common assumption that the tallest trees are the most likely to employ wind pollination

Paragraph 1 is marked with an arrow [→]

Paragraph 2

Since wind-pollinated flowers have no need to attract insects or other animals, they have dispensed with bright petals, nectar, and scent. These are at best a waste and **at worst an impediment**阻碍 to the transfer of pollen in the air. The result is insignificant-looking flowers and catkins (dense cylindrical clusters of small, petalless flowers).

6. Paragraph 2 suggests that wind-pollinated plants do not have bright petals, nectar, and scent for which TWO of the following reasons? To receive credit, you must select TWO answers.

- ☐ They interfere with pollination by wind. 它们干扰风授粉
- ☐ They are easily damaged by wind.
- ☐ They are unnecessary.
- ☐ They reduce the amount of pollen that can be produced.

Paragraph 3

Wind pollination does, of course, require a lot of pollen. Birch and hazel trees can produce 5.5 and 4 million grains per catkin, **respectively**. There are various adaptations to help as much of the pollen go as far as possible. Most **deciduous**落叶性的, 非永久性的 wind-pollinated trees (which shed their leaves every fall) produce their pollen in the spring while the branches are bare of leaves to reduce the surrounding surfaces that “compete” with the **stigmas** (the part of the flower that receives the pollen) for pollen. Evergreen conifers, which do not shed their leaves, have less to gain from

spring flowering, and, indeed, some flower in the autumn or winter.

7. The word “respectively” in the passage is closest in meaning to
- ☐ over time
 - ☐ separately 独立地
 - ☒ in that order 以那样的顺序，分别地
 - ☐ consistently
8. According to paragraph 3, why do most deciduous wind-pollinated trees produce their pollen in the spring?
- ☐ To avoid competing with evergreen conifers, which flower in the fall or winter
 - ☐ So that the leaves of the trees receiving the pollen will not prevent the pollen from reaching the trees’ stigmas
 - ☐ Because they do not have enough energy to produce new leaves and pollen at the same time
 - ☐ In order to take advantage of the windiest time of year

Paragraph 3 is marked with an arrow [→]

Paragraph 4

Pollen produced higher in the top branches is likely to go farther, it is windier (and **gustier**) and the pollen can be blown farther before hitting the ground. Moreover, **dangling** catkins like hazel hold the pollen in until the wind is strong enough to **ben** them, ensuring that pollen is only shed into the air when the wind is blowing hard. Weather is also important. Pollen is shed primarily when the air is dry to prevent too much sticking to wet surfaces or being knocked out of the air by rain. Despite these adaptations, much of the pollen fails to leave the top branches, and only between 0.5 percent and 40 percent gets more than 100 meters away from the parent. But once this far, significant quantities can go a kilometer or more. Indeed, pollen can travel many thousands of kilometers at high altitudes. Since all this pollen is floating around in the air, it is no **wonder** that wind-pollinated trees are a major source of allergies.

9. According to paragraph 4, which of the following is NOT an adaptation that helps ensure that pollen travels as far as possible?
- ☐ Pollen-producing flowers and catkins are located at or near the top of the tree.
 - ☐ Trees grow at least 100 meters away from each other.
 - ☐ Dangling catkins release pollen only when the wind is blowing hard.
 - ☐ Pollen is not released during rain storms or when the air is damp.

Paragraph 4 is marked with an arrow [→]

10. The word “significant” in the passage is closest in meaning to
- ☐ sufficient
 - ☐ considerable
 - ☐ increasing
 - ☐ small

11. The phrase “no wonder” in the passage is closest in meaning to

- ☐ unsurprising
- ☐ understandable
- ☐ well-known
- ☐ unfortunate

Paragraph 5

Once the pollen has been snatched by the wind, but not everything is left to chance. Windborne pollen is dry, rounded, smooth, and generally smaller than that of insect-pollinated plants. But size is a two-edged sword. Small grains may be blown farther but they are also more prone to be whisked past the waiting stigma because smaller particles tend to stay trapped in the fast-moving air that flows around the stigma. But stigmas create turbulence, which slows the air speed around them and may help pollen stick to them.

whisk 扫，搅动；stigma 柱头

12. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Because smaller particles tend to stay trapped in the fast-moving air, they are blown much farther than other grains.
- ☐ Smaller particles are trapped by the stigma when fast-moving air flows past it.
- ☐ Small particles that are whisked past the waiting stigma gain speed and are often trapped in the fast-moving air.
- ☐ While smallness helps pollen travel farther, it also makes it more likely to be blown past the stigma. 之所以选D不选C，是因为D有因果关系while，这是核心，和原文句子契合

Paragraph 3

Wind pollination does, of course, require a lot of pollen. ■ Birch and hazel trees can produce 5.5 and 4 million grains per catkin, respectively. ■ There are various adaptations to help as much of the pollen go as far as possible. ■ Most deciduous wind-pollinated trees (which shed their leaves every fall) produce their pollen in the spring while the branches are bare of leaves to reduce the surrounding surfaces that “compete” with the stigmas (the part of the flower that receives the pollen) for pollen. ■ Evergreen conifers, which do not shed their leaves, have less to gain from spring flowering, and, indeed, some flower in the autumn or winter.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

This level of volume is important to ensure that at least some of the pollen reaches a target tree, but dispersing the pollen is crucial as well.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

15. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because

they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Some plants depend on the wind to carry their pollen.

-
-
-

Answer choices

Because there are few trees in temperate forests, it is safer to transport pollen by insects or birds.

Wind pollination is a safe reproductive strategy for trees in temperate forests where there are only a few dominant species and, therefore, many individuals of the same species.

Wind pollination requires production of large amount of pollen, which must be released at the right time and under the right conditions to extend its range.

Most wind-pollinated trees are deciduous because evergreen needles compete with the stigma for pollen, making wind pollination uncertain.

Wind-pollinated plants usually have small petalless flowers which often grow in catkins that produce a very fine-grained pollen.

Wind-pollinated trees must grow in regions that are only moderately windy because strong winds will blow the tiny pollen grains past the stigma.

Feeding Strategies in the Ocean

In the open sea, animals can often find food reliably available in particular regions or seasons (e.g., in coastal areas in springtime). In these circumstances, animals are neither constrained to get the last calorie out of their diet nor is energy conservation a high priority. In contrast, the food levels in the deeper layers of the ocean are greatly reduced, and the energy constraints on the animals are much more severe. To survive at those levels, animals must maximize their energy input, finding and eating whatever potential food source may be present.

In the near-surface layers, there are many large, fast carnivores as well as an immense variety of planktonic animals, which feed on plankton (small, free-floating plants or animals) by filtering them from currents of water that pass through a specialized anatomical structure. These filter-feeders thrive in the well-illuminated surface waters because oceans have so many very small organisms, from bacteria to large algae to larval crustaceans. Even fishes can become successful filter-feeders in some circumstances. Although the vast majority marine fishes are carnivores, in near-surface regions of high productivity the concentrations of larger phytoplankton (the plant component of plankton) are sufficient to support huge populations of filter-feeding sardines and anchovies. These small fishes use their gill filaments to strain out the algae that dominate such areas. Sardines and anchovies provide the basis for huge commercial fisheries as well as a food resource for large numbers of local carnivores, particularly seabirds. At a much larger scale, baleen whales and whale sharks are also efficient filter-feeders in productive coastal or polar waters, although their filtered particles comprise small animals such as copepods and krill rather than phytoplankton.

Filtering seawater for its particulate nutritional content can be an energetically demanding method of feeding, particularly when the current of water to be filtered has to be generated by the organism itself, as is the case for all planktonic animals. Particulate organic matter of at least 2.5 micrograms per cubic liter is required to provide a filter-feeding planktonic organism with a net energy gain. This value is easily exceeded in most coastal waters, but in the deep sea, the levels of organic matter range from next to nothing to around 7 micrograms per cubic liter. Even though mean levels may mask much higher local concentrations, it is still the case that many deep-sea animals are exposed to conditions in which a normal filter-feeder would starve.

There are, therefore, fewer successful filter-feeders in deep water, and some of those that are there have larger filtering systems to cope with the scarcity of particles. Another solution for such animals is to forage in particular layers of water where the particles may be more concentrated. Many of the groups of animals that typify the filter-feeding lifestyle in shallow water have deep-sea representatives that have become predatory. Their filtering systems, which reach such a high degree of development in shallow-water species, are greatly reduced. Alternative methods of active or passive prey capture have been evolved, including trapping and seizing prey, entangling prey, and sticky tentacles.

In the deeper waters of the oceans, there is a much greater tendency for animals to await the

arrival of food particles or prey rather than to search them out actively (thus minimizing energy expenditure). This has resulted in a more stealthy style of feeding, with the consequent emphasis on lures and/or the evolution of elongated appendages that increase the active volume of water controlled or monitored by the animal. Another consequence of the limited availability of prey is that many animals have developed ways of coping with much larger food particles, relative to their own body size, than the equivalent shallower species can process. Among the fishes there is a tendency for the teeth and jaws to become appreciably enlarged. In such creatures, not only are the teeth hugely enlarged and/or the jaws elongated but the size of the mouth opening may be greatly increased by making the jaw articulations so flexible that they can be effectively dislocated. Very large or long teeth provide almost no room for cutting the prey into a convenient size for swallowing; the fish must gulp the prey down whole.

Paragraph 1

In the open sea, animals can often find food reliably available in particular regions or seasons (e.g., in coastal areas in springtime). In these circumstances, animals are neither constrained to get the last calorie out of their diet nor is energy conservation a high priority. In contrast, the food levels in the deeper layers of the ocean are greatly reduced, and the energy constraints on the animals are much more severe. To survive at those levels, animals must maximize their energy input, finding and eating whatever potential food source may be present.

1. The word “severe” in the passage is closest in meaning to
 - ☐ extreme
 - ☐ complex
 - ☐ basic
 - ☐ immediate
2. Which can be inferred from paragraph 1 about why energy conservation is not a high priority for ocean animals in coastal waters during the spring?
 - ☐ Those animals are least active during the spring.
 - ☐ Those animals have a plentiful supply of food.
 - ☐ Those animals have to expend energy to avoid predators.
 - ☐ Those animals store energy during the colder seasons.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

In the near-surface layers, there are many large, fast carnivores as well as an immense variety of planktonic animals, which feed on plankton (small, free-floating plants or animals) by **filtering** them from currents of water that pass through a specialized anatomical structure. These filter-feeders thrive in the well-illuminated surface waters because oceans have so many very small organisms, from bacteria to large algae to larval crustaceans. Even fishes can become successful filter-feeders in some circumstances. Although the vast majority marine fishes are carnivores, in near-surface regions of high productivity the concentrations of larger phytoplankton (the plant component of plankton) are sufficient to support huge populations of

filter-feeding sardines and anchovies. These small fishes use their gill filaments to strain out the algae that dominate such areas. Sardines and anchovies provide the basis for huge commercial fisheries as well as a food resource for large numbers of local carnivores, particularly seabirds. At a much larger scale, baleen whales and whale sharks are also efficient filter-feeders in productive coastal or polar waters, although their filtered particles comprise small animals such as copepods and krill rather than phytoplankton.

3. What can be inferred from paragraph 2 about fish?

- ☐ Most fish feed on plankton.
- ☐ Fish tend to avoid well-illuminated areas.
- ☐ Most fish species are not filter-feeders.
- ☐ Few fish species are successful in the near-surface layers.

Paragraph 2 is marked with an arrow [→]

4. According to paragraph 2, how do sardines and anchovies obtain food near the surface of the ocean?

- ☐ They rely on the large quantities of food resources also available to local carnivores.
- ☐ They capture the larvae of some crustaceans.
- ☐ They feed on the organisms left over by commercial fisheries.
- ☐ They obtain algae by using their gills as filters.

Paragraph 2 is marked with an arrow [→]

5. According to paragraph 2, in which of the following ways are baleen whales and whale sharks different from smaller filter-feeders like sardines and anchovies?

- ☐ They are not found in coastal waters.
- ☐ They are caught by commercial fisheries.
- ☐ They filter small animals instead of phytoplankton.
- ☐ They are not carnivores.

Paragraph 2 is marked with an arrow [→]

Paragraph 3

Filtering seawater for its particulate nutritional content can be an energetically demanding method of feeding, particularly when the current of water to be filtered has to be generated by the organism itself, as is the case for all planktonic animals. Particulate organic matter of at least 2.5 micrograms per cubic liter is required to provide a filter-feeding planktonic organism with a net energy gain. This value is easily exceeded in most coastal waters, but in the deep sea, the levels of organic matter range from next to nothing to around 7 micrograms per cubic liter. Even though mean levels may mask much higher local concentrations, it is still the case that many deep-sea animals are exposed to conditions in which a normal filter-feeder would starve.

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ An organism may obtain food particles from seawater when currents generated by the

organism cause water to pass through its filter.

- The nutritional content of the food particles in seawater is sufficient to meet the energy demands of most planktonic animals.
- As planktonic animals require a lot of energy, they have to generate a current strong enough to bring them sufficient particles of food.
- Organisms that filter seawater to meet their nutritional needs expend a lot of energy, especially those organisms that have to create their own currents.

Paragraph 4

There are, therefore, fewer successful filter-feeders in deep water, and some of those that are there have larger filtering systems to cope with the **scarcity** of particles. Another solution for such animals is to forage in particular layers of water where the particles may be more concentrated. Many of the groups of animals that typify the filter-feeding lifestyle in shallow water have deep-sea representatives that have become predatory. Their filtering systems, which reach such a high degree of development in shallow-water species, are greatly reduced. Alternative methods of active or passive prey capture have been evolved, including trapping and seizing prey, **entangling** prey, and sticky tentacles.

7. The word "**scarcity**" in the passage is closest in meaning to
 - speed
 - variety
 - lack
 - size
8. According to paragraph 4, deep-water filter-feeders have adopted all of the following ways to obtain food EXCEPT
 - developing larger filtering systems
 - capturing prey using sticky tentacles
 - swimming up to the surface at feeding time
 - searching in ocean layers that contain a substantial amount of particles

Paragraph 4 is marked with an arrow [→]

Paragraph 5

■ In the deeper waters of the oceans, there is a much greater tendency for animals to await the arrival of food particles or prey rather than to search them out actively (thus minimizing energy expenditure). ■ This has resulted in a more **stealthy** style of feeding, with the consequent **emphasis on lures** and/or the evolution of **elongated appendages** that increase the active volume of water controlled or monitored by the animal. ■ Another consequence of the limited availability of prey is that many animals have developed ways of **copied with** much larger food particles, relative to their own body size, than the equivalent shallower species can process. ■ Among the fishes there is a tendency for the teeth and jaws to become appreciably enlarged. In such creatures, not only are the teeth hugely enlarged and/or the jaws elongated but the size of the mouth opening may be greatly increased by making the jaw articulations so **flexible** that they can be effectively dislocated. Very large or long teeth provide almost no room for cutting the prey

into a convenient size for swallowing; the fish must **gulp**狼吞虎咽地吃 the prey down whole.

9. Why does the author include the information that animals in the deep ocean place an “**emphasis on lures**” and have evolved “**elongated appendages**”? 附气看BC
- ☐ To argue against the view that animals in the deep ocean use more energy to find food than do animals in shallow waters
 - ☐ To emphasize the importance of an animal’s ability to control a large volume of water
 - ☐ To identify some feeding strategies that animals have developed to minimize their energy expenditure
 - ☐ To give examples of body structures that help those animals move quickly in deep ocean waters
10. The phrase “**copied with**” in the passage is closest in meaning to
- ☐ absorbing
 - ☐ finding
 - ☐ approaching
 - ☐ managing
11. The word “**flexible**” in the passage is closest in meaning to
- ☐ huge
 - ☐ adaptable
 - ☐ powerful
 - ☐ precise
12. According to paragraph 5, why do some fish swallow their prey whole?
- ☐ Their **teeth** are too large to allow for cutting prey.
 - ☐ They have no jaw muscles to allow chewing.
 - ☐ Swallowing prey whole results in a higher net energy gain.
 - ☐ Chewing can cause their jaws to dislocate.
- Paragraph 5 is marked with an arrow [→]

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Such a passive approach requires those animals to develop specialized methods of acquiring food.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

16. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Ocean animals have developed various strategies for maximizing energy input from food.

-
-
-

Answer choices

Large carnivores near the ocean surface feed mainly on organic matter left over after it passes through the filters of the filter-feeders.

Even in deep ocean layers where prey is relatively hard to find, filter feeding is still the least energetically-demanding method of obtaining food.

Animals in deeper water have evolved strategies and body structures that allow them to use as little energy as possible in obtaining food.

Near the surface of the water, many animals obtain food by using specialized body parts to filter plankton from the water.

Filter-feeding is more common in shallow water, where there is a higher concentration of organic matter than there is in deeper water.

At deeper ocean levels plankton is relatively rare, requiring animals at those levels to actively search for their food sources.

2014-03-23

From Fish to Terrestrial Vertebrates

One of the most significant evolutionary events that occurred on Earth was the transition of water-dwelling fish to terrestrial tetrapods (four-limbed organisms with backbones). Fish probably originated in the oceans, and our first records of them are in marine rocks. However, by the Devonian Period (408 million to 362 million years ago), they had radiated into almost all available aquatic habitats, including freshwater settings. One of the groups whose fossils are especially common in rocks deposited in fresh water is the lobe-finned fish.

The freshwater Devonian lobe-finned fish rhipidistian crossopterygian is of particular interest to biologists studying tetrapod evolution. These fish lived in river channels and lakes on large deltas. The delta rocks in which these fossils are found are commonly red due to oxidized iron minerals, indicating that the deltas formed in a climate that had alternate wet and dry periods. If there were periods of drought, any adaptations allowing the fish to survive the dry conditions would have been advantageous. In these rhipidistians, several such adaptations existed. It is known that they had lungs as well as gills for breathing. Cross sections cut through some of the fossils reveal that the mud filling the interior of the carcass differed in consistency and texture depending on its location inside the fish. These differences suggest a saclike cavity below the front end of the gut that can only be interpreted as a lung. Gills were undoubtedly the main source of oxygen for these fish, but the lungs served as an auxiliary breathing device for gulping air when the water became oxygen depleted, such as during extended periods of drought. So, these fish had already evolved one of the prime requisites for living on land: the ability to use air as a source of oxygen.

A second adaptation of these fish was in the structure of the lobe fins. The fins were thick, fleshy, and quite sturdy, with a median axis of bone down the center. They could have been used as feeble locomotor devices on land, perhaps good enough to allow a fish to flop its way from one pool of water that was almost dry to an adjacent pond that had enough water and oxygen for survival. These fins eventually changed into short, stubby legs. The bones of the fins of a Devonian rhipidistian exactly match in number and position the limb bones of the earliest known tetrapods, the amphibians. It should be emphasized that the evolution of lungs and limbs was in no sense an anticipation of future life on land. These adaptations developed because they helped fish to survive in their existing aquatic environment.

What ecological pressures might have caused fishes to gradually abandon their watery habitat and become increasingly land-dwelling creatures? Changes in climate

during the Devonian may have had something to do with this if freshwater areas became progressively more restricted. Another impetus may have been new sources of food. The edges of ponds and streams surely had scattered dead fish and other water-dwelling creatures. In addition, plants had emerged into terrestrial habitats in areas near streams and ponds, and crabs and other arthropods were also members of this earliest terrestrial community. Thus, by the Devonian the land habitat marginal to freshwater was probably a rich source of protein that could be exploited by an animal that could easily climb out of water. Evidence from teeth suggests that these earliest tetrapods did not utilize land plants as food; they were presumably carnivorous and had not developed the ability to feed on plants.

How did the first tetrapods make the transition to a terrestrial habitat? Like early land plants such as rhyniophytes, they made only a partial transition; they were still quite tied to water. However, many problems that faced early land plants were not applicable to the first tetrapods. The ancestors of these animals already had a circulation system, and they were mobile, so that they could move to water to drink. Furthermore, they already had lungs, which rhipidistians presumably used for auxiliary breathing. The principal changes for the earliest tetrapods were in the skeletal system—changes in the bones of the fins, the vertebral column, pelvic girdle, and pectoral girdle.

Paragraph 1

One of the most significant evolutionary events that occurred on Earth was the transition of water-dwelling fish to terrestrial tetrapods (four-limbed organisms with backbones). Fish probably originated in the oceans, and our first records of them are in marine rocks. However, by the Devonian Period (408 million to 362 million years ago), they had radiated into almost all available aquatic habitats, including freshwater settings. 然而，到了泥盆纪（4.08亿至3.62亿年前），鱼类的栖息地已经扩展到几乎所有可能的水生环境，包括淡水水域。One of the groups whose fossils are especially common in rocks deposited in fresh water is the lobe-finned fish. 淡水沉积岩石中十分常见的是一种叶状鳍鱼化石。

1.Paragraph 1 supports which of the following statements about fish evolution?

- ☐ Lobe-finned fish were among the earliest types of fish to appear.
- ☒ Fish began living in freshwater habitats only after originating elsewhere.
- ☐ Lobe-finned fish radiated into almost all available aquatic habitats.
- ☐ During the Devonian, lobe-finned fish were more common in marine than in freshwater habitats.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

The freshwater Devonian lobe-finned fish rhipidistian crossopterygian is of particular interest to biologists studying tetrapod evolution. These fish lived in river channels and lakes on large deltas. The delta rocks in which these fossils are found are commonly red due to oxidized iron minerals, indicating that the deltas formed in a

climate that had alternate wet and dry periods. If there were periods of drought, any adaptations allowing the fish to survive the dry conditions would have been **advantageous**. In these rhipidistians, several such adaptations existed. It is known that they had lungs as well as gills for breathing. Cross sections cut through some of the fossils reveal that the mud filling the interior of the carcass differed in consistency and texture depending on its location inside the fish. These differences suggest a saclike cavity below the front end of the gut that can only be interpreted as a lung. **Gills were undoubtedly the main source of oxygen for these fish, but the lungs served as an auxiliary breathing device for gulping air when the water became oxygen depleted, such as during extended periods of drought.** So, these fish had already evolved one of the prime requisites for living on land: the ability to use air as a source of oxygen.

2. The word “**advantageous**” in the passage is closest in meaning to

- ☐ beneficial
- ☐ necessary
- ☐ available
- ☐ common

3.

4. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Because the lungs of these fish were able to provide only a small amount of oxygen, these fish obtained most of their oxygen through their gills during periods of drought.
- ☐ During periods of extended drought, these fish used their lungs to increase their intake of oxygen beyond the levels absorbed by the gills in normal times.
- ☐ Although these fish primarily used their gills to obtain oxygen, they used their lungs to obtain oxygen from the air when there was not enough in the water.
- ☐ During periods of extended drought, the gills became an auxiliary breathing device and the lungs became the main source of oxygen for these fish.

Paragraph 3

A second adaptation of these fish was in the structure of the lobe fins. The fins were thick, fleshy, and quite sturdy, with a median axis of bone down the center. They could have been used as feeble locomotor devices on land, perhaps good enough to allow a fish to flop its way from one pool of water that was almost dry to an **adjacent** pond that had enough water and oxygen for survival. These fins eventually changed into short, stubby legs. The bones of the fins of a Devonian rhipidistian exactly match in number and position the limb bones of the earliest known tetrapods, the amphibians. It should be emphasized that the evolution of lungs and limbs was in no sense an anticipation of future life on land. These adaptations developed because

they helped fish to survive in their existing aquatic environment.

5. The word “adjacent” in the passage is closest in meaning to

- ☐ nearby
- ☐ available
- ☐ temporary
- ☐ fresh

Paragraph 4

What ecological pressures might have caused fishes to gradually abandon their watery habitat and become increasingly land-dwelling creatures? Changes in climate during the Devonian may have had something to do with this if freshwater areas became progressively more restricted. Another impetus may have been new sources of food. The edges of ponds and streams surely had scattered dead fish and other water-dwelling creatures. In addition, plants had emerged into terrestrial habitats in areas near streams and ponds, and crabs and other arthropods were also members of this earliest terrestrial community. Thus, by the Devonian the land habitat marginal to freshwater was probably a rich source of protein that could be exploited by an animal that could easily climb out of water. Evidence from teeth suggests that these earliest tetrapods did not utilize land plants as food; they were presumably carnivorous and had not developed the ability to feed on plants.

6. The word “increasingly” in the passage is closest in meaning to

- ☐
- ☐
- ☐
- ☐

7. In paragraph 4, why does the author point out that crabs and other arthropods were already living on land when the ancestors of the first tetrapods began living there?

- ☐ To account for the presence of dead fish along the edges of ponds and streams during the Devonian
- ☐ To support the claim that climate change caused freshwater habitats to become more restricted during the Devonian
- ☐ To identify a consequence of the emergence of plants into terrestrial habitats near ponds and streams
- ☐ To identify a possible reason for why certain fish gradually became terrestrial organisms

Paragraph 4 is marked with an arrow [→]

Paragraph 5

How did the first tetrapods make the transition to a terrestrial habitat? Like early land plants such as rhyniophytes, they made only a partial transition; they were still

quite tied to water. However, many problems that faced early land plants were not applicable 合适的, 适用的 to the first tetrapods. 不过, 早期陆生植物面临的许多问题并不会困扰到它们。 The ancestors of these animals already had a circulation system, and they were mobile, so that they could move to water to drink. Furthermore, they already had lungs, which rhipidistians presumably used for auxiliary breathing. The principal changes for the earliest tetrapods were in the skeletal system—changes in the bones of the fins, the vertebral column, pelvicadj. (靠近) 骨盆的【解剖】 girdle, and pectoral adj. 胸部的 girdlen. 腰带, 围绕物 vt. 以腰带束缚, 围绕.

8. According to paragraph 5, which of the following was true of the first tetrapods?

- ☐ They became dependent for food on organisms already living on land.
- ☐ They needed to develop new mechanisms for obtaining nutrients.
- ☐ They continued to live in close association with aquatic environments.
- ☐ They were evolutionarily far removed from their rhipidistian ancestors.

far cry from

Paragraph 5 is marked with an arrow [→]

9. According to paragraph 5, what was the main way that the earliest tetrapods differed from their immediate fish ancestors?

- ☐ The tetrapods had a different skeletal structure.
- ☐ The tetrapods had more sources of food available.
- ☐ The tetrapods had a circulation system.
- ☐ The tetrapods could move to new pools of water.

Paragraph 5 is marked with an arrow [→]

Paragraph 4

What ecological pressures might have caused fishes to gradually abandon their watery habitat and become increasingly land-dwelling creatures? Changes in climate during the Devonian may have had something to do with this if freshwater areas became progressively more restricted. Another impetus may have been new sources of food. The edges of ponds and streams surely had scattered dead fish and other water-dwelling creatures. ■ In addition, plants had emerged into terrestrial habitats in areas near streams and ponds, and crabs and other arthropods were also members of this earliest terrestrial community. ■ Thus, by the Devonian the land habitat marginal to freshwater was probably a rich source of protein that could be exploited by an animal that could easily climb out of water. ■ Evidence from teeth suggests that these earliest tetrapods did not utilize land plants as food; they were presumably carnivorous and had not developed the ability to feed on plants. ■

10. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

These would have been deposited by the receding waters of droughts, during which many aquatic animals must have died.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

11. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Freshwater lobe-finned fish may be the direct ancestors of terrestrial tetrapods.

-
-
-

Answer choices

Rhipidistian crossopterygian had features such as primitive lungs and thick fins that could have helped it survive dry periods.

During the Devonian, the number of bones increased in the fins of rhipidistians, improving such animals' ability to swim and move over land.

Shortly after the earliest tetrapods developed lungs, plants and other animals began to flourish on land.

By the Devonian period, lobe-finned fish preferred freshwater habitats to life in the ocean.选项 D 中的细节信息错误, 原文并没有对 lobe-finned fish 的栖息环境进行喜好比较

A drier climate and new sources of food on land may have encouraged the lobe-finned fish's move to a terrestrial existence.

Early tetrapods remained closely connected to water, but several of their body structures were adapted for life on land.

Early Modern Industrialization

Industrial output increased smartly across nearly all of Europe between 1450 and 1575. Although trade with the Americas had something to do with this, the main determinants of this industrial advance lay within Europe itself.

Population grew from 61 million in 1500 to 78 million a century later, and the proportion of Europeans living in cities of 10,000 or more—and thus dependent on the market for what they consumed—expanded from less than 6 percent to nearly 8 percent during the same period. More important than sheer numbers, many Europeans' incomes rose. This was especially true among more fully employed urban groups, farmers who benefited from higher prices and the intensifying commercialization and specialization in agriculture (which also led them to shed much non-agricultural production in favor of purchased goods), and landlords and other property owners who collected mounting rents. Government activities to build and strengthen the state were a stimulus to numerous industries, notably shipbuilding, textiles, and metallurgy. To cite just one example, France hastened to develop its own iron industry when the Hapsburgs—the family that governed much of Europe, and whom France fought repeatedly in the sixteenth century—came to dominate the manufacture of weapons in Germany and the cities of Liege and Milan, which boasted Europe's most advanced technology.

The supply of goods was also significantly modified. Migration had long been critical for the diffusion of knowledge that spawned new trades or revived others. Now thousands of workers, and sizeable amounts of capital, moved from one region to another. At the same time, new commodities appeared on the market, often broadening and deepening demand. Most were inexpensive items destined for individual consumers. Knitted stockings, ribbon and lace, buttons, starch, soap, vinegar brewed from beer, knives and tools, pots and ovens, and many more goods, formerly made only for local sale, now entered into channels of national or international trade. The best-known and most widely adopted new industry was printing with movable type, which spread swiftly throughout Europe after Johannes Gutenberg perfected his innovation in 1453. Despite isolated cases of resistance—the scribes' guild (an association of book copiers) delayed printing's introduction into Paris for twenty years, for example—more than 380 working presses had sprung up by 1480, and 1,000 (in nearly 250 towns) by 1500. Between 1453 and 1500, all the presses of Europe together turned out some 40,000 editions (known as incunabula), but from 1501 to 1600, that same quantity was produced in Lyon and Paris alone.

In metals and mining, technical improvements were available that saved substantially on raw materials and fuel, causing prices to drop. The construction of ever-larger furnaces capable of higher temperatures culminated in the blast furnace, which used

cheaper ores and economized on scarce and expensive wood, cutting costs per ton by 20 percent while boosting output substantially. A new technique for separating silver from copper allowed formerly worthless ores to be exploited. Better drainage channels, pumps, and other devices made it possible to tunnel more deeply into the earth as surface deposits began to be exhausted. In most established industries, however, technological change played little role, as in the past, new customers were sought by developing novel products based on existing technologies, such as a new type of woolen cloth with the texture of silk.

Sharply declining transaction costs (the direct and indirect expenses associated with transporting, distributing, and marketing goods and services) were more influential. On a general level, the decrease was due to greater security thanks to the lessening of wartime disruptions and to the economies of scale achieved when selling to large, concentrated urban populations. More specifically, it can be traced to transport innovations such as the carrack, a large ship that reduced rates for oceanborne freight by up to 25 percent, and big four-wheeled Hesse carts for overland routes. The spread of efficient organizational forms further contributed to declining costs, as did falling interest rates, which dropped from 20 percent or 25 percent in the mid-fifteenth century to 10 percent 100 years later.

Paragraph 1

Industrial output increased smartly across nearly all of Europe between 1450 and 1575. Although trade with the Americas had something to do with this, the main determinants of this industrial advance lay within Europe itself.

1. The word “determinants” in the passage is closest in meaning to
 - ☐ origins
 - ☐ long-term benefits
 - ☐ causes
 - ☐ effects

Paragraph 2

Population grew from 61 million in 1500 to 78 million a century later, and the proportion of Europeans living in cities of 10,000 or more—and thus dependent on the market for what they consumed—expanded from less than 6 percent to nearly 8 percent during the same period. More important than sheer numbers, many Europeans’ incomes rose. This was especially true among more fully employed urban groups, farmers who benefited from higher prices and the intensifying commercialization and specialization in agriculture (which also led them to shed much non-agricultural production in favor of purchased goods), and landlords and other property owners who collected mounting rents. Government activities to build and strengthen the state were a stimulus to numerous industries, notably shipbuilding, textiles, and metallurgy. To cite just one example, France hastened to

develop its own iron industry when the Hapsburgs—the family that governed much of Europe, and whom France fought repeatedly in the sixteenth century—came to dominate the manufacture of weapons in Germany and the cities of Liege and Milan, which boasted Europe’s most advanced technology.

2. The word “hastened” in the passage is closest in meaning to
- ☐ needed
 - ☐ rushed
 - ☐ decided
 - ☐ attempted
3. According to paragraph 2, the fact that more people lived in European cities meant that
- ☐ more people had to purchase food and other basic necessities rather than producing these things themselves
 - ☐ industrial output increased because more people were available for employment in manufacturing
 - ☐ fewer people were available for agricultural work and thus farmers were forced to pay higher wages
 - ☐ more people competed for full-time urban employment, driving wages down

Paragraph 2 is marked with an arrow [→]

4. Which of the following is discussed in paragraph 2 as contributing to the growth in the market for manufactured goods that occurred in Europe after 1500?
- ☐ Lower costs for food and housing
 - ☐ Advancements in manufacturing technology
 - ☐ Higher incomes
 - ☐ Increased property ownership by farmers

Paragraph 2 is marked with an arrow [→]

Paragraph 3

The supply of goods was also significantly modified. Migration had long been critical for the diffusion of knowledge that spawned new trades or revived others. Now thousands of workers, and sizeable amounts of capital, moved from one region to another. At the same time, new commodities appeared on the market, often broadening and deepening demand. Most were inexpensive items destined for individual consumers. Knitted stockings, ribbon and lace, buttons, starch, soap, vinegar brewed from beer, knives and tools, pots and ovens, and many more goods, formerly made only for local sale, now entered into channels of national or international trade. The best-known and most widely adopted new industry was printing with movable type, which spread swiftly throughout Europe after Johannes Gutenberg perfected his innovation in 1453. Despite isolated cases of resistance—the scribes’ guild (an association of book copiers) delayed printing’s introduction into Paris for twenty years, for example—more than 380 working

presses had sprung up by 1480, and 1,000 (in nearly 250 towns) by 1500. Between 1453 and 1500, all the presses of Europe together turned out some 40,000 editions (known as incunabula), but from 1501 to 1600, that same quantity was produced in Lyon and Paris alone.

5. The word “diffusion” in the passage is closest in meaning to

- ☐
- ☐
- ☐
- ☐

6. The word “perfected” in the passage is closest in meaning to

- ☐ marketed
- ☐ completed
- ☐ announced
- ☐ exhibited

7. According to paragraph 3, which of the following was true about the new technology for printing with movable type?

- ☐ It met with opposition wherever attempts were made to introduce it.
- ☒ It spread with increasing rapidity throughout Europe after 1453.
- ☐ It rapidly turned printing into the most important industry in Paris.
- ☐ It was controlled in most places by the local scribes' guild.

Paragraph 3 is marked with an arrow [→]

Paragraph 4

In metals and mining, technical improvements were available that saved substantially on raw materials and fuel, causing prices to drop. The construction of ever-larger furnaces capable of higher temperatures culminated in the blast furnace, which used cheaper ores and economized on scarce and expensive wood, cutting costs per ton by 20 percent while boosting output substantially. A new technique for separating silver from copper allowed formerly worthless ores to be exploited. Better drainage channels, pumps, and other devices made it possible to tunnel more deeply into the earth as surface deposits began to be exhausted. In most established industries, however, technological change played little role, as in the past, new customers were sought by developing novel products based on existing technologies, such as a new type of woolen cloth with the texture of silk.

culminate in以...告终 economize on节约

8. Which of the following is NOT identified in paragraph 4 as an improvement made possible by technological developments?

- ☐ The mining of ores that had previously been too deep to reach
- ☐ The use of previously worthless ores
- ☒ A reduction in cost of expensive wood用料少了，而不是木材本身价格下降了
- ☐ The construction of furnaces that cost less to operate

Paragraph 4 is marked with an arrow [→]

9. Which of the sentences below expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Although most established industries continued operating with existing technologies, some novel products required the development of new technologies.
- ☐ In the past, technological change had been unimportant in most established industries because the products that customers wanted could be produced using existing technologies.
- ☐ In most established industries, technological change played a role only when it made possible the production of novel products that attracted new customers.
- ☐ Most established industries attracted new customers by developing new products based on existing technologies rather than by applying new technologies.

Paragraph 5

Sharply declining transaction costs (the direct and indirect expenses associated with transporting, distributing, and marketing goods and services) were more influential. On a general level, the decrease was due to greater security thanks to the lessening of wartime disruptions and to the economies of scale achieved when selling to large, concentrated urban populations. More specifically, it can be traced to transport innovations such as the carrack, a large ship that reduced rates for oceanborne freight by up to 25 percent, and big four-wheeled Hesse carts for overland routes. The spread of efficient organizational forms further contributed to declining costs, as did falling interest rates, which dropped from 20 percent or 25 percent in the mid-fifteenth century to 10 percent 100 years later.

10. According to paragraph 5, what was one reason for the increase in security?

- ☐ Reductions in transaction costs
- ☐ Improvements to overland routes
- ☐ Less war
- ☐ More sophisticated technologies for distributing goods

Paragraph 5 is marked with an arrow [→]

11. In paragraph 5, the author mentions the transport innovations of the carrack and Hesse carts in order to

- ☐ provide examples of wartime inventions that were adapted for use in industry
- ☐ explain how knowledge of more efficient organizational forms was spread
- ☐ provide reasons for the decline in transaction costs
- ☐ identify innovations that led to falling interest rates

Paragraph 5 is marked with an arrow [→]

Paragraph 3

The supply of goods was also significantly modified. ■ Migration had long been critical for the diffusion of knowledge that spawned new trades or revived others. Now thousands of workers, and sizeable amounts of capital, moved from one region to another. ■ At the same time, new commodities appeared on the market, often broadening and deepening demand. ■ Most were inexpensive items destined for individual consumers. ■ Knitted stockings, ribbon and lace, buttons, starch, soap, vinegar brewed from beer, knives and tools, pots and ovens, and many more goods, formerly made only for local sale, now entered into channels of national or international trade. The best-known and most widely adopted new industry was printing with movable type, which spread swiftly throughout Europe after Johannes Gutenberg perfected his innovation in 1453. Despite isolated cases of resistance—the scribes' guild (an association of book copiers) delayed printing's introduction into Paris for twenty years, for example—more than 380 working presses had sprung up by 1480, and 1,000 (in nearly 250 towns) by 1500. Between 1453 and 1500, all the presses of Europe together turned out some 40,000 editions (known as incunabula), but from 1501 to 1600, that same quantity was produced in Lyon and Paris alone.

12. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

For example, during this period, international investment in Switzerland rapidly expanded after the country was flooded by skilled workers fleeing religious persecution in Italy and France.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

13. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

The sharp rise in industrial output across most of Europe between 1450 and 1575 was mainly due to changes within Europe.

-
-
-

Answer choices

Industrial development was stimulated by rising incomes and growing populations—especially in market-dependent urban centers—and by government support for certain industries.

Numerous technological innovations that could be applied to the manufacture of consumer goods grew out of the weapons industry that had developed in Germany, Liege, and Milan.

New industries that manufactured goods for individual consumers were able to keep prices low in large part because of new technologies designed to accommodate economies of scale.

With increased mobility of labor and capital, more and more goods—especially inexpensive consumer goods—were produced for wide distribution rather than being limited to local markets

Industrial development was limited in some areas due to the high costs of transportation, labor, and rents, and because of the localized nature of industrial knowledge.

Lower prices for manufactured goods were the result of lower interest rates, of improvements in transportation, security, and organization, and of innovations in metal-making, mining, and printing.



Environmental Impact of the Anasazi

A major question in the archaeology of the southwestern region of the United States is why so many impressive settlements, and even entire regions, were abandoned in prehistoric times. Archaeologist Tim Kohler has suggested that the nature of human-environmental interaction was an important reason in the case of the Anasazi people. The actual case study that Kohler relies on is from the Dolores River basin of southwest Colorado, where the Anasazi seem to have moved in about A.D. 600. Over the following couple of centuries, the population increased, and they aggregated (or gathered) into villages, but by about A.D. 900 the area began to be abandoned. Other archaeologists have identified the immediate cause of this abandonment to be a series of short growing seasons that would have put pressure on corn production at that high an altitude. Kohler, however, asserts that a growing population led to human-environmental interactions that caused people to live in villages, intensify agrarian food production, deforest the region, deplete the local soils, and ultimately abandon the area.

Kohler uses several kinds of evidence to show that human effects, not solely climatic factors, were important factors in the abandonment of settlements. One key indicator of change in the environment surrounding these prehistoric settlements is the wood that was used there. Archaeological study of wood charcoal found in hearths dating to the various episodes of occupation indicated that the species use changed in a patterned way. Over time there was a decline in the use of juniper and pinon (native, slow-growing species of trees) and an increase in woody shrubs and fast-growing cottonwood. The species of wood used in the construction of buildings also changed. Fewer pinon were being used, and those that were used seem to be from increasingly old trees, while juniper continued to be from young trees. The implication is that the forest that did remain was changing to relatively more junipers, a tree that is more fire resistant, better able to reproduce in open settings, and less desirable for construction than pinon. Kohler argues that pinon was disappearing from the locale of settlements and that this put an additional nutritional strain on the population, which used nuts from the tree as well as its wood. The relative proportion of different species of animals hunted by people in the region also changed progressively. A final source of evidence was the seeds found in the archaeological deposits, which had blown or been brought to the settlement. As time went on, there was a substantial increase in seeds from pioneer plants, attesting both to agricultural intensification and to an increasingly disturbed local environment.

This evidence has convinced Kohler of the importance of human impact in degrading the local environment. His interpretation of the situation is that by about A.D. 840, people had aggregated into villages in favorable settings because of their competitive organizational advantages over smaller units in the face of growing population and depletion of local wild resources. Hence, the very nature of the initial slash-and-burn

agriculture encouraged a further dependence on agriculture and the aggregation of people into denser settlements. However, there are costs to aggregation, such as the increasing distance to usable fields, the heavier pressure on local soils, and the accompanying increase in agricultural risk. The Anasazi responded to this by further intensification, such as water-control mechanisms, to feed the increasing population. Such a trajectory is fraught with risks, but it is also pushed forward by advantages it bestows on its participants who organize and cooperate. Advantages might include sharing food across groups in a village, investment in facilities to improve the processing and storage of food, and cooperative labor pools and social groupings larger than villages, which would enable organized long-distance hunts and participation in trading networks. Larger and larger villages became possible, but this also made the system vulnerable to collapse. A reliance on the management of resources through cooperative action reduced their flexibility of action, so that when poor seasons occurred, people were seriously hurt. Thus an expectable aberration in the climatic regime may have been enough to cause the collapse of the village system in the Dolores area.

Paragraph 1

A major question in the archaeology of the southwestern region of the United States is why so many impressive settlements, and even entire regions, were abandoned in prehistoric times. Archaeologist Tim Kohler has suggested that the nature of human-environmental interaction was an important reason in the case of the Anasazi people. The actual case study that Kohler relies on is from the Dolores River basin of southwest Colorado, where the Anasazi seem to have moved in about A.D. 600. Over the following couple of centuries, the population increased, and they aggregated (or gathered) into villages, but by about A.D. 900 the area began to be abandoned. Other archaeologists have identified the immediate cause of this abandonment to be a series of short growing seasons that would have put pressure on corn production at that high an altitude. Kohler, however, asserts that a growing population led to human-environmental interactions that caused people to live in villages, intensify agrarian food production, deforest the region, deplete the local soils, and ultimately abandon the area.

1. The word “ultimately” in the passage is closest in meaning to
 - ☐ quietly
 - ☐ gradually
 - ☐ eventually
 - ☐ simply
2. According to paragraph 1, other archaeologists differ from Tim Kohler in giving which of the following as the reason for the abandonment of settlements by the Anasazi?
 - ☐ The nature of human interaction with the environment

- A large increase in population over a short period of time
- The way in which people gathered together in villages
- A limited production of corn due to one short growing season after another

Paragraph 1 is marked with an arrow [→]

3. According to paragraph 1, Kohler views all of the following as changes that occurred as a result of increased population growth EXCEPT

- the organization of the people into villages
- the improvement of local soils
- increased food production
- a decrease in the number of trees in the area

Paragraph 1 is marked with an arrow [→]

Paragraph 2

Kohler uses several kinds of evidence to show that human effects, not solely climatic factors, were important factors in the abandonment of settlements. One key indicator of change in the environment surrounding these prehistoric settlements is the wood that was used there. Archaeological study of wood charcoal found in hearths dating to the various episodes of occupation indicated that the species use changed in a patterned way. Over time there was a decline in the use of juniper and pinon (native, slow-growing species of trees) and an increase in woody shrubs and fast-growing cottonwood. The species of wood used in the construction of buildings also changed. Fewer pinon were being used, and those that were used seem to be from increasingly old trees, while juniper continued to be from young trees. The implication is that the forest that did remain was changing to relatively more junipers, a tree that is more fire resistant, better able to reproduce in open settings, and less desirable for construction than pinon. Kohler argues that pinon was disappearing from the locale of settlements and that this put an additional nutritional strain on the population, which used nuts from the tree as well as its wood. The relative proportion of different species of animals hunted by people in the region also changed progressively. A final source of evidence was the seeds found in the archaeological deposits, which had blown or been brought to the settlement. As time went on, there was a **substantial** increase in seeds from pioneer plants, attesting both to agricultural intensification and to an increasingly disturbed local environment.

4. The word “**substantial**” in the passage is closest in meaning to

- gradual
- appropriate
- apparent
- significant

5. All of the following are mentioned in paragraph 2 as changes over time in the pattern of wood use in prehistoric settlements EXCEPT:

- Cottonwood was increasingly used in hearths.
- Fewer pinon trees were used in building construction.
- Juniper wood was increasingly used in hearths.
- The pinon wood used in construction came increasingly from older trees.

Paragraph 2 is marked with an arrow [→]

6. It can be inferred from paragraph 2 that juniper
- is less valuable nutritionally than pinon is
 - is easily destroyed by fire
 - produces fewer seeds per plant than pinon does
 - reproduces easily in the presence of pinon

Paragraph 2 is marked with an arrow [→]

7. In paragraph 2, why does the author compare the use of juniper with pinon in the construction of buildings?
- To prove that juniper is more suitable for building construction than pinon is
 - To indicate that the choice of wood for building construction depended on the age of the tree species
 - To support the claim that wood use changed over time in a patterned way
 - To identify the features of juniper that made the Anasazi use it more often than pinon for constructing buildings

Paragraph 2 is marked with an arrow [→]

Paragraph 3

This evidence has convinced Kohler of the importance of human impact in degrading the local environment. His interpretation of the situation is that by about A.D. 840, people had aggregated into villages in favorable settings because of their competitive organizational advantages over smaller units in the face of growing population and depletion of local wild resources. Hence, the very nature of the initial slash-and-burn agriculture encouraged a further dependence on agriculture and the aggregation of people into denser settlements. However, there are costs to aggregation, such as the increasing distance to usable fields, the heavier pressure on local soils, and the accompanying increase in agricultural risk. The Anasazi responded to this by further intensification, such as water-control mechanisms, to feed the increasing population. Such a trajectory is fraught with risks, but it is also pushed forward by advantages it bestows on its participants who organize and cooperate. Advantages might include sharing food across groups in a village, investment in facilities to improve the processing and storage of food, and cooperative labor pools and social groupings larger than villages, which would enable organized long-distance hunts and participation in trading networks. Larger and larger villages became possible, but this also made the system vulnerable to collapse. A reliance on the management of resources through cooperative action reduced their flexibility of action, so that when poor seasons occurred, people were seriously hurt. Thus an expectable aberration in the climatic regime may have been enough to cause the collapse of the village

system in the Dolores area.

8. The word “denser” in the passage is closest in meaning to
- ☐ more distant
 - ☐ more crowded
 - ☐ newer
 - ☐ more permanent
9. The word “reliance” in the passage is closest in meaning to
- ☐ dependence
 - ☐ disagreement
 - ☐ policy
 - ☐ limit
10. According to paragraph 3, which of the following is Kohler’s explanation of the fact that people came together to form villages?
- ☐ Combining the population into large villages reduced the negative environmental effects of small social groups.
 - ☐ As the populations increased, the size of the small social units expanded so much that by A.D. 840 they had grown together forming villages.
 - ☐ Villages provided the efficient social organization needed to deal with population growth and reduced resources.
 - ☐ Villages formed on the land cleared by the slash and burn agriculture practiced by small social groups.

Paragraph 3 is marked with an arrow [→]

11. According to paragraph 3, why did the apparent advantages of larger settlements in fact lead to the failure of the Anasazi village system?
- ☐ The large size of the settlements made improving processing and storage of necessary food difficult.
 - ☐ The trading networks could no longer support the growing needs of large villages.
 - ☐ The long-distance hunts that had to be organized took too many resources away from the labor pool.
 - ☐ The management system the people practiced made it difficult to deal appropriately with periods of bad harvests.

Paragraph 3 is marked with an arrow [→]

12. According to paragraph 3, which of the following was a disadvantage of aggregation?
- ☐ People lived farther from the fields in which they worked.
 - ☐ Facilities for food processing and food storage became inadequate.
 - ☐ Cooperation between the various groups in the management of labor pools became difficult to maintain.
 - ☐ Networks set up to facilitate trade between the larger villages broke down over

time.

Paragraph 3 is marked with an arrow [→]

Paragraph 2

Kohler uses several kinds of evidence to show that human effects, not solely climatic factors, were important factors in the abandonment of settlements. One key indicator of change in the environment surrounding these prehistoric settlements is the wood that was used there. Archaeological study of wood charcoal found in hearths dating to the various episodes of occupation indicated that the species use changed in a patterned way. Over time there was a decline in the use of juniper and pinon (native, slow-growing species of trees) and an increase in woody shrubs and fast-growing cottonwood. The species of wood used in the construction of buildings also changed. Fewer pinon were being used, and those that were used seem to be from increasingly old trees, while juniper continued to be from young trees. The implication is that the forest that did remain was changing to relatively more junipers, a tree that is more fire resistant, better able to reproduce in open settings, and less desirable for construction than pinon. Kohler argues that pinon was disappearing from the locale of settlements and that this put an additional nutritional strain on the population, which used nuts from the tree as well as its wood. ■ The relative proportion of different species of animals hunted by people in the region also changed progressively. ■ A final source of evidence was the seeds found in the archaeological deposits, which had blown or been brought to the settlement. ■ As time went on, there was a substantial increase in seeds from [pioneer plants](#), attesting both to agricultural intensification and to an increasingly disturbed local environment. ■

14. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Earlier, they had pursued animals native to woodlands such as deer and rabbit and later, those more at home in open or disturbed environments such as antelope and jackrabbit.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

15. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Archaeologist T. Kohler has attempted to explain why villages and areas in the southwestern part of the United States were abandoned around A.D. 900.

-
-
-

Answer choices

Kohler attributes the immediate cause of the abandonment to problems with corn production at a high altitude during short growing seasons.

Kohler maintains that the Anasazi's transition to living together in villages was a key factor in the process of the degradation of the environment.

The development of intensive agricultural methods depleted the soil and resource management strategies made it difficult to cope with poor growing seasons.

Kohler's research indicates that in addition to agriculture, the Anasazi lived on the pinon nuts they grew and the animals they hunted in the area.

Increases in seeds from pioneer plants and systematic changes in the animals hunted and the trees used for construction and fuel are evidence of environmental degradation.

The dependence of the Anasazi on food supplies from nearby villages with better systems of water control and food storage facilities resulted in the Anasazi abandoning the larger villages.

2014-04-12

Surface Fluids on Venus and Earth

A fluid is a substance, such as a liquid or gas, in which the component particles (usually molecules) can move past one another. Fluids flow easily and conform to the shape of their containers. The geologic processes related to the movement of fluids on a planet's surface can completely resurface a planet many times. These processes derive their energy from the Sun and the gravitational forces of the planet itself. As these fluids interact with surface materials, they move particles about or react chemically with them to modify or produce materials. On a solid planet with a [hydrosphere](#) and an atmosphere, only a tiny fraction of the planetary mass flows as surface fluids. Yet the movements of these fluids can drastically alter a planet. Consider Venus and Earth, both terrestrial planets with atmosphere.

Venus and Earth are commonly regarded as twin planets but not identical twins. They are about the same size, are composed of roughly the same mix of materials, and may have been comparably endowed at their beginning with carbon dioxide and water. However, the twins evolved differently, largely because of differences in their distance from the Sun. With a significant amount of internal heat, Venus may continue to be geologically active with volcanoes, rifting, and folding. However, it lacks any sign of a hydrologic system (water circulation and distribution): there are no streams, lakes, oceans, or glaciers. Space probes suggest that Venus may have started with as much water as Earth, but it was unable to keep its water in liquid form. Because Venus receives more heat from the Sun, water released from the interior evaporated and rose to the upper atmosphere where the Sun's ultraviolet rays broke the molecules apart. Much of the freed hydrogen escaped into space, and Venus lost its water. Without water, Venus became less and less like Earth and kept an atmosphere filled with carbon dioxide. The carbon dioxide acts as a blanket, creating an intense greenhouse effect and driving surface temperatures high enough to melt lead and to prohibit the formation of carbonate minerals. Volcanoes continually vented more carbon dioxide into the atmosphere. On Earth, liquid water removes carbon dioxide from the atmosphere and combines it with calcium, from rock weathering, to form carbonate sedimentary rocks. Without liquid water to remove carbon from the atmosphere, the level of carbon dioxide in the atmosphere of Venus remains high.

Like Venus, Earth is large enough to be geologically active and for its gravitational field to hold an atmosphere. Unlike Venus, it is just the right distance from the Sun so that temperature ranges allow water to exist as a liquid, a solid, and a gas. Water is thus extremely mobile and moves rapidly over the planet in a continuous hydrologic cycle. Heated by the Sun, the water moves in great cycles from the oceans to the atmosphere, over the landscape in river systems, and ultimately back to the oceans. As a result, Earth's surface has been continually changed and eroded into delicate systems of river valleys—a remarkable contrast to the surfaces of other

planetary bodies where impact craters dominate. Few areas on Earth have been untouched by flowing water. As a result, river valleys are the dominant feature of its landscape. Similarly, wind action has scoured fine particles away from large areas, depositing them elsewhere as vast sand seas dominated by dunes or in sheets of loess (fine-grained soil deposits). These fluid movements are caused by gravity flow systems energized by heat from the Sun. Other geologic changes occur when the gases in the atmosphere or water react with rocks at the surface to form new chemical components with different properties. An important example of this process was the removal of most of Earth's carbon dioxide from its atmosphere to form carbonate rocks. However, if Earth were a little closer to the Sun, its oceans would evaporate. If it were farther from the Sun, the oceans would freeze solid. Because liquid water was present, self-replicating molecules of carbon, hydrogen, and oxygen developed life early in Earth's history and have rapidly modified its surface, blanketing huge parts of the continents with greenery. Life thrives on this planet, and it helped create the planet's oxygen- and nitrogen-rich atmosphere and moderate temperature.

Paragraph 1

A fluid is a substance, such as a liquid or gas, in which the component particles (usually molecules) can move past one another. Fluids flow easily and conform to the shape of their containers. The geologic processes related to the movement of fluids on a planet's surface can completely resurface a planet many times. These processes derive their energy from the Sun and the gravitational forces of the planet itself. As these fluids interact with surface materials, they move particles about or react chemically with them to modify or produce materials. On a solid planet with a [hydrosphere](#) and an atmosphere, only a tiny fraction of the planetary mass flows as surface fluids. Yet the movements of these fluids can drastically alter a planet. Consider Venus and Earth, both terrestrial planets with atmosphere.

1. The word "modify" in the passage is closest in meaning to
 - ☐ obtain
 - ☐ change
 - ☐ replace
 - ☐ absorb
2. The word "drastically" in the passage is closest in meaning to
 - ☐ gradually
 - ☐ permanently
 - ☐ extensively
 - ☐ possibly
3. Paragraph 1 supports all of the following statements about fluids EXCEPT:
 - ☐ They can chemically react with particles on a planet's surface.
 - ☐ Most of their mass does not flow but remains in place.
 - ☐ Their movements can reshape the surface of certain kinds of planets.
 - ☐ Their movements is driven by the Sun and by gravity.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

Venus and Earth are commonly regarded as twin planets but not identical twins. They are about the same size, are composed of roughly the same mix of materials, and may have been comparably endowed at their beginning with carbon dioxide and water. However, the twins evolved differently, largely because of differences in their distance from the Sun. With a significant amount of internal heat, Venus may continue to be geologically active with volcanoes, rifting, and folding. However, it lacks any sign of a hydrologic system (water circulation and distribution): there are no streams, lakes, oceans, or glaciers. Space probes suggest that Venus may have started with as much water as Earth, but it was unable to keep its water in liquid form. Because Venus receives more heat from the Sun, water released from the interior evaporated and rose to the upper atmosphere where the Sun's ultraviolet rays broke the molecules apart. Much of the freed hydrogen escaped into space, and Venus lost its water. Without water, Venus became less and less like Earth and kept an atmosphere filled with carbon dioxide. The carbon dioxide acts as a blanket, creating an intense greenhouse effect and driving surface temperatures high enough to melt lead and to **prohibit** the formation of carbonate minerals. Volcanoes continually vented more carbon dioxide into the atmosphere. On Earth, liquid water removes carbon dioxide from the atmosphere and combines it with calcium, from rock weathering, to form carbonate sedimentary rocks. Without liquid water to remove carbon from the atmosphere, the level of carbon dioxide in the atmosphere of Venus remains high.

4. The word "prohibit" in the passage is closest in meaning to
- ☐ prevent
 - ☐ speed up
 - ☐ affect
 - ☐ encourage
5. According to paragraph 2, what is one difference between Earth and Venus?
- ☐ Earth has less water in its atmosphere than Venus does.
 - ☐ Earth has a hydrologic system, but Venus does not.
 - ☐ Earth is less geologically active than Venus is.
 - ☐ Earth has more carbon dioxide than Venus does.

Paragraph 2 is marked with an arrow [→]

6. It can be inferred from paragraph 2 that Earth evolved differently than Venus did in part because
- ☐ there was more volcanic activity on early Venus than on early Earth
 - ☐ they received different amounts of solar energy
 - ☐ their interiors contained different amounts of heat
 - ☐ their early atmosphere contained different levels of oxygen and nitrogen

Paragraph 2 is marked with an arrow [→]

7. According to paragraph 2, all of the following played a role in keeping carbon dioxide levels high in the atmosphere of Venus EXCEPT

- the breaking apart of water molecules by ultraviolet rays
- the evaporation of water released from the planet's interior
- the escape of hydrogen into space
- the release of molecules from melting metals such as lead

Paragraph 2 is marked with an arrow [→]

Paragraph 3

Like Venus, Earth is large enough to be geologically active and for its gravitational field to hold an atmosphere. Unlike Venus, it is just the right distance from the Sun so that temperature ranges allow water to exist as a liquid, a solid, and a gas. Water is thus extremely mobile and moves rapidly over the planet in a continuous hydrologic cycle. Heated by the Sun, the water moves in great cycles from the oceans to the atmosphere, over the landscape in river systems, and ultimately back to the oceans. As a result, Earth's surface has been continually changed and eroded into delicate systems of river valleys—a remarkable contrast to the surfaces of other planetary bodies where impact craters dominate. Few areas on Earth have been untouched by flowing water. As a result, river valleys are the dominant feature of its landscape. Similarly, wind action has scoured fine particles away from large areas, depositing them elsewhere as vast sand seas dominated by dunes or in sheets of loess (fine-grained soil deposits). These fluid movements are caused by gravity flow systems energized by heat from the Sun. Other geologic changes occur when the gases in the atmosphere or water react with rocks at the surface to form new chemical components with different properties. An important example of this process was the removal of most of Earth's carbon dioxide from its atmosphere to form carbonate rocks. However, if Earth were a little closer to the Sun, its oceans would evaporate. If it were farther from the Sun, the oceans would freeze solid. Because liquid water was present, self-replicating molecules of carbon, hydrogen, and oxygen developed life early in Earth's history and have rapidly modified its surface, blanketing huge parts of the continents with greenery. Life thrives on this planet, and it helped create the planet's oxygen- and nitrogen-rich atmosphere and moderate temperature.

8. The word “ultimately” in the passage is closest in meaning to
- finally
 - slowly
 - repeatedly
 - constantly
9. According to paragraph 3, Earth's surface is different from the surfaces of many other planetary bodies in which of the following ways?
- It is more strongly marked by river valleys and erosion.
 - It is more geologically active.
 - It is covered by impact craters.
 - It has an atmosphere.

Paragraph 3 is marked with an arrow [→]

10. According to paragraph 3, why is water able to move so freely on Earth?
- Earth's temperatures are such that water exists in solid, liquid, and gas forms.

- Earth is large enough to be geologically active and for its gravitational field to hold an atmosphere.
- Earth's surface allows river valleys to develop across the landscape.
- Earth has active winds that blow across seas and oceans, causing fluid movements.

Paragraph 3 is marked with an arrow [→]

11. Why does the author point out that on Earth “gases in the atmosphere or water react with rocks at the surface to form new chemical components”?
- To explain why scientists believe that few areas on Earth have been untouched by flowing water
 - To identify one of several ways in which the movement of fluids can affect the surface of a planet
 - To provide evidence that fluid movements are caused by gravity flow systems energized by the Sun
 - To identify an effect of wind scouring fine particles away from large areas
12. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- Life on Earth is responsible for many changes to the planet's surface, including blankets of greenery.
 - Self-replicating molecules of carbon, hydrogen, and oxygen led to the development of life early in Earth's history.
 - The presence of water made it possible for life to develop early in Earth's history and to significantly change its surface.
 - Early in life's history on Earth, self-replicating molecules of carbon, hydrogen, and oxygen began blanketing the surface in greenery.

Paragraph 2

Venus and Earth are commonly regarded as twin planets but not identical twins. They are about the same size, are composed of roughly the same mix of materials, and may have been comparably endowed at their beginning with carbon dioxide and water. However, the twins evolved differently, largely because of differences in their distance from the Sun. With a significant amount of internal heat, Venus may continue to be geologically active with volcanoes, rifting, and folding. ■ However, it lacks any sign of a hydrologic system (water circulation and distribution): there are no streams, lakes, oceans, or glaciers. ■ Space probes suggest that Venus may have started with as much water as Earth, but it was unable to keep its water in liquid form. ■ Because Venus receives more heat from the Sun, water released from the interior evaporated and rose to the upper atmosphere where the Sun's ultraviolet rays broke the molecules apart. ■ Much of the freed hydrogen escaped into space, and Venus lost its water. Without water, Venus became less and less like Earth and kept an atmosphere filled with carbon dioxide. The carbon dioxide acts as a blanket, creating an intense greenhouse effect and driving surface temperatures high enough to melt lead and to prohibit the formation of carbonate minerals. Volcanoes continually vented more carbon dioxide into the atmosphere. On Earth, liquid water removes

carbon dioxide from the atmosphere and combines it with calcium, from rock weathering, to form carbonate sedimentary rocks. Without liquid water to remove carbon from the atmosphere, the level of carbon dioxide in the atmosphere of Venus remains high.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Venus may not have always been this way.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Over time, the movement of surface fluids has greatly changed Venus and Earth.

-
-
-

Answer Choices

Although Venus is about the same size as Earth, its greater volcanic activity has added considerably to carbon dioxide levels in its atmosphere.

Like Venus, Earth has an atmosphere, but Earth's atmosphere has far more oxygen and nitrogen than does the atmosphere of Venus.

On Earth, chemical reactions involving fluids remove carbon dioxide from the atmosphere by giving rise to carbonate rocks, and winds energized by gravity flow systems move fine particles from one place to another.

Because Venus lost the water it originally had, most of its carbon dioxide remained in its atmosphere, causing the planet to become very warm.

On Earth, the dominance of river valley landscapes and the existence of life are due to the planet's hydrologic cycle.

The evaporation of liquid water from Earth's surface is largely limited by the life forms that have developed, particularly the vegetation.

Water Management in Early Agriculture

As the first cities formed in Mesopotamia in the Middle East, probably around 3000 B.C., it became necessary to provide food for larger populations, and thus to find ways of increasing agricultural production. This, in turn, led to the problem of obtaining sufficient water.

Irrigation must have started on a small scale with rather simple constructions, but as its value became apparent, more effort was invested in new construction to divert more water into the canals and to extend the canal system to reach greater areas of potential farmland. Because of changing water levels and clogging by waterborne particles, canals and their intakes required additional labor to maintain, besides the normal labor required to guide water from field to field. Beyond this, some personnel had to be devoted to making decisions about the allocation of available water among the users and ensuring that these directions were carried out. With irrigation water also came potential problems, the most obvious being the susceptibility of low-lying farmlands to disastrous flooding and the longer-term problem of salinization (elevated levels of salt in the soil). To combat flooding from rivers, people from early historic times until today have constructed protective levees (raised **barriers** of earth) between the river and the settlement or fields to be protected. This, of course, is effective up to a certain level of flooding but changes the basic water patterns of the area and can multiply the damage when the flood level exceeds the height of the levee.

Salinization is caused by an accumulation of salt in the soil near its surface. This salt is carried by river water from the sedimentary rocks in the mountains and deposited on the Mesopotamian fields during natural flooding or purposeful irrigation. Evaporation of water sitting on the surface in hot climates is rapid, concentrating the salts in the remaining water that then descends through the soil to the underlying water table. In southern Mesopotamia, for example, the natural water table comes to within roughly six feet of the surface. Conditions of excessive irrigation bring the water table to eighteen inches, and water can rise further to the root zone, where the high concentration of salts would kill most plants.

Solutions for salinization were not as straightforward as for flooding, but even in ancient times it was understood that the deleterious effects of salinization could be minimized by removing harmful elements through leaching the fields with additional fresh water, digging deep wells to lower the water table, or instituting a system of leaving fields uncultivated. The first two cures would have required considerable labor, and the third solution would have led to diminished productivity, not often viewed as a likely decision in periods of growing population. An effective irrigation system laid the foundation for many of the world's early civilizations, but it also required a great deal of labor input.

Growing agrarian societies often tried to meet their food-producing needs by farming less-desirable hill slopes surrounding the favored low-lying valley bottoms. Since bringing irrigation water to a hill slope is usually impractical, the key is effective utilization of rainfall. Rainfall either soaks into the soil or runs off of it due to gravity. A soil that is deep, well-structured, and covered by protective vegetation and much will normally absorb almost all of the rain that

falls on it, provided that the slope is not too steep. However, soils that have lost their vegetative cover and surface mulch will absorb much less, with almost half the water being carried away by runoff in more extreme conditions. This runoff carries with it topsoil particles, nutrients, and humus (decayed vegetable matter) that are concentrated in the topsoil. The loss of this material reduces the thickness of the rooting zone and its capacity to absorb moisture for crop needs.

The most direct solution to this problem of slope runoff was to lay lines of stones along the contours of the slope and hence, perpendicular to the probable flow of water and sediment. These stones could then act as small dams, slowing the downhill flow of water and allowing more water to infiltrate and soil particles to collect behind the dam. This provided a buildup of sediments for plants and improved the landscape's water-retention properties.

Paragraph 2

Irrigation must have started on a small scale with rather simple constructions, but as its value became apparent, more effort was invested in new construction to divert more water into the canals and to extend the canal system to reach greater areas of potential farmland. Because of changing water levels and clogging by waterborne particles, canals and their intakes required additional labor to maintain, besides the normal labor required to guide water from field to field. Beyond this, some personnel had to be devoted to making decisions about the allocation of available water among the users and ensuring that these directions were carried out. With irrigation water also came potential problems, the most obvious being the susceptibility of low-lying farmlands to disastrous flooding and the longer-term problem of salinization (elevated levels of salt in the soil). To combat flooding from rivers, people from early historic times until today have constructed protective levees (raised barriers of earth) between the river and the settlement or fields to be protected. This, of course, is effective up to a certain level of flooding but changes the basic water patterns of the area and can multiply the damage when the flood level exceeds the height of the levee.

1. All of the following are mentioned in paragraph 2 as operations involved in the Mesopotamian irrigation system EXCEPT

- ☐ determining how much irrigation water should be distributed to various farmers
- ☐ widening existing canals so they could hold more water
- ☐ removing undesirable materials from the intakes of irrigation canals
- ☐ building new canals so irrigation water could be transported to distant areas

Paragraph 2 is marked with an arrow [→]

2. According to paragraph 2, protective levees can have which of the following disadvantages?

- ☐ They can greatly increase the destruction caused by floodwaters when floodwaters are higher than the levee.
- ☐ They can fail even when the flood level remains below the height of the levee.
- ☐ They can lead over time to a serious salinization problem.
- ☐ They can cause damaging floods to occur more frequently by changing basic water patterns.

Paragraph 2 is marked with an arrow [→]

3. Paragraph 2 suggests that irrigation increased the likelihood of destructive floods because

- ☐ irrigated fields were often in locations that tended to flood naturally
- ☐ the canal intakes for irrigation water often did not work
- ☐ most irrigation canals were too narrow and thus overflowed
- ☐ levees built to protect irrigation systems required maintenance

Paragraph 2 is marked with an arrow [→]

4. The word “potential” in the passage is closest in meaning to

- ☐ serious
- ☐ basic
- ☐ new
- ☐ possible

Paragraph 3

Salinization is caused by an accumulation of salt in the soil near its surface. This salt is carried by river water from the sedimentary rocks in the mountains and deposited on the Mesopotamian fields during natural flooding or purposeful irrigation. Evaporation of water sitting on the surface in hot climates is rapid, concentrating the salts in the remaining water that then descends through the soil to the underlying water table. In southern Mesopotamia, for example, the natural water table comes to within roughly six feet of the surface. Conditions of excessive irrigation bring the water table to eighteen inches, and water can rise further to the root zone, where the high concentration of salts would kill most plants.

5. The word “accumulation” in the passage is closest in meaning to

- ☐ distribution
- ☐ mixture
- ☐ buildup
- ☐ exchange

6. According to paragraph 3, excessive irrigation can destroy crops by

- ☐ raising salty water to the level of the roots
- ☐ forcing the roots of plants to grow close to the surface
- ☐ taking the place of some natural flooding
- ☐ creating salt deposits on the surface of the soil

Paragraph 3 is marked with an arrow [→]

Paragraph 4

Solutions for salinization were not as straightforward as for flooding, but even in ancient times it was understood that the deleterious effects of salinization could be minimized by removing harmful elements through leaching the fields with additional fresh water, digging deep wells to lower the water table, or instituting a system of leaving fields uncultivated. The first two cures would have required considerable labor, and the third solution would have led to diminished productivity, not often viewed as a likely decision in periods of growing population. An effective irrigation system laid the foundation for many of the world’s early civilizations, but it also

required a great deal of labor input.

7. The word “straightforward” in the passage is closest in meaning to
- ☐ successful
 - ☐ simple
 - ☐ common
 - ☐ complex
8. According to paragraph 4, which of the following is true of the more-likely-used solutions to the problem of salinization?
- ☐ They resulted in a decrease in the amount of food that was produced.
 - ☐ They succeeded only on areas where the natural water table was especially low.
 - ☐ They often demanded much time and effort on the part of their users.
 - ☐ They often led to other technological advances.

Paragraph 4 is marked with an arrow [→]

Paragraph 5

Growing agrarian societies often tried to meet their food-producing needs by farming less-desirable hill slopes surrounding the favored low-lying valley bottoms. Since bringing irrigation water to a hill slope is usually impractical, the key is effective utilization of rainfall. Rainfall either soaks into the soil or runs off of it due to gravity. A soil that is deep, well-structured, and covered by protective vegetation and much will normally absorb almost all of the rain that falls on it, provided that the slope is not too steep. However, soils that have lost their vegetative cover and surface mulch will absorb much less, with almost half the water being carried away by runoff in more extreme conditions. This runoff carries with it topsoil particles, nutrients, and humus (decayed vegetable matter) that are concentrated in the topsoil. The loss of this material reduces the thickness of the rooting zone and its capacity to absorb moisture for crop needs.

9. According to paragraph 5, which of the following was the main challenge faced by early agricultural societies that wanted to grow crops on hill slopes?
- ☐ Getting enough irrigation water to the hill slope
 - ☐ Growing crops without disturbing the natural vegetative cover
 - ☐ Retaining rainwater and thus preventing excessive runoff
 - ☐ Identifying crops that do not need a thick rooting zone

Paragraph 5 is marked with an arrow [→]

10. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ However, soils that are unable to absorb much water experience massive runoff during heavy rains.
 - ☐ However, where neither protective vegetation nor mulch covers the soil, much rainwater can be lost to runoff.
 - ☐ However, on extremely steep slopes there is no vegetative cover or mulch to prevent runoff.

- ☐ However, in more extreme conditions water that runs off can carry away the vegetative cover and the surface mulch

11. The word “impractical” in the passage is closest in meaning to

- ☐ unnecessary
- ☐ unsafe
- ☐ unrealistic
- ☐ unpredictable

Paragraph 6

The most direct solution to this problem of slope runoff was to lay lines of stones along the contours of the slope and hence, perpendicular to the probable flow of water and sediment. These stones could then act as small dams, slowing the downhill flow of water and allowing more water to infiltrate and soil particles to collect behind the dam. This provided a buildup of sediments for plants and improved the landscape’s water-retention properties.

12. Which of the following best describes how paragraph 6 relates to paragraph 5?

- ☐ Paragraph 6 describes how the solution to a problem identified in paragraph 5 created unexpected benefits.
- ☐ Paragraph 6 compares two possible solutions to a problem described in paragraph 5.
- ☐ Paragraph 6 explains how the attempt to solve a problem introduced in paragraph 5 led to more difficult problems.
- ☐ Paragraph 6 explains one way in which a difficulty described in paragraph 5 was resolved.

Paragraphs 5 and 6 is marked with an arrow [→]

Paragraph 3

Salinization is caused by an accumulation of salt in the soil near its surface. ■ This salt is carried by river water from the sedimentary rocks in the mountains and deposited on the Mesopotamian fields during natural flooding or purposeful irrigation. Evaporation of water sitting on the surface in hot climates is rapid, concentrating the salts in the remaining water that then descends through the soil to the underlying water table. ■ In southern Mesopotamia, for example, the natural water table comes to within roughly six feet of the surface. ■ Conditions of excessive irrigation bring the water table to eighteen inches, and water can rise further to the root zone, where the high concentration of salts would kill most plants. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Natural flooding, however, does not raise the water table nearly as much and thus does not have these sorts of consequences.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most

important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

As cities emerged and populations grew in Mesopotamia, more water had to be provided to increase agricultural production.

-
-
-

Answer Choices

Early on, irrigation was recognized as a valuable practice, even though it was labor-intensive and brought with it problems of salinization and damaging floods.

The mountain water that was used to irrigate farmland in Mesopotamia was exceptionally high in salt, causing rapid salinization of the soil.

Levees were the major means of protection against flooding, while leaching with added water and lowering the water table helped to control salinization.

The practice of leaving fields uncultivated periodically was used primarily by societies lacking a large labor force.

Because of the enormous amount of labor involved in irrigating fields, farming was increasingly moved to hill slopes, where irrigation systems required less labor.

As cultivation was extended to hill slopes, methods were developed to better retain water from rainfall for crops growing on hillsides.

Plant and Animal Life of the Pacific Islands

There are both great similarities and considerable diversity in the ecosystems that evolved on the islands of Oceania in and around the Pacific Ocean. The islands, such as New Zealand, that were originally parts of continents still carry some small plant and animal remnants of their earlier biota (animal and plant life), and they also have been extensively modified by evolution, adaptation, and the arrival of new species. By contrast, the other islands, which emerged via geological processes such as volcanism, possessed no terrestrial life, but over long periods, winds, ocean currents, and the feet, feathers, and digestive tracts of birds brought the seeds of plants and a few species of animals. Only those species with ways of spreading to these islands were able to undertake the long journeys, and the various factors at play resulted in diverse combinations of new colonists on the islands. One estimate is that the distribution of plants was 75 percent by birds, 23 percent by floating, and 2 percent by wind.

The migration of Oceanic biota was generally from west to east, with four major factors influencing their distribution and establishment. The first was the size and fertility of the islands on which they landed, with larger islands able to provide hospitality for a wider range of species. Second, the further east the islands, generally the less the species diversity, largely because of the distance that had to be crossed and because the eastern islands tended to be smaller, more scattered, and remote. This easterly decline in species diversity is well demonstrated by birds and coral fish. It is estimated that there were over 550 species of birds in New Guinea, 127 in the Solomon Islands, 54 in Fiji, and 17 in the Society Islands. From the west across the Pacific, the Bismarck Archipelago and the Solomon Islands have more than 90 families of shore fish (with many species within the families), Fiji has 50 families, and the Society Islands have 30. Third, the latitude of the islands also influenced the biotic mix, as those islands in relatively cooler latitudes, notably New Zealand, were unsuited to supporting some of the tropical plants with which Pacific islands are generally associated.

Finally, a fourth major factor in species distribution, and indeed in the shaping of Pacific ecosystems, was wind. It takes little experience on Pacific islands to be aware that there are prevailing winds. To the north of the equator these are called north-easterlies, while to the south they are called south-easterlies. Further south, from about 30° south, the winds are generally from the west. As a result on nearly every island of significant size there is an ecological difference between its windward and leeward (away from the wind) sides. Apart from the wind action itself on plants and soils, wind has a major effect on rain distribution. The Big Island of Hawaii offers a prime example; one can leave Kona on the leeward side in brilliant sunshine and drive across to the windward side where the city of Hilo is blanketed in mist and rain.

While such localized plant life and climatic conditions are very noticeable, over Oceania as a whole there is relatively little biodiversity, and the smaller the island and the further east it lies, the less there is likely to be. When humans moved beyond the islands of Near Oceania (Australia, New Guinea, and the Solomon Islands), they encountered no indigenous mammals except for flying foxes, fruit bats, and seals on some islands. Other vertebrate species were restricted to flying animals and a few small reptiles. However, local adaptations and evolution over long

periods of isolation promoted fascinating species adaptations to local conditions. Perhaps most notable, in the absence of mammals and other predators, are the many species of flightless and ground-nesting birds. Another consequence of evolution was that many small environments boasted their own endemic (native) species, often small in number, unused to serious predation, limited in range, and therefore vulnerable to disruption. In Hawaii, for example, the highly adapted 39 species and subspecies of honeycreepers, several hundred species of fruit flies, and more than 750 species of tree snails are often cited to epitomize the extent of localized Oceanic endemism (species being native to the area).

Paragraph 1

There are both great similarities and considerable diversity in the ecosystems that evolved on the islands of Oceania in and around the Pacific Ocean. The islands, such as New Zealand, that were originally parts of continents still carry some small plant and animal **remnants** of their earlier biota (animal and plant life), and they also have been extensively modified by evolution, adaptation, and the arrival of new species. By contrast, the other islands, which emerged via geological processes such as volcanism, possessed no terrestrial life, but over long periods, winds, ocean currents, and the feet, feathers, and digestive tracts of birds brought the seeds of plants and a few species of animals. Only those species with ways of spreading to these islands were able to undertake the long journeys, and the various factors at play resulted in diverse combinations of new colonists on the islands. One estimate is that the distribution of plants was 75 percent by birds, 23 percent by floating, and 2 percent by wind.

1. The word “remnants” in the passage is closest in meaning to
 - ☐ remainders
 - ☐ reminders
 - ☐ reproductions
 - ☐ resemblances
2. The passage supports which of the following statements about species on volcanic islands?
 - ☐ Volcanic island species are unlike the species found in other Pacific Ocean locations.
 - ☐ Volcanic islands lack the diversity of species found elsewhere in the Pacific.
 - ☐ Volcanic island species are all transplants from distant locations and exist in combinations not found elsewhere.
 - ☐ Volcanic island species differ from those on other islands in that animal species have greater diversity than plant species do.
3. According to paragraph 1, how did the majority of plant species arrive on islands created by geological processes such as volcanism?
 - ☐ They were transported by ocean currents.
 - ☐ They were carried to the islands by birds.
 - ☐ They were brought to the islands by humans.
 - ☐ They were transported by winds.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

The migration of Oceanic biota was generally from west to east, with four major factors influencing their distribution and establishment. The first was the size and fertility of the islands on which they landed, with larger islands able to provide hospitality for a wider range of species. Second, the further east the islands, generally the less the species diversity, largely because of the distance that had to be crossed and because the eastern islands tended to be smaller, more scattered, and **remote**. This easterly decline in species diversity is well demonstrated by birds and coral fish. It is estimated that there were over 550 species of birds in New Guinea, 127 in the Solomon Islands, 54 in Fiji, and 17 in the Society Islands. From the west across the Pacific, the Bismarck Archipelago and the Solomon Islands have more than 90 families of shore fish (with many species within the families), Fiji has 50 families, and the Society Islands have 30. **Third, the latitude of the islands also influenced the biotic mix, as those islands in relatively cooler latitudes, notably New Zealand, were unsuited to supporting some of the tropical plants with which Pacific islands are generally associated.**

4. The word "**remote**" in the passage is closest in meaning to
 - ☐ unknown
 - ☐ isolated
 - ☐ hostile
 - ☐ infertile
5. In paragraph 2, what is the author's purpose in mentioning the estimated numbers of birds and coral fish species on various Oceanic islands?
 - ☐ To give examples of the wide range of species that can be found on Oceanic islands
 - ☐ To demonstrate how much knowledge about Oceanic species has been accumulated
 - ☐ To illustrate the decline in species diversity from west to east on Oceanic islands
 - ☐ To identify the influence of latitude upon Oceanic plants and animals

Paragraph 1 is marked with an arrow [→]

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - ☐ Because of its latitude, New Zealand had a relatively cooler climate than other Pacific islands.
 - ☐ New Zealand, like other Pacific islands, showed the effects of latitudes on its rich tropical plants.
 - ☐ Because the latitudinal position of an island also affected its biotic mix, islands in cooler latitudes did not support some tropical species typical of the Pacific islands.
 - ☐ Pacific islands were notable for their impressive biotic mix and association with tropical plants.
7. According to paragraph 2, all of the following types of islands are associated with higher species diversity EXCEPT
 - ☐ islands that are large in size
 - ☐ islands located in cool latitudes

- islands located in the western part of Oceania
- islands located near other landmasses

Paragraph 2 is marked with an arrow [→]

Paragraph 3

Finally, a fourth major factor in species distribution, and indeed in the shaping of Pacific ecosystems, was wind. It takes little experience on Pacific islands to be aware that there are prevailing winds. To the north of the equator these are called north-easterlies, while to the south they are called south-easterlies. Further south, from about 30° south, the winds are generally from the west. As a result on nearly every island of significant size there is an ecological difference between its windward and leeward (away from the wind) sides. Apart from the wind action itself on plants and soils, wind has a major effect on rain distribution. The Big Island of Hawaii offers a prime example; one can leave Kona on the leeward side in brilliant sunshine and drive across to the windward side where the city of Hilo is blanketed in mist and rain.

8. The Big Island of Hawaii is discussed in the passage as an example of
 - the relationship between latitude and wind
 - how prevailing winds influence rainfall patterns
 - the relationship between rainfall and species distribution
 - the effects of wind action upon plants and soils
9. What can be inferred from paragraph 3 about Kona and Hilo?
 - The ecosystems of Kona and Hilo differ from each other.
 - Kona and Hilo have approximately the same rainfall in a given year.
 - Kona receives northeasterly winds while Hilo receives southeasterly winds.
 - Both Kona and Hilo have plants and soils that are often damaged by winds.

Paragraph 3 is marked with an arrow [→]

Paragraph 4

While such localized plant life and climatic conditions are very noticeable, over Oceania as a whole there is relatively little biodiversity, and the smaller the island and the further east it lies, the less there is likely to be. When humans moved beyond the islands of Near Oceania (Australia, New Guinea, and the Solomon Islands), they encountered no indigenous mammals except for flying foxes, fruit bats, and seals on some islands. Other vertebrate species were restricted to flying animals and a few small reptiles. However, local adaptations and evolution over long periods of isolation promoted fascinating species adaptations to local conditions. Perhaps most notable, in the absence of mammals and other predators, are the many species of flightless and ground-nesting birds. Another consequence of evolution was that many small environments boasted their own endemic (native) species, often small in number, unused to serious predation, limited in range, and therefore vulnerable to disruption. In Hawaii, for example, the highly adapted 39 species and subspecies of honeycreepers, several hundred species of fruit flies, and more than 750 species of tree snails are often cited to epitomize the extent of localized Oceanic endemism (species being native to the area).

10. The word “cited” in the passage is closest in meaning to
- ☐ expected
 - ☐ believed
 - ☐ compared
 - ☐ mentioned
11. According to paragraph 4, why have species of flightless and ground-nesting birds become so numerous on Oceanic islands?
- ☐ They have no predators on the islands.
 - ☐ They were some of the strongest species to arrive on the islands.
 - ☐ They live closer to their food sources than other species do.
 - ☐ They are affected less by climatic changes than other animals are.

Paragraph 4 is marked with an arrow [→]

12. Which of the following is NOT mentioned in paragraph 4 about the species that live on Oceanic islands?
- ☐ Certain species are native only to particular islands.
 - ☐ Species that are native to Oceanic islands include relatively few mammals.
 - ☐ Populations of most species are small in number.
 - ☐ Some species have evolved over time to become predators.

Paragraph 4 is marked with an arrow [→]

Paragraph 1

There are both great similarities and considerable diversity in the ecosystems that evolved on the islands of Oceania in and around the Pacific Ocean. ■ The islands, such as New Zealand, that were originally parts of continents still carry some small plant and animal remnants of their earlier biota (animal and plant life), and they also have been extensively modified by evolution, adaptation, and the arrival of new species. ■ By contrast, the other islands, which emerged via geological processes such as volcanism, possessed no terrestrial life, but over long periods, winds, ocean currents, and the feet, feathers, and digestive tracts of birds brought the seeds of plants and a few species of animals. ■ Only those species with ways of spreading to these islands were able to undertake the long journeys, and the various factors at play resulted in diverse combinations of new colonists on the islands. ■ One estimate is that the distribution of plants was 75 percent by birds, 23 percent by floating, and 2 percent by wind.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

When varied ecosystems are present, they can be explained as resulting in part from the process that formed the islands.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most

important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Biodiversity on Oceanic islands is dependent on a number of factors.

-
-
-

Answer Choices

Unlike Oceanic islands that were once part of continental landmasses, islands formed by such geological processes as volcanism contain only plants and animals that could be transported there.

An island's size is less important than its latitude in determining species diversity.

Though biodiversity is low on many Oceanic islands, many native species have evolved that are uniquely adapted to their local environments.

Species distribution in Oceania is determined by the location of islands, their size, and the direction of the wind.

Most Oceanic islands are similar to one another in latitude and contain plants and animals typical of tropical islands.

The absence of natural predators on the eastern Oceanic islands allowed many species of large mammals to evolve that were capable of inhabiting a wide range of territory.

The Cosmological Principle

Cosmologists attempt to understand the origin and structure of the universe as a whole. They begin their search with an assumption about the nature of the universe—namely, that in looking out from our vantage point in the cosmos, we see essentially the same kind of universe that an observer stationed in any other part of it, no matter how remote, would see. As far as our telescopes can reach, we see galaxies and clusters of galaxies distributed in more or less the same way in every direction. This assumption that the universe is uniform on a large scale is called “the cosmological principle.”

One thing that is certain is that the universe is expanding. In every direction we look, distant galaxies are moving away from each other. Until the 1960s, the expansion of the universe was the primary fact of cosmological significance that cosmological theories had to accommodate. There were two general classes of cosmological theories that fit with the expanding universe: the evolutionary (Big Bang) theory and the steady-state theory.

The essential idea of the evolutionary cosmology is that there was a beginning—a moment of creation at which the universe came into existence in a hot, violent explosion—the Big Bang. In the beginning, the universe was very hot, very dense, and very tiny. As the explosion evolved, the temperature dropped, the distribution of matter and energy thinned, and the universe expanded. From the current observed rate of expansion, we conclude that the creation event occurred between ten and twenty billion years ago.

The steady-state theory is based on an idea called the “perfect cosmological principle.” It is “perfect” in that it maintains that the universe is uniform not only in space but in time. Thus it is the hypothesis that the large-scale universe has always been the way it is now and will be this way forever in the future. This view is consistent with philosophical approaches that reject the notion of an absolute beginning of the universe as unacceptable. The steady-state universe would have no beginning and no end.

In an expanding universe, the galaxies move away from each other, spreading matter more thinly over space. On the other hand, the perfect cosmological principle requires that the density of matter in the universe remain constant over time. To make the steady-state theory compatible with the expanding universe, its

proponents introduced the notion of continuous creation. As the universe expands and the galaxies move farther apart, new matter—in the form of hydrogen—is introduced into the universe. The rate at which the hypothesized new matter is created is far too small for this creation to be detected with available instruments, but continuous creation provides just enough matter to form new stars and galaxies that fill in the space left by the old ones. Thus in the steady-state universe there is evolution of stars and galaxies, but the general character and the overall density of the universe remains unchanged over time. In this special sense, the steady-state universe itself does not evolve.

Both of these views—steady-state and Big Bang—allow for cosmic expansion. However, the discovery in the 1960s of a comparatively small star-like objects called quasars tipped the scales in favor of the Big Bang cosmology. Astronomers determined that almost all quasars are very distant. Given how bright quasars appear even at such great distances, astronomers concluded that quasars typically have an output of light that is 1,000 times greater than that of a whole spiral galaxy composed of billions of stars.

Quasars are such distant objects that the light now reaching us from quasars left them billions of years ago. This means that when we observe quasars today we are seeing that state of the universe billions of years ago. Thus the fact that almost all quasars are very far away implies that earlier in the history of the universe quasars were developing more frequently than they are now. This evolution is consistent with the Big Bang theory. But it violates the perfect cosmological principle, and so it is inconsistent with the steady-state view.

Paragraph 1

Cosmologists attempt to understand the origin and structure of the universe as a whole. They begin their search with an assumption about the nature of the universe—namely, that in looking out from our vantage point in the cosmos, we see essentially the same kind of universe that an observer stationed in any other part of it, no matter how remote, would see. As far as our telescopes can reach, we see galaxies and clusters of galaxies distributed in more or less the same way in every direction. This assumption that the universe is uniform on a large scale is called “the cosmological principle.”

1. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - ☐ Cosmologists make the assumption that the universe would appear basically the same from anywhere within it.
 - ☐ Cosmologists simplify their assumptions because they are basically trying to understand the whole universe.
 - ☐ Cosmologists make an assumption about the universe basically in an effort to avoid stationing observers in remote places.

- Cosmologists incorrectly assumed the universe to be far simpler than it really is when viewed on a larger scale.

Paragraph 2

One thing that is certain is that the universe is expanding. In every direction we look, distant galaxies are moving away from each other. Until the 1960s, the expansion of the universe was the primary fact of cosmological **significance** that cosmological theories had to accommodate. There were two general classes of cosmological theories that fit with the expanding universe: the evolutionary (Big Bang) theory and the steady-state theory.

2. The word “**significance**” in the passage is closest in meaning to

- ☐ Change
- ☐ Expansion
- ☐ activity
- ☐ importance

3. According to paragraph 2, what is true of both the evolutionary and steady-state theories?

- ☐ Both were based on the work of the same observer of the galaxies.
- ☐ Both were used to overturn the theories of the early twentieth century.
- ☐ Both took 30 years to become widely accepted by cosmologists.
- ☐ Both accommodate the fact that the galaxies are moving apart.

Paragraph 2 is marked with an arrow [→].

Paragraph 3

The **essential** idea of the evolutionary cosmology is that there was a beginning—a moment of creation at which the universe came into existence in a hot, violent explosion—the Big Bang. In the beginning, the universe was very hot, very dense, and very tiny. As the explosion evolved, the temperature dropped, the distribution of matter and energy thinned, and the universe expanded. From the current observed rate of expansion, we conclude that the creation event occurred between ten and twenty billion years ago.

4. The word “**essential**” in the passage is closest in meaning to

- ☐ familiar
- ☐ new
- ☐ fundamental
- ☐ controversial

5. According to paragraph 3, which of the following is NOT a feature of the Big Bang cosmology?

- ☐ A sudden, hot explosion of the universe

- A reduction in the density of matter over time
- A significant rise in temperature over time
- An expansion of the universe

Paragraph 3 is marked with an arrow [→].

Paragraph 4

The steady-state theory is based on an idea called the “perfect cosmological principle.” It is “perfect” in that it maintains that the universe is uniform not only in space but in time. Thus it is the hypothesis that the large-scale universe has always been the way it is now and will be this way forever in the future. This view is consistent with philosophical approaches that reject the notion of an absolute beginning of the universe as unacceptable. The steady-state universe would have no beginning and no end.

6. The word “**notion**” in the passage is closest in meaning to
 - nature
 - argument
 - question
 - idea
7. Why does the author state that “The steady-state universe would have no beginning and no end.”?
 - To present evidence against the steady-state view of the universe
 - To clarify why the steady-state view is attractive to certain philosophical approaches
 - To contrast cosmology with philosophy
 - To indicate that the steady-state view is the most accepted view among cosmologists

Paragraph 5

In an expanding universe, the galaxies move away from each other, spreading matter more thinly over space. On the other hand, the perfect cosmological principle requires that the density of matter in the universe remain constant over time. To make the steady-state theory **compatible** with the expanding universe, its proponents introduced the notion of continuous creation. As the universe expands and the galaxies move farther apart, new matter—in the form of hydrogen—is introduced into the universe. The rate at which the hypothesized new matter is created is far too small for this creation to be detected with available instruments, but **continuous creation** provides just enough matter to form new stars and galaxies that fill in the space left by the old ones. Thus in the steady-state universe there is evolution of stars and galaxies, but the general character and the overall density of the universe remains unchanged over time. In this special sense, the steady-state universe itself does not evolve.

8. The word “compatible” in the passage is closest in meaning to
- ☐ respected
 - ☐ equivalent
 - ☐ consistent
 - ☐ competitive
9. Which of the following is true of the concept of “continuous creation”?
- ☐ It suggests that hydrogen for new stars is created as galaxies move farther apart.
 - ☐ It asserts that matter in the universe becomes denser as hydrogen is created.
 - ☐ It explains why the galaxies are moving away from each other.
 - ☐ It predicts a change in density of matter in the universe over time.
10. According to paragraph 5, which of the following is characteristic of the steady-state theory?
- ☐ The rate of expansion it predicts is too slow to be measured by current instruments.
 - ☐ New stars and galaxies are formed, but the large-scale properties of the universe remain the same.
 - ☐ The density of matter changes over time as the universe evolves.
 - ☐ The creation of new galaxies will eventually stop the universe from expanding.
- Paragraph 5 is marked with an arrow [→].

Paragraph 6

Both of these views—steady-state and Big Bang—allow for cosmic expansion. However, the discovery in the 1960s of a comparatively small star-like objects called quasars tipped the scales in favor of the Big Bang cosmology. Astronomers determined that almost all quasars are very distant. Given how bright quasars appear even at such great distances, astronomers concluded that quasars typically have an output of light that is 1,000 times greater than that of a whole spiral galaxy composed of billions of stars.

11. Paragraph 6 answers which of the following questions about quasars?
- ☐ What is the ratio of the number of quasars to the number of spiral galaxies?
 - ☐ Why was the discovery of quasars of importance for cosmologists?
 - ☐ Why were quasars not discovered before the 1960s?
 - ☐ How do quasars produce so much light?

Paragraph 6 is marked with an arrow [→].

Paragraph 7

Quasars are such distant objects that the light now reaching us from quasars left them billions of years ago. This means that when we observe quasars today we are seeing that state of the universe billions of years ago. ■ Thus the fact that almost all quasars are very far away implies that earlier in the history of the universe quasars were developing more frequently than they are now. ■ This evolution is consistent

with the Big Bang theory. ■ But it violates the perfect cosmological principle, and so it is inconsistent with the steady-state view. ■

12. Which of the following best describes the relationship between paragraphs 6 and 7?

- ☐ Paragraph 6 makes a claim about support for the Big Bang cosmology and paragraph 7 describes how the distribution of quasars provides that support.
- ☐ Paragraph 6 describes the similarities between the steady-state and Big Bang theories, while paragraph 7 explains the differences between them.
- ☐ Paragraph 6 presents a question about quasars while paragraph 7 provides several possible answers to that question.
- ☐ Paragraph 6 describes the chemical composition of quasars while paragraph 7 describes the locations of quasars.

Paragraphs 6 and 7 are marked with arrows [→].

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

In turn this means that over a period of billions of years the large-scale distribution of the kinds of galaxies the universe contains has fundamentally changed.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Drag your choices to the spaces where they belong. To review the passage, click on

[View Text.](#)

Evolutionary and steady-state theories have been proposed to explain the large-scale structure of the universe.

-
-
-

Answer Choices

The Big Bang theory maintains that the continuous creation of matter is the explanation for why the universe is expanding at a constant rate. While the steady-state cosmology

rejects the belief that the universe has an end, it accepts the possibility that the universe had a beginning.

The distribution of quasars suggests that the large-scale structure of the universe has changed over time and thus makes the evolutionary theory more plausible than the steady-state theory.

The evolutionary theory maintains that the universe had a beginning with a high density explosion and has been expanding to yield a less dense

distribution of matter ever since.

The steady-state theory maintains that the expanding universe has existed forever, with new matter being continuously created to keep the large-scale density of matter the same as we observe it today.

The extreme brightness of quasars is proof of an explosion that marked the beginning of the universe as hypothesized by the evolutionary cosmology.



Distribution of Seaweeds

Seaweeds are multicellular algae that inhabit the oceans. Despite their evolutionary distance from each other, seaweeds—such as brown algae, red algae, and green algae—have in common many aspects of their biology and contributions to the ecology of the seas.

Most species of seaweed are benthic (living on the seafloor); they grow on rock, sand, mud, and coral on the sea bottom. Other species live on other organisms and as part of fouling communities (plants and animals that live on pilings, boat bottoms, and other artificial surfaces). Some seaweeds attach to very specific surfaces, whereas other seaweeds are rather nonselective. In general, seaweeds inhabit about 2 percent of the seafloor. The presence of benthic seaweeds defines the inner continental shelf, where the marine community largely depends on the food and protection that seaweeds provide. Life on the outer continental shelf and in the deep sea is quite different in the absence of seaweeds. The distinction between the inner and outer shelves is based on the compensation depth of algae. The compensation depth is the depth of water at which there is just enough light for algae to survive. At that depth all the oxygen produced by photosynthesis is consumed by the algae's respiration, so that no further growth can occur.

The environmental factors that are most influential in governing the distribution of seaweeds are light and temperature. Some other abiotic (nonliving) factors critical in governing the distribution of seaweeds are duration of tidal exposure and desiccation (drying out), wave action and surge, salinity, and availability of mineral nutrients. The areas of the world most favorable to seaweed diversity include both sides of the North Pacific Ocean, Australia, southwestern Africa, and the Mediterranean Sea.

The vertical and horizontal distributions of seaweeds are limited in part by the availability of sunlight and, therefore, vary by depth, latitude, sea conditions, and season. It was once thought that the vertical distribution of red, brown, and green algae could be explained by their accessory photosynthetic pigments (photosynthetic pigments other than chlorophyll a), the presence of which gives the seaweeds their characteristic colors, a concept known as chromatic adaptation. Because green light penetrates deepest in coastal waters and the accessory pigments of red algae absorb mostly green wavelengths, red algae were thought to extend to the greatest depth. It followed that green algae, which have pigments absorbing mostly blue and red wavelengths that are diminished rapidly in seawater, should be found at the shallowest depths. Because accessory pigments of brown algae absorb intermediate wavelengths of light, brown algae would be expected to be most abundant at intermediate depths. Indeed, some recent evidence would seem to support the

hypothesis of chromatic adaptation because the depth record (295 meters, or 973 feet) for seaweeds is held by a yet undescribed species of red algae from the Bahamas. However, the green alga *Rhipiliopsis profunda* is close behind this record at 268 meters (884 feet).

The concept of chromatic adaptation was proposed in 1883, and the hypothesis was accepted for about 100 years, until it was realized that such zonation did not necessarily occur and that the distribution of seaweeds depended more on herbivory (the consumption of plant material), competition, varying concentration of the specialized pigments, and the ability of seaweeds to alter their forms of growth.

Temperature affects the distribution of seaweeds. The greatest diversity of algal species is in tropical waters. Farther north or south of the equator, the number of species decreases, and the species themselves are different. Many marine algae in colder latitudes are perennials, meaning that they live longer than two years. During the colder seasons only part of the alga remains alive, sometimes only a few cells, but most often a mass of stemlike structures. When the temperature warms up in the spring, this body part initiates new growth. Temperature is not usually a limiting factor for algae that live in tropical and subtropical seas, although temperatures in intertidal areas (those areas between high and low tides) may become too warm and contribute to seasonal mass mortality of many seaweeds and the animals they shelter. At high latitudes, freezing and scouring by ice may eliminate seaweeds from the intertidal and shallow subtidal zones.

Paragraph 2

Most species of seaweed are benthic (living on the seafloor); they grow on rock, sand, mud, and coral on the sea bottom. Other species live on other organisms and as part of fouling communities (plants and animals that live on pilings, boat bottoms, and other artificial surfaces). Some seaweeds attach to very specific surfaces, whereas other seaweeds are rather nonselective. In general, seaweeds inhabit about 2 percent of the seafloor. The presence of benthic seaweeds defines the inner continental shelf, where the marine community largely depends on the food and protection that seaweeds provide. Life on the outer continental shelf and in the deep sea is quite different in the absence of seaweeds. The distinction between the inner and outer shelves is based on the compensation depth of algae. The compensation depth is the depth of water at which there is just enough light for algae to survive. At that depth all the oxygen produced by photosynthesis is consumed by the algae's respiration, so that no further growth can occur.

1. The word "artificial" in the passage is closest in meaning to
 - ☐ suitable
 - ☐ human-made
 - ☐ uneven
 - ☐ available

2. What can be inferred from paragraph 2 about the outer continental shelf?
- ☐ The outer continental shelf contains more benthic seaweeds than the deep sea does.
 - ☐ The outer continental shelf lacks enough sunlight to support the growth of algae.
 - ☐ The outer continental shelf has a compensation depth that is about the same as that of the inner continental shelf.
 - ☐ The outer continental shelf has a greater variety of marine life than does the inner continental shelf.

Paragraph 2 is marked with an arrow [→].

3. According to paragraph 2, all of the following are true of seaweeds EXCEPT:
- ☐ Seaweeds provide food and protection for other organisms.
 - ☐ Seaweeds can grow on both living and nonliving things.
 - ☐ Some seaweeds select specific surfaces to attach to, while others are not selective.
 - ☐ Two percent of seaweed species live on the ocean floor.

Paragraph 2 is marked with an arrow [→].

Paragraph 3

The environmental factors that are most influential in governing the distribution of seaweeds are light and temperature. Some other abiotic (nonliving) factors critical in governing the distribution of seaweeds are duration of tidal exposure and desiccation (drying out), wave action and surge, salinity, and availability of mineral nutrients. The areas of the world most favorable to seaweed diversity include both sides of the North Pacific Ocean, Australia, southwestern Africa, and the Mediterranean Sea.

4. The word “duration” in the passage is closest in meaning to
- ☐ frequency
 - ☐ force of impact
 - ☐ length of time
 - ☐ occurrence

Paragraph 4

The vertical and horizontal distributions of seaweeds are limited in part by the availability of sunlight and, therefore, vary by depth, latitude, sea conditions, and season. It was once thought that the vertical distribution of red, brown, and green algae could be explained by their accessory photosynthetic pigments (photosynthetic pigments other than chlorophyll a), the presence of which gives the seaweeds their characteristic colors, a concept known as chromatic adaptation. Because green light penetrates deepest in coastal waters and the accessory pigments of red algae absorb mostly green wavelengths, red algae were thought to extend to the greatest depth. It followed that green algae, which have pigments absorbing mostly blue and red

wavelengths that are diminished rapidly in seawater, should be found at the shallowest depths. Because accessory pigments of brown algae absorb intermediate wavelengths of light, brown algae would be expected to be most abundant at intermediate depths. Indeed, some recent evidence would seem to support the hypothesis of chromatic adaptation because the depth record (295 meters, or 973 feet) for seaweeds is held by a yet undescribed species of red algae from the Bahamas. However, the green alga *Rhipiliopsis profunda* is close behind this record at 268 meters (884 feet).

5. The word “diminished” in the passage is closest in meaning to
- ☐ reduced
 - ☐ scattered
 - ☐ transmitted
 - ☐ altered
6. According to the concept of chromatic adaptation explained in paragraph 4, which of the following types of algae should be found closest to the ocean’s surface?
- ☐ Red algae
 - ☐ Blue algae
 - ☐ Green algae
 - ☐ Brown algae

Paragraph 4 is marked with an arrow [→].

7. Why does the author mention “a yet undescribed species of red algae from the Bahamas”?
- ☐ To suggest that the concept of chromatic adaptation applies only to algae living in deep waters
 - ☐ To argue that location is more important than chromatic adaptation in determining the vertical distribution of algae
 - ☐ To emphasize the difficulty of testing the concept of chromatic adaptation in deep waters
 - ☐ To provide information that may confirm the idea that chromatic adaptation is responsible for the vertical distribution of algae
8. What can be inferred from the author’s discussion of the green algae *Rhipiliopsis profunda* in paragraph 4?
- ☐ Chromatic adaptation may not be an adequate explanation for the vertical distribution of algae.
 - ☐ All types of algae share the same accessory photosynthetic pigments.
 - ☐ Chromatic adaptation is responsible for the distribution of red algae but not for the distribution of green algae.
 - ☐ Both red and green algae absorb red wavelengths.

Paragraph 4 is marked with an arrow [→].

Paragraph 5

The concept of chromatic adaptation was proposed in 1883, and the hypothesis was accepted for about 100 years, until it was realized that such zonation did not necessarily occur and that the distribution of seaweeds depended more on herbivory (the consumption of plant material), competition, varying concentration of the specialized pigments, and the ability of seaweeds to alter their forms of growth.

9. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ For about 100 years, chromatic adaptation was thought to have a greater impact on seaweed distribution than various other factors such as herbivory and competition.
 - ☐ The hypothesis of chromatic adaptation was accepted for about 100 years until it was concluded that other factors could explain the distribution of seaweeds.
 - ☐ About 100 years after the concept of chromatic adaptation was developed, it was realized that seaweed zonation occurred even in the absence of chromatic adaptation.
 - ☐ Until the hypothesis of chromatic adaptation became widely accepted 100 years after it was proposed, various other factors were thought to determine the distribution of seaweeds.

Paragraph 6

Temperature affects the distribution of seaweeds. The greatest diversity of algal species is in tropical waters. Farther north or south of the equator, the number of species decreases, and the species themselves are different. Many marine algae in colder latitudes are perennials, meaning that they live longer than two years. During the colder seasons only part of the alga remains alive, sometimes only a few cells, but most often a mass of stemlike structures. When the temperature warms up in the spring, this body part initiates new growth. Temperature is not usually a limiting factor for algae that live in tropical and subtropical seas, although temperatures in intertidal areas (those areas between high and low tides) may become too warm and contribute to seasonal mass mortality of many seaweeds and the animals they shelter. At high latitudes, freezing and scouring by ice may eliminate seaweeds from the intertidal and shallow subtidal zones. 刨蚀

10. What is the author's purpose in discussing "temperatures in intertidal areas"?
- ☐ To emphasize the idea that warm temperatures can be just as dangerous to algae as cold temperatures
 - ☐ To identify a limiting factor for algae common to both tropical seas and waters at higher latitudes
 - ☐ To compare the causes of seaweed mortality in tropical seas and in seas at higher latitudes

- ☐ To note an exception to the idea that temperature does not limit the growth of algae in tropical and subtropical waters

11. The word “mortality” in the passage is closest in meaning to

- ☐ movement
- ☐ exposure
- ☐ death
- ☐ injury

12. According to paragraph 6, all of the following are true of perennials EXCEPT:

- ☐ Perennials live for more than two years.
- ☐ Only part of a perennial remains alive during the winter.
- ☐ A large number of the marine algae found in cold latitudes are perennials.
- ☐ Species in the tropics are more sensitive to temperature than are species in colder latitudes.

Paragraph 4 is marked with an arrow [→].

Paragraph 2

Most species of seaweed are benthic (living on the seafloor); they grow on rock, sand, mud, and coral on the sea bottom. Other species live on other organisms and as part of fouling communities (plants and animals that live on pilings, boat bottoms, and other artificial surfaces). ■ Some seaweeds attach to very specific surfaces, whereas other seaweeds are rather nonselective. In general, seaweeds inhabit about 2 percent of the seafloor. ■ The presence of benthic seaweeds defines the inner continental shelf, where the marine community largely depends on the food and protection that seaweeds provide. ■ Life on the outer continental shelf and in the deep sea is quite different in the absence of seaweeds. ■ The distinction between the inner and outer shelves is based on the compensation depth of algae. The compensation depth is the depth of water at which there is just enough light for algae to survive. At that depth all the oxygen produced by photosynthesis is consumed by the algae’s respiration, so that no further growth can occur.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

They are most prevalent on the part of the sea bottom known as the continental shelf.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in

the passage or are minor ideas in the passage. **This question is worth 2 points.**
 Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Seaweeds are multicellular algae living in the oceans.

-
-
-

Answer Choices

Seaweeds are limited to areas with enough light for them to survive, with most living on the seafloor and more species inhabiting warm waters than colder waters.

Some recent evidence shows that the color of any species of seaweed is determined by the amount of chlorophyll a contained in its accessory pigments.

The greatest diversity of seaweed is found in warm waters, and seaweed cannot survive when the temperature is either too hot or too cold.

Seaweeds are distributed most evenly over areas of ocean that experience changing tides, strong waves, a dry climate, and high concentrations of salt and minerals.

The chromatic-adaptation hypothesis was long thought to explain the vertical distribution of seaweed, but current belief is that distribution is better explained by other factors.

Temperature extremes in tropical and subtropical areas as well as at high latitudes cause seaweeds to abandon the intertidal areas for the subtidal zones of the seafloor.

Bird Colonies

About 13 percent of bird species, including most seabirds, nest in colonies. Colonial nesting evolves in response to a combination of two environmental conditions: (1) a shortage of nesting sites that are safe from predators and (2) abundant or unpredictable food that is distant from safe nest sites. First and foremost, individual birds are safer in colonies that are inaccessible to predators, as on small rocky islands. In addition, colonial birds detect predators more quickly than do small groups or pairs and can drive the predators from the vicinity of the nesting area. Because nests at the edges of breeding colonies are more vulnerable to predators than those in the centers, the preference for advantageous central sites promotes dense centralized packing of nests.

The yellow-rumped cacique, which nests in colonies in Amazonian Peru, demonstrates how colonial birds prevent predation. These tropical blackbirds defend their closed, pouchlike nests against predators in three ways. First, by nesting on islands and near wasp nests, caciques are safe from arboreal mammals such as primates. Second, caciques mob predators (work together as a group to attack predators). The effectiveness of mobbing increases with group size, which increases with colony size. Third, caciques hide their nests from predators by mixing active nests with abandoned nests. Overall, nests in cluster on islands and near wasp nests suffer the least predation.

Coordinated social interactions tend to be weak when a colony is first forming, but true colonies provide extra benefits. Synchronized nesting, for example, produces a sudden abundance of eggs and chicks that exceeds the daily needs of local predators. Additionally, colonial neighbors can improve their foraging by watching others. This behavior is especially valuable when the off-site food supplies are restricted or variable in location, as are swarms of aerial insects harvested by swallows. The colonies of American cliff swallows, for example, serve as information centers from which unsuccessful individual birds follow successful neighbors to good feeding sites. Cliff swallows that are unable to find food return to their colony, locate a neighbor that has been successful, and then follow that neighbor to its food source. All birds in the colony are equally likely to follow or to be followed and thus contribute to the sharing of information that helps to ensure their reproductive success. As a result of their enhanced foraging efficiency, parent swallows in large colonies return with food for their nestlings more often and bring more food each trip than do parents in small colonies.

To support large congregations of birds, suitable colony sites must be near rich, clumped food supplies. Colonies of pinyon jays and red crossbills settle near seed-rich conifer forests, and wattled starlings nest in large colonies near locust

outbreaks. The huge colonies of guanay cormorants and other seabirds that nest on the coast of Peru depend on the productive cold waters of the Humboldt Current. The combination of abundant food in the Humboldt Current and the vastness of oceanic habitat can support enormous populations of seabirds, which concentrate at the few available nesting locations. The populations crash when their food supplies decline during [El Nino](#) years.

Among the costs, colonial nesting leads to increased competition for nest sites and mates, the stealing of nest materials, and increased physical interference among other effects. In spite of food abundance, large colonies sometimes exhaust their local food supplies and abandon their nests. Large groups also attract predators, especially raptors, and facilitate the spread of parasites and diseases. The globular mud nests in large colonies of the American cliff swallow, for example, are more likely to be infested by fleas or other bloodsucking parasites than are nests in small colonies. Experiments in which some burrows were fumigated to kill the parasites showed that these parasites lowered survivorship by as much as 50 percent in large colonies but not significantly in small ones. The swallows inspect and then select parasite-free nests in large colonies, they tend to build new nests rather than use old, infested ones. On balance, the advantages of colonial nesting clearly outweigh the disadvantages, given the many times at which colonial nesting has evolved independently among different groups of birds. Still lacking, however, is a general framework for testing different hypothesis for the evolution of coloniality.

Paragraph 1

About 13 percent of bird species, including most seabirds, nest in colonies. Colonial nesting evolves in response to a combination of two environmental conditions: (1) a shortage of nesting sites that are safe from predators and (2) abundant or unpredictable food that is distant from safe nest sites. First and foremost, individual birds are safer in colonies that are inaccessible to predators, as on small rocky islands. In addition, colonial birds detect predators more quickly than do small groups or pairs and can drive the predators from the vicinity of the nesting area. Because nests at the edges of breeding colonies are more vulnerable to predators than those in the centers, the preference for advantageous central sites promotes dense centralized packing of nests.

1. The word “[vicinity](#)” in the passage is closest in meaning to
 - ☐ protection
 - ☐ region
 - ☐ population
 - ☐ resources
2. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- It is more advantageous for birds to choose central locations for their colonies rather than locations near the edges of their territory.
- Compared to nests at the edges of colonies, centrally located nests are preferred for their safety from predators and therefore are more densely packed together.
- Predators generally prefer the densely packed, central portion of nesting colonies, which can make this part of the colony more vulnerable to predators.
- Birds nesting in colonies that are vulnerable to predators tend to prefer more densely packed nests to those less densely packed.

Paragraph 2

The yellow-rumped cacique, which nests in colonies in Amazonian Peru, demonstrates how colonial birds prevent predation. These tropical blackbirds defend their closed, pouchlike nests against predators in three ways. First, by nesting on islands and near wasp nests, caciques are safe from arboreal mammals such as primates. Second, caciques mob predators (work together as a group to attack predators). The effectiveness of mobbing increases with group size, which increases with colony size. Third, caciques hide their nests from predators by mixing active nests with abandoned nests. Overall, nests in cluster on islands and near wasp nests suffer the least predation.

3. Paragraph 2 implies which of the following about yellow-rumped caciques?
- They are comparatively unlikely to be harmed by the wasps that attack their predators.
 - They are able to protect their nests without using colonies.
 - Mixing active nests with abandoned nests is the least useful way of defending their nests.
 - Most of their predators are members of other bird species.

Paragraph 2 is marked with an arrow [→].

4. Paragraph 2 claims that yellow-rumped cacique colonies defend themselves from predators in all of the following ways EXCEPT:
- They establish colonies in hard-to-reach places.
 - They physically attack invading predators.
 - They hide active nests among previously used ones.
 - They limit the size of their colonies so they are hard to find.

Paragraph 2 is marked with an arrow [→].

Paragraph 3 simultaneous grazing feeding

Coordinated social interactions tend to be weak when a colony is first forming, but true colonies provide extra benefits. Synchronized nesting, for example, produces a sudden abundance of eggs and chicks that exceeds the daily needs of local predators. Additionally, colonial neighbors can improve their foraging by watching others. This behavior is especially valuable when the off-site food supplies are restricted or variable in location, as are swarms of aerial insects harvested by swallows. The

colonies of American cliff swallows, for example, serve as information centers from which unsuccessful individual birds follow successful neighbors to good feeding sites. Cliff swallows that are unable to find food return to their colony, locate a neighbor that has been successful, and then follow that neighbor to its food source. All birds in the colony are equally likely to follow or to be followed and thus contribute to the sharing of information that helps to ensure their reproductive success. As a result of their enhanced foraging efficiency, parent swallows in large colonies return with food for their nestlings more often and bring more food each trip than do parents in small colonies.

5. The phrase “contribute to” in the passage is closest in meaning to
- ☐ learn from
 - ☐ depend upon
 - ☐ take advantage of
 - ☐ add to
6. According to paragraph 3, what advantage do birds gain by hatching all the colony’s eggs at the same time?
- ☐ They are able to time the hatching of their chicks for when predators are not likely to be around.
 - ☐ Chicks hatch when food is abundant, which is generally only a few times a year.
 - ☐ Even if predators eliminate some of the newly hatched birds, many others will survive.
 - ☐ Weaker birds gain protection for their young by synchronizing their nesting behavior with that of the more dominant birds of the colony.

Paragraph 3 is marked with an arrow [→].

7. According to paragraph 3, cliff swallows closely observe other members of their colony in order to
- ☐ follow successful birds to safe nesting sites
 - ☐ learn the location of local predators
 - ☐ place their eggs near other birds in the colony
 - ☐ find good sources of food

Paragraph 3 is marked with an arrow [→].

Paragraph 4

To support large congregations of birds, suitable colony sites must be near rich, clumped food supplies. Colonies of pinyon jays and red crossbills settle near seed-rich conifer forests, and wattled starlings nest in large colonies near locust outbreaks. The huge colonies of guanay cormorants and other seabirds that nest on the coast of Peru depend on the productive cold waters of the Humboldt Current. The combination of abundant food in the Humboldt Current and the vastness of oceanic habitat can support enormous populations of seabirds, which concentrate at the few available nesting locations. The populations crash when their food supplies

decline during **El Nino** years.

8. Which of the following is a probable effect of the fact mentioned in paragraph 4 that there are few available nesting locations near the Humboldt Current?

- ☐ Seabirds compete with each other for a limited supply of food.
- ☐ The number of seabirds at any one site is extremely large.
- ☐ Some seabirds nest in conifer forests near locust outbreaks.
- ☐ Colonies near the Humboldt Current contain small numbers of seabirds.

Paragraph 4 is marked with an arrow [→].

Paragraph 5

Among the costs, colonial nesting leads to increased competition for nest sites and mates, the stealing of nest materials, and increased physical interference among other effects. In spite of food abundance, large colonies sometimes **exhaust** their local food supplies and abandon their nests. Large groups also attract predators, especially raptors, and facilitate the spread of parasites and diseases. The globular mud nests in large colonies of the American cliff swallow, for example, are more likely to be infested by fleas or other bloodsucking parasites than are nests in small colonies. Experiments in which some burrows were fumigated to kill the parasites showed that these parasites lowered survivorship by as much as 50 percent in large colonies but not significantly in small ones. The swallows inspect and then select parasite-free nests in large colonies, they tend to build new nests rather than use old, infested ones. **On balance**, the advantages of colonial nesting clearly outweigh the disadvantages, given the many times at which colonial nesting has evolved independently among different groups of birds. Still lacking, however, is a general framework for testing different hypothesis for the evolution of coloniality.

9. The word “**exhaust**” in the passage is closet in meaning to

- ☐ use up
- ☐ leave
- ☐ avoid
- ☐ hide

10. The phrase “**On balance**” in the passage is closet in meaning to

- ☐ Nevertheless
- ☐ Overall
- ☐ Therefore
- ☐ Periodically

11. In paragraph 5, why does the author discuss experiments in which some burrows were fumigated?

- ☐ To demonstrate that parasites have a very negative effect on large colonies
- ☐ To show that attacks by predators are a worse problem than the spread of parasites

- ☐ To explain how swallows inspect nests for parasites
- ☐ To prove that the benefits of colonial nesting outweigh the disadvantages

Paragraph 5 is marked with an arrow [→].

12. Which of the following questions is NOT answered by paragraph 5?

- ☐ What causes colonies to have problems with their food supplies?
- ☐ What are the disadvantages of colonial nesting?
- ☐ What percentage of cliff swallow nests are infected by parasites?
- ☐ How can one test the different hypotheses explaining the evolution of bird colonies?

Paragraph 5 is marked with an arrow [→].

Paragraph 3

Coordinated social interactions tend to be weak when a colony is first forming, but true colonies provide extra benefits. ■ Synchronized nesting, for example, produces a sudden abundance of eggs and chicks that exceeds the daily needs of local predators. ■ Additionally, colonial neighbors can improve their foraging by watching others. ■ This behavior is especially valuable when the off-site food supplies are restricted or variable in location, as are swarms of aerial insects harvested by swallows. ■ The colonies of American cliff swallows, for example, serve as information centers from which unsuccessful individual birds follow successful neighbors to good feeding sites. Cliff swallows that are unable to find food return to their colony, locate a neighbor that has been successful, and then follow that neighbor to its food source. All birds in the colony are equally likely to follow or to be followed and thus contribute to the sharing of information that helps to ensure their reproductive success. As a result of their enhanced foraging efficiency, parent swallows in large colonies return with food for their nestlings more often and bring more food each trip than do parents in small colonies.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

The overall survival of the next generation is thus enhanced.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

In response to environmental pressures, many species of birds have developed the practice of nesting in colonies.

-
-
-

Answer Choices

The abundance of food on rocky islands makes them an attractive environment for bird species that nest in colonies, including the yellow-rumped cacique.

Because they are not able to physically defend themselves from predators, yellow-rumped caciques rely on hiding their active nests among empty nests to confuse predators.

The concentration of bird colonies along coasts makes their inhabitants especially vulnerable to food shortages and weather-related destruction.

The challenge of protecting nests and newborn chicks from predators is a major contributor to the evolution of large colonies.

Colonial life provides birds with benefits besides protection from predators including help with finding food.

There are some disadvantages associated with life in large colonies like the increased rate of infestation by parasites.

2014-04-27

Urban Development in the United States in the Nineteenth Century

In discussing the growth of cities in the United States in the nineteenth century, one cannot really use the term “urban planning,” as it suggests modern concerns for spatial and service organization which, in most instances, did not exist before the planning revolution called the City Beautiful Movement that began in the 1890s. While there certainly were urban areas that were “planned” in the comprehensive contemporary sense of the word before that date, most notably Washington, D.C., these were the exception. Most “planned” in the nineteenth century was limited to areas much smaller than a city and was closely associated with developers trying to make a profit from a piece of land. Even when these small-scale plans were well designed, the developers made only those improvements that were absolutely necessary to attract the wealthy segment of the market. Indeed, it was the absence of true urban planning that allowed other factors to play such an important role in shaping the nineteenth-century American city.

Three forces particularly affected the configuration of urban and suburban areas in the nineteenth century: economics, transportation technology, and demographics. Added to these was the characteristic American preference both for independent living, usually associated with having an individual, free-standing home for one’s family, and for rural living. Economics affected urbanization in two ways. First, economic considerations influenced location decisions for business and industry, which often preempted choice sites. Second, industrial growth generated higher incomes for large segments of the population, which in turn provided more money for larger homes and commuter transportation. Related to economics (since costs to individuals always played a role) were improvements in transportation, from the first horse-drawn buses of the 1820s to electrified street railways at the end of the century. Each transport innovation extended the distance that a person could reasonably travel as a commuter or shopper, while constant system improvements and increased ridership lessened costs.

Demographic patterns also affected urbanization in two ways: first, urban populations grew steadily throughout the century due to immigration from rural areas, principally by those seeking factory work, and emigration from abroad. Therefore cities expanded as new housing had to be provided. Secondly, at the same time that new residents were surging into cities, many urbanites, particularly those of the middle classes, began to leave. While a preference for rural living explained part of this exodus, it was also due to the perception that various urban problems were becoming worse.

Many nineteenth-century urban problems were those that continue to plague cities today—crime, pollution, noise—but others were the direct result of lack of planning and regulation, such as threat of fire, poor sanitation, and shoddy building construction. Fire was a significant problem in urban areas of North America from the time of the first European settlement. Construction with combustible materials coupled with close placement of buildings and the use of open flames in heating, cooking, and lighting meant that the potential for raging

fires was ever present. Lack of sanitation, and the ensuing public health problems it created, was a more constant, if less dramatic, urban issue. It was not until the 1860s that any serious, concerted effort was made to develop proper systems for water delivery and sewage removal. In spite of remarkable strides made in the 1870s and 1880s by the newly established profession of sanitary engineering, the common nineteenth-century pattern of individual unprofessionally planned and installed cesspools (underground tanks for holding household sewage) continued. This led to water contamination and the spread of disease by rodents and insects.

Problems of fire and poor sanitation were inextricably linked with the last major urban problem of the nineteenth century—lack of coordination in the physical expansion of cities and their infrastructure systems (systems for providing services such as water, gas, electricity, and sewage). Typically, development was both unplanned and unrestricted, with landowners making all choices of lot size, services, and street arrangement based only on their individual needs in the marketplace. Distortions of streets and abrupt changes in the distance of houses from the street in urban areas, which so clearly delineate where one development ended and another began, were just the most obvious problems that this lack of coordination created.

Paragraph 1

In discussing the growth of cities in the United States in the nineteenth century, one cannot really use the term “urban planning,” as it suggests modern concerns for spatial and service organization which, in most instances, did not exist before the planning revolution called the City Beautiful Movement that began in the 1890s. While there certainly were urban areas that were “planned” in the comprehensive contemporary sense of the word before that date, most notably Washington, D.C., these were the exception. Most “planned” in the nineteenth century was limited to areas much smaller than a city and was closely associated with developers trying to make a profit from a piece of land. Even when these small-scale plans were well designed, the developers made only those improvements that were absolutely necessary to attract the wealthy segment of the market. Indeed, it was the absence of true urban planning that allowed other factors to play such an important role in shaping the nineteenth-century American city.

1. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - ☐ Understanding the growth of cities in nineteenth-century America requires recognizing how the City Beautiful Movement of the 1890s changed “urban planning.”
 - ☐ For the most part, there was no “urban planning,” as that term is understood today, before the beginning of the City Beautiful Movement in the 1890s.
 - ☐ Concerns for spatial and service organization had little impact on the growth of cities before the 1890s when the City Beautiful Movement began.
 - ☐ The growth of cities in nineteenth-century America resulted in the creation of the City Beautiful Movement in the 1890s and the rise of the term “urban planning.”
2. According to paragraph 1, Washington, D.C. was
 - ☐ a typical nineteenth-century American city

- a city that was planned in separate sections by land developers
- the very first city in America to be described as “planned”
- one of the few cases of true urban planning in America before the 1890s

Paragraph 1 is marked with an arrow [→].

3. Select the TWO answer choices that, according to the paragraph 1, best describe most urban plans in the nineteenth century. To receive credit, you must select TWO answers.

- ☐ They were created to profit land developers.
- ☐ They typically affected only part of a city rather than the whole city.
- ☐ Their success sometimes directly led to other necessary improvements in cities.
- ☐ They were carefully reviewed by city governments.

Paragraph 1 is marked with an arrow [→].

Paragraph 2

Three forces particularly affected the configuration of urban and suburban areas in the nineteenth century: economics, transportation technology, and demographics. Added to these was the characteristic American preference both for independent living, usually associated with having an individual, free-standing home for one’s family, and for rural living. Economics affected urbanization in two ways. First, economic considerations influenced location decisions for business and industry, which often preempted choice sites. Second, industrial growth generated higher incomes for large segments of the population, which in turn provided more money for larger homes and commuter transportation. Related to economics (since costs to individuals always played a role) were improvements in transportation, from the first horse-drawn buses of the 1820s to electrified street railways at the end of the century. Each transport innovation extended the distance that a person could reasonably travel as a commuter or shopper, while constant system improvements and increased ridership lessened costs.

4. In paragraph 2, the author mentions the characteristic American preference both for independent living and for rural living to
- identify one of the factors that affected the configuration of urban and suburban areas in America in the nineteenth century
 - explain which of the three forces mentioned—economics, transportation technology, and demographics—was the most important in shaping American cities and suburbs
 - explain how decisions were made in American nineteenth-century cities and suburbs about locations for businesses and factories
 - provide evidence showing that industrial growth was able to generate higher incomes for much of the population in nineteenth-century cities and suburbs in America

Paragraph 2 is marked with an arrow [→].

Paragraph 3

Demographic patterns also affected urbanization in two ways: first, urban populations grew **steadily** throughout the century due to immigration from rural areas, principally by those seeking factory work, and emigration from abroad. Therefore cities expanded as new housing had to be provided. Secondly, at the same time that new residents were surging into cities, many urbanites,

particularly those of the middle classes, began to leave. While a preference for rural living explained part of this exodus, it was also due to the perception that various urban problems were becoming worse.

5. The word “steadily” in the passage is closest in meaning to
- ☐ significantly
 - ☐ rapidly
 - ☐ continuously
 - ☐ unevenly
6. Which of the following can be inferred from paragraph 3 about changes in the demographics of cities during the nineteenth century?
- ☐ The reason most people left the city was to take jobs in the country.
 - ☐ The middle class population increased.
 - ☐ The population became more ethnically diverse.
 - ☐ The working class population tended to live in the oldest housing.

Paragraph 3 is marked with an arrow [→].

Paragraph 4

Many nineteenth-century urban problems were those that continue to plague cities today—crime, pollution, noise—but others were the direct result of lack of planning and regulation, such as threat of fire, poor sanitation, and shoddy building construction. Fire was a significant problem in urban areas of North America from the time of the first European settlement. Construction with combustible materials coupled with close placement of buildings and the use of open flames in heating, cooking, and lighting meant that the potential for raging fires was ever present. Lack of sanitation, and the ensuing public health problems it created, was a more constant, if less dramatic, urban issue. It was not until the 1860s that any serious, concerted effort was made to develop proper systems for water delivery and sewage removal. In spite of remarkable strides made in the 1870s and 1880s by the newly established profession of sanitary engineering, the common nineteenth-century pattern of individual unprofessionally planned and installed cesspools (underground tanks for holding household sewage) continued. This led to water contamination and the spread of disease by rodents and insects.

7. The word “plague” in the passage is closest in meaning to
- ☐ be discussed in
 - ☐ be found in
 - ☐ isolate
 - ☐ cause trouble for
8. According to paragraph 4, lack of adequate systems for removing sewage led to
- ☐ the appearance of new types of rodents and insects
 - ☐ the development of new, more effective cesspools
 - ☐ disruptions of water-delivery systems
 - ☐ contamination of water supplies

Paragraph 4 is marked with an arrow [→].

Paragraph 5

Problems of fire and poor sanitation were inextricably linked with the last major urban problem of the nineteenth century—lack of coordination in the physical expansion of cities and their infrastructure systems (systems for providing services such as water, gas, electricity, and sewage). Typically, development was both unplanned and unrestricted, with landowners making all choices of lot size, services, and street arrangement based only on their individual needs in the marketplace. Distortions of streets and abrupt changes in the distance of houses from the street in urban areas, which so clearly delineate where one development ended and another began, were just the most obvious problems that this lack of coordination created.

9. The word “plague” in the passage is closest in meaning to
- ☐ noticeable
 - ☐ random
 - ☐ variable
 - ☐ sudden
10. According to paragraph 5, one of the major consequences of the lack of restrictions and comprehensive urban planning was that
- ☐ infrastructure systems were coordinated by local groups rather than by city governments
 - ☐ cities became smaller over time because people began leaving
 - ☐ landowners developed urban lots however they wished
 - ☐ some housing developments that were begun were never completed

Paragraph 5 is marked with an arrow [→].

11.

12.

13.

The Climate of Japan

At the most general level, two major climatic forces determine Japan's weather. Prevailing westerly winds move across Eurasia, sweep over the Japanese islands, and continue eastward across the Pacific Ocean. In addition, great cyclonic airflows (masses of rapidly circulating air) that arise over the western equatorial Pacific move in a wheel-like fashion northeastward across Japan and nearby regions. During winter months heavy masses of cold air from Siberia dominate the weather around Japan. Persistent cold winds skim across the Sea of Japan from the northwest, picking up moisture that they deposit as several feet of snow on the western side of the mountain ranges on Honshu Island. As the cold air drops its moisture, it flows over high ridges and down eastern slopes to bring cold, relatively dry weather to valleys and coastal plains and cities.

In spring the Siberian air mass warms and loses density, enabling atmospheric currents over the Pacific to steer warmer air into northeast Asia. This warm, moisture-laden air covers most of southern Japan during June and July. The resulting late spring rains then give way to a drier summer that is sufficiently hot and muggy, despite the island chain's northerly latitude, to allow widespread rice cultivation.

Summer heat is followed by the highly unpredictable autumn rains that accompany the violent tropical windstorms known as typhoons. These cyclonic storms originate over the western Pacific and travel in great clockwise arcs, initially heading west toward the Philippines and southern China, curving northward later in the season. Cold weather drives these storms eastward across Japan through early autumn, revitalizing the Siberian air mass and ushering in a new annual weather cycle.

This yearly cycle has played a key role in shaping Japanese civilization. It has assured the islands ample precipitation, ranging irregularly from more than 200 centimeters annually in parts of the southwest to about 100 in the northeast and averaging 180 for the country as a whole. The moisture enables the islands to support uncommonly lush forest cover, but the combination of precipitous slopes and heavy rainfall also gives the islands one of the world's highest rates of natural erosion, intensified by both human activity and the natural shocks of earthquakes and volcanism. These factors have given Japan its wealth of sedimentary basins, but they have also made mountainsides extremely susceptible to erosion and landslides and hence generally unsuitable for agricultural manipulation.

The island chain's mountains backbone and great length from north to south produce climatic diversity that has contributed to regional differences. Generally sunny winters along the Pacific seaboard have made habitation there relatively pleasant. Along the Sea of Japan, on the other hand, cold, snowy winters have discouraged settlement. Furthermore, although annual precipitation is high in that region, much of it comes as snow and rushes to the sea as spring runoff, leaving little moisture for farming.

Summer weather patterns in northern Honshu, and especially along the Sea of Japan, have also discouraged agriculture. The area is subject to the *yamase* effect, when cool air from the north sometimes lowers temperatures sharply and damages farm production. The impact of this effect has been especially great on rice cultivation because, if it is to grow well, the rice grown in Japan requires a mean summer temperature of 20° centigrade or higher. A drop of 2°-3° can lead to a 30-50 percent drop in rice yield, and the *yamase* effect is capable of exceeding that level. This *yamase* effect does not, however, extend very far south, where most precipitation comes in the form of rain and the bulk of it in spring, summer, and fall, when most useful for cultivation. Even the autumn typhoons, which deposit most of their moisture along the southern seaboard, are beneficial because they promote the start of the winter crops that for centuries have been grown in southern Japan.

In short, for the past two millennia, the climate in general and patterns of precipitation in particular have encouraged the Japanese to cluster their settlements along the southern coast, most densely along the sheltered Inland Sea, moving into the northeast. There the limits that topography imposed on production have been tightened by climate, with the result that agricultural output has been more modest and less reliable, making the risk of crop failure and hardship commensurately greater.

Paragraph 1

At the most general level, two major climatic forces determine Japan's weather. Prevailing westerly winds move across Eurasia, sweep over the Japanese islands, and continue eastward across the Pacific Ocean. In addition, great cyclonic airflows (masses of rapidly circulating air) that arise over the western equatorial Pacific move in a wheel-like fashion northeastward across Japan and nearby regions. During winter months heavy masses of cold air from Siberia dominate the weather around Japan. Persistent cold winds skim across the Sea of Japan from the northwest, picking up moisture that they deposit as several feet of snow on the western side of the mountain ranges on Honshu Island. As the cold air drops its moisture, it flows over high ridges and down eastern slopes to bring cold, relatively dry weather to valleys and coastal plains and cities.

1. According to paragraph 1, all of the following are true of the cold air from Siberia EXCEPT:
 - ☐ It gathers moisture as it moves across the Sea of Japan.
 - ☐ It is responsible for the snow that falls on the western side of Honshu Island.
 - ☐ It is warmed by the cyclonic airflows from the south that mix with it.
 - ☐ It is responsible for the cold, dry weather of the eastern valleys and coastal plains and cities.

Paragraph 1 is marked with an arrow [→].

Paragraph 2

In spring the Siberian air mass warms and loses density, enabling atmospheric currents over the Pacific to steer warmer air into northeast Asia. This warm, moisture-laden air covers most of southern Japan during June and July. The resulting late spring rains then give way to a drier summer that is sufficiently hot and muggy, despite the island chain's northerly latitude, to allow

widespread rice cultivation.

2. The word “enabling” in the passage is closest in meaning to
 - ☐ preparing
 - ☐ requiring
 - ☐ allowing
 - ☐ distributing
3. Why does the author include the phrase “despite the island chain’s northerly latitude” in the paragraph?
 - ☐ To indicate that one would not expect such hot, muggy weather at Japan’s latitude
 - ☐ To compare Japan’s climate to the climates of more northerly latitudes
 - ☐ To give a reason for the hot, muggy weather experienced in Japan during the summer
 - ☐ To explain why Japan’s climate is only suitable for rice cultivation

Paragraph 3

Summer heat is followed by the highly unpredictable autumn rains that accompany the violent tropical windstorms known as typhoons. These cyclonic storms originate over the western Pacific and travel in great clockwise arcs, initially heading west toward the Philippines and southern China, curving northward later in the season. Cold weather drives these storms eastward across Japan through early autumn, revitalizing the Siberian air mass and ushering in a new annual weather cycle.

4. According to paragraph 3, all of the following are true of autumn storms EXCEPT:
 - ☐ They involve rain combined with tropical windstorms
 - ☐ Cyclonic storms have a predictable pattern of travel.
 - ☐ Their movement creates a weather cycle that repeats itself.
 - ☐ They begin as northern Siberian air masses with consistent rains following the summer heat.

Paragraph 3 is marked with an arrow [→].

Paragraph 4

This yearly cycle has played a key role in shaping Japanese civilization. It has assured the islands ample precipitation, ranging irregularly from more than 200 centimeters annually in parts of the southwest to about 100 in the northeast and averaging 180 for the country as a whole. The moisture enables the islands to support uncommonly lush forest cover, but the combination of precipitous slopes and heavy rainfall also gives the islands one of the world’s highest rates of natural erosion, intensified by both human activity and the natural shocks of earthquakes and volcanism. These factors have given Japan its wealth of sedimentary basins, but they have also made mountainsides extremely susceptible to erosion and landslides and hence generally unsuitable for agricultural manipulation.

5. All of the following are mentioned in paragraph 4 as contributing to the high rate of erosion in the Japanese islands EXCEPT:
 - ☐ very steep slopes and heavy rainfall

- ☐ intense agricultural manipulation
- ☐ earthquakes and volcanic activities
- ☐ human activity

Paragraph 4 is marked with an arrow [→].

6. The word “susceptible to” in the passage is closest in meaning to

- ☐ slow to replace losses from
- ☐ likely to be affected by
- ☐ unable to benefit from
- ☐ well-known for

Paragraph 5

The island chain’s mountains backbone and great length from north to south produce climatic diversity that has contributed to regional differences. Generally sunny winters along the Pacific seaboard have made habitation there relatively pleasant. Along the Sea of Japan, on the other hand, cold, snowy winters have discouraged settlement. Furthermore, although annual precipitation is high in that region, much of it comes as snow and rushes to the sea as spring runoff, leaving little moisture for farming.

7. According to paragraph 5, which of the following is a major factor in the limited habitation in the area along the Sea of Japan?

- ☐ It has too many mountains.
- ☐ It is vulnerable to floods during spring runoff.
- ☐ Its climate is highly irregular and unpredictable.
- ☐ It is cold and snowy during winter.

Paragraph 5 is marked with an arrow [→].

Paragraph 6

Summer weather patterns in northern Honshu, and especially along the Sea of Japan, have also discouraged agriculture. The area is subject to the *yamase* effect, when cool air from the north sometimes lowers temperatures sharply and damages farm production. The impact of this effect has been especially great on rice cultivation because, if it is to grow well, the rice grown in Japan requires a mean summer temperature of 20° centigrade or higher. A drop of 2°-3° can lead to a 30-50 percent drop in rice yield, and the *yamase* effect is capable of exceeding that level. This *yamase* effect does not, however, extend very far south, where most precipitation comes in the form of rain and the bulk of it in spring, summer, and fall, when most useful for cultivation. Even the autumn typhoons, which deposit most of their moisture along the southern seaboard, are beneficial because they promote the start of the winter crops that for centuries have been grown in southern Japan.

8. According to paragraph 6, how can the *yamase* effect lead to lower rice production in northern Honshu?

- ☐ It can cause temperatures to drop below the level required for rice to grow well.
- ☐ It can limit the amount of summer rainfall, resulting in less water for cultivation.

- ☐ It can damage a large portion of the land on which rice is grown.
- ☐ It can prevent rice cultivation during seasons other than summer.

Paragraph 6 is marked with an arrow [→].

9. Which of the following can be inferred from paragraph 6 about farming in southern Japan?

- ☐ Farming is limited to rice cultivation.
- ☐ Farming is difficult because of the *yamase* effect.
- ☐ Farming takes place throughout the year.
- ☐ Farming suffers from the effects of autumn typhoons.

Paragraph 6 is marked with an arrow [→].

10. The word “**exceeding**” in the passage is closest in meaning to

- ☐ almost reaching
- ☐ going beyond
- ☐ maintaining
- ☐ reducing

Paragraph 7

In short, for the past two millennia, the climate in general and patterns of precipitation in particular have encouraged the Japanese to **cluster** their settlements along the southern coast, most densely along the sheltered Inland Sea, moving into the northeast. There the limits that **topography imposed on production have been tightened by climate, with the result that agricultural output has been more modest and less reliable, making the risk of crop failure and hardship commensurately greater.**

11. The word “**cluster**” in the passage is closest in meaning to

- ☐ build
- ☐ group
- ☐ move
- ☐ expand

12. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Agricultural production has been more successful in northeastern Japan than along the Inland Sea, where topography and climate make life difficult for people.
- ☐ Topography and climate have combined to limit agricultural production in northeastern Japan, resulting in an increased risk of crop failure and hardship.
- ☐ Along the Inland Sea, where topography makes the climate more severe, decreased agricultural output has resulted from crop failure and hardship.
- ☐ The risk of crop failure in northeastern Japan has caused greater hardship than have climate and topography.

Paragraph 4

This yearly cycle has played a key role in shaping Japanese civilization. ■ It has assured the islands ample precipitation, ranging irregularly from more than 200 centimeters annually in parts of the southwest to about 100 in the northeast and averaging 180 for the country as a whole. ■ The moisture enables the islands to support uncommonly lush forest cover, but the combination of precipitous slopes and heavy rainfall also gives the islands one of the world’s highest rates of natural erosion, intensified by both human activity and the natural shocks of earthquakes and volcanism. ■ These factors have given Japan its wealth of sedimentary basins, but they have also made mountainsides extremely susceptible to erosion and landslides and hence generally unsuitable for agricultural manipulation. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.
- Such a large amount of rainfall has both positive and negative effects on the environment of the Japanese islands.**

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.
- This question is worth 2 points.**
- Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Japan’s yearly weather cycle influences settlement patterns and agriculture across the islands.

Answer Choices

- | | |
|--|--|
| Cold, westerly winds from Siberia and cyclonic airflows from the Pacific Ocean provide ample rainfall for farming but contribute to high rates of erosion. | Japan’s yearly weather cycle makes farming possible only in the summer, as the effects of the Siberian air mass result in winters that are too cold and snowy for agriculture. |
| Settlements are most concentrated along the Pacific seaboard to the south, where climate and topography are more suitable for crop cultivation than along the Sea of Japan. | Agricultural practices that stabilize sediments have reduced erosion and landslides and allowed the growth of lush forests in Japan. |
| The <i>yamase</i> effect has a great impact on rice growing in northern Japan but does not affect cultivation in southern Japan, where precipitation comes in the form of rain rather than snow. | Climate changes during the last two millennia have caused the Japanese to move their settlements toward the northeast, where the climate is more favorable to agriculture. |

Economic Decline in Europe During the Fourteenth Century

After three hundred years of impressive gains in wealth and population, Europe's economy began to slow around 1300. Several factors accounted for the decline. One the most important, though perhaps the least dramatic to relate, was a shift in climate. The remarkably fair weather of the twelfth and thirteenth centuries took a decided turn for the worse in the fourteenth. Chronicler's comments, tree-ring examination, and pollen analysis all indicate that over the course of the fourteenth century Europe's average annual temperature declined approximately two degrees Celsius—which may sound like very little at first, but if one considers current projections about the possible effects of global warming, in which the average annual temperature shift is only one degree Celsius, a rather different impression emerges. As the temperature dropped, shortening the summer growing season and affecting the resilience of certain vegetable species, the wind and rain increased. This meant that crop yields declined precipitously and the agricultural economy began to contract. As food supplies dwindled, costs rose accordingly and cut into the amount of capital that people had available for other purchases or investments. This in turn added to the gradual construction of the commercial economy.

Just as significant were changes in the geopolitics of the Mediterranean world. The decline of the Byzantine Empire, which had dominated the eastern Mediterranean, meant the interruption or trade routes to central and eastern Asia. The rise of new political powers signaled a new era in Mediterranean connections, one in which religious loyalty and ethnic fidelity mattered more than commercial ties. Consequently the movement of goods and services between east and west began to slow. European interest in circumnavigating Africa and exploring westward into the Atlantic Ocean, in fact, originated in the desire to avoid the roadblock in the eastern Mediterranean and to tap directly into the trade with eastern Asia that had long sustained Europe's economic growth.

A more immediate cause of the sputtering economy was an observable absence: since the eleventh century there had been few significant changes in the technology of agriculture. Developments like the wheeled plow, the rotation of crops, and the use of natural fertilizer that had made possible the agricultural revolution of the past two hundred years had had no follow-up. Farming was still conducted in 1300 roughly the same way it had been done in 1100, but with a considerably larger population to feed, there was little surplus left to generate fresh capital. As a consequence, food production fell perilously close to subsistence level. Although the failure of agriculture to keep up with the growing population did not become a crisis until the fourteenth century, clear signs of the problem had already emerged by the middle of the thirteenth century, when occasionally low yields due to bad weather or social disruption revealed how perilous the balance between Europe's population and its food supply had become. Apart from territories beset by war, the tentativeness of the food supply became evident first on the farmlands most recently brought under cultivation during the economic depression of the twelfth century. The less established farmers of these lands frequently did not have the means to survive successive poor harvests. Tenant farmers unable to pay their rents thus began to slip into debt,

and landlords who depended on rents for their income began to rely increasingly on urban financiers for credit.

Even whole governments became entangled in the credit crisis, England being the most notable example. The cycle of indebtedness was hardly inevitable, but the string of bank failures and commercial collapses in the first half of the fourteenth century was striking. The famed Bardi and Peruzzi banks of Florence (the two largest financial houses of Europe) collapsed spectacularly in the 1340's. They were soon followed by the Riccardi bank of Lucca, whose massive loans had kept the English government afloat for years. Many more houses collapsed in turn.

An important demographic trend resulted from and contributed to the economic malaise: large-scale migration of rural populations into the cities. Europe's overall population growth from 1050 to 1300 had been primarily due to an increase in the number of rural folk. But as economic forces made agrarian life more perilous around 1300, hard-pressed farmers and their families began to migrate to the cities in large numbers in search of work. Many cities doubled in size, and some even tripled, over the course of just one or two generations. Few were capable of absorbing such large numbers of people.

Paragraph 1

After three hundred years of impressive gains in wealth and population, Europe's economy began to slow around 1300. Several factors accounted for the decline. One the most important, though perhaps the least dramatic to relate, was a shift in climate. The remarkably fair weather of the twelfth and thirteenth centuries took a decided turn for the worse in the fourteenth. Chronicler's comments, tree-ring examination, and pollen analysis all indicate that over the course of the fourteenth century Europe's average annual temperature declined approximately two degrees Celsius—which may sound like very little at first, but if one considers current projections about the possible effects of global warming, in which the average annual temperature shift is only one degree Celsius, a rather different impression emerges. As the temperature dropped, shortening the summer growing season and affecting the resilience of certain vegetable species, the wind and rain increased. This meant that crop yields declined precipitously and the agricultural economy began to contract. As food supplies dwindled, costs rose accordingly and cut into the amount of capital that people had available for other purchases or investments. This in turn added to the gradual construction of the commercial economy.

1. The word “accounted for” in the passage is closet in meaning to
 - ☐ predicted
 - ☐ explained
 - ☐ typified
 - ☐ worsened
2. Why does the author mention “current projections about the possible effects of global warming” in the passage?
 - ☐ To argue that global warning was a factor in the climate shift of the fourteenth century in Europe

- To suggest that the current climate change is greater than the climate change in the fourteenth century in Europe
- To show the direct connection between temperature changes in the fourteenth century and changes that are currently occurring in Europe
- To emphasize the impact of the temperature change that occurred during the fourteenth century in Europe

3. In paragraph 1, all of the following are mentioned as factors that contributed to the decline of the agricultural economy in the fourteenth century EXCEPT

- an increase in rain and wind
- a shortened growing season
- the investment of capital in areas other than agriculture
- a significant drop in temperature

Paragraph 1 is marked with an arrow [→].

Paragraph 2

Just as significant were changes in the geopolitics of the Mediterranean world. The decline of the Byzantine Empire, which had dominated the eastern Mediterranean, meant the interruption or trade routes to central and eastern Asia. The rise of new political powers signaled a new era in Mediterranean connections, one in which religious loyalty and ethnic fidelity mattered more than commercial ties. Consequently the movement of goods and services between east and west began to slow. European interest in circumnavigating Africa and exploring westward into the Atlantic Ocean, in fact, originated in the desire to avoid the roadblock in the eastern Mediterranean and to tap directly into the trade with eastern Asia that had long sustained Europe's economic growth.

4. In paragraph 2, the author implies that the Byzantine Empire
- allowed international trade from which Europe benefited
 - became a dominant force during the fourteenth century
 - centered its rule on religion and ethnic ties
 - interrupted trade routes to Asia that had already been established

Paragraph 2 is marked with an arrow [→].

5. According to paragraph 2, European interest in exploring the coast of Africa and the Atlantic Ocean grew out of a wish to

- build a roadblock against Asian powers
- restore valuable trade with eastern Asia
- create faster trade routes to eastern Asia
- connect trade between the eastern Mediterranean and the Atlantic Ocean

Paragraph 2 is marked with an arrow [→].

Paragraph 3

A more immediate cause of the sputtering economy was an observable absence: since the eleventh century there had been few significant changes in the technology of agriculture.

Developments like the wheeled plow, the rotation of crops, and the use of natural fertilizer that had made possible the agricultural revolution of the past two hundred years had had no follow-up. Farming was still conducted in 1300 roughly the same way it had been done in 1100, but with a considerably larger population to feed, there was little surplus left to generate fresh capital. As a consequence, food production fell perilously close to subsistence level. Although the failure of agriculture to keep up with the growing population did not become a crisis until the fourteenth century, clear signs of the problem had already emerged by the middle of the thirteenth century, when occasionally low yields due to bad weather or social disruption revealed how perilous the balance between Europe's population and its food supply had become. Apart from territories beset by war, the tentativeness of the food supply became evident first on the farmlands most recently brought under cultivation during the economic depression of the twelfth century. The less established farmers of these lands frequently did not have the means to survive successive poor harvests. Tenant farmers unable to pay their rents thus began to slip into debt, and landlords who depended on rents for their income began to rely increasingly on urban financiers for credit.

6. According to paragraph 3, what was one cause of the economic problems in Europe of the fourteenth century?
- ☐ Farming techniques produced insufficient amounts of food.
 - ☐ Territories that farmers had begun to use for agriculture for the first time were disrupted by war.
 - ☐ The technological improvements in farming made in earlier centuries were abandoned after 1300.
 - ☐ Farming techniques used capital that was needed for investment in the development of technology.

Paragraph 3 is marked with an arrow [→].

7. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Maintaining the population of Europe with existing food supplies continued to be a problem after the middle of the thirteenth century.
 - ☐ The delicate balance between population and food supply in Europe was apparent in years of poor harvest half a century before it became a crisis in the fourteenth century.
 - ☐ Clear signals of the emerging crisis in Europe appeared in the thirteenth century in the form of bad weather, social unrest, and insufficient food.
 - ☐ In the thirteenth century, a problem emerged in Europe when the food supply was sufficient to feed the population only occasionally.
8. The word “considerably” in the passage is closest in meaning to
- ☐ significantly
 - ☐ increasingly
 - ☐ constantly
 - ☐ naturally

9. It can be inferred from paragraph 3 that people who farmed on land recently brought under cultivation were at a bigger disadvantage than well-established farmers in fourteenth-century Europe because

- ☐ their land was located in areas that were engaged in war
- ☐ they relied on urban financiers for credit
- ☐ they had no surplus on which to live or money to pay their rent
- ☐ they did not use wheeled plows, rotate their crops, or use natural fertilizer

Paragraph 3 is marked with an arrow [→].

Paragraph 4

Even whole governments became entangled in the credit crisis, England being the most notable example. The cycle of indebtedness was hardly inevitable, but the string of bank failures and commercial collapses in the first half of the fourteenth century was striking. The famed Bardi and Peruzzi banks of Florence (the two largest financial houses of Europe) collapsed spectacularly in the 1340's. They were soon followed by the Riccardi bank of Lucca, whose massive loans had kept the English government afloat for years. Many more houses collapsed in turn.

10. The word “striking” in the passage is closest in meaning to

- ☐ understandable
- ☐ necessary
- ☐ limiting
- ☐ noteworthy

11. Why does the author mention in the passage that the Bardi and Peruzzi banks were “the two largest financial houses of Europe”?

- ☐ To indicate the connection between Florence banks and the English government
- ☐ To emphasize the great impact that these bank failures had on the economy
- ☐ To compare the Bardi and Peruzzi banks with the Riccardi bank
- ☐ To indicate the success that these banks had previously achieved

Paragraph 5

An important demographic trend resulted from and contributed to the economic malaise: large-scale migration of rural populations into the cities. Europe's overall population growth from 1050 to 1300 had been primarily due to an increase in the number of rural folk. ■ But as economic forces made agrarian life more perilous around 1300, hard-pressed farmers and their families began to migrate to the cities in large numbers in search of work. ■ Many cities doubled in size, and some even tripled, over the course of just one or two generations. ■ Few were capable of absorbing such large numbers of people. ■

12. Paragraph 5 suggests that the large-scale migration to cities resulted in which of the following?

- ☐ After two generations in the cities, migrants returned to agricultural life.
- ☐ The overall population in Europe declined.

- Farmers worked in cities, and their families worked the land.
- Cities contained large numbers of people who were unemployed.

Paragraph 5 is marked with an arrow [→].

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Farms, villages, and entire regions were abandoned.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The economic decline in Europe during the fourteenth century resulted from several factors.

-
-
-

Answer Choices

Climate changes affected agricultural production, which led to food shortages.

The loss of trade with central and east Asia negatively impacted economic growth.

England was among the nations that suffered a credit loss.

The performance of the commercial economy could not keep pace with the agricultural economy.

The lack of innovation in agricultural technology affected food production.

Migrations from city to rural areas led to an increase in the number of rural folk.

Determining the Ages of the Planets and the Universe

The planets of our solar system all revolve around the Sun in the same direction and in orbits that lie in nearly the same plane. This is strong evidence that the planets formed simultaneously from a single disk of material that rotated in the same direction as the modern planets.

Precisely when the planets came into being has been a difficult issue to resolve. While Earth's water is necessary for life, its abundance near the planet's surface makes rapid erosion inevitable. Continuous alteration of the crust by erosion and also by igneous (volcanic) and metamorphic (pressure and heat within Earth) processes makes unlikely any discovery of rocks nearly as old as Earth. Thus geologists have had to look beyond this planet in their efforts to date Earth's origin. Fortunately, we do have samples of rock that appear to represent the primitive material of the solar system. These samples are meteorites, which originate as extraterrestrial objects, called meteors, that have been captured in Earth's gravitational field and have then crashed into our planet.

Some meteorites consist of rocky material and, accordingly, are called stony meteorites. Others are metallic and have been designated iron meteorites even though they contain lesser amounts of elements other than iron. Still others consist of mixtures of rocky and metallic material and thus are called stony-iron meteorites. Meteors come in all sizes, from small particles to the small planets known as asteroids; no asteroid, however, has struck Earth during recorded human history. Many meteorites appear to be fragments of larger bodies that have undergone collisions and broken into pieces. Iron meteorites are fragments of the interiors of these bodies, comparable to Earth's core, and stony meteorites are from outer portions of these bodies, comparable to Earth's mantle (the layer between the core and outer crust).

Meteorites have been radiometrically dated by means of several decay systems, including rubidium-strontium, potassium-argon, and uranium-thorium. The dates thus derived tend to cluster around 4.6 billion years, which suggests that this is the approximate age of the solar system. After many meteorites had been dated, it was gratifying to find that the oldest ages obtained for rocks gathered on the surface of the Moon also were approximately 4.6 billion years. This must, indeed, be the age of the solar system. Ancient rocks can be found on the Moon because the lunar surface, unlike that of Earth, has no water to weather and erode rocks and is characterized by only weak movements of its crust.

Determining the age of the universe has been more complicated. Most stars in the universe are clustered into enormous disk-like galaxies. The distance between our galaxy, known as the Milky Way, and all others is increasing. In fact, all galaxies are moving away from one another, evidence that the universe is expanding. It is not the galaxies themselves that are expanding but the space between them. What is happening is analogous to inflating a balloon with small coins attached to its surface. The coins behave like galaxies: although they do not expand, the space between them does. Before the galaxies formed, matter that they contain was concentrated with infinite density

at a single point from which it exploded in an event called the big bang. Even after it assembled into galaxies, matter continued to spread in all directions from the site of the big bang.

The evidence that the universe is expanding makes it possible to estimate its age. This evidence, called the redshift, is an increase in the wavelengths of light waves traveling through space—a shift toward the red end of the visible spectrum of wavelengths. Expansion of the space between galaxies causes this shift by stretching light waves as they pass through it. The farther these light waves have traveled through space, the greater the redshift they have undergone. For this reason, light waves that reach Earth from distant galaxies have larger redshifts than those from nearby galaxies. Calculations based on these redshifts indicate that about 13.7 billion years ago all of the galaxies would have been at one spot, the site of the big bang. This, then, is the approximate date of the big bang and the age of the universe.

Paragraph 1

The planets of our solar system all revolve around the Sun in the same direction and in orbits that lie in nearly the same plane. This is strong evidence that the planets formed simultaneously from a single disk of material that rotated in the same direction as the modern planets.

Paragraph 2

Precisely when the planets came into being has been a difficult issue to resolve. While Earth's water is necessary for life, its abundance near the planet's surface makes rapid erosion inevitable. Continuous alteration of the crust by erosion and also by igneous (volcanic) and metamorphic (pressure and heat within Earth) processes makes unlikely any discovery of rocks nearly as old as Earth. Thus geologists have had to look beyond this planet in their efforts to date Earth's origin. Fortunately, we do have samples of rock that appear to represent the primitive material of the solar system. These samples are meteorites, which originate as extraterrestrial objects, called meteors, that have been captured in Earth's gravitational field and have then crashed into our planet.

1. According to paragraphs 1 and 2, what evidence leads astronomers to believe that all the planets formed at approximately the same time?
 - ☐ Samples of rocks from all the planets are the same age.
 - ☐ All the planets orbit the Sun in the same direction and in about the same plane.
 - ☐ All planets have the same igneous and metamorphic processes.
 - ☐ The gravitational field of each planet is about the same strength.

Paragraphs 1 and 2 are marked with arrows [→]

2. The word "inevitable" in the passage is closest in meaning to
 - ☐ continuous
 - ☐ obvious
 - ☐ certain
 - ☐ easy
3. Which of the following is NOT mentioned in paragraph 2 as a cause of constant change to

Earth's crust?

- ☐ Water
- ☐ Igneous processes
- ☐ Metamorphic processes
- ☐ Meteorites

Paragraph 2 is marked with an arrow [→]

Paragraph 3

Some meteorites consist of rocky material and, **accordingly**, are called stony meteorites. Others are metallic and have been designated iron meteorites even though they contain lesser amounts of elements other than iron. Still others consist of mixtures of rocky and metallic material and thus are called stony-iron meteorites. Meteors come in all sizes, from small particles to the small planets known as asteroids; no asteroid, however, has struck Earth during recorded human history. Many meteorites appear to be fragments of larger bodies that have undergone collisions and broken into pieces. Iron meteorites are fragments of the interiors of these bodies, comparable to Earth's core, and stony meteorites are from outer portions of these bodies, comparable to Earth's mantle (the layer between the core and outer crust).

4. The word " **accordingly** " in the passage is closest in meaning to
- ☐ correspondingly
 - ☐ frequently
 - ☐ interestingly
 - ☐ informally
5. Which of the following can be inferred from paragraph 3 about meteorites?
- ☐ Their composition can help determine the part of the larger body from which they broke off.
 - ☐ They are difficult to distinguish from rocks in Earth's mantle.
 - ☐ Their collisions with Earth have become more frequent than in the past.
 - ☐ They are older than the rest of the solar system.

Paragraph 3 is marked with an arrow [→]

6. According to paragraph 3, which of the following is a characteristic of asteroids?
- ☐ They are the largest meteors.
 - ☐ They are made mostly of iron and other metals.
 - ☐ They often collide with Earth.
 - ☐ They are the oldest meteors.

Paragraph 3 is marked with an arrow [→]

Paragraph 4

Meteorites have been radiometrically dated by means of several decay systems, including rubidium-strontium, potassium-argon, and uranium-thorium. The dates thus derived tend to **cluster** around 4.6 billion years, which suggests that this is the approximate age of the solar system. After many meteorites had been dated, it was gratifying to find that the oldest ages obtained for rocks gathered on the surface of the Moon also were approximately 4.6 billion years.

This must, indeed, be the age of the solar system. Ancient rocks can be found on the Moon because the lunar surface, unlike that of Earth, has no water to weather and erode rocks and is characterized by only weak movements of its crust.

7. Which of the following can be inferred from paragraph 4 about the radiometric dating of meteorites?

- ☐ Scientists tried several different radiometric systems before finding one that worked.
- ☐ The radiometric dating of different meteorites produced similar results.
- ☐ Many meteorites were damaged by the radiometric dating.
- ☐ Radiometric dating was not as accurate as scientists expected.

Paragraph 4 is marked with an arrow [→]

8. The word "cluster" in the passage is closest in meaning to

- ☐ approach
- ☐ spread
- ☐ group
- ☐ vary

9. According to paragraph 4, why are scientists confident that the age of the oldest meteorites they studied is also the age of the solar system?

- ☐ Radiometric dating has been proven to be reliable.
- ☐ The oldest rocks found on the surface of the Moon are the same age as the meteorites.
- ☐ No meteorites have been found that are younger than 4.6 billion years old.
- ☐ Meteorites on the Moon are the same age as those on other planets.

Paragraph 4 is marked with an arrow [→]

Paragraph 5

Determining the age of the universe has been more complicated. Most stars in the universe are clustered into enormous disk-like galaxies. The distance between our galaxy, known as the Milky Way, and all others is increasing. In fact, all galaxies are moving away from one another, evidence that the universe is expanding. It is not the galaxies themselves that are expanding but the space between them. What is happening is analogous to inflating a balloon with small coins attached to its surface. The coins behave like galaxies: although they do not expand, the space between them does. Before the galaxies formed, matter that they contain was concentrated with infinite density at a single point from which it exploded in an event called the big bang. Even after it assembled into galaxies, matter continued to spread in all directions from the site of the big bang.

10. The word "enormous" in the passage is closest in meaning to

- ☐ expanding
- ☐ very bright
- ☐ distant
- ☐ huge

11. Why does the author refer to "inflating a balloon with small coins attached to its surface"?

- ☐ To help explain how the universe can expand while the galaxies remain the same size
- ☐ To imply that the universe must eventually stop expanding
- ☐ To support the statement that most stars are found in disk-shaped galaxies
- ☐ To help explain how the universe began as a single point of dense matter

Paragraph 6

The evidence that the universe is expanding makes it possible to estimate its age. This evidence, called the redshift, is an increase in the wavelengths of light waves traveling through space—a shift toward the red end of the visible spectrum of wavelengths. Expansion of the space between galaxies causes this shift by stretching light waves as they pass through it. The farther these light waves have traveled through space, the greater the redshift they have undergone. For this reason, light waves that reach Earth from distant galaxies have larger redshifts than those from nearby galaxies. Calculations based on these redshifts indicate that about 13.7 billion years ago all of the galaxies would have been at one spot, the site of the big bang. This, then, is the approximate date of the big bang and the age of the universe.

12. According to paragraph 6, how did astronomers learn that the universe is expanding?

- ☐ By measuring the distance between galaxies
- ☐ By observing the movement of stars within galaxies
- ☐ By studying the wavelengths of light from distant galaxies
- ☐ By comparing the sizes of different galaxies

Paragraph 6 is marked with an arrow [→]

Paragraph 3

■ Some meteorites consist of rocky material and, accordingly, are called stony meteorites. ■ Others are metallic and have been designated iron meteorites even though they contain lesser amounts of elements other than iron. ■ Still others consist of mixtures of rocky and metallic material and thus are called stony-iron meteorites. ■ Meteors come in all sizes, from small particles to the small planets known as asteroids; no asteroid, however, has struck Earth during recorded human history. Many meteorites appear to be fragments of larger bodies that have undergone collisions and broken into pieces. Iron meteorites are fragments of the interiors of these bodies, comparable to Earth's core, and stony meteorites are from outer portions of these bodies, comparable to Earth's mantle (the layer between the core and outer crust).

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Several varieties of meteorites have been observed.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because

they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

There is strong evidence that all the planets of our solar system were formed from the same disk of matter at the same time.

-
-
-

Answer Choices

Determining the age of the planets by examining rocks on Earth is difficult because Earth's crust is constantly changing.

All three kinds of meteorites are similar in composition to Earth in that they have an inner core, a rocky mantle, and an outer crust.

Meteorites found on Earth and rocks from the Moon's surface are the best evidence for estimating the age of the planets.

Radiometric dating of meteorites recovered from the Moon shows that they are older than those that are found on Earth.

The expansion of the universe makes it possible to estimate its age by measuring the amount of the redshift of light coming from distant galaxies.

The redshift phenomenon measures the rate at which the stars in a galaxy are moving away from each other.

Costs and Benefits of Dispersal

In order to move from one home base to another, animals must expend calories not only while moving but even before the dispersal when they invest in the development of the muscles needed to move. For example, if a cricket is to leave a deteriorating environment and move to a new and better place, it will need large flight muscles to fly away. Presumably, the calories and materials that go into flight muscle development and maintenance have to come out of the general energy budget of the animal. This means that other organ systems cannot develop as rapidly as they could otherwise, which may mean that the flight-capable individual is, in some other respects, less fit to survive.

Dispersing individuals not only have to pay energetic, developmental, and travel costs but are also more often exposed to predators—all of which raises the question, why are animals so often willing to leave home even when this means leaving a familiar, resource-rich location? This question is particularly pertinent for species in which some individuals disperse while others do not or do not disperse as far. One species in which some individuals travel farther than others is Belding's ground squirrel. Young male squirrels travel about 150 meters from the burrow in which they were born, whereas young females usually settle down only 50 meters or so from where they were born. Why should young Belding's ground squirrels disperse at all, and why should the males disperse farther than their sisters?

According to one argument, dispersal by juvenile animals of many species may be an adaptation against problems associated with inbreeding. When two closely related individuals mate, their offspring are more likely to manifest genetic diseases than are the offspring of genetically unrelated individuals, and as a result, inbreeding tends to produce animals that are less likely to survive to adulthood and reproduce. Dispersal of juveniles makes inbreeding less likely.

If avoidance of inbreeding is the point of dispersing, then one might expect as many female ground squirrels as males to travel 150 meters from their natal burrow. In fact females do not disperse as far as males, perhaps because the costs and benefits of dispersal differ for the two sexes. It has been suggested that the reproductive success of female Belding's ground squirrels depends on their possession of a territory in which to rear their young. Female ground squirrels that remain near their birthplace enjoy assistance from their mothers in the defense of their burrows against rival females. Thus, the benefits of remaining on familiar ground are greater for females than for males.

There may, however, be another reason why male mammals disperse greater distances than females. The usual rule is that males, not females, fight with one another for access to mates, and, therefore, males that lose such conflicts may find it advantageous to move away from same-sex rivals that they cannot subdue. Although this hypothesis probably does not apply to Belding's ground squirrels, since young males have not been seen fighting with older ones around the time of dispersal, the idea is more plausible with respect to some other species, such as lions.

Lions live in large groups, or prides, from which young males disperse. In contrast, the daughters

of the resident lionesses usually spend their entire lives close to where they were born. The sedentary females benefit from their familiarity with good hunting grounds and safe breeding dens in their natal territory, among other things. The departure of many young male lions coincides with the arrival of new mature males that violently displace the previous masters of the pride and chase off the males that are not yet adults in the pride as well. These observations support the mate-competition hypothesis for male dispersal. However, if young males are not evicted after a pride takeover, they often leave anyway without any coercion from adult males and without ever having attempted to mate with their female relatives. Moreover, mature males that have claimed a pride sometimes disperse again, expanding their range to add a second pride of females, at a time when their daughters in the first pride are becoming sexually mature. Inhibitions against inbreeding apparently exist in lions and cause males to leave home.

Paragraph 1

In order to move from one home base to another, animals must expend calories not only while moving but even before the dispersal when they invest in the development of the muscles needed to move. For example, if a cricket is to leave a deteriorating environment and move to a new and better place, it will need large flight muscles to fly away. Presumably, the calories and materials that go into flight muscle development and maintenance have to come out of the general energy budget of the animal. This means that other organ systems cannot develop as rapidly as they could otherwise, which may mean that the flight-capable individual is, in some other respects, less fit to survive.

1. The word "Presumably" in the passage is closest in meaning to
 - ☐ It is reasonable to assume
 - ☐ It is possible
 - ☐ It can be argued
 - ☐ It is certainly true

2. In paragraph 1, the discussion of cricket dispersal is used to illustrate which of the following principles?
 - ☐ For dispersing animals, the benefits of dispersal exceed the costs.
 - ☐ Dispersal always involves costs for the dispersing animal.
 - ☐ Only animals with the greatest fitness for survival are able to disperse.
 - ☐ Dispersal, as much as organ development, requires energy.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

Dispersing individuals not only have to pay energetic, developmental, and travel costs but are also more often exposed to predators—all of which raises the question, why are animals so often willing to leave home even when this means leaving a familiar, resource-rich location? This question is particularly pertinent for species in which some individuals disperse while others do not or do not disperse as far. One species in which some individuals travel farther than others is Belding's ground squirrel. Young male squirrels travel about 150 meters from the burrow in which they were born, whereas young females usually settle down only 50 meters or so from where

they were born. Why should young Belding's ground squirrels disperse at all, and why should the males disperse farther than their sisters?

3. The word "pertinent" in the passage is closest in meaning to
- ☐ puzzling
 - ☐ important
 - ☐ complex
 - ☐ relevant
4. According to paragraph 2, what is of particular interest about the dispersal of Belding's ground squirrels?
- ☐ Young squirrels disperse farther than adults.
 - ☐ Young squirrels disperse even from resource-rich areas.
 - ☐ The males and females have different patterns of dispersal.
 - ☐ They leave the mother's burrow at an earlier age than other squirrels.

Paragraph 2 is marked with an arrow [→]

5. It can be inferred from paragraph 2 that the phenomenon of dispersal
- ☐ is difficult to observe in the wild
 - ☐ occurs for reasons that are not always immediately apparent
 - ☐ is motivated by the desire to obtain more or better resources
 - ☐ has few benefits for most species

Paragraph 2 is marked with an arrow [→]

Paragraph 3

According to one argument, dispersal by juvenile animals of many species may be an adaptation against problems associated with inbreeding. When two closely related individuals mate, their offspring are more likely to manifest genetic diseases than are the offspring of genetically unrelated individuals, and as a result, inbreeding tends to produce animals that are less likely to survive to adulthood and reproduce. Dispersal of juveniles makes inbreeding less likely.

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ When closely related individuals mate, their offspring are likely to manifest genetic diseases that will probably be passed on to the next generation.
 - ☐ Genetic diseases inherited from parents that are not related make it less likely that the affected offspring will survive to adulthood and reproduce.
 - ☐ When two related individuals mate, their offspring may survive to adulthood, but they are unlikely to reproduce.
 - ☐ There is a tendency for the offspring of closely related parents to have genetic diseases that make it less likely they will survive and reproduce.

Paragraph 4

If avoidance of inbreeding is the point of dispersing, then one might expect as many female ground squirrels as males to travel 150 meters from their natal burrow. In fact females do not disperse as far as males, perhaps because the costs and benefits of dispersal differ for the two sexes. It has been suggested that the reproductive success of female Belding's ground squirrels depends on their possession of a territory in which to rear their young. Female ground squirrels that remain near their birthplace enjoy assistance from their mothers in the defense of their burrows against rival females. Thus, the benefits of remaining on familiar ground are greater for females than for males.

7. According to paragraph 4, young female Belding's ground squirrels may remain closer to where they were born than do males in order to
- ☐ avoid inbreeding with their male siblings
 - ☐ avoid competition with other females for territory
 - ☐ save energy they need to defend their burrows
 - ☐ benefit from their mothers' help

Paragraph 4 is marked with an arrow [→]

Paragraph 5

There may, however, be another reason why male mammals disperse greater distances than females. The usual rule is that males, not females, fight with one another for access to mates, and, therefore, males that lose such conflicts may find it advantageous to move away from same-sex rivals that they cannot subdue. Although this hypothesis probably does not apply to Belding's ground squirrels, since young males have not been seen fighting with older ones around the time of dispersal, the idea is more plausible with respect to some other species, such as lions.

8. According to paragraph 5, the hypothesis that males disperse as a result of conflicts with other males is not supported in the case of Belding's ground squirrels because
- ☐ young male squirrels that lose fights to other males do not disperse
 - ☐ unlike most species, it is the females, not the males, that fight with each other
 - ☐ there is no evidence that young male squirrels fight with other males before dispersing
 - ☐ fights between males generally take place after the males have dispersed

Paragraph 5 is marked with an arrow [→]

9. The word "subdue" in the passage is closest in meaning to
- ☐ remove
 - ☐ fight
 - ☐ defeat
 - ☐ accept

Paragraph 5

There may, however, be another reason why male mammals disperse greater distances than females. The usual rule is that males, not females, fight with one another for access to mates, and, therefore, males that lose such conflicts may find it advantageous to move away from same-sex rivals that they cannot subdue. Although this hypothesis probably does not apply to

Belding's ground squirrels, since young males have not been seen fighting with older ones around the time of dispersal, the idea is more plausible with respect to some other species, such as lions.

Paragraph 6

Lions live in large groups, or prides, from which young males disperse. In contrast, the daughters of the resident lionesses usually spend their entire lives close to where they were born. The sedentary females benefit from their familiarity with good hunting grounds and safe breeding dens in their natal territory, among other things. The departure of many young male lions coincides with the arrival of new mature males that violently displace the previous masters of the pride and chase off the males that are not yet adults in the pride as well. These observations support the mate-competition hypothesis for male dispersal. However, if young males are not evicted after a pride takeover, they often leave anyway without any coercion from adult males and without ever having attempted to mate with their female relatives. Moreover, mature males that have claimed a pride sometimes disperse again, expanding their range to add a second pride of females, at a time when their daughters in the first pride are becoming sexually mature. Inhibitions against inbreeding apparently exist in lions and cause males to leave home.

10. According to paragraphs 5 and 6, the patterns of dispersal in Belding's ground squirrels and lions are similar in which of the following ways?

- ☐ Young males are forced to disperse by older males.
- ☐ Avoiding inbreeding is probably not a factor in the explanation of the pattern of dispersal.
- ☐ Males disperse when new siblings are born.
- ☐ The males disperse farther than the females.

Paragraphs 5 and 6 are marked with arrows [→]

11. The phrase "coincides with" in the passage is closest in meaning to

- ☐ is driven by
- ☐ occurs at the same time as
- ☐ makes possible
- ☐ is a sign of

12. Why does the author provide the information that "if young males are not evicted after a pride takeover, they often leave anyway without any coercion from adult males and without ever having attempted to mate with their female relatives"?

- ☐ To support the conclusion that male lions disperse to find mates to which they are not closely related
- ☐ To indicate that the mate-competition hypothesis does not fully account for the dispersal of young male lions
- ☐ To imply that young male lions voluntarily leave a pride that has been taken over because the adult males prevent them from mating within the pride
- ☐ To indicate that males, not females, leave a pride after it has been taken over by adult males

Paragraph 4

■ If avoidance of inbreeding is the point of dispersing, then one might expect as many female

ground squirrels as males to travel 150 meters from their natal burrow. ■ In fact females do not disperse as far as males, perhaps because the costs and benefits of dispersal differ for the two sexes. ■ It has been suggested that the reproductive success of female Belding's ground squirrels depends on their possession of a territory in which to rear their young. ■ Female ground squirrels that remain near their birthplace enjoy assistance from their mothers in the defense of their burrows against rival females. Thus, the benefits of remaining on familiar ground are greater for females than for males.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

However, inbreeding can be avoided without both males and their female relatives dispersing the same distance.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Many young animals disperse from where they are born.

-
-
-

Answer Choices

There are a number of competing explanations of why animals disperse, but none of them is consistent with all the observed patterns of dispersal.

The reason there is little inbreeding among lions is that young males are driven away from their female relatives by the adult males who forcibly take over a lion pride.

Inbreeding significantly lowers the likelihood that offspring will survive, and so avoiding inbreeding is a significant benefit of dispersal.

Since males of most species must establish a territory for mating, dispersing of the males makes it more likely that most males will be able to reproduce.

There need to be significant benefits for a species to disperse because the requirements of dispersal involve energy costs and dangers for the dispersing individuals.

In mammal species, young males often leave their family group while related females benefit from remaining in close association with each other and their birthplace.

The Multiplier Effect

The causes behind the rapid development of the Minoan and Mycenaean civilizations in the Aegean during the late third and second millennia B.C.E. have intrigued scholars for years. Until recently, most explanations attributed Aegean development to outside influence. Civilization had emerged in Mesopotamia by 3000 B.C.E, and, some archeologists argued, Mesopotamian trade introduced civilized ideas and technological innovations into nearby, less advanced areas. Others hypothesized that civilization was brought to the Aegean by invasion from some adjacent region, of which Anatolia in modern Turkey seemed the most plausible.

In a work published in 1972, Professor Colin Renfrew approached the problem from a different viewpoint. He argued that the scanty available evidence for invasion or immigration from Anatolia into Greece in the early Bronze Age (about 3300-2200 B.C.E.) showed that, at most, such incursion was limited, and that it could not be regarded as responsible for the transformation of society there. Trade, though clearly documented, was also an inadequate explanation in itself. To understand the major changes in social organization and complexity that took place, it was necessary, said Renfrew, to determine the impact that new variables emerging in the early Bronze Age may have had on every interrelated aspect of the local social system. The two new major developments he considered were changes in the subsistence economy and the introduction of bronze metallurgy.

The economy of Neolithic Greece was based on growing grains and raising sheep. Early in the third millennium B.C.E., cultivation of grape vines and olive trees also became important in southern Greece and the Aegean Islands. Both crops were eminently suitable for trade and storage in the form of olive oil and wine. They were grown on land that was not suitable for grain farming. Their cultivation required work at a different time of year from that needed by grain crops, and much of this work, such as harvesting, was relatively light. As a result, agricultural yields were substantially increased without disrupting established agricultural practice. That increase in turn allowed, or stimulated, population growth. For the first time there was enough demand for specialized crafts and services to justify the existence of full-time craftspeople, who could be supported from the extra agricultural output.

Some copper artifacts were made during the fourth millennium B.C.E, but there were not many of them and they had little economic or social significance. When, in the third millennium, copper began to be mixed with tin to produce the relatively hard alloy bronze, demand for metal goods grew. Bronze could be used to make a range of useful new tools and weapons and a variety of impressive ornaments. The demand for metalwork stimulated further specialization in crafts such as toolmaking and jewelry making. The new tools promoted the development of other crafts, like carpentry and shipbuilding. Competition for prestigious or useful craft products and for control of their producers helped to heighten both social differences within communities and conflicts between them, resulting in the emergence of local chieftains, who were also in many instances warriors. These chieftains regulated agricultural and craft production, operating a distribution system through which the farmers could obtain tools or ornaments they needed or

wanted. The organizational demands of controlled distribution made it necessary to develop methods of measurement and recording, which led to the development of writing.

Renfrew argued that any single innovation would have had a limited or negligible effect on social organization because the inherently conservative nature of societies acts to minimize change. However, the interaction of several simultaneous developments created a multiplier effect. In the Aegean, increased agricultural productivity provided the means to support craft specialization, while bronze metallurgy provided the technology for producing highly valued new products. These factors set in motion a series of changes in other subsystems of society. Those changes in turn resulted in what, in a term borrowed from electronics, are called positive feedback loops—alterations in the workings of a social system that serve to reinforce themselves. Thus Aegean society was transformed from one consisting of basically self-sufficient and egalitarian farming villages to one of prosperous, hierarchical chiefdoms, with palace-dwelling rulers, actively competing with one another both at home and in international trade.

Paragraph 1

The causes behind the rapid development of the Minoan and Mycenaean civilizations in the Aegean during the late third and second millennia B.C.E. have intrigued scholars for years. Until recently, most explanations attributed Aegean development to outside influence. Civilization had emerged in Mesopotamia by 3000 B.C.E. and, some archeologists argued, Mesopotamian trade introduced civilized ideas and technological innovations into nearby, less advanced areas. Others hypothesized that civilization was brought to the Aegean by invasion from some adjacent region, of which Anatolia in modern Turkey seemed the most plausible.

1. Paragraph 1 implies which of the following about traditional scholarship concerning the development of Aegean civilizations?
 - ☐ It focused on the influence of Aegean trade on more advanced civilization rather than the invasion of the Aegean by a technologically advanced culture.
 - ☐ The development of Minoan civilization was thought to require a different explanation from that for the development of Mycenaean civilization.
 - ☐ It maintained that the rapid development of Aegean civilization could be explained only through external influence.
 - ☐ There was general agreement that Anatolia and Mesopotamia were both responsible for the development of civilization in the Aegean.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

In a work published in 1972, Professor Colin Renfrew approached the problem from a different viewpoint. He argued that the scanty available evidence for invasion or immigration from Anatolia into Greece in the early Bronze Age (about 3300-2200 B.C.E.) showed that, at most, such incursion was limited, and that it could not be regarded as responsible for the transformation of society there. Trade, though clearly documented, was also an inadequate explanation in itself. To understand the major changes in social organization and complexity that took place, it was necessary, said Renfrew, to determine the impact that new variables emerging in the early Bronze

Age may have had on every interrelated aspect of the local social system. The two new major developments he considered were changes in the subsistence economy and the introduction of bronze metallurgy.

2. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Renfrew said that local variables affected the complex social organization of the early Bronze Age.
 - ☐ Renfrew said that the major changes in social organization could be understood only by examining the effects of the new developments that took place in the early Bronze Age.
 - ☐ The early Bronze Age, Renfrew said, was a time in which there were major changes to many interrelated aspects of local social systems.
 - ☐ To understand the social organization and complexity of the early Bronze Age, it was necessary, said Renfrew, first to determine what new variables had emerged.

Paragraph 3

The economy of Neolithic Greece was based on growing grains and raising sheep. Early in the third millennium B.C.E., cultivation of grape vines and olive trees also became important in southern Greece and the Aegean Islands. Both crops were eminently suitable for trade and storage in the form of olive oil and wine. They were grown on land that was not suitable for grain farming. Their cultivation required work at a different time of year from that needed by grain crops, and much of this work, such as harvesting, was relatively light. As a result, agricultural yields were substantially increased without disrupting established agricultural practice. That increase in turn allowed, or stimulated, population growth. For the first time there was enough demand for specialized crafts and services to justify the existence of full-time craftspeople, who could be supported from the extra agricultural output.

3. The word "justify" in the passage is closest in meaning to
- ☐ include within the social system
 - ☐ provide a rational basis for
 - ☐ employ
 - ☐ encourage
4. According to paragraph 3, growing grape vines and olive trees increased agricultural yields because
- ☐ grapes and olives were mostly used for trade rather than for competing with established crops in local markets
 - ☐ grape vines and olive trees could be cultivated on land that was also used for raising sheep
 - ☐ growing grapes and olives did not require the services of specialized workers
 - ☐ grape vines and olive trees did not compete with other crops for land or labor

Paragraph 3 is marked with an arrow [→]

5. According to paragraph 3, the increased importance of grape and olive cultivation led to

which of the following changes in Greece?

- ☐ An increase in the number of inhabitants
- ☐ An increase in the amount of grain traded
- ☐ An increase in the percentage of farmers in the population
- ☐ The development of new methods of storing foods

Paragraph 3 is marked with an arrow [→]

Paragraph 4

Some copper artifacts were made during the fourth millennium B.C.E, but there were not many of them and they had little economic or social significance. When, in the third millennium, copper began to be mixed with tin to produce the relatively hard alloy bronze, demand for metal goods grew. Bronze could be used to make a range of useful new tools and weapons and a variety of impressive ornaments. The demand for metalwork stimulated further specialization in crafts such as toolmaking and jewelry making. The new tools **promoted** the development of other crafts, like carpentry and shipbuilding. Competition for prestigious or useful craft products and for control of their producers helped to heighten both social differences within communities and conflicts between them, resulting in the emergence of local chieftains, who were also in many instances warriors. These chieftains regulated agricultural and craft production, operating a distribution system through which the farmers could obtain tools or ornaments they needed or wanted. The organizational demands of controlled distribution made it necessary to develop methods of measurement and recording, which led to the development of writing.

6. The word "promoted" in the passage is closest in meaning to
- ☐ required
 - ☐ benefited from
 - ☐ encouraged
 - ☐ speeded up
7. According to paragraph 4, why did the invention of bronze lead to increased conflict?
- ☐ It resulted in more effective weapons for warriors.
 - ☐ It led to competition for the control of skilled craftspeople and their products.
 - ☐ It left the craftspeople who had been involved in making copper products without an occupation.
 - ☐ It led to competition for available supplies of tin to be used in the production of bronze.

Paragraph 4 is marked with an arrow [→]

8. All of the following are discussed in paragraph 4 as changes in Greek society after bronze was introduced EXCEPT:
- ☐ Social differentiation within communities increased.
 - ☐ Metalcraft workers became more specialized.
 - ☐ Methods were invented for documenting the distribution of goods.
 - ☐ Craft workers gained greater independence from local chieftains.

Paragraph 4 is marked with an arrow [→]

Paragraph 5

Renfrew argued that any single innovation would have had a limited or negligible effect on social organization because the inherently conservative nature of societies acts to minimize change. However, the interaction of several simultaneous developments created a multiplier effect. In the Aegean, increased agricultural productivity provided the means to support craft specialization, while bronze metallurgy provided the technology for producing highly valued new products. These factors set in motion a series of changes in other subsystems of society. Those changes in turn resulted in what, in a term borrowed from electronics, are called positive feedback loops—alterations in the workings of a social system that serve to reinforce themselves. Thus Aegean society was transformed from one consisting of basically self-sufficient and egalitarian farming villages to one of prosperous, hierarchical chiefdoms, with palace-dwelling rulers, actively competing with one another both at home and in international trade.

9. The phrase “several simultaneous developments” in the passage is closest in meaning to
- ☐ independent developments
 - ☐ developments that reinforced each other
 - ☐ developments that occurred at the same time
 - ☐ developments that followed one another in a sequence
10. The word “hierarchical” in the passage is closest in meaning to
- ☐ having several levels of authority
 - ☐ dependent on military support
 - ☐ hereditary
 - ☐ nondemocratic
11. What role does the idea of a “multiplier effect” serve in Renfrew’s explanation of the development of civilization in the Aegean?
- ☐ It allows him to argue that the combined effect of individual developments can lead to a major transformation of a social organization.
 - ☐ It allows him to explain the idea of a positive feedback loop.
 - ☐ It allows him to explain the inherently conservative nature of societies.
 - ☐ It allows him to explain how the changes in agriculture, bronze metallurgy, and craft specialization came about.
12. According to paragraph 5, the transformation of Aegean society involved all the following EXCEPT
- ☐ interdependence of the society’s various social and economic structures
 - ☐ economic success
 - ☐ competition between chiefdoms for economic dominance
 - ☐ social and political equality

Paragraph 5 is marked with an arrow [→]

Paragraph 2

In a work published in 1972, Professor Colin Renfrew approached the problem from a different viewpoint. ■ He argued that the scanty available evidence for invasion or immigration from Anatolia into Greece in the early Bronze Age (about 3300-2200 B.C.E.) showed that, at most, such incursion was limited, and that it could not be regarded as responsible for the transformation of society there. ■ Trade, though clearly documented, was also an inadequate explanation in itself. ■ To understand the major changes in social organization and complexity that took place, it was necessary, said Renfrew, to determine the impact that new variables emerging in the early Bronze Age may have had on every interrelated aspect of the local social system. ■ The two new major developments he considered were changes in the subsistence economy and the introduction of bronze metallurgy.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Instead, Renfrew emphasized the complex internal dynamics of change.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.



14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Colin Renfrew's explanation of the rise of Aegean civilization rejected traditional views that the cause was one or another external influence on the developing society.

-
-
-

Answer Choices

The emergence of Aegean civilization was the result of internal factors interacting with each other in a way that multiplied each other's effects.

chieftains who regulated the distribution of economic goods.

An increase in population created a shortage of agricultural land that increased social tensions and forced many farmers to become warriors for local chieftains.

More effective bronze weapons increased the power of local chieftains, who imposed a system in which they collected farm and craft products in exchange for protecting their subjects.

Once people began producing bronze, it surpassed wine and olive oil as the most important good traded in the Aegean.

Agricultural surpluses from the production of olive oil and wine and the introduction of bronze technology together supported population growth, the development of crafts, and economic expansion.

As the economy diversified and grew, a higher level of social organization was required, a need that encouraged the emergence of

Cognitive Maps in Animals

A central hypothesis of animal cognition is that many animals make use of cognitive maps—internal representations or codes—of the spatial relationships among objects in their surroundings. In a broad sense, most animals that migrate probably have some internal map of the way external objects are arranged.

The most extensive studies of cognitive maps have been made for animals exhibit seasonal migration, the regular back-and-forth movement of animals between two geographic areas at particular times of the year. Seasonal migration enables many species to access rich food resources throughout the year and to breed or winter in areas that favor survival. One long-distance traveler is the gray whale. During summer, these giant aquatic mammals feast on small, bottom-dwelling invertebrates that abound in northern oceans. In the autumn, they leave their northern feeding grounds and begin a long trip south along the North American coastline. Arriving in warm, shallow lagoons off Baja California (Mexico) in the winter months, they breed, and pregnant females give birth to young before migrating back north. The yearly round-trip, some 20,000 kilometers, is the longest for any mammal.

Among insects, the monarch butterfly has one of the most remarkable seasonal migrations. During winter, these insects decorate certain trees at the western tip of Cuba, in a few mountain valleys of central Mexico, and at a few sites along the California coast. All monarchs alive at the end of summer fly south to reach their wintering sites. With the onset of spring, monarchs mate at the wintering sites and begin migrating northward. As they arrive at summer destinations, they lay eggs and then die. Two or more generations are produced during the summer, repopulating the United States and southern Canada. With the approach of fall, the summer's last generation of monarchs flies south to the wintering grounds. They migrate as far as 4,000 kilometers and end up at a specific site, although they have not flown the route before.

Researchers have found that migrating animals stay on course by using a variety of environmental cues. Gray whales, for instance, seem to use coastal landmarks to pilot their way north and south. Migrating south in the autumn, they orient with the North American coastline on their left. Migrating north in the spring, they keep the coast on their right. Whale watchers sometimes see gray whales stick their heads straight up out of the water, perhaps to obtain a visual fix on land. Many birds migrate at night, navigating by the stars the way ancient human soldiers did. In contrast, monarch butterflies migrate during the day, resting in trees and bushes at night, genetic programming may enable them to use the Sun as a compass.

Navigating by the Sun or by stars requires an ability to keep track of compass direction. Many migrators also must have an internal timing mechanism that compensates for the continuous daily movement of Earth relative to celestial objects. The timing mechanism must also allow for the apparent change in position of celestial objects as the animal moves over its migration route. Almost nothing is known yet about the nature of these timing mechanisms. At least one

night-migrating bird, the indigo bunting, seems to avoid the need for a timing mechanism by fixing on the North Star, the one bright star in northern skies that appears almost stationary. Researchers have found that buntings learn a star map and fix on a stationary star when navigating at night.

Another interesting, and more or less open, question about migration is how birds continue navigating when the Sun or stars are obscured by clouds. There is strong evidence that some birds can orient to Earth's magnetic field. Magnetite, the iron-containing mineral once used by sailors as a crude compass, is probably involved in sensing the field. The mineral has been found in the heads of pigeons, in the abdomens of bees, and in certain bacteria that orient to a magnetic field. Future research may show that magnetic sensing is a widespread, important part of a complex navigation mechanism in many animals.

Paragraph 2

The most extensive studies of cognitive maps have been made for animals **exhibit** seasonal migration, the regular back-and-forth movement of animals between two geographic areas at particular times of the year. Seasonal migration enables many species to access rich food resources throughout the year and to breed or winter in areas that favor survival. One long-distance traveler is the gray whale. During summer, these giant aquatic mammals feast on small, bottom-dwelling invertebrates that abound in northern oceans. In the autumn, they leave their northern feeding grounds and begin a long trip south along the North American coastline. Arriving in warm, shallow lagoons off Baja California (Mexico) in the winter months, they breed, and pregnant females give birth to young before migrating back north. The yearly round-trip, some 20,000 kilometers, is the longest for any mammal.

1. The word "exhibit" in the passage is closest in meaning to
 - ☐ display
 - ☐ prefer
 - ☐ initiate
 - ☐ enjoy
2. Paragraph 2 mentions all of the following as reasons that animals engage in seasonal migrations EXCEPT
 - ☐ to have access to rich food resources throughout the year
 - ☐ to spend the winter months in more favorable locations
 - ☐ to escape from other animals that feast on them
 - ☐ to reproduce in areas that make the survival of their young possible

Paragraph 2 is marked with an arrow [→].

Paragraph 3

Among insects, the monarch butterfly has one of the most remarkable seasonal migrations. During winter, these insects decorate certain trees at the western tip of Cuba, in a few mountain valleys of central Mexico, and at a few sites along the California coast. All monarchs alive at the end of summer fly south to reach their wintering sites. With the onset of spring, monarchs mate

at the wintering sites and begin migrating northward. As they arrive at summer destinations, they lay eggs and then die. Two or more generations are produced during the summer, repopulating the United States and southern Canada. With the approach of fall, the summer's last generation of monarchs flies south to the wintering grounds. They migrate as far as 4,000 kilometers and end up at a specific site, although they have not flown the route before.

3. According to paragraph 3, what is remarkable about the monarch butterfly's seasonal migrations?

- ☐ The ability of monarchs born 4,000 kilometers away from the species' wintering sites to migrate to those sites
- ☐ The ability to come back from wintering sites to summer destinations before laying eggs
- ☐ The ability to locate wintering sites in various parts of North America
- ☐ The ability to fly 4,000 kilometers without stopping to reach their wintering sites

Paragraph 3 is marked with an arrow [→].

Paragraph 4

Researchers have found that migrating animals stay on course by using a variety of environmental cues. Gray whales, for instance, seem to use coastal landmarks to pilot their way north and south. Migrating south in the autumn, they orient with the North American coastline on their left. Migrating north in the spring, they keep the coast on their right. Whale watchers sometimes see gray whales stick their heads straight up out of the water, perhaps to obtain a visual fix on land. Many birds migrate at night, navigating by the stars the way ancient human soldiers did. In contrast, monarch butterflies migrate during the day, resting in trees and bushes at night, genetic programming may enable them to use the Sun as a compass.

4. All of the following are mentioned in paragraph 4 as possible ways that animals stay on course during migration EXCEPT

- ☐ observing the position of the coast relative to the direction of migration
- ☐ observing the migration patterns of other animals
- ☐ navigating by the stars
- ☐ relying on their genetic programming

Paragraph 4 is marked with an arrow [→].

5. Which of the following is a "contrast" implied in paragraph 4?

- ☐ Many birds, unlike monarchs, migrate over water.
- ☐ Ancient sailors, unlike monarchs, were unable to navigate during the day.
- ☐ Butterflies, unlike many birds, need to rest during migrations.
- ☐ Monarchs, unlike many birds, are unable to navigate by the stars.

Paragraph 5

Navigating by the Sun or by stars requires an ability to keep track of compass direction. Many migrators also must have an internal timing mechanism that compensates for the continuous daily movement of Earth relative to celestial objects. The timing mechanism must also allow for the apparent change in position of celestial objects as the animal moves over its migration route.

Almost nothing is known yet about the nature of these timing mechanisms. At least one night-migrating bird, the indigo bunting, seems to avoid the need for a timing mechanism by fixing on the North Star, the one bright star in northern skies that appears almost stationary. Researcher have found that buntings learn a star map and fix on a stationary star when navigating at night.

6. The word “compensates for” in the passage is closet in meaning to
- ☐ follows
 - ☐ recognizes
 - ☐ makes up for
 - ☐ is consistent with
7. Which of the following facts demonstrate that migrating birds need a “timing mechanism”?
- ☐ Sometimes a migrating bird’s internal compass does not function correctly.
 - ☐ The Sun is visible to migrating birds in the east in the morning, but in the west in the evening.
 - ☐ Most birds migrate either during the day or during the night, but not both.
 - ☐ Some stars, for example the North Star, do not appear to change position during a single night or even during the course of a long migration.

Paragraph 6

Another interesting, and more or less open, question about migration is how birds continue navigating when the Sun or stars are obscured by clouds. There is strong evidence that some birds can orient to Earth’s magnetic field. Magnetite, the iron-containing mineral once used by sailors as a crude compass, is probably involved in sensing the field. The mineral has been found in the heads of pigeons, in the abdomens of bees, and in certain bacteria that orient to a magnetic field. Future research may show that magnetic sensing is a widespread, important part of a complex navigation mechanism in many animals.

8. The word “crude” in the passage is closet in meaning to
- ☐ effective
 - ☐ temporary
 - ☐ reliable
 - ☐ primitive
9. The discussion of bird migration in paragraph 6 implies that
- ☐ birds that can navigate by sensing Earth’s magnetic field cannot navigate using the Sun or stars
 - ☐ pigeons can navigate when the Sun or stars are obscured by clouds
 - ☐ magnetite is likely to be part of the navigational system of only a few species of birds
 - ☐ birds’ navigation using magnetite is not as accurate as their navigation based on the Sun or stars

Paragraph 6 is marked with an arrow [→].

Paragraph 2

The most extensive studies of cognitive maps have been made for animals exhibit seasonal migration, the regular back-and-forth movement of animals between two geographic areas at particular times of the year. ■Seasonal migration enables many species to access rich food resources throughout the year and to breed or winter in areas that favor survival. ■One long-distance traveler is the gray whale. ■During summer, these giant aquatic mammals feast on small, bottom-dwelling invertebrates that abound in northern oceans. ■In the autumn, they leave their northern feeding grounds and begin a long trip south along the North American coastline. Arriving in warm, shallow lagoons off Baja California (Mexico) in the winter months, they breed, and pregnant females give birth to young before migrating back north. The yearly round-trip, some 20,000 kilometers, is the longest for any mammal.

10. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

For a variety of animals, there are two clear purposes for this behavior.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

11. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Animals that migrate seasonally seem to have formed cognitive maps.

-
-
-

Answer Choices

Important aspects of animals' use of cognitive maps are not understood since the operation of cognitive timing mechanisms remains a mystery.

When navigating by the Sun and the stars, animals need to keep track of compass direction as well as adjust to other relative changes in Earth's movement and changes in the position of objects in the sky.

It has been suggested that under cloudy conditions, certain animals that navigate using the Sun and stars are able to sense

Earth's magnetic field.

During seasonal migrations, animals use environmental signs, the Sun, and stars to navigate to wintering sites and back.

Some animals migrate during the day while others migrate during the night, depending on their genetic programming.

Future research is likely to show that magnetic sensing alone does not explain how birds navigate when objects in the sky are not visible.

The Use of the Camera Obscura

The precursor of the modern camera, the camera obscura is a darkened enclosure into which light is admitted through a lens in a small hole. The image of the illuminated area outside the enclosure is thrown upside down as if by magic onto a surface in the darkened enclosure. This technique was known as long ago as the fifth century B.C., and Leonardo da Vinci described it in his notebooks in 1490. In 1558 Giovanni Battista Della Porta wrote in his twenty-volume work *Magia naturalis* (meaning “natural magic”) instructions for adding a convex lens to improve the quality of the image thrown against a canvas or panel in the darkened area where its outlines could be traced. Later, portable camera obscures were developed, with interior mirrors and drawing tablets on which the artist could trace the image. For the artist, this technique allows forms and linear perspective to be drawn precisely as they would be seen from a single viewpoint. Mirrors were also used to reverse the projected images to their original positions.

Did some of the great masters of painting, then, trace their images using a camera obscura? Some art historians are now looking for clues of artists’ use of such devices. One of the artists whose paintings are being analyzed from this point of view is the great Dutch master, Jan Vermeer, who lived from 1632 to 1675 during the flowering of art and science in the Netherlands, including the science of optics. Vermeer produced only about 30 known paintings, including his famous *The Art of Painting*. The room shown in it closely resembles the room in other Vermeer paintings, with lighting coming from a window on the left, the same roof beams, and similar floor tiles, suggesting that the room was fitted with a camera obscura on the side in the foreground. The map hung on the opposite wall was a real map in Vermeer’s possession, reproduced in such faithful detail that some kind of tracery is suspected. When one of Vermeer’s paintings was X-rayed, it did not have any preliminary sketches on the canvas beneath the paint, but rather the complete image drawn in black and white without any trial sketches. Vermeer did not have any students, did not keep any records, and did not encourage anyone to visit his studio, facts that can be interpreted as protecting his secret use of a camera obscura.

In recent times the British artist David Hockney has published his investigations into the secret use of the camera obscura, claiming that for up to 400 years, many of Western art’s great masters probably used the device to produce almost photographically realistic details in their paintings. He includes in this group Caravaggio, Hans Holbein, Leonardo da Vinci, Diego Velazquez, Jean-Auguste-Dominique Ingres, Agnolo Bronzino, and Jan van Eyck. From an artist’s point of view, Hockney observed that a camera obscura compresses the complicated forms of a three-dimensional scene into two-dimensional shapes that can easily be traced and also increases the contrast between light and dark, leading to the [chiaroscuro](#) effect seen in many of these paintings. In Jan van Eyck’s *the Marriage of Giovanni Arnolfini and Goivanna Genami*, the complicated [foreshortening](#) in the chandelier and the intricate detail in the bride’s garments are among the clues that Hockney thinks point to the use of the camera obscura.

So what are we to conclude? If these artists did use a camera obscura, does that diminish their

statue? Hockney argues that the camera obscura does not replace artistic skill in drawing and painting. In experimenting with it, he found that it is actually quite difficult to use for drawing, and he speculates that the artists probably combined their observations from life with tracing of shapes.

Chiaroscuro: artistic term for a contrast between light and dark

Foreshortening: a technique for representing an image in art that makes it appear to recede in space

Paragraph 1

The precursor of the modern camera, the camera obscura is a darkened enclosure into which light is admitted through a lens in a small hole. The image of the illuminated area outside the enclosure is thrown upside down as if by magic onto a surface in the darkened enclosure. This technique was known as long ago as the fifth century B.C., and Leonardo da Vinci described it in his notebooks in 1490. In 1558 Giovanni Battista Della Porta wrote in his twenty-volume work *Magia naturalis* (meaning “natural magic”) instructions for adding a convex lens to improve the quality of the image thrown against a canvas or panel in the darkened area where its outlines could be traced. Later, **portable** camera obscures were developed, with interior mirrors and drawing tablets on which the artist could trace the image. For the artist, this technique allows forms and linear perspective to be drawn precisely as they would be seen from a single viewpoint. Mirrors were also used to reverse the **projected** images to their original positions.

1. What can be inferred from paragraph 1 about Giovanni Battista Della Porta’s contribution to the camera obscura?
 - ☐ He translated a Chinese description of the use of the camera obscura and made the technique available to artists.
 - ☐ His convex lens made the projected image easier to trace.
 - ☐ His version of the camera obscura allowed for the later addition of a mirror.
 - ☐ His improvements relied heavily on design changes proposed earlier by Leonardo da Vinci.

Paragraph 1 is marked with an arrow [→].

2. The word “**portable**” in the passage is closest in meaning to
 - ☐ valuable
 - ☐ practical
 - ☐ moveable
 - ☐ popular
3. The word “**projected**” in the passage is closest in meaning to
 - ☐ whole
 - ☐ corrected
 - ☐ enlarged
 - ☐ shown

Paragraph 2

Did some of the great masters of painting, then, trace their images using a camera obscura? Some art historians are now looking for clues of artists' use of such devices. One of the artists whose paintings are being analyzed from this point of view is the great Dutch master, Jan Vermeer, who lived from 1632 to 1675 during the flowering of art and science in the Netherlands, including the science of optics. Vermeer produced only about 30 known paintings, including his famous *The Art of Painting*. The room shown in it closely resembles the room in other Vermeer paintings, with lighting coming from a window on the left, the same roof beams, and similar floor tiles, suggesting that the room was fitted with a camera obscura on the side in the foreground. The map hung on the opposite wall was a real map in Vermeer's possession, reproduced in such faithful detail that some kind of tracery is suspected. When one of Vermeer's paintings was X-rayed, it did not have any preliminary sketches on the canvas beneath the paint, but rather the complete image drawn in black and white without any trial sketches. Vermeer did not have any students, did not keep any records, and did not encourage anyone to visit his studio, facts that can be interpreted as protecting his secret use of a camera obscura.

4. Paragraph 2 answers which of the following questions about paintings by Vermeer?
- ☐ What characteristics of Vermeer's paintings suggest that he may have used a camera obscura?
 - ☐ Why did Vermeer produce only about 30 paintings?
 - ☐ Do Vermeer's paintings in general suggest that he was unable to paint accurately without using a camera obscura?
 - ☐ Why did Vermeer need to draw an image on the canvas of the painting that was X-rayed if he was using a camera obscura?

Paragraph 2 is marked with an arrow [→].

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information:
- ☐ One artist with a particularly interesting point of view about the use of the camera obscura in painting was Jan Vermeer, who lived in the Netherlands from 1632 to 1675.
 - ☐ Historical analyses suggest that Dutch masters were interested in the science of optics, so they may have used the camera obscura to trace images.
 - ☐ The use of the camera obscura is being analyzed in the paintings of Jan Vermeer, who lived in the Netherlands when art and science were flourishing there.
 - ☐ One view held by historians is that most Dutch masters were as interested in art as they were in science, and that provides clues about the techniques used in their paintings.
6. What does paragraph 2 indicate about Vermeer's *The Art of Painting*?
- ☐ It is the first in a series of about 30 paintings that he created.
 - ☐ It may have been painted by one of his students.
 - ☐ It was in his possession until his death in 1675.
 - ☐ It has the same setting as several other works of his.

Paragraph 2 is marked with an arrow [→].

7. The word “faithful” in the passage is closest in meaning to
- ☐ unusual
 - ☐ extensive
 - ☐ exact
 - ☐ historical
8. Why does the author provide the information on that “When one of Vermeer’s paintings was X-rayed, it did not have any preliminary sketches on the canvas beneath the paint, but rather the complete image drawn in black and white without any trial sketches”?
- ☐ To provide an example of a way to learn about the practices of artists who did not keep good records
 - ☐ To emphasize Vermeer’s confidence and skill as an artist
 - ☐ To provide evidence that Vermeer may have traced the image using a camera obscura
 - ☐ To argue that Vermeer did his preliminary sketching on paper, rather than on canvas

Paragraph 3

In recent times the British artist David Hockney has published his investigations into the secret use of the camera obscura, claiming that for up to 400 years, many of Western art’s great masters probably used the device to produce almost photographically realistic details in their paintings. He includes in this group Caravaggio, Hans Holbein, Leonardo da Vinci, Diego Velazquez, Jean-Auguste-Dominique Ingres, Agnolo Bronzino, and Jan van Eyck. From an artist’s point of view, Hockney observed that a camera obscura compresses the complicated forms of a three-dimensional scene into two-dimensional shapes that can easily be traced and also increases the contrast between light and dark, leading to the chiaroscuro effect seen in many of these paintings. In Jan van Eyck’s *the Marriage of Giovanni Arnolfini and Goivanna Genami*, the complicated foreshortening in the chandelier and the intricate detail in the bride’s garments are among the clues that Hockney thinks point to the use of the camera obscura.

9. According to paragraph 3, Hockney believes that all of the following indicate use of a camera obscura EXCEPT
- ☐ very detailed, realistic work
 - ☐ increased contrast between light and dark
 - ☐ oversimplification of forms when the image is traced
 - ☐ complicated foreshortening of objects

Paragraph 3 is marked with an arrow [→].

10. The word “intricate” in the passage is closest in meaning to
- ☐ surprising
 - ☐ complex
 - ☐ beautiful
 - ☐ clear

Paragraph 4

So what are we to conclude? If these artists did use a camera obscura, does that diminish their

statue? Hockney argues that the camera obscura does not replace artistic skill in drawing and painting. In experimenting with it, he found that it is actually quite difficult to use for drawing, and he speculates that the artists probably combined their observations from life with tracing of shapes.

11. The word “diminish” in the passage is closest in meaning to
- ☐ reduce
 - ☐ affect
 - ☐ reflect
 - ☐ determine
12. According to paragraph 4, what does Hockney argue about the use of the camera obscura in producing art?
- ☐ Works produced using a camera obscura do not deserve as much respect as those produced without it.
 - ☐ The camera obscura was probably used primarily as a training device, rather than used in producing finished works.
 - ☐ Use of the camera obscura by Western art’s great masters was probably relatively rare.
 - ☐ While the use of the camera obscura may have helped artists, they still needed to have significant artistic ability.

Paragraph 4 is marked with an arrow [→].

Paragraph 1

The precursor of the modern camera, the camera obscura is a darkened enclosure into which light is admitted through a lens in a small hole. The image of the illuminated area outside the enclosure is thrown upside down as if by magic onto a surface in the darkened enclosure. This technique was known as long ago as the fifth century B.C., and Leonardo da Vinci described it in his notebooks in 1490. In 1558 Giovanni Battista Della Porta wrote in his twenty-volume work *Magia naturalis* (meaning “natural magic”) instructions for adding a convex lens to improve the quality of the image thrown against a canvas or panel in the darkened area where its outlines could be traced. ■ Later, portable camera obscures were developed, with interior mirrors and drawing tablets on which the artist could trace the image. ■ For the artist, this technique allows forms and linear perspective to be drawn precisely as they would be seen from a single viewpoint. ■ Mirrors were also used to reverse the projected images to their original positions. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

All these developments helped artists to create accurate images of objects, people, and scenes.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below.

Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The camera obscura is a darkened enclosure into which light is admitted through a lens in a small hole.

-
-
-

Answer Choices

Evidence that the use of the camera obscura has long been known is provided by its description in many sources, including works dating back to Chinese writers from the fifth century B.C.

Some historians who have studied paintings by Western masters have found clues indicating that the masters may have secretly used the camera obscura in their works.

It is now widely believed that the use of the camera obscura led to the development of a style of photographic realism in Western art.

The camera obscura was most widely used by artists in seventeenth-century Netherlands, a period when art and science thrived.

The unique features of Vermeer's *The Art of Painting* make it unlikely that it was made with a camera obscura, as opposed to his other works.

The artist David Hockney has speculated that artists probably combined the use of the camera obscura with their own original observations from life.

Habitat Selection

Researchers who study habitat selection have proposed various models for the process. Marine biologist Peter F. Sale hypothesized the existence of a simple mechanism of habitat selection in fish that is based on levels of exploratory behavior. Sense organs monitor specific stimuli in the environment and send a summation of pertinent stimuli back to central-nervous-system centers, which regulate the amount of exploration. As the constellation of cues approaches some optimum level, exploratory behavior ceases and the animal stays where it is.

An alternative hypothesis is that an animal has a cognitive map of the ideal habitat and that its behavior is goal directed. However, working with a species of surgeonfish, Sale tested juveniles in laboratory tanks with various water depths and bottom covers under which fish could hide. Exploration time was least in the tank with shallow water and bottom cover and highest in the tank with shallow water and no bottom cover. In choice tests and field observations, most fish preferred shallow areas with bottom cover. Thus, Sale concluded, there is no need to suggest the inheritance of complex cognitive maps and goal-directed behaviors, rather, the animal simply moves around more in an unsuitable habitat and less in a suitable one.

Sale's model still does not explain how the animal "knows" what is suitable and what is not, or how stimuli from multiple cues are integrated. Nor does it explain the role of photoperiod (the duration of the animal's daily exposure to sunlight) in the response of dark-eyed juncos to photographs of their natural habitat. These wild-caught birds were presented a choice of viewing one of two 35-millimeter color slides showing different habitats. Birds kept in the lab under a winter photoperiod of nine hours of light and fifteen hours of darkness preferred (spent more time in front of) slides of their southern winter habitat. After day length was increased to fifteen hours of light and nine hours of darkness, the birds' viewing preferences shifted to the northern summer habitat.

Social cues may also affect choice of habitat. Large juncos (usually males) dominate smaller individuals (usually females and juveniles) in wintering flocks. Biologist Ellen Ketterson explained the finding that females usually migrate farther south than males by hypothesizing that subordinate birds are forced to migrate farther to avoid competing with dominants. In their lab study, researchers E. Roberts and Peter Weigl found that during the short days (stimulating winter), small subordinate juncos showed the strongest preference for winter scenes.

Risk of predation and competition are other factors that may affect habitat use. Hairy-footed gerbils live in vegetated islands in a sea of sand in the Namib Desert of southern Africa. Habitat use was determined by tracks in the sand and by how quickly they gave up feeding at stations containing seeds mixed with sand. Gerbils preferred sites around bushes or grass clumps to open areas and were more active on new-Moon nights than on full-Moon nights. They also gave up feeding at seed trays sooner in open areas and on full-Moon nights. These differences were likely caused by greater risk of predation in open areas and when the Moon was full. When striped mice, a close competitor of the gerbil, were removed, gerbils increased foraging activity, especially in the grass clumps.

The immediate cues to which animals respond when selecting a habitat may not be the same as the ultimate factors that have brought about the evolution of the response. For example, the blue tit, a European bird, lives in oak woodlands where most of its preferred food is found. But the blue tit establishes its territory each year before leaves and caterpillars (its staple food) have even appeared, so it must be using some other cue, such as the shape of the trees, to select its habitat. In fact, we know little about the signals that animals respond to when choosing their habitat. And in migratory species, it is not even clear when in the life cycle a choice of habitat is made. One study found that breeding sites may be selected in late summer or fall before migration, rather than in the spring, as is usually assumed.

Paragraph 1

Researchers who study habitat selection have proposed various models for the process. Marine biologist Peter F. Sale hypothesized the existence of a simple mechanism of habitat selection in fish that is based on levels of exploratory behavior. Sense organs monitor specific stimuli in the environment and send a summation of **pertinent** stimuli back to central-nervous-system centers, which regulate the amount of exploration. As the constellation of cues approaches some optimum level, exploratory behavior **ceases** and the animal stays where it is.

1. The word "**pertinent**" in the passage is closest in meaning to
 - ☐ important
 - ☐ selected
 - ☐ strong
 - ☐ relevant
2. According to paragraph 1, all of the following are processes that occur during exploratory behavior EXCEPT:
 - ☐ A favorable combination of cues are gathered.
 - ☐ Sense organs detect and record stimuli in the surrounding area.
 - ☐ Sense organs receive further directions from the central nervous system.
 - ☐ The central nervous system receives a summary of stimuli from the sense organs.

Paragraph 1 is marked with an arrow [→].

3. The word "**ceases**" in the passage is closest in meaning to
 - ☐ stops
 - ☐ decreases
 - ☐ succeeds
 - ☐ occurs

Paragraph 2

An alternative hypothesis is that an animal has a cognitive map of the ideal habitat and that its behavior is goal directed. However, working with a species of surgeonfish, Sale tested juveniles in laboratory tanks with various water depths and bottom covers under which fish could hide. Exploration time was least in the tank with shallow water and bottom cover and highest in the

tank with shallow water and no bottom cover. In choice tests and field observations, most fish preferred shallow areas with bottom cover. Thus, Sale concluded, there is no need to suggest the inheritance of complex cognitive maps and goal-directed behaviors, rather, the animal simply moves around more in an unsuitable habitat and less in a suitable one.

4. In paragraph 2, why does the author discuss Sale's research with a species of surgeonfish?
- ☐ To demonstrate how animals distinguish suitable habitats from unsuitable ones
 - ☐ To give an example of an animal that shows little or no exploratory behavior during habitat selection
 - ☐ To challenge the hypothesis that animals have complex cognitive maps of their ideal habitat that guide their habitat selection
 - ☐ To provide evidence that fish prefer areas that are shallow and have covering under which to hide

Paragraph 2 is marked with an arrow [→].

Paragraph 3

Sale's model still does not explain how the animal "knows" what is suitable and what is not, or how stimuli from multiple cues are integrated. Nor does it explain the role of photoperiod (the duration of the animal's daily exposure to sunlight) in the response of dark-eyed juncos to photographs of their natural habitat. These wild-caught birds were presented a choice of viewing one of two 35-millimeter color slides showing different habitats. Birds kept in the lab under a winter photoperiod of nine hours of light and fifteen hours of darkness preferred (spent more time in front of) slides of their southern winter habitat. After day length was increased to fifteen hours of light and nine hours of darkness, the birds' viewing preferences shifted to the northern summer habitat.

5. The word "integrated" in the passage is closest in meaning to
- ☐ interpreted
 - ☐ produced
 - ☐ found
 - ☐ combined
6. According to paragraph 3, Sale's model of habitat selection is unable to explain which of the following?
- ☐ Why dark-eyed juncos' interest in a particular habitat changes with the duration of daylight they are exposed to
 - ☐ Why the environment suitable for dark-eyed juncos in the winter differs from the environment suitable for them in the summer
 - ☐ Why only some species of animals respond to photographs of their natural habitats
 - ☐ Why birds studied in the lab behave differently than birds in the wild do

Paragraph 3 is marked with an arrow [→].

7. Paragraph 3 supports which of the following statements about dark-eyed juncos?
- ☐ The habitat preference of dark-eyed juncos changes with the amount of daylight present.

- Wild-caught dark-eyed juncos do not recognize photographs of their own habitat.
- Dark-eyed juncos choose their habitat based on seasonal temperatures.
- Artificial light affects dark-eyed juncos' choice of habitat differently than does natural sunlight.

Paragraph 3 is marked with an arrow [→].

Paragraph 4

Social cues may also affect choice of habitat. Large juncos (usually males) dominate smaller individuals (usually females and juveniles) in wintering flocks. Biologist Ellen Ketterson explained the finding that females usually migrate farther south than males by hypothesizing that subordinate birds are forced to migrate farther to avoid competing with dominants. In their lab study, researchers E. Roberts and Peter Weigl found that during the short days (stimulating winter), small subordinate juncos showed the strongest preference for winter scenes.

8. According to paragraph 4, what reason has been suggested for why female juncos often migrate farther south than do male juncos?
- To avoid the colder temperatures farther north
 - To provide safer habitats for their juveniles
 - To avoid having to compete with males
 - To find habitats having the longest days

Paragraph 4 is marked with an arrow [→].

Paragraph 5

Risk of predation and competition are other factors that may affect habitat use. Hairy-footed gerbils live in vegetated islands in a sea of sand in the Namib Desert of southern Africa. Habitat use was determined by tracks in the sand and by how quickly they gave up feeding at stations containing seeds mixed with sand. Gerbils preferred sites around bushes or grass clumps to open areas and were more active on new-Moon nights than on full-Moon nights. They also gave up feeding at seed trays sooner in open areas and on full-Moon nights. These differences were likely caused by greater risk of predation in open areas and when the Moon was full. When striped mice, a close competitor of the gerbil, were removed, gerbils increased foraging activity, especially in the grass clumps.

9. According to paragraph 5, why do hairy-footed gerbils prefer to feed in bushes and grass clumps?
- There is more food available there.
 - There is less competition there from striped mice.
 - There is more shade there from the heat of the desert.
 - There is less danger there of being harmed by a predator.

Paragraph 5 is marked with an arrow [→].

Paragraph 6

The immediate cues to which animals respond when selecting a habitat may not be the same as the ultimate factors that have brought about the evolution of the response. For example, the

blue tit, a European bird, lives in oak woodlands where most of its preferred food is found. But the blue tit establishes its territory each year before leaves and caterpillars (its staple food) have even appeared, so it must be using some other cue, such as the shape of the trees, to select its habitat. In fact, we know little about the signals that animals respond to when choosing their habitat. And in migratory species, it is not even clear when in the life cycle a choice of habitat is made. One study found that breeding sites may be selected in late summer or fall before migration, rather than in the spring, as is usually assumed.

10. According to paragraph 6, which of the following is true about habitat selection by the blue tit?

- ☐ It selects a habitat having the tallest trees.
- ☐ It selects a habitat where there are no caterpillars to eat the leaves.
- ☐ It selects its territory each year in spring.
- ☐ It selects its habitat before its preferred food appears in the area.

Paragraph 6 is marked with an arrow [→].

11. Paragraph 6 supports which of the following statements about scientists' knowledge of habitat selection?

- ☐ Scientists have successfully identified most of the immediate factors that animals respond to in choosing their habitats.
- ☐ Scientists have determined the seasons in which most animal species choose their habitats.
- ☐ Scientists know that evolution has led animals to ignore cues in their environment when choosing habitats.
- ☐ Scientists still have much to learn about the cues to which animals respond in choosing their habitats.

Paragraph 6 is marked with an arrow [→].

Paragraphs 3 and 4

Sale's model still does not explain how the animal "knows" what is suitable and what is not, or how stimuli from multiple cues are integrated. Nor does it explain the role of photoperiod (the duration of the animal's daily exposure to sunlight) in the response of dark-eyed juncos to photographs of their natural habitat. These wild-caught birds were presented a choice of viewing one of two 35-millimeter color slides showing different habitats. Birds kept in the lab under a winter photoperiod of nine hours of light and fifteen hours of darkness preferred (spent more time in front of) slides of their southern winter habitat. ■ After day length was increased to fifteen hours of light and nine hours of darkness, the birds' viewing preferences shifted to the northern summer habitat.

■ Social cues may also affect choice of habitat. ■ Large juncos (usually males) dominate smaller individuals (usually females and juveniles) in wintering flocks. ■ Biologist Ellen Ketterson explained the finding that females usually migrate farther south than males by hypothesizing that subordinate birds are forced to migrate farther to avoid competing with dominants. In their lab study, researchers E. Roberts and Peter Weigl found that during the short days (stimulating winter), small subordinate juncos showed the strongest preference for winter scenes.

12. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

However, photoperiod is not the only factor in the habitat selection of this bird species.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

13. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Researchers who study habitat selection among animals have proposed various models for the process.

-
-
-

Answer Choices

Peter F. Sale argued against the hypothesis that animals have cognitive maps of their ideal habitat by demonstrating that animals explore less in suitable habitats than in unsuitable ones.

Animals kept in the lab easily recognized images of their natural habitats during different seasons of the year.

Researchers have demonstrated that migratory species of birds select their breeding habitats at a point in their life cycle when they are ready to reproduce.

Peter F. Sale's model of habitat selection explained how animals distinguish suitable habitats from unsuitable ones as they integrate stimuli from multiple cues in the environment.

Laboratory experiments and field observations showed that day length, dominance relation within species, risk of predation, and competition for food play a role in habitat selection.

The fundamental elements that determine habitat selection may differ from the immediate cues to which the animals respond.

Sumerian Contributions

Before about 4500 B.C., lower Mesopotamia, the region between the Tigris and Euphrates rivers just north of the Persian Gulf, was much less densely populated than other inhabited regions of the Near and Middle East. Its marshy soil, subject to annual inundations (floods) from the rivers, was not suited to the primitive hoe culture of early agriculture, in which land was cultivated without domestic animals or beasts. Moreover, the land was virtually treeless and lacked building stone and mineral resources. During the next thousand years, however, this unpromising area became the seat of Sumer, the first great civilization known to history, with large concentrations of people, bustling cities, monumental architecture, and a wealth of religious, artistic, and literary traditions that influenced other ancient civilizations for thousands of years. The exact sequence of events that led to this culmination is unknown, but it is clear that the economic basis of this first civilization lay in its highly productive agriculture.

The natural fertility of the rich black soil was renewed annually by the silt left from the spring floods of the Tigris and Euphrates rivers. Harnessing its full productive power, however, required an elaborate system of drainage and irrigation, which in turn required a large and well-disciplined workforce as well as skilled management and supervision. The latter were supplied by a class of priests and warriors who ruled a large population of peasants and artisans. Through taxation and other means the rulers extracted wealth from the population and then used it to construct temples and other public buildings and to create works of art. That gave them (or some of them) the leisure to perfect the other refinements of civilization.

The rise of civilization brought with it a far more complex division of labor and system of economic organization. Full-time artisans specialized in the manufacture of textiles and pottery, metalworking, and other crafts. The professions of architecture, engineering, and medicine, among others, were born. Weights and measures were systematized, mathematics was invented, and primitive forms of science emerged. Since Sumer was virtually devoid of natural resources other than its rich soil, it traded with other people, thereby contributing to the diffusion of Sumerian civilization. The scarcity of stone, for tools as well as for buildings, probably hastened the adoption of copper and bronze. Copper, at least, was already known before the rise of Sumerian civilization, but lack of demand for it among the Stone Age peasant villages inhibited its widespread use. In Sumerian cities, on the other hand, stone imported by sea through the Persian Gulf from Oman and downriver from the mountains of Anatolia and the Caucasus had to compete with imported copper, and the latter proved more economical and effective for a variety of uses. Thereafter metallurgy, the technology of separating metals from their ores and purifying them,

was regarded as one of the hallmarks of civilization.

Sumer's greatest contribution to subsequent civilizations, the invention of writing, likewise grew out of economic necessity. The early cities—Eridu, Ur, Uruk, and Lagash—were temple cities: both economic and religious organizations centered on the temple of the local patron deity, represented by a priestly hierarchy. Members of the hierarchy directed the construction and maintenance of irrigation and drainage systems, oversaw agricultural activities, and supervised the collection of produce as taxation or tribute (money or other wealth given as a sign of submission or in return for protection). The need to keep records of the sources and uses of this tribute led to the use of simple pictographs on clay tablets sometime before 3000 B.C. By about 2800 B.C. the pictographs had been stylized into the system of writing known as cuneiform (using wedge-shaped marks on clay), a distinctive characteristic of Mesopotamian civilization. It is one of the few examples in history of a significant innovation issuing from a bureaucratic organization.

Although writing originated in response to the need for administrative bookkeeping, it soon found multiple religious, literary, and economic uses. In a later phase of development, after the strict temple-centered organization of the economy had given way to greater freedom of enterprise, clay tablets were used for recording the details of contracts, debts, and other commercial and financial transactions.

Paragraph 1

Before about 4500 B.C., lower Mesopotamia, the region between the Tigris and Euphrates rivers just north of the Persian Gulf, was much less densely populated than other inhabited regions of the Near and Middle East. Its marshy soil, subject to annual inundations (floods) from the rivers, was not suited to the primitive hoe culture of early agriculture, in which land was cultivated without domestic animals or beasts. Moreover, the land was virtually treeless and lacked building stone and mineral resources. During the next thousand years, however, this unpromising area became the seat of Sumer, the first great civilization known to history, with large concentrations of people, bustling cities, monumental architecture, and a wealth of religious, artistic, and literary traditions that influenced other ancient civilizations for thousands of years. The exact sequence of events that led to this culmination is unknown, but it is clear that the economic basis of this first civilization lay in its highly productive agriculture.

1. The word "unpromising" in the passage is closest in meaning to
 - ☐ unfavorable
 - ☐ underdeveloped
 - ☐ distant
 - ☐ expansive
2. The phrase "a wealth of" in the passage is closest in meaning to

- ☐ a strong competition among
- ☐ a valuable source of
- ☐ a deep respect for
- ☐ an abundance of

3. According to paragraph 1, which of the following was NOT true of lower Mesopotamia before 4500 B.C.?

- ☐ It was flooded every year by rivers.
- ☐ Its soil was unsuitable for Stone Age hoe culture agriculture.
- ☐ It was comparatively dense in population.
- ☐ It had few trees

Paragraph 1 is marked with an arrow [→]

Paragraph 2

The natural fertility of the rich black soil was renewed annually by the silt left from the spring floods of the Tigris and Euphrates rivers. Harnessing its full productive power, however, required an elaborate system of drainage and irrigation, which in turn required a large and well-disciplined workforce as well as skilled management and supervision. The latter were supplied by a class of priests and warriors who ruled a large population of peasants and artisans. Through taxation and other means the rulers extracted wealth from the population and then used it to construct temples and other public buildings and to create works of art. That gave them (or some of them) the leisure to perfect the other refinements of civilization.

4. According to paragraph 2, which of the following was true of the soil in lower Mesopotamia?

- ☐ It was relatively unfertile until humans added other, richer types of soil to it.
- ☐ It reached full productivity only when elaborate drainage and irrigation systems were added.
- ☐ Its most valuable nutrients were washed away by the spring floods of the Tigris and Euphrates rivers.
- ☐ Its silt was removed by a large, well-disciplined workforce.

Paragraph 2 is marked with an arrow [→]

5. According to paragraph 2, which of the following was true of priests and warriors in Sumerian society?

- ☐ Their technical expertise was essential in the development of new irrigation and drainage systems.
- ☐ They encouraged peasants and artisans to perfect the refinements of civilization.
- ☐ They were responsible for managing and supervising the workforce.
- ☐ They alone paid the taxes that funded the construction of temples.

Paragraph 2 is marked with an arrow [→]

Paragraph 3

The rise of civilization brought with it a far more complex division of labor and system of economic organization. Full-time artisans specialized in the manufacture of textiles and pottery, metalworking, and other crafts. The professions of architecture, engineering, and medicine, among others, were born. Weights and measures were systematized, mathematics was invented, and primitive forms of science emerged. Since Sumer was virtually devoid of natural resources other than its rich soil, it traded with other people, thereby contributing to the diffusion of Sumerian civilization. The scarcity of stone, for tools as well as for buildings, probably hastened the adoption of copper and bronze. Copper, at least, was already known before the rise of Sumerian civilization, but lack of demand for it among the Stone Age peasant villages inhibited its widespread use. In Sumerian cities, on the other hand, stone imported by sea through the Persian Gulf from Oman and downriver from the mountains of Anatolia and the Caucasus had to compete with imported copper, and the latter proved more economical and effective for a variety of uses. Thereafter metallurgy, the technology of separating metals from their ores and purifying them, was regarded as one of the hallmarks of civilization.

6. The phrase "scarcity of" in the passage is closest in meaning to
- ☐ using up
 - ☐ looking for
 - ☐ lacking in
 - ☐ uninterested in
7. The word "diffusion" in the passage is closest in meaning to
- ☐ stability
 - ☐ spread
 - ☐ prosperity
 - ☐ productivity
8. Which of the following is mentioned in paragraph 3 as an effect of Sumerian trade?
- ☐ The spread of Sumerian civilization to people outside Sumer
 - ☐ The discovery of copper
 - ☐ A rise in the price of textiles, pottery, and other goods sold within Sumer
 - ☐ An increase in the scarcity of stone within Sumer
- Paragraph 3 is marked with an arrow [→]
9. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Sumerian cities developed a trade in copper and stone that extended as far as Oman in the Persian Gulf and the mountains of Anatolia and the Caucasus.
 - ☐ Sumerian cities competed with traders elsewhere to produce copper that was more economical and useful than stone.

- In Sumerian cities, imported copper often proved more economical and useful than imported stone.
- The copper and stone imported from Oman, Anatolia, and the Caucasus proved more economical and effective than the copper and stone found in Sumerian cities.

Paragraph 4

Sumer's greatest contribution to subsequent civilizations, the invention of writing, likewise grew out of economic necessity. The early cities—Eridu, Ur, Uruk, and Lagash—were temple cities: both economic and religious organizations centered on the temple of the local patron deity, represented by a priestly hierarchy. Members of the hierarchy directed the construction and maintenance of irrigation and drainage systems, oversaw agricultural activities, and supervised the collection of produce as taxation or tribute (money or other wealth given as a sign of submission or in return for protection). The need to keep records of the sources and uses of this tribute led to the use of simple pictographs on clay tablets sometime before 3000 B.C. By about 2800 B.C. the pictographs had been stylized into the system of writing known as cuneiform (using wedge-shaped marks on clay), a distinctive characteristic of Mesopotamian civilization. It is one of the few examples in history of a significant innovation issuing from a bureaucratic organization.

10. In paragraph 4, why does the author point out that the economic and religious organizations of early Sumerian cities centered on local temples?
- To identify the bureaucratic needs that led to the development of writing
 - To identify factors responsible for the significant influence of Sumerian writing on subsequent civilizations
 - To explain why few examples of significant innovations have issued from a bureaucratic organization
 - To explain why the use of simple pictographs eventually became stylized into the system of writing known as cuneiform

Paragraph 4 is marked with an arrow [→]

11. Paragraph 4 implies all the following about cuneiform EXCEPT:

- It arose after 3000 B.C.
- It involved stylized pictographs.
- It was designed to further the purposes of the priestly bureaucracy.
- It was developed outside of the early temple cities.

Paragraph 4 is marked with an arrow [→]

Paragraph 5

Although writing originated in response to the need for administrative bookkeeping, it soon found multiple religious, literary, and economic uses. In a later phase of development, after the strict temple-centered organization of the economy had given way to greater freedom of enterprise, clay tablets were used for recording the

details of contracts, debts, and other commercial and financial transactions.

12. According to paragraph 5, which of the following was true of Sumerian writing during the period of greater freedom of enterprise?

- ☐ Knowledge of it was limited to the temple-centered administration.
- ☐ It was used for purposes beyond those for which it was first created.
- ☐ It was used primarily to help keep the traditional leadership in power.
- ☐ It led to further technological developments by 2800 B.C.

Paragraph 5 is marked with an arrow [→]



Impacts and Mass Extinctions

Meteorites and impact craters bear witness to the fact that large impacts occasionally occur on Earth. Meteor Crater in the northern Arizona desert of the United States formed about 50,000 years ago when a metallic impactor roughly 50 meters across crashed to Earth with the explosive power of a 20-megaton hydrogen bomb. Although the crater is only slightly more than one kilometer across, an area covering hundreds of square kilometers was probably battered by the blast and ejecta—the debris ejected or displaced during the formation of an impact crater. Far bigger impacts have occurred, sometimes with catastrophic consequences for life on Earth.

While collecting geological samples in Italy in 1978, the father-son team of Luis and Walter Alvarez discovered a thin layer of dark sediment that had apparently been deposited 65 million years ago—at about the same time that the dinosaurs and many other organisms suddenly became extinct. Subsequent studies found similar sediment deposited at the same time at many sites around the world. Careful analysis showed this worldwide sediment layer to be rich in iridium, an element that is rare on Earth's surface. But iridium is common in primitive meteorites, which led the Alvarazes to a stunning conclusion: the extinction of the dinosaurs was caused by the impact of an asteroid or comet. This conclusion was not immediately accepted and still generates some controversy, but it now seems clear that a major impact coincided with the death of the dinosaurs. While the dinosaurs were the most famous victims of this mass extinction, it seems that up to 99 percent of all living things were killed and that 75 percent of all species living on Earth were wiped out at that time.

How could an impact lead to mass extinction? The amount of iridium deposited worldwide suggests that the impactor must have been about 10 kilometers across. After a decade-long search, scientists identified what appears to be the impact crater from the event. Located off the coast of Mexico's Yucatan peninsula, it is 200 kilometers across, which is close to what one would expect for a 10-kilometer impactor, and dates to 65 million years ago. Further evidence that the Yucatan crater is the right one comes from the distribution of small glassy spheres that formed when the molten impact ejecta solidified as it rained back to Earth. More of these glassy spheres are found in regions near the crater, and careful study of their distribution suggests that the impactor crashed to Earth at a slight angle. These pieces of once molten rock are evidence of an explosion powerful enough to instantly melt bedrock and propel it far from its origin. The impact almost immediately sent a shower of debris raining across much of North and South America and generated huge waves that may have sloshed more than 1,000 kilometers inland. Many North American species thus may have been wiped out

shortly after impact. For the rest of the world, death may have come more slowly. Heat from the impact and returning ejecta probably ignited wildfires in forests around the world. Evidence of wildfires is found in the large amount of soot (a black powdery form of carbon produced when coal, wood, or oil is burned) that is also present in the iridium-rich sediment from 65 million years ago. The impact also sent huge quantities of dust high into the stratosphere, where it remained for several years, blocking out sunlight, cooling the surface, and affecting atmospheric chemistry. Plants died for lack of sunlight, and effects propagated throughout the food chain.

Perhaps the most astonishing fact is not that 75 percent of all species died, but that 25 percent survived. Among the survivors were a few small, rodent-like mammals. These mammals may have survived because they lived in underground burrows and managed to store enough food to outlast the long spell of cold, dark days. Small mammals had first arisen at about the same time as the dinosaurs, more than 100 million years earlier. But the sudden disappearance of the dominant dinosaurs made these mammals dominant.

Paragraph 1

Meteorites and impact craters bear witness to the fact that large impacts occasionally occur on Earth. Meteor Crater in the northern Arizona desert of the United States formed about 50,000 years ago when a metallic impactor roughly 50 meters across crashed to Earth with the explosive power of a 20-megaton hydrogen bomb. Although the crater is only slightly more than one kilometer across, an area covering hundreds of square kilometers was probably battered by the blast and ejecta—the debris ejected or displaced during the formation of an impact crater. Far bigger impacts have occurred, sometimes with catastrophic consequences for life on Earth.

1. The word "catastrophic" in the passage is closest in meaning to
 - ☐ widespread
 - ☐ disastrous
 - ☐ important
 - ☐ immediate

Paragraph 2

While collecting geological samples in Italy in 1978, the father-son team of Luis and Walter Alvarez discovered a thin layer of dark sediment that had apparently been deposited 65 million years ago—at about the same time that the dinosaurs and many other organisms suddenly became extinct. Subsequent studies found similar sediment deposited at the same time at many sites around the world. Careful analysis showed this worldwide sediment layer to be rich in iridium, and element that is rare on Earth's surface. But iridium is common in primitive meteorites, which led the Alverezes to a stunning conclusion: the extinction of the dinosaurs was caused by the impact of an asteroid or comet. This conclusion was not immediately

accepted and still generates some controversy, but it now seems clear that a major impact coincided with the death of the dinosaurs. While the dinosaurs were the most famous victims of this mass extinction, it seems that up to 99 percent of all living things were killed and that 75 percent of all species living on Earth were wiped out at that time.

2. The word "Subsequent" in the passage is closest in meaning to
 - ☐ Detailed
 - ☐ Later
 - ☐ Recent
 - ☐ Additional
3. The phrase "coincided with" in the passage is closest in meaning to
 - ☐ occurred before
 - ☐ happened at the same time as
 - ☐ speeded up
 - ☐ caused
4. In paragraph 2, the author discusses the presence of iridium in the sediment layer discovered by the Alvarezes because
 - ☐ it supported their claim that toxic levels of iridium could have killed the dinosaurs
 - ☐ it led some scientists to question the validity of the Alvarezes' study
 - ☐ it led the Alvarezes to conclude that a space impactor killed the dinosaurs
 - ☐ it showed that new discoveries were still being made about the extinction of the dinosaurs

Paragraph 2 is marked with an arrow [→]

5. Which of the following statements about scientists' theories on the extinction of the dinosaurs can be inferred from paragraph 2?
 - ☐ Some scientists do not believe that an asteroid or comet impact killed the dinosaurs.
 - ☐ Some scientists think that iridium was abundant on Earth before the extinction of the dinosaurs.
 - ☐ Some scientists doubt that all of the dinosaurs died out 65 million years ago.
 - ☐ Some scientists believe that the dinosaurs became extinct only after the plants and animals they ate disappeared.

Paragraph 2 is marked with an arrow [→]

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - ☐ A large percentage of Earth's living things were killed along with the dinosaurs.
 - ☐ Seventy-five percent of all species, including 75 percent of dinosaur species, were victims of a single mass extinction.

- In the extinction that eliminated the dinosaurs, up to 99 percent of all living things and 75 percent of all species were wiped out.
- Dinosaurs were the most famous victims of a mass extinction, but there have also been mass extinctions that wiped out most of Earth's other species.

Paragraph 3

How could an impact lead to mass extinction? The amount of iridium deposited worldwide suggests that the impactor must have been about 10 kilometers across. After a decade-long search, scientists identified what appears to be the impact crater from the event. Located off the coast of Mexico's Yucatan peninsula, it is 200 kilometers across, which is close to what one would expect for a 10-kilometer impactor, and dates to 65 million years ago. Further evidence that the Yucatan crater is the right one comes from the distribution of small glassy spheres that formed when the molten impact ejecta solidified as it rained back to Earth. More of these glassy spheres are found in regions near the crater, and careful study of their distribution suggests that the impactor crashed to Earth at a slight angle. These pieces of once molten rock are evidence of an explosion powerful enough to instantly melt bedrock and propel it far from its origin. The impact almost immediately sent a shower of debris raining across much of North and South America and generated huge waves that may have sloshed more than 1,000 kilometers inland. Many North American species thus may have been wiped out shortly after impact. For the rest of the world, death may have come more slowly. Heat from the impact and returning ejecta probably ignited wildfires in forests around the world. Evidence of wildfires is found in the large amount of soot (a black powdery form of carbon produced when coal, wood, or oil is burned) that is also present in the iridium-rich sediment from 65 million years ago. The impact also sent huge quantities of dust high into the stratosphere, where it remained for several years, blocking out sunlight, cooling the surface, and affecting atmospheric chemistry. Plants died for lack of sunlight, and effects propagated throughout the food chain.

7. According to paragraph 3, what information did scientists use to predict the size of the impactor that they believed struck Earth 65 million years ago?
- The size of the debris found in geological samples from that time
 - The distance inland that traces of tidal waves were found
 - The amount of iridium found in sediment from that time
 - The number of species that became extinct at that time
- Paragraph 3 is marked with an arrow [→]

8. According to paragraph 3, the shower of debris created by the impact suggests that the impactor
- was 200 kilometers across
 - landed about 1,000 kilometers inland
 - had a molten core that solidified immediately as the impactor hit Earth
 - was probably responsible for the immediate extinction of many North American

species

Paragraph 3 is marked with an arrow [→]

9. In paragraph 3, why does the author include the information that a large amount of soot is found in sediment from 65 million years ago?
- ☐ To identify the largest component of the dust sent into the atmosphere by the impactor's crash
 - ☐ To support the idea that the impactor's crash probably caused widespread wildfires
 - ☐ To emphasize the large variety of substances found in the sediment
 - ☐ To cast doubt on the claim that the impact affected only the area near the crash site

Paragraph 3 is marked with an arrow [→]

10. Which of the following is NOT mentioned in paragraph 3 as an effect of the dust raised by the impact 65 million years ago?
- ☐ It cooled the surface of Earth.
 - ☐ It released toxic chemicals into the stratosphere.
 - ☐ It caused changes in the chemical composition of the atmosphere.
 - ☐ It prevented sunlight from reaching the surface of Earth.

Paragraph 3 is marked with an arrow [→]

11. The word "propagated" in the passage is closest in meaning to
- ☐ were seen
 - ☐ occurred
 - ☐ caused harm
 - ☐ spread

Paragraph 4

Perhaps the most astonishing fact is not that 75 percent of all species died, but that 25 percent survived. Among the survivors were a few small, rodent-like mammals. These mammals may have survived because they lived in underground burrows and managed to store enough food to outlast the long spell of cold, dark days. Small mammals had first arisen at about the same time as the dinosaurs, more than 100 million years earlier. But the sudden disappearance of the dominant dinosaurs made these mammals dominant.

12. According to paragraph 4, all of the following are true of the animals that survived the impact EXCEPT:
- ☐ They probably had a lot of food stored up.
 - ☐ Many lived underground.
 - ☐ Most were relatively new species.
 - ☐ They tended to be small.

Paragraph 4 is marked with an arrow [→]

Paragraph3

How could an impact lead to mass extinction? The amount of iridium deposited worldwide suggests that the impactor must have been about 10 kilometers across. After a decade-long search, scientists identified what appears to be the impact crater from the event. Located off the coast of Mexico's Yucatan peninsula, it is 200 kilometers across, which is close to what one would expect for a 10-kilometer impactor, and dates to 65 million years ago. Further evidence that the Yucatan crater is the right one comes from the distribution of small glassy spheres that formed when the molten impact ejecta solidified as it rained back to Earth. ■ More of these glassy spheres are found in regions near the crater, and careful study of their distribution suggests that the impactor crashed to Earth at a slight angle. ■ These pieces of once molten rock are evidence of an explosion powerful enough to instantly melt bedrock and propel it far from its origin. ■ The impact almost immediately sent a shower of debris raining across much of North and South America and generated huge waves that may have sloshed more than 1,000 kilometers inland. ■ Many North American species thus may have been wiped out shortly after impact. For the rest of the world, death may have come more slowly. Heat from the impact and returning ejecta probably ignited wildfires in forests around the world. Evidence of wildfires is found in the large amount of soot (a black powdery form of carbon produced when coal, wood, or oil is burned) that is also present in the iridium-rich sediment from 65 million years ago. The impact also sent huge quantities of dust high into the stratosphere, where it remained for several years, blocking out sunlight, cooling the surface, and affecting atmospheric chemistry. Plants died for lack of sunlight, and effects propagated throughout the food chain.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

As a result, molten material was pushed out toward the north with small glass spheres found more than 2,500 kilometers to the northwest.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Asteroids and other large bodies from space have occasionally struck Earth, causing mass destruction and extinction.

-
-
-

Answer Choices

The large impactor that crashed in what is now the Arizona desert exploded with the force of a 20-megaton hydrogen bomb.

Although the crash of a large impactor would have killed all living things near the site of impact, most animals living farther away adapted to changes in the environment.

Scientists believe the reason the impactor that struck Earth 65 million years ago was so deadly was because it contained many toxic compounds, such as iridium.

A worldwide sediment layer of iridium suggests that an asteroid or comet impacted Earth 65 million years ago, possibly causing the extinction of the dinosaurs.

The large impact that caused the Yucatan crater probably produced both immediate natural disasters and a long-lasting layer of dust in the atmosphere.

Despite the destructive effects of the large impact 65 million years ago, around 25 percent of Earth's species survived and mammals came to be dominant.

Milankovitch Cycles and Glaciation

Although the history of glaciation during the Pleistocene epoch (2 million to 10,000 years ago) is well established, we do not know with complete certainty why glaciation takes place. For over a century, geologists and climatologists have struggled with this problem, but it remains unsolved.

It is long known that Earth's orbit around the Sun changes periodically, cyclically affecting the way solar radiation strikes the Earth, but the idea that these changes affect climate was first advanced by James Croll in the late 1800s. Later, Milutin Milankovitch elaborated the theory with calculations that convincingly argued that the cycles, now known as Milankovitch cycles, could cause climatic variations.

The Milankovitch cycles emerge from the way three cyclic changes in Earth's orbit combine. One characteristic of Earth's orbit is its *eccentricity*, the degree to which the orbit is an ellipse rather than a circle. Changes in the eccentricity of Earth's orbit occur in a cycle of about 96,000 years. The *inclination*, or tilt, of Earth's axis also varies periodically, moving between 22 degrees and 24.5 degrees. The tilt of Earth's axis, toward the Sun at some times of the year and away from the Sun at other times, is responsible for the annual cycle of seasons. The greater the tilt, the greater the contrast between summer and winter temperatures. Changes in the tilt occur in a cycle 41,000 years long. Also, Earth *wobbles* as it spins, like a slightly unsteady top. The wobble cycle is completed once every 21,700 years. Changes in eccentricity, tilt and wobble do not affect the total amount of solar radiation Earth receives in a year, but they do affect how evenly or unevenly this radiation is disturbed over the course of a year. According to the Milankovitch theory, about every 40,000 years the three separate cycles combine in such a way that the *difference* between summer and winter temperatures is at a minimum. At this point winter temperatures are milder but so too are summer temperatures. As a result, less ice is melted in the summer than is formed in the winter, so glaciers build up and a period of glaciation results.

Milankovitch worked out the ideas of climatic cycles in the 1920s and 1930s, but it was not until the 1970s that a detailed chronology of the Pleistocene temperature changes was determined that could test the predictions of this theory. A correspondence between Milankovitch cycles and climate fluctuations of the last 65 million years seems clear. Furthermore, studies of rock samples drilled from the deep-sea floor and the fossils contained in them indicate that the fluctuation of climate during the past few hundred thousand years is remarkably close to that predicted by Milankovitch.

A problem with Milankovitch's explanation of glaciation arises from the fact that the

variations in Earth's orbit, and hence the Milankovitch cycles, have existed for billions of years. Thus we might expect that glaciation would have been a cyclic event throughout geologic time. In fact, periods of glaciation are rare. So there must be another factor acting together with the Milankovitch cycles that causes periods of glaciation. Once this additional factor makes the temperature low enough, the cyclic variations of the Milankovitch cycles will force the planet into and out of glacial epochs with a fixed regularity.

Many hypotheses have been proposed for the additional cooling factor. Some suggest that variations in the Sun's energy output could account for the ice ages. However, our present understanding of the Sun's luminosity holds that it should have progressively increased, not decreased, over the course of Earth's history. Still others argue that volcanic dust injected into the atmosphere shields Earth from the Sun's rays and initiates an ice age. However, no correlation has been found between volcanic activity and the start of the last ice age. An increasingly attractive theory holds that decreases in atmospheric carbon dioxide starts the cooling trend that leads to glaciation. Carbon dioxide traps solar energy reflected from the Earth's surface. If carbon dioxide levels decrease, less heat is trapped and Earth's surface cools. Recent studies of the carbon dioxide content of gas bubbles preserved in the Greenland ice cap do in fact show that high carbon dioxide levels are associated with warm interglacial periods, and low levels with cold glacial periods.

Paragraph 2

It is long known that Earth's orbit around the Sun changes periodically, cyclically affecting the way solar radiation strikes the Earth, but the idea that these changes affect climate was first advanced by James Croll in the late 1800s. Later, Milutin Milankovitch elaborated the theory with calculations that convincingly argued that the cycles, now known as Milankovitch cycles, could cause climatic variations.

1. The word "elaborated" in the passage is closest in meaning to
 - ☐ corrected
 - ☐ defended
 - ☐ studied
 - ☐ developed
2. According to paragraph 2, Milankovitch was the first to do which of the following?
 - ☐ Introduce the idea that periodic changes in Earth's orbit could influence Earth's climate
 - ☐ Show that periodic changes in Earth's orbit could be mathematically calculated
 - ☐ Provide calculations to show that periodic changes in Earth's orbit could affect Earth's climate
 - ☐ Propose that Earth's orbit is not fixed but undergoes periodic changes

Paragraph 2 is marked with an arrow [→]

Paragraph 3

The Milankovitch cycles emerge from the way three cyclic changes in Earth's orbit combine. One characteristic of Earth's orbit is its *eccentricity*, the degree to which the orbit is an ellipse rather than a circle. Changes in the eccentricity of Earth's orbit occur in a cycle of about 96,000 years. The *inclination*, or tilt, of Earth's axis also varies periodically, moving between 22 degrees and 24.5 degrees. The tilt of Earth's axis, toward the Sun at some times of the year and away from the Sun at other times, is responsible for the annual cycle of seasons. The greater the tilt, the greater the contrast between summer and winter temperatures. Changes in the tilt occur in a cycle 41,000 years long. Also, Earth *wobbles* as it spins, like a slightly unsteady top. The wobble cycle is completed once every 21,700 years. Changes in eccentricity, tilt and wobble do not affect the total amount of solar radiation Earth receives in a year, but they do affect how evenly or unevenly this radiation is disturbed over the course of a year. According to the Milankovitch theory, about every 40,000 years the three separate cycles combine in such a way that the *difference* between summer and winter temperatures is at a minimum. At this point winter temperatures are milder but so too are summer temperatures. As a result, less ice is melted in the summer than is formed in the winter, so glaciers build up and a period of glaciation results.

3. According to paragraph 3, Milankovitch's theory holds that periods of glaciation result from a particular combination of changes in all the following EXCEPT
- ☐ the shape of Earth's orbit
 - ☐ the inclination of Earth's axis
 - ☐ the wobble of Earth as it spins
 - ☐ the amount of time required for Earth's rotation around the Sun

Paragraph 3 is marked with an arrow [→]

4. According to paragraph 3, Milankovitch's theory predicts that glaciers build up most when
- ☐ the least amount of solar heat is being delivered to Earth
 - ☐ winter temperatures are lowest
 - ☐ the difference between winter temperatures and summer temperatures is greatest
 - ☐ both winter temperatures and summer temperatures are relatively mild

Paragraph 3 is marked with an arrow [→]

Paragraph 4

Milankovitch worked out the ideas of climatic cycles in the 1920s and 1930s, but it was not until the 1970s that a detailed **chronology** of the Pleistocene temperature changes was determined that could test the predictions of this theory. A correspondence between Milankovitch cycles and climate fluctuations of the last 65 million years seems clear. Furthermore, studies of rock samples drilled from the deep-sea floor and the fossils contained in them indicate that the fluctuation of

climate during the past few hundred thousand years is remarkably close to that predicted by Milankovitch.

5. A "chronology" is

- ☐ a study of past events
- ☐ a list that pairs past events with dates
- ☐ an explanation of why past events happened when they did
- ☐ an ordering of past events according to how long they lasted

6. Which of the following can be inferred from paragraph 4 about rock samples taken from the sea floor and the fossils they contain?

- ☐ There is a correspondence between the fossils in the samples and climate fluctuations.
- ☐ Milankovitch's theory predicts when the samples will contain fossils.
- ☐ It was not known until the 1970s that fossils were present in rocks taken from the sea floor.
- ☐ There is no fossil record in the samples older than a few hundred thousand years.

Paragraph 4 is marked with an arrow [→]

7. The phrase "remarkably close" in the passage is closest in meaning to

- ☐ sufficiently close
- ☐ relatively close
- ☐ extremely close
- ☐ frequently close

Paragraph 5

A problem with Milankovitch's explanation of glaciation arises from the fact that the variations in Earth's orbit, and hence the Milankovitch cycles, have existed for billions of years. Thus we might expect that glaciation would have been a cyclic event throughout geologic time. In fact, periods of glaciation are rare. So there must be another factor acting together with the Milankovitch cycles that causes periods of glaciation. Once this additional factor makes the temperature low enough, the cyclic variations of the Milankovitch cycles will force the planet into and out of glacial epochs with a fixed regularity.

8. According to paragraph 5, which of the following is a problem with Milankovitch's theory?

- ☐ It assures that the astronomical cycles have been in existence for billions of years.
- ☐ It cannot explain why glaciation has been a relatively rare occurrence in Earth's history.
- ☐ It cannot predict periods of glaciation in Earth's distant past.
- ☐ It assures that astronomical cycles have an effect on Earth's climate even during periods when there is no glaciation.

Paragraph 5 is marked with an arrow [→]

Paragraph 6

Many hypotheses have been proposed for the additional cooling factor. Some suggest that variations in the Sun's energy output could account for the ice ages. However, our present understanding of the Sun's luminosity holds that it should have progressively increased, not decreased, over the course of Earth's history. Still others argue that volcanic dust injected into the atmosphere shields Earth from the Sun's rays and initiates an ice age. However, no correlation has been found between volcanic activity and the start of the last ice age. An increasingly attractive theory holds that decreases in atmospheric carbon dioxide starts the cooling trend that leads to glaciation. Carbon dioxide traps solar energy reflected from the Earth's surface. If carbon dioxide levels decrease, less heat is trapped and Earth's surface cools. Recent studies of the carbon dioxide content of gas bubbles preserved in the Greenland ice cap do in fact show that high carbon dioxide levels are associated with warm interglacial periods, and low levels with cold glacial periods.

9. The author discusses "our present understanding of the Sun's luminosity" in order to
- ☐ provide evidence that Milankovitch's astronomical cycles cannot explain the occurrence of the ice ages
 - ☐ present an objection to a proposed explanation of the cause of ice ages
 - ☐ challenge the claim that long-term cooling can account for glaciation
 - ☐ introduce a problem that none of the proposed hypotheses about the causes of glaciation can answer
10. The word "initiates" in the passage is closest in meaning to
- ☐ begins
 - ☐ accelerates
 - ☐ ends
 - ☐ avoids
11. According to paragraph 6, in addition to Milankovitch cycles, each of the following has been proposed as a contributing cause of ice ages EXCEPT
- ☐ variations in the composition of the Greenland ice cap
 - ☐ variations in the Sun's energy output
 - ☐ volcanic dust injected into the atmosphere
 - ☐ decreases in atmospheric carbon dioxide

Paragraph 6 is marked with an arrow [→]

12. It follows from the theory of the role of carbon dioxide discussed in paragraph 6, that
- ☐ the decrease in temperature during the last ice age caused a decrease in atmospheric carbon dioxide
 - ☐ the atmosphere had higher carbon dioxide content during the last ice age than it

- had during the warm period immediately before it
- ☐ the cooling of temperatures that led to the last ice age was brought about by a decrease in atmospheric carbon dioxide
 - ☐ there was less carbon dioxide in the atmosphere toward the end of the last ice age than there was at the beginning

Paragraph 6 is marked with an arrow [→]

Paragraph 5

A problem with Milankovitch's explanation of glaciation arises from the fact that the variations in Earth's orbit, and hence the Milankovitch cycles, have existed for billions of years. ■ Thus we might expect that glaciation would have been a cyclic event throughout geologic time. In fact, periods of glaciation are rare. ■ So there must be another factor acting together with the Milankovitch cycles that causes periods of glaciation. ■ Once this additional factor makes the temperature low enough, the cyclic variations of the Milankovitch cycles will force the planet into and out of glacial epochs with a fixed regularity. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

This factor must precede an ice age and have the effect of slightly lowering Earth's temperature.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

In the 1920s and 1930s, Milutin Milankovitch worked out an account of the cause of Earth's glaciations.

-
-
-

Answer Choices

Milankovitch argued that glaciations would result when three cycles that affect characteristics of Earth's orbit combined in a certain way.

Milankovitch's predictions have been shown to be in agreement with periods of glaciation over the past 65 million years.

Since ice ages are much less frequent than Milankovitch's explanation predicts, some factor, such as low levels of atmospheric carbon dioxide, must also be involved in triggering glaciation.

Milankovitch showed that changes in Earth's orbit periodically lengthen cold seasons and shorten warm ones so that more ice forms than melts and glaciers build up.

Dust from volcanic eruptions that blocks the Sun's warming radiation has been proposed as an explanation alternative to Milankovitch cycles for the cause of ice ages.

Studies of gas bubbles preserved in the Greenland ice cap strongly suggest that Milankovitch cycles may also influence atmospheric carbon dioxide levels.

The Long History of Overexploitation

Overexploitation is the overuse by humans of a population of organisms to an extent that threatens the viability of the population or radically alters the natural community in which it lives. There is a tendency to think that overexploitation is a relatively new phenomenon. However, that view is a bit naïve, as we will see in some examples of past overexploitation.

After the most recent glaciation, which was at its height between about 20,000 and 14,000 years ago, the grasslands in central North America harbored an extraordinary array of large animals. The diversity of antelope, horses, cheetahs, giant ground sloths, mammoths, mastodons, and other animals easily rivaled that of the large animal fauna of Africa today. However, about 11,000 years ago, at the end of the Pleistocene epoch, many disappeared; 34 genera of large mammals became extinct in fewer than 1,000 years, while 40 more became extinct in South America. This was a massive die-off when you consider that only 20 large mammal genera had become extinct in North America over the previous three million years.

Is it a coincidence that so many large mammals became extinct shortly after the time that humans, crossing from Siberia to Alaska, probably first arrived in the Western Hemisphere? Paul Martin, an anthropologist, thinks not and has argued in many articles and books that overhunting was primarily responsible for the extinctions. Martin's critics have argued that the extinctions were primarily the result of significant climate change. If climate change was responsible, we would expect many small mammals, primarily rodents, to have become extinct in the region, too; however, only four North American genera of small animals became extinct at the end of the Pleistocene, compared with 46 genera over the preceding three million years. On the other hand, perhaps small mammals are less susceptible to climate change than large mammals are. We cannot definitely say which answer is correct, and the truth may lie in the middle, but many scientists believe that the end of the Pleistocene saw a massive overkill by the first human inhabitants of the Americas.

The best evidence that overhunting by early people eliminated some species comes from islands. On many remote islands, birds evolved in the absence of mammalian predators, sometimes losing their ability to fly in the process. When people arrived on these islands, they found easy prey. For example, when Polynesians, now known as Maori, arrived in New Zealand about A.D. 1200, the islands had 11 species of moas: flightless birds that ranged in size from as small as a turkey to larger than an ostrich. By the time Europeans colonized the islands in the 1700s, the moas were gone, along with five species of rail and six waterfowl species. The demise of the moas and other birds undoubtedly was hastened by forest clearing and other changes brought about by the Maori, but the abundance of moa remains at Maori village sites makes it clear that hunting was a major factor.

On small islands throughout the Pacific, scores of birds are known to have become extinct after the arrival of Polynesians. In the Hawaiian Islands, 44 species of endemic land birds out of 82

became extinct between the arrival of Polynesians and the arrival of Europeans. Again, habitat changes were undoubtedly important, but it is likely that overhunting was a major problem, especially for various species of flightless geese, ibis, and rail. On Madagascar, the loss was not limited to birds. The arrival of people 1,500 to 2,000 years ago caused the extinctions of two giant tortoises, a bear-sized giant lemur, a small species of hippopotamus, many other mammals, and elephant birds, some of which rivaled the largest moas in size.

Currently, the worst overexploitation may be happening through global overfishing. This is partially disguised by the fact that we are still able to harvest huge quantities of marine species; only on closer inspection does one notice that the predatory fish that used to dominate catches are being replaced by species further down the food chain.

Paragraph 1

Overexploitation is the overuse by humans of a population of organisms to an extent that threatens the **viability** of the population or radically alters the natural community in which it lives. There is a tendency to think that overexploitation is a relatively new phenomenon. However, that view is a bit naïve, as we will see in some examples of past overexploitation.

1. The word "**viability**" in the passage is closest in meaning to

- ☐ ability to exist
- ☐ mobility
- ☐ rate of growth
- ☐ ability to change

Paragraph 2

After the most recent glaciation, which was at its height between about 20,000 and 14,000 years ago, the grasslands in central North America harbored an extraordinary array of large animals. The diversity of antelope, horses, cheetahs, giant ground sloths, mammoths, mastodons, and other animals easily rivaled that of the large animal fauna of Africa today. However, about 11,000 years ago, at the end of the Pleistocene epoch, many disappeared; 34 genera of large mammals became extinct in fewer than 1,000 years, while 40 more became extinct in South America. This was a massive die-off when you consider that only 20 large mammal genera had become extinct in North America over the previous three million years.

Paragraph 3

Is it a coincidence that so many large mammals became extinct shortly after the time that humans, crossing from Siberia to Alaska, probably first arrived in the Western Hemisphere? Paul Martin, an anthropologist, thinks not and has argued in many articles and books that overhunting was primarily responsible for the extinctions. Martin's critics have argued that the extinctions were primarily the result of significant climate change. If climate change was responsible, we would expect many small mammals, primarily rodents, to have become extinct in the region, too; however, only four North American genera of small animals became extinct at the end of the Pleistocene, compared with 46 genera over the preceding three million years. On the other hand, perhaps small mammals are less susceptible to climate change than large mammals are. **We**

cannot definitely say which answer is correct, and the truth may lie in the middle, but many scientists believe that the end of the Pleistocene saw a massive overkill by the first human inhabitants of the Americas.

2. According to paragraphs 2 and 3, all of the following are true of the diversity of large mammals in the North American grasslands after the most recent glaciation EXCEPT:
- ☐ It was easily as great as the diversity of large animals in Africa today.
 - ☐ It diminished significantly over a period of fewer than 1,000 years.
 - ☐ It was probably reduced by both climate change and hunting at the end of the Pleistocene epoch.
 - ☐ It was much greater than the diversity of small animal populations during the Pleistocene epoch.

Paragraphs 2 and 3 are marked with arrows [→]

3. Why does the author mention that “20 large mammal genera had become extinct in North America over the previous three million years”?
- ☐ To give an example of extinction that was caused by overexploitation by humans
 - ☐ To cast doubt on the idea that the extinctions at the end of the Pleistocene epoch were unique or notable
 - ☐ To provide evidence for the idea that large mammal species were able to flourish in central North America after the glaciation
 - ☐ To emphasize the significance of the number of extinctions at the end of the Pleistocene epoch
4. Which of the following statements best describes anthropologist Paul Martin’s views on Pleistocene extinction as explained in paragraph 3?
- ☐ It is a coincidence that so many large mammals became extinct shortly after humans first arrived in the Western Hemisphere.
 - ☐ Overhunting was primarily responsible for the extinction of large mammals after the arrival of humans in the Western Hemisphere.
 - ☐ The massive extinction of large animal species was due to the severe climate changes at the end of the Pleistocene epoch.
 - ☐ The extinction of large mammals was caused by the extinction of small rodents that large animals hunted for food.

Paragraph 3 is marked with an arrow [→]

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ We cannot definitely say whether or not a massive overkill happened at the end of the Pleistocene, despite what many scientists believe.
 - ☐ We cannot definitely say whether the massive overkill by the first human inhabitants of the Americas occurred in the middle of the end of the Pleistocene.
 - ☐ Although it is difficult to know for sure, many scientists believe that the first human

inhabitants were responsible for an overkill of mammals at the end of the Pleistocene.

- Many scientists believe that many of the first human inhabitants of the Americas did not survive the climate changes at the end of the Pleistocene.

Paragraph 5

On small islands throughout the Pacific, scores of birds are known to have become extinct after the arrival of Polynesians. In the Hawaiian Islands, 44 species of endemic land birds out of 82 became extinct between the arrival of Polynesians and the arrival of Europeans. Again, habitat changes were undoubtedly important, but it is likely that overhunting was a major problem, especially for various species of flightless geese, ibis, and rail. On Madagascar, the loss was not limited to birds. The arrival of people 1,500 to 2,000 years ago caused the extinctions of two giant tortoises, a bear-sized giant lemur, a small species of hippopotamus, many other mammals, and elephant birds, some of which rivaled the largest moas in size.

10. According to paragraph 5, which of the following did NOT contribute to the extinction of many bird species on Pacific islands?

- Habitat changes
- The inability to fly
- Overhunting
- New species of mammals introduced by the Polynesians

Paragraph 5 is marked with an arrow [→]

Paragraph 6

Currently, the worst overexploitation may be happening through global overfishing. This is partially disguised by the fact that we are still able to harvest huge quantities of marine species; only on closer inspection does one notice that the predatory fish that used to dominate catches are being replaced by species further down the food chain.

11. The word "disguised" in the passage is closest in meaning to

- hidden
- contradicted
- balanced
- made worse

12. Which of the following statements is supported by the discussion of fish catches in paragraph 6?

- Global overfishing has been difficult to recognize because there are so many different species of predatory fish in different areas around the world.
- Closer inspection of the global fishing industry was undertaken recently because we have not been able to harvest enough fish.
- Reduced numbers of predatory fish that used to be caught most often are evidence that humans have overexploited these fish.
- Fish species further down the food chain now prey on species that used to be harvested.

Paragraph 6 is marked with an arrow [→]

Paragraph 4

The best evidence that overhunting by early people eliminated some species comes from islands. On many remote islands, birds evolved in the absence of mammalian predators, sometimes losing their ability to fly in the process. When people arrived on these islands, they found easy prey. ■ For example, when Polynesians, now known as Maori, arrived in New Zealand about A.D. 1200, the islands had 11 species of moas: flightless birds that ranged in size from as small as a turkey to larger than an ostrich. ■ By the time Europeans colonized the islands in the 1700s, the moas were gone, along with five species of rail and six waterfowl species. ■ The demise of the moas and other birds undoubtedly was hastened by forest clearing and other changes brought about by the Maori, but the abundance of moa remains at Maori village sites makes it clear that hunting was a major factor. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Indeed, some evidence suggests that these extinctions occurred long before, perhaps as early as 100 years after Polynesian colonization.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions

Although it is sometimes thought to be a new phenomenon, overexploitation has many examples in human history.

Answer Choices

Strong evidence for overexploitation comes from the extinction of many species of flightless birds on Pacific islands after these places were colonized by humans.

The fact that only flightless-bird species became extinct on Pacific islands after the arrival of humans could indicate that overexploitation of forest habitats rather than hunting caused the massive overkill.

Presently, overexploitation may be worst in the sea, where global overfishing has significantly reduced some predatory-fish population.

Although some people attribute the mass extinctions of large animals at the end of the Pleistocene epoch to climate change, many scientists believe them to result from overexploitation by humans.

Because of their comparative diversity, marine species are less susceptible to the overexploitation that has affected mammals and birds on land.

The fact that most small-rodent species died out during the Pleistocene epoch provides evidence that humans were responsible for the mass extinction of animals during this period.

How Animals in Rain Forests Make Themselves Heard

Scientists have discovered that animals are experts at exploiting weather conditions and the physical conditions of their environments so that they are heard or not heard, and seen or not seen. The species living in rain forests must engineer their calls to accommodate all of the obstacles, such as leaf cover, that can deflect and degrade the sounds intended for a potential receiver. Over, short, loud bursts of sound tend to be more effective than longer calls at cutting through the dense foliage.

There is no natural environment on Earth noisier than a virgin rain forest. In the Peruvian rain forest, every species has developed clever or remarkably sophisticated strategies to ensure that its voice is heard. The noise creates a real challenge for the smaller residents, such as male tree crickets, which need to get the attention of females, often from a relatively long distance. Some species of crickets maximize the volume of their calls by chewing a hole in the middle of a leaf to create a sound baffle, similar to a stereo speaker. The leaf functions as a speaker cabinet, with the cricket in the center acting as the speaker.

A species of tree frog in Borneo has an inventive approach to getting its mating call heard over the noise. *Mataphrenella sudana*, which is only an inch long, has learned to exploit the sound properties of a water-filled hole in a tree in the same way that a person uses resonance, the intensification and enrichment of a sound by added vibration, in the shower to sing like a professional performer. The frog searches for a suitable hole and then partially submerges itself in the water. Its forte is the ability to adjust the frequency of its call to the size of the hole and play the tree like a musical instrument. As it sits in the hole, it begins vocalizing at different frequencies until it hits the one note that makes the hole and tree resonate.

The time of day affects how sound travels in any environment, and this fact is not lost on animals and insects. Early morning and late evening produce conditions that allow sound to travel greater distances than during the middle parts of the day. Sound travels best at night, which is why the rain forest is so wonderfully noisy between dusk and dawn. For species that sleep at night, dusk and dawn are their windows of opportunity to get the best resonance and distance out of a signal. This is why animals, especially birds, tend to be more active and noisy in the early morning and late evening. The British call the phenomenon of birds singing in the early morning the dawn chorus. Because of the superior sound conditions, dusk and dawn are the times to conduct the serious business of attracting mates and defending territory. For predators, it is the best time to track down their noisy prey.

Another way animals and insects ensure that their calls connect with the intended receivers is by developing their own specialized frequencies, which are determined primarily by the size of their bodies. Recently, a scientist visiting the Peruvian rain forest made an audiotape of a little of the night's music. When he took the tape back to his lab and analyzed it, he discovered that this seemingly chaotic banquet of sound was actually highly ordered. Each animal and insect is tuned

to and calling on its own species-specific frequency, in the same way that radio stations use different signals so that many stations can broadcast at the same time.

Bernard Krause, a professor at the University of Oregon in Eugene, has found that in older tropical rain forests some species, such as the Asian paradise flycatcher, have become so specialized that their voices occupy several niches of the sound spectrum at the same time, thus laying territorial claim to several audio channels. His recordings from undisturbed rain forests around the world demonstrate a remarkable stability in the combined voices of the residents from year to year. The stability of the ambient sound gives each region a unique sound signature, or fingerprint.

Paragraph 1

Scientists have discovered that animals are experts at exploiting weather conditions and the physical conditions of their environments so that they are heard or not heard, and seen or not seen. The species living in rain forests must engineer their calls to accommodate all of the obstacles, such as leaf cover, that can **deflect** and degrade the sounds intended for a **potential** receiver. Over, short, loud bursts of sound tend to be more effective than longer calls at cutting through the dense foliage.

1. To "**deflect**" sounds means to change their
 - ☐ direction
 - ☐ volume
 - ☐ frequency
 - ☐ clarity
2. The word "**potential**" in the passage is closet in meaning to
 - ☐ favorable
 - ☐ possible
 - ☐ expected
 - ☐ chosen

Paragraph 3

A species of tree frog in Borneo has an **inventive** approach to getting its mating call heard over the noise. *Mataphrenella sudana*, which is only an inch long, has learned to **exploit** the sound properties of a water-filled hole in a tree in the same way that a person uses resonance, the intensification and enrichment of a sound by added vibration, in the shower to sing like a professional performer. The frog searches for a suitable hole and then partially submerges itself in the water. Its forte is the ability to adjust the frequency of its call to the size of the hole and play the tree like a musical instrument. As it sits in the hole, it begins vocalizing at different frequencies until it hits the one note that makes the hole and tree resonate.

3. The word "**inventive**" in the passage is closet in meaning to
 - ☐ simple
 - ☐ alternate

- ☐ relative
 - ☐ creative
4. The word "exploit" in the passage is closest in meaning to
- ☐ identify
 - ☐ be aware of
 - ☐ take advantage of
 - ☐ examine

Paragraph 2

There is no natural environment on Earth noisier than a virgin rain forest. In the Peruvian rain forest, every species has developed clever or remarkably sophisticated strategies to ensure that its voice is heard. The noise creates a real challenge for the smaller residents, such as male tree crickets, which need to get the attention of females, often from a relatively long distance. Some species of crickets maximize the volume of their calls by chewing a hole in the middle of a leaf to create a sound baffle, similar to a stereo speaker. The leaf functions as a speaker cabinet, with the cricket in the center acting as the speaker.

5. Why does the author describe how some cricket species "maximize the volume of their calls"?
- ☐ To argue that crickets are a major source of noise in virgin rain forests
 - ☐ To help explain why it is difficult for many smaller animals to be heard in rain forests
 - ☐ To help explain why rain forests are noisier than other natural environments
 - ☐ To illustrate a sophisticated way of making a call heard in a rain forest
6. The author mentions a "stereo speaker" in the passage in order to
- ☐ contrast the ways in which humans and insects magnify their sounds
 - ☐ compare the ranges of sounds produced by humans and insects
 - ☐ support the claim that small size is a disadvantage for insects that produce calls
 - ☐ help explain how small insects magnify the sounds of their calls

Paragraph 3

A species of tree frog in Borneo has an inventive approach to getting its mating call heard over the noise. *Mataphrenella sudana*, which is only an inch long, has learned to exploit the sound properties of a water-filled hole in a tree in the same way that a person uses resonance, the intensification and enrichment of a sound by added vibration, in the shower to sing like a professional performer. The frog searches for a suitable hole and then partially submerges itself in the water. Its forte is the ability to adjust the frequency of its call to the size of the hole and play the tree like a musical instrument. As it sits in the hole, it begins vocalizing at different frequencies until it hits the one note that makes the hole and tree resonate.

7. According to paragraph 3, which of the following is NOT part of the process that the Borneo tree frog used to make its mating call heard?
- ☐ Finding a tree that has a hole of adequate size

- Immersing part of its body in a water-filled hole
- Aiming its call at a particular frog in a nearby tree
- Trying out a number of distinct call frequencies

Paragraph 3 is marked with an arrow [→]

Paragraph 4

The time of day affects how sound travels in any environment, and this fact is not lost on animals and insects. Early morning and late evening produce conditions that allow sound to travel greater distances than during the middle parts of the day. Sound travels best at night, which is why the rain forest is so wonderfully noisy between dusk and dawn. For species that sleep at night, dusk and dawn are their windows of opportunity to get the best resonance and distance out of a signal. This is why animals, especially birds, tend to be more active and noisy in the early morning and late evening. The British call the phenomenon of birds singing in the early morning the dawn chorus. Because of the superior sound conditions, dusk and dawn are the times to conduct the serious business of attracting mates and defending territory. For predators, it is the best time to track down their noisy prey.

8. According to paragraph 4, why do birds prefer to sing at the beginning and end of the day?
- Their calls are less likely to produce echoes.
 - Their calls do not have to compete with the sounds of other animals.
 - They are less vulnerable to predators then.
 - The sound of their voices travels farther.

Paragraph 4 is marked with an arrow [→]

9. According to paragraph 4, what is a disadvantage of many birds all singing at dusk and dawn?
- There is more competition from other birds at these times.
 - These are the safest times for birds to sleep.
 - It is easier for predators to locate birds when so many are making noise.
 - There are the best times for birds to find prey.

Paragraph 4 is marked with an arrow [→]

10. It can be inferred from paragraph 4 that a tropical rain forest is most quiet at
- dawn
 - the middle of the day
 - dusk
 - night

Paragraph 4 is marked with an arrow [→]

Paragraph 5

Another way animals and insects ensure that their calls connect with the intended receivers is by developing their own specialized frequencies, which are determined primarily by the size of their bodies. Recently, a scientist visiting the Peruvian rain forest made an audiotape of a little of the night's music. When he took the tape back to his lab and analyzed it, he discovered that this seemingly chaotic banquet of sound was actually highly ordered. Each animal and insect is tuned

to and calling on its own species-specific frequency, in the same way that radio stations use different signals so that many stations can broadcast at the same time.

11. According to paragraph 5, a scientist recently visiting the Peruvian rain forest discovered that
- ☐ the specialized frequencies of the sounds made by animals are determined by their body size
 - ☐ the frequencies of sounds made by animals at night are generally random
 - ☐ animals and insects of particular species make sounds at specific frequencies
 - ☐ animal and insect calls differ depending on whether they are heard in the laboratory or outside in nature

Paragraph 5 is marked with an arrow [→]

Paragraph 6

Bernard Krause, a professor at the University of Oregon in Eugene, has found that in older tropical rain forests some species, such as the Asian paradise flycatcher, have become so specialized that their voices occupy several niches of the sound spectrum at the same time, thus laying territorial claim to several audio channels. His recordings from undisturbed rain forests around the world demonstrate a remarkable stability in the combined voices of the residents from year to year. The stability of the ambient sound gives each region a unique sound signature, or fingerprint.

12. According to paragraph 6, what did the research of Professor Krause indicate about sound in rain forests?
- ☐ Older rain forests have more varied sound frequencies than younger rain forests.
 - ☐ The patterns of sounds in rain forests are consistent over time.
 - ☐ The speed of sound movement can be affected by the location of the rain forest.
 - ☐ It is difficult to distinguish the atmospheric sound patterns of rain forests in one region from those in another.

Paragraph 6 is marked with an arrow [→]

Paragraph 5

Another way animals and insects ensure that their calls connect with the intended receivers is by developing their own specialized frequencies, which are determined primarily by the size of their bodies. ■ Recently, a scientist visiting the Peruvian rain forest made an audiotape of a little of the night's music. ■ When he took the tape back to his lab and analyzed it, he discovered that this seemingly chaotic banquet of sound was actually highly ordered. ■ Each animal and insect is tuned to and calling on its own species-specific frequency, in the same way that radio stations use different signals so that many stations can broadcast at the same time. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Indeed, there were numerous layers of sound, each clearly distinct from the others.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Animal species in rain forests have developed a series of adaptations to make themselves heard.

-
-
-

Answer Choices

Scientists have discovered that animals in rain forests adapt the length of their calls to overcome the barriers created by dense plant growth.

Many species of predators in rain forests prefer dawn rather than dusk as the best time to hear and pursue their prey.

Specialized frequencies facilitate sound transmission within species and also create regional sound environments that are highly stable.

Some smaller animals that live in rain forests use special strategies to increase the volume of their calls and the distance that their calls travel.

Because time of day affects how sound travels, many animals restrict their calls to either nighttime, when sound travels best, or dusk and dawn.

Research conducted in the Peruvian rain forest indicates that well-ordered periods of sound alternate with more chaotic sound periods.

Pest Control

Many pest species that are native to North America, such as white-footed mice and ground moles, are more nuisance pests and are usually regulated by native predators and parasites. This situation is not true for nonindigenous pests in North America, such as brown rats and cockroaches. After centuries, it is evident that these pests cannot be eradicated. The best that can be done is to introduce pest control measures that will control their numbers.

And ancient and popular means of pest control is chemical. For example, the Sumerians used sulfur to combat crop pests, and by the early 1800s such chemicals as arsenic were used to combat insect and fungal pests.

However, chemical control has its dark side. Chemical pesticides have many unintended consequences through their effects not just on the target species but on a wide array of nontarget species as well, often eliminating them and thereby upsetting the existing food webs, especially through the suppression of native predator species. The surviving pests then rebound in greater numbers than ever.

Perhaps more insidious is that a pesticide loses its effectiveness because the target species evolves resistance to it. As one pesticide replaces another, the pests acquire a resistance to them all. Some species, notably certain mosquitoes, have overcome the toxic effects of every pesticide to which they have been exposed. Insect pests need not only about five years to evolve pesticide resistance, their predators do so much more slowly. So after the pest develops resistance, pest outbreaks become even more disastrous.

Farmer long ago observed that enemies of pests act as controls. As early as 300 C.E., the Chinese were introducing predatory ants into their citrus orchards to control leaf-eating caterpillars. Insect pests have their own array of enemies in their native habitats. When an animal or plant is introduced, intentionally or unintentionally, into a new habitat outside of its natural range, it may adapt to the new environment and leave its enemies behind. Freed from predation and finding an abundance of resources, the species quickly becomes a pest or a weed. This fact had led to the search for natural enemies to introduce into populations of pests to reduce their populations.

Because the serious pest is usually a nonnative species, biological control involves the introduction of a nonindigenous predator or parasite to control the pest. The introduction of the cactus-eating moth, a native of Argentina, into Australia effectively reduced and controlled the rapidly spreading prickly pear, which had been introduced into Australia in 1901.

But biological control, like chemical control, can backfire. The success of the cactus-feeding moth in controlling prickly pear in Australia encouraged its introduction to several West Indies islands to control prickly pear there. In time the moth made its way to Florida, where it now threatens the existence of several native prickly pear species. The moral is that although using nonindigenous predators as biological controls can be effective, these species possess their own

inherent dangers that must be assessed before they are released. They, too, can become alien invaders.

Because chemical, biological, and other methods used individually are obviously not the solution to pest control, entomologists have developed a holistic approach to pest control, called integrated pest management (IPM). IPM considers the biological, ecological, economic, social, and even aesthetic aspects of pest control and employs a variety of techniques. The objective of IPM is to control the pest not at the time a major outbreak but at an earlier time, when the size of the population is easier to control. The approach is to rely first on natural mortality caused by weather and natural enemies, with as little disruption of the natural system as possible, and to use other methods only if they are needed to hold the pest below the economic injury level.

Successful IPM requires the knowledge of the population ecology of each pest and its associated species and the dynamics of the host species. It involves considerable field work monitoring the pest species and its natural enemies by such techniques as egg counts and the trapping of adults to acquire information to determine the necessity, timing, and intensity of control measures. These control measures must be adjusted to the situation, which may vary from one location to another. The intensity of control or no control is based on the degree of pest damage that can be tolerated, the costs of control, and the benefits to be derived.

Paragraph 1

Many pest species that are native to North America, such as white-footed mice and ground moles, are more nuisance pests and are usually regulated by native predators and parasites. This situation is not true for nonindigenous pests in North America, such as brown rats and cockroaches. After centuries, it is evident that these pests cannot be eradicated. The best that can be done is to introduce pest control measures that will control their numbers.

1. The word "evident" in the passage is closest in meaning to
 - ☐ surprising
 - ☐ obvious
 - ☐ believed
 - ☐ frustrating
2. What can be inferred from paragraph 1 about nonindigenous pests such as brown rats and cockroaches?
 - ☐ Attempts limit the size of their populations have been unsuccessful.
 - ☐ They have inhabited North America longer than white-footed mice and ground moles.
 - ☐ Their numbers cannot usually be controlled by native predators and parasites.
 - ☐ They do not pose as many problems for humans as do white-footed mice and ground moles.

Paragraph 1 is marked with an arrow [→]

Paragraph 3

However, chemical control has its dark side. Chemical pesticides have many unintended consequences through their effects not just on the target species but on a wide array of

nontarget species as well, often eliminating them and thereby upsetting the existing food webs, especially through the suppression of native predator species. The surviving pests then rebound in greater numbers than ever.

3. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Chemical pesticides often eliminate species other than the intended target and thereby upset food webs, especially by suppressing native predator species.
 - ☐ Native predator species are often eliminated by chemical pesticides that are intended to have consequences for other pests.
 - ☐ Chemical pesticides upset existing food webs by eliminating native species and by increasing the number of nonnative predators.
 - ☐ The effects of chemical pesticides on a wide array of food webs and native predators are often unintended.

Paragraph 5

Farmer long ago observed that enemies of pests act as controls. As early as 300 C.E., the Chinese were introducing predatory ants into their citrus orchards to control leaf-eating caterpillars. Insect pests have their own array of enemies in their native habitats. When an animal or plant is introduced, intentionally or unintentionally, into a new habitat outside of its natural range, it may adapt to the new environment and leave its enemies behind. Freed from predation and finding abundance of resources, the species quickly becomes a pest or a weed. This fact had led to the search for natural enemies to introduce into populations of pests to reduce their populations.

4. In paragraph 5, the author mentions the Chinese use of predatory ants to control pests in order to
- ☐ support the claim that using pests' natural enemies is a pest control technique that has been known for a long time
 - ☐ show that pests' enemies introduced unintentionally have proved more dangerous than those introduced intentionally
 - ☐ help explain that when pests' enemies find enough resources, they can become pests themselves
 - ☐ argue that a pest insect in its native habitat always has a predator in that habitat

Paragraph 5 is marked with an arrow [→]

5. According to paragraph 5, why is a species likely to become a pest when it is introduced into a new habitat?
- ☐ The species becomes more effective at escaping from its enemies.
 - ☐ The species has no natural predators in its new habitat.
 - ☐ The species adapts to habitats outside its natural range.
 - ☐ The species does not have to compete for resources with other plants and animals.

Paragraph 5 is marked with an arrow [→]

Paragraph 7

But biological control, like chemical control, can backfire. The success of the cactus-feeding moth in controlling prickly pear in Australia encouraged its introduction to several West Indies islands to control prickly pear there. In time the moth made its way to Florida, where it now threatens the existence of several native prickly pear species. The **moral** is that although using nonindigenous predators as biological controls can be effective, these species possess their own inherent dangers that must be **assessed** before they are released. They, too, can become alien invaders.

6. The word "**assessed**" in the passage is closest in meaning to

- ☐ minimized
- ☐ identified
- ☐ evaluated
- ☐ dealt with

7. The word "**moral**" in the passage is closest in meaning to

- ☐ situation
- ☐ argument
- ☐ fact
- ☐ lesson

Paragraph 6

Because the serious pest is usually a nonnative species, biological control involves the introduction of a nonindigenous predator or parasite to control the pest. The introduction of the cactus-eating moth, a native of Argentina, into Australia effectively reduced and controlled the rapidly spreading prickly pear, which had been introduced into Australia in 1901.

8. In paragraph 6, the discussion of the cactus-eating moth and the prickly pear in Australia illustrates which of the following about biological control?

- ☐ Nonnative pests cannot be controlled through biological means once they have begun to spread rapidly.
- ☐ A nonnative pest can sometimes be controlled by the introduction of a nonnative predator.
- ☐ A nonindigenous pest can be controlled only by a predator that comes from the same original habitat as the pest.
- ☐ A native pest can be controlled by either a native or a nonnative predator..

Paragraph 6 is marked with an arrow [→]

Paragraph 7

But biological control, like chemical control, can backfire. The success of the cactus-feeding moth in controlling prickly pear in Australia encouraged its introduction to several West Indies islands to control prickly pear there. In time the moth made its way to **Florida**, where it now threatens the existence of several native prickly pear species. The **moral** is that although using nonindigenous predators as biological controls can be effective, these species possess their own inherent dangers that must be assessed before they are released. They, too, can become alien

invaders.

9. The author discusses the cactus-feeding moth in **Florida** in order to
- ☐ explain why the prickly pear species that are native to Florida have no indigenous predators
 - ☐ show how a predator spreads more rapidly in alien environments than it does in its native environment
 - ☐ indicate that a single nonindigenous predator species can be effective against a wide array of nonindigenous pest species
 - ☐ argue that controlling pests with nonindigenous predators can have unintended consequences

Paragraph 8

Because chemical, biological, and other methods used individually are obviously not the solution to pest control, entomologists have developed a holistic approach to pest control, called integrated pest management (IPM). IPM considers the biological, ecological, economic, social, and even aesthetic aspects of pest control and employs a variety of techniques. The objective of IPM is to control the pest not at the time a major outbreak but at an earlier time, when the size of the population is easier to control. The approach is to rely first on natural mortality caused by weather and natural enemies, with as little disruption of the natural system as possible, and to use other methods only if they are needed to hold the pest below the economic injury level.

10. According to paragraph 8, each of the following is a principle of integrated pest management EXCEPT
- ☐ to control pest populations before a major outbreak occurs
 - ☐ to first determine if weather and natural enemies are able to control a pest
 - ☐ to increase the populations of the pest's natural enemies during certain seasons of the year
 - ☐ to use artificial methods of pest control only when pests begin to cause economic injury

Paragraph 8 is marked with an arrow [→]

Paragraph 9

Successful IPM requires the knowledge of the population ecology of each pest and its associated species and the dynamics of the host species. It involves **considerable** field work monitoring the pest species and its natural enemies by such techniques as egg counts and the trapping of adults to acquire information to determine the necessity, timing, and intensity of control measures. These control measures must be adjusted to the situation, which may vary from one location to another. The intensity of control or no control is based on the degree of pest damage that can be tolerated, the costs of control, and the benefits to be derived.

11. The word "**considerable**" in the passage is closest in meaning to
- ☐ tiring
 - ☐ a large amount of
 - ☐ complicated
 - ☐ carefully planned

12. According to paragraph 9, each of the following helps to determine how intensely to apply pest control measures EXCEPT

- ☐ how much pest damage can be tolerated
- ☐ the cost of pest control measures
- ☐ what can be gained through pest control measures
- ☐ whether pest control measures have been used before

Paragraph 9 is marked with an arrow [→]

Paragraph 4

Perhaps more insidious is that a pesticide loses its effectiveness because the target species evolves resistance to it. As one pesticide replaces another, the pests acquire a resistance to them all. ■ Some species, notably certain mosquitoes, have overcome the toxic effects of every pesticide to which they have been exposed. ■ Insect pests need not only about five years to evolve pesticide resistance, their predators do so much more slowly. ■ So after the pest develops resistance, pest outbreaks become even more disastrous. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

And the damage will continue until a new pesticide is developed, at which time the cycle will begin anew.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Pest control measures vary in their approach and overall degree of success.

-
-
-

Answer Choices

Biological methods of pest control were introduced by the ancient Sumerians, and chemical control was first used in ancient China.

Biological control, for example, the use of natural enemies of pests, has been effective at regulating nonnative pests, though it can also threaten the existence of native species.

Integrated pest management is a holistic approach that has been successful at controlling major pest outbreaks in locations where chemical and biological control have already failed.

Pesticides are limited in their usefulness because pests quickly become resistant to them, and because they can harm species for which they were not intended.

The success of biological and chemical approaches to pest control has been difficult to measure because situations vary significantly from one location to another.

Integrated pest management, an approach that consider biological, ecological, economic, and aesthetic aspects of pest control, uses a variety of techniques adjusted to specific situations.

Passage 40 2014-06-15-1

Early life-forms and Earth's Atmosphere

Why has life flourished on Earth? This question has a two-part answer. First, Earth has been a cradle for life because of its position relative to the Sun. second, once life began on Earth, simple early life-forms (photosynthetic bacteria) slowly but inexorably altered the environment in a manner that not only maintained life but also paved the way for later, complex life-forms. These changes allowed later organisms to evolve and thrive. Humans and other higher organisms owe their life-supporting environment to these early life-forms.

Earth's earliest atmosphere contained several gases: hydrogen, water vapor, ammonia, nitrogen, methane, and carbon dioxide, but no oxygen. Gas mixtures emitted from present-day volcanoes resemble this early atmosphere, suggesting its origin from volcanic eruptions. In Earth's earliest atmosphere, methane and carbon dioxide occurred at much higher levels than at present—a circumstance that was favorable for early life. Methane and carbon dioxide are greenhouse gases that warm atmospheres by retarding loss of heat to space. These two gases kept Earth warm during the Sun's early history, when the Sun did not burn as brightly as it now does. (An early dim period, with later brightening, is normal for stars of our Sun's type.)

Earth's modern atmosphere, which is 78 percent nitrogen gas, 21 percent oxygen, and about 1 percent argon, water vapor, ozone, and carbon dioxide, differs dramatically from the earliest atmosphere just described. The modern atmosphere supports many forms of complex life that would not have been able to exist in Earth's first atmosphere because the oxygen level was too low. Also, if atmospheric methane and carbon dioxide were as abundant now as they were in Earth's earliest atmosphere, the planet's temperature would likely be too hot for most species living today. How and when did the atmosphere change?

The answer to this riddle lies in the metabolic activity of early photosynthetic life-forms that slowly but surely transformed the chemical composition of Earth's atmosphere. Some of these early organisms were photosynthetic relatives of modern cyanobacteria (blue-green bacteria). In the process of photosynthesis, carbon dioxide gas combined with water yields oxygen. In Earth's early days, all over the planet countless photosynthetic bacteria performed photosynthesis. Together, these ancient bacteria removed massive amounts of carbon dioxide from Earth's atmosphere by converting it to solid organic carbon. These ancient bacteria also released huge quantities of oxygen into the atmosphere. Other ancient bacteria consumed methane, greatly reducing its amount in the atmosphere. When our Sun later became hotter, the continued removal of atmospheric carbon dioxide and methane by early bacteria kept Earth's climate from becoming too hot to sustain life. Modern cyanobacteria still provide these valuable services today.

The bacterial oxygen release improved conditions for life in two ways. First, oxygen is essential for

the metabolic process known as cell respiration that allows cells to efficiently harvest energy from organic food. Second, oxygen in the upper atmosphere reacts to form a protective shield of ozone. Earth is constantly bombarded by harmful ultraviolet (UV) radiation from the Sun. Today, Earth's upper-atmosphere ozone shield absorbs enough UV to allow diverse forms of life to survive. But because early Earth lacked oxygen in its atmosphere, it also lacked a protective ozone barrier. As a result, early life on Earth was confined to the oceans, where the water absorbed the UV radiation. Only after oxygen released by ancient bacteria drifted up into the upper atmosphere and reacted with other oxygen molecules to form a protective layer of ozone could life flourish at the surface and on the land. The absence of an oxygen atmosphere on Mars and other planets in our solar system means that these planets also lack an ozone shield that would protect surface-dwelling life from UV radiation. The surface of Mars is bombarded with deadly radiation; if any life exists on Mars, it would almost certainly be subterranean.

Paragraph 1

Why has life flourished on Earth? This question has a two-part answer. First, Earth has been a cradle for life because of its position relative to the Sun. second, once life began on Earth, simple early life-forms (photosynthetic bacteria) slowly but inexorably **altered** the environment in a manner that not only maintained life but also paved the way for later, complex life-forms. These changes allowed later organisms to evolve and thrive. Humans and other higher organisms owe their life-supporting environment to these early life-forms.

1. The word "altered" in the passage is closest in meaning to
 - ☐ transformed
 - ☐ protected
 - ☐ made use of
 - ☐ adapted to
2. Paragraph 1 supports which of the following inferences?
 - ☐ After complex life forms arose on Earth, bacteria quickly evolved in order to maintain favorable conditions for life.
 - ☐ Early life forms on Earth arose from the process of photosynthesis.
 - ☐ The development of complex life on Earth depended on the presence of bacteria that could be consumed by larger organisms.
 - ☐ Complex life forms on Earth may not have evolved if Earth had been farther from or closer to the Sun.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

Earth's earliest atmosphere contained several gases: hydrogen, water vapor, ammonia, nitrogen, methane, and carbon dioxide, but no oxygen. Gas mixtures **emitted** from present-day volcanoes resemble this early atmosphere, suggesting its origin from volcanic eruptions. In Earth's earliest atmosphere, methane and carbon dioxide occurred at much higher levels than at present—a circumstance that was favorable for early life. Methane and carbon dioxide are greenhouse gases that warm atmospheres by **retarding** loss of heat to space. These two gases kept Earth warm

during the Sun's early history, when the Sun did not burn as brightly as it now does. (An early dim period, with later brightening, is normal for stars of our Sun's type.)

3. The word "emitted" in the passage is closest in meaning to
 - ☐ disappearing
 - ☐ examined
 - ☐ released
 - ☐ taken
 4. The word "retarding" in the passage is closest in meaning to
 - ☐ regulating
 - ☐ slowing
 - ☐ restoring
 - ☐ directing
 5. In paragraph 2, why does the author provide the information that methane and carbon dioxide kept the Earth warm during the Sun's early history?
 - ☐ To explain how the early Earth and the early Sun were related
 - ☐ To support the claim that methane and carbon dioxide are greenhouse gases
 - ☐ To explain why the high levels of methane and carbon dioxide in Earth's early atmosphere were favorable for early life
 - ☐ To suggest that these gases affect how brightly the Sun burns
- Paragraph 2 is marked with an arrow [→]
6. Paragraph 2 supports which of the following statements about methane and carbon dioxide present in Earth's earliest atmosphere?
 - ☐ They slowed down the loss of heat from Earth's atmosphere.
 - ☐ They caused the sunlight to be less bright than it currently is.
 - ☐ They occurred in smaller amounts than they currently do.
 - ☐ They prevented the development of early life-forms.

Paragraph 2 is marked with an arrow [→]

Paragraph 3

Earth's modern atmosphere, which is 78 percent nitrogen gas, 21 percent oxygen, and about 1 percent argon, water vapor, ozone, and carbon dioxide, differs dramatically from the earliest atmosphere just described. The modern atmosphere supports many forms of complex life that would not have been able to exist in Earth's first atmosphere because the oxygen level was too low. Also, if atmospheric methane and carbon dioxide were as abundant now as they were in Earth's earliest atmosphere, the planet's temperature would likely be too hot for most species living today. How and when did the atmosphere change?

7. The word "abundant" in the passage is closest in meaning to
 - ☐ pure
 - ☐ balanced

- ☐ plentiful
- ☐ warm

8. According to paragraph 2 and 3, which of the following is one way in which Earth's early atmosphere differed from its current atmosphere?

- ☐ It had fewer greenhouse gases.
- ☐ It was warmer.
- ☐ It had lower levels of carbon dioxide.
- ☐ It did not contain methane.

Paragraphs 2 and 3 are marked with arrows [→]

Paragraph 4

The answer to this riddle lies in the metabolic activity of early photosynthetic life-forms that slowly but surely transformed the chemical composition of Earth's atmosphere. Some of these early organisms were photosynthetic relatives of modern cyanobacteria (blue-green bacteria). In the process of photosynthesis, carbon dioxide gas combined with water yields oxygen. In Earth's early days, all over the planet countless photosynthetic bacteria performed photosynthesis. Together, these ancient bacteria removed massive amounts of carbon dioxide from Earth's atmosphere by converting it to solid organic carbon. These ancient bacteria also released huge quantities of oxygen into the atmosphere. Other ancient bacteria consumed methane, greatly reducing its amount in the atmosphere. When our Sun later became hotter, the continued removal of atmospheric carbon dioxide and methane by early bacteria kept Earth's climate from becoming too hot to sustain life. Modern cyanobacteria still provide these valuable services today.

9. The word "sustain" in the passage is closest in meaning to

- ☐ change
- ☐ generate
- ☐ destroy
- ☐ support

10. According to paragraph 4, ancient bacteria changed the chemical composition of Earth's atmosphere by performing all of the following activities EXCEPT

- ☐ raising the temperature of the atmosphere
- ☐ removing methane and carbon dioxide
- ☐ creating organic carbon
- ☐ producing oxygen

Paragraph 4 is marked with an arrow [→]

Paragraph 5

The bacterial oxygen release improved conditions for life in two ways. First, oxygen is essential for the metabolic process known as cell respiration that allows cells to efficiently harvest energy from organic food. Second, oxygen in the upper atmosphere reacts to form a protective shield of ozone. Earth is constantly bombarded by harmful ultraviolet (UV) radiation from the Sun. Today,

Earth's upper-atmosphere ozone shield absorbs enough UV to allow diverse forms of life to survive. But because early Earth lacked oxygen in its atmosphere, it also lacked a protective ozone barrier. As a result, early life on Earth was confined to the oceans, where the water absorbed the UV radiation. Only after oxygen released by ancient bacteria drifted up into the upper atmosphere and reacted with other oxygen molecules to form a protective layer of ozone could life flourish at the surface and on the land. The absence of an oxygen atmosphere on Mars and other planets in our solar system means that these planets also lack an ozone shield that would protect surface-dwelling life from UV radiation. The surface of Mars is bombarded with deadly radiation; if any life exists on Mars, it would almost certainly be subterranean.

11. It can be inferred from paragraph 5 that early life-forms on Earth were confined to the oceans because

- ☐ the thick ozone layer at the time would have made it difficult for them to survive on land
- ☐ water was the only available protection they had against ultraviolet radiation
- ☐ land provided them with only limited amounts of water needed for survival
- ☐ their metabolic systems were inefficient

Paragraph 5 is marked with an arrow [→]

12. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Any life forms that may have existed on other planets probably were destroyed by UV radiation.
- ☐ Other planets in our solar system lack the oxygen atmosphere that helps explain why life exists on Earth.
- ☐ The absence of oxygen on other planets means that those planets lack an ozone shield to protect life forms against UV radiation.
- ☐ Life forms cannot survive UV radiation without the protection of an ozone shield.

Paragraph 5

The bacterial oxygen release improved conditions for life in two ways. First, oxygen is essential for the metabolic process known as cell respiration that allows cells to efficiently harvest energy from organic food. Second, oxygen in the upper atmosphere reacts to form a protective shield of ozone. ■ Earth is constantly bombarded by harmful ultraviolet (UV) radiation from the Sun. ■ Today, Earth's upper-atmosphere ozone shield absorbs enough UV to allow diverse forms of life to survive. ■ But because early Earth lacked oxygen in its atmosphere, it also lacked a protective ozone barrier. ■ As a result, early life on Earth was confined to the oceans, where the water absorbed the UV radiation. Only after oxygen released by ancient bacteria drifted up into the upper atmosphere and reacted with other oxygen molecules to form a protective layer of ozone could life flourish at the surface and on the land. The absence of an oxygen atmosphere on Mars and other planets in our solar system means that these planets also lack an ozone shield that would protect surface-dwelling life from UV radiation. The surface of Mars is bombarded with deadly radiation; if any life exists on Mars, it would almost certainly be subterranean.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.
But protection against what?

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Earth’s earliest atmosphere was transformed in ways that allowed the development of complex life forms.

Answer Choices

Volcanoes changed Earth’s earliest atmosphere in ways that allowed life to develop, and bacteria that used oxygen produced by photosynthesis further altered the atmosphere to what we find today.

When the sun became brighter, early bacteria removed methane and carbon dioxide from the atmosphere, preventing Earth from becoming too hot for life to survive.

Early bacteria provided the oxygen that was needed to support the metabolism of complex life forms and to form an ozone shield against deadly UV radiation.

Volcanic carbon dioxide in Earth’s early atmosphere kept Earth warm enough for life to begin during the time when the Sun was too dim to provide much warmth.

Earth has been able to support life because its position relative to the Sun provided it with enough heat, but not too much heat for early bacteria to evolve.

Complex life evolved on the Earth’s surface, but not on Mars or other planets in the solar system because on those planets, early surface life was killed by UV radiation.

Constraints on Natural Selection

Natural selection is the process in which organisms with certain traits survive and reproduce while organisms that are less able to adapt to their environment die off. As Darwin pointed out, natural selection does not necessarily produce evolutionary progress, much less perfection. The limits to the effectiveness of natural selection are most clearly revealed by the universality of extinction. More than 99.9 percent of all evolutionary lines that once existed on Earth have become extinct. Mass extinctions remind us forcefully that evolution is not a steady approach to an ever-higher perfection but an unpredictable process in which the best-adapted organisms may be suddenly exterminated by a catastrophe and their place taken by lineages that prior to the catastrophe seemed to be without distinction or prospects.

There are numerous constraints, or limits, on the power of natural selection to bring about change. First, the genetic variation needed to perfect a characteristic may not be forthcoming. Second, during evolution, the adoption of one among several possible solutions to a new environmental opportunity may greatly restrict the possibilities for subsequent evolution. For instance, when a selective advantage for a skeleton developed among the ancestors of the vertebrates and the arthropods, the ancestors of the arthropods had the prerequisites for developing an external skeleton, and those of the vertebrates had the prerequisites for acquiring an internal skeleton. The entire subsequent history of these two large groups of organisms was affected by the two different paths taken by their remote ancestors. The vertebrates were able to develop such huge creatures as dinosaurs, elephants, and whales. A large crab is the largest type that the arthropods were able to achieve.

Another constraint on natural selection is developmental interaction. The different components of an individual organism—its structures and organs—are not independent of one another, and none of them responds to selection without interacting with the others. The whole developmental machinery is a single interacting system. Organisms are compromises among competing demands. How far a particular structure or organ can respond to the forces of selection depends, to a considerable extent, on the resistance offered by other structures and organs, as well as components of the genotype (the totality of an individual's genes).

The structure of the genotype itself imposes limits on the power of natural selection. The classical metaphor of the genotype was that of a beaded string on which the genes were lined up like pearls in a necklace. According to this view, each gene was more or less independent of the others. Not much is left of this previously accepted image. It is now known that there are different functional classes of genes, some charged to produce material, others to regulate it, and still others that are apparently not functioning at all. There are single coding genes, moderately repetitive DNA, highly repetitive DNA, and many other kinds of DNA. Discovering exactly how they all interact with one another is still a rather poorly understood area of genetics.

A further constraint on natural selection is the capacity for nongenetic modification. The more plastic the organism's body characteristics are (owing to developmental flexibility), the more this

reduces the force of adverse selection pressures. Plants, and particularly microorganisms, have a far greater capacity for individual modification than do animals. Natural selection is involved even in this phenomenon, since the capacity for nongenetic adaptation is under strict genetic control. When a population shifts to a new specialized environment, genes will be selected during the following generations that reinforce and may eventually largely replace the capacity for nongenetic adaptation.

Finally, which organisms survive and reproduce in a population is partly the result of chance, and this also limits the power of natural selection. Chance operates at every level of the process of reproduction, from the transmission of parental chromosomes to the survival of the newly formed individual. Furthermore, potentially favorable gene combinations are often destroyed by indiscriminate environmental forces such as storms, floods, earthquakes, or volcanic eruptions, without natural selection being given the opportunity to favor these genotypes. Yet over time, in the survival of those few individuals that become the ancestors of subsequent generations, relative fitness always plays a major role.

Paragraph 1

Natural selection is the process in which organisms with certain traits survive and reproduce while organisms that are less able to adapt to their environment die off. As Darwin pointed out, natural selection does not necessarily produce evolutionary progress, much less perfection. The limits to the effectiveness of natural selection are most clearly revealed by the universality of extinction. More than 99.9 percent of all evolutionary lines that once existed on Earth have become extinct. Mass extinctions remind us forcefully that evolution is not a steady approach to an ever-higher perfection but an unpredictable process in which the best-adapted organisms may be suddenly exterminated by a catastrophe and their place taken by lineages that prior to the catastrophe seemed to be without distinction or prospects.

1. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - ☐ Evolution is an unpredictable process because in mass extinctions highly-evolved organisms are exterminated.
 - ☐ Evolution does not progress steadily to ever-higher levels of perfection because, as shown by mass extinctions, lineages favored by evolution can be suddenly replaced by those not favored previously when circumstances change.
 - ☐ Catastrophes remind us that evolution is a process in which the best-adapted organisms are exterminated and their place taken by lineages shown to be poorly adapted.
 - ☐ When mass extinctions exterminate the best-adapted organisms, less important lineages suddenly become better adapted and take their place.
2. According to paragraph 1, which of the following provides evidence that natural selection does not always lead to evolutionary progress?
 - ☐ Most evolutionary lines that once existed on Earth have become extinct.
 - ☐ Evolutionary lines usually weaken as they increase in age.

- The history of evolution shows that many evolutionary lines can become extinct at the same time.
- So far, less than one percent of evolutionary lines have achieved such perfect adaptation to their environment that they will never become extinct.

Paragraph 1 is marked with an arrow [→]

Paragraph 2

There are numerous constraints, or limits, on the power of natural selection to bring about change. First, the genetic variation needed to perfect a characteristic may not be forthcoming. Second, during evolution, the adoption of one among several possible solutions to a new environmental opportunity may greatly restrict the possibilities for subsequent evolution. For instance, when a selective advantage for a skeleton developed among the ancestors of the vertebrates and the arthropods, the ancestors of the arthropods had the prerequisites for developing an external skeleton, and those of the vertebrates had the prerequisites for acquiring an internal skeleton. The entire subsequent history of these two large groups of organisms was affected by the two different paths taken by their **remote** ancestors. The vertebrates were able to develop such huge creatures as dinosaurs, elephants, and whales. A large crab is the largest type that the arthropods were able to achieve.

3. The word "**remote**" in the passage is closest in meaning to
 - corresponding
 - distant
 - separate
 - direct
4. In paragraph 2, why does the author discuss the ancestors of vertebrates and arthropods?
 - To explain how a single feature can cause the extinction of entire groups of organisms
 - To identify some factors that determine how large an organism can become
 - To illustrate the point that earlier developments influence the possibilities for future development
 - To emphasize the role of the environment in the development of organisms

Paragraph 2 is marked with an arrow [→]

5. Which of the following can be inferred from paragraph 2 about arthropods?
 - There are fewer of them than there are of vertebrates.
 - Their ancestors had a selective advantage over the ancestors of vertebrates.
 - Their ancestors once possessed internal skeletons.
 - Their body features prevent them from becoming large organisms.

Paragraph 2 is marked with an arrow [→]

Paragraph 3

Another constraint on natural selection is developmental interaction. The different components of an individual organism—its structures and organs—are not independent of one another, and none of them responds to selection without interacting with the others. The whole

developmental machinery is a single interacting system. Organisms are compromises among competing demands. How far a particular structure or organ can respond to the forces of selection depends, to a considerable extent, on the **resistance** offered by other structures and organs, as well as components of the genotype (the totality of an individual's genes).

6. The word "**resistance**" in the passage is closest in meaning to
- ☐ opposition
 - ☐ pressure
 - ☐ interaction
 - ☐ competition
7. According to paragraph 3, why must organisms compromise between competing demands?
- ☐ A particular organ or structure may be unable to respond to selection pressures due to the needs of other parts of the organism.
 - ☐ An organism's ability to respond to the forces of selection depends on the demands of other organisms within its environment.
 - ☐ An organism's environment and its genotype try at the same time to influence its ability to respond to natural selection.
 - ☐ Different elements of the environment call for adaptations that are often incompatible with one another.

Paragraph 3 is marked with an arrow [→]

Paragraph 4

The structure of the genotype itself imposes limits on the power of natural selection. The classical metaphor of the genotype was that of a beaded string on which the genes were lined up like pearls in a necklace. According to this view, each gene was more or less independent of the others. Not much is left of this previously accepted image. It is now known that there are different functional classes of genes, some charged to produce material, others to regulate it, and still others that are apparently not functioning at all. There are single coding genes, moderately repetitive DNA, highly repetitive DNA, and many other kinds of DNA. Discovering exactly how they all interact with one another is still a rather poorly understood area of genetics.

8. Paragraph 4 supports all of the statements about genes EXCEPT:
- ☐ Our understanding of the extent to which genes act independently has changed over time.
 - ☐ Genes are classified into groups on the basis of their function.
 - ☐ Some genes seem to have no function.
 - ☐ Studies to identify ways that genes interact have been largely successful.

Paragraph 4 is marked with an arrow [→]

Paragraph 5

A further constraint on natural selection is the capacity for nongenetic modification. The more plastic the organism's body characteristics are (owing to developmental flexibility), the more this reduces the force of **adverse** selection pressures. Plants, and particularly microorganisms, have a far greater capacity for individual modification than do animals. Natural selection is involved even

in this phenomenon, since the capacity for nongenetic adaptation is under strict genetic control. When a population shifts to a new specialized environment, genes will be selected during the following generations that reinforce and may eventually largely replace the capacity for nongenetic adaptation.

9. The word "adverse" in the passage is closest in meaning to

- ☐ harmful
- ☐ continuous
- ☐ strong
- ☐ excessive

Paragraph 6

Finally, which organisms survive and reproduce in a population is partly the result of chance, and this also limits the power of natural selection. Chance operates at every level of the process of reproduction, from the transmission of parental chromosomes to the survival of the newly formed individual. Furthermore, potentially favorable gene combinations are often destroyed by indiscriminate environmental forces such as storms, floods, earthquakes, or volcanic eruptions, without natural selection being given the opportunity to favor these genotypes. Yet over time, in the survival of those few individuals that become the ancestors of subsequent generations, relative fitness always plays a major role.

10. What point does paragraph 6 make about the individuals that survive a particular natural disaster?

- ☐ They were the small number of organisms that happened to be well-adapted to survive that kind of natural disaster.
- ☐ Their descendants will likely be shaped by natural selection to evolve genes that will make them fit to survive the next natural disaster.
- ☐ Their survival may have been due to chance, but fitness also contributes to their survival over time.
- ☐ They will probably have comparatively weak offspring because the natural disaster eliminated the individuals that had more favorable gene combinations.

Paragraph 6 is marked with an arrow [→]

11. According to paragraph 6, environmental forces limit the power of natural selection in which of the following ways?

- ☐ They change the reproductive process in significant ways.
- ☐ They destroy potentially favorable gene combinations before they can be selected.
- ☐ They interfere with the transmission of chromosomes from parent to offspring.
- ☐ They weaken the ability of individuals to maintain high fitness levels.

Paragraph 6 is marked with an arrow [→]

12. The word "indiscriminate" in the passage is closest in meaning to

- ☐ dangerous
- ☐ random

- ☐ common
- ☐ powerful

Paragraph 4

The structure of the genotype itself imposes limits on the power of natural selection. The classical metaphor of the genotype was that of a beaded string on which the genes were lined up like pearls in a necklace. ■ According to this view, each gene was more or less independent of the others. ■ Not much is left of this previously accepted image. ■ It is now known that there are different functional classes of genes, some charged to produce material, others to regulate it, and still others that are apparently not functioning at all. ■ There are single coding genes, moderately repetitive DNA, highly repetitive DNA, and many other kinds of DNA. Discovering exactly how they all interact with one another is still a rather poorly understood area of genetics.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

New models of the genotype depict a much more complicated relationship among genes.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Genetics and environmental factors often prevent organisms from achieving evolutionary perfection.

-
-
-

Answer Choices

Darwin first challenged the idea of evolutionary progress by observing that more than 99.9 percent of all evolutionary lines that once existed on Earth have become extinct.

A single adaptation to an organism's environment may determine the way in which the organism's subsequent ancestors are able to evolve.

The structure of the genotype itself restricts natural selection, since genes must line up like pearls on a necklace and cannot be moved out of their proper order.

The development of vertebrates and arthropods from a single ancestor demonstrates the power of the environment to limit natural selection.

A single structure or gene in an organism cannot respond to the forces of natural selection without affecting the functioning of other structures or genes.

Both environmental and genetic chance may prevent even the fittest organisms from surviving and reproducing.



2014-06-21

Passage 42 2014-06-15-3

Climate Change and the Natufian People

The so-called Natufian culture inhabited what is now the Middle East between approximately 14,000 and 11,500 years ago. This period is commonly split into two subperiods, Early Natufian (14,000 to 13,000 years ago) and Late Natufian (13,000 to 11,500). The Natufians were hunter-gatherers who relied primarily on gazelle, although they also cultivated some cereal grains. During the early period at least, they lived year-round in villages in built stone houses. Like all human beings, their way of life depended on the climate. Around 13,000 years ago, their climate began to change, becoming colder and drier, a period known as the Younger Dryas.

We know that times were hard in the increasingly arid landscapes of the Younger Dryas, but quite how hard remains unclear. The droughts certainly caused many ponds and rivers to disappear completely and the larger lakes to shrink in size. The people who lived in the south, in today's deserts of the Negev and the Sinai, were most likely hit the hardest. They returned to a completely transient hunter-gatherer way of life, moving from place to place. Survival required improved hunting weapons: game (animals hunted for food) had become scarce, and consequently, success had become essential when a kill was possible. And so we see the invention of the Harif point, a new kind of arrowhead.

Further north, the impact of the Younger Dryas may have been less severe. Yet survival still required more than just a return to the ancient mobile hunter-gatherer lifestyle, especially as there were now many more people needing food than had been the case during earlier periods, when the Natufians lived in permanent dwellings. One response was to hunt a much wider range of animals than before, and hence we find in Late Natufian settlements the bones of many small-game species as well as larger, ever-present gazelle.

Another response to the changing climate was to continue, and perhaps expand the cultivation of plants. Wild cereals were particularly hard hit by the Younger Dryas owing to a decrease in the concentration of carbon dioxide (CO₂) in the atmosphere. This diminution, carefully documented from air bubbles trapped in Antarctic ice, inhibited their photosynthesis and markedly reduced their yields. Consequently, whatever cultivation practices had begun during the Early Natufian period—weeding, transplanting, watering, pest control—may now have become essential to secure sufficient food. And these may have created the first domesticated strains.

This appears to be what happened at the village of Abu Hureyra just before its abandonment. When the archaeologist Gordon Hillman studied the cereal grains from the site, he found a few grains of rye from plants that had undergone the transition into domestic forms. When dated, they were shown to lie between 11,000 and 10,500 B.C.—the oldest domesticated cereal grain from anywhere in the world. Along with these grains, Hillman found seeds from the weeds that typically grow in cultivated soil. And so it appears that, as the availability of wild plant foods declined due to the onset of the Younger Dryas, the Abu Hureyra people invested an ever greater amount of time and effort in caring for the wild rye and by doing so unintentionally transformed

it into a domestic crop. But even this could not support the village—it was abandoned as people were forced to return to a mobile lifestyle, perhaps carrying pouches of cereal grain. The domesticated rye of Abu Hureyra reverted to its wild state.

The geographical range of the Late Natufians also changed. With their increased interest in plant cultivation, the Late Natufians drifted away from the depleted woodlands where their forebears once flourished. They were drawn to the alluvial soils (soils deposited by rivers) of the valleys, not only those of the River Jordan, but also those found by the great rivers of the Mesopotamian plain and in the vicinity of lakes and rivers throughout the Middle East. Large expanses of these rich, fertile soils became available as the rivers and lakes struck during the Younger Dryas Wild, but cultivated, cereals grew well in such soil, especially when close to the meager springs, ponds, and streams that survived the arid conditions.

Paragraph 2

We know that times were hard in the increasingly arid landscapes of the Younger Dryas, but quite how hard remains unclear. The droughts certainly caused many ponds and rivers to disappear completely and the larger lakes to shrink in size. The people who lived in the south, in today's deserts of the Negev and the Sinai, were most likely hit the hardest. They returned to a completely transient hunter-gatherer way of life, moving from place to place. Survival required improved hunting weapons: game (animals hunted for food) had become scarce, and consequently, success had become essential when a kill was possible. And so we see the invention of the Harif point, a new kind of arrowhead.

1. Which of the following can be inferred from paragraph 2 about why game became scarce during the Younger Dryas?
 - ☐ The development of new weapons in the south improved the Natufians' hunting successes.
 - ☐ Human settlement in the south destroyed the resources that allowed game animals to survive.
 - ☐ Animals died or moved away as their sources of drinking water dried up.
 - ☐ Animals were scared away by the Natufians' rapid movement from place to place.

Paragraph 2 is marked with an arrow [→]

2. In paragraph 2, why does the author mention the Harif point?
 - ☐ To support the idea that the Younger Dryas required Natufians to hunt in new ways
 - ☐ To illustrate how humans had begun to protect their settlements during the Younger Dryas
 - ☐ To contrast the settled lifestyle of villages with the transient lifestyle of hunter-gathers
 - ☐ To explain the process by which ancient people invented arrowheads

Paragraph 3

Further north, the impact of the Younger Dryas may have been less severe. Yet survival still required more than just a return to the ancient mobile hunter-gatherer lifestyle, especially as there were now many more people needing food than had been the case during earlier periods, when the Natufians lived in permanent dwellings. One response was to hunt a much wider range of animals than before, and hence we find in Late Natufian settlements the bones of many

small-game species as well as larger, ever-present gazelle.

3. According to paragraph 3, what did the Natufians do to feed their increasing population during the Younger Dryas?

- ☐ They began to trade food products with other people in neighboring areas.
- ☐ They began hunting a wider range of animals.
- ☐ They tried to hunt mostly the larger animals.
- ☐ They moved south where there were more animals.

Paragraph 3 is marked with an arrow [→]

Paragraph 4

Another response to the changing climate was to continue, and perhaps expand the cultivation of plants. Wild cereals were particularly hard hit by the Younger Dryas owing to a decrease in the concentration of carbon dioxide (CO₂) in the atmosphere. This diminution, carefully documented from air bubbles trapped in Antarctic ice, **inhibited** their photosynthesis and markedly reduced their yields. Consequently, whatever cultivation practices had begun during the Early Natufian period—weeding, transplanting, watering, pest control—may now have become essential to secure **sufficient** food. And these may have created the first domesticated strains.

4. According to paragraph 4, air bubbles in Antarctic ice are evidence of which of the following during the Younger Dryas period?

- ☐ A reduction in atmospheric carbon dioxide
- ☐ An increase in the number of weeds
- ☐ Improvements in cultivation of plants
- ☐ Greater use of pest controls

Paragraph 4 is marked with an arrow [→]

5. The word “**inhibited**” in the passage is closest in meaning to

- ☐ altered
- ☐ required
- ☐ restricted
- ☐ allowed

6. The word “**sufficient**” in the passage is closest in meaning to

- ☐ enough
- ☐ fresh
- ☐ valuable
- ☐ nutritious

Paragraph 5

This appears to be what happened at the village of Abu Hureyra just before its abandonment. When the archaeologist Gordon Hillman studied the cereal grains from the site, he found a few grains of rye from plants that had undergone the transition into domestic forms. When dated, they were shown to lie between 11,000 and 10,500 B.C.—the oldest domesticated cereal grain

from anywhere in the world. Along with these grains, Hillman found seeds from the weeds that typically grow in cultivated soil. And so it appears that, as the availability of wild plant foods declined due to the onset of the Younger Dryas, the Abu Hureyra people invested an ever greater amount of time and effort in caring for the wild rye and by doing so unintentionally transformed it into a domestic crop. But even this could not support the village—it was abandoned as people were forced to return to a mobile lifestyle, perhaps carrying pouches of cereal grain. The domesticated rye of Abu Hureyra reverted to its wild state.

7. In paragraph 5, why does the author described the archaeologist's findings at Abu Hureyra?
- ☐ To identify the best-known Natufian village
 - ☐ To question the idea that wild grains were first domesticated more than 11,000 years ago
 - ☐ To challenge the idea presented in paragraph 4 that earlier cultivation practices became essential for Natufian survival during the Younger Dryas
 - ☐ To provide evidence supporting the theory presented in paragraph 4 about how wild plants became domesticated

Paragraph 5 is marked with an arrow [→]

8. According to paragraph 5, what was one response of the people of Abu Hureyra to the onset of the Younger Dryas?
- ☐ They began collecting a wider variety of wild grains.
 - ☐ They began cultivating crops at sites outside Abu Hureyra.
 - ☐ They developed special pouches for storing and carrying cereal grain.
 - ☐ They spent more energy and time cultivating their rye crop.

Paragraph 5 is marked with an arrow [→]

9. According to paragraph 5, what happened to the domesticated rye left at Abu Hureyra?
- ☐ It developed a number of different strains.
 - ☐ It became wild again.
 - ☐ It became extinct.
 - ☐ It could no longer be used as food.

Paragraph 5 is marked with an arrow [→]

Paragraph 6

The geographical range of the Late Natufians also changed. With their increased interest in plant cultivation, the Late Natufians drifted away from the depleted woodlands where their forebears once flourished. They were drawn to the alluvial soils (soils deposited by rivers) of the valleys, not only those of the River Jordan, but also those found by the great rivers of the Mesopotamian plain and in the vicinity of lakes and rivers throughout the Middle East. Large expanses of these rich, fertile soils became available as the rivers and lakes struck during the Younger Dryas Wild, but cultivated, cereals grew well in such soil, especially when close to the meager springs, ponds, and streams that survived the arid conditions.

10. The word "flourished" in the passage is closet in meaning to
- ☐ settled

- ☐ struggled
- ☐ gathered
- ☐ prospered

11. The phrase “drawn to” in the passage is closest in meaning to

- ☐ concerned by
- ☐ attracted toward
- ☐ impressed by
- ☐ surrounded by

12. According to paragraph 6, why did the Natufians move to river valleys during the Younger Dryas?

- ☐ Plants grew better in the fertile soil next to bodies of water.
- ☐ The Natufians followed the game animals there.
- ☐ The Natufians used the rivers as their means of transportation.
- ☐ The valleys had more woodlands, which the Natufians needed for home building.

Paragraph 6 is marked with an arrow [→]

Paragraphs 1 and 2

The so-called Natufian culture inhabited what is now the Middle East between approximately 14,000 and 11,500 years ago. This period is commonly split into two subperiods, Early Natufian (14,000 to 13,000 years ago) and Late Natufian (13,000 to 11,500). The Natufians were hunter-gatherers who relied primarily on gazelle, although they also cultivated some cereal grains. ■ During the early period at least, they lived year-round in villages in built stone houses. Like all human beings, their way of life depended on the climate. ■ Around 13,000 years ago, their climate began to change, becoming colder and drier, a period known as the Younger Dryas. ■

We know that times were hard in the increasingly arid landscapes of the Younger Dryas, but quite how hard remains unclear. ■ The droughts certainly caused many ponds and rivers to disappear completely and the larger lakes to shrink in size. The people who lived in the south, in today’s deserts of the Negev and the Sinai, were most likely hit the hardest. They returned to a completely transient hunter-gatherer way of life, moving from place to place. Survival required improved hunting weapons: game (animals hunted for food) had become scarce, and consequently, success had become essential when a kill was possible. And so we see the invention of the Harif point, a new kind of arrowhead.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

As long as the climate remained moderate, the Natufians were able to thrive by remaining in their villages.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

About 13,000 years ago, a colder and drier period known as the Younger Dryas caused the Natufian people to alter their lifestyle.

-
-
-

Answer Choices

As lakes and rivers dried up, Natufians abandoned their settlements in some areas and became mobile hunters, developing new, more efficient hunting weapons to improve their success in the hunt.

Cereal yields dropped when the air became excessively concentrated with carbon dioxide, which impaired photosynthesis.

In one Natufian village, an archaeologist found the oldest-known domesticated cereal grain, dating from more than 14,000 years ago.

A major contribution of Natufian culture to early civilization was the invention of a new arrowhead called the Harif point.

With wild grains no longer plentiful, the Natufians had to invent better cultivation techniques, resulting in new domesticated varieties.

The climate change forced some Late Natufians to move away from their former woodland homes and inhabit the fertile lands left by the receding rivers and lakes.

Metamorphosis

Organisms that metamorphose undergo radical changes over the course of their life cycle. A frog egg hatches a tadpole that metamorphoses into an adult frog within a few days or weeks. A fruit fly egg hatches a larva that feeds for a few hours or days and then enters the pupal stage during which it develops a protective covering. The changes that occur during the metamorphosis of a single species may be so great that the species occupies two separate and very different niches or places in an environment at different times. In fact, the larvae of two species may be more similar to each other than to the corresponding adult forms of their own species.

Organisms that utilize different resources at different stages of their life cycle face an unusual evolutionary problem, exploiting different niches may be difficult with a single body plan. The solution is a juvenile (immature) form specialized for one niche, followed by metamorphosis to an entirely new body plan, adapted to a different niche in the adult. Clearly, species that metamorphose must undertake complex genetic and physiological processes in the transformation. These changes require complex regulatory mechanisms that involve turning on and off many genes at appropriate times. In addition, the reorganization of the body plan in a metamorphic species entails considerable energy costs. What sorts of ecological advantages could outweigh these complications?

One prevailing hypothesis is that metamorphic species specialize so as to exploit habitats with high but transient (short term) productivity----and hence high potential for growth. Part of this strategy is that specializations for feeding, dispersal, and reproduction are separated across stages. A frog tadpole occupies an aquatic environment (such as a pond) with extremely high potential for growth. The existence of the pond or its high production may be transient, however. Whereas an aquatic larva is not capable of dispersal to new ponds if its habitat becomes unsuitable, the adult frog is. In this case rapid growth in the larva is separated from dispersal and reproduction in the adult. Although the adult feeds, its growth rate is far less than that of the tadpole. The energy adults obtain from feeding is dedicated to dispersal and reproduction.

Many insects benefit from the same strategy. Although a butterfly larva feeds voraciously, often on a very specific set of host plant species, the adult does not grow. If it feeds, it does so only to maintain energy reserves required for dispersal and reproduction. The monarch butterfly exemplifies this strategy. Its larvae feed specifically on milkweeds. Monarch pupae also develop on this host plant. The emerging adults migrate long distances----from all over eastern North America to nine small sites in the Sierra Madre mountains of Mexico. There, females become sexually mature and migrate north, mating along the way and feeding only to maintain energy reserves. In this example, the feeding specialist stage is again separated from the dispersal and reproduction stages.

In the previous examples, the reproductive function is delegated to the adult. Under certain

ecological conditions, however, it is apparently advantageous for reproduction to occur in the larval stage. Thus, even the reproductive function typically fulfilled by the adult can apparently be modified under certain circumstances. Species that show this modification of a metamorphic life cycle are said to demonstrate neoteny, a life cycle in which the larvae of some populations or races become sexually mature and no longer metamorphose into adult. Some populations of the salamander *Ambystoma maculatum* show this trait. In fact, the larvae of this species were originally classified as a separate species before it was recognized that they are neotenic forms.

The selective factors leading to neoteny are not well understood. We know, however, that neotenic forms are more frequently found in extreme environments, often high altitudes or latitudes. High-altitude populations of certain salamanders have higher frequencies of neoteny than do low-elevation population of these species. If the larval environment is rich compared to the harsh adult environment, selection may favor neoteny. One research study, has ruled out simple food effects, supplemental food did not increase the frequency with which organisms reached the adult stage. This suggests that neoteny may be a genetically determined feature of some amphibian life histories.

Paragraph 1

Organisms that metamorphose undergo radical changes over the course of their life cycle. A frog egg hatches a tadpole that metamorphoses into an adult frog within a few days or weeks. A fruit fly egg hatches a larva that feeds for a few hours or days and then enters the pupal stage during which it develops a protective covering. The changes that occur during the metamorphosis of a single species may be so great that the species occupies two separate and very different niches or places in an environment at different times. In fact, the larvae of two species may be more similar to each other than to the corresponding adult forms of their own species.

1. The word “radical” in the passage is closest in meaning to
 - ☐ extreme
 - ☐ distinctive
 - ☐ periodic
 - ☐ structural
2. According to paragraph 1, which of the following is true of the organisms that become fruit flies?
 - ☐ They feed during the pupal stage.
 - ☐ They become winged insects several days after the larvae hatch from eggs.
 - ☐ They remain in the larval stage longer than they remain in the pupal stage.
 - ☐ They need to be protected during the larval stage in order to reach the pupal stage.

Paragraph 2

Organisms that utilize different resources at different stages of their life cycle face an unusual evolutionary problem, exploiting different niches may be difficult with a single body plan. The solution is a juvenile (immature) form specialized for one niche, followed by metamorphosis to

an entirely new body plan, adapted to a different niche in the adult. Clearly, species that metamorphose must undertake complex genetic and physiological processes in the transformation. These changes require complex regulatory mechanisms that involve turning on and off many genes at appropriate times. In addition, the reorganization of the body plan in a metamorphic species entails considerable energy costs. What sorts of ecological advantages could outweigh these complications?

3. It can be inferred from paragraph 2 that an advantage of
 - ☐ develop regulatory mechanisms for turning genes on and off
 - ☐ occupy different niches at different parts of the life cycle
 - ☐ undertake complex genetic and physiological processes
 - ☐ reduce their energy costs
4. Why does the author ask the question “What sorts of ecological advantages could outweigh these complications?”
 - ☐ To suggest that there is no single advantage but many advantages
 - ☐ To challenge the idea that metamorphic transformations are always beneficial
 - ☐ To prepare readers for a discussion that may explain why metamorphosis occurs
 - ☐ To identify a particular aspect of metamorphosis that is poorly understood

Paragraph 3

One prevailing hypothesis is that metamorphic species specialize so as to exploit habitats with high but transient (short term) productivity—and hence high potential for growth. Part of this strategy is that specializations for feeding, dispersal, and reproduction are separated across stages. A frog tadpole occupies an aquatic environment (such as a pond) with extremely high potential for growth. The existence of the pond or its high production may be transient, however. Whereas an aquatic larva is not capable of dispersal to new ponds if its habitat becomes unsuitable, the adult frog is. In this case rapid growth in the larva is separated from dispersal and reproduction in the adult. Although the adult feeds, its growth rate is far less than that of the tadpole. The energy adults obtain from feeding is dedicated to dispersal and reproduction.

5. The word “exploit” in the passage is closest in meaning to
 - ☐ identify
 - ☐ adapt to
 - ☐ use to advantage
 - ☐ become established in
6. According to paragraph 3, by changing from a tadpole into an adult frog, an adult frog is able to
 - ☐ maintain a high growth rate
 - ☐ provide a safer habitat for its offspring
 - ☐ obtain more food
 - ☐ move to a more suitable location

Paragraph 4

Many insects benefit from the same strategy. Although a butterfly larva feeds voraciously, often on a very specific set of host plant species, the adult does not grow. If it feeds, it does so only to maintain energy reserves required for dispersal and reproduction. The monarch butterfly exemplifies this strategy. Its larvae feed specifically on milkweeds. Monarch pupae also develop on this host plant. The emerging adults migrate long distances----from all over eastern North America to nine small sites in the Sierra Madre mountains of Mexico. There, females become sexually mature and migrate north, mating along the way and feeding only to maintain energy reserves. In this example, the feeding specialist stage is again separated from the dispersal and reproduction stages.

7. According to paragraph 3 and 4, what do frogs and butterflies have in common?
- ☐ Adults of both reproduce only when there is enough food in a habitat to sustain their offspring.
 - ☐ Adults of both eat only enough to supply the energy needed for dispersal and reproduction.
 - ☐ Immature forms of both disperse to new habitats when the habitats they are in can no longer sustain them.
 - ☐ Immature forms of both depend on aquatic environments.
8. The phrase “the same strategy” in the passage refers to
- ☐ dispersing to new habitats as adults
 - ☐ spending most of the life cycle on dispersal and reproduction
 - ☐ feeding on a specific set of host plant species
 - ☐ separating specialization for feeding from dispersal and reproduction

Paragraph 5

In the previous examples, the reproductive function is delegated to the adult. Under certain ecological conditions, however, it is apparently advantageous for reproduction to occur in the larval stage. Thus, even the reproductive function typically fulfilled by the adult can apparently be modified under certain circumstances. Species that show this modification of a metamorphic life cycle are said to demonstrate neoteny, a life cycle in which the larvae of some populations or races become sexually mature and no longer metamorphose into adult. Some populations of the salamander *Ambystoma maculatum* show this trait. In fact, the larvae of this species were originally classified as a separate species before it was recognized that they are neotenic forms.

9. The word “apparently” in the passage is closest in meaning to
- ☐ usually
 - ☐ especially
 - ☐ seemingly
 - ☐ certainly
10. According to paragraph 5, why were the larvae of some populations of the salamander species *Ambystoma maculatum* once thought to be members of a separate species?
- ☐ Because they were shaped differently than other larvae of *Ambystoma maculatum*.

- ☐ Because they were discovered far away from other members of *Ambystoma maculatum*.
- ☐ Because they were sexually mature and could reproduce.
- ☐ Because the neotenic form of *Ambystoma maculatum* looks very different from the neotenic forms of other salamanders.

Paragraph 6

The selective factors leading to neoteny are not well understood. We know, however, that neotenic forms are more frequently found in extreme environments, often high altitudes or latitudes. High-altitude populations of certain salamanders have higher frequencies of neoteny than do low-elevation population of these species. If the larval environment is rich compared to the **harsh** adult environment, selection may favor neoteny. One research study, has ruled out simple food effects, supplemental food did not increase the frequency with which organisms reached the adult stage. This suggests that neoteny may be a genetically determined feature of some amphibian life histories.

11. The word “**harsh**” in the passage is closest in meaning to
- ☐ severe
 - ☐ typical
 - ☐ restricted
 - ☐ available
12. Paragraph 6 indicates that greater frequency of neoteny in salamanders may be associated with all of the following EXCEPT
- ☐ an environment that is richer for larvae than for adults
 - ☐ an inadequate food supply for larvae
 - ☐ a high-altitude location
 - ☐ a genetic makeup favoring neoteny

Paragraph 2

Organisms that utilize different resources at different stages of their life cycle face an unusual evolutionary problem, exploiting different niches may be difficult with a single body plan. ■ The solution is a juvenile (immature) form specialized for one niche, followed by metamorphosis to an entirely new body plan, adapted to a different niche in the adult. ■ Clearly, species that metamorphose must undertake complex genetic and physiological processes in the transformation. ■ These changes require complex regulatory mechanisms that involve turning on and off many genes at appropriate times. ■ In addition, the reorganization of the body plan in a metamorphic species entails considerable energy costs. What sorts of ecological advantages could outweigh these complications?

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Metamorphosis, however, comes with its own set of difficulties.

Where would the sentence best fit? Click on a square [■] to add the sentence to the

passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

In metamorphic species, an organism's body plan changes at different points in its life cycle.

-
-
-

Answer Choices

Organisms that metamorphose use resources less efficiently than do organisms with a single body plan.

Because of the high energy costs associated with metamorphosis, only a few types of niches are rich enough to support metamorphic species.

The larvae of some population found in extreme environments reproduce at the juvenile stage and never achieve the adult form.

Metamorphosis enables an organism to make use of environments that are highly productive for a short period of time but not for the organism's entire life span.

In many metamorphic species, the juvenile stage is focused on eating and growing and the adult stage is focused on dispersal and reproduction.

Some population remain at the larval stage in part because they lack sufficient nutrition to metamorphose into adults.

The Origin of Coral Reefs

Coral reefs are natural structures formed from deposits of the calcium carbonate secretions of coral, a marine animal that lives in colonies. In general, coral reefs are grouped into one of three categories, atolls, barrier reefs, and fringing reefs. Atolls are usually easily distinguished because they are modified horseshoe-shaped reefs that rise out of very deep water far from land and enclose a lagoon (a body of certain water surrounded by a coral reef). With few exceptions, atolls are found only in the Indo-Pacific area. Barrier reefs and fringing reefs, on the other hand, tend to grade into each other and are not readily separable. Some scientists would prefer to group them into a single category. Both types occur adjacent to a landmass, with a barrier reef being separated from the landmass by a greater distance and deeper water channel than the fringing reef. Fringing reefs and barrier reefs are common throughout the coral reef zones in all oceans.

Different types of reefs and reefs in different oceans may have diverse origins and histories. The greatest interest in the origin of reefs has centered on atolls. For many years, humans speculated as to how such reefs could develop in such deep water, miles from the nearest emergent land. This interest was heightened when it was discovered that reef corals could not live deeper than 50-70 meters. This led to the development of several theories concerning the origin of atolls. Only one need be discussed here—thegrow on the shores of newly formed volcanic islands that have pushed to the surface from deep water. These islands often begin to subside, and if the subsidence is not too fast, reef growth will keep up with the subsidence. The reef growth will then form a barrier reef and, ultimately, an atoll as the island disappears beneath the sea. When the island has disappeared, corals continue to grow on the outside and keep the reef at the surface. On the inside, where the island used to be, quiet water conditions and high sedimentation prevail. These conditions prevent continued vigorous coral growth, hence, a lagoon develops. This theory links all three reef types into evolutionary sequence, but is not an explanation for all fringing and barrier reef types.

Since the current surface features of atolls give no evidence of a volcanic base, in the years after the development of Darwin's theory other explanations were offered, and the whole concept of the origin of atolls became embroiled in the controversy over the origin of coral reefs. If Darwin's theory was correct, it must be assumed that drilling down through the current atoll reefs would yield layer after layer of reef limestone until, finally, volcanic rock would be encountered. The ability to drill to the base of atoll reefs and resolve the problem had to wait until the mid-twentieth century in 1953. Ladd and other geologists reported borings at Eriwetok atoll in the Marshall Islands that penetrated 1,283 meters of reef limestone and then hit volcanic rock. This was the evidence that Darwin's theory was substantially correct. The correctness of this theory has been strengthened by the discovery of flat-topped mountains or guyots that, at present, have their tops many hundreds or thousands of meters below the ocean surface, but have on their surface the remains of shallow water corals. Evidently, these mountains sank too fast for reef growth to keep above the ocean surface.

Although the subsidence theory links all three reef types in a successional sequence, not all barrier reefs and fringing reefs can be explained by this mechanism. Indeed, the reasons barrier and fringing reef types occur around continental margins and high non-volcanic islands are simply that these areas offer suitable environmental conditions for the growth of reefs and a suitable substrate (surface) on which to begin growth. The extensive reefs around the Indonesian Islands, the Philippines, New Guinea, Fiji and most of the Caribbean Islands are there because a suitable substrate in shallow water existed on which they could initiate growth. In none of these areas are large land areas subsiding, not will these reefs ultimately become atolls.

Paragraph 1

Coral reefs are natural structures formed from deposits of the calcium carbonate secretions of coral, a marine animal that lives in colonies. In general, coral reefs are grouped into one of three categories, atolls, barrier reefs, and fringing reefs. Atolls are usually easily distinguished because they are modified horseshoe-shaped reefs that rise out of very deep water far from land and enclose a lagoon (a body of certain water surrounded by a coral reef). With few exceptions, atolls are found only in the Indo-Pacific area. Barrier reefs and fringing reefs, on the other hand, tend to grade into each other and are not readily separable. Some scientists would prefer to group them into a single category. Both types occur adjacent to a landmass, with a barrier reef being separated from the landmass by a greater distance and deeper water channel than the fringing reef. Fringing reefs and barrier reefs are common throughout the coral reef zones in all oceans.

1. According to paragraph 1, all of the following are true of atolls EXCEPT
 - ☐ Most atolls occur in the Indo-Pacific area.
 - ☐ Atolls occur only in very deep waters.
 - ☐ Atolls are more common than other types of reefs.
 - ☐ Atolls occur far from a landmass.
2. According to paragraph 1, fringing reefs and barrier reefs can be identified by
 - ☐ their similarities to atolls
 - ☐ their organization into groups of separable parts
 - ☐ their distance from the nearest landmass
 - ☐ whether or not they enclose a lagoon

Paragraph 2

Different types of reefs and reefs in different oceans may have diverse origins and histories. The greatest interest in the origin of reefs has centered on atolls. For many years, humans speculated as to how such reefs could develop in such deep water, miles from the nearest emergent land. This interest was heightened when it was discovered that reef corals could not live deeper than 50-70 meters. This led to the development of several theories concerning the origin of atolls. Only one need be discussed here—thegrow on the shores of newly formed volcanic islands that have pushed to the surface from deep water. These islands often begin to subside, and if the subsidence is not too fast, reef growth will keep up with the subsidence. The reef growth will then form a barrier reef and, ultimately, an atoll as the island disappears beneath the

sea. When the island has disappeared, corals continue to grow on the outside and keep the reef at the surface. On the inside, where the island used to be, quiet water conditions and high sedimentation prevail. These conditions prevent continued vigorous coral growth, hence, a lagoon develops. This theory links all three reef types into evolutionary sequence, but is not an explanation for all fringing and barrier reef types.

3. The phrase “was heightened” in the passage is closest in meaning to
 - ☐ developed
 - ☐ varied
 - ☐ continued
 - ☐ increased
4. The word “prevail” in the passage is closest in meaning to
 - ☐ dominate
 - ☐ combine
 - ☐ spread
 - ☐ compete
5. Which of the following can be inferred from paragraph 2 as the reason that the origin of atolls has generated more interest than has the origin of other types of reefs?
 - ☐ Most reefs have a predictable shape, but atolls do not.
 - ☐ Atolls appear to form more rapidly than other kinds of reefs.
 - ☐ The animals that form coral reefs cannot live in water below 70 meters, but atolls only form in very deep water.
 - ☐ Atolls have more diverse origins and histories than other types of reefs.
6. According to paragraph 2, Darwin’s subsidence theory provided an answer to each of the following questions EXCEPT
 - ☐ How does a lagoon develop?
 - ☐ Why do volcanic islands subside?
 - ☐ How does a fringing reef become a barrier reef?
 - ☐ What happens to coral growth when a volcanic island subsides slowly?
7. What can be inferred from paragraph 2 about the development of atolls according to Darwin’s theory?
 - ☐ All atolls begin as fringing reefs.
 - ☐ An atoll will not develop if an island subsides too slowly.
 - ☐ Most atolls will eventually become barrier reefs.
 - ☐ The formation of a lagoon accelerates the overall development of an atoll.

Paragraph 3

Since the current surface features of atolls give no evidence of a volcanic base, in the years after the development of Darwin’s theory other explanations were offered, and the whole concept of the origin of atolls became embroiled in the controversy over the origin of coral reefs. If Darwin’s

theory was correct, it must be assumed that drilling down through the current atoll reefs would yield layer after layer of reef limestone until, finally, volcanic rock would be encountered. The ability to drill to the base of atoll reefs and resolved the problem had to wait until the mid-twentieth century in 1953. Ladd and other geologists reported borings at Eriwetok atoll in the Marshall Islands that penetrated 1,283 meters of reef limestone and then hit volcanic rock. This was the evidence that Darwin's theory was substantially correct. The correctness of this theory has been strengthened by the discovery of flat-topped mountains or guyots that, at present, have their tops many hundreds or thousands of meters below the ocean surface, but have on their surface the remains of shallow water corals. Evidently, these mountains sank too fast for reef growth to keep above the ocean surface.

8. The word "substantially" in the passage is closest in meaning to
- ☐ often
 - ☐ largely
 - ☐ probably
 - ☐ undoubtedly
9. Which of the following is mentioned in paragraph 3 as the reason many scientists doubted Darwin's theory of how atolls form?
- ☐ After Darwin's five-year voyage, different kinds of atolls than those he observed were discovered.
 - ☐ Scientists discovered guyots, or flat-topped mountains, below the ocean's surface.
 - ☐ Surface features of atolls do not show any signs of a volcanic base.
 - ☐ Drilling through layers of atoll reef did not produce the results Darwin predicted.
10. What can be inferred from paragraph 2 and 3 about the volcanic rock found by the geologists in the Marshall Islands?
- ☐ It extends from the surface to 1,283 meters deep.
 - ☐ It was the first guyot ever discovered by researchers.
 - ☐ It formed after the limestone layers of the Eriwetok atoll.
 - ☐ It was once an island existing above the ocean surface.

Paragraph 4

Although the subsidence theory links all three reef types in a successional sequence, not all barrier reefs and fringing reefs can be explained by this mechanism. Indeed, the reasons barrier and fringing reef types occur around continental margins and high non-volcanic islands are simply that these areas offer suitable environmental conditions for the growth of reefs and a suitable substrate (surface) on which to begin growth. The extensive reefs around the Indonesian Islands, the Philippines, New Guinea, Fiji and most of the Caribbean Islands are there because a suitable substrate in shallow water existed on which they could initiate growth. In none of these areas are large land areas subsiding, not will these reefs ultimately become atolls.

11. The word "initiate" in the passage is closest in meaning to
- ☐ begin

- ☐ encourage
- ☐ establish
- ☐ maintain

12. In paragraph 4, why does the author discuss the reefs around the Indonesian Islands, the Philippines, New Guinea, Fiji, and the Caribbean Islands?

- ☐ To argue that these islands have suitable environmental conditions for subsidence
- ☐ To support the claim that the subsidence theory cannot explain the formation of all barrier and fringing reefs.
- ☐ To indicate how different barrier and fringing reefs could be from atolls
- ☐ To identify reefs that grow around non-volcanic islands

Paragraph 3

Since the current surface features of atolls give no evidence of a volcanic base, in the years after the development of Darwin's theory other explanations were offered, and the whole concept of the origin of atolls became embroiled in the controversy over the origin of coral reefs. If Darwin's theory was correct, it must be assumed that drilling down through the current atoll reefs would yield layer after layer of reef limestone until, finally, volcanic rock would be encountered. The ability to drill to the base of atoll reefs and resolved the problem had to wait until the mid-twentieth century in 1953. Ladd and other geologists reported borings at Eriwetok atoll in the Marshall Islands that penetrated 1,283 meters of reef limestone and then hit volcanic rock. ■ This was the evidence that Darwin's theory was substantially correct. ■ The correctness of this theory has been strengthened by the discovery of flat-topped mountains or guyots that, at present, have their tops many hundreds or thousands of meters below the ocean surface, but have on their surface the remains of shallow water corals. ■ Evidently, these mountains sank too fast for reef growth to keep above the ocean surface. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Had subsidence occurred more slowly, the reefs would have eventually become atolls.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Atolls have the most distinctive features of all coral reef types.

-
-
-

Answer Choices

Because of its many marine animal colonies, the Indo-Pacific area contains the world's largest number of atolls, barrier reefs, and fringing reefs.

In the years after Darwin's voyage, researchers discovered that Darwin was incorrect in believing that all fringing reefs and barrier reefs will eventually become atolls.

The discovery of guyots suggested to researchers that atolls would never form in non-volcanic islands where fringing and barrier reefs are common. Darwin argued that atolls are formed when volcanic islands surrounded by fringing reefs begin to slowly sink, forming first a barrier reef and finally an atoll.

Darwin's theory was confirmed by the discovery of volcanic rock underneath the limestone layers of an atoll in the Marshall Islands.

The fringing reefs and barrier reefs that grow in the waters of continental margins and around non-volcanic islands will never become atolls.

Southwest Agriculture

After the arrival of hunter-gatherers in the southwestern region of North America, several alternative types of agriculture emerged, all involving different solutions to the Southwest's fundamental problem: how to obtain enough water to grow crops in an environment in which rainfall is so low and unpredictable that little or no farming is practiced there today. People experimented with alternative strategies for almost a thousand years in different locations, and many experiments succeeded for centuries, but eventually all except one succumbed to environmental problems caused by human impact or climate change.

One strategy was to live at higher elevations where rainfall was higher, as did the Mogollon, the people at Mesa Verde, and the people of the early agricultural phase at Chaco Canyon known as the Pueblo I phase. But that carried a risk, because it is cooler at high than at low elevations, and in an especially cool year, it might be too cold to grow crops at all. An opposite extreme was to farm at the warmer low elevations, but there the rainfall is insufficient even for dryland agriculture. The Hohokam got around that problem by constructing the most extreme irrigation system in the Americas outside Peru. But irrigation entailed the risk that human digging of ditches and canals could lead to sudden heavy water runoff from rainstorms, digging further down into the ditches and canals and carving out deep channels called arroyos. In that case, the water level would drop below the field level, making irrigation impossible for people without pumps.

A more conservative strategy was to plant crops only in areas with reliable springs and groundwater tables. That was the solution initially adopted by the Mimbres and by people in the phase known as Pueblo II. However, it then became dangerously tempting to expand agriculture during wet decades with favorable growing conditions into marginal areas with less reliable springs and groundwater. The population multiplying in those marginal areas might then find itself unable to grow crops and might starve when the unpredictable climate turned dry again. That fate actually befell the Mimbres, who started by farming the floodplain and then began to farm adjacent land above the floodplain as their population came to exceed the floodplain's capacity to support it. They got away with their gamble during a wet climate phase, when they were able to obtain half their food outside the floodplain. However, when drought conditions returned, that gamble left them with a population double what the floodplain could support, and Mimbres society collapsed suddenly under the stress.

Still another solution was to occupy an area only for a few decades, until the area's soil became exhausted, then to move to another area. That method worked when people were living at low population densities, when there were

many unoccupied areas to move to, and when each occupied area could be left unoccupied again for sufficiently long after occupation so that its vegetation and soil nutrients had time to recover. However, the method of shifting sites after a short occupation became impossible at high population densities, when people filled up the whole landscape and there was nowhere left empty to move to.

One more strategy was to plant crops at many sites even though rainfall was locally unpredictable and then to harvest crops at whichever sites did get enough rain to produce a good harvest and to redistribute some of the harvest to the people still living at all the sites that did not happen to receive enough rain that year. But redistribution was not without risks because it involved a complex political and social system to integrate activities between different sites, so when that complex system collapsed, lots of people ended up starving.

The remaining strategy was to plant crops and live near permanent or dependable sources of water, but on landscape benches above the main floodways, so as to avoid the risk of a heavy flood washing out fields and villages, and to practice a diverse economy, exploiting ecologically diverse zones so that each settlement would be self-sufficient. That solution, adopted by people whose descendents live today in the Southwest's Hopi and Zuni villages, has succeeded for more than a thousand years.

Paragraph 1

After the arrival of hunter-gatherers in the southwestern region of North America, several alternative types of agriculture emerged, all involving different solutions to the Southwest's fundamental problem: how to obtain enough water to grow crops in an environment in which rainfall is so low and unpredictable that little or no farming is practiced there today. People experimented with alternative strategies for almost a thousand years in different locations, and many experiments succeeded for centuries, but **eventually** all except one succumbed to environmental problems caused by human impact or climate change.

1. The word "**eventually**" in the passage is closest in meaning to
 - ☐ to some degree
 - ☐ unfortunately
 - ☐ in the end
 - ☐ gradually
2. Paragraph 1 supports which of the following inferences about the North American Southwest?
 - ☐ Its sources of water were plentiful when the hunter-gathers first arrived.
 - ☐ It has always lacked the large population needed to perform successful experiments in agriculture.
 - ☐ Its climatic conditions today are essentially similar to those existing when the

hunter-gathers first arrived.

- It was more seriously affected by climate change than it was by human impact.

Paragraph 2

One strategy was to live at higher elevations where rainfall was higher, as did the Mogollon, the people at Mesa Verde, and the people of the early agricultural phase at Chaco Canyon known as the Pueblo I phase. But that carried a risk, because it is cooler at high than at low elevations, and in an especially cool year, it might be too cold to grow crops at all. An opposite extreme was to farm at the warmer low elevations, but there the rainfall is insufficient even for dryland agriculture. The Hohokam got around that problem by constructing the most extreme irrigation system in the Americas outside Peru. But irrigation entailed the risk that human digging of ditches and canals could lead to sudden heavy water runoff from rainstorms, digging further down into the ditches and canals and carving out deep channels called arroyos. In that case, the water level would drop below the field level, making irrigation impossible for people without pumps.

3. All of the following are mentioned in paragraph 2 as potential problems faced by the Hohokam **EXCEPT**
 - insufficient rainfall to allow crops to grow
 - rainstorms leading to destructive water runoff
 - insufficient workers to dig ditches and canals
 - irrigation water levels in channels too low to be used

Paragraph 3

A more conservative strategy was to plant crops only in areas with reliable springs and groundwater tables. That was the solution **initially** adopted by the Mimbres and by people in the phase known as Pueblo II. However, it then became dangerously tempting to expand agriculture during wet decades with favorable growing conditions into marginal areas with less reliable springs and groundwater. The population multiplying in those marginal areas might then find itself unable to grow crops and might starve when the unpredictable climate turned dry again. That fate actually befell the Mimbres, who started by farming the floodplain and then began to farm **adjacent** land above the floodplain as their population came to exceed the floodplain's capacity to support it. They got away with their gamble during a wet climate phase, when they were able to obtain half their food outside the floodplain. However, when drought conditions returned, that gamble left them with a population double what the floodplain could support, and Mimbres society collapsed suddenly under the stress.

4. The word "**initially**" in the passage is closest in meaning to
 - first
 - possibly

- ☐ sometimes
 - ☐ reluctantly
5. The word “adjacent” in the passage is closest in meaning to
- ☐ neighboring
 - ☐ higher
 - ☐ unused
 - ☐ additional
6. According to paragraph 3, which of the following was the cause of the collapse of the Mimbres society?
- ☐ They could not overcome the stress of moving to a new area with each change in the climate.
 - ☐ The flooding of their farmland during a wet climate phase prevented them from growing enough to feed their population.
 - ☐ A decline in population during dry periods prevented them from expanding their farming into nearby areas.
 - ☐ Their population became too large to survive when the climate entered a dry period.

Paragraph 4

Still another solution was to occupy an area only for a few decades, until the area's soil became exhausted, then to move to another area. That method worked when people were living at low population densities, when there were many unoccupied areas to move to, and when each occupied area could be left unoccupied again for sufficiently long after occupation so that its vegetation and soil nutrients had time to recover. However, the method of shifting sites after a short occupation became impossible at high population densities, when people filled up the whole landscape and there was nowhere left empty to move to.

7. According to paragraph 4, the solution of moving to other, previously occupied areas was effective only when
- ☐ the soil on the previously occupied land had been used for farming at least once
 - ☐ the vegetation and nutrients of the previously occupied area had time to recover
 - ☐ the reoccupations did not last a long time
 - ☐ the reoccupied areas became more densely populated

Paragraph 5

One more strategy was to plant crops at many sites even though rainfall was locally unpredictable and then to harvest crops at whichever sites did get enough rain to produce a good harvest and to redistribute some of the harvest to the people still living at all the sites that did not happen to receive enough rain that

year. But redistribution was not without risks because it involved a complex political and social system to integrate activities between different sites, so when that complex system collapsed, lots of people ended up starving.

8. The solution for unpredictable rainfall described in paragraph 5 involved all of the following EXCEPT
- ☐ planting crops in many different places, regardless of predicted rainfall
 - ☐ using a simple political and social system for the coordination of agricultural activities
 - ☐ harvesting crops from those sites that did receive adequate rain
 - ☐ sharing harvested crops with people whose crops did not grow well

Paragraph 6

The remaining strategy was to plant crops and live near permanent or **dependable** sources of water, but on landscape benches above the main floodways, so as to avoid the risk of a heavy flood washing out fields and villages, and to practice a diverse economy, exploiting ecologically diverse zones so that each settlement would be self-sufficient. That solution, adopted by people whose descendents live today in the Southwest's Hopi and Zuni villages, has succeeded for more than a thousand years.

9. The word "dependable" in the passage is closest in meaning to
- ☐ adequate
 - ☐ familiar
 - ☐ reliable
 - ☐ plentiful
10. Which of the following contributed to the success of the strategy discussed in paragraph 6?
- ☐ Crops were planted high enough to avoid being carried away by floods.
 - ☐ Crops were planted wherever there were sources of water, regardless of their proximity to where people actually lived.
 - ☐ Farming was limited to ecological zones that were similar to each other.
 - ☐ Instead of aiming at becoming self-sufficient, each village became part of a network of diverse economic activities.
11. Which of the following best describes the organization of the passage?
- ☐ A historical account of the various people that have lived in southwestern North America followed by conflicting views about the accuracy of the account
 - ☐ A comparison and contrast of the historical role of agriculture in different parts of the southwestern region of North America
 - ☐ An explanation of how the solutions developed to improve living conditions in southwestern North America spread to other neighboring regions

- A description of a long-standing problem faced by the people in southwestern North America followed by a discussion on strategies that have been used to overcome the problem

Paragraph 2

One strategy was to live at higher elevations where rainfall was higher, as did the Mogollon, the people at Mesa Verde, and the people of the early agricultural phase at Chaco Canyon known as the Pueblo I phase. But that carried a risk, because it is cooler at high than at low elevations, and in an especially cool year, it might be too cold to grow crops at all. An opposite extreme was to farm at the warmer low elevations, but there the rainfall is insufficient even for dryland agriculture. ■ The Hohokam got around that problem by constructing the most extreme irrigation system in the Americas outside Peru. ■ But irrigation entailed the risk that human digging of ditches and canals could lead to sudden heavy water runoff from rainstorms, digging further down into the ditches and canals and carving out deep channels called arroyos. ■ In that case, the water level would drop below the field level, making irrigation impossible for people without pumps. ■

12. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Another risk of irrigation was that floods could simply wash away the dams and channels, as indeed may have happened eventually to the Hohokam.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

13. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

People trying to grow crops in the dry southwestern area of North America experimented with various strategies for almost a thousand years.

-
-
-

Answer Choices

Experiments in agriculture succeeded in those areas that experienced climate changes that increased the amount of rainfall gradually over many centuries.

In spite of usually dry conditions, water runoff from rainstorms was used by the Hohokam to flood fields and deepen channels in irrigation systems.

Shifting sites or planting in areas with good sources of water was successful only in areas with low population densities.

Farming at higher elevations meant risking failure from cold weather, while extensive irrigation at the lower elevations risked destructive floods from sudden rainstorms.

Farming in dry climate phases was successful only at higher elevations.

Sharing food produced at different sites had limited success, while self-sufficient and ecologically sound farming continued to succeed.



The Development of Printing

Printing with movable type, a revolutionary departure from the old practice of copying by hand, was invented in the 1440s by Johannes Gutenberg, a German goldsmith. Mass production of identical books and pamphlets made the world of letters more accessible to a literate audience. Two preconditions proved essential for the advent of printing: the industrial production of paper and the commercial production of manuscripts.

Increased paper production in the fourteenth and fifteenth centuries was the first stage in the rapid growth of manuscript books—hand-copied works bound as books—which in turn led to the invention of mechanical printing. Papermaking came to Europe from China via Arab intermediaries. By the fourteenth century, paper mills were operating in Italy, producing paper that was much more fragile but much cheaper than parchment or vellum, animal skins that Europeans had previously used for writing. To produce paper, old rags were soaked in a chemical solution, beaten by mallets into a pulp, washed with water, treated, and dried in sheets—a method that still produces good-quality paper today.

By the fifteenth century, a brisk industry in manuscript books was flourishing in Europe's university towns and major cities. Production was in the hands of merchants called stationers, who supplied materials, arranged contracts for book production, and organized workshops known as scriptoria, where the manuscripts were copied, and acted as retail booksellers. The largest stationers, in Paris and Florence, were extensive operations by fifteenth-century standards. The Florentine Vespasiano da Bisticci, for example, created a library for Cosimo de' Medici, the head of Florence's leading family, by employing 45 copyists to complete 200 volumes in 22 months. Nonetheless, bookmaking in scriptoria was slow and expensive.

The invention of movable type was an enormous technological breakthrough that took bookmaking out of the hands of human copyists. Printing was not new: the Chinese had been printing by woodblock since the tenth century, and woodcut pictures (in which an image is cut on wood and then transferred to paper) made their appearance in Europe in the early fifteenth century. Movable type, however, allowed entire manuscripts to be printed. The process involved casting durable metal molds to represent the letters of the alphabet. The letters were arranged to represent the text on a page and then pressed in ink against a sheet of paper. The imprint could be repeated numerous times with only a small amount of human labor. In 1467 two German printers established the first press in Rome and produced 12,000 volumes in five years, a feat that in the past would have required one thousand scribes working full time for the same number of years.

After the 1440s, printing spread rapidly from Germany to other European countries. The cities of Cologne, Strasbourg, Nuremberg, Basel, and Augsburg had major presses, many Italian cities had established their own by 1480. In the 1490s, the German city of Frankfurt became an international meeting place for printers and booksellers. The Frankfurt book fair, where printers from different nations exhibited their newest titles, represented a major international cultural event and remains an unbroken tradition to this day. Early books from other presses were still rather exclusive and inaccessible, especially to a largely illiterate population. Perhaps the most famous early book, Gutenberg's two-volume edition of the Latin Bible, was unmistakably a luxury item. Altogether 185 copies were printed. First priced at well over what a fifteenth-century professor could earn in a year, the Gutenberg Bible has always been one of the most expensive books in history, both for its rarity and its exquisite crafting.

Some historians argue that the invention of mechanical printing gave rise to a communications revolution as significant as, for example, the widespread use of the personal computer today. The multiplication of standardized texts altered the thinking habits of Europeans by freeing individuals from having to memorize everything they learned; it certainly made possible the speedy and inexpensive dissemination of knowledge. It created a wider community of scholars, no longer dependent on personal patronage or church sponsorship for texts. Printing facilitated the free expression and exchange of ideas, and its disruptive potential did not go unnoticed by political and church authorities. Emperors and bishops in Germany, the homeland of the printing industry, moved quickly to issue censorship regulations.

Paragraph 1

Printing with movable type, a revolutionary departure from the old practice of copying by hand, was invented in the 1440s by Johannes Gutenberg, a German goldsmith. Mass production of identical books and pamphlets made the world of letters more accessible to a literate audience. Two preconditions proved essential for the **advent** of printing: the industrial production of paper and the commercial production of manuscripts.

1. The word "**advent**" in the passage is closest in meaning to
- ☐ success
 - ☐ increase
 - ☐ arrival
 - ☐ practice

Paragraph 2

Increased paper production in the fourteenth and fifteenth centuries was the first stage in the rapid growth of manuscript books—hand-copied works bound as books—which in turn led to the invention of mechanical printing. Papermaking

came to Europe from China via Arab intermediaries. By the fourteenth century, paper mills were operating in Italy, producing paper that was much more **fragile** but much cheaper than parchment or vellum, animal skins that Europeans had previously used for writing. To produce paper, old rags were soaked in a chemical solution, beaten by mallets into a pulp, washed with water, treated, and dried in sheets—a method that still produces good-quality paper today.

2. In describing the paper produced by Italian paper mills in the fourteenth century as comparatively "**fragile**" the author means that this paper was
 - ☐ variable in quality
 - ☐ limited in how it could be used
 - ☐ easily damaged
 - ☐ dark in color
3. According to paragraph 2, which of the following is true of papermaking?
 - ☐ In fourteenth- and fifteenth-century Europe, it was restricted to Italy, which was the only country with the technology to build paper mills.
 - ☐ It was brought to Europe from China.
 - ☐ Its development ended the practice of copying books by hand.
 - ☐ It produces a superior writing material that is stronger than parchment.
4. Which of the following can be inferred from paragraph 2 about the process of making paper in fourteenth- and fifteenth-century Europe?
 - ☐ It was based on the process previously used to make parchment and vellum.
 - ☐ It was replaced by a process that added steps to create a better quality of paper.
 - ☐ It was dependent on the availability of discarded or used cloth.
 - ☐ It produced paper that was more fragile than paper made in China.

Paragraph 3

By the fifteenth century, a brisk industry in manuscript books was flourishing in Europe's university towns and major cities. Production was in the hands of merchants called stationers, who supplied materials, arranged contracts for book production, and organized workshops known as scriptoria, where the manuscripts were copied, and acted as retail booksellers. The largest stationers, in Paris and Florence, were extensive operations by fifteenth-century standards. The Florentine **Vespasiano da Bisticci**, for example, created a library for Cosimo de' Medici, the head of Florence's leading family, by employing 45 copyists to complete 200 volumes in 22 months. Nonetheless, bookmaking in scriptoria was slow and expensive.

5. The author mentions "**Vespasiano da Bisticci**" in order to emphasize
 - ☐ the process by which stationers obtained the necessary materials for book production

- the equal importance of Florence and Paris in the rapidly developing book industry
 - the superiority of Florentine libraries to those in other European cities
 - that making books was a commercial enterprise in Europe before the invention of printing
6. All of the following were mentioned in paragraph 3 as functions of stationers in fifteenth-century Europe **EXCEPT**
- the creation of contracts for production of books
 - the provision of materials for copying books
 - the hiring of authors to compose new books
 - the organization of scriptoria

Paragraph 4

The invention of movable type was an enormous technological breakthrough that took bookmaking out of the hands of human copyists. Printing was not new: the Chinese had been printing by woodblock since the tenth century, and woodcut pictures (in which an image is cut on wood and then transferred to paper) made their appearance in Europe in the early fifteenth century. Movable type, however, allowed entire manuscripts to be printed. The process involved casting **durable** metal molds to represent the letters of the alphabet. The letters were arranged to represent the text on a page and then pressed in ink against a sheet of paper. The imprint could be repeated numerous times with only a small amount of human labor. In 1467 two German printers established the first press in Rome and produced 12,000 volumes in five years, a feat that in the past would have required one thousand scribes working full time for the same number of years.

7. The word "**durable**" in the passage is closest in meaning to
- light
 - movable
 - long lasting
 - flexible
8. According to paragraph 4, which of the following occurred because of the invention of movable type?
- An increase in the cost of book production
 - An increase in the popularity of the techniques of woodblock and woodcut
 - The creation of a large number of printing jobs
 - A decline in the importance of human copyists

Paragraph 5

After the 1440s, printing spread rapidly from Germany to other European countries. The cities of Cologne, Strasbourg, Nuremberg, Basel, and Augsburg had major presses, many Italian cities had established their own by 1480. In the 1490s, the

German city of Frankfurt became an international meeting place for printers and booksellers. The Frankfurt book fair, where printers from different nations exhibited their newest titles, represented a major international cultural event and remains an unbroken tradition to this day. Early books from other presses were still rather exclusive and inaccessible, especially to a largely illiterate population. Perhaps the most famous early book, Gutenberg's two-volume edition of the Latin Bible, was unmistakably a luxury item. Altogether 185 copies were printed. First priced at well over what a fifteenth-century professor could earn in a year, the Gutenberg Bible has always been one of the most expensive books in history, both for its rarity and its exquisite crafting.

9. Which of the following can be inferred from paragraph 5 about the Gutenberg Bible?

- ☐ It was printed in Frankfurt.
- ☐ People with average incomes were not likely to own it.
- ☐ Numerous copies of it were sold at the Frankfurt book fair.
- ☐ It was one of the first expensive books to be sold to an international population.

Paragraph 6

Some historians argue that the invention of mechanical printing gave rise to a communications revolution as significant as, for example, the widespread use of the personal computer today. The multiplication of standardized texts altered the thinking habits of Europeans by freeing individuals from having to memorize everything they learned; it certainly made possible the speedy and inexpensive dissemination of knowledge. It created a wider community of scholars, no longer dependent on personal patronage or church sponsorship for texts. Printing facilitated the free expression and exchange of ideas, and its disruptive potential did not go unnoticed by political and church authorities. Emperors and bishops in Germany, the homeland of the printing industry, moved quickly to issue censorship regulations.

10. The author mentions "the personal computer" in the passage in order to

- ☐ compare the importance of the development of mechanical printing to the development of the computer
- ☐ contrast the superiority of communications technology in the modern era to that of the fifteenth century
- ☐ emphasize the sophisticated social and economic growth of fifteenth-century society
- ☐ emphasize the conflicting views of historians about the significance of new technologies

11. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning

in important ways of leave out essential information.

- ☐ European ways of thinking were affected by the new multiple sources of knowledge that became available through the standardization of texts.
- ☐ Once standardized texts became widespread, people could acquire knowledge in faster, more affordable ways because they no longer had to memorize texts.
- ☐ Individuals were increasingly free to take advantage of the large number of standardized texts that became available.
- ☐ It became easier for people to increase their knowledge in many areas because books were significantly cheaper than they had previously been.

12. The word "facilitated" in the passage is closest in meaning to

- ☐ introduced
- ☐ standardized
- ☐ made easier
- ☐ transformed

Paragraph 6

Some historians argue that the invention of mechanical printing gave rise to a communications revolution as significant as, for example, the widespread use of the personal computer today. ■ The multiplication of standardized texts altered the thinking habits of Europeans by freeing individuals from having to memorize everything they learned; it certainly made possible the speedy and inexpensive dissemination of knowledge. ■ It created a wider community of scholars, no longer dependent on personal patronage or church sponsorship for texts. ■ Printing facilitated the free expression and exchange of ideas, and its disruptive potential did not go unnoticed by political and church authorities. ■ Emperors and bishops in Germany, the homeland of the printing industry, moved quickly to issue censorship regulations.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

For such leaders, the uncontrolled expression of ideas was clearly unwelcome.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click

on **View Text.**

The invention of printing was a revolutionary development that brought about profound social change in Europe in the fourteenth and fifteenth centuries.

-
-
-

Answer Choices

Printing with metal type rather than woodblocks was the invention of a German goldsmith living in Rome, the center of printing in the fifteenth century.

A precondition for the successful production and distribution of mechanically printed books was the earlier commercial organization of hand-copied book production by merchants known as stationers.

Because movable type made it more economical to produce many copies of a single book, the invention of mechanical printing had an immense effect on the spread of knowledge.

The expansion of the paper making industry in the fourteenth and fifteenth centuries made it possible to fully and economically use the invention of movable type in book production.

In the fifteenth century, printed books were internationally distributed through the Frankfurt book fair, although few people in the fifteenth century could afford books or were able to read them.

The most important and expensive early mechanically printed book was Gutenberg's two-volume Latin Bible, produced in an edition of only 185 copies.

Earth's Energy Cycle

To understand most of the processes at work on Earth, it is useful to envisage interactions within the Earth system as a series of interrelated cycles. One of these is the energy cycle, which encompasses the great "engines"—the external and internal energy sources—that drive the Earth system and all its cycles. We can think of Earth's energy cycle as a "budget": energy may be added to or subtracted from the budget and may be transferred from one storage place to another, but overall the additions and subtractions and transfers must balance each other. If a balance did not exist, Earth would either heat up or cool down until a balance was reached.

The total amount of energy flowing into Earth's energy budget is more than 174,000 terawatts (or $174,000 \times 10^{12}$ watts). This quantity completely dwarfs the 10 terawatts of energy that humans use per year. There are three main sources from which energy flows into the Earth system.

Incoming short-wavelength solar radiation overwhelmingly dominates the flow of energy in Earth's energy budget, accounting for about 99.986 percent of the total. An estimated 174,000 terawatts of solar radiation is intercepted by Earth. Some of this vast influx powers the winds, rainfall, ocean currents, waves, and other processes in the hydrologic (or water) cycle. Some is used for photosynthesis and is temporarily stored in the biosphere in the form of plant and animal life. When plants die and are buried, some of the solar energy is stored in rocks, when we burn coal, oil, or natural gas, we release stored solar energy.

The second most powerful source of energy, at 23 terawatts or 0.013 percent of the total, is geothermal energy, Earth's internal heat energy. Geothermal energy eventually finds its way to Earth's surface, primarily via volcanic pathways. It drives the rock cycle and is therefore the source of the energy that uplifts mountains, causes earthquakes and volcanic eruptions, and generally shapes the face of the Earth.

The smallest source of energy for Earth is the kinetic (motion) energy of Earth's rotation. The Moon's gravitational pull lifts a tidal bulge in the ocean; as Earth spins on its axis, this bulge remains essentially stationary. As Earth rotates, the tidal bulge runs into the coastlines of continents and islands, causing high tides. The force of the tidal bulge *piling up* against land masses acts as a very slow brake, actually causing Earth's rate of rotation to decrease slightly. The transfer of tidal energy accounts for approximately 3 terawatts, or 0.002 percent of the tidal energy budget.

Earth loses energy from the cycle in two main ways: reflection, and degradation

and reradiation. About 40 percent of incoming solar radiation is simply reflected, unchanged, back into space by the clouds, the sea, and other surfaces. For any planetary body, the percentage of incoming radiation that is reflected is called the “albedo.” Each different material has a characteristic reflectivity. For example, ice is more reflectant than rocks or pavement; water is more highly reflectant than vegetation; and forested land reflects light differently than agricultural land. Thus, if large expanses of land are converted from forest to plowed land, or from forest to city, the actual reflectivity of Earth’s surface, and hence its albedo, may be altered. Any change in albedo will, of course, have an effect on Earth’s energy budget.

The portion of incoming solar energy that is not reflected back into space, along with tidal and geothermal energy, is absorbed by materials at Earth’s surface, in particular the atmosphere and hydrosphere. This energy undergoes a series of irreversible degradations in which it is transferred from one reservoir to another and converted from one form to another. The energy that is absorbed, utilized, transferred, and degraded eventually ends up as heat, in which form it is reradiated back into space as long-wavelength (infrared) radiation. Weather patterns are a manifestation of energy transfer and degradation.

Paragraph 1

To understand most of the processes at work on Earth, it is useful to envisage interactions within the Earth system as a series of interrelated cycles. One of these is the energy cycle, which encompasses the great “engines”—the external and internal energy sources—that drive the Earth system and all its cycles. We can think of Earth’s energy cycle as a “budget”: energy may be added to or subtracted from the budget and may be transferred from one storage place to another, but overall the additions and subtractions and transfers must balance each other. If a balance did not exist, Earth would either heat up or cool down until a balance was reached.

1. The word “encompasses” in the passage is closest in meaning to
explains
includes
combines
creates
2. In paragraph 1, the author introduces the concept of a budget in order to
indicate how different cycles in the Earth system relate to each other
illustrate how Earth’s energy cycle must maintain an overall balance
show that Earth gains energy from both external and internal sources
explain how energy is transferred from one storage place to another

Paragraph 2

The total amount of energy flowing into Earth's energy budget is more than 174,000 terawatts (or $174,000 \times 10^{12}$ watts). This quantity completely dwarfs the 10 terawatts of energy that humans use per year. There are three main sources from which energy flows into the Earth system.

3. Why does the author include information about the “energy that humans use per year” in the discussion?

To call into question the idea that humans can use up all the energy available in Earth's energy budget

To provide a comparison that established how huge the amount of energy flowing into Earth's energy budget is

To explain why there must be more than one source of energy for the Earth system

To argue that the use of energy by humans amounts to such a small part of Earth's energy budget that it cannot have significant effects

Paragraph 3

Incoming short-wavelength solar radiation overwhelmingly dominates the flow of energy in Earth's energy budget, accounting for about 99.986 percent of the total.

An estimated 174,000 terawatts of solar radiation is intercepted by Earth. Some of this vast influx powers the winds, rainfall, ocean currents, waves, and other processes in the hydrologic (or water) cycle. Some is used for photosynthesis and is temporarily stored in the biosphere in the form of plant and animal life. When plants die and are buried, some of the solar energy is stored in rocks, when we burn coal, oil, or natural gas, we release stored solar energy.

4. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

Almost all of the short-wavelength energy in Earth's energy budget comes from solar radiation.

Short-wavelength radiation is by far the largest part of the total energy that the Sun radiates to Earth.

The amount of short-wavelength radiation received from the Sun is huge by comparison to Earth's own energy production.

Almost the entire amount of energy that flows into Earth's energy budget is short-wavelength radiation from the Sun.

5. According to paragraph 3, solar radiation powers all of the following EXCEPT photosynthesis in plants

winds

formation of rocks

processes in the hydrologic cycle

Paragraph 4

The second most powerful source of energy, at 23 terawatts or 0.013 percent of the total, is geothermal energy, Earth's internal heat energy. Geothermal energy eventually finds its way to Earth's surface, primarily via volcanic pathways. It drives the rock cycle and is therefore the source of the energy that uplifts mountains, causes earthquakes and volcanic eruptions, and generally shapes the face of the Earth.

6. According to paragraph 4, all of the following statements about geothermal energy are true EXCEPT

It is the main source of heat for the surface of the Earth.

It is responsible for earthquakes.

It causes the eruptions of volcanoes.

It causes mountains to rise high above the rest of Earth's surface.

Paragraph 5

The smallest source of energy for Earth is the kinetic (motion) energy of Earth's rotation. The Moon's gravitational pull lifts a tidal bulge in the ocean; as Earth spins on its axis, this bulge remains essentially stationary. As Earth rotates, the tidal bulge runs into the coastlines of continents and islands, causing high tides. The force of the tidal bulge *piling up* against land masses acts as a very slow brake, actually causing Earth's rate of rotation to decrease slightly. The transfer of tidal energy accounts for approximately 3 terawatts, or 0.002 percent of the tidal energy budget.

7. The word "stationary" in the passage is closest in meaning to

isolated

visible

raised

unmoving

8. Paragraph 5 mentions which of the following as an effect of the Moon's gravitation on Earth?

It causes high tides that reshape the coastlines of continents and islands.

It causes Earth to rotate on its axis at a somewhat faster speed than it would otherwise.

It pulls ocean water into a bulge that runs into land masses as Earth rotates on its axis.

It reduces the force with which the tidal bulge would otherwise pile up against continents.

Paragraph 6

Earth loses energy from the cycle in two main ways: reflection, and degradation and reradiation. About 40 percent of incoming solar radiation is simply reflected, unchanged, back into space by the clouds, the sea, and other surfaces. For any

planetary body, the percentage of incoming radiation that is reflected is called the "albedo." Each different material has a characteristic reflectivity. For example, ice is more reflectant than rocks or pavement; water is more highly reflectant than vegetation; and forested land reflects light differently than agricultural land. Thus, if large expanses of land are converted from forest to plowed land, or from forest to city, the actual reflectivity of Earth's surface, and hence its albedo, may be altered. Any change in albedo will, of course, have an effect on Earth's energy budget.

9. The word "hence" in the passage is closest in meaning to
 therefore
 perhaps
 sometimes
 obviously

10. What can be inferred from paragraph 6 about what would likely occur if cloud cover increased worldwide?

Different materials would become more similar to each other in their reflectivity.
 It would become a greater necessity to convert forests into plowed land and cities.

A large percentage of incoming solar radiation would be reflected back into space.

The reflectivity of ice and water would change and become greater over time.

Paragraph 7

The portion of incoming solar energy that is not reflected back into space, along with tidal and geothermal energy, is absorbed by materials at Earth's surface, in particular the atmosphere and hydrosphere. This energy undergoes a series of irreversible degradations in which it is transferred from one reservoir to another and converted from one form to another. The energy that is absorbed, utilized, transferred, and degraded eventually ends up as heat, in which form it is reradiated back into space as long-wavelength (infrared) radiation. Weather patterns are a manifestation of energy transfer and degradation.

11. The word "irreversible" in the passage is closest in meaning to
 severe
 permanent
 complex
 poorly understood

12. According to paragraph 7, weather patterns are produced as part of the cycle in which

incoming solar energy becomes reflected back into space
 solar energy is converted into geothermal and tidal energy

the atmosphere and hydrosphere absorb long-wavelength radiation energy that has been absorbed near Earth's surface undergoes transfer and conversion of form

Paragraph 6

Earth loses energy from the cycle in two main ways: reflection, and degradation and reradiation. About 40 percent of incoming solar radiation is simply reflected, unchanged, back into space by the clouds, the sea, and other surfaces. ■ For any planetary body, the percentage of incoming radiation that is reflected is called the "albedo." ■ Each different material has a characteristic reflectivity. ■ For example, ice is more reflectant than rocks or pavement; water is more highly reflectant than vegetation; and forested land reflects light differently than agricultural land. ■ Thus, if large expanses of land are converted from forest to plowed land, or from forest to city, the actual reflectivity of Earth's surface, and hence its albedo, may be altered. Any change in albedo will, of course, have an effect on Earth's energy budget.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

How reflective a material is depends on how light or dark it is, among other things.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Earth's energy cycle consists of all the energy inputs, outputs, and conversions within the Earth system, which must maintain an overall balance.

-
-
-

Answer Choices

Incoming short-wavelength solar radiation provides Earth with nearly all its energy and powers the hydrologic cycle as well as biological processes.

Heat energy from Earth's interior, which powers the rock cycle, and the kinetic energy of Earth's rotation provide small additions to solar energy.

Some of the incoming solar radiation is reflected, and the rest, after being absorbed, undergoes a series of conversions until it is reradiated into space as heat.

Humans use only a small amount of the available solar energy for heat, satisfying most of their energy needs by burning coal, oil, and natural gas.

Solar energy stored in rocks on Earth's surface is the primary source of geothermal energy and tidal energy.

Earth's atmosphere and hydrosphere absorb most of the incoming solar radiation, using up much of the energy to power weather patterns, with only the remainder radiated out as heat.

Removing Dams

For nearly a century, two United States governmental agencies, the United States Army Corps of Engineers and the Bureau of Reclamation, have constructed dams to store water and to generate electricity. Building these dams provided cheap electricity, created jobs for workers, stimulated regional economic development, and allowed farming on lands that would otherwise be too dry. But not everyone agrees that big dam projects are entirely beneficial. Their storage reservoirs stop the flow of rivers and often submerge towns, farms, and historic sites. They prevent fish migrations and change aquatic habitats essential for native species.

The tide may have turned, in fact, against dam building. In 1998 the Army Corps announced that it would no longer be building large dams. In the few remaining sites where dams might be built, public opposition is so great that getting approval for projects is unlikely. Instead, the new focus may be on removing existing dams and restoring natural habitats. In 1999 Bruce Babbitt, then the United States interior secretary, said, "Of the 75,000 large dams in the United States, most were built a long time ago and are now obsolete, expensive, and unsafe. They were built with no consideration of the environmental costs. As operating licenses come up for renewal, dam removal and habitat restoration to original stream flows will be among the options considered."

The first active hydroelectric dam in the United States to be removed against the wishes of its owners was the 162-year-old Edwards Dam, on the Kennebec River in Augusta, Maine. For many years, the United States Fish and Wildlife Service had advocated the removal of this dam, which prevented migration of salmon, shad, sturgeon, and other fish species up the river. In a precedent-setting decision, the Federal Energy Regulatory Commission ordered the dam removed after concluding that the environmental and economic benefits of a free-flowing river outweighed the electricity generated by the dam. In July 1999 the dam was removed and restoration work began on wetlands and stream banks long underwater.

The next dams likely to be taken down are the Elwha and Glines Dams on the Elwha River in Olympic National Park in the state of Washington. Built nearly a century ago to provide power to lumber and paper mills in the town of Port Angeles, these dams blocked access to upstream spawning beds for six species of salmon on what once was one of the most productive salmon rivers in the world. Simply removing the dams will not restore the salmon, however. Where 50-kilogram king salmon once fought their way up waterfalls to lay their eggs in

gravel beds, there now are only concrete walls holding back still water and deep beds of muddy deposits. Removing the mud, uncovering gravel beds where fish spawn, and finding suitable salmon types to rebuild the population is a daunting task. Congress will have to appropriate somewhere around \$300 to \$400 million to remove these two relatively small dams and rehabilitate the area.

Environmental groups, encouraged by these examples, have begun to talk about much more ambitious projects. Four giant dams on the Snake River in Washington State, for example, might be removed to restore salmon and steelhead fish runs to the headwaters of the Columbia River. The Hetch Hetchy Dam in Yosemite National Park might be taken down to reveal what John Muir, the founder of the prestigious environmental organization Sierra Club, called a valley “just as beautiful and worthy of preservation as the majestic Yosemite.” Some groups have even suggested removing the Glen Canyon Dam on the Colorado River. In each of these cases, powerful interests stand in opposition. These dams generate low-cost electricity and store water that is needed for agriculture and industry. Local economies, domestic water supplies, and certain types of recreation all would be severely impacted by destruction of these dams.

How does one weigh the many different economic, cultural, and aesthetic considerations for removing or not removing these dams? Do certain interests, such as the rights of native people or the continued existence of native species of fish or wildlife, take precedence over economic factors, or should this be a utilitarian calculation of the greatest good for the greatest number? And does that number include only humans or do other species count as well?

Paragraph 1

For nearly a century, two United States governmental agencies, the United States Army Corps of Engineers and the Bureau of Reclamation, have constructed dams to store water and to generate electricity. Building these dams provided cheap electricity, created jobs for workers, stimulated regional economic development, and allowed farming on lands that would otherwise be too dry. But not everyone agrees that big dam projects are entirely beneficial. Their storage reservoirs stop the flow of rivers and often submerge towns, farms, and historic sites. They prevent fish migrations and change aquatic habitats essential for native species.

1. According to paragraph 1, building dams was beneficial in each of the following ways **EXCEPT**
 - ☐ increasing the amount of land that could be used for farming
 - ☐ strengthening local economies
 - ☐ increasing the availability of low-cost electricity
 - ☐ expanding the aquatic habitats of native species

Paragraph 2

The tide may have turned, in fact, against dam building. In 1998 the Army Corps announced that it would no longer be building large dams. In the few remaining sites where dams might be built, public opposition is so great that getting approval for projects is unlikely. Instead, the new focus may be on removing existing dams and restoring natural habitats. In 1999 Bruce Babbitt, then the United States interior secretary, said, "Of the 75,000 large dams in the United States, most were built a long time ago and are now **obsolete**, expensive, and unsafe. They were built with no consideration of the environmental costs. As operating licenses come up for renewal, dam removal and habitat restoration to original stream flows will be among the options considered."

2. According to paragraph 2, the likelihood that new dams will be built has decreased because
 - ☐ construction costs have increased enormously
 - ☐ safety standards have become much higher
 - ☐ public opposition to dam construction has increased
 - ☐ at most suitable sites an existing dam would have to be removed first
3. The word "**obsolete**" in the passage is closest in meaning to
 - ☐ unpopular
 - ☐ inefficient
 - ☐ out of date
 - ☐ unnecessary
4. Paragraph 2 supports which of the following ideas about operating licenses for large dams?
 - ☐ Since 1999 licenses have been renewed only for small dams.
 - ☐ Before 1999, owners applying for a license renewal were more likely to have their applications than they were after that date.
 - ☐ Strong public opposition to their renewal was common even before 1999, but it was based on safety considerations, not on environmental ones.
 - ☐ The environmental cost of dams has been a minor consideration in license renewal applications since 1999.

Paragraph 3

The first active hydroelectric dam in the United States to be removed against the wishes of its owners was the 162-year-old Edwards Dam, on the Kennebec River in Augusta, Maine. For many years, the United States Fish and Wildlife Service had advocated the removal of this dam, which prevented migration of salmon, shad, sturgeon, and other fish species up the river. In a precedent-setting decision, the Federal Energy Regulatory Commission ordered the dam removed after concluding that the environmental and economic benefits of a free-flowing river outweighed the electricity generated by the dam. In July 1999 the dam was removed and restoration work began on wetlands and stream banks long

underwater.

5. According to paragraph 3, why did the United States Fish and Wildlife Service want the Edwards Dam removed?
 - ☐ Because the age of the dam made it unsafe
 - ☐ Because the dam was negatively affecting various species of fish
 - ☐ Because the dam had caused wetlands to form
 - ☐ Because the dam no longer provided economic benefits
6. Paragraph 3 suggests that one main consideration for keeping the Edwards Dam was
 - ☐ the electricity it generated
 - ☐ the length of time it had been in operation
 - ☐ the high cost of removing it
 - ☐ the fact that removing it would set a bad example

Paragraph 4

The next dams likely to be taken down are the Elwha and Glines Dams on the Elwha River in Olympic National Park in the state of Washington. Built nearly a century ago to provide power to lumber and paper mills in the town of Port Angeles, these dams blocked access to upstream spawning beds for six species of salmon on what once was one of the most productive salmon rivers in the world. Simply removing the dams will not restore the salmon, however. Where 50-kilogram king salmon once fought their way up waterfalls to lay their eggs in gravel beds, there now are only concrete walls holding back still water and deep beds of muddy deposits. Removing the mud, uncovering gravel beds where fish spawn, and finding suitable salmon types to rebuild the population is a daunting task. Congress will have to appropriate somewhere around \$300 to \$400 million to remove these two relatively small dams and rehabilitate the area.

7. According to paragraph 4, why would removing the Elwha and Glines dams not be enough to restore salmon to the Elwha River?
 - ☐ They are not the only dams on the Elwha River.
 - ☐ The lumber and paper mills in Port Angeles also block access to upstream spawning beds.
 - ☐ Too many species of salmon are competing for survival in one river.
 - ☐ The dams have left the river's spawning beds in an unusable condition.
8. The word "suitable" in the passage is closest in meaning to
 - ☐ new
 - ☐ healthy
 - ☐ appropriate
 - ☐ similar

Paragraph 5

Environmental groups, encouraged by these examples, have begun to talk about much more **ambitious** projects. Four giant dams on the Snake River in Washington State, for example, might be removed to restore salmon and steelhead fish runs to the headwaters of the Columbia River. The Hetch Hetchy Dam in Yosemite National Park might be taken down to reveal what John Muir, the founder of the prestigious environmental organization Sierra Club, called a valley "just as beautiful and worthy of preservation as the majestic Yosemite." Some groups have even suggested removing the Glen Canyon Dam on the Colorado River. In each of these cases, powerful interests stand in opposition. These dams generate low-cost electricity and store water that is needed for agriculture and industry. Local economies, domestic water supplies, and certain types of recreation all would be severely impacted by destruction of these dams.

9. The word "**ambitious**" in the passage is closest in meaning to
- ☐ impressive but difficult to achieve
 - ☐ dangerous and require considerable planning
 - ☐ complex and unlikely to be complete
 - ☐ greatly needed
10. According to paragraph 5, why do environmental groups want the Hetch Hetchy Dam removed?
- ☐ To restore salmon and steelhead runs to the Snake River
 - ☐ To allow access to the headwaters of the Columbia River
 - ☐ To increase the size of Yosemite National Park
 - ☐ To restore a valley to its original beauty

Paragraph 6

How does one weigh the many different economic, cultural, and aesthetic considerations for removing or not removing these dams? Do certain interests, such as the rights of native people or the continued existence of native species of fish or wildlife, **take precedence over** economic factors, or should this be a utilitarian calculation of the greatest good for the greatest number? And does that number include only humans or do other species count as well?

11. The phrase "**take precedence over**" in the passage is closest in meaning to
- ☐ affect
 - ☐ have greater importance than
 - ☐ get included among
 - ☐ minimize
12. What is the role of paragraph 6 in the passage?
- ☐ To propose a method for deciding whether a given dam should be removed
 - ☐ To emphasize the complexity of the issues involved in deciding what should

be done about dams

- ☐ To suggest that the recent tendency not to build new dams may be wrong
- ☐ To sum up the points made earlier in the passage about the advantages and disadvantages of removing dams

Paragraph 4

The next dams likely to be taken down are the Elwha and Glines Dams on the Elwha River in Olympic National Park in the state of Washington. Built nearly a century ago to provide power to lumber and paper mills in the town of Port Angeles, these dams blocked access to upstream spawning beds for six species of salmon on what once was one of the most productive salmon rivers in the world. Simply removing the dams will not restore the salmon, however. ■ Where 50-kilogram king salmon once fought their way up waterfalls to lay their eggs in gravel beds, there now are only concrete walls holding back still water and deep beds of muddy deposits. ■ Removing the mud, uncovering gravel beds where fish spawn, and finding suitable salmon types to rebuild the population is a daunting task. ■ Congress will have to appropriate somewhere around \$300 to \$400 million to remove these two relatively small dams and rehabilitate the area. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

But aside from the technical challenges, the project will also pose a serious financial challenge.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Many dams were built in the United States during the last century, and they provided a broad range of economic benefits.

-
-
-

Answer Choices

Until recently, the emphasis in dam building was on the economic benefits

of low-cost energy and water that dams provided, but more attention is now being paid to the damage they cause.

Environmental groups now have a very good chance of forcing the removal of two major dams, the Glen Canyon Dam on the Colorado and the Hetch Hetchy Dam in Yosemite.

The removal of dams remains controversial because of high restoration costs, loss of low-cost electricity, and the loss of water storage facilities.

Since the late 1990s, the government has stopped building large dams, instead focusing on removing existing dams and restoring natural habitats.

Until recently, the main reason for removing dams was to restore salmon runs, but it is now recognized that a more important reason to remove dams is that they are no longer safe.

Although the U. S. government originally planned to remove the Elwha and Glines Dams in Washington, the enormous expense of removal has resulted in a postponement of this effort.



Coral Reef Communities

Coral reefs are massive underwater structures made from the hard limestone exoskeletons of thousands of tiny living organisms (coral polyps) produced one on top of another in warm, clear, shallow ocean waters. Living polyps extend upward and outward from the coral colony center and live on top of the old dead exoskeletons. Coral reef communities are crowded with other animals representing virtually every major animal phylum. Space is at a premium on reefs, corals, seaweeds (various forms of algae), sponges, or other organisms cover virtually every surface. Because both corals and algae require light to survive, access to light, like space, is also a resource subject to competition.

Fast-growing, branching corals can grow over slower-growing, encrusting, or massive corals and deny them light. In response, the slower-growing forms can extend stinging filaments from their digestive cavity and kill their competitor's polyps. Undamaged polyps on the faster-growing, branching coral, however, may grow very long sweeper tentacles, containing powerful nematocysts (stingers) that kill polyps on the slower-growing form. The faster-growing form repairs the damage and continues to overgrow its competitor. In addition to sweeper tentacles and stinging filaments, corals have several other mechanisms available for attack or defense.

In general, slower-growing corals are more aggressive than fast-growing species. In cases where a competitor cannot be overcome, however, corals may survive by taking advantage of differences in local habitats. Massive corals are generally more shade tolerant and able to survive at greater depths. Therefore, on many reefs it is the fast-growing, branching corals that ultimately dominate at the upper, shallower portion of the reef, whereas more massive forms dominate in deeper areas.

Corals also must compete with other reef organisms, each with its own strategies for survival. Sponges, soft corals, and seaweeds (algae) can overgrow stony corals and smother them. Algae are competitively superior to corals in shallow water but less so at depth. Survival of coral in shallow water, therefore, may depend on grazing by plant-eating echinoderms (starfish and sea urchins) and fishes. In Jamaica, overfishing removed most of the plant-eating fish from coral reefs. Initially, algal growth was kept in check by grazing sea urchins, but in 1982, a pathogen reduced the population by 99 percent. Without grazers, the algae were able to completely overgrow the coral.

Competition may occur among other reef communities. Grazing by urchins and fishes is important in preventing seaweeds from overgrowing the reef. The

dominant algae on a healthy reef are usually fast-growing filamentous forms or coralline algae, well protected by calcification (hardening) and the production of noxious chemicals. These algae are inferior competitors to larger, fleshier seaweeds, so grazing by urchins and fishes on the larger seaweeds allows these algae to persist. Grazing on plants is greatest in the shallow reef areas but decreases with depth, where lower temperatures and light reduce algal growth. The reef is, therefore, a mosaic of microhabitats with different levels of grazing and different algal communities.

An additional complexity arises from the activity of damselfish. Because they are territorial, many damselfish species exclude grazers and other species from certain areas of the reef. Algae grow rapidly in these territories, providing habitat for many small invertebrates but overgrowing the corals. Branching corals tend to dominate in damselfish territories because they are upright and faster growing than the more massive or encrusting forms.

Although less studied than on rocky shores, predation almost certainly has a significant influence on the community structure of coral reefs. Fish and other predators may preferentially prey on such competitors of corals as sponges and gorgonians, giving competitively inferior reef corals an advantage in securing space. Many species of fish, mollusks, and crustaceans also feed directly on coral polyps. Several surgeonfish and parrotfish may actually pass coral skeletons through their digestive tracts and add sediment to the reef. Both fish and invertebrate corallivores (coral-feeding organisms) seem to attack faster-growing, branching species preferentially, perhaps preventing slower-growing forms from being overgrown. Corallivores, however, rarely ever completely destroy a coral colony except in cases where tropical storms or humans have already done severe damage. The fact that almost all small invertebrates on reefs are so well hidden or highly camouflaged is another indicator of how prevalent predation is on reefs and its importance in determining reef structure.

Paragraph 1

Coral reefs are massive underwater structures made from the hard limestone exoskeletons of thousands of tiny living organisms (coral polyps) produced one on top of another in warm, clear, shallow ocean waters. Living polyps extend upward and outward from the coral colony center and live on top of the old dead exoskeletons. Coral reef communities are crowded with other animals representing virtually every major animal phylum. Space is at a premium on reefs, corals, seaweeds (various forms of algae), sponges, or other organisms cover virtually every surface. Because both corals and algae require light to survive, access to light, like space, is also a resource subject to competition.

1. According to paragraph 1, all of the following are true of coral reefs **EXCEPT**:
☐ Coral reefs grow biggest in the deepest waters of the ocean.

- The organisms living around coral reefs compete for limited resources.
- There are many different organisms in coral reef communities.
- Coral reefs consist of the outer skeletons of small living organisms.

Paragraph 2

Fast-growing, branching corals can grow over slower-growing, encrusting, or massive corals and deny them light. In response, the slower-growing forms can extend stinging filaments from their digestive cavity and kill their competitor's polyps. Undamaged polyps on the faster-growing, branching coral, however, may grow very long sweeper tentacles, containing powerful nematocysts (stingers) that kill polyps on the slower-growing form. The faster-growing form repairs the damage and continues to overgrow its competitor. In addition to sweeper tentacles and stinging filaments, corals have several other mechanisms available for attack or defense.

2. According to paragraph 2, how do fast-growing branching corals defend themselves from attacks by slower-growing corals?
 - By producing stinging sweeper tentacles
 - By growing on top of the slower-growing corals
 - By blocking the light to the slower-growing corals
 - By destroying the stinging filament of the slower-growing corals
3. The word "mechanisms" in the passage is closest in meaning to
 - adaptations
 - weapons
 - parts
 - means

Paragraph 3

In general, slower-growing corals are more aggressive than fast-growing species. In cases where a competitor cannot be overcome, however, corals may survive by taking advantage of differences in local habitats. Massive corals are generally more shade tolerant and able to survive at greater depths. Therefore, on many reefs it is the fast-growing, branching corals that ultimately dominate at the upper, shallower portion of the reef, whereas more massive forms dominate in deeper areas.

4. The word "ultimately" in the passage is closest in meaning to
 - naturally
 - eventually
 - quickly
 - clearly

Paragraph 4

Corals also must compete with other reef organisms, each with its own strategies for survival. Sponges, soft corals, and seaweeds (algae) can overgrow stony corals and smother them. Algae are competitively superior to corals in shallow water but less so at depth. Survival of coral in shallow water, therefore, may depend on grazing by plant-eating echinoderms (starfish and sea urchins) and fishes. In Jamaica, overfishing removed most of the plant-eating fish from coral reefs. Initially, algal growth was kept in check by grazing sea urchins, but in 1982, a pathogen reduced the population by 99 percent. Without grazers, the algae were able to completely overgrow the coral.

5. The phrase "kept in check" in the passage is closest in meaning to
 - ☐ limited
 - ☐ prevented
 - ☐ allowed
 - ☐ stimulated

6. In paragraph 4, why does the author discuss the effects of removing plant-eating fish and sea urchins from coral reefs?
 - ☐ To identify a situation that contributes to the dominance of corals in shallow waters
 - ☐ To demonstrate the importance of grazing on seaweeds for the survival of some corals
 - ☐ To provide evidence that seaweeds are better competitors than coral at depth
 - ☐ To argue that sea urchin pathogens also attack corals

7. According to paragraph 4, all of these pairs of organisms are in competition **EXCEPT**
 - ☐ corals and sponges
 - ☐ algae and corals
 - ☐ echinoderms and corals
 - ☐ sea urchins and algae

Paragraph 5

Competition may occur among other reef communities. Grazing by urchins and fishes is important in preventing seaweeds from overgrowing the reef. The dominant algae on a healthy reef are usually fast-growing filamentous forms or coralline algae, well protected by calcification (hardening) and the production of noxious chemicals. These algae are inferior competitors to larger, fleshier seaweeds, so grazing by urchins and fishes on the larger seaweeds allows these algae to persist. Grazing on plants is greatest in the shallow reef areas but decreases with depth, where lower temperatures and light reduce algal growth. The reef is, therefore, a mosaic of microhabitats with different levels of grazing and different algal communities.

8. According to paragraph 5, fast-growing filamentous or coralline algae are usually the dominant algae on healthy coral reefs in part because they
- ☐ are not affected by noxious chemicals produced by other organisms
 - ☐ are less attractive as food for sea urchins and fishes than bigger seaweeds are
 - ☐ occupy the areas of coral reefs that have lower temperatures and less light
 - ☐ can live in a wider variety of microhabitats than their competitors can

Paragraph 6

An additional complexity arises from the activity of damselfish. Because they are territorial, many damselfish species exclude grazers and other species from certain areas of the reef. Algae grow rapidly in these territories, providing habitat for many small invertebrates but overgrowing the corals. Branching corals tend to dominate in damselfish territories because they are upright and faster growing than the more massive or encrusting forms.

9. According to paragraph 6, which of the following is an effect of the activity of damselfish on the reef environment?
- ☐ Encrusting corals dominate in damselfish territories.
 - ☐ The damselfish attract many species of grazers.
 - ☐ Algae grow rapidly.
 - ☐ Algal diversity is greater than in other parts of the reef.

Paragraph 7

Although less studied than on rocky shores, predation almost certainly has a significant influence on the community structure of coral reefs. Fish and other predators may preferentially prey on such competitors of corals as sponges and gorgonians, giving competitively inferior reef corals an advantage in securing space. Many species of fish, mollusks, and crustaceans also feed directly on coral polyps. Several surgeonfish and parrotfish may actually pass coral skeletons through their digestive tracts and add sediment to the reef. Both fish and invertebrate corallivores (coral-feeding organisms) seem to attack faster-growing, branching species preferentially, perhaps preventing slower-growing forms from being overgrown. Corallivores, however, rarely ever completely destroy a coral colony except in cases where tropical storms or humans have already done severe damage. The fact that almost all small invertebrates on reefs are so well hidden or highly camouflaged is another indicator of how prevalent predation is on reefs and its importance in determining reef structure.

10. The word "securing" in the passage is closest in meaning to
- ☐ looking for
 - ☐ getting
 - ☐ maximizing
 - ☐ sharing

11. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ The presence of only very small invertebrates on a reef is an indication of widespread predation.
 - ☐ Most reefs are structured in a way which allows small invertebrates to remain hidden or camouflaged.
 - ☐ Almost all small invertebrates are hidden or camouflaged, indicating the extent and importance of predation to reef structure.
 - ☐ Almost all of the small invertebrates on reefs are difficult to find because they are so highly camouflaged.
12. Paragraph 7 mentions all of the following as effects of predation on the community structure of coral reefs **EXCEPT**:
- ☐ Corals are advantaged when predators prefer to attack competitors or corals.
 - ☐ Faster-growing corals are prevented from overgrowing slower-growing corals when faster-growing species are preferred by competitors.
 - ☐ Predation contributes to the sediment deposit of the reef.
 - ☐ Small invertebrates are exposed to competitively superior organisms.

Paragraph 3

In general, slower-growing corals are more aggressive than fast-growing species. ■ In cases where a competitor cannot be overcome, however, corals may survive by taking advantage of differences in local habitats. ■ Massive corals are generally more shade tolerant and able to survive at greater depths. ■ Therefore, on many reefs it is the fast-growing, branching corals that ultimately dominate at the upper, shallower portion of the reef, whereas more massive forms dominate in deeper areas. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

For example, different species of corals have different needs for light.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Coral reef communities include many different organisms that must compete for resources such as space and light.

-
-
-

Answer Choices

In order to keep from being overgrown, the different species of coral kill each other's polyps, or live in different local habitats within the reef community.

Competition among the different species of corals is more intense than that between corals and other coral reef inhabitants.

Predation shapes reef structure by getting rid of competitors of corals, but coral polyps themselves are also eaten, as are many reef inhabitants.

Coral reefs are divided into a shallow upper portion and a deeper lower portion with branching corals dominating in the lower portion.

Grazing by fishes and urchins prevents algae and seaweeds from overgrowing the corals, although damselfish exclude grazers from some areas.

Fish and invertebrate corallivores are the most common cause of coral colony destruction, followed by tropical storms and damage by humans.



Sociality in Animals

Social insects represent the high point of invertebrate evolution. Some species live in communities of millions, coordinating their building and foraging, their reproduction, and their offspring care. Yet sociality is found in only a few species of insects, and is rare among vertebrates as well: wildebeest (large antelope) and lions are the exception rather than the rule. Nearly all fish, amphibians, reptiles, birds, and mammals are solitary, except when courting and mating. Birds and mammals usually rear their young, but year-round family groups are almost unknown, though they are intensely studied where they do exist. The same is true for insects.

We know, or think we know, that social groups are good. Wolves are better predators when they hunt in packs, and pigeons escape from falcons far more often when feeding in flocks. Group building projects—the dams beavers build to block a body of water that provides them with relative safety from predators and the lodges they build for shelter, for instance—can provide a high level of protection and comfort. Why, then, are social species so very rare? In fact, living socially presents inevitable problems that transcend habitat needs so that only when these costs are offset by corresponding benefits is group living a plus.

The most obvious cost is competition. All the members of a species share the same habitat; when they live together, they are trying to eat the same food and occupy the same nesting sites. In general, there is far less competition away from a group, and selection should favor any individual who (all things being equal) sets off on its own, leaving the members of its group behind to compete among themselves for limited resources. Another difficulty is that concentrations of individuals facilitate disease and parasite transmission. On the whole, social animals carry more parasites and species-specific diseases than do solitary animals. Parasites and diseases diminish the strength and limit the growth of animals, and among highly social creatures, epidemics can devastate whole populations. Distemper (a viral disease) has been known to wipe out entire colonies of seals, for instance. So the penalty of social life is potentially huge.

But in some instances, the payoffs can be even greater. Two have already been mentioned: cooperative hunting and defensive groups. Social hunting is likely to evolve where prey is too large to be taken by individuals operating alone. To capture wildebeest some members of a group of lions follow their prey and herd them toward others lying in ambush. In other species, individuals forage or hunt simultaneously and share the food. Vampire bats that have had a bad day, for instance, are fed by more successful members of the community, but they are expected to return the favor in the future. Cooperation can even involve sharing

information about the location of food. Some colonial birds, such as bank swallows, use the departure direction of a successful forager (food hunter) to locate concentrations of prey. Information transfer can be unintentional though some species make use of special assembly calls or behavior.

Cooperation in group defense, such as we see in circles of musk oxen or elephants, is quite rare among vertebrates but is prevalent among the social insects. The strategy of employing many eyes to watch for danger, on the other hand, is widespread in birds and mammals. A herd of gazelles (small antelope) is far more likely to spot a lurking lion or a concealed cheetah than is a lone individual, and at a greater distance. In fact, a group enters into a kind of time-sharing arrangement in which individual antelope alternate biting off a mouthful of grass with a period of erect and watchful chewing. A larger group can afford more bites per individual per minute, there being more eyes to scan for danger. For a small antelope living in a forest where visibility is limited, however, remaining hidden is probably a better bet than assembling into noisy herds.

Among the millions of species of insects, only a few thousand are social. Those rarities are generally confined to termites and Hymenoptera. All termites are social: their diet (cellulose) requires that each generation feed a special kind of bacteria or fungi to the next generation to aid in its digestion. Of the numerous hymenopterans, some are social—including all ants and a few bees and wasps—but many are solitary.

Paragraph 1

Social insects represent the high point of invertebrate evolution. Some species live in communities of millions, coordinating their building and foraging, their reproduction, and their offspring care. Yet sociality is found in only a few species of insects, and is rare among vertebrates as well: wildebeest (large antelope) and lions are the exception rather than the rule. Nearly all fish, amphibians, reptiles, birds, and mammals are solitary, except when courting and mating. Birds and mammals usually rear their young, but year-round family groups are almost unknown, though they are intensely studied where they do exist. The same is true for insects.

1. According to paragraph 1, which of the following is true of sociality among animal species?
 - ☐ Sociality is much more common among invertebrates than among vertebrates.
 - ☐ Very few animals are considered social because most spend the majority of their lives alone.
 - ☐ An animal group must contain more than one family unit for the species to be considered social.
 - ☐ All animals that rear their young are considered to be social.

Paragraph 2

We know, or think we know, that social groups are good. Wolves are better predators when they hunt in packs, and pigeons escape from falcons far more often when feeding in flocks. Group building projects—the dams beavers build to block a body of water that provides them with relative safety from predators and the lodges they build for shelter, for instance—can provide a high level of protection and comfort. Why, then, are social species so very rare? In fact, living socially presents inevitable problems that transcend habitat needs so that only when these costs are offset by corresponding benefits is group living a plus.

2. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Group living allows some animals to transcend problems of habitat.
 - ☐ Group living is only advantageous when benefits outweigh inevitable problems.
 - ☐ Problems resulting from living socially extend beyond the habitat.
 - ☐ It is difficult to determine whether the benefits of social living offset its costs.

Paragraph 3

The most obvious cost is competition. All the members of a species share the same habitat; when they live together, they are trying to eat the same food and occupy the same nesting sites. In general, there is far less competition away from a group, and selection should favor any individual who (all things being equal) sets off on its own, leaving the members of its group behind to compete among themselves for limited resources. Another difficulty is that concentrations of individuals facilitate disease and parasite transmission. On the whole, social animals carry more parasites and species-specific diseases than do solitary animals. Parasites and diseases diminish the strength and limit the growth of animals, and among highly social creatures, epidemics can devastate whole populations. Distemper (a viral disease) has been known to wipe out entire colonies of seals, for instance. So the penalty of social life is potentially huge.

3. What does paragraph 3 say about the relationship between natural selection and animals that live apart from other members of their species?
- ☐ Natural selection does not favor living apart because individuals are unable to obtain resources equal to those of group members.
 - ☐ Natural selection does not favor living apart because of the intense competition for resources among individuals apart from a group.
 - ☐ Natural selection favors living apart because individuals are unlikely to attract the attention of predators.
 - ☐ Natural selection favors living apart because group living increases competition for limited resources of food and nesting sites.

4. The word "devastate" in the passage is closest in meaning to
- ☐ alter
 - ☐ spread to
 - ☐ destroy
 - ☐ involve
5. In paragraph 3, why does the author mention distemper and its effect on seals?
- ☐ To explain why sociality is now relatively rare in seals
 - ☐ To prove that parasites prevent animals from growing healthy and strong
 - ☐ To show how easily social animals transmit potentially deadly diseases
 - ☐ To provide an example of a species-specific disease that affects solitary animals

Paragraph 4

But in some instances, the payoffs can be even greater. Two have already been mentioned: cooperative hunting and defensive groups. Social hunting is likely to evolve where prey is too large to be taken by individuals operating alone. To capture wildebeest some members of a group of lions follow their prey and herd them toward others lying in ambush. In other species, individuals forage or hunt simultaneously and share the food. Vampire bats that have had a bad day, for instance, are fed by more successful members of the community, but they are expected to return the favor in the future. Cooperation can even involve sharing information about the location of food. Some colonial birds, such as bank swallows, use the departure direction of a successful forager (food hunter) to locate concentrations of prey. Information transfer can be unintentional though some species make use of special assembly calls or behavior.

6. The word "simultaneously" in the passage is closest in meaning to
- ☐ at the same time
 - ☐ in the same way
 - ☐ for the same reason
 - ☐ on the same scale
7. The word "unintentional" in the passage is closest in meaning to
- ☐ unsuccessful
 - ☐ unplanned
 - ☐ inaccurate
 - ☐ impractical
8. According to paragraph 4, how do lions increase their chances of successfully hunting wildebeest?
- ☐ They work together as a team to chase and capture the animals.

- They wait until groups of the animals come close to their hiding place.
 - They follow and catch the animals that are operating alone.
 - They focus on the smallest members of the species.
9. Which of the following can be inferred from paragraph 4 about how bank swallows find food cooperatively?
- They give special favors to members of their community that have been successful foragers.
 - They use special calls and signals to indicate to other swallows where food is located.
 - They observe the direction a successful forager took to locate prey.
 - They locate concentrations of prey by setting off in different directions.

Paragraph 5

Cooperation in group defense, such as we see in circles of musk oxen or elephants, is quite rare among vertebrates but is prevalent among the social insects. The strategy of employing many eyes to watch for danger, on the other hand, is widespread in birds and mammals. A herd of gazelles (small antelope) is far more likely to spot a lurking lion or a concealed cheetah than is a lone individual, and at a greater distance. In fact, a group enters into a kind of time-sharing arrangement in which individual antelope alternate biting off a mouthful of grass with a period of erect and watchful chewing. A larger group can afford more bites per individual per minute, there being more eyes to scan for danger. For a small antelope living in a forest where visibility is limited, however, remaining hidden is probably a better bet than assembling into noisy herds.

10. The word "alternate" in the passage is closest in meaning to
- make a habit of
 - have conflicts over
 - show a preference for
 - take turns at
11. According to paragraph 5, how does cooperation help gazelles avoid predation?
- The herd forms a defensive circle around the weakest members of the group.
 - Because they share food, all of the animals are well nourished and ready to run.
 - The herd makes a lot of noise, which scares off predators.
 - Some animals watch for danger while others concentrate on feeding.

Paragraph 6

Among the millions of species of insects, only a few thousand are social. Those rarities are generally confined to termites and Hymenoptera. All termites are social: their diet (cellulose) requires that each generation feed a special kind of bacteria

or fungi to the next generation to aid in its digestion. Of the numerous hymenopterans, some are social—including all ants and a few bees and wasps—but many are solitary.

12. Which of the following can be inferred from paragraph 6 about termites?

- ☐ They are the only insect species whose members are all social.
- ☐ They have smaller communities than hymenopterans.
- ☐ They are more independent than hymenopterans.
- ☐ They have to live together to pass on digestive bacteria to their young.

Paragraph 2

We know, or think we know, that social groups are good. ■ Wolves are better predators when they hunt in packs, and pigeons escape from falcons far more often when feeding in flocks. ■ Group building projects—the dams beavers build to block a body of water that provides them with relative safety from predators and the lodges they build for shelter, for instance—can provide a high level of protection and comfort. ■ Why, then, are social species so very rare? ■ In fact, living socially presents inevitable problems that transcend habitat needs so that only when these costs are offset by corresponding benefits is group living a plus.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

The engineering of structures of such size and complexity is unknown among solitary animals.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Some animal species demonstrate sociability, but solitary behaviors are far more typical among animals.

-
-
-

Answer Choices

Because birds and mammals engage in courting, mating, and establishing year-round family groups to rear young, they are most likely to be social species.

Some social animals obtain food through cooperation and share food and information about food sources with each other.

Social animals help each other watch for danger more effectively, and some species work together to combat predators.

Living in communities presents disadvantages, such as the competition for resources and the easy transmission of diseases.

Some entire species of social animals have become extinct because of disease epidemics.

Although some species of ants, bees, and wasps are famous for their sociality, most social insects are not hymenopterans.

Understanding Ancient Mesoamerican Art

Starting at the end of the eighteenth century and continuing up to the present, explorers have searched for the ruins of ancient Mesoamerica, a region that includes Central America and central and southern Mexico. With the progress of time, archaeologists have unearthed civilizations increasingly remote in age. It is as if with each new century in the modern era an earlier stratum of antiquity has been revealed. Nineteenth-century explorers, particularly John Lloyd Stephens and Frederick Catherwood, came upon Maya cities in the jungle, as well as evidence of other Classic cultures. Twentieth-century research revealed a much earlier high civilization, the Olmec. It now scarcely seems possible that the frontiers of early Mesoamerican civilization can be pushed back any further, although new work—such as in Oaxaca, southern Mexico—will continue to fill in details of the picture.

The process of discovery often shapes what we know about the history of Mesoamerican art. New finds are just as often made accidentally as intentionally. In 1971 workers installing sound and light equipment under the Pyramid of the Sun at Teotihuacan stumbled upon a remarkable cave that has since been interpreted by some scholars as a royal burial chamber. Archaeology has its own fashions too: the isolation of new sites may be the prime goal in one decade and the excavation of pyramids the focus in the next. In a third decade, outlying structures rather than principal buildings may absorb archaeologists' energies. Nor should one forget that excavators are vulnerable to local interests. At one point, reconstruction of pyramids to attract tourism may be desired; at another, archaeologists may be precluded from working at what has already become a tourist attraction. Also, modern construction often determines which ancient sites can be excavated. In Mexico City, for example, the building of the subway initiated the excavations there and renewed interest in the old Aztec capital.

But the study of Mesoamerican art is not based exclusively on archaeology. Much useful information about the native populations was written down in the sixteenth century, particularly in central Mexico, and it can help us unravel the pre-Columbian past (the time prior to the arrival of Columbus in the Americas in 1492). Although many sources exist, the single most important one to the art historian is Bernardino de Sahagún's *General History of the Things of New Spain*. A Franciscan friar (member of the Roman Catholic religious order), Sahagún recorded for posterity many aspects of pre-Hispanic life in his encyclopedia of twelve books, including history, ideology, and cosmogony (theories of the origin of the universe), as well as detailed information on the materials and methods of the

skilled native craft workers. Furthermore, traditional ways of life survive among the native peoples of Mesoamerica, and scholars have increasingly found that modern practice and belief can decode the past. Remarkably, some scholars have been turned this process around, teaching ancient writing to modern peoples who may use it to articulate their identity in the twenty-first century.

During the past 40 years, scholars also have made great progress in deciphering and interpreting ancient Mesoamerican writing systems, a breakthrough that has transformed our understanding of the pre-Columbian mind. Classic Maya inscriptions, for example—long thought to record only calendrical information and astrological incantations—can now be read, and we find that most of them glorify family and ancestry by displaying the right of individual sovereigns to rule. The carvings can thus be seen as portraits or public records of dynastic power. Although scholars long believed that Mesoamerican artists did not sign their works, Mayanist scholar David Stuart's 1986 deciphering of the Maya glyphs (written symbols) for "scribe" and "to write" opened a window on Maya practice; now we know at least one painter of ceramic vessels was the son of a king. Knowledge of the minor arts has also come in large part through an active art market. Thousands more small-scale objects are known now than in the twentieth century, although at a terrible cost to the ancient ruins from which they have been plundered.

Paragraph 1

Starting at the end of the eighteenth century and continuing up to the present, explorers have searched for the ruins of ancient Mesoamerica, a region that includes Central America and central and southern Mexico. With the progress of time, archaeologists have unearthed civilizations increasingly remote in age. It is as if with each new century in the modern era an earlier stratum of antiquity has been revealed. Nineteenth-century explorers, particularly John Lloyd Stephens and Frederick Catherwood, came upon Maya cities in the jungle, as well as evidence of other Classic cultures. Twentieth-century research revealed a much earlier high civilization, the Olmec. It now scarcely seems possible that the frontiers of early Mesoamerican civilization can be pushed back any further, although new work—such as in Oaxaca, southern Mexico—will continue to fill in details of the picture.

15. According to paragraph 1, which of the following best describes the progress made in the search for ruins of ancient Mesoamerica?

- ☐ The oldest remains were discovered immediately after Columbus arrived, and later findings correspond to more recent ruins.
- ☐ From the late eighteenth century through the twentieth century, increasingly older remains were found.
- ☐ All of the archaeological work was completed in the twentieth century.
- ☐ Technological developments made it progressively easier for explorers to

discover Mayan cities in the jungle.

16. Paragraph 1 supports which of the following statements about the revealing of early Mesoamerican civilization?

- ☐ The Maya and Olmec civilizations were discovered by explorers at approximately the same time.
- ☐ Most of our understanding of Mesoamerican civilization comes from discoveries made in the twentieth century.
- ☐ The discoveries made at Oaxaca in southern Mexico show that the Olmec civilization had its origins there.
- ☐ Evidence still to be found in Oaxaca, Mexico, is likely to provide additional information about the high civilization of early Mesoamerica.

Paragraph 2

The process of discovery often shapes what we know about the history of Mesoamerican art. New finds are just as often made accidentally as intentionally. In 1971 workers installing sound and light equipment under the Pyramid of the Sun at Teotihuacan stumbled upon a remarkable cave that has since been interpreted by some scholars as a royal burial chamber. Archaeology has its own fashions too: the isolation of new sites may be the prime goal in one decade and the excavation of pyramids the focus in the next. In a third decade, outlying structures rather than principal buildings may absorb archaeologists' energies. Nor should one forget that excavators are vulnerable to local interests. At one point, reconstruction of pyramids to attract tourism may be desired; at another, archaeologists may be precluded from working at what has already become a tourist attraction. Also, modern construction often determines which ancient sites can be excavated. In Mexico City, for example, the building of the subway initiated the excavations there and renewed interest in the old Aztec capital.

17. In paragraph 2, why does the author discuss the Pyramid of the Sun at Teotihuacan?

- ☐ To introduce the discussion of a specific style of Mesoamerican art
- ☐ To illustrate the importance of an accidental discovery
- ☐ To emphasize the necessity of systematic study
- ☐ To argue against the use of modern equipment in archaeology

18. The word "outlying" in the passage is closest in meaning to

- ☐ ceremonial
- ☐ temporary
- ☐ far from the center
- ☐ simple

19. Which of the following is NOT mentioned in paragraph 2 as a factor affecting the process of discovery in archaeology?

- ☐ Fashions that change over time
- ☐ The popularity of archaeology as a field of study
- ☐ The interests of local inhabitants
- ☐ The building of modern structures

20. The word "initiated" in the passage is closest in meaning to

- ☐ destroyed
- ☐ interfered with
- ☐ occurred at the same time as
- ☐ started

Paragraph 3

But the study of Mesoamerican art is not based **exclusively** on archaeology. Much useful information about the native populations was written down in the sixteenth century, particularly in central Mexico, and it can help us unravel the pre-Columbian past (the time prior to the arrival of Columbus in the Americas in 1492). Although many sources exist, the single most important one to the art historian is Bernardino de Sahagún's *General History of the Things of New Spain*. A Franciscan friar (member of the Roman Catholic religious order), Sahagún recorded for **posterity** many aspects of pre-Hispanic life in his encyclopedia of twelve books, including history, ideology, and cosmogony (theories of the origin of the universe), as well as detailed information on the materials and methods of the skilled native craft workers. Furthermore, traditional ways of life survive among the native peoples of Mesoamerica, and scholars have increasingly found that modern practice and belief can decode the past. Remarkably, some scholars have been turned this process around, teaching ancient writing to modern peoples who may use it to articulate their identity in the twenty-first century.

21. The word "exclusively" in the passage is closest in meaning to

- ☐ exactly
- ☐ solely
- ☐ traditionally
- ☐ primarily

22. According to paragraph 3, why was Bernardino de Sahagún important to the study of Mesoamerican art?

- ☐ He recorded detailed information about native populations in the sixteenth century.
- ☐ He made important archaeological discoveries in central Mexico.
- ☐ He studied the methods and materials of skilled Mesoamerican craft workers as a basis for his own art.
- ☐ He encouraged native people to preserve their traditional ways of life.

23. The word "posterity" in the passage is closest in meaning to

- the religious leadership
- publication
- the native people
- future generations

24. According to paragraph 3, which of the following is true of ancient Mesoamerican writing?

- It is very similar to the modern writing of Mesoamerican peoples.
- It is not as old as originally thought.
- It can be used by modern peoples to express their own identities.
- It is most thoroughly understood by skilled native craft workers.

Paragraph 4

During the past 40 years, scholars also have made great progress in deciphering and interpreting ancient Mesoamerican writing systems, a breakthrough that has transformed our understanding of the pre-Columbian mind. Classic Maya inscriptions, for example—long thought to record only calendrical information and astrological incantations—can now be read, and we find that most of them glorify family and ancestry by displaying the right of individual sovereigns to rule. The carvings can thus be seen as portraits or public records of dynastic power. Although scholars long believed that Mesoamerican artists did not sign their works, Mayanist scholar David Stuart's 1986 deciphering of the Maya glyphs (written symbols) for "scribe" and "to write" opened a window on Maya practice; now we know at least one painter of ceramic vessels was the son of a king. Knowledge of the minor arts has also come in large part through an active art market. Thousands more small-scale objects are known now than in the twentieth century, although at a terrible cost to the ancient ruins from which they have been plundered.

25. According to paragraph 4, how did the deciphering of ancient Mesoamerican inscriptions transform our understanding of the pre-Columbian mind?

- It showed that most attention was given to the prediction and magical control of the future.
- It revealed a much greater concern with the identity and ancestry of particular individuals than had been suspected.
- It demonstrated that painting was regarded as a royal art.
- It disproved the idea that the Maya devoted much thought to record keeping.

26. According to paragraph 4, all of the following are true of Mayanist scholar David Stuart's 1986 discovery **EXCEPT**:

- It reversed a belief held by earlier scholars.
- It uncovered long lists of royal dynastic records.

- It allowed scholars to understand two glyphs for the first time.
- It revealed that at least one Maya painter was a king's son.

Paragraph 2

The process of discovery often shapes what we know about the history of Mesoamerican art. New finds are just as often made accidentally as intentionally. ■ In 1971 workers installing sound and light equipment under the Pyramid of the Sun at Teotihuacan stumbled upon a remarkable cave that has since been interpreted by some scholars as a royal burial chamber. ■ Archaeology has its own fashions too: the isolation of new sites may be the prime goal in one decade and the excavation of pyramids the focus in the next. In a third decade, outlying structures rather than principal buildings may absorb archaeologists' energies. ■ Nor should one forget that excavators are vulnerable to local interests. ■ At one point, reconstruction of pyramids to attract tourism may be desired; at another, archaeologists may be precluded from working at what has already become a tourist attraction. Also, modern construction often determines which ancient sites can be excavated. In Mexico City, for example, the building of the subway initiated the excavations there and renewed interest in the old Aztec capital.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

The chance discovery has done as much for our understanding of the pyramid as any systematic study would have.

Where would the sentence best fit?

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Our knowledge of Mesoamerican art has grown since explorers first began searching for ruins of ancient Mesoamerica.

Answer Choices

John Lloyd Stephens and Frederick Catherwood used the ruins of Maya cities in the jungle as evidence that the Maya civilization was older than the Olmec civilization.

Classic Maya inscriptions primarily recorded calendrical information, astrological incantations, and signature ...

Often discoveries accidentally, Mesoamerican pyramids not only contain numerous samples of native art but also attract tourism.

How Soil is Formed

Soil formation is a dynamic process that takes place in different environments. It is strongly influenced by the parent material, climate (largely vegetation and temperature and water exchanges), topography (the elevations, depressions, directions and angles of slopes, and other surface features of the landscape), and time.

The parent material is the unconsolidated mass on which soil formation takes place. This material may or may not be derived from the on-site geological substrate or bedrock on which it rests. Parent materials can be transported by wind, water, glaciers, and gravity and deposited on top of bedrock. Because of the diversity of materials involved, soils derived from transported parent materials are commonly more fertile than soils from parent materials derived in place. Whatever the parent material, whether derived in place from bedrock or from transported material, it ultimately comes from geological materials, such as igneous, sedimentary, and metamorphic rocks, and the composition of the rocks largely determines the chemical composition of the soil.

Climate is most influential in determining the nature and intensity of weathering and the type of vegetation that further affects soil formation. The soil material experiences daily and seasonal variations in heating and cooling. Open surfaces exposed to thermal radiation undergo the greatest daily fluctuations in heating and cooling, soils covered with vegetation the least. Hill slopes facing the sun absorb more heat than those facing away from the sun. Radiant energy has a pronounced effect on the moisture regime, especially the evaporative process and dryness. Temperature can stimulate or inhibit biogeochemical reactions in soil material.

Water is involved in all biogeochemical reactions in the soil because it is the carrier of the acids that influence the weathering process. Water enters the soil material as a liquid and leaves it as a liquid by percolation (the slow movement of water through the soil's pores) and as a gas through evaporation. The water regime—the water flow over a given time—in soil material is sporadic, and in many parts of the Earth is highly seasonal. Water that enters the soil during heavy rainfall and snowmelt moves down through the soil. As it moves, it leaves behind suspended material and may carry away mineral matter in solution, a process called leaching. On sloping land, water distributes material laterally (sideways) through the soil.

Topography is a major factor in soil development. More water runs off and less enters the soil on steep slopes than on relatively level land. Water draining from slopes enters the soil on low and flat land. Thus soils and soil material tend to be dry on slopes and moist on wet on the low land. Steep slopes are subject to surface erosion and soil creep—the downslope movement of soil material, which accumulates on lower slopes and lowlands.

Vegetation, animals, bacteria, and fungi all contribute to the formation of soil. Vegetation, in particular, is responsible for organic material in the soil and influences its nutrient content. For example, forests store most of their organic matter on the surface, whereas in grasslands most of the organic matter added to the soil comes from the deep fibrous root systems. Organic acids produced by vegetation accelerate the weathering process.

The weathering of rock material and the accumulation, decomposition, and mineralization of organic material require considerable time. Well-developed soils in equilibrium with weathering, erosion, and biotic influences may require 2,000 to 20,000 years for their formation, but soil differentiation from parent material may take place in as short a time as 30 years. Certain acid soils in humid regions develop in 2,000 years because the leaching process is speeded by acidic materials. Parent materials heavy in texture require a much longer time to develop into soils because of an impeded downward flow of water. Soils develop more slowly in dry regions than in humid ones. Soils on steep slopes often remain poorly developed regardless of geological age because rapid erosion removes soil nearly as fast as it is formed. Floodplain soils age little through time because of the continuous accumulation of new materials. Such soils are not deeply weathered and are more fertile than geologically old soils because they have not been exposed to the leaching process as long. The latter soils tend to be infertile because of long-time leaching of nutrients without replacement from fresh material.

Paragraph 2

The parent material is the unconsolidated mass on which soil formation takes place. This material may or may not be derived from the on-site geological substrate or bedrock on which it rests. Parent materials can be transported by wind, water, glaciers, and gravity and deposited on top of bedrock. Because of the diversity of materials involved, soils derived from transported parent materials are commonly more fertile than soils from parent materials derived in place. Whatever the parent material, whether derived in place from bedrock or from transported material, it ultimately comes from geological materials, such as igneous, sedimentary, and metamorphic rocks, and the composition of the rocks largely determines the chemical composition of the soil.

15. According to paragraph 2, which of the following is true about parent material

found at a given site?

- ☐ It was most likely deposited there by glaciers rather than by wind and water.
 - ☐ It is formed and consolidated entirely from the bedrock on which it rests.
 - ☐ It can vary in its richness and fertility according to where it originated.
 - ☐ The speed with which it is transported determines its location.
16. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Transported parent materials, parent materials derived in place from bedrock, and various kinds of rocks all contribute to the composition of a soil.
 - ☐ The chemical composition of a soil is ultimately determined by the geological materials from which it forms, regardless of where or how the parent material originates.
 - ☐ The parent material from which a soil is formed ultimately comes from igneous, sedimentary, and metamorphic rocks.
 - ☐ The composition of the geological materials from which the parent material was originally derived can be determined by studying the chemical composition of the soil.

Paragraph 3

Climate is most influential in determining the nature and intensity of weathering and the type of vegetation that further affects soil formation. The soil material experiences daily and seasonal variations in heating and cooling. Open surfaces exposed to thermal radiation undergo the greatest daily fluctuations in heating and cooling, soils covered with vegetation the least. Hill slopes facing the sun absorb more heat than those facing away from the sun. Radiant energy has a pronounced effect on the moisture regime, especially the evaporative process and dryness. Temperature can stimulate or inhibit biogeochemical reactions in soil material.

17. The word "further" in the passage is closest in meaning to
- ☐ as a result
 - ☐ additionally
 - ☐ potentially
 - ☐ eventually
18. The word "fluctuations" in the passage is closest in meaning to
- ☐ extremes
 - ☐ increases
 - ☐ variations
 - ☐ effects
19. The word "inhibit" in the passage is closest in meaning to

- ☐ initiate
- ☐ remove
- ☐ slow down
- ☐ damage

Paragraph 4

Water is involved in all biogeochemical reactions in the soil because it is the carrier of the acids that influence the weathering process. Water enters the soil material as a liquid and leaves it as a liquid by percolation (the slow movement of water through the soil's pores) and as a gas through evaporation. The water regime—the water flow over a given time—in soil material is sporadic, and in many parts of the Earth is highly seasonal. Water that enters the soil during heavy rainfall and snowmelt moves down through the soil. As it moves, it leaves behind suspended material and may carry away mineral matter in solution, a process called leaching. On sloping land, water distributes material laterally (sideways) through the soil.

20. According to paragraph 4, water passing through the soil can affect the soil's composition in which of the following ways?
- ☐ By stopping certain chemical reactions from taking place in the soil
 - ☐ By carrying away minerals
 - ☐ By reducing the acidity of the soil
 - ☐ By preventing leaching of minerals

Paragraph 5

Topography is a major factor in soil development. More water runs off and less enters the soil on steep slopes than on relatively level land. Water draining from slopes enters the soil on low and flat land. Thus soils and soil material tend to be dry on slopes and moist on wet on the low land. Steep slopes are subject to surface erosion and soil creep—the downslope movement of soil material, which accumulates on lower slopes and lowlands.

21. According to paragraph 5, why does soil material tend to be drier on steep slopes than on flatter land?
- ☐ Steep slopes are less likely to experience soil creep.
 - ☐ Soil material on steep slopes tends to be thicker.
 - ☐ Water that falls on steep slopes tends to erode the soil there and deposit it on level ground.
 - ☐ More water runs off steep slopes before it can enter the soil.

Paragraph 6

Vegetation, animals, bacteria, and fungi all contribute to the formation of soil. Vegetation, in particular, is responsible for organic material in the soil and influences its nutrient content. For example, forests store most of their organic

matter on the surface, whereas in grasslands most of the organic matter added to the soil comes from the deep fibrous root systems. Organic acids produced by vegetation accelerate the weathering process.

22. Paragraph 6 supports which of the following ideas about organic material soil?

- ☐ Most of the organic material in soil comes from fibrous root systems.
- ☐ How organic material is distributed in soil depends on the type of vegetation the soil supports.
- ☐ The organic materials in soil prevent acids from forming during the weathering process.
- ☐ Animals, bacteria, and fungi consume much of the nutrient content in soil.

Paragraph 7

The weathering of rock material and the accumulation, decomposition, and mineralization of organic material require considerable time. Well-developed soils in equilibrium with weathering, erosion, and biotic influences may require 2,000 to 20,000 years for their formation, but soil differentiation from parent material may take place in as short a time as 30 years. Certain acid soils in humid regions develop in 2,000 years because the leaching process is speeded by acidic materials. Parent materials heavy in texture require a much longer time to develop into soils because of an impeded downward flow of water. Soils develop more slowly in dry regions than in humid ones. Soils on steep slopes often remain poorly developed regardless of geological age because rapid erosion removes soil nearly as fast as it is formed. Floodplain soils age little through time because of the continuous accumulation of new materials. Such soils are not deeply weathered and are more fertile than geologically old soils because they have not been exposed to the leaching process as long. The latter soils tend to be infertile because of long-time leaching of nutrients without replacement from fresh material.

23. The word "impeded" in the passage is closest in meaning to

- ☐ excessive
- ☐ restricted
- ☐ diverted
- ☐ uneven

24. According to paragraph 7, soil on steep slopes tends to remain poorly developed because

- ☐ soil on steep slopes tends to be very acidic
- ☐ soil that forms on steep slopes tends to be heavy in texture
- ☐ so much of the soil that forms on steep slopes becomes eroded quickly
- ☐ soil differentiates very quickly from parent materials

25. According to paragraph 7, why does floodplain soil tend to be especially

fertile?

- ☐ Because floodplain soil tends to be deeply weathered
- ☐ Because floodplain soil is not continually replaced by the addition of new material
- ☐ Because floodplain soil has not had many nutrients removed
- ☐ Because most floodplains are in geologically new regions

26. The purpose of paragraph 7 in the passage is to

- ☐ provide evidence that soils in equilibrium maintain nutrients for longer periods of time than other soils
- ☐ explain why geologically younger soils are more fertile than older soils, regardless of where they are found
- ☐ explain how long it takes for soils to develop and what elements affect their fertility
- ☐ summarize the various stages of soil formation discussed earlier in the passage

Paragraph 5

Topography is a major factor in soil development. ■ More water runs off and less enters the soil on steep slopes than on relatively level land. ■ Water draining from slopes enters the soil on low and flat land. ■ Thus soils and soil material tend to be dry on slopes and moist on wet on the low land. ■ Steep slopes are subject to surface erosion and soil creep—the downslope movement of soil material, which accumulates on lower slopes and lowlands.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Its role is demonstrated by the contrast between what happens on steep slopes and what happens on the lower-lying land at the bottom of those slopes.

Where would the sentence best fit?

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

There are a variety of elements that influence soil development.

-
-
-

Answer Choices

By analyzing the chemical composition of developed soil, it is possible to determine the location of the geological substrate from which the parent material was originally derived.

Both topography and the types of organisms present in the soil as it develops determine its nutrient level.

The length of time that it takes to develop high-quality soil is determined by humidity levels, the makeup of parent materials, and patterns of water flow and erosion.

The weathering process is strongly influenced by temperature and its variation, the nature of the water flow, and by the acids carried in the water.

Where soil material is regularly subject to rapid and extreme variations in temperature and water content, certain biogeochemical reactions necessary for soil formation cannot take place.

Because high humidity speeds up both leaching processes and erosion, soils in humid regions take much longer to reach equilibrium than soils in dry regions, and are seldom as fertile.

 pdfelement

The Emergence of Civilization

Starting around 8000 B.C.E., the most extensive exploitation of agriculture occurred in river valleys, where there were both good soil and a dependable water supply regardless of the amount of rainfall. In the Near East, this happened in the Fertile Crescent, the region extending up the Nile Valley in Egypt, north through the Levant (Palestine, Lebanon, and Syria), and southeast into the Tigris and Euphrates river valleys of Mesopotamia. The richest soil was located in the deltas at the mouths of the rivers, but the deltas were swampy and subject to flooding. Before they could be farmed, they needed to be drained and irrigated, and flood-control systems had to be constructed. These activities required administrative organization and the ability to mobilize large pools of labor. In Mesopotamia, perhaps as a consequence of a period of drought, massive land-use projects were undertaken after 4000 B.C.E. to cultivate the rich delta soils of the Tigris and Euphrates Rivers. The land was so productive that many more people could be fed, and a great population explosion resulted. Villages grew into cities of tens of thousands of persons.

These large cities needed some form of centralized administration. Archaeological evidence indicates that the organization initially was provided by religion, for the largest building in each city was a massive temple honoring one of the Mesopotamian gods. In Uruk, for example, a 60-foot-long temple known as the White House was built before 3000 B.C.E. There were no other large public buildings, suggesting that the priests who were in charge of the temples also were responsible for governing the city and organizing people to work in the fields and on irrigation projects building and maintaining systems of ditches and dams.

The great concentration of wealth and resources in the river valleys brought with it further technological advances, such as wheeled vehicles, multicolored pottery and the pottery wheel, and the weaving of wool garments. Advances in metal technology just before 2000 B.C.E. resulted in the creation of bronze, a durable alloy (or mixture) of about 90 percent copper and 10 percent tin that provided a sharp cutting edge for weapons.

By 3000 B.C.E., the economies and administrations of Mesopotamia and Egypt had become so complex that some form of record keeping was needed. As a result, writing was invented. Once a society became literate, it passed from the period known as prehistory into the historic period. In fact, the word "history" comes from a Greek word meaning "narrative"—people could not provide a detailed permanent account of their past until they were able to write.

The totality of these developments resulted in the appearance, around 3000 B.C.E., of a new form of culture called civilization. The first civilizations had several defining characteristics. They had economies based on agriculture. They had cities that functioned as administrative centers and usually had large populations. They had different social classes, such as free persons and slaves. They had specialization of labor, that is, different people serving, for example, as rulers, priests, craft workers, merchants, soldiers, and farmers. And they had metal technology and a system of writing. As of 3000 B.C.E., civilization in these terms existed in Mesopotamia, Egypt, India, and China.

This first phase of civilization is called the Bronze Age because of the importance of metal technology. The most characteristic Near Eastern Bronze Age civilizations, those of Mesopotamia and Egypt, were located in river valleys, were based on the extensive exploitation of agriculture, and supported large populations. Bronze was a valuable commodity in these civilizations, the copper and tin needed for its manufacture did not exist in river valleys and had to be imported. Bronze was therefore used mainly for luxury items, such as jewelry or weapons, not for everyday domestic items, which were made from pottery, animal products, wood, and stone. In particular, bronze was not used for farming tools. Thus, early civilizations based on large-scale agriculture, such as those of Mesopotamia and Egypt, were feasible only in soils that could be worked by wooden plows pulled by people or draft animals such as oxen. Other Bronze Age civilizations, however, such as those that arose in the Levant and eastern Mediterranean took advantage of their location on communication routes to pursue economies based on trade.

Paragraph 1

Starting around 8000 B.C.E., the most extensive exploitation of agriculture occurred in river valleys, where there were both good soil and a dependable water supply regardless of the amount of rainfall. In the Near East, this happened in the Fertile Crescent, the region extending up the Nile Valley in Egypt, north through the Levant (Palestine, Lebanon, and Syria), and southeast into the Tigris and Euphrates river valleys of Mesopotamia. The richest soil was located in the deltas at the mouths of the rivers, but the deltas were swampy and subject to flooding. Before they could be farmed, they needed to be drained and irrigated, and flood-control systems had to be constructed. These activities required administrative organization and the ability to mobilize large pools of labor. In Mesopotamia, perhaps as a consequence of a period of drought, massive land-use projects were undertaken after 4000 B.C.E. to cultivate the rich delta soils of the Tigris and Euphrates Rivers. The land was so productive that many more people could be fed, and a great population explosion resulted. Villages grew into cities of tens of thousands of persons.

15. Which of the following helps explain why "the most extensive exploitation of agriculture occurred in river valleys"?

- ☐ In river valleys farmers did not have to depend on rain for water.
- ☐ The soil in river valleys did not require irrigation.
- ☐ Swampy areas in river valleys were easy to drain.
- ☐ The expanding populations in river valleys provided large pools of labor.

16. Why does the author mention "a period of drought"?

- ☐ To help explain why the richest soils in the Near East were located in the deltas at the mouths of the Tigris and Euphrates Rivers
- ☐ To suggest a reason for undertaking the massive effort to make the deltas of the Tigris and Euphrates Rivers farmable
- ☐ To identify a condition that often affected agricultural production in Mesopotamia
- ☐ To support the idea that mobilizing large pools of labor after 4000 B.C.E. required significant administrative organization

17. According to paragraph 1, what was one result of the farming systems developed in river deltas in the Near East?

- ☐ There was a large increase in the overall amount of food produced.
- ☐ Large pools of labor became available to perform administrative tasks.
- ☐ The soil in these deltas grew much richer.
- ☐ The number of farming villages surrounding cities increased.

Paragraph 2

These large cities needed some form of centralized administration. Archaeological evidence indicates that the organization initially was provided by religion, for the largest building in each city was a massive temple honoring one of the Mesopotamian gods. In Uruk, for example, a 60-foot-long temple known as the White House was built before 3000 B.C.E. There were no other large public buildings, suggesting that the priests who were in charge of the temples also were responsible for governing the city and organizing people to work in the fields and on irrigation projects building and maintaining systems of ditches and dams.

18. According to paragraph 2, the fact that temples appear to have been the only large public buildings in Mesopotamian cities has been interpreted as evidence that these cities

- ☐ needed some form of central administration
- ☐ were initially administered by priests
- ☐ were all governed from Uruk
- ☐ had difficulty organizing workers for building projects

Paragraph 3

The great concentration of wealth and resources in the river valleys brought with it

further technological advances, such as wheeled vehicles, multicolored pottery and the pottery wheel, and the weaving of wool garments. Advances in metal technology just before 2000 B.C.E. resulted in the creation of bronze, a **durable** alloy (or mixture) of about 90 percent copper and 10 percent tin that provided a sharp cutting edge for weapons.

19. The word "**durable**" in the passage is closest in meaning to

- ☐ existing for a long time without significant damage
- ☐ difficult to produce
- ☐ recently discovered
- ☐ extremely useful

20. Paragraph 3 indicates that technological advances affected all of the following **EXCEPT**

- ☐ transportation
- ☐ clothing manufacture
- ☐ warfare and hunting
- ☐ the distribution of wealth and resources

Paragraph 4

By 3000 B.C.E., the economies and administrations of Mesopotamia and Egypt had become so complex that some form of record keeping was needed. As a result, writing was invented. Once a society became literate, it passed from the period known as prehistory into the historic period. In fact, the word "history" comes from a Greek word meaning "narrative"—people could not provide a detailed permanent account of their past until they were able to write.

21. According to paragraph 4, why was writing invented?

- ☐ To reduce unnecessary social complexity
- ☐ To keep economic and administrative records
- ☐ To record oral historical narratives
- ☐ To help people better understand their own past

Paragraph 5

The totality of these developments resulted in the appearance, around 3000 B.C.E., of a new form of culture called civilization. The first civilizations had several **defining** characteristics. They had economies based on agriculture. They had cities that functioned as administrative centers and usually had large populations. They had different social classes, such as free persons and slaves. They had specialization of labor, that is, different people serving, for example, as rulers, priests, craft workers, merchants, soldiers, and farmers. And they had metal technology and a system of writing. As of 3000 B.C.E., civilization in these terms existed in Mesopotamia, Egypt, India, and China.

22. The word "defining" in the passage is closest in meaning to
- ☐ important
 - ☐ obvious
 - ☐ identifying
 - ☐ interesting
23. According to paragraph 5, all of the following are true of the first civilizations **EXCEPT**:
- ☐ Their soldiers and priests also worked as farmers.
 - ☐ Their populations were divided into different social classes.
 - ☐ They had developed technologies for working with metals.
 - ☐ They were typically administered from large cities.

Paragraph 6

This first phase of civilization is called the Bronze Age because of the importance of metal technology. The most characteristic Near Eastern Bronze Age civilizations, those of Mesopotamia and Egypt, were located in river valleys, were based on the extensive exploitation of agriculture, and supported large populations. Bronze was a valuable commodity in these civilizations, the copper and tin needed for its manufacture did not exist in river valleys and had to be imported. Bronze was therefore used mainly for luxury items, such as jewelry or weapons, not for everyday domestic items, which were made from pottery, animal products, wood, and stone. In particular, bronze was not used for farming tools. Thus, early civilizations based on large-scale agriculture, such as those of Mesopotamia and Egypt, were feasible only in soils that could be worked by wooden plows pulled by people or draft animals such as oxen. Other Bronze Age civilizations, however, such as those that arose in the Levant and eastern Mediterranean took advantage of their location on communication routes to pursue economies based on trade.

24. The word "domestic" in the passage is closest in meaning to
- ☐ practical
 - ☐ household
 - ☐ standard
 - ☐ necessary
25. According to paragraph 6, why was bronze not used for farming tools in Bronze Age civilizations of the Near East?
- ☐ Wooden farming tools were more effective in the soils of the region.
 - ☐ Bronze farming tools would have deteriorated quickly in the climate.
 - ☐ Bronze was too expensive to use for farming tools.
 - ☐ People had not yet discovered how to make farming tools out of bronze.
26. According to paragraph 6, which of the following was true of at least some

civilizations of the Bronze Age?

- ☐ They did not develop urban centers with large populations.
- ☐ They did not use metals to make bronze.
- ☐ They had an economy that was not based on agriculture.
- ☐ They did not use bronze for luxury items such as jewelry.

Paragraph 6

This first phase of civilization is called the Bronze Age because of the importance of metal technology. The most characteristic Near Eastern Bronze Age civilizations, those of Mesopotamia and Egypt, were located in river valleys, were based on the extensive exploitation of agriculture, and supported large populations. ■ Bronze was a valuable commodity in these civilizations, the copper and tin needed for its manufacture did not exist in river valleys and had to be imported. ■ Bronze was therefore used mainly for luxury items, such as jewelry or weapons, not for everyday domestic items, which were made from pottery, animal products, wood, and stone. ■ In particular, bronze was not used for farming tools. ■ Thus, early civilizations based on large-scale agriculture, such as those of Mesopotamia and Egypt, were feasible only in soils that could be worked by wooden plows pulled by people or draft animals such as oxen. Other Bronze Age civilizations, however, such as those that arose in the Levant and eastern Mediterranean took advantage of their location on communication routes to pursue economies based on trade.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

This significantly limited the availability of bronze.

Where would the sentence best fit?

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

After 4000 B.C.E., organized drainage and flood-control projects in the Fertile Crescent allowed cultivation of river deltas and led to large population increases.

-
-
-

Answer Choices

Before the rise of large cities with complex economies, there had been no need for any kind of administrative structure to organize workers, and thus religion tended to be the only source of authority.

By 3000 B.C.E., a number of agricultural societies had emerged as civilization characterized by large cities, centralized administrations, specialization of labor, class divisions, metal technology, and writing.

Although river-valley societies had to import the metals for making bronze, the first phase of civilization is known as the Bronze Age because of the importance of metal technology in these societies.

Large cities functioned as administrative centers, creating a concentration of wealth and resources that stimulated technological advances, such as the invention of writing and the creation of bronze.

For a culture to be considered a civilization, it must have independently invented its own form of writing and become generally literate, thereby moving from prehistory into the historic period.

The creation of bronze made it possible for civilizations based on large-scale agriculture to be located far away from the river valleys.

Passage 55 2014-08-16-1

Hunting and the Setting of Inner Eurasia

Inner Eurasia refers to the large continental area extending from Russia in the west to the Pacific Ocean, and to the north of Iran, India, and most of China. The first systematic colonization of parts of Inner Eurasia occurred about 80,000 to 90,000 years ago, which is relatively late in human history compared with Africa, Europe, and southern Asia. Why was it difficult to settle?

The long, cold, arid winters of this region's steppes (grass covered plains) posed two distinctive problems for human settlers. The first was how to keep warm. Humans may have used fire even a million years ago. Presumably their ability to scavenge animal carcasses meant that they could use skins or furs for warmth. However, there are no signs of hearths before about 200,000 years ago. This suggests that humans used fire opportunistically and had not yet domesticated it enough to survive the harsh winters of Ice Age Inner Eurasia.

The second, even trickier problem was getting food during the long winters. It was not that Inner Eurasia lacked sources of food. The problem was that the food was of the wrong kind, and it was not always available. Humans could not exploit the abundant grasses of the steppes, and most of the edible plants died off in winter. So, for long periods of each year, it was necessary to rely mainly on meat. However, hunting is a more difficult, dangerous, and unreliable way of life than gathering. Animals, unlike plants, can evade predators and may even fight back. Hunters must also cover more ground than gatherers.

Setting Inner Eurasia meant overcoming these difficulties. Systematic and reliable hunting methods meant more than the development of new technologies, they also demanded new social structures. According to the formulation of archaeologist Lewis Binford, in a typical hunter/collector food-gathering strategy parties of hunters leave camps with very specific goals in mind, based on intimate knowledge of their intended prey. They may be away for days or weeks at a time and will often store their kill at specific storage sites, from which they will bring food back to a base camp when needed. As a result, they move their base camps less often than in forager societies, but they range more widely, their movements are more carefully planned, and so are their methods of storage.

Thus, hunters have to plan in advance and in great detail. They need reliable information about the movements and habits of animal prey over large areas, which can be secured only by maintaining regular contacts with neighboring groups. Finally, they need reliable methods of storage because, where plant

foods cannot provide a dietary safety net, planning has to be precise and detailed to ensure that there is enough to tide them over in periods of shortage. Such planning appears in the choice of hunting gear, in the selection of routes and prey, in the choice of companions and timing, in the maintenance of communications with neighbors, and in the methods of storage. Failure at any point can be fatal for the entire group.

Hunting strategies also imply greater social complexity. The regular exchange of information and sometimes of material goods is critical not only within groups, but also between groups scattered over large distances. This increases the importance of symbolic exchanges of both goods and information, and makes it necessary to clarify group identity. Internally, groups may split for long periods as hunting parties travel over great distances. All in all, each group has to exist and survive in several distinct configurations.

For these reasons, archaeologist Clive Gamble has argued that the difficulties of setting the Eurasian heartland arose less from the technological than from the social and organizational features of human communities before 120,000 years ago. There is little or no archaeological evidence that these communities engaged in such practices as detailed planning or widespread contacts. Nor is there any physical evidence for storage, raw materials all come from within a radius of 50 kilometers—and usually less than 5 kilometers—of the sites where they were used.

Paragraph 2

The long, cold, arid winters of this region's steppes (grass covered plains) poised two distinctive problems for human settlers. The first was hot to keep warm. Humans may have used fire even a million years ago. Presumably their ability to scavenge animal carcasses meant that they could use skins or furs for warmth. However, there are no signs of hearths before about 200,000 years ago. This suggests that humans used fire opportunistically and had not yet domesticated it enough to survive the **harsh** winters of Ice Age Inner Eurasia.

29. According to paragraph 2, why would it have been difficult for humans to stay warm in Inner Eurasia before 200,000 years ago?

- ☐ Fire had not yet been discovered.
- ☐ Humans did not have access to skins or furs to help keep them warm.
- ☐ Humans had not yet learned how to bring fire into daily use.
- ☐ Lack of fuel made building fires on the steppes almost impossible.

30. The word "**harsh**" in the passage is closest in meaning to

- ☐ severe
- ☐ lengthy
- ☐ exceptional

- ☐ dark

Paragraph 3

The second, even trickier problem was getting food during the long winters. It was not that Inner Eurasia lacked sources of food. The problem was that the food was of the wrong kind, and it was not always available. Humans could not exploit the abundant grasses of the steppes, and most of the edible plants died off in winter. So, for long periods of each year, it was necessary to rely mainly on meat. However, hunting is a more difficult, dangerous, and unreliable way of life than gathering. Animals, unlike plants, can evade predators and may even fight back. Hunters must also cover more ground than gatherers.

31. The word "evade" in the passage is closest in meaning to

- ☐ escape
- ☐ trick
- ☐ capture
- ☐ threaten

32. According to paragraph 3, getting food during the long winters was a problem for humans in Inner Eurasia because

- ☐ the area lacked sources of food
- ☐ steppe animals were not suitable for humans to hunt
- ☐ the animals migrated when the edible plants died off each year
- ☐ the lack of edible plants in the winter forced humans to depend on meat

Paragraph 4

Setting Inner Eurasia meant overcoming these difficulties. Systematic and reliable hunting methods meant more than the development of new technologies, they also demanded new social structures. According to the formulation of archaeologist Lewis Binford, in a typical hunter/collector food-gathering strategy parties of hunters leave camps with very specific goals in mind, based on intimate knowledge of their intended prey. They may be away for days or weeks at a time and will often store their kill at specific storage sites, from which they will bring food back to a base camp when needed. As a result, they move their base camps less often than in forager societies, but they range more widely, their movements are more carefully planned, and so are their methods of storage.

33. Why does the author present an extended discussion of Lewis Binford's formulation of a typical "hunter/collector food-gathering strategy"?

- ☐ To introduce recent evidence from Inner Eurasia that has changed the archaeological understanding of the daily lives of human hunters
- ☐ To support the claim that setting Inner Eurasia must have required new social structures
- ☐ To challenge the claim from the previous paragraph that hunting is a difficult,

- dangerous, and unreliable way of life
- ☐ To present a theory about the life of humans in Inner Eurasia that will be contradicted later in the passage

34. All of the following are mentioned in paragraph 4 as behaviors that Lewis Binford considered typical of the hunter/collector food-gathering strategy

EXCEPT

- ☐ having a detailed knowledge of the animals being hunted
- ☐ ranging over a relatively wide area in search of food
- ☐ storing extra food at places other than the campsite
- ☐ hunting away from the campsite for one day or less

Paragraph 5

Thus, hunters have to plan in advance and in great detail. They need reliable information about the movements and **habits** of animal prey over large areas, which can be secured only by maintaining regular contacts with neighboring groups. Finally, they need reliable methods of storage because, where plant foods cannot provide a dietary safety net, planning has to be precise and detailed to ensure that there is enough to tide them over in periods of shortage. Such planning appears in the choice of hunting gear, in the selection of routes and prey, in the choice of companions and timing, in the maintenance of communications with neighbors, and in the methods of storage. Failure at any point can be fatal for the entire group.

35. The word "**habits**" in the passage is closest in meaning to

- ☐ likes and dislikes
- ☐ biological instincts
- ☐ usual behaviors
- ☐ home environments

36. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Precise and detailed diet planning is needed for times when neither fresh plants nor animals are available.
- ☐ Careful planning for storage is necessary to ensure that there is sufficient food during periods when plant foods are not available.
- ☐ Planning must be precise and detailed in order to ensure that their supply of plant food is safe.
- ☐ To survive periods of shortage, they need either reliable methods of storage or precise and detailed planning.

37. According to paragraph 5, hunting trips require precise and detailed planning in terms of each of the following **EXCEPT**

- when to leave and where to go
- what animals will be hunted
- how the captured prey will be divided among neighboring groups
- who will make up the hunting party and what gear they will bring

Paragraph 6

Hunting strategies also imply greater social complexity. The regular exchange of information and sometimes of material goods is critical not only within groups, but also between groups scattered over large distances. This increases the importance of symbolic exchanges of both goods and information, and makes it necessary to clarify group identity. Internally, groups may split for long periods as hunting parties travel over great distances. All in all, each group has to exist and survive in several distinct configurations.

38. It can be inferred from paragraph 6 that hunting groups differ from other groups in that hunting groups

- tend to have the same individuals for longer periods of time
- have a greater need to establish a clear identity
- generally have social connections only with other hunting groups
- are less likely to exchange information with other groups

39. The word "configurations" in the passage is closest in meaning to

- environments
- arrangements
- situations
- conditions

Paragraph 7

For these reasons, archaeologist Clive Gamble has argued that the difficulties of setting the Eurasian heartland arose less from the technological than from the social and organizational features of human communities before 120,000 years ago. There is little or no archaeological evidence that these communities engaged in such practices as detailed planning or widespread contacts. Nor is there any physical evidence for storage, raw materials all come from within a radius of 50 kilometers—and usually less than 5 kilometers—of the sites where they were used.

40. According to paragraph 7, which of the following was true of human communities before 120,000 years ago?

- They obtained their raw materials from the area in which these materials were used.
- They left little in the way of archaeological evidence that can be used to understand their technologies.
- They were usually located less than 5 kilometers from other human

- communities.
- ☐ They stored raw materials at multiple locations.

Paragraph 5

Thus, hunters have to plan in advance and in great detail. ■ They need reliable information about the movements and habits of animal prey over large areas, which can be secured only by maintaining regular contacts with neighboring groups. ■ Finally, they need reliable methods of storage because, where plant foods cannot provide a dietary safety net, planning has to be precise and detailed to ensure that there is enough to tide them over in periods of shortage. ■ Such planning appears in the choice of hunting gear, in the selection of routes and prey, in the choice of companions and timing, in the maintenance of communications with neighbors, and in the methods of storage. ■ Failure at any point can be fatal for the entire group.

41. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

The consequences of inadequate planning are serious.

Where would the sentence best fit?

42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The long, cold winters of Inner Eurasia made the setting of the region difficult for humans.

-
-
-

Answer Choices

Although humans had fire and animal skins for warmth, they lacked the technology that would have allowed them to hunt animals over a large territory.

Although humans had sufficient means of storage, they could not gather enough edible plants to last them through the year.

Hunting requires social complexity, since information and goods must be exchanged among groups that are scattered across a large territory and that have different members at different times.

Heavy snowfall and extreme temperatures made hunting impossible in Inner Eurasia for much of the year and forced humans to depend on grasses for survival.

Humans would have needed to survive the winters by hunting, which would have required them to be expert planners and organizers.

The absence of certain kinds of archaeological evidence of / sites suggests that before 120,000 years ago, humans weren't socially sophisticated enough to survive in Inner Eurasia.



Written Records

For those ancient civilizations that used writing—for instance, all the great civilizations in Mesoamerica, China, Egypt, and the Near East—written historical records can answer many social questions. A prime goal of the archaeologist studying these societies is therefore to find appropriate texts. Many of the early excavations of the great sites of the Near East had the recovery of clay writing tablets as the main goal. Major finds of this kind are still being made—for example, at the ancient city of Ebla (Tell Mardikh) in Syria, where an archive of 5,000 clay tablets written in an early dialect of Akkadian (Babylonian) was discovered in the 1970s.

In each early literate society, writing had its own function and purpose. For instance, the clay tablets of Mycenaean Greece, dating from around 1200 B.C., were all, without exception, primarily records of commercial transactions (goods coming in or going out) at the Mycenaean palaces. This discovery gives us an impression of many aspects of the Mycenaean economy and a glimpse into craft organization (through the names for the different kinds of craftspeople), as well as introducing the names of the offices of state. But here, as in other cases, accidents of preservation may be important. It could be that the Mycenaeans wrote on clay only for their commercial records and used other perishable materials for literary or historical texts now lost to us. It is certainly true that for the Classical Greek and Roman civilizations, it is mainly official decrees inscribed on marble that have survived. Fragile rolls of papyrus—the predecessor of modern paper—with literary texts on them, have usually remained intact only when retained in the dry air of Egypt, or when buried beneath the volcanic ash covering Pompeii.

Coins also provide a valuable source of written records: they can reveal information about the location where they are found, which can provide evidence about trade practices there, and their inscriptions can be informative about the issuing authority, whether they were city-states (as in ancient Greece) or sole rulers (as in Imperial Rome or in the kingdoms of medieval Europe).

In recent years, one of the most significant advances in Mesoamerican archaeology has come from deciphering many of the inscribed symbols (glyphs) on the stone stelae (pillars or columns) at the largest centers. It had been widely assumed that the inscriptions were exclusively of a calendrical nature or that they dealt with purely religious matters, notably the deeds of the gods. But the inscriptions can now in many cases be interpreted as relating to real historical events, mainly the deeds of the Maya kings. We can now also begin to deduce the likely territories belonging to individual Maya centers. Maya history has thus taken on a new dimension.

Written records undoubtedly contribute greatly to our knowledge of the society in question. But one should not accept them uncritically at face value. Nor should one forget the bias introduced by the accidents of preservation and the particular uses of literacy in a society. The great risk with historical records is that they can impose their own perspective so that they begin not only to supply the answers to our questions but subtly to determine the nature of those questions and even our concepts and terminology. A good example is the question of kingship in Anglo-Saxon England. Most anthropologists and historians tend to think of a king as the leader of a state society. Therefore, when the earliest records for Anglo-Saxon England, found in *the Anglo-Saxon Chronicle*, which took final shape in about A.D. 1155, refer to kings around A.D. 500, it is easy for the historian to think of kings and states at that period. But the archaeology strongly suggests that a full state society did not emerge until the time of King Offa of Mercia in around A.D. 780, or perhaps King Alfred of Wessex in A.D. 871. It is fairly clear that the earlier so-called kings were generally less significant figures than some of the rulers in either Africa or Polynesia in recent times, whom anthropologists would term "chiefs."

Paragraph 1

For those ancient civilizations that used writing—for instance, all the great civilizations in Mesoamerica, China, Egypt, and the Near East—written historical records can answer many social questions. A prime goal of the archaeologist studying these societies is therefore to find appropriate texts. Many of the early excavations of the great sites of the Near East had the recovery of clay writing tablets as the main goal. Major finds of this kind are still being made—for example, at the ancient city of Ebla (Tell Mardikh) in Syria, where an archive of 5,000 clay tablets written in an early dialect of Akkadian (Babylonian) was discovered in the 1970s.

29. The word "appropriate" in the passage is closest in meaning to
- ☐ ancient
 - ☐ suitable
 - ☐ new
 - ☐ complete
30. According to paragraph 1, why did many early excavations of sites of the great civilizations of the Near East have the recovery of clay writing tablets as the main goal?
- ☐ Archaeologists wanted to determine the writing systems used by the ancient societies that once inhabited those sites.
 - ☐ Archaeologists wanted to show that early literate civilizations used clay tablets for their historical records.
 - ☐ Archaeologists hoped that the clay tablets would answer many of their

questions about the ancient societies that once inhabited those sites.

- Archaeologists hoped to find evidence that languages other than early Akkadian had been used by the ancient societies that once inhabited those sites.

Paragraph 2

In each early literate society, writing had its own function and purpose. For instance, the clay tablets of Mycenaean Greece, dating from around 1200 B.C., were all, without exception, primarily records of commercial transactions (goods coming in or going out) at the Mycenaean palaces. This discovery gives us an impression of many aspects of the Mycenaean economy and a glimpse into craft organization (through the names for the different kinds of craftspeople), as well as introducing the names of the offices of state. But here, as in other cases, accidents of preservation may be important. It could be that the Mycenaeans wrote on clay only for their commercial records and used other perishable materials for literary or historical texts now lost to us. It is certainly true that for the Classical Greek and Roman civilizations, it is mainly official decrees inscribed on marble that have survived. Fragile rolls of papyrus—the predecessor of modern paper—with literary texts on them, have usually remained intact only when retained in the dry air of Egypt, or when buried beneath the volcanic ash covering Pompeii.

31. According to paragraph 2, the writing on Mycenaean clay tablets helped to reveal all of the following about Mycenaean society **EXCEPT**:
- the flow of goods entering and leaving palaces
 - the names for various types of craftspeople
 - the names of government offices
 - the kinds of materials used to build Mycenaean palaces
32. The phrase "a glimpse into" in the passage is closest in meaning to
- new evidence of
 - surprising information about
 - a brief view of
 - a complete picture of
33. The word "perishable" in the passage is closest in meaning to
- likely to decay
 - difficult to repair
 - valuable
 - hard to find
34. In paragraph 2, why does the author discuss writing from the Classical Greek and Roman civilizations?
- To help explain why early civilizations wrote mainly on stone and clay tablets

- To explain how the role of physical material in the preservation of texts can affect our understanding of ancient societies
- To compare the function of writing in Classical Greek and Roman civilizations
- To show that some texts on papyrus survived as long as texts inscribed on marble

Paragraph 3

Coins also provide a valuable source of written records: they can reveal information about the location where they are found, which can provide evidence about trade practices there, and their inscriptions can be informative about the issuing authority, whether they were city-states (as in ancient Greece) or sole rulers (as in Imperial Rome or in the kingdoms of medieval Europe).

35. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- The locations where coins are found can provide information about local trade practices, and the writing on the coins can indicate what kind of government issued them.
 - Coins issued in ancient Greece and Rome and in medieval Europe are important sources of information about the role of writing at those locations.
 - Kings and other rulers often used coins to record information about trade with other governments and about their personal lives.
 - Coins became important in conducting trade transactions, and their inscriptions often indicate the locations where they were issued.

Paragraph 4

In recent years, one of the most significant advances in Mesoamerican archaeology has come from deciphering many of the inscribed symbols (glyphs) on the stone stelae (pillars or columns) at the largest centers. It had been widely assumed that the inscriptions were exclusively of a calendrical nature or that they dealt with purely religious matters, notably the deeds of the gods. But the inscriptions can now in many cases be interpreted as relating to real historical events, mainly the deeds of the Maya kings. We can now also begin to deduce the likely territories belonging to individual Maya centers. Maya history has thus taken on a new dimension.

36. The word "notably" in the passage is closest in meaning to
- particularly
 - probably
 - usually
 - understandably
37. According to paragraph 4, one result of understanding the symbols on stone

stelae in Mesoamerica was that archaeologists could start to determine

- ☐ what types of calendars had been developed in Mesoamerica
- ☐ what the Mayans believed about the actions of the gods
- ☐ when texts on stone stelae were used for the first time
- ☐ what territories likely belonged to individual Mayan centers

Paragraph 5

Written records undoubtedly contribute greatly to our knowledge of the society in question. But one should not accept them uncritically at face value. Nor should one forget the bias introduced by the accidents of preservation and the particular uses of literacy in a society. The great risk with historical records is that they can impose their own perspective so that they begin not only to supply the answers to our questions but subtly to determine the nature of those questions and even our concepts and terminology. A good example is the question of kingship in Anglo-Saxon England. Most anthropologists and historians tend to think of a king as the leader of a state society. Therefore, when the earliest records for Anglo-Saxon England, found in *the Anglo-Saxon Chronicle*, which took final shape in about A.D. 1155, refer to kings around A.D. 500, it is easy for the historian to think of kings and states at that period. But the archaeology strongly suggests that a full state society did not emerge until the time of King Offa of Mercia in around A.D. 780, or perhaps King Alfred of Wessex in A.D. 871. It is fairly clear that the earlier so-called kings were generally less significant figures than some of the rulers in either Africa or Polynesia in recent times, whom anthropologists would term "chiefs."

38. In paragraph 5, why does the author discuss *The Anglo-Saxon Chronicle*?

- ☐ To explain why there are only a few written records from Anglo-Saxon England
- ☐ To explain the uses of literacy in Anglo-Saxon England
- ☐ To demonstrate how archaeologists and anthropologists differ in their understanding of kingship
- ☐ To illustrate the risk associated with written records

39. According to paragraph 5, all of the following are true about written records

EXCEPT

- ☐ They help us understand a lot about the society that produced them.
- ☐ They can influence the kinds of questions we ask about the society that produced them.
- ☐ They provide archaeologists with reliable evidence about the uses of literacy in most ancient societies.
- ☐ They can lead to inaccurate interpretations about the organization of ancient societies.

40. Paragraph 5 suggests which of the following about written records from ancient societies?

- They are of far more interest to historians of ancient societies than to the archaeologists studying those societies.
- They should be interpreted in light of other archaeological evidence about the society in question.
- They provide the only reliable sources of evidence about the society that produced them.
- They have been interpreted differently by archaeologists and anthropologists.

Paragraph 4

■ In recent years, one of the most significant advances in Mesoamerican archaeology has come from deciphering many of the inscribed symbols (glyphs) on the stone stelae (pillars or columns) at the largest centers. ■ It had been widely assumed that the inscriptions were exclusively of a calendrical nature or that they dealt with purely religious matters, notably the deeds of the gods. ■ But the inscriptions can now in many cases be interpreted as relating to real historical events, mainly the deeds of the Maya kings. ■ We can now also begin to deduce the likely territories belonging to individual Maya centers. Maya history has thus taken on a new dimension.

41. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Being able to accurately interpret an ancient language has in many cases transformed our knowledge about the society that used it.

Where would the sentence best fit?

42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Written records of ancient civilizations preserved in several forms have added to our knowledge of early literate societies.

-
-
-

Answer Choices

Clay tablets from Mycenaean Greece provide insights into its economy and state structures, and interpreting Mayan symbols on stone stelae has led to a new understanding of Mayan history.

Accidents of preservation of ancient texts can lead to a misunderstanding of the functions and purposes of writing in early literate societies.

The earliest written record of Anglo-Saxon England, *The Anglo-Saxon Chronicle*, describes the early kings of England and provides valuable insights into its emergence as a full state society.

Literary texts written on marble by the Classical Greek and Roman civilizations have survived, while Egyptian literary texts that were written on papyrus are now very rare.

As a result of reading ancient historical texts, historians have come to understand that many early European kings were as powerful as the chiefs of Africa and Polynesia in recent times.

Ancient historical and literary texts present their societies from particular points of view that can lead researcher to develop mistaken assumptions about these societies.

Temperature Regulation in Marine Organisms

There are two extremes of temperature regulation in organisms. Homeotherms are organisms that regulate body temperature to a constant level, usually above that of the ambient (surrounding) environment. A constant and relatively high body temperature enables biochemical reactions to occur in a relatively constant internal environment and at a relatively high rate. Most birds have a body temperature of about 40°C , whereas the temperature of most marine mammals is about 38°C . Because such temperatures are much higher than that of most seawater, marine homeotherms lose heat rapidly to the surrounding environment.

There is another completely different style of living. Poikilotherms are organisms whose body temperature conforms to that of the ambient environment. All subtidal marine invertebrates and most fishes fit into this category. There is an interesting intermediate status in which body temperature is usually somewhat higher than ambient temperature. Strong-swimming fishes, such as skipjack tuna and yellowfin tuna, have this intermediate status. Their rise in temperature above ambient conditions stems from metabolic heat generated by muscular activity (swimming) combined with a heat retention mechanism. The temperature rise is probably necessary to generate the increased biochemical reaction rates that are needed for sustained activity. In contrast, some intertidal animals are not true Poikilotherms, they maintain themselves at lower-than-ambient body temperature, using both evaporation and circulation of body fluids to avoid being heated at low tide by the Sun. Their body temperatures, therefore, differ from that of an inanimate object of the same size and shape that might be placed on the shore. Intertidal organisms absorb and lose heat directly to the air. Darker-colored forms can absorb more heat than can light-colored forms, therefore, variation in color can reflect differences in adaptation to the capture of solar energy at different latitudes.

Ocean temperatures are usually less than 27°C and may be less than 0°C in some locations and during some seasons. Therefore, most homeothermic mammals and birds must lose heat continuously to the environment. Their skin is the main pathway of heat loss, especially by direct conduct of heat from the skin to the contacting colder water. Because animals have a circulatory system, heat loss from the body surface also occurs as warm interior blood is transferred and moves into contact with the periphery of the body. Their bodies also radiate heat, usually in the infrared part of the spectrum. Finally, as animals exhale, the resulting evaporation of water involves a considerable loss of heat.

The first line of defense against heat loss is a well-insulated body surface. Marine birds deal with this problem by means of specially adapted feathers. A series of interlocking contour feathers encloses a thick layer of down feathers that traps

stationary air, which in turn acts as an insulating layer. Whales, porpoises, and seals are insulated against the lower sea temperatures by a thick layer of subcutaneous fat. Sea otters lack such a layer, but they constantly preen and fluff up a relatively thick layer of fur. Such mechanisms are only partly successful, however, and to generate more body heat to maintain a constant temperature, marine mammals usually must have a higher metabolic rate than similarly sized terrestrial (land) animals.

In marine mammals that have limbs, the limbs are the principal sources of heat loss because they expose a relatively greater amount of body surface area per unit volume to cold water. However, warm arterial blood must be supplied to limbs, such as the flipper of a porpoise. Heat loss in porpoises is minimized by a countercurrent heat exchanger. The arteries are surrounded by veins, within which blood is returning to the core of the animal. At any contact point, the artery, which is on the inside, is warmer than a surrounding vein, so heat is lost to the returning venous blood flow. Heat is thus reabsorbed and returned to the porpoise's body core. This spatial relationship of circulatory vessels minimizes heat loss to the flipper and thence to the water. Although the anatomical details are quite different, fishes such as skipjack tuna have a circulatory anatomy based on the same overall design. Arteries and veins in the near-surface musculature are in contact, and in arteries and veins, respectively, blood flows in opposite directions.

Paragraph 2

There is another completely different style of living. Poikilotherms are organisms whose body temperature conforms to that of the ambient environment. All subtidal marine invertebrates and most fishes fit into this category. There is an interesting intermediate status in which body temperature is usually somewhat higher than ambient temperature. Strong-swimming fishes, such as skipjack tuna and yellowfin tuna, have this intermediate status. Their rise in temperature above ambient conditions stems from metabolic heat generated by muscular activity (swimming) combined with a heat retention mechanism. The temperature rise is probably necessary to generate the increased biochemical reaction rates that are needed for **sustained** activity. In contrast, some intertidal animals are not true Poikilotherms, they maintain themselves at lower-than-ambient body temperature, using both evaporation and circulation of body fluids to avoid being heated at low tide by the Sun. Their body temperatures, therefore, differ from that of an inanimate object of the same size and shape that might be placed on the shore. Intertidal organisms absorb and lose heat directly to the air. Darker-colored forms can absorb more heat than can light-colored forms, therefore, variation in color can reflect differences in adaptation to the capture of solar energy at different latitudes.

29. The word "**sustained**" in the passage is closest in meaning to
- ☐ intense

- ☐ necessary
- ☐ continued
- ☐ physical

30. According to paragraph 2, the body temperature of strong-swimming fishes is usually above that of their surroundings probably so they can

- ☐ generate heat to warm themselves in cold waters
- ☐ have enough energy for prolonged activity
- ☐ generate metabolic heat for muscular activity
- ☐ retain heat for later use when the surrounding conditions change

31. According to paragraph 2, some intertidal animals are not considered true poikilotherms because they

- ☐ can change color and therefore the amount of heat that is absorbed
- ☐ keep their body temperature lower than that of the ambient environment
- ☐ use various techniques to lower their body temperature when overheated by the Sun
- ☐ match their body temperature directly with the air temperature

Paragraph 3

Ocean temperatures are usually less than 27°C and may be less than 0°C in some locations and during some seasons. Therefore, most homeothermic mammals and birds must lose heat continuously to the environment. Their skin is the main pathway of heat loss, especially by direct conduct of heat from the skin to the contacting colder water. Because animals have a circulatory system, heat loss from the body surface also occurs as warm interior blood is transferred and moves into contact with the periphery of the body. Their bodies also radiate heat, usually in the infrared part of the spectrum. Finally, as animals exhale, the resulting evaporation of water involves a considerable loss of heat.

32. The word "considerable" in the passage is closest in meaning to

- ☐ gradual
- ☐ unusual
- ☐ temporary
- ☐ significant

33. According to paragraph 3, all of the following contribute to heat loss in homeothermic animals **EXCEPT**

- ☐ heat transfer from the skin to the water
- ☐ the movement of blood to the animal's periphery
- ☐ an increase in their activity during certain seasons
- ☐ the evaporation of water during breathing

Paragraph 4

The first line of defense against heat loss is a well-insulated body surface. Marine birds deal with this problem by means of specially adapted feathers. A series of interlocking contour feathers encloses a thick layer of down feathers that traps **stationary** air, which in turn acts as an insulating layer. Whales, porpoises, and seals are insulated against the lower sea temperatures by a thick layer of subcutaneous fat. Sea otters lack such a layer, but they constantly preen and fluff up a relatively thick layer of fur. Such mechanisms are only partly successful, however, and to generate more body heat to maintain a constant temperature, marine mammals usually must have a higher metabolic rate than similarly sized terrestrial (land) animals.

34. The word "**stationary**" in the passage is closest in meaning to
- ☐ warm
 - ☐ surface
 - ☐ nonmoving
 - ☐ nearby
35. All of the following are mentioned in paragraph 4 as defenses against heat loss **EXCEPT**
- ☐ subcutaneous fat
 - ☐ layers of feathers
 - ☐ a thick layer of fur
 - ☐ constant metabolic rates
36. What can be inferred from the comparison of terrestrial animals to marine animals in the last sentence of paragraph 4?
- ☐ An animal's size is not the only factor affecting its metabolic rate.
 - ☐ An animal's size determines what mechanism can be successfully used to prevent heat loss.
 - ☐ Smaller animals are more successful than larger ones at preventing heat loss.
 - ☐ Terrestrial animals have a wider variety of mechanisms for preventing heat loss than marine animals do.
37. Paragraph 4 expands on paragraph 3 by
- ☐ presenting various reasons why first-time defenses are inadequate to deal with the problem described in paragraph 3
 - ☐ discussing mechanisms that marine animals use to reduce the problem described in paragraph 3
 - ☐ identifying specific ways that the problem described in paragraph 3 harms marine animals
 - ☐ explaining why the problem described in paragraph 3 harms some marine animals more than others

Paragraph 5

In marine mammals that have limbs, the limbs are the principal sources of heat loss because they expose a relatively greater amount of body surface area per unit volume to cold water. However, warm arterial blood must be supplied to limbs, such as the flipper of a porpoise. Heat loss in porpoises is minimized by a countercurrent heat exchanger. The arteries are surrounded by veins, within which blood is returning to the core of the animal. At any contact point, the artery, which is on the inside, is warmer than a surrounding vein, so heat is lost to the returning venous blood flow. Heat is thus reabsorbed and returned to the porpoise's body core. This spatial relationship of circulatory vessels minimizes heat loss to the flipper and thence to the water. Although the anatomical details are quite different, fishes such as skipjack tuna have a circulatory anatomy based on the same overall design. Arteries and veins in the near-surface musculature are in contact, and in arteries and veins, respectively, blood flows in opposite directions.

38. The word "overall" in the passage is closest in meaning to
- ☐ simple
 - ☐ effective
 - ☐ well-planned
 - ☐ general
39. According to paragraph 5, some marine mammals that have limbs minimize heat loss by using a system in which
- ☐ vessels that return blood to the animal's core absorb heat from warm interior arteries
 - ☐ blood returning to the core is warmer than blood flowing from the core
 - ☐ the placement of the arteries reduces blood flow to the veins
 - ☐ both arteries and veins are in contact with near-surface musculature
40. Why does the author discuss "fishes such as skipjack tuna"?
- ☐ To explain by contrast why the circulatory anatomy of porpoises is efficient
 - ☐ To show that marine animals other than mammals use a countercurrent exchange system to minimize heat loss
 - ☐ To identify and illustrate a type of circulatory anatomy that is common in fishes
 - ☐ To provide evidence that the amount of heat marine animals lose increases with increased body surface area

Paragraph 2

There is another completely different style of living. ■ Poikilotherms are organisms whose body temperature conforms to that of the ambient environment. ■ All subtidal marine invertebrates and most fishes fit into this category. ■ There is an interesting intermediate status in which body temperature is usually somewhat higher than ambient temperature. ■ Strong-swimming fishes, such as skipjack tuna and yellowfin tuna, have this intermediate status. Their rise in temperature above ambient conditions stems from metabolic heat generated by muscular

activity (swimming) combined with a heat retention mechanism. The temperature rise is probably necessary to generate the increased biochemical reaction rates that are needed for sustained activity. In contrast, some intertidal animals are not true Poikilotherms, they maintain themselves at lower-than-ambient body temperature, using both evaporation and circulation of body fluids to avoid being heated at low tide by the Sun. Their body temperatures, therefore, differ from that of an inanimate object of the same size and shape that might be placed on the shore. Intertidal organisms absorb and lose heat directly to the air. Darker-colored forms can absorb more heat than can light-colored forms, therefore, variation in color can reflect differences in adaptation to the capture of solar energy at different latitudes.

41. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

However, not all marine organisms can be easily classified as either homeotherms or poikilotherms.

Where would the sentence best fit?



42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Marine and other organisms are classified as homeotherms if they maintain a constant body temperature and as poikilotherms if their body temperature matches that of the environment.

-
-
-

Answer Choices

Although strong-swimming fishes and intertidal organisms have body temperatures that are higher or lower than ambient temperatures, most fishes and subtidal marine organisms are poikilotherms.

There are no homeotherms in cold ocean waters because it is too difficult for homeotherms to prevent heat loss in these environments.

Features such as fur help reduce heat loss, but marine homeotherms require a higher metabolism than do terrestrial homeotherms of similar size to maintain a constant body temperature.

Some organisms are thought to have various methods of heat regulation in order to maintain different body temperatures at different times of year.

Marine homeotherms lose heat to the environment through respiration and other means, the most important

being skin contact with colder seawater.

Animals with limbs are at a disadvantage regarding body temperature because their circulatory anatomy is poorly designed for managing heat loss.

Mesolithic Complexity in Scandinavia

The European Mesolithic (roughly the period from 8000 B.C to 2700 B.C) testifies to a continuity in human culture from the times of the Ice Age. This continuity, however, was based on continuous adjustment to environmental changes following the end of the last glacial period (about 12,500 years ago). Three broad subdivisions within the northern Mesolithic are known in Scandinavia. The Maglemose Period (7500-5700 B.C.) was a time of seasonal exploitation of rivers and lakes, combined with terrestrial hunting and foraging. The sites from the Kongemose Period (5700-4600 B.C.) are mainly on the Baltic Sea coasts, along bays and near lagoons, where the people exploited both marine and terrestrial resources. Many Kongemose sites are somewhat larger than Maglemose ones. The Ertebolle Period (4600-3200 B.C) was the culmination of Mesolithic culture in southern Scandinavia.

By the Ertebolle Period, the Scandinavia were occupying coastal settlements year-round and subsisting off a very wide range of food sources. These included forest game and waterfowl, shellfish, sea mammals, and both shallow-water and deepwater fish. There were smaller, seasonal coastal sites, too, for specific activities such as deepwater fishing, sealing, or hunting of migratory birds. One such site, the Aggersund site in Denmark, was occupied for short periods of time in the autumn, when the inhabitants collected oysters and hunted some game, especially migratory swans. Ertebolle technology was far more elaborate than that of its Mesolithic predecessors, a wide variety of antler, bone, and wood tools for socialized purposes such as fowling and sea-mammal hunting were developed, including dugout canoes up to ten meters long.

With sedentary settlement comes evidence of greater social complexity in the use of cemeteries for burials and changes in burial practices. The trend toward more sedentary settlement, the cemeteries, and the occasional social differentiation revealed by elaborate burials are all reflections of an intensified use of resources among these relatively affluent hunter-gatherers of 3000 B.C Mesolithic societies intensified the food quest by exploiting many more species, making productive use of migratory waterfowl and their breeding grounds, and collecting shellfish in enormous numbers. This intensification is also reflected in a much more elaborate and diverse technology, more exchange of goods and materials between neighbors, greater variety in settlement types, and a slowly rising population throughout southern Scandinavia. These phenomena may, in part, be a reflection of rising sea levels throughout the Mesolithic that flooded many cherished territories. There are signs, too, of regional variations in artifact forms and styles,

indicative of culture differences between people living in well-delineated territories and competing for resources.

Mesolithic cultures are much less well-defined elsewhere in Europe, partly because the climatic changes were less extreme than in southern Scandinavia and because there were fewer opportunities for coastal adaptation. In much of central Europe, settlement was confined to lakeside and riverside locations, widely separated from one another by dense forests. Many Mesolithic lakeside sites were located in transitional zones between different environments so that the inhabitants could return to a central base location, where for much of the year they lived close to predictable resources such as lake fish. However, they would exploit both forest game and other seasonal resources from satellite camps. For example, the archaeologist Michael Jochim believes that some groups lived during most of the year in camps along the Danube River in central Europe, moving to summer encampments on the shores of neighboring lakes. In areas like Spain, there appears to have been intensified exploitation of marine and forest resources. There was a trend nearly everywhere toward greater variety in the diet, with more attention being paid to less obvious foods and to those that require more complex processing methods than do game and other such resources.

Thus, in part of Europe, there was a long-term trend among hunter-gatherer societies toward a more extensive exploitation of food resources, often within the context of a strategy that sought ways to minimize the impact of environmental uncertainty. In more favored southern Scandinavia, such societies achieved a new level of social complexity that was to become commonplace among later farming peoples, and this preadaptation proved an important catalyst for rapid economic and social change when farming did come to Europe.

Paragraph 1

The European Mesolithic (roughly the period from 8000 B.C to 2700 B.C) testifies to a continuity in human culture from the times of the Ice Age. This continuity, however, was based on continuous adjustment to environmental changes following the end of the last glacial period (about 12,500 years ago). Three broad subdivisions within the northern Mesolithic are known in Scandinavia. The Maglemose Period (7500-5700 B.C.) was a time of seasonal exploitation of rivers and lakes, combined with terrestrial hunting and foraging. The sites from the Kongemose Period (5700-4600 B.C.) are mainly on the Baltic Sea coasts, along bays and near lagoons, where the people exploited both marine and terrestrial resources. Many Kongemose sites are somewhat larger than Maglemose ones. The Ertebølle Period (4600-3200 B.C) was the culmination of Mesolithic culture in southern Scandinavia.

43. What can be infer from paragraph 1 about human life in Mesolithic Scandinavia?

- People tend to live in smaller group during the Ertebolle Period than during earlier Mesolithic Period.
- The areas where it was advantages to live changed over time as a result of environmental changes.
- Human groups were less affected by environmental change during the Maglemose Period than during the Kongemose Period.
- During most of the Mesolithic, people were more dependent on terrestrial food sources than other food sources.

Paragraph 2

By the Ertebolle Period, the Scandinavia were occupying coastal settlements year-round and subsisting off a very wide range of food sources. These included forest game and waterfowl, shellfish, sea mammals, and both shallow-water and deepwater fish. There were smaller, seasonal coastal sites, too, for specific activities such as deepwater fishing, sealing, or hunting of migratory birds. One such site, the Aggersund site in Denmark, was occupied for short periods of time in the autumn, when the inhabitants collected oysters and hunted some game, especially migratory swans. Ertebolle technology was far more elaborate than that of its Mesolithic predecessors, a wide variety of antler, bone, and wood tools for socialized purposes such as fowling and sea-mammal hunting were developed, including dugout canoes up to ten meters long.

44. Why does the author mention "Aggersund site in Denmark" and its brief periods of occupation?
- To suggest that supply of year round food sources near earlier settlement sites had nearly disappeared
 - To give an example of a small, temporary coastal site that took advantage of seasonal food sources
 - To illustrate how small coastal settlements could not last as long as large forest settlements
 - To high the fact that none of the Denmark camps were able to be occupied year-round.
45. Paragraph 2 suggests that before the Ertebolle Period, hunting tools and other Mesolithic technologies
- were available only in small coastal sites
 - were developed mainly in Denmark
 - were made mainly from animal bones
 - were somewhat simple

Paragraph 3

With sedentary settlement comes evidence of greater social complexity in the use of cemeteries for burials and changes in burial practices. The trend toward more sedentary settlement, the cemeteries, and the occasional social differentiation

revealed by elaborate burials are all reflections of an intensified use of resources among these relatively affluent hunter-gatherers of 3000 B.C. Mesolithic societies intensified the food quest by **exploiting** many more species, making productive use of migratory waterfowl and their breeding grounds, and collecting shellfish in enormous numbers. This intensification is also reflected in a much more elaborate and diverse technology, more exchange of goods and materials between neighbors, greater variety in settlement types, and a slowly rising population throughout southern Scandinavia. These phenomena may, in part, be a reflection of rising sea levels throughout the Mesolithic that flooded many cherished territories. There are signs, too, of regional variations in artifact forms and styles, **indicative of** culture differences between people living in well-delineated territories and competing for resources.

46. The word "**exploiting**" in the passage is closest in meaning to
- ☐ calling attention to
 - ☐ focusing on
 - ☐ taking advantage of
 - ☐ searching for
47. The word "**indicative of**" in the passage is closest in meaning to
- ☐ suggesting
 - ☐ leading to
 - ☐ resulting from
 - ☐ decreasing
48. According to paragraph 3, the existence of cemeteries in M Mesolithic Scandinavia is associated with
- ☐ increased social complexity
 - ☐ problems in obtaining sufficient food
 - ☐ a dramatically increasing population
 - ☐ a trend toward reduced dependence on the food sources provided by hunter-gatherers
49. According to paragraph 3, each of the following is linked to the intensified use of resources that occurred in southern Scandinavia **EXCEPT**
- ☐ greater complexity and diversity of technology
 - ☐ greater similarity between artifacts from different regions
 - ☐ more exchanges of goods between groups
 - ☐ more diversity in types of settlements

Paragraph 4

Mesolithic cultures are much less well-defined elsewhere in Europe, partly because the climatic changes were less extreme than in southern Scandinavia and because there were fewer opportunities for coastal adaptation. In much of

central Europe, settlement was **confined** to lakeside and riverside locations, widely separated from one another by dense forests. Many Mesolithic lakeside sites were located in transitional zones between different environments so that the inhabitants could return to a central base location, where for much of the year they lived close to predictable resources such as lake fish. However, they would exploit both forest game and other seasonal resources from satellite camps. For example, the archaeologist **Michael Jochim** believes that some groups lived during most of the year in camps along the Danube River in central Europe, moving to summer encampments on the shores of neighboring lakes. In areas like Spain, there appears to have been intensified exploitation of marine and forest resources. There was a trend nearly everywhere toward greater variety in the diet, with more attention being paid to less obvious foods and to those that require more complex processing methods than do game and other such resources.

50. The word "**confined**" in the passage is closest in meaning to
- ☐ adapted to
 - ☐ limited to
 - ☐ expanded to
 - ☐ located next to
51. Select the **TWO** answer choices that are discussed in paragraph 4 as factors that explain why Mesolithic culture in Scandinavia developed differently from Mesolithic culture in other parts of Europe. To receive credit, you must choose **TWO** answers from the answer choices.
- ☐ Climate
 - ☐ Population numbers
 - ☐ To size of satellite camps
 - ☐ The location of sites of human habitation
52. According to paragraph 4, how were Mesolithic societies in central Europe able to meet their food needs for much of the year?
- ☐ By finding new opportunities for coastal adaptations wherever they could
 - ☐ By keeping their base camps in dense forests with plenty of forest game
 - ☐ By settling in areas that provided both predictable food resources and access to different kinds of environments
 - ☐ By hunting farther and farther from their central base camps each day
53. The author includes the views of **Michael Jochim** for which of the following reasons?
- ☐ To show that the Mesolithic cultures in other parts of Europe were less well-defined than Mesolithic cultures in southern Scandinavia
 - ☐ To help explain why it became necessary for Mesolithic populations to pay more attention to less obvious foods and foods requiring more complex processing

- To provide evidence that opportunities for coastal development were generally much greater in Central Europe than elsewhere in Europe
- To support the idea that populations in Central Europe stayed in a central location much of the year but relocated to be closer to seasonal resources

Paragraph 5

Thus, in part of Europe, there was a long-term trend among hunter-gatherer societies toward a more extensive exploitation of food resources, often within the context of a strategy that sought ways to minimize the impact of environmental uncertainty. In more favored southern Scandinavia, such societies achieved a new level of social complexity that was to become commonplace among later farming peoples, and this preadaptation proved an important catalyst for rapid economic and social change when farming did come to Europe.

54. Which of the sentences below best expresses the essential information in the highlight sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- Because of favorable conditions in southern Scandinavia and the social complexity of their societies, hunter-gatherer societies did not adapt to farming until economic change required it.
 - When farming came to Europe and became common, hunter-gatherer societies finally achieved high levels of social complexity.
 - Social complexity was common in the societies of southern Scandinavia but was less common in other areas where farming came later.
 - Hunter-gatherer societies in southern Scandinavia achieved a new level of social complexity, and this allowed them to quickly achieve economic and social change when farming was introduced.

Paragraph 1

The European Mesolithic (roughly the period from 8000 B.C to 2700 B.C) testifies to a continuity in human culture from the times of the Ice Age. ■ This continuity, however, was based on continuous adjustment to environmental change following the end of the last glacial period (about 12,500 years ago). ■ Three broad subdivisions within the northern Mesolithic are known in Scandinavia. ■ The Maglemose Period (7500-5700 B.C.) was a time of seasonal exploitation of rivers and lakes, combined with terrestrial hunting and foraging. ■ The sites from the Kongemose Period (5700-4600 B.C) are mainly on the Baltic Sea coasts, along bays and near lagoons, where the people exploited both marine and terrestrial resources. Many Kongemose sites are somewhat larger than Maglemose ones. The Ertebølle Period (4600-3200 B.C) was the culmination of Mesolithic culture in southern Scandinavia.

55. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Such adaptation involved changes in hunting and food gathering that make it possible to identify distinct Mesolithic periods and cultures in some regions

Where would the sentence best fit?

56. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Throughout the European Mesolithic, the trend was toward greater social complexity and greater exploitation of food resources.

-
-
-

Answer Choices

Early in Mesolithic, Maglemose hunter-gatherers arrived in Scandinavia from regions beyond the Baltic Sea and began to establish seasonal sites along bays and near lagoons.

In Scandinavia, populations were growing and becoming more sedentary, leading to more elaborate burials and to exchanges of goods with other groups.

Because most Mesolithic populations outside of Scandinavia were widely separated by deep forest, they tended to develop more economically and socially complex

societies.

By the end of the Mesolithic, Scandinavians had developed elaborate and specialized tools for exploiting a very wide range of food sources.

Climate change forced southern Scandinavia to leave coastal areas and relocate to Central Europe, where they took advantage of food resources in dense forests and lakes.

The stages of Mesolithic culture in Scandinavia, unlike those in other parts of Europe, are more distinct in part because climate changes were more extreme in Scandinavia.

Evolution of the Flowering Plants

Many aspects of the history of flowering plants (angiosperms) remain mysterious. Evidence of the earliest angiosperms comes from fossilized leaves, stems, fruits, pollen, and, very rarely, flowers. In addition, there has been much study of modern plant morphology (structure) and genetics in order to determine which living species might be most closely related to the ancient ancestors of angiosperms. Despite intensive efforts for over 200 years, scientists have still not reached consensus on which type of plant was the ancestor to the angiosperms, and when and where the angiosperms first evolved. Indeed, Charles Darwin himself called the origin of the flowering plants an "abominable mystery."

What type of plant was the ancestor to the angiosperms? Most botanists now agree that the flowering plants are monophyletic in origin, meaning that they evolved from a common ancestor. Some paleontologists have suggested that the common ancestor may have been a type of cycad (palmlike tropical plants). Other paleontologists maintain that the angiosperms may have evolved from seed-bearing ferns. Finally, analysis of the morphological traits of some primitive living plants suggests that the ancestor may have been related to the modern pines. The question of angiosperm ancestry remains unresolved.

The time and place of the first appearance of flowering plants have long been a topic of great interest. There is good fossil evidence that early angiosperms, including a number resembling modern magnolias, were present in the Early Cretaceous geologic period (more than 100 million years ago). Angiosperms became increasingly abundant during this period. Between 100 million and 65 million years ago, a period known as the Late Cretaceous, angiosperms increased from less than 1 percent of flora (plant life) to well over 50 percent. Many of the modern plant families appeared during this time period. In the Early Tertiary period which followed, angiosperms increased to comprise 90 percent or more of Earth's total flora. Where did these successful plants first originate and spread from?

Analysis of the fossil leaf structure and geographic distribution of the earliest Cretaceous angiosperms has led many biogeographers to conclude that they evolved in the tropics and then migrated poleward. It is known that angiosperms did not become dominant in the high latitudes until the Late Cretaceous. Paleontologists have recovered fossil angiosperm leaves, stems, and pollen from Early Cretaceous deposits in eastern South America and western Africa. These two continents were joined together as part of Gondwanaland, one of two supercontinents that existed at that time. The locations of these early angiosperm finds would have been close to the equator during the Early Cretaceous and are comfortable with a model by which angiosperms spread from the tropics poleward.

Not all botanists agree with an African-South American center for the evolution and dispersal of the angiosperms, pointing out that many of the most primitive forms of flowering plants are found in the South Pacific, including portions of Fiji, New Caledonia, New Guinea, eastern Australia, and the Malay Archipelago. Recent genetic research has identified the rare tropical shrub *Amborella* as being the living plant most closely related to the ancient ancestor of all the angiosperms. This small shrub, which has tiny yellow-white flowers and red fruit, is found only on New Caledonia, a group of islands in the South Pacific. Many botanists conclude that the best explanation for the large numbers of primitive living angiosperms in the South Pacific region is that this is where the flowering plants first evolved and these modern species are relics of this early evolution. Comparisons of the DNA of *Amborella* and many hundreds of species of flowering plants suggest that the first angiosperm arose and the development of separate species occurred about 135 million years ago.

Recently discovered fossils complicate our understanding of the origin of the angiosperms even further. Paleontologists from China have found beautifully preserved fossils of an angiosperm plant, including flowers and seeds, in Jurassic period deposits from China. The site, which is about 130 million years old, is near modern Beijing. The new fossil plant found at the site is now the oldest known angiosperm. The age of the fossils and the very primitive features of the flowers have led the discoverers to suggest that the earliest flowering plants may have evolved in northern Asia.

Paragraph 1

Many aspects of the history of flowering plants (angiosperms) remain mysterious. Evidence of the earliest angiosperms comes from fossilized leaves, stems, fruits, pollen, and, very rarely, flowers. In addition, there has been much study of modern plant morphology (structure) and genetics in order to determine which living species might be most closely related to the ancient ancestors of angiosperms. Despite **intensive** efforts for over 200 years, scientists have still **not reached consensus on** which type of plant was the ancestor to the angiosperms, and when and where the angiosperms first evolved. Indeed, Charles Darwin himself called the origin of the flowering plants an "abominable mystery."

43. The word "**intensive**" in the passage is closest in meaning to

- ☐ periodic
- ☐ various
- ☐ strong
- ☐ promising

44. **not reached consensus on**

- ☐ not formulated a theory about

- ☐ not agreed on
- ☐ not been able to explain
- ☐ not found evidence for

45. According to paragraph 1, all of the following types of evidence have been important for studying the history of angiosperms **EXCEPT**

- ☐ fossils of angiosperm stems and leaves
- ☐ knowledge of the structure of modern angiosperms
- ☐ large collections of well-preserved angiosperm flowers
- ☐ genetic studies of modern angiosperms

Paragraph 2

What type of plant was the ancestor to the angiosperms? Most botanists now agree that the flowering plants are monophyletic in origin, meaning that they evolved from a common ancestor. Some paleontologists have suggested that the common ancestor may have been a type of cycad (palmlike tropical plants). Other paleontologists **maintain** that the angiosperms may have evolved from seed-bearing ferns. Finally, analysis of the morphological traits of some primitive living plants suggests that the ancestor may have been related to the modern pines. The question of angiosperm ancestry remains unresolved.

46. The word "maintain" in the passage is closest in meaning to

- ☐ reply
- ☐ assume
- ☐ suspect
- ☐ claim

47. According to paragraph 2, most botanists now believe which of the following about angiosperms?

- ☐ The ancestor of today's angiosperms was most likely a palmlike tropical plant.
- ☐ Today's angiosperms can be traced back to a single ancestor.
- ☐ Angiosperms likely evolved from three very different plants.
- ☐ Angiosperms share a common ancestor with modern pines.

Paragraph 3

The time and place of the first appearance of flowering plants have long been a topic of great interest. There is good fossil evidence that early angiosperms, including a number resembling modern magnolias, were present in the Early Cretaceous geologic period (more than 100 million years ago). Angiosperms became increasingly abundant during this period. Between 100 million and 65 million years ago, a period known as the Late Cretaceous, angiosperms increased from less than 1 percent of flora (plant life) to well over 50 percent. Many of the modern plant families appeared during this time period. In the Early Tertiary period which followed, angiosperms increased to comprise 90 percent or more of Earth's

total flora. Where did these successful plants first originate and spread from?

48. Which of the following can be inferred from paragraph 3 about angiosperms during the Early Cretaceous?

- ☐ They represented less than 1 percent of plant life.
- ☐ Their numbers remained stable during this period.
- ☐ They existed in forms that are very unlike plants living today.
- ☐ They only rarely left traces in the fossil record.

Paragraph 4

Analysis of the fossil leaf structure and geographic distribution of the earliest Cretaceous angiosperms has led many biogeographers to conclude that they evolved in the tropics and then migrated poleward. It is known that angiosperms did not become dominant in the high latitudes until the Late Cretaceous. Paleontologists have recovered fossil angiosperm leaves, stems, and pollen from Early Cretaceous deposits in eastern South America and western Africa. These two continents were joined together as part of Gondwanaland, one of two supercontinents that existed at that time. The locations of these early angiosperm finds would have been close to the equator during the Early Cretaceous and are comfortable with a model by which angiosperms spread from the tropics poleward.

49. fossil angiosperm leaves, stems, and pollen that were once ... ?

- ☐ To provide support for the claim that angiosperms first appeared in the tropics
- ☐ To explain how scientists have determined the geographical location of supercontinents during Earth's early history
- ☐ To provide evidence that angiosperms evolved in similar ways in different parts of the world
- ☐ To challenge the idea that angiosperms spread toward the poles during the Cretaceous

50.

- ☐ They ... the poles and began spreading there.
- ☐ They spread from east to west on the Gondwanaland supercontinent.
- ☐ They were the main type of plant in high-latitude ecosystems.
- ☐ They were growing in some areas with a tropical climate.

Paragraph 5

Not all botanists agree with an African-South American center for the evolution and dispersal of the angiosperms, pointing out that many of the most primitive forms of flowering plants are found in the South Pacific, including portions of Fiji, New Caledonia, New Guinea, eastern Australia, and the Malay Archipelago. Recent genetic research has identified the rare tropical shrub Amborella as being the living plant most closely related to the ancient ancestor of all the angiosperms.

This small shrub, which has tiny yellow-white flowers and red fruit, is found only on New Caledonia, a group of islands in the South Pacific. Many botanists conclude that the best explanation for the large numbers of primitive living angiosperms in the South Pacific region is that this is where the flowering plants first evolved and these modern species are relics of this early evolution. Comparisons of the DNA of Amborella and many hundreds of species of flowering plants suggest that the first angiosperm arose and the development of separate species occurred about 135 million years ago.

51. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Some botanists who agree with the idea of an African-South American center for the dispersal of angiosperms point to primitive South Pacific angiosperms as evidence.
- ☐ Some botanists argue that the very primitive forms of angiosperms found in the South Pacific call into question the theory that angiosperm evolution began in Africa and South America.
- ☐ Angiosperms in Fiji, New Caledonia, New Guinea, eastern Australia, and the Malay Archipelago are more primitive than angiosperms in other parts of the world.
- ☐ Botanists disagree over whether the most primitive forms of living plants are located in parts of Africa and South America or in the South Pacific.

52. What can be inferred from paragraph 5 about primitive living angiosperms in the South Pacific?

- ☐ They more closely resemble the earliest angiosperms than do most other living angiosperms.
- ☐ They are very similar to angiosperm plants in South America and Africa.
- ☐ They are only found on the group of islands called New Caledonia.
- ☐ They provide evidence that angiosperms evolved in several sites simultaneously.

Paragraph 6

Recently discovered fossils complicate our understanding of the origin of the angiosperms even further. Paleontologists from China have found **beautifully preserved** fossils of an angiosperm plant, including flowers and seeds, in Jurassic period deposits from China. The site, which is about 130 million years old, is near modern Beijing. The new fossil plant found at the site is now the oldest known angiosperm. The age of the fossils and the very primitive features of the flowers have led the discoverers to suggest that the earliest flowering plants may have evolved in northern Asia.

53. In describing Jurassic-age fossils found in China as "**beautifully preserved**" the

author means that these fossils

- ☐ seem much younger than they actually are
- ☐ were found as a result of unusually good luck
- ☐ have suffered little damage over time
- ☐ were comparatively easy to dig up

54. According to paragraph 6, which of the following is true of angiosperm fossils discovered in China?

- ☐ They have made it even more challenging to understand the origin of angiosperms.
- ☐ They contain flowers that appear modern for the age of the fossils.
- ☐ They convinced most botanists that angiosperms did not originate in northern Asia.
- ☐ They are not as old as angiosperm fossils discovered in the South Pacific.

Paragraph 2

What type of plant was the ancestor to the angiosperms? Most botanists now agree that the flowering plants are monophyletic in origin, meaning that they evolved from a common ancestor. ■ Some paleontologists have suggested that the common ancestor may have been a type of cycad (palmlike tropical plants). ■ Other paleontologists maintain that the angiosperms may have evolved from seed-bearing ferns. ■ Finally, analysis of the morphological traits of some primitive living plants suggests that the ancestor may have been related to the modern pines. ■ The question of angiosperm ancestry remains unresolved.

55. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

But their origin and evolution have been under considerable dispute.

Where would the sentence best fit?

56. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Despite many years of research, some basic questions about the history of angiosperms remain unanswered.

-
-
-

Answer Choices

One of the major difficulties of studying angiosperm evolution is that fossils from angiosperms are very rarely discovered.

Fossil evidence led some botanists to conclude that angiosperms first evolved in the tropics and spread outward toward the poles.

Emerging research from fossils in South America, the South Pacific, and Asia suggests that the first angiosperms appeared more recently than was once believed.

Though the original ancestor of angiosperms is unknown, they existed at least by the Early Cretaceous and eventually made up most of the plants on Earth.

Early Cretaceous fossil deposits have confirmed that the ancestor to the angiosperms was a plant resembling the modern magnolia.

The study of living angiosperms in the South Pacific and new fossil discoveries in China have led to new theories about where angiosperms first evolved.

How Herding Can Provide Safety

In open grasslands there is no place for a large animal to hide. Thus a watchful grazing animal will see the slight movement that betrays the presence of a predator long before it is close enough to launch an attack. It sounds as though the hunters (predators) stand no chance at all. Unfortunately for the grazers, life is not so simple, however. A grazing animal must lower its head and look at the ground to feed. Its attention may be occupied for only a few seconds before it raises its head and resumes its watch while chewing the food it took, but hunters are patient and skillful and are concentrating intensely. Those few seconds provide time enough to advance a few steps and then freeze, body flattened against the ground. It may take hours, but eventually these repeated small advances will put the hunter within range—close enough to outrun its prey—and the long time the hunt has taken will have been worthwhile, because the resulting feast will be highly nutritious.

Clearly the grazers are at a disadvantage, because while they eat they are vulnerable to attack. The hunters also have a weakness, however, and it is one that allows the grazers to survive. Hunters can attack only one prey animal at a time. This applies even to the predators that hunt as a team, such as lionesses, wolves, and hunting dogs. Their hunt involves running down or ambushing an individual. Teamwork allows them to hunt animals much bigger and stronger than themselves and to hunt more successfully, but it does not allow them to attack more than one individual at a time.

The grazers exploit this weakness by making it as difficult as they can for the predators to choose an individual as a target. They do not graze alone, scattered widely across the landscape, but together, as a herd. The approaching hunter sees not a solitary animal, but a crowd of animals, all of them moving, so they are constantly crossing and recrossing each other's paths. No sooner does the hunter choose an individual than another animal has crossed in front of it and the target has disappeared into the herd. From the hunter's point of view this is highly confusing behavior—as, indeed, it is meant to be.

There is another advantage to the grazers. A herd is much more alert than a solitary animal. An animal has to relax its guard while it is taking food, but in a herd there are at any time some animals with their heads down, biting, and others, with their heads up, watching. What is more, those with their heads up are looking in different directions so that together they are alert to any movement anywhere on the landscape around them. There is no way for a hunter to approach a herd unobserved. When a member of the herd spots trouble, it starts to move away.

Other members of the herd move with it and the entire herd starts to move. If the trouble is serious and close, the herd will run. The individual raising the alarm is simply protecting itself, but in doing so it is warning all of the others.

Herding is highly successful, provided members of the herd stay together in a tight bunch. The hunter moves with the herd, watching for an individual to wander away from the others. When that happens, it tries to move between that individual and the rest of the herd, preventing it from rejoining. Once it has done that the hunter has a good chance of making a kill. If the herd starts to run, a solitary hunter may abandon the chase, but a pack of wolves or hunting dogs will regard the running herd of animals as an opportunity and set off in pursuit. As the herd runs, one or two old or sick animals, or young animals that become separated from their mothers, may fall behind.

As soon as the hunter or hunters seize their prey, they lose interest in all other grazers since then they, too, must concentrate on eating, at which point the herd stops running, those who were left behind rejoin the group, and they all resume grazing.

Paragraph 1

In open grasslands there is no place for a large animal to hide. Thus a watchful grazing animal will see the slight movement that betrays the presence of a predator long before it is close enough to launch an attack. It sounds as though the hunters (predators) stand no chance at all. Unfortunately for the grazers, life is not so simple, however. A grazing animal must lower its head and look at the ground to feed. Its attention may be occupied for only a few seconds before it raises its head and resumes its watch while chewing the food it took, but hunters are patient and skillful and are concentrating intensely. Those few seconds provide time enough to advance a few steps and then freeze, body flattened against the ground. It may take hours, but eventually these repeated small advances will put the hunter within range—close enough to outrun its prey—and the long time the hunt has taken will have been worthwhile, because the resulting feast will be highly nutritious.

43. The word "slight" in the passage is closest in meaning to
- ☐ hidden
 - ☐ slow
 - ☐ distant
 - ☐ small
44. According to paragraph 1, which of the following presents a serious difficulty for the predators that hunt animals grazing in grasslands?
- ☐ It is difficult to find grazers in grasslands when they are feeding with their heads down.

- Predators lack places to hide from prey in grasslands.
 - It is difficult for predators to outrun grazers on grasslands.
 - Grazers in open grasslands tend to be very large animals.
45. Which of the following can be inferred from paragraph 1 about grazing animals?
- They spend significantly more time looking down at the ground than they do with their heads raised.
 - They tend to remain within a relatively small area for extended periods of time.
 - They are extremely difficult for predators to see as long as they remain motionless.
 - They can run significantly faster than most of their predators.
46. Which of the following can be inferred from paragraph 1 about grazing animals?
- They spend significantly more time looking down at the ground than they do with their heads raised.
 - They tend to remain within a relatively small area for extended periods of time.
 - They are extremely difficult for predators to see as long as they remain motionless.
 - They can run significantly faster than most of their predators.

Paragraph 2

Clearly the grazers are at a disadvantage, because while they eat they are vulnerable to attack. The hunters also have a weakness, however, and it is one that allows the grazers to survive. Hunters can attack only one prey animal at a time. This applies even to the predators that hunt as a team, such as lionesses, wolves, and hunting dogs. Their hunt involves running down or ambushing an individual. Teamwork allows them to hunt animals much bigger and stronger than themselves and to hunt more successfully, but it does not allow them to attack more than one individual at a time.

47. According to paragraph 2, what is an advantage to predators that hunt as a team?
- A team can kill animals that are much bigger and stronger than any of the predators could take on by itself.
 - A team can ambush their prey without having to run them down as an individual predator has to do.
 - A team can attack more prey animals at one time than an individual predator could.
 - A team is less vulnerable to an aggressive response from grazers.

48. In paragraph 2, why does the author describe the hunting technique used by predators that hunt as a team?
- ☐ To explain why predators that hunt as a team wait until grazers start eating to attack
 - ☐ To argue that species that hunt as a team do not have the same weakness as individual predators
 - ☐ To emphasize that even predators that hunt in groups cannot hunt and attack more than one prey at a time
 - ☐ To help explain why grazers are at a particular disadvantage with certain predatory animals

Paragraph 3

The grazers **exploit** this weakness by making it as difficult as they can for the predators to choose an individual as a target. They do not graze alone, scattered widely across the landscape, but together, as a herd. The approaching hunter sees not a solitary animal, but a crowd of animals, all of them moving, so they are constantly crossing and recrossing each other's paths. No sooner does the hunter choose an individual than another animal has crossed in front of it and the target has disappeared into the herd. From the hunter's point of view this is highly confusing behavior—as, indeed, it is meant to be.

49. The word "exploit" in the passage is closest in meaning to
- ☐ react to
 - ☐ spread out
 - ☐ cover up
 - ☐ take advantage of

Paragraph 4

There is another advantage to the grazers. A herd is much more alert than a solitary animal. An animal has to relax its guard while it is taking food, but in a herd there are at any time some animals with their heads down, biting, and others, with their heads up, watching. What is more, those with their heads up are looking in different directions so that together they are alert to any movement anywhere on the landscape around them. There is no way for a hunter to **approach** a herd unobserved. When a member of the herd spots trouble, it starts to move away. Other members of the herd move with it and the entire herd starts to move. If the trouble is serious and close, the herd will run. The individual raising the alarm is simply protecting itself, but in doing so it is warning all of the others.

50. The word "approach" in the passage is closest in meaning to
- ☐ come nearer to
 - ☐ drive apart
 - ☐ attack
 - ☐ pursue

51. Which of the following is **NOT** mentioned in paragraph 4 as a way in which animals that graze as a herd benefit?
- ☐ Some animals in the herd will be in a position to observe danger while others have their heads down.
 - ☐ Each animal in a herd tries to protect the others from danger.
 - ☐ As a herd, animals can observe the landscape in a variety of directions at once.
 - ☐ The movement of any animal away from potential trouble warns all of the other animals within the herd.

Paragraph 5

Herding is highly successful, **provided** members of the herd stay together in a tight bunch. The hunter moves with the herd, watching for an individual to wander away from the others. When that happens, it tries to move between that individual and the rest of the herd, preventing it from rejoining. Once it has done that the hunter has a good chance of making a kill. If the herd starts to run, a solitary hunter may abandon the chase, but a pack of wolves or hunting dogs will regard the running herd of animals as an opportunity and set off in pursuit. As the herd runs, one or two old or sick animals, or young animals that become separated from their mothers, may fall behind.

52. The word "provided" in the passage is closest in meaning to
- ☐ if
 - ☐ because
 - ☐ unless
 - ☐ therefore
53. According to paragraph 5, any member of a herd is most at risk from predators when
- ☐ it is very young
 - ☐ it becomes separated from the others
 - ☐ the herd includes relatively few old or sick animals
 - ☐ the herd begins to run

Paragraph 6

As soon as the hunter or hunters seize their prey, they lose interest in all other grazers since then they, too, must concentrate on eating, at which point the herd stops running, those who were left behind rejoin the group, and they all resume grazing.

54. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- The hunter or hunters are interested only in seizing their prey and not in any of the rest of the grazers in the herd.
- Because the hunters must concentrate on eating once they make a kill, the rest of the herd can then stop running and resume grazing.
- Once the hunters finish eating their prey, they lose interest in the rest of the herd that was left behind.
- Because they must also concentrate on eating, herd animals resume grazing as soon as the hunter or hunters seize their prey.

Paragraph 1

In open grasslands there is no place for a large animal to hide. ■ Thus a watchful grazing animal will see the slight movement that betrays the presence of a predator long before it is close enough to launch an attack. It sounds as though the hunters (predators) stand no chance at all. ■ Unfortunately for the grazers, life is not so simple, however. ■ A grazing animal must lower its head and look at the ground to feed. ■ Its attention may be occupied for only a few seconds before it raises its head and resumes its watch while chewing the food it took, but hunters are patient and skillful and are concentrating intensely. Those few seconds provide time enough to advance a few steps and then freeze, body flattened against the ground. It may take hours, but eventually these repeated small advances will put the hunter within range—close enough to outrun its prey—and the long time the hunt has taken will have been worthwhile, because the resulting feast will be highly nutritious.

55. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

This is because grasslands lack the trees and heavy bush to provide cover for anything but very small animals.

Where would the sentence best fit?

56. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Grazing in a herd rather than as solitary individuals decreases the risk to grazers from predators.

-
-
-

Answer Choices

If a grazer can spot a predator before it actually attacks, the grazer has a very good chance of getting away and since grasslands provide little cover for predators, the grazer always has the advantage.

Hunters can attack only one grazer at a time, and where there are many animals moving back and forth and grazing together in a herd, it is difficult for hunters to maintain their focus on a single target.

When a member of a herd wanders away from the others or falls behind as the herd runs from danger, a hunter who manages to get between it and the herd stands a good chance of making a kill.

A grazer cannot watch for danger while feeding, so predators can use those moments gradually to approach a solitary animal unobserved, but in a herd there will always be some animals on alert.

When a herd of grazers spots the

approach of a solitary hunter, it may run toward the hunter, which generally causes it to abandon the chase, but predators hunting as a pack are not so easily driven away.

After a herd of running animals sees that the hunters pursuing it have stopped, the herd tries to help members of the group that were left behind.

The Origins of Writing

It was in Egypt and Mesopotamia (modern-day Iraq) that civilization arose, and it is there that we find the earliest examples of the key feature of civilization, writing. These examples, in the form of inscribed clay tablets that date to shortly before 3000 B.C.E., have been discovered among the archaeological remains of the Sumerians, a gifted people settled in southern Mesopotamia.

The Egyptians were not far behind in developing writing, but we cannot follow the history of their writing in detail because they used a perishable writing material. In ancient times the banks of the Nile were lined with papyrus plants, and from the papyrus reeds the Egyptians made a form of paper, it was excellent in quality but, like any paper, fragile. Mesopotamia's rivers boasted no such useful reeds, but its land did provide good clay, and as a consequence the clay tablet became the standard material. Though clumsy and bulky it has a virtue dear to archaeologists: it is durable. Fire, for example, which is death to papyrus paper or other writing materials such as leather and wood, simply bakes it hard, thereby making it even more durable. So when a conqueror set a Mesopotamian palace ablaze, he helped ensure the survival of any clay tablets in it. Clay, moreover, is cheap, and forming it into tablets is easy, factors that helped the clay tablet become the preferred writing material not only throughout Mesopotamia but far outside it as well, in Syria, Asia Minor, Persia, and even for a while in Crete and Greece. Excavators have unearthed clay tablets in all these lands. In the Near East they remained in use for more than two and a half millennia, and in certain areas they lasted down to the beginning of the common era until finally yielding, once and for all, to more convenient alternatives.

The Sumerians perfected a style of writing suited to clay. This script consists of simple shapes, basically just wedge shapes and lines that could easily be incised in soft clay with a reed or wooden stylus; scholars have dubbed it cuneiform from the wedge-shaped marks (*cunei* in Latin) that are its hallmark. Although the ingredients are merely wedges and lines, there are hundreds of combinations of these basic forms that stand for different sounds or words. Learning these complex signs required long training and much practice, inevitably, literacy was largely limited to a small professional class, the scribes.

The Akkadians conquered the Sumerians around the middle of the third millennium B.C.E., and they took over the various cuneiform signs used for writing Sumerian and gave them sound and word values that fit their own language. The Babylonians and Assyrians did the same, and so did peoples in Syria and Asia Minor. The literature of the Sumerians was treasured throughout the Near East, and

long after Sumerian ceased to be spoken, the Babylonians and Assyrians and others kept it alive as a literary language, the way Europeans kept Latin alive after the fall of Rome. For the scribes of these non-Sumerian languages, training was doubly demanding since they had to know the values of the various cuneiform signs for Sumerian as well as their own language.

The contents of the earliest clay tablets are simple notations of numbers of commodities—animals, jars, baskets, etc. Writing, it would appear, started as a primitive form of bookkeeping. Its use soon widened to document the multitudinous things and acts that are involved in daily life, from simple inventories of commodities to complicated governmental rules and regulations.

Archaeologists frequently find clay tablets in batches. The batches, some of which contain thousands of tablets, consist for the most part of documents of the types just mentioned: bills, deliveries, receipts, inventories, loans, marriage contracts, divorce settlements, court judgments, and so on. These records of factual matters were kept in storage to be available for reference—they were, in effect, files, or, to use the term preferred by specialists in the ancient Near East, archives. Now and then these files include pieces of writing that are of a distinctly different order, writings that do not merely record some matter of fact but involve creative intellectual activity. They range from simple textbook material to literature—and they make an appearance very clearly, even from the third millennium B.C.E.

Paragraph 1

It was in Egypt and Mesopotamia (modern-day Iraq) that civilization arose, and it is there that we find the earliest examples of the **key** feature of civilization, writing. These examples, in the form of inscribed clay tablets that date to shortly before 3000 B.C.E., have been discovered among the archaeological remains of the Sumerians, a gifted people settled in southern Mesopotamia.

57. The word "**key**" in the passage is closest in meaning to

- ☐ frequent
- ☐ essential
- ☐ original
- ☐ familiar

Paragraph 2

The Egyptians were not far behind in developing writing, but we cannot follow the history of their writing in detail because they used a perishable writing material. In ancient times the banks of the Nile were lined with papyrus plants, and from the papyrus reeds the Egyptians made a form of paper, it was excellent in quality but, like any paper, fragile. Mesopotamia's rivers boasted no such useful reeds, but its land did provide good clay, and as a consequence the clay tablet became the standard material. Though clumsy and bulky it has a **virtue** dear to archaeologists:

it is durable. Fire, for example, which is death to papyrus paper or other writing materials such as leather and wood, simply bakes it hard, thereby making it even more durable. So when a conqueror set a Mesopotamian palace ablaze, he helped ensure the survival of any clay tablets in it. Clay, moreover, is cheap, and forming it into tablets is easy, factors that helped the clay tablet become the preferred writing material not only throughout Mesopotamia but far outside it as well, in Syria, Asia Minor, Persia, and even for a while in Crete and Greece. Excavators have unearthed clay tablets in all these lands. In the Near East they remained in use for more than two and a half millennia, and in certain areas they lasted down to the beginning of the common era until finally yielding, once and for all, to more convenient alternatives.

58. The word "virtue" in the passage is closest in meaning to
- ☐ price
 - ☐ design
 - ☐ desirable quality
 - ☐ physical characteristic
59. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ In part because of its low cost and ease of use, clay became the preferred writing material throughout Mesopotamia and well beyond it.
 - ☐ Clay was cheap throughout Mesopotamia, so clay tablets from Mesopotamia became the preferred writing material as far as the Mediterranean.
 - ☐ For a while, the clay tablet was the preferred writing material in Crete and Greece.
 - ☐ Moreover, because clay was used as the writing material of choice in Mesopotamia, Syria, Asia Minor, Persia, and the Mediterranean, it was cheap and popular.
60. What can be inferred from paragraph 2 about clay as a writing material?
- ☐ It had to be baked before it could be written on.
 - ☐ Its good points outweighed its bad points.
 - ☐ Its durability was its most important feature for its users.
 - ☐ It was not available in Egypt.
61. In paragraph 2, why does the author discuss the Egyptian use of papyrus as a writing material?
- ☐ To describe the superiority of papyrus over leather and wood as a writing material
 - ☐ To explain why writing in Egypt did not develop as quickly as it did Mesopotamia
 - ☐ To explain why archaeologists' knowledge of the early history of writing relies

mainly on Sumerian cuneiform

- To explain why the Sumerians preferred clay tablets for writing over papyrus

Paragraph 3

The Sumerians perfected a style of writing suited to clay. This script consists of simple shapes, basically just wedge shapes and lines that could easily be incised in soft clay with a reed or wooden stylus; scholars have dubbed it cuneiform from the wedge-shaped marks (*cunei* in Latin) that are its hallmark. Although the ingredients are merely wedges and lines, there are hundreds of combinations of these basic forms that stand for different sounds or words. Learning these complex signs required long training and much practice, inevitably, literacy was largely limited to a small professional class, the scribes.

62. According to paragraph 3, all of the following are true of cuneiform writing

EXCEPT:

- It was composed of very simple shapes.
- It was perfected by the ancient Sumerians.
- It influenced the choice of material on which it was written.
- It was understood by very few Sumerians.

Paragraph 4

The Akkadians conquered the Sumerians around the middle of the third millennium B.C.E., and they took over the various cuneiform signs used for writing Sumerian and gave them sound and word values that fit their own language. The Babylonians and Assyrians did the same, and so did peoples in Syria and Asia Minor. The literature of the Sumerians was treasured throughout the Near East, and long after Sumerian ceased to be spoken, the Babylonians and Assyrians and others kept it alive as a literary language, the way Europeans kept Latin alive after the fall of Rome. For the scribes of these non-Sumerian languages, training was doubly demanding since they had to know the values of the various cuneiform signs for Sumerian as well as their own language.

63. According to paragraph 4, how did the Akkadians use the Sumerian language?

- They used Sumerian for speaking but used their own national language for writing.
- They used the complex cuneiform signs developed by the Babylonians and Assyrians rather than the Sumerian signs.
- They developed their own cuneiform shapes on clay tablets to replace those used by the Sumerians.
- They assigned new sound and word values to the signs of Sumerian cuneiform.

64. Paragraph 4 answers all the following questions about Sumerian writing in the period after the Sumerians were conquered **EXCEPT:**

- ☐ Did Sumerian literature continue to be read?
- ☐ Did Sumerian continue to be spoken?
- ☐ Did scribes compose new texts in Sumerian?
- ☐ Did Sumerian have the same fate as Latin had after the fall of Rome?

Paragraph 5

The contents of the earliest clay tablets are simple notations of numbers of commodities—animals, jars, baskets, etc. Writing, it would appear, started as a primitive form of bookkeeping. Its use soon widened to **document** the multitudinous things and acts that are involved in daily life, from simple inventories of commodities to complicated governmental rules and regulations.

65. The word "**document**" in the passage is closest in meaning to

- ☐ include
- ☐ influence
- ☐ organize
- ☐ record

66. According to paragraph 5, writing was first used for

- ☐ simple bookkeeping
- ☐ descriptions of daily events
- ☐ counting the contents of clay tablets
- ☐ government reports

Paragraph 6

Archaeologists frequently find clay tablets in batches. The batches, some of which contain thousands of tablets, consist for the most part of documents of the types just mentioned: bills, deliveries, receipts, inventories, loans, marriage contracts, divorce settlements, court judgments, and so on. These records of factual matters were kept in storage to be available for reference—they were, in effect, files, or, to use the term preferred by specialists in the ancient Near East, archives. Now and then these files include pieces of writing that are of a distinctly different order, writings that do not merely record some matter of fact but involve creative intellectual activity. They range from simple textbook material to literature—and they make an appearance very clearly, even from the third millennium B.C.E.

67. The phrase "**document**" in the passage is closest in meaning to

- ☐ always
- ☐ occasionally
- ☐ sooner or later
- ☐ first and last

68. According to paragraph 6, large batches of clay writing tablets were stored because the tablets

- were being produced quickly and in large quantities
- did not serve any practical purpose for most Mesopotamians
- contained information that needed to be available for future reference
- could not be used again once they had been written on

Paragraph 4

The Akkadians conquered the Sumerians around the middle of the third millennium B.C.E., and they took over the various cuneiform signs used for writing Sumerian and gave them sound and word values that fit their own language. ■ The Babylonians and Assyrians did the same, and so did peoples in Syria and Asia Minor. ■ The literature of the Sumerians was treasured throughout the Near East, and long after Sumerian ceased to be spoken, the Babylonians and Assyrians and others kept it alive as a literary language, the way Europeans kept Latin alive after the fall of Rome. ■ For the scribes of these non-Sumerian languages, training was doubly demanding since they had to know the values of the various cuneiform signs for Sumerian as well as their own language. ■

69. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

However, the Sumerian language did not entirely disappear.

Where would the sentence best fit?

70. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The earliest examples of writing have been found in Mesopotamia and date to shortly before 3000 B.C.E.

-
-
-

Answer Choices

Writing was invented in the same areas in which civilization began by the ancient civilization of Mesopotamia, Asia Minor and the Mediterranean.

The development of cuneiform is known because it was written on a long-lasting material and because it was long and widely used throughout the ancient Near East.

Cuneiform tablets generally dealt with business and factual matters, but other topics, including literature, were also recorded and valued.

Writing was developed first by the Sumerians using wedge-shaped marks (cuneiform) on clay tablets and then by the Egyptians using papyrus paper.

Scribes using cuneiform in Assyria, Babylon, Syria and Asia Minor had to learn all the languages that used the cuneiform script.

Batches of clay tablets, sometimes with as many as a thousand tablets each, are often found by archaeologists.



Soil Fertilization

Fertilizers partially restore plant nutrients lost by erosion, crop harvesting, and leaching. Farmers can use either organic fertilizer from plant and animal materials or commercial inorganic fertilizer produced from various minerals. Three basic types of organic fertilizer are animal manure, green manure, and compost. Animal manure includes the waste matter of cattle, horses, poultry, and other farm animals. It improves soil structure, adds organic nitrogen, and stimulates beneficial soil bacteria and fungi.

Despite its effectiveness, the use of animal manure in the United States has decreased. There are three reasons for this: the replacement of most mixed animal-raising and crop-farming operations with separate operations for growing crops and raising animals, the high costs of transporting animal manure from feedlots near urban areas to distant rural crop-growing areas, and the replacement of horses and other draft animals that added manure to the soil with tractors and other motorized farm machinery.

Green manure is fresh or growing green vegetation plowed into the soil to increase the organic matter and humus (degraded organic matter) available to the next crop. Compost is a sweet-smelling, dark-brown, humuslike material that is rich in organic matter and soil nutrients. It is produced when microorganisms in soil (mostly fungi and aerobic bacteria) break down organic matter such as leaves, food wastes, paper, and wood in the presence of oxygen. Compost is a rich natural fertilizer and soil conditioner that aerates soil, improves its ability to retain water and nutrients, helps prevent erosion, and prevents nutrients from being wasted by being dumped in landfills. Compost is produced by piling up alternating layers of nitrogen-rich wastes (such as grass clippings, weeds, animal manure, and vegetable kitchen scraps), carbon-rich plant wastes (dead leaves, hay, straw, sawdust), and topsoil. Compost provides a home for microorganisms that help decompose plant and manure layers and reduces the amount of plant wastes taken to landfills and incinerators.

Another form of organic fertilizer is the spores of mushrooms, puffballs, and truffles. Rapidly growing and spreading mycorrhizae fungi in the spores attach to plant roots and help them take in moisture and nutrients from the soil. Unlike typical fertilizers that must be applied every few weeks, one application of mushroom fungi lasts all year and costs just pennies per plant. The fungi also produce a bigger root system, which makes plants more disease resistant.

Corn, tobacco, and cotton can deplete the topsoil of nutrients, especially nitrogen, if planted on the same land several years in a row. One way to reduce such losses is crop rotation. Farmers plant areas or strips with nutrient-depleting

crops one year. In the next year they plant the same areas with legumes, whose root nodules add nitrogen to the soil. In addition to helping restore soil nutrients, this method reduces erosion by keeping the soil covered with vegetation and helps reduce crop losses to insects by presenting them with a changing target.

Today, many farmers rely on commercial inorganic fertilizers containing nitrogen (as ammonium ions, nitrate ions, or urea), phosphorus (as phosphate ions), and potassium (as potassium ions). Inorganic commercial fertilizers are easily transported, stored, and applied. Worldwide, their use increased about tenfold between 1950 and 1989 but declined by 12% between 1990 and 1999. Today, the additional food they help produce feeds one of every three people in the world, without them, world food output, would drop an estimated 40%.

Commercial inorganic fertilizers have some disadvantages, however. These include (1) not adding humus to the soil, (2) reducing the soil's content of organic matter and thus its ability to hold water (unless animal manure and green manure are also added to the soil), (3) lowering the oxygen content of soil and keeping fertilizer from being taken up as efficiently, (4) typically supplying only two or three of the twenty or so nutrients needed by plants, and (5) releasing nitrous oxide, a greenhouse gas that can enhance global warming. The widespread use of commercial inorganic fertilizers, especially on sloped land near streams and lakes, also causes water pollution as nitrate and phosphate fertilizer nutrients are washed into nearby bodies of water. The resulting plant nutrient enrichment causes algae blooms that use up oxygen dissolved in the water, thereby killing fish.

Paragraph 1

Fertilizers partially restore plant nutrients lost by erosion, crop harvesting, and leaching. Farmers can use either organic fertilizer from plant and animal materials or commercial inorganic fertilizer produced from various minerals. Three basic types of organic fertilizer are animal manure, green manure, and compost. Animal manure includes the waste matter of cattle, horses, poultry, and other farm animals. It improves soil structure, adds organic nitrogen, and stimulates **beneficial** soil bacteria and fungi.

57. The word "**beneficial**" in the passage is closest in meaning to

- ☐ necessary
- ☐ helpful
- ☐ several
- ☐ hidden

Paragraph 2

Despite its effectiveness, the use of animal manure in the United States has decreased. There are three reasons for this: the replacement of most mixed

animal-raising and crop-farming operations with separate operations for growing crops and raising animals, the high costs of transporting animal manure from feedlots near urban areas to distant rural crop-growing areas, and the replacement of horses and other draft animals that added manure to the soil with tractors and other motorized farm machinery.

58. According to paragraph 2, all of the following contributed to the decrease in the use of animal manure in the United States **EXCEPT**

- ☐ changes in crop-farming and animal-raising operations
- ☐ high transportation costs
- ☐ the movement of large numbers of feedlots to distant rural areas
- ☐ the introduction of motorized farm machinery

Paragraph 3

Green manure is fresh or growing green vegetation plowed into the soil to increase the organic matter and humus (degraded organic matter) available to the next crop. Compost is a sweet-smelling, dark-brown, humuslike material that is rich in organic matter and soil nutrients. It is produced when microorganisms in soil (mostly fungi and aerobic bacteria) break down organic matter such as leaves, food wastes, paper, and wood in the presence of oxygen. Compost is a rich natural fertilizer and soil conditioner that aerates soil, improves its ability to retain water and nutrients, helps prevent erosion, and prevents nutrients from being wasted by being dumped in landfills. Compost is produced by piling up alternating layers of nitrogen-rich wastes (such as grass clippings, weeds, animal manure, and vegetable kitchen scraps), carbon-rich plant wastes (dead leaves, hay, straw, sawdust), and topsoil. Compost provides a home for microorganisms that help decompose plant and manure layers and reduces the amount of plant wastes taken to landfills and incinerators.

59. According to paragraph 3, what is one advantage of using compost in raising crops?

- ☐ It makes the soil easier to plow in preparation for planting.
- ☐ Water and nutrients stay in the soil and enable plant growth.
- ☐ The damage to soil from microorganisms is reduced.
- ☐ Excess oxygen is removed from the soil.

60. According to paragraph 3, one important physical effect of compost is that it

- ☐ adds an oxygen-rich layer to the layers rich in nitrogen and carbon
- ☐ decreases the quantity of plant material that needs to be discarded
- ☐ helps keep the same amount of moisture in each layer of soil
- ☐ reduces the amount of fungi and aerobic bacteria in the pile

Paragraph 4

Another form of organic fertilizer is the spores of mushrooms, puffballs, and truffles.

Rapidly growing and spreading mycorrhizae fungi in the spores attach to plant roots and help them take in moisture and nutrients from the soil. Unlike typical fertilizers that must be applied every few weeks, one application of mushroom fungi lasts all year and costs just pennies per plant. The fungi also produce a bigger root system, which makes plants more disease resistant.

61. According to paragraph 4, each of the following is an advantage of using mushroom spores as fertilizer **EXCEPT**:

- ☐ The cost of using mushroom spores is relatively low.
- ☐ A single application of mushroom spores is enough for a whole year.
- ☐ Mushroom fungi are nutrients that plants can take in easily.
- ☐ Mushroom fungi help protect plants from disease.

Paragraph 5

Corn, tobacco, and cotton can deplete the topsoil of nutrients, especially nitrogen, if planted on the same land several years in a row. One way to reduce such losses is crop rotation. Farmers plant areas or strips with nutrient-depleting crops one year. In the next year they plant the same areas with legumes, whose root nodules add nitrogen to the soil. In addition to helping restore soil nutrients, this method reduces erosion by keeping the soil covered with vegetation and helps reduce crop losses to insects by presenting them with a changing target.

62. What is the "target" referred to in the passage?

- ☐ Crop losses that are caused by insects
- ☐ The crop being grown at any given time
- ☐ The areas that are planted with legumes
- ☐ The various insects that cause crop losses

63. According to paragraph 5, one of the main reasons for using legumes in crop rotation is that legumes

- ☐ have no serious insect pests
- ☐ cover the soil more completely than any other crop does
- ☐ build up the nitrogen content of the soil
- ☐ are very easy to plant in strips

Paragraph 6

Today, many farmers **rely** on commercial inorganic fertilizers containing nitrogen (as ammonium ions, nitrate ions, or urea), phosphorus (as phosphate ions), and potassium (as potassium ions). Inorganic commercial fertilizers are easily transported, stored, and applied. Worldwide, their use increased about tenfold between 1950 and 1989 but declined by 12% between 1990 and 1999. Today, the additional food they help produce feeds one of every three people in the world, without them, world food output, would drop an estimated 40%.

64. The word "rely" in the passage is closest in meaning to
- ☐ save
 - ☐ depend
 - ☐ pile
 - ☐ waste money
65. According to paragraph 6, what is true about the quantities of commercial inorganic fertilizers used globally?
- ☐ They are likely to drop by about 40% in the next ten years.
 - ☐ They have increased at a steady rate since 1950.
 - ☐ They increased during the 1990s, but at a slower rate than over the previous 40 years.
 - ☐ They dropped during the 1990s after having risen sharply over the previous 40 years.

Paragraph 7

Commercial inorganic fertilizers have some disadvantages, however. These include (1) not adding humus to the soil, (2) reducing the soil's content of organic matter and thus its ability to hold water (unless animal manure and green manure are also added to the soil), (3) lowering the oxygen content of soil and keeping fertilizer from being taken up as efficiently, (4) typically supplying only two or three of the twenty or so nutrients needed by plants, and (5) releasing nitrous oxide, a greenhouse gas that can enhance global warming. The widespread use of commercial inorganic fertilizers, especially on sloped land near streams and lakes, also causes water pollution as nitrate and phosphate fertilizer nutrients are washed into nearby bodies of water. The resulting plant nutrient enrichment causes algae blooms that use up oxygen dissolved in the water, thereby killing fish.

66. How is paragraph 7 related to paragraph 6?
- ☐ Paragraph 7 presents the drawbacks of a practice that paragraph 6 presents as an advantage.
 - ☐ Paragraph 7 argues that the viewpoint presented in paragraph 6 is based on several factual errors.
 - ☐ Paragraph 7 provides supporting evidence for some of the claims made in paragraph 6.
 - ☐ Paragraph 7 contrasts recently developed practices with the more traditional, established ones discussed in paragraph 6.
67. The phrase "twenty or so" in the passage is closest in meaning to
- ☐ not quite twenty
 - ☐ roughly twenty
 - ☐ no more than twenty
 - ☐ a total of twenty

68. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ The main cause of water pollution is the widespread use of commercial inorganic fertilizers on sloped land near streams and lakes.
- ☐ In addition, the widespread use of commercial inorganic fertilizers causes water pollution when nitrates and phosphates are washed into streams and lakes.
- ☐ Also, the widespread commercial production of inorganic fertilizers has caused water pollution in bodies of water such as streams and lakes.
- ☐ The most commonly used commercial inorganic fertilizers are nitrate and phosphate fertilizers, which cause water pollution if they enter streams or lakes.

Paragraph 1

Fertilizers partially restore plant nutrients lost by erosion, crop harvesting, and leaching. ■ Farmers can use either organic fertilizer from plant and animal materials or commercial inorganic fertilizer produced from various minerals. ■ Three basic types of organic fertilizer are animal manure, green manure, and compost. ■ Animal manure includes the waste matter of cattle, horses, poultry, and other farm animals. ■ It improves soil structure, adds organic nitrogen, and stimulates beneficial soil bacteria and fungi.

69. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Thus, they help keep farmland productive over the long term.

Where would the sentence best fit?

70. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Both organic and inorganic fertilizers are used to keep croplands productive.

-
-
-

Answer Choices

Animal manures are effective organic fertilizers, but their use in the United States has decreased because of changes in modern farming.

Many economically important crops, including corn, tobacco, and cotton, can only be grown in the same fields year after year if large amounts of inorganic fertilizers are added to the soil.

Since 1950, farmers worldwide have begun to replace organic fertilizers with inorganic commercial ones, because the latter helps soil to retain oxygen.

Compost, a humuslike material that is rich in organic material, is often used together with green manure, since the combination of the two helps to prevent soil erosion.

Green manures, compost, mushroom spores, and crop rotation, if used correctly, all have positive impacts on growing conditions for crops that go beyond simply providing nutrients.

Commercial inorganic fertilizers play a key role in feeding the world's population, but they can also cause serious environmental damage.

The Collapse of the Mays

The Mayan society of Central America (2000 B.C. – A.D. 1500), like other ancient states, was characterized by populations unprecedented both in their size and density. It was not just the number of people that lived in the Mayan city-states but also the relatively small area into which they were concentrated. To support such populations, societies developed various intensive agricultural methods, including large-scale irrigation and hill-slope terracing (the cutting of horizontal ridges into hillsides so they can be farmed). These were designed both to increase yields from a given area and to increase the absolute amount of land under cultivation. These strategies were in essence very successful: they made it possible to feed larger populations than ever before and supported the growth of cities. But they also placed considerable strains on the environment and rendered it increasingly fragile and vulnerable to unexpected climatic events, and even to short-term fluctuations. Thus, the argument is that because of their size and ever more intensive agriculture, the Mayan and other ancient state societies were fundamentally unsustainable.

Claims about environmental degradation and disaster have figured prominently in discussions of the collapse of the Mayan city-states of the Central American lowlands. When two explorers came upon the Mayan cities in the 1830s, they were struck by the sight of tall pyramids and elaborately carved stones among luxuriant forest growth. Here was the archetypal picture of a great lost civilization: abandoned cities submerged in vegetation. Theories of catastrophic collapse or apocalyptic overthrow came naturally to mind to explain these dramatic scenes.

Recent studies of the Mayan collapse (beginning around A.D. 900) have emphasized the gradual and progressive nature of the process, beginning in earnest in the South and advancing northward. It was not a single, sudden event, as had once been thought. Warfare and social unrest are thought to have played a part, but these may well have arisen through pressure from other causes. The Mayan cities had, after all, flourished for over 500 years and had frequently been at war with each other.

But what about the possibility of food shortages? These could have come about through either natural or humanly induced changes in the environment. Increasingly fierce competition between Mayan cities led to an upsurge of monument construction during the eighth and ninth centuries A.D., which would have placed added strain on agricultural production and expansion. Interstate rivalry may hence have pushed the Maya toward overexploitation of their fragile ecosystem. Deforestation and soil erosion might ultimately have destroyed the capacity of the land to support the high population levels of the Mayan cities, leading to famine, social unrest, and the collapse of the major Mayan centers.

Yet it may be incorrect to lay the blame entirely on human action. Several of the lowland cities, such as Tikal, appear to have depended heavily on the cultivation of raised fields set in the marshy depressions known as *bajos*, which today flood intermittently in the rainy season but may originally have been permanent lakes. The raised-field system of intensive cultivation (created by digging surrounding canals and using the soil removed to elevate the fields for planting) allows year-round food production through the constant supply of soil nutrients that erode into the drainage ditches dug around the raised fields, nutrients that are then collected and replaced. Stable water levels were essential to this subsistence system, but evidence from Lake Chichancanab in Yucatán shows that between A.D. 800 and A.D. 1000 this region suffered its driest period of climate in several thousand years. We may expect that as a result water levels fell, and the raised fields in many areas became unusable. But the human response must be viewed through the lens of the social, political, and cultural circumstances. These exerted a powerful mediating effect on the way the Maya endeavored to cope with their difficulties. Had population levels been lower, the impact of the drought may not have been catastrophic, as it was, the Maya were already reaching the limits of the available subsistence capacity, and Mayan elites had espoused certain social and political agendas (including expensive warfare and competition with each other). It was against this specific background that a period of drought led quickly to crisis and collapse.

Paragraph 1

The Mayan society of Central America (2000 B.C. – A.D. 1500), like other ancient states, was characterized by populations unprecedented both in their size and density. It was not just the number of people that lived in the Mayan city-states but also the relatively small area into which they were concentrated. To support such populations, societies developed various intensive agricultural methods, including large-scale irrigation and hill-slope terracing (the cutting of horizontal ridges into hillsides so they can be farmed). These were designed both to increase yields from a given area and to increase the absolute amount of land under cultivation. These strategies were in essence very successful: they made it possible to feed larger populations than ever before and supported the growth of cities. But they also placed considerable strains on the environment and rendered it increasingly fragile and vulnerable to unexpected climatic events, and even to short-term fluctuations. Thus, the argument is that because of their size and ever more intensive agriculture, the Mayan and other ancient state societies were fundamentally unsustainable.

57. According to paragraph 1, ancient societies increased their agricultural output by

- ☐ increasing the percentage of the population that worked as farmers
- ☐ creating large irrigation systems

- being highly selective of the fields they would farm
- moving more people into the city to free up farmland

58. Which of the following can be inferred from paragraph 1 about the intensive agricultural methods of the Maya?

- They helped the Maya overcome short-term fluctuations in the climate.
- They could not supply all of the food required for the growth of Mayan cities.
- They strained the environment more than the Maya's previous farming techniques did.
- They were invented by the Maya to help them grow new kinds of crops.

Paragraph 2

Claims about environmental degradation and disaster have figured prominently in discussions of the collapse of the Mayan city-states of the Central American lowlands. When two explorers came upon the Mayan cities in the 1830s, they were struck by the sight of tall pyramids and **elaborately** carved stones among luxuriant forest growth. Here was the archetypal picture of a great lost civilization: **abandoned** cities submerged in vegetation. Theories of catastrophic collapse or apocalyptic overthrow came naturally to mind to explain these dramatic scenes.

59. The word "elaborately" in the passage is closest in meaning to

- with great detail
- artistically
- mysteriously
- gently hand-made

60. The word "abandoned" in the passage is closest in meaning to

- carefully hidden
- destroyed
- enormous
- no longer occupied

61. In paragraph 2, the author implies which of the following about the collapse of the Mayan city-states?

- The fact that vegetation had grown over the ruins of Mayan buildings indicates that environmental degradation did not contribute to the Mayan collapse.
- Early explorers supposed that there was a catastrophic collapse of the Mayan city states largely because this view fit their preconceived ideas about lost civilizations.
- The condition of the tall pyramids and carved stones discovered by early explorers proves that Mayan city-states were violently overthrown.
- The Mayan cities were abandoned because they became submerged in vegetation.

Paragraph 3

Recent studies of the Mayan collapse (beginning around A.D. 900) have emphasized the gradual and progressive nature of the process, beginning in earnest in the South and advancing northward. It was not a single, sudden event, as had once been thought. Warfare and social unrest are thought to have played a part, but these may well have arisen through pressure from other causes. The Mayan cities had, after all, flourished for over 500 years and had frequently been at war with each other.

62. Why does the author include the information that Mayan cities had “flourished for over 500 years and had frequently been at war with each other”?
- ☐ To identify a possible reason for the eventual collapse of Mayan society
 - ☐ To make the point that war and social unrest alone do not account for the Mayan collapse
 - ☐ To explain why recent studies argue that human actions were responsible for the Mayan collapse
 - ☐ To provide evidence that frequent wars weakened Mayan society only very gradually
63. According to paragraph 3, recent studies claim which of the following about the Mayan collapse?
- ☐ It was caused primarily by frequent wars between rival city-states.
 - ☐ It was caused by a single, sudden event.
 - ☐ It was preceded by social unrest in northern city-states.
 - ☐ It began in southern city-states and spread to others.

Paragraph 4

But what about the possibility of food shortages? These could have come about through either natural or humanly induced changes in the environment. Increasingly fierce competition between Mayan cities led to an upsurge of monument construction during the eighth and ninth centuries A.D., which would have placed added strain on agricultural production and expansion. Interstate rivalry may hence have pushed the Maya toward overexploitation of their fragile ecosystem. Deforestation and soil erosion might ultimately have destroyed the capacity of the land to support the high population levels of the Mayan cities, leading to famine, social unrest, and the collapse of the major Mayan centers.

64. All of the following are mentioned in paragraph 4 as possible direct or indirect causes of food shortages **EXCEPT**
- ☐ increased monument construction
 - ☐ rivalries between states
 - ☐ deforestation and erosion
 - ☐ introduction of new crops

Paragraph 5

Yet it may be incorrect to lay the blame **entirely** on human action. Several of the lowland cities, such as Tikal, appear to have depended heavily on the cultivation of raised fields set in the marshy depressions known as *bajos*, which today flood **intermittently** in the rainy season but may originally have been permanent lakes. The raised-field system of intensive cultivation (created by digging surrounding canals and using the soil removed to elevate the fields for planting) allows year-round food production through the constant supply of soil nutrients that erode into the drainage ditches dug around the raised fields, nutrients that are then collected and replaced. Stable water levels were essential to this subsistence system, but evidence from Lake Chichancanab in Yucatán shows that between A.D. 800 and A.D. 1000 this region suffered its driest period of climate in several thousand years. We may expect that as a result water levels fell, and the raised fields in many areas became unusable. But the human response must be viewed through the lens of the social, political, and cultural circumstances. These exerted a powerful mediating effect on the way the Maya endeavored to cope with their difficulties. Had population levels been lower, the impact of the drought may not have been catastrophic, as it was, the Maya were already reaching the limits of the available subsistence capacity, and Mayan elites had espoused certain social and political agendas (including expensive warfare and competition with each other). It was against this specific background that a period of drought led quickly to crisis and collapse.

65. The word "entirely" in the passage is closest in meaning to
- ☐ generally
 - ☐ clearly
 - ☐ completely
 - ☐ specifically
66. The word "intermittently" in the passage is closest in meaning to
- ☐ constantly
 - ☐ periodically
 - ☐ usually
 - ☐ especially
67. According to paragraph 5, why did the raised fields in many areas become unusable?
- ☐ The marshy depressions around the fields flooded in the rainy season.
 - ☐ Intensive cultivation of the fields drained the soil of nutrients.
 - ☐ The area where the fields were located experienced a drop in water levels.
 - ☐ Unstable design caused the failure of the drainage ditches.
68. According to paragraph 5, all of the following made it more difficult for the

Maya to cope with the effects of the drought **EXCEPT**

- ☐ failure to properly cultivate the fields
- ☐ high population levels
- ☐ competition between Mayan groups
- ☐ warfare

Paragraph 5

Yet it may be incorrect to lay the blame entirely on human action. ■ Several of the lowland cities, such as Tikal, appear to have depended heavily on the cultivation of raised fields set in the marshy depressions known as *bajos*, which today flood intermittently in the rainy season but may originally have been permanent lakes. ■ The raised-field system of intensive cultivation (created by digging surrounding canals and using the soil removed to elevate the fields for planting) allows year-round food production through the constant supply of soil nutrients that erode into the drainage ditches dug around the raised fields, nutrients that are then collected and replaced. ■ Stable water levels were essential to this subsistence system, but evidence from Lake Chichancanab in Yucatán shows that between A.D. 800 and A.D. 1000 this region suffered its driest period of climate in several thousand years. ■ We may expect that as a result water levels fell, and the raised fields in many areas became unusable. But the human response must be viewed through the lens of the social, political, and cultural circumstances. These exerted a powerful mediating effect on the way the Maya endeavored to cope with their difficulties. Had population levels been lower, the impact of the drought may not have been catastrophic, as it was, the Maya were already reaching the limits of the available subsistence capacity, and Mayan elites had espoused certain social and political agendas (including expensive warfare and competition with each other). It was against this specific background that a period of drought led quickly to crisis and collapse.

69. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Nature apparently also contributed to the food shortages.

Where would the sentence best fit?

70. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Since the discovery of the Mayan ruins in the 1830s, researchers have tried to explain the collapse of Mayan civilizations.

-
-
-

Answer Choices

The Mayan attempt to develop intensive agricultural methods to support large populations in relatively small areas probably was unsuccessful and could have caused the Mayan collapse.

The discovery of intact pyramids submerged in vegetation among the Mayan ruins led researchers to believe that Mayan cities were simply overgrown rather than catastrophically destroyed.

Warfare and rivalry between Mayan states may have caused food shortages that contributed to the eventual collapse of Mayan civilization.

Early theories that the Mayan collapse was a sudden, catastrophic event were followed by views that treated the collapse as a gradual process.

The continuing warfare and social unrest that started in the North and spread to the South provided researchers with evidence that the Mayan collapse took hundreds of years to occur.

Drought between A.D. 800 and A.D. 1000 likely caused the Mayan systems of intensive irrigated agriculture to fall, which could have brought about a rapid collapse of the Mayan states.

Forms of Locomotion

Using metabolic energy as “currency” to measure the “cost” of locomotion—that is, the amount of energy that must be spent to move from one place to another—we can compare the costs of different types of locomotion. Terrestrial locomotion—walking or running—is the most expensive form of locomotion. Given that humans are naturally terrestrial, many people may be surprised to learn that walking is so costly. The cost per kilogram of locomotion for human running is about five times higher than for the flight of a typical bird, and ten times more expensive than for fish swimming.

Just why is locomotion so cheap for a fish? The main reason is that the water supports most of the body weight of such a swimmer, so all the animal needs to do to swim is to produce enough force to overcome the drag of its own body. Most aquatic animals have nearly the same density as the water in which they swim, so they do almost no work to support their weight against gravity. However, swimming is cheap only for those animals well adapted to swimming completely submerged. When animals such as ducks and muskrat swim on the surface, they use two or three times more energy to swim on the surface than when submerged, and as much as twenty times more energy than fish of a similar size. This is because of what is called the “bow wave” any object moving on the surface of water pushes up a bow wave at the front, which streams alongside and trails back. Boat designers have long known that the bigger the bow wave, the harder it is to push a boat through the water. The bow wave produces extra drag on any body moving on the surface of water. An animal swimming on the surface of the water uses extra energy in order to overcome drag. Thus, for our purposes, efficient “swimming” means underwater locomotion by animals with streamlined bodies, not the exhausting, inefficient locomotion of humans in swimming pools.

Flying animals move through air that is less dense and less viscous than water, so why does flying cost more than swimming? First, most flying animals move much faster than a swimmer in order to produce enough lift (the upward force necessary to overcome gravity). This higher speed increases the drag that a flyer must overcome. Furthermore, a flyer has an extra source of drag that a swimmer does not have: the extra drag that comes from lift production. In a way, the extra drag represents the cost of supporting the flyer's weight in air.

Walking (or running or galloping) is so costly because it involves at least three processes that require muscular work. The first is simply supporting the body's weight. The second is overcoming the friction in joints and muscles, and the third is constantly producing accelerations (speeding up) and decelerations (slowing down). The exact proportion of muscular effort that goes into these three processes depends on the anatomy of a given animal, but the third process probably accounts for most of the energy used by the muscles. When a person takes a step, first one foot pushes off, which accelerates the body. Then the other foot swings forward and hits the ground, and as the weight shifts onto that foot, the body decelerates. Some of the leg muscles actively tense to act as shock absorbers during this deceleration. Momentum carries the body over the grounded foot, at which time that foot pushes off to accelerate the body, and the cycle repeats.

In terms of energy, walking is inefficient because of the acceleration and deceleration required with every step. Both the decelerations and accelerations need muscular effort and thus energy use. In swimming and flying, animals accelerate and decelerate relatively little over the course of a tail stroke or a wingbeat, so less energy is consumed by this process. As an analogy, consider riding a bicycle. When a person rides a bicycle, the bicycle does not accelerate or decelerate much with each turn of the pedal. Thus, a person can ride a bicycle much faster than he or she could run using the same amount of effort.

Paragraph 1

Using metabolic energy as “currency” to measure the “cost” of locomotion—that is, the amount of energy that must be spent to move from one place to another—we can compare the costs of different types of locomotion. Terrestrial locomotion—walking or running—is the most expensive form of locomotion. Given that humans are naturally terrestrial, many people may be surprised to learn that walking is so costly. The cost per kilogram of locomotion for human running is about five times higher than for the flight of a typical bird, and ten times more expensive than for fish swimming.

1. According to paragraph 1, which of the following is true of the cost of locomotion?
 - ☐ For humans, the cost of running is lower than the cost of swimming.
 - ☐ The type of locomotion that comes most naturally to an animal is also the type of locomotion that is least expensive.
 - ☐ The more metabolic energy required for locomotion, the more expensive the locomotion is considered to be.
 - ☐ The faster a particular form of locomotion is, the more metabolic energy it requires.

Paragraph 2

Just why is locomotion so cheap for a fish? The main reason is that the water supports most of the body weight of such a swimmer, so all the animal needs to do to swim is to produce enough force to overcome the drag of its own body. Most aquatic animals have nearly the same density as the water in which they swim, so they do almost no work to support their weight against gravity. However, swimming is cheap only for those animals well adapted to swimming completely submerged. When animals such as ducks and muskrat swim on the surface, they use two or three times more energy to swim on the surface than when submerged, and as much as twenty times more energy than fish of a similar size. This is because of what is called the “bow wave” any object moving on the surface of water pushes up a bow wave at the front, which streams alongside and trails back. Boat designers have long known that the bigger the bow wave, the harder it is to push a boat through the water. The bow wave produces extra drag on any body moving on the surface of water. An animal swimming on the surface of the water uses extra energy in order to overcome drag. Thus, for our purposes, efficient “swimming” means underwater locomotion by animals with streamlined bodies, not the exhausting, inefficient locomotion of humans in swimming pools.

2. Paragraph 2 suggests which of the following about the drag faced by aquatic animals?
 - ☐ The closer an aquatic animal's body density is to the density of the water, the less drag the animal must overcome.
 - ☐ Aquatic animals that swim on the surface must overcome fewer sources of drag than animals that swim underwater.
 - ☐ The drag faced by most fish swimming underwater is much greater than previously thought.
 - ☐ All fish must overcome at least two forces of drag—the drag produced by their own bodies and by the bow wave.
3. According to paragraph 2, which of the following animal types is the most efficient swimmer?
 - ☐ Animals that are well suited to swimming completely submerged
 - ☐ Animals such as ducks and muskrat
 - ☐ Animals that spend their time both on the surface of the water and completely underwater
 - ☐ Animals that produce only a small bow wave

4. Select the **TWO** answer choices that, according to paragraph 2, are true of the swimming behavior of ducks and muskrat. To receive credit you must select TWO answer choices.
- ☐ Ducks and muskrat require far more energy to swim on the surface than fish of a similar size require to swim underwater.
 - ☐ Ducks and muskrat must overcome drag produced by the bow wave when swimming both on the surface and submerged.
 - ☐ Ducks and muskrat spend as much time underwater as possible in order to use less energy in swimming.
 - ☐ Ducks and muskrat are more efficient when swimming underwater than when swimming on the surface.
5. What is the author's purpose in mentioning "Boat designers" in a discussion of the amount of energy used in swimming?
- ☐ To illustrate the point that large animals are not efficient swimmers because they produce very large bow waves
 - ☐ To make a comparison between the type of drag produced by the bow wave and the type produced by an animal's weight
 - ☐ To suggest that boat designers could design better boats if they studied locomotion in animals
 - ☐ To apply a principle used in boat design to further illustrate why surface swimming is inefficient
6. The word "exhausting" in the passage is closest in meaning to
- ☐ enjoyable
 - ☐ corresponding
 - ☐ extremely tiring
 - ☐ very complicated

Paragraph 3

Flying animals move through air that is less dense and less viscous than water, so why does flying cost more than swimming? First, most flying animals move much faster than a swimmer in order to produce enough lift (the upward force necessary to overcome gravity). This higher speed increases the drag that a flyer must overcome. Furthermore, a flyer has an extra source of drag that a swimmer does not have: the extra drag that comes from lift production. In a way, the extra drag represents the cost of supporting the flyer's weight in air.

7. The word "Furthermore" in the passage is closest in meaning to
- ☐ Unfortunately
 - ☐ In addition
 - ☐ However
 - ☐ Consequently
8. According to paragraph 3, flying requires more energy than swimming for all of the following reasons **EXCEPT**:
- ☐ A flying animal must overcome more drag because it moves more quickly than a swimming animal.
 - ☐ A flying animal must expend more energy to create lift than a swimming animal does.
 - ☐ More energy is required for a flying animal to support its weight in air than for a swimming animal to support its weight in water.
 - ☐ Because air has a lower density than water does, more energy is required for an animal to move through air than through water.

Paragraph 4

Walking (or running or galloping) is so costly because it involves at least three processes that require muscular work. The first is simply supporting the body's weight. The second is overcoming the friction in joints and muscles, and the third is constantly producing accelerations (speeding up) and decelerations (slowing down). The exact proportion of muscular effort that goes into these three processes depends on the anatomy of a given animal, but the third process probably accounts for most of the energy used by the muscles. When a person takes a step, first one foot pushes off, which accelerates the body. Then the other foot swings forward and hits the ground, and as the weight shifts onto that foot, the body decelerates. Some of the leg muscles actively tense to act as shock absorbers during this deceleration. Momentum carries the body over the grounded foot, at which time that foot pushes off to accelerate the body, and the cycle repeats.

9. According to paragraph 4, which of the following statements is true about the energy expended by animals in walking?
- ☐ It requires more energy for a walking animal to overcome the friction in its joints and muscles than to support its body weight.
 - ☐ The process involved in walking that usually requires the most energy is producing accelerations and decelerations.
 - ☐ Whether or not walking requires more energy than other types of locomotion depends on the anatomy of a given animal.
 - ☐ The energy generated by momentum during walking is greater than the amount of energy expended in accelerating and decelerating.

Paragraph 5

In terms of energy, walking is inefficient because of the acceleration and deceleration required with every step. Both the decelerations and accelerations need muscular effort and thus energy use. In swimming and flying, animals accelerate and decelerate relatively little over the course of a tail stroke or a wingbeat, so less energy is consumed by this process. As an analogy, consider riding a bicycle. When a person rides a bicycle, the bicycle does not accelerate or decelerate much with each turn of the pedal. Thus, a person can ride a bicycle much faster than he or she could run using the same amount of effort.

10. The word "consumed" in the passage is closest in meaning to
- ☐ saved
 - ☐ depended on
 - ☐ transferred
 - ☐ used up
11. Paragraph 5 states that swimming and flying are more efficient than walking because
- ☐ swimming and flying animals make tail strokes and wingbeats much less often than walking animals take steps
 - ☐ swimming and flying animals use more muscles than walking animals do
 - ☐ swimming and flying do not require as much acceleration and deceleration with every movement
 - ☐ animals that are swimming or flying can accelerate and decelerate more quickly than can animals that are walking

12. What is the author's purpose in discussing "riding a bicycle"?

- ☐ To explain why walking requires more energy than swimming and flying
- ☐ To contrast the amount of energy used in riding a bicycle with the amount of energy used in swimming and flying
- ☐ To suggest that humans should ride a bicycle instead of running or walking in order to conserve energy
- ☐ To provide an example of an activity for which accelerations and decelerations require a great deal of muscular effort

Paragraph 3

Flying animals move through air that is less dense and less viscous than water, so why does flying cost more than swimming? First, most flying animals move much faster than a swimmer in order to produce enough lift (the upward force necessary to overcome gravity). ■ This higher speed increases the drag that a flyer must overcome. ■ Furthermore, a flyer has an extra source of drag that a swimmer does not have: the extra drag that comes from lift production. ■ In a way, the extra drag represents the cost of supporting the flyer's weight in air. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Thus, the need to overcome resistance to fast forward motion plus the muscular effort needed just to keep from falling makes flying more energy intensive than swimming.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The amount of energy required for movement depends on the type of locomotion.

-
-
-

Answer Choices

Swimming is cheaper than flying for animals that swim completely submerged but not for animals that swim on the surface of the water.

The demands of creating lift and overcoming drag make flying a more costly form of locomotion than swimming.

The costs of lift and drag are greater for walking and running than they are for swimming.

For animals that are well adapted to underwater

swimming, locomotion requires very little energy because they do not have to support their own body weight.

Running and walking require more energy than other types of locomotion mainly because of the amount of muscular effort involved in constantly slowing the body down and speeding it up.

Riding a bicycle at a fast pace requires much less energy than either running or walking.

How Plants and Animals Arrived in the Hawaiian Islands

Scientists have attempted to explain how living things that are not native to the Hawaiian Islands were able to reach the islands from distant places. The way in which birds reached the Hawaiian Islands is obvious enough. Some of the plants that probably came with them had seeds that readily attached to feathers, about 7 percent of the Hawaiian nonendemic (nonnative) seed plants probably arrived in this way. The Hawaiian insects, too, arrived by air. Entomologists have used airplanes and ships to trail fine nets over the Pacific at different heights and have trapped a variety of insects, most of which, as would be expected, are light-bodied. These types also predominate in the Hawaiian Islands (an indication of their airborne arrival), although heavier dragonflies, sphinx moths, and butterflies are also found there.

The influence of the winds in providing colonists is shown by the fact that, although flowering plants are far more common than ferns in the world as a whole, their diversity in Hawaii is more evenly balanced: 225 immigrant flowering plants and 135 immigrant ferns. The relatively greater success of the ferns is probably due to the fact that their spores (reproductive structures) are much smaller and lighter than the seeds of flowering plants. Of the nonendemic seed plants of the Hawaiian Islands, about 7.5 percent almost certainly arrived carried by the wind, while another 30.5 percent have small seeds (up to three millimeters in diameter) and thus may also have arrived this way.

One of the most interesting plants that probably arrived as a wind-borne seed is the tree *Metrosideros*. It is unusual because its seeds are relatively tiny, and this has allowed it to become widely dispersed through the Pacific islands. It is able to form forests on lowland lava with virtually no soil—a great advantage on a volcanic island. *Metrosideros* shows great variability in its appearance in different environments, from a large tree in the wet rain forest, to a shrub on windswept ridges, to as little as 15 centimeters high in peatlands, and it is therefore the dominant tree of the Hawaiian forest. The different forms are not distinct species, and intermediates are found where two different types are adjacent to one another.

Probably the single most important method of entry of seed plants to the Hawaiian Islands has been as seeds within the digestive systems of birds that have eaten their fruit (e.g. blueberry, sandalwood), about 37 percent of the nonendemic seed plants of the islands probably arrived in this way. Significantly, many plants that succeeded in reaching the islands are those that, unlike the rest of their families, bear fleshy fruits instead of dry seeds, such as the species of mint, lily, and nightshade found in Hawaii.

Dispersed by sea accounts for only about 5 percent of the nonendemic Hawaiian seed plants. As well as the widespread coconut, the islands also contain *Scaevola toccata*, this shrub has white, buoyant fruits and forms dense hedges along the edge of the beach. Another seaborne migrant is *Erythrina*, most species of this plant have buoyant, beanlike seeds. On Hawaii, after its arrival on the beach, *Erythrina* was unusual in adapting to an island environment, and a new endemic species, the coral tree *E. sandwichensis*, has evolved on the island. Unlike those of its ancestors, the seeds of the coral tree do not float—an example of the loss of its dispersal mechanism often characteristic of an island species.

The successful colonists of the Hawaiian Islands are the exceptions, many groups, both plants and animals, have failed to reach the islands by natural processes. There are no truly freshwater fish and no native amphibians, reptiles, or mammals (except for one species of bat), while 21 orders of insect are completely absent. As might be expected, most of these are types that seem in general to have very limited powers of dispersal. For example, the ants, which are an important part of the insect fauna in other tropical parts of the world, were originally absent. They have, however, since been introduced by humans, and 36 different species have now established themselves and filled their usual dominant role

in insect faunas. This proves that the obstacle was reaching the islands, not the nature of the Hawaiian environment.

Paragraph 1

Scientists have attempted to explain how living things that are not native to the Hawaiian Islands were able to reach the islands from distant places. The way in which birds reached the Hawaiian Islands is obvious enough. Some of the plants that probably came with them had seeds that readily attached to feathers, about 7 percent of the Hawaiian nonendemic (nonnative) seed plants probably arrived in this way. The Hawaiian insects, too, arrived by air. Entomologists have used airplanes and ships to trail fine nets over the Pacific at different heights and have trapped a variety of insects, most of which, as would be expected, are light-bodied. These types also predominate in the Hawaiian Islands (an indication of their airborne arrival), although heavier dragonflies, sphinx moths, and butterflies are also found there.

15. The word "obvious" in the passage is closest in meaning to

- ☐ unusual
- ☐ natural
- ☐ clear
- ☐ simple

16. According to paragraph 1, how are dragonflies, sphinx moths, and butterflies different from other insects in Hawaii?

- ☐ They are heavier.
- ☐ They are native to Hawaii.
- ☐ They fly at much higher levels above the land.
- ☐ They were brought to the islands by birds.

Paragraph 2

The influence of the winds in providing colonists is shown by the fact that, although flowering plants are far more common than ferns in the world as a whole, their diversity in Hawaii is more evenly balanced: 225 immigrant flowering plants and 135 immigrant ferns. The relatively greater success of the ferns is probably due to the fact that their spores (reproductive structures) are much smaller and lighter than the seeds of flowering plants. Of the nonendemic seed plants of the Hawaiian Islands, about 7.5 percent almost certainly arrived carried by the wind, while another 30.5 percent have small seeds (up to three millimeters in diameter) and thus may also have arrived this way.

17. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ In the world as a whole, flowering plants are much more common than ferns, but in Hawaii the opposite is true.
- ☐ A wide variety of plants, including 225 flowering plants and 135 ferns, were probably brought to Hawaii by wind.
- ☐ Ferns are well suited to the windy conditions in Hawaii, as demonstrated by their high diversity compared to flowering plants.
- ☐ Hawaii has a higher proportion of ferns to flowering plants than the world as a whole because many of its colonists arrived by wind.

18. According to paragraph 2, Hawaiian plants whose seeds are no more than 3 millimeters in diameter
- ☐ are more likely to be flowering plants than ferns
 - ☐ make up about 7.5 percent of all nonendemic plants on the islands
 - ☐ make up the majority of seed-bearing plants on the islands
 - ☐ may have been transported to the islands by wind

Paragraph 3

One of the most interesting plants that probably arrived as a wind-borne seed is the tree *Metrosideros*. It is unusual because its seeds are relatively tiny, and this has allowed it to become widely dispersed through the Pacific islands. It is able to form forests on lowland lava with virtually no soil—a great advantage on a volcanic island. *Metrosideros* shows great variability in its appearance in different environments, from a large tree in the wet rain forest, to a shrub on windswept ridges, to as little as 15 centimeters high in peatlands, and it is therefore the dominant tree of the Hawaiian forest. The different forms are not distinct species, and intermediates are found where two different types are adjacent to one another.

19. According to paragraph 3, all of the following are true of the tree *Metrosideros* **EXCEPT**:

- ☐ It has very small seeds that were probably brought to the islands by wind.
- ☐ It needs almost no soil to grow.
- ☐ It has evolved into at least three distinct species since reaching the islands.
- ☐ It adapts its size to fit environmental conditions.

20. The phrase "adjacent to" in the passage is closest in meaning to

- ☐ next to
- ☐ mixed with
- ☐ competing with
- ☐ related to

Paragraph 4

Probably the single most important method of entry of seed plants to the Hawaiian Islands has been as seeds within the digestive systems of birds that have eaten their fruit (e.g. blueberry, sandalwood), about 37 percent of the nonendemic seed plants of the islands probably arrived in this way. Significantly, many plants that succeeded in reaching the islands are those that, unlike the rest of their families, bear fleshy fruits instead of dry seeds, such as the species of mint, lily, and nightshade found in Hawaii.

21. In paragraph 4, what point is the author making about the particular species of mint, lily and nightshade found in Hawaii?

- ☐ These species have close relatives that, because they produce dry seeds, are unable to grow in the Hawaiian Islands.
- ☐ If these species did not produce edible fruits, they probably would not have succeeded in reaching the islands.
- ☐ These species, in addition to Hawaii's blueberry and sandalwood, provide further examples of fruiting plants that now grow in Hawaii but that are not native to the islands.
- ☐ These species have close relatives in the Hawaiian Islands that are able to attract birds without producing fleshy fruit.

Paragraph 5

Dispersed by sea accounts for only about 5 percent of the nonendemic Hawaiian seed plants. As well as the **widespread** coconut, the islands also contain *Scaevola toccata*, this shrub has white, buoyant fruits and forms **dense** hedges along the edge of the beach. Another seaborne migrant is *Erythrina*, most species of this plant have buoyant, beanlike seeds. On Hawaii, after its arrival on the beach, *Erythrina* was unusual in adapting to an island environment, and a new endemic species, the coral tree *E sandwichensis*, has evolved on the island. Unlike those of its ancestors, the seeds of the coral tree do not float—an example of the loss of its dispersal mechanism often characteristic of an island species.

22. The word "**widespread**" in the passage is closest in meaning to

- ☐ impressive
- ☐ common
- ☐ adaptable
- ☐ hardy

23. The word "**dense**" in the passage is closest in meaning to

- ☐ long
- ☐ tall
- ☐ beautiful
- ☐ thick

24. According to paragraph 5, which of the following plants does **NOT** have seeds that float?

- ☐ *Scaevola toccata*
- ☐ *E. sandwichensis*
- ☐ *Erythrina*
- ☐ Coconut

Paragraph 6

The successful colonists of the Hawaiian Islands are the exceptions, many groups, both plants and animals, have failed to reach the islands by natural processes. There are no truly freshwater fish and no native amphibians, reptiles, or mammals (except for one species of bat), while 21 orders of insect are completely absent. As might be expected, most of these are types that seem in general to have very limited powers of dispersal. For example, the ants, which are an important part of the insect fauna in other tropical parts of the world, were originally absent. They have, however, since been introduced by humans, and 36 different species **have now established themselves and filled their usual dominant role in insect faunas**. This proves that the obstacle was reaching the islands, not the nature of the Hawaiian environment.

25. Which of the following can be inferred from paragraph 6 about the animals currently on the Hawaiian Islands?

- ☐ Except one species of bat, all of the amphibians, reptiles, and mammals on the islands were introduced by people.
- ☐ Except for one species of bat, there are no amphibians, reptiles, or mammals on the islands
- ☐ Except for one species of bat, none of the amphibians, reptiles, or mammals on the islands have spread to other islands.
- ☐ Except for one species of bat, all of the amphibians, reptiles, and mammals on the islands died out after people arrived.

26. Why does the author provide the information that a number of species of ants “have now established themselves and filled their usual dominant role in insect faunas”?
- ☐ To demonstrate that species introduced by humans often disrupt the island ecosystem by replacing endemic species
 - ☐ To provide an example of some of the most successful colonists of the Hawaiian Islands
 - ☐ To support the claim that many types of organisms are absent from Hawaii because they could not get there, not because they are unsuited to its environment
 - ☐ To argue that ants have better powers of dispersal than the 21 orders of insects that are absent from the islands

Paragraph 1

Scientists have attempted to explain how living things that are not native to the Hawaiian Islands were able to reach the islands from distant places. The way in which birds reached the Hawaiian Islands is obvious enough. ■ Some of the plants that probably came with them had seeds that readily attached to feathers, about 7 percent of the Hawaiian nonendemic (nonnative) seed plants probably arrived in this way. ■ The Hawaiian insects, too, arrived by air. ■ Entomologists have used airplanes and ships to trail fine nets over the Pacific at different heights and have trapped a variety of insects, most of which, as would be expected, are light-bodied. ■ These types also predominate in the Hawaiian Islands (an indication of their airborne arrival), although heavier dragonflies, sphinx moths, and butterflies are also found there.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Flying over from the American mainland or from other Pacific Islands, they often brought additional colonists with them.

Where would the sentence best fit?

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The nonnative plants and animals of Hawaii arrived on the islands by several different means.

-
-
-

Mexican Mural Art

The first major modern art movement in Latin America was Mexican muralism, which featured large-scale murals painted on the wall surfaces of public buildings. One of the most persistent strands in Latin American art in the last 80 years has been an engagement with political and social issues, including the struggle for social justice. This in turn has been accompanied by a desire for authentic forms of self-expression and freedom from cultural dependency. Although these preoccupations have taken many different forms, Mexican muralism was the first, and its influence was the most far-reaching. Muralism flourished in Mexico in the years immediately following the Mexican Revolution (1910-1920) as a result of a combination of circumstances: a climate of revolutionary optimism and cultural experimentation that challenged traditional Eurocentrism, a small but strong group of relatively mature artists of energy, ideas, and ability, and a visionary minister of education, Jose Vasconcelos. Vasconcelos believed that Mexico was destined to play a central role on the international stage. He understood that ideas could be more quickly assimilated through images than through any other medium, and he had the courage to allocate the funds, and the walls of public buildings, to the artists to do with as they liked.

The muralists shared a belief in the power of art to transform society for the better, to challenge social, political, economic, and cultural stereotypes, and to enrich the intellectual life of their country. During the 1920s and 1930s, they covered miles of wall with paintings representing aspects of Mexico's past and present and the future to which all aspired. Although Mexican muralism is representational and often narrative in form, it should be recognized as a modern movement, it was modernizing in intent, in that it challenged the old order—culturally, socially, and politically. By definition, it was a public, accessible form of art—not a commodity that could be bought and sold by the wealthy elite. Its purpose was to educate, inform, enlighten, politicize and thus empower the general public, in particular the working classes.

The muralist movement was not a unified force, however. The painters who were its leaders took different directions and did not always see eye to eye. Diego Rivera (1886-1957) sought to promote a pluralistic vision of Mexican society by drawing on the rich heritage of the pre-Columbian past (before Christopher Columbus arrived in the Americas in 1492) and contemporary popular culture, and he investigated pre-Columbian styles and techniques in an effort to create an aesthetic language that was new and Mexican. He was deeply influenced by native pictographic traditions of communication in which pictures represent written words and ideas, and he sought to develop a modern equivalent, a visual language that could be read like a book. The art of Jose Clemente Orozco (1883-1949) is less optimistic: he saw both the pre-Columbian past and the revolutionary present in a more negative light, the former as barbarous, the latter often tarnished by corruption and cruelty. He offers no comforting narratives and his expressive, aggressive technique serves as a metaphor of Mexico's harsh, contradictory reality. David Alfaro Siqueiros (1898-1976) was the most politically active of the three and was an internationalist both ideologically and artistically. In his art he deliberately avoided traditional materials and methods, preferring to use modern industrial paints and spray guns. His works look forward to a fully socialist future where the workers will have won the right to the benefits of the modern industrial era, and his often fragmented, complex imagery does not patronize or make concessions to his audience.

The Mexican muralist movement is undoubtedly one of the most important manifestations of twentieth-century Mexican culture. Its impact elsewhere in the region, as well as in the United States and Europe, has been enormous. The work of Rivera, Orozco, and Siqueiros triggered a homegrown muralist movement in the United States in cities like New York City, Detroit, Los Angeles, and San Francisco. The influence of the Mexicans on the modern Spanish painter Picasso's first mural and almost his only major explicitly propagandist work of art—his famous *Guernica* of 1937—is unmistakable even though the artist

himself would have derived it. In Latin America, Mexican-influenced muralism has recurred whenever artists have felt the need to make a clear, public statement in a language that has not been borrowed from outside.

Paragraph 1

The first major modern art movement in Latin America was Mexican muralism, which featured large-scale murals painted on the wall surfaces of public buildings. One of the most **persistent** strands in Latin American art in the last 80 years has been an engagement with political and social issues, including the struggle for social justice. This in turn has been accompanied by a desire for authentic forms of self-expression and freedom from cultural dependency. Although these preoccupations have taken many different forms, Mexican muralism was the first, and its influence was the most far-reaching. Muralism flourished in Mexico in the years immediately following the **Mexican Revolution** (1910-1920) as a result of a combination of circumstances: a climate of revolutionary optimism and cultural experimentation that challenged traditional Eurocentrism, a small but strong group of relatively mature artists of energy, ideas, and ability, and a visionary minister of education, Jose Vasconcelos. Vasconcelos believed that Mexico was destined to play a central role on the international stage. He understood that ideas could be more quickly assimilated through images than through any other medium, and he had the courage to allocate the funds, and the walls of public buildings, to the artists to do with as they liked.

29. The word "**persistent**" in the passage is closest in meaning to
- ☐ important
 - ☐ fascinating
 - ☐ lasting
 - ☐ powerful
30. According to paragraph 1, Mexican muralism is concerned with
- ☐ the attempt to make art a more important subject in the Latin American educational system
 - ☐ the combination of European art traditions with authentic Latin American art forms
 - ☐ the creation of a just society and an independent form of cultural expression
 - ☐ the use of art to raise funds for the construction of new public buildings
31. The author mentions the "**Mexican Revolution**" in the passage in order to
- ☐ explain how the Mexican government used muralism to challenge European political beliefs
 - ☐ emphasize an important reason that Mexican muralism thrived
 - ☐ give an example of one the most popular subjects of muralism
 - ☐ emphasize the success of Mexican artists who participated in political conflicts
32. It can be inferred from paragraph 1 that the muralists got most of their financial support from
- ☐ opponents of traditional Europe art
 - ☐ wealthy art lovers
 - ☐ other muralists from around the world
 - ☐ the Mexican government
33. According to paragraph 2, in what way can Mexican muralism be regarded as a characteristically modern art movement?
- ☐ It was representational and often narrative in form.
 - ☐ It was supported by a small but enlightened artistic elite.
 - ☐ It questioned traditional ideas.
 - ☐ It emphasized the future rather than dwelling on the past.

Paragraph 3

The muralist movement was not a unified force, however. The painters who were its leaders took different directions and did not always see eye to eye. Diego Rivera (1886-1957) sought to **promote** a pluralistic vision of Mexican society by drawing on the rich heritage of the pre-Columbian past (before Christopher Columbus arrived in the Americas in 1492) and contemporary popular culture, and he investigated pre-Columbian styles and techniques in an effort to create an aesthetic language that was new and Mexican. He was deeply influenced by native pictographic traditions of communication in which pictures represent written words and ideas, and he sought to develop a modern equivalent, a visual language that could be read like a book. The art of Jose Clemente Orozco (1883-1949) is less optimistic: he saw both the pre-Columbian past and the revolutionary present in a more negative light, the former as barbarous, the latter often tarnished by corruption and cruelty. He offers no comforting narratives and his expressive, aggressive technique serves as a metaphor of Mexico's harsh, **contradictory** reality. David Alfaro Siqueiros (1898-1976) was the most politically active of the three and was an internationalist both ideologically and artistically. In his art he deliberately avoided traditional materials and methods, preferring to use modern industrial paints and spray guns. His works look forward to a fully socialist future where the workers will have won the right to the benefits of the modern industrial era, and his often fragmented, complex imagery does not patronize or make concessions to his audience.

34. The word "**promote**" in the passage is closest in meaning to

- ☐ express
- ☐ create
- ☐ emphasize
- ☐ encourage

35. The word "**contradictory**" in the passage is closest in meaning to

- ☐ discouraging
- ☐ conflicting
- ☐ unchanging
- ☐ unusual

36. Paragraph 3 makes all of the following points about artist Diego Rivera **EXCEPT**:

- ☐ He used elements of pre-Columbian art to help make a new, modern art.
- ☐ He tried to develop a visual language that communicated as clearly as native pictographs had.
- ☐ He used art to express his ideas of what Mexican society should be like.
- ☐ He tried but failed to unify the muralist movement.

37. According to paragraph 3, which of the following was true of Orozco's art?

- ☐ It was concerned with Mexican problems of the past and the present.
- ☐ It presented the pre-Columbian past favorably.
- ☐ Its images were intended to be pleasing to viewers.
- ☐ Its technique was more typical of international artists than Mexican artists.

38. According to paragraph 3, which of the following is NOT true of David Alfaro Siqueiros?

- ☐ He used modern industrial materials rather than traditional materials in his art.
- ☐ He designed images that were intentionally meant to please his audience.
- ☐ He believed in socialism and viewed the future of workers in the modern industrial era favorably.
- ☐ He took an international approach to both politics and art.

Paragraph 4

The Mexican muralist movement is undoubtedly one of the most important manifestations of twentieth-century Mexican culture. Its impact elsewhere in the region, as well as in the United States and Europe, has been enormous. The work of Rivera, Orozco, and Siqueiros triggered a homegrown muralist movement in the United States in cities like New York City, Detroit, Los Angeles, and San Francisco. The influence of the Mexicans on the modern Spanish painter Picasso's first mural and almost his only major explicitly propagandist work of art—his famous *Guernica* of 1937—is unmistakable even though the artist himself would have derived it. In Latin America, Mexican-influenced muralism has recurred whenever artists have felt the need to make a clear, public statement in a language that has not been borrowed from outside.

39. The word "manifestations" in the passage is closest in meaning to

- ☐ expressions
- ☐ modifications
- ☐ contributions
- ☐ components

40. The author mentions Picasso's mural "*Guernica*" in order to

- ☐ provide an example of one of the biggest European influences on Mexican muralism
- ☐ indicate that politically motivated murals were as popular in Europe as they were in Mexico
- ☐ explain why the influence of Mexican muralism was especially strong among Spanish artists
- ☐ provide evidence that the Mexican muralists had a significant impact on the international art world

Paragraph 4

■ The Mexican muralist movement is undoubtedly one of the most important manifestations of twentieth-century Mexican culture. ■ Its impact elsewhere in the region, as well as in the United States and Europe, has been enormous. ■ The work of Rivera, Orozco, and Siqueiros triggered a homegrown muralist movement in the United States in cities like New York City, Detroit, Los Angeles, and San Francisco. ■ The influence of the Mexicans on the modern Spanish painter Picasso's first mural and almost his only major explicitly propagandist work of art—his famous *Guernica* of 1937—is unmistakable even though the artist himself would have derived it. In Latin America, Mexican-influenced muralism has recurred whenever artists have felt the need to make a clear, public statement in a language that has not been borrowed from outside.

41. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

However, its influence was not limited to Mexico itself.

Where would the sentence best fit?

42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Mexican muralism, the first major modern art movement in Latin America, has been highly influential throughout the Americas and internationally.

-
-
-

Answer Choices

The Mexican Revolution resulted in a new respect for traditional culture, leading the muralists to paint scenes depicting the everyday lives of poor Mexicans.

Jose Vasconcelos made Mexico an important international player by promoting the art and ideas of the revolutionary muralists.

The leaders of the muralist movement—Rivera, Orozco, and Siqueiros—all believed in the transformative power of art but differed in terms of their artistic methods and political beliefs.

The muralists challenged cultural and economic stereotypes and experimented with both pre-Columbian and industrial themes, styles, and techniques.

Mexican muralism was a traditional representational art focused on Mexico's pre-Columbian society and culture.

In the 1920s and 1930s, following the Mexican Revolution, a talented group of artists painted many large-scale, politically motivated murals on public buildings.



2015-01-10

Artisans in Sixteenth-Century Europe

For centuries European artisans had operated in small, autonomous handcraft businesses, but by the sixteenth century an evolving economic system—moving toward modern capitalism, with its free-market pricing, new organization of production, investments, and so on—had started to erode their stable and relatively prosperous position. What forces contributed to the decline of the artisan?

In a few industries there appeared technological innovations that cost more to install and operate than artisans—even associations of artisans—could afford. For example, in iron production, such specialized equipment as blast furnaces, tilt hammers, wire-drawing machines, and stamping, rolling, and slitting mills became more familiar components of the industry. Thus the need for fixed capital (equipment and buildings used in production) soared. Besides these items, expensive in their own right, facilities for water, storage, and deliveries were needed. In addition, pig (raw) iron turned out by blast furnaces could not be forged until refined further in a new intermediate stage. In late sixteenth-century Antwerp, where a skilled worker earned 125 to 250 guilders a year, a large blast furnace alone cost 3,000 guilders, and other industrial equipment was equally or more expensive.

Raw materials, not equipment, constituted artisans' major expense in most traders, however. Whereas in 1583 an Antwerp silk weaver paid 12 guilders for a loom (and made small payments over many years to pay off the debt for purchasing the loom), every six weeks he or she had to lay out 24 guilders for the 2 pounds of raw silk required to make a piece of cloth. Thus access to cheap and plentiful primary materials was a constant preoccupation for independent producers. Using local materials might allow even the poorest among them to avoid reliance on merchant suppliers. The loss of nearby sources could therefore be devastating. As silk cultivation waned around the Spanish cities of Cordoba and Toledo, weavers in these cities were forced to become employees of merchants who put out raw silk from Valencia and Murcia provinces. In the Dutch Republic, merchants who imported unprocessed salt from France, Portugal, and Spain gained control of the salt-refining industry once exploitation of local salt marshes was halted for fear that dikes (which held back the sea from the low-lying Dutch land) would be undermined.

Credit was necessary for production but created additional vulnerabilities for artisans. Prices for industrial products lagged behind those of raw materials and foodstuffs, and this, coupled with rising taxes, made it difficult for many producers to repay their creditors. Periodic downturns, when food prices shot up and demand for manufactures fell off, drove them further into debt or even into bankruptcy, from which they might emerge only by agreeing to sell their products exclusively to

merchants or fellow artisans who extended them loans. Frequent enough during periods of growth, such credit crises became deeper and lasted longer after about 1570, as did war-related disruptions of raw-material supplies and markets.

Artisans' autonomy was imperiled, too, by restrictions on their access to markets. During the sixteenth century, a situation like this often resulted from the concentration of export trade in a few great storage and distribution centers. The disappearance of regional markets where weavers in Flanders (what is now northern Belgium) had previously bought flax and sold linen left them at the mercy of big-city middlemen, who quickly turned them into domestic workers. In a similar fashion, formerly independent producers in southern Wiltshire in England, who had bought yarn from spinners or local brokers and sold their cloth to merchants in nearby Salisbury, became subject to London merchants who monopolized both wool supplies and woollens exports.

With good reason, finally, urban artisans feared the growth of industries in the countryside. For one thing, they worried that the spread of village crafts would reduce their supply of raw materials, driving up prices. City producers also knew that rural locations enjoyed lower living costs, wages, and taxes, and often employed fewer or simplified processes. These advantages became a major preoccupation as competition intensified in the 1570s and 1580s

Paragraph 1

For centuries European artisans had operated in small, autonomous handcraft businesses, but by the sixteenth century an evolving economic system—moving toward modern capitalism, with its free-market pricing, new organization of production, investments, and so on—had started to erode their stable and relatively prosperous position. What forces contributed to the decline of the artisan?

1. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information. C (句子主干)
 - ☐ In the sixteenth century, the European economy moved toward a system of free-market pricing, new ways of production, and investments.
 - ☐ Before the sixteenth century, European makers of handcrafts enjoyed stability, autonomy, and relative prosperity.
 - ☐ By the sixteenth century, the rise of capitalism began to weaken the autonomy and relative prosperity of European artisans.
 - ☐ European artisans operated small, autonomous businesses before modern capitalism emerged in the sixteenth century.

Paragraph 2

In a few industries there appeared technological innovations that cost more to install

and operate than artisans—even associations of artisans—could afford. For example, in iron production, such specialized equipment as blast furnaces, tilt hammers, wire-drawing machines, and stamping, rolling, and slitting mills became more familiar components of the industry. Thus the need for fixed capital (equipment and buildings used in production) soared. Besides these items, expensive in their own right, facilities for water, storage, and deliveries were needed. In addition, pig (raw) iron turned out by blast furnaces could not be forged until refined further in a new intermediate stage. In late sixteenth-century Antwerp, where a skilled worker earned 125 to 250 guilders a year, a large blast furnace alone cost 3,000 guilders, and other industrial equipment was equally or more expensive.

2. The word “Besides” in the passage is closest in meaning to A
- ☐ In addition to
 - ☐ More important than
 - ☐ Different from
 - ☐ Together with
3. According to paragraph 2, how did technological advances contribute to the economic decline of artisans?
- ☐ Artisans had no place to store or use the new machines.
 - ☐ Goods produced by the new technology were cheaper than those produced by artisans.
 - ☐ The fixed costs of remaining in business became very high.
 - ☐ Artisans did not know how to use the new machines.

Paragraph 3

Raw materials, not equipment, constituted artisans’ major expense in most traders, however. Whereas in 1583 an Antwerp silk weaver paid 12 guilders for a loom (and made small payments over many years to pay off the debt for purchasing the loom), every six weeks he or she had to lay out 24 guilders for the 2 pounds of raw silk required to make a piece of cloth. Thus access to cheap and plentiful primary materials was a constant preoccupation for independent producers. Using local materials might allow even the poorest among them to avoid reliance on merchant suppliers. The loss of nearby sources could therefore be devastating. As silk cultivation waned around the Spanish cities of Cordoba and Toledo, weavers in these cities were forced to become employees of merchants who put out raw silk from Valencia and Murcia provinces. In the Dutch Republic, merchants who imported unprocessed salt from France, Portugal, and Spain gained control of the salt-refining industry once exploitation of local salt marshes was halted for fear that dikes (which held back the sea from the low-lying Dutch land) would be undermined.

4. The word “preoccupation” in the passage is closest in meaning to
- ☐ necessity
 - ☐ concern

- ☐ struggle
 - ☐ uncertainty
5. In paragraph 3, why does the author provide the information about an Antwerp silk weaver's costs in 1583?
- ☐ To describe some typical costs in the silk-weaving industry
 - ☐ To support the statement that artisans' main expense was materials, not equipment
 - ☐ To argue against the view that artisans did not have to borrow money to buy equipment
 - ☐ To show that materials were cheap and plentiful for most artisans
6. What can be inferred from paragraph 3 about local materials?
- ☐ They were of higher quality than imported materials.
 - ☐ They were usually more plentiful than imported materials.
 - ☐ They remained available even after merchants began to control the industries.
 - ☐ They tended to be more affordable than materials supplied by merchants.
7. According to paragraph 3, which of the following was sometimes an effect on artisans of the loss of local sources of their primary materials?
- ☐ They had to sell their products to merchants.
 - ☐ They needed to take loans in order to buy the materials from merchants.
 - ☐ They could no longer afford to be independent producers.
 - ☐ They imported the materials from distant sources.

Paragraph 4

Credit was necessary for production but created additional vulnerabilities for artisans. Prices for industrial products lagged behind those of raw materials and foodstuffs, and this, coupled with rising taxes, made it difficult for many producers to repay their creditors. Periodic downturns, when food prices shot up and demand for manufactures fell off, drove them further into debt or even into bankruptcy, from which they might emerge only by agreeing to sell their products exclusively to merchants or fellow artisans who extended them loans. Frequent enough during periods of growth, such credit crises became deeper and lasted longer after about 1570, as did war-related disruptions of raw-material supplies and markets.

8. The phrase "coupled with" in the passage is closest in meaning to
- ☐ caused by
 - ☐ compared with
 - ☐ affected by
 - ☐ combined with
9. According to paragraph 4, all of the following caused economic difficulties for artisans EXCEPT

- ☐ decreasing availability of credit
- ☐ decreased demand for manufactured goods
- ☐ increased taxes
- ☐ problems caused by wars

Paragraph 5

Artisans' **autonomy** was imperiled, too, by restrictions on their access to markets. During the sixteenth century, a situation like this often resulted from the concentration of export trade in a few great storage and distribution centers. The disappearance of regional markets where weavers in Flanders (what is now northern Belgium) had previously bought flax and sold linen left them at the mercy of big-city middlemen, who quickly turned them into domestic workers. In a similar fashion, formerly independent producers in southern Wiltshire in England, who had bought yarn from spinners or local brokers and sold their cloth to merchants in nearby Salisbury, became subject to London merchants who monopolized both wool supplies and woolens exports.

10. The word "**autonomy**" in the passage is closest in meaning to
- ☐ independence
 - ☐ influence
 - ☐ ability to make a living
 - ☐ ability to adapt
11. Paragraph 5 supports which of the following statements about artisans during the sixteenth century?
- ☐ They had difficulty transporting their goods to the best markets.
 - ☐ They were at a disadvantage because the concentration of supplies and exports was in the hands of big-city merchants.
 - ☐ They received higher wages as employees of big-city merchants.
 - ☐ They were able to obtain raw materials from local merchants.

Paragraph 6

With good reason, finally, urban artisans feared the growth of industries in the countryside. For one thing, they worried that the spread of village crafts would reduce their supply of raw materials, driving up prices. City producers also knew that rural locations enjoyed lower living costs, wages, and taxes, and often employed fewer or simplified processes. These advantages became a major preoccupation as competition intensified in the 1570s and 1580s

12. All of the following are identified in paragraph 6 as concerns that urban artisans had about the growth of industry in the countryside EXCEPT
- ☐ a decrease in the supply of raw materials
 - ☐ a cheaper cost of living in the countryside
 - ☐ a more manageable level of competition

- less complex production processes in the countryside

Paragraph 3

Raw materials, not equipment, constituted artisans' major expense in most traders, however. ■ Whereas in 1583 an Antwerp silk weaver paid 12 guilders for a loom (and made small payments over many years to pay off the debt for purchasing the loom), every six weeks he or she had to lay out 24 guilders for the 2 pounds of raw silk required to make a piece of cloth. ■ Thus access to cheap and plentiful primary materials was a constant preoccupation for independent producers. ■ Using local materials might allow even the poorest among them to avoid reliance on merchant suppliers. ■ The loss of nearby sources could therefore be devastating. As silk cultivation waned around the Spanish cities of Cordoba and Toledo, weavers in these cities were forced to become employees of merchants who put out raw silk from Valencia and Murcia provinces. In the Dutch Republic, merchants who imported unprocessed salt from France, Portugal, and Spain gained control of the salt-refining industry once exploitation of local salt marshes was halted for fear that dikes (which held back the sea from the low-lying Dutch land) would be undermined.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

This was possible because when transportation costs were low, the price of raw materials was generally also low.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

European artisans experienced a decline during the sixteenth century as result of a number of developments.

-
-
-

Answer choices

Artisans' ability to earn profits was restricted by their dependence on creditors, to whom they were forced to sell their goods, and by competition from rural areas.

Advances in technology in some industries increased the cost of the equipment, buildings, and facilities that artisans needed for producing and selling their goods.

As industries came under the control of merchants, artisans lost access to cheap raw materials, and they had to borrow money to buy the materials they needed from merchant suppliers.

Iron production in the sixteenth century depended on new inventions such as blast furnaces, tilt hammers, wire-drawing machines, and stamping, rolling, and silting mills.

In the new industrial system from which skilled artisans were progressively excluded, the quality of manufactured items gradually declined.

The rising prices of food and other necessities often left artisans without enough money to pay their taxes and other business expenses.

Disease and History

What is the relationship between disease and the evolution of human societies?

Epidemiology is the study of the causes, distribution, and control of diseases in populations. Throughout history, there have been general trends in the relationship between diseases and the human species. Anthropologist George Armelagos has outlined these trends and refers to them as three “epidemiological transitions.”

For most of our species’ history, we lived in small, widely dispersed, nomadic groups. Our ancestors certainly experienced diseases of various sorts and would have come into contact with new diseases as they migrated to new environments. But infectious disease may not have had serious effects on large numbers of people or many different populations, since diseases would have had little chance of being passed on to many other humans.

When some people began to settle down and produce their food through farming and animal domestication—starting about 10,000 years ago—the *first epidemiological transition* occurred. Infectious diseases increased in impact, as larger and denser concentrations of people provided greater opportunity for disease to be passed from host to host. Animal domestication may have brought people into contact with new diseases previously limited to other species. Working the soil would have exposed farmers to insects and other pathogens. Irrigation in some areas provided breeding places for mosquitoes, increasing the incidence of malaria and other mosquito-borne diseases. Sanitation problems caused by larger, more sedentary populations would have helped transmit diseases in human waste, as would the use of animal dung for fertilizer. In addition, agriculture also led to a narrowing of food sources, as compared to the varied diets of hunters and gatherers. This could have resulted in nutritional deficiencies; moreover, the storage of food surpluses attracted new disease carriers such as insects and rats. Trade between settled communities helped spread diseases over large geographic areas, as in the case of the Black Death in Europe. Epidemics, in the sense of diseases that affect a large number of populations at the same time, were essentially nonexistent until the development of agricultural economies.

Beginning in the last years of the nineteenth century and continuing into the twentieth, we experienced the *second epidemiological transition*. With modern medical science providing immunizations and antibiotics and with better public health measures and improved nutrition, many infectious diseases were brought under control, or even eliminated. In terms of what ailed and killed us, there was a shift to chronic diseases such as heart and lung diseases. The increase in many of these came not only from the fact that fewer people were dying from infectious disease and were living longer but also from the results of modern lifestyles in

developed countries and among the upper classes of developing countries—a more sedentary life leading to less physical activity, more stress, environmental pollution, and high-fat diets. But at least, we thought, many of these problems were things we could potentially control; all those infectious epidemics were of the past.

But on the heels of the second transition had come the *third epidemiological transition*, and we are in it now. New diseases are emerging, and old ones are returning. Both of these phenomena can be understood in terms of evolutionary theory.

The return of old diseases is the result of the fact that microorganisms are evolving species themselves. For example, new and serious antibiotic-resistant strains of tuberculosis have recently appeared. This evolution may have been encouraged by what some authorities consider our overuse of antibiotics, giving microorganisms a greater chance to evolve resistance by exposing them to a constant barrage of selective challenges. Some bacteria reproduce hourly, and so the processes of genetic mutation and natural selection are speeded up in these species.

Emerging diseases are also the result of human activity in the modern world, which brings more people into contact with more diseases, some of which were unheard of even a few decades ago. As people and their products became more mobile, and as our populations spread into previously little-inhabited areas, cutting down forests and otherwise altering ecological conditions, we contact other species that may carry diseases to which they are immune but that prove deadly to us.

Paragraph 2

For most of our species' history, we lived in small, widely dispersed, nomadic groups. Our ancestors certainly experienced diseases of various sorts and would have come into contact with new diseases as they migrated to new environments. But infectious disease may not have had serious effects on large numbers of people or many different populations, since diseases would have had little chance of being passed on to many other humans.

15. The word “dispersed” in the passage is closest in meaning to

- ☐ active
- ☐ scattered
- ☐ varied
- ☐ linked

16. According to paragraph 2, why were infectious diseases not a serious problem for most of human history?

- ☐ There were very few infectious diseases early in human history.
- ☐ Population groups did not move around enough to be exposed to new diseases.
- ☐ Many disease-causing organisms had features that made them difficult to pass

on to other humans.

- Population groups did not have enough contact with each other to spread diseases widely.

Paragraph 3

When some people began to settle down and produce their food through farming and animal domestication—starting about 10,000 years ago—the *first epidemiological transition* occurred. Infectious diseases increased in impact, as larger and denser concentrations of people provided greater opportunity for disease to be passed from host to host. Animal domestication may have brought people into contact with new diseases previously limited to other species. Working the soil would have exposed farmers to insects and other pathogens. Irrigation in some areas provided breeding places for mosquitoes, increasing the incidence of malaria and other mosquito-borne diseases. Sanitation problems caused by larger, more sedentary populations would have helped transmit diseases in human waste, as would the use of animal dung for fertilizer. In addition, agriculture also led to a narrowing of food sources, as compared to the varied diets of hunters and gatherers. This could have resulted in nutritional deficiencies; moreover, the storage of food surpluses attracted new disease carriers such as insects and rats. Trade between settled communities helped spread diseases over large geographic areas, as in the case of the Black Death in Europe. Epidemics, in the sense of diseases that affect a large number of populations at the same time, were essentially nonexistent until the development of agricultural economies.

17. The word “deficiencies” in the passage is closest in meaning to
- complications
 - illnesses
 - shortages
 - irregularities
18. According to paragraph 3, which of the following contributed to the use of epidemics?
- The development of more deadly forms of human disease
 - The spread of ineffective treatments for infectious disease
 - The spread of mosquito-borne diseases to other disease carriers
 - The practice of exchanging goods between settled areas
19. Which of the following is NOT mentioned in paragraph 3 as a reason that agriculture led to greater exposure to disease?
- Irrigation created areas where disease-carrying mosquitoes could reproduce.
 - People increasingly came into contact with disease-carrying animals attracted to food storage areas.
 - Agricultural products spoiled more readily, leading to more frequent episodes of disease.

- Farming exposed humans to disease-carrying insects in the soil.
20. According to paragraph 3, how did sanitation problems in early farming societies lead to the spread of infectious diseases?
- Water used for irrigation crops was not always clean.
 - Larger populations were increasingly exposed to human and animal waste.
 - Farm products that spoiled in fields attracted insects and animals with diseases.
 - Lack of varied food sources occasionally forced communities to eat food that carried diseases.
21. According to paragraph 3, how did sanitation problems in early farming societies lead to the spread of infectious diseases? (重复了)
- Water used for irrigation crops was not always clean.
 - Larger populations were increasingly exposed to human and animal waste.
 - Farm products that spoiled in fields attracted insects and animals with diseases.
 - Lack of varied food sources occasionally forced communities to eat food that carried diseases.

Paragraph 4

Beginning in the last years of the nineteenth century and continuing into the twentieth, we experienced the *second epidemiological transition*. With modern medical science providing immunizations and antibiotics and with better public health measures and improved nutrition, many infectious diseases were brought under control, or even eliminated. In terms of what ailed and killed us, there was a shift to *chronic* diseases such as heart and lung diseases. The increase in many of these came not only from the fact that fewer people were dying from infectious disease and were living longer but also from the results of modern lifestyles in developed countries and among the upper classes of developing countries—a more sedentary life leading to less physical activity, more stress, environmental pollution, and high-fat diets. But at least, we thought, many of these problems were things we could potentially control; all those infectious epidemics were of the past.

22. According to paragraph 4, which of the following best describes the second epidemiological transition?
- Modern medicine made it possible for people to live longer even if they had an infectious disease.
 - Infectious diseases were harder to cure due to factors like stress and pollution.
 - New infectious diseases appeared as quickly as modern medical science was able to control old ones.
 - Chronic diseases replaced infectious diseases as the major cause of human sickness and death.
23. According to paragraph 4, all of the following likely contributed to chronic disease

EXCEPT

- ☐ longer life expectations
- ☐ decreased physical activity
- ☐ higher rates of poverty
- ☐ changes in diet

Paragraph 5

But on the heels of the second transition had come the *third epidemiological transition*, and we are in it now. New diseases are **emerging**, and old ones are returning. Both of these phenomena can be understood in terms of evolutionary theory.

24. The word “**emerging**” in the passage is closest in meaning to

- ☐ appearing
- ☐ spreading
- ☐ becoming more serious
- ☐ replacing others

Paragraph 6

The return of old diseases is the result of the fact that microorganisms are evolving species themselves. For example, new and serious antibiotic-resistant strains of tuberculosis have recently appeared. This evolution may have been encouraged by what some authorities consider our overuse of antibiotics, giving microorganisms a greater chance to evolve resistance by exposing them to a constant barrage of selective challenges. Some bacteria reproduce hourly, and so the processes of genetic mutation and natural selection are speeded up in these species.

25. What can be inferred from the discussion of antibiotic-resistant strains of bacteria in paragraph 6?

- ☐ Most microorganisms cannot survive multiple exposures to antibiotics.
- ☐ Tuberculosis strains are much more likely to be antibiotic-resistant than are other microorganisms.
- ☐ Bacteria that reproduce quickly are more likely to become resistant to antibiotics.
- ☐ Exposing microorganisms to a constant barrage of antibiotics prevents them from evolving resistance.

Paragraph 7

Emerging diseases are also the result of human activity in the modern world, which brings more people into contact with more diseases, some of which were unheard of even a few decades ago. As people and their products became more mobile, and as our populations spread into previously little-inhabited areas, cutting down forests and otherwise altering ecological conditions, we contact other species that may carry diseases to which they are immune but that prove deadly to us.

26. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Humans contact other species during population growth into previously little-inhabited areas.
- ☐ Species that carry diseases deadly to humans live in areas with small, mobile populations.
- ☐ Increased mobility and population expansion into new areas exposes humans to new, deadly diseases carried by other species.
- ☐ Some species that humans contact in little-inhabited areas are immune to diseases that are deadly to humans.

Paragraph 3

When some people began to settle down and produce their food through farming and animal domestication—starting about 10,000 years ago—the *first epidemiological transition* occurred. ■ Infectious diseases increased in impact, as larger and denser concentrations of people provided greater opportunity for disease to be passed from host to host. ■ Animal domestication may have brought people into contact with new diseases previously limited to other species. ■ Working the soil would have exposed farmers to insects and other pathogens. ■ Irrigation in some areas provided breeding places for mosquitoes, increasing the incidence of malaria and other mosquito-borne diseases. Sanitation problems caused by larger, more sedentary populations would have helped transmit diseases in human waste, as would the use of animal dung for fertilizer. In addition, agriculture also led to a narrowing of food sources, as compared to the varied diets of hunters and gatherers. This could have resulted in nutritional deficiencies; moreover, the storage of food surpluses attracted new disease carriers such as insects and rats. Trade between settled communities helped spread diseases over large geographic areas, as in the case of the Black Death in Europe. Epidemics, in the sense of diseases that affect a large number of populations at the same time, were essentially nonexistent until the development of agricultural economies.

27. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

But the denser populations of agricultural communities were only one of many factors contributing to the increased risk of disease.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do

not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Throughout history, the impact of diseases on humans has changed as human societies have developed.

-
-
-

Answer choices

Once advances in medical science resulted in better control or elimination of many infectious diseases, diseases resulting from the modern lifestyle became a major problem.

Humans today are at risk for contracting both new diseases and old diseases that have reemerged and, in some cases, have become resistant to antibiotics.

The transition to farming meant that humans had both more contact with one another and with other species that carried disease, leading to disease epidemics.

In early human history, nomadic groups started encountering diseases when they moved to new environments.

Infectious diseases have increased steadily in impact and severity from the agricultural revolution through today as a result of increasing contact between human societies.

During the second epidemiological transition, better public health measures and improved nutrition helped control chronic diseases.

Earth's Atmosphere

Earth's atmosphere has changed through time. Compared to the Sun, whose composition is representative of the raw materials from which Earth and other planets in our solar system formed, Earth contains less of some volatile elements, such as nitrogen, argon, hydrogen, and helium. These elements were lost when the envelope of gases, or primary atmosphere, that surrounded early Earth, was stripped away by the solar wind or by meteorite impacts, or both. Little by little, the planet generated a new, secondary atmosphere by volcanic outgassing of volatile materials from its interior.

Volcanic outgassing continues to be the main process by which volatile materials are released from Earth—although it is now going on at a much slower rate. The main chemical constituent of volcanic gases (as much as 97 percent of volume) is water vapor, with varying amounts of nitrogen, carbon dioxide, and other gases. In fact, the total volume of volcanic gases released over the past 4 billion years or so is believed to account for the present composition of the atmosphere with one important exception: oxygen. Earth had virtually no oxygen in its atmosphere more than 4 billion years ago, but the atmosphere is now approximately 21 percent oxygen.

Traces of oxygen were probably generated in the early atmosphere by the breakdown of water molecules into oxygen and hydrogen by ultraviolet light (a process called photodissociation). Although this is an important process, it cannot begin to account for the present high levels of oxygen in the atmosphere. Almost all of the free oxygen now in the atmosphere originated through photosynthesis, the process whereby plants use light energy to induce carbon dioxide to react with water, producing carbohydrates and oxygen.

Oxygen is a very reactive chemical, so at first most of the free oxygen produced by photosynthesis was combined with iron in ocean water to form iron oxide-bearing minerals. The evidence of the gradual transition from oxygen-poor to oxygen-rich water is preserved in seafloor sediments. The minerals in seafloor sedimentary rocks that are more than about 2.5 billion years old contain reduced (oxygen-poor) iron compounds. In rocks that are less than 1.8 billion years old, oxidized (oxygen-rich) compounds predominate. The sediments that were precipitated during the transition contain alternating bands of red (oxidized iron) and black (reduced iron) minerals. These rocks are called banded-iron formations. Because ocean water is in constant contact with the atmosphere, and the two systems function together in a state of dynamic equilibrium, the transition from an oxygen-poor to an oxygen-rich atmosphere also must have occurred during this period.

Along with the buildup of molecular oxygen (O₂) came an eventual increase in ozone (O₃) levels in the atmosphere. Because ozone filters out harmful ultraviolet radiation,

this made it possible for life to flourish in shallow water and finally on land. This critical state in the evolution of the atmosphere was replaced between 1100 and 542 million years ago. Interestingly, the fossil record shows an explosion of life forms 542 million years ago.

Oxygen has continued to play a key role in the evolution and form of life. Over the last 200 million years, the concentration of oxygen has risen from 10 percent to as much as 25 percent of the atmosphere, before setting (probably not permanently) at its current value of 21 percent. This increase has benefited mammals, which are voracious oxygen consumers. Not only do we require oxygen to fuel our high-energy, warm-blooded metabolism, our unique reproductive system demands even more. An expectant mother's used (venous) blood must still have enough oxygen in it to diffuse through the placenta into her unborn child's bloodstream. It would be very difficult for any mammal species to survive in an atmosphere of only 10 percent oxygen.

Geologists cannot yet be certain why the atmospheric oxygen levels increased, but they have a hypothesis. First, photosynthesis is only one part of the oxygen cycle. The cycle is completed by decomposition, in which organic carbon combines with oxygen and forms carbon dioxide. But if organic matter is buried as sediment before it fully decomposes, its carbon is no longer available to react with the free oxygen. Thus there will be a net accumulation of carbon in sediments and of oxygen in the atmosphere.

Paragraph 1

Earth's atmosphere has changed through time. Compared to the Sun, whose composition is representative of the raw materials from which Earth and other planets in our solar system formed, Earth contains less of some volatile elements, such as nitrogen, argon, hydrogen, and helium. These elements were lost when the envelope of gases, or primary atmosphere, that surrounded early Earth, was stripped away by the solar wind or by meteorite impacts, or both. Little by little, the planet generated a new, secondary atmosphere by volcanic outgassing of volatile materials from its interior.

29. In paragraph 1, why does the author state that Earth has less nitrogen, argon, hydrogen, and helium than the Sun?

- ☐ To argue that these elements were once part of an early atmosphere, which disappeared
- ☐ To suggest that these elements were drawn into the Sun's atmosphere
- ☐ To provide evidence that Earth's original atmosphere came primarily from meteorites
- ☐ To support the claim that Earth's atmosphere would have changed even more if it had contained more volatile elements

Paragraph 2

Volcanic outgassing continues to be the main process by which volatile materials are released from Earth—although it is now going on at a much slower rate. The main chemical **constituent** of volcanic gases (as much as 97 percent of volume) is water vapor, with varying amounts of nitrogen, carbon dioxide, and other gases. In fact, the total volume of volcanic gases released over the past 4 billion years or so is believed to account for the present composition of the atmosphere with one important exception: oxygen. Earth had virtually no oxygen in its atmosphere more than 4 billion years ago, but the atmosphere is now approximately 21 percent oxygen.

30. The word “**constituent**” in the passage is closest in meaning to
- ☐ source
 - ☐ equivalent
 - ☐ component
 - ☐ product
31. According to paragraph 2, the history of volcanic outgassing cannot explain which of the following?
- ☐ The lack of oxygen in the atmosphere 4 billion years ago
 - ☐ The amount of water vapor in the atmosphere today
 - ☐ The proportions of nitrogen and carbon dioxide in the atmosphere today
 - ☐ The present abundance of oxygen in the atmosphere

Paragraph 3

Traces of oxygen were probably generated in the early atmosphere by the breakdown of water molecules into oxygen and hydrogen by ultraviolet light (a process called photodissociation). Although this is an important process, it cannot begin to account for the present high levels of oxygen in the atmosphere. Almost all of the free oxygen now in the atmosphere originated through photosynthesis, the process whereby plants use light energy to induce carbon dioxide to react with water, producing carbohydrates and oxygen.

32. Paragraph 3 suggests which of the following about the process of photodissociation?
- ☐ It is more common today than it was in the early history of the atmosphere.
 - ☐ It is responsible for only a small amount of the oxygen in the atmosphere today.
 - ☐ It removes trace amounts of oxygen from the atmosphere.
 - ☐ It produces more free oxygen than photosynthesis does.

Paragraph 4

Oxygen is a very reactive chemical, so at first most of the free oxygen produced by photosynthesis was combined with iron in ocean water to form iron oxide-bearing minerals. The evidence of the **gradual** transition from oxygen-poor to oxygen-rich

water is preserved in seafloor sediments. The minerals in seafloor sedimentary rocks that are more than about 2.5 billion years old contain reduced (oxygen-poor) iron compounds. In rocks that are less than 1.8 billion years old, oxidized (oxygen-rich) compounds **predominate**. The sediments that were precipitated during the transition contain alternating bands of red (oxidized iron) and black (reduced iron) minerals. These rocks are called banded-iron formations. **Because ocean water is in constant contact with the atmosphere, and the two systems function together in a state of dynamic equilibrium, the transition from an oxygen-poor to an oxygen-rich atmosphere also must have occurred during this period.**

33. The word “**gradual**” in the passage is closest in meaning to
- ☐ critical
 - ☐ original
 - ☐ beneficial
 - ☐ slow
34. The word “**predominate**” in the passage is closest in meaning to
- ☐ are in the majority
 - ☐ are present
 - ☐ are increasing
 - ☐ first appear
35. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Since the oceans and the atmosphere function together, the atmosphere must have become oxygen rich during this period.
 - ☐ Because ocean water is in constant contact with the atmosphere, the two systems maintain a dynamic equilibrium.
 - ☐ The transition to an oxygen-rich atmosphere could not have happened without constant contact with the oceans.
 - ☐ Much of the oxygen in the oceans must have been pulled out of the atmosphere during this period.
36. According to paragraph 4, what can be learned from the type of iron compounds in seafloor rocks?
- ☐ How the process of photosynthesis has changed over time
 - ☐ The level of oxygen in the water at a certain time in history
 - ☐ How levels of iron in ocean water decreased over time
 - ☐ The overall mineral content of the ocean water
37. According to paragraph 3, banded-iron formations are found in what kind of rocks?
- ☐ Those that are more than 2.5 billion years old

- Those that do not contain oxidized compounds
- Those that are from a transitional period in terms of oxygen richness
- Those that are less than 1.8 billion years old

Paragraph 5

Along with the buildup of molecular oxygen (O₂) came an eventual increase in ozone (O₃) levels in the atmosphere. Because ozone filters out harmful ultraviolet radiation, this made it possible for life to flourish in shallow water and finally on land. This critical state in the evolution of the atmosphere was replaced between 1100 and 542 million years ago. Interestingly, the fossil record shows an explosion of life forms 542 million years ago.

38. According to paragraph 5, which of the following happened sometime between 1100 and 542 million years ago?

- A sudden explosion of life forms on land occurred together with a sharp decline of life in the water.
- Ultraviolet radiation became more harmful to living organisms.
- Molecular oxygen levels in the atmosphere stabilized, and ozone levels began to rise.
- Ozone reduced ultraviolet radiation to a level acceptable for life on land.

Paragraph 6

Oxygen has continued to play a key role in the evolution and form of life. Over the last 200 million years, the concentration of oxygen has risen from 10 percent to as much as 25 percent of the atmosphere, before setting (probably not permanently) at its current value of 21 percent. This increase has benefited mammals, which are voracious oxygen consumers. Not only do we require oxygen to fuel our high-energy, warm-blooded metabolism, our unique reproductive system demands even more. An expectant mother's used (venous) blood must still have enough oxygen in it to **diffuse** through the placenta into her unborn child's bloodstream. It would be very difficult for any mammal species to survive in an atmosphere of only 10 percent oxygen.

39. The word "**diffuse**" in the passage is closest in meaning to

- spread
- break
- squeeze
- speed

40. According to paragraph 6, which of the following is NOT true of the connection between mammals and oxygen?

- Mammals are able to survive only because oxygen levels are relatively high.
- Mammals first emerged when atmospheric oxygen levels reached 10 percent.
- A mammal's unborn child receives oxygen through the mother's placenta.

- ☐ Mammals use a lot of oxygen partly because they are warm-blooded.

Paragraph 5

Along with the buildup of molecular oxygen (O₂) came an eventual increase in ozone (O₃) levels in the atmosphere. ■ Because ozone filters out harmful ultraviolet radiation, this made it possible for life to flourish in shallow water and finally on land. ■ This critical state in the evolution of the atmosphere was replaced between 1100 and 542 million years ago. ■ Interestingly, the fossil record shows an explosion of life forms 542 million years ago. ■

41. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

The timing strongly suggests that atmospheric changes were responsible for this sudden increase in new life.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.



42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Internal and external forces on Earth's atmosphere have changed its chemical composition over time.

-
-
-

Answer choices

Mammals could not have survived without an oxygen-rich atmosphere, and land-based life would not be possible without the ozone layer to filter solar radiation.

Although they are currently at about 21 percent, oxygen levels will probably not always remain this high.

The breakdown of organic matter removes free oxygen, but if this process is interrupted, extra oxygen may accumulate in the atmosphere.

Over the last 4 billion years, outgassing destroyed Earth's primary atmosphere of volatile elements and replaced it with nonvolatile materials including carbon dioxide.

The small amount of oxygen in Earth's early atmosphere was due to photo dissociation and, later, photosynthesis created free oxygen.

When oxygen levels in the ocean water reached a critical level about 542 million years ago, life emerged in the oceans, as shown by sedimentary rocks

Birdsong

Birdsong is the classic example of how genes (hereditary information) and environment both have a crucial role to play in the behavioral development of animals. Since the pioneering work of W. H. Thorpe on chaffinches (a common European bird), many species have been studied, and it has become clear both that learning plays an important role for all species and also that there are constraints on what they are able to learn.

Thorpe was able to show that learning from others was involved in chaffinch birds through a series of experiments on hand-reared chicks (young birds). As in most other species, only the males sing. Thorpe found that, if he raised young males in total isolation from all others, the song they produced was quite different from that of a normal adult. It was about the right length and in the correct frequency range. It was also split up into a series of notes as it should be. But these notes lacked the detailed structure found in wild birds, nor was the song split up into distinct phrases as it usually is. This suggested that song development requires some social influence. Later experiments in which researchers played recordings of songs to young birds showed just how precise this influence was: many of them would learn the exact pattern of the recording they had heard. A remarkable feature here was that birds were able to copy precisely songs that they only heard in the first few weeks of life, yet they did not sing themselves until about eight months old. They are thus able to store a memory of the sound within their brain and then match their own output to their recollection of it when they mature.

Young chaffinches normally learn only chaffinch song, though Thorpe found they could be trained to sing the song of a tree pipit (another type of bird), which is very similar to that of their own species. In general, however, the constraints on learning which birds have ensure that they only learn songs appropriate to the species to which they themselves belong. These constraints may be in their brain's circuitry, the young bird hatching with a rough idea of the sounds that it should copy. The crude song of a bird reared in isolation gives some clues as to what this rough idea may be: the length, the frequency range and the breaking up into notes are all aspects of chaffinch song

shared between normal birds and those reared in isolation. In other cases the constraints are more social, young birds only being prepared to learn from individuals with whom they have social interactions. Thus, in a number of species, it has been found that they will not copy from recordings, but will do so from a live tutor. In some cases this may occur when they are young birds, but in others the main learning period is when they set up their territories and interact with neighbors for the first time, enabling them to match their neighbor's songs and so countersing with them. Whatever the nature of the learning rules in a particular species, there is no doubt that they are effective; it is very unusual to hear a wild bird singing a song which is not typical of its own species despite the many different songs which often occur in a small patch of woodland.

However, not all birds show the same learning pattern as do chaffinches. There are some species which produce normal sounds even if deaf, so that they cannot hear their own efforts, much less copy those of others. The cooing of doves and the crowing of cocks are examples here. In other cases, such as parrots and hill mynahs, birds can be trained to copy a huge variety of sounds, though those they learn in the wild are usually more restricted. The amazing capability of mynahs has apparently arisen simply because birds in an area learn a small number of their calls from each other, males from males and females from females, and these calls are highly varied in structure. The ability to master them has led the birds, incidentally, to be capable of saying "hello" and mimicking a wide variety of other sounds.

Paragraph 1

Birdsong is the classic example of how genes (hereditary information) and environment both have a crucial role to play in the behavioral development of animals. Since the **pioneering** work of W. H. Thorpe on chaffinches (a common European bird), many species have been studied, and it has become clear both that learning plays an important role for all species and also that there are constraints on what they are able to learn.

1. The word "**pioneering**" in the passage is closest in meaning to

- ☐ recent

- ☐ famous
- ☐ original
- ☐ controversial

Paragraph 2

Thorpe was able to show that learning from others was involved in chaffinch birds through a series of experiments on hand-reared chicks (young birds). As in most other species, only the males sing. Thorpe found that, if he raised young males in total isolation from all others, the song they produced was quite different from that of a normal adult. It was about the right length and in the correct frequency range. It was also split up into a series of notes as it should be. But these notes lacked the detailed structure found in wild birds, nor was the song split up into **distinct** phrases as it usually is. This suggested that song development requires some social influence. Later experiments in which researchers played recordings of songs to young birds showed just how precise this influence was: many of them would learn the exact pattern of the recording they had heard. A remarkable feature here was that birds were able to copy precisely songs that they only heard in the first few weeks of life, yet they did not sing themselves until about eight months old. They are thus able to store a memory of the sound within their brain and then match their own output to their recollection of it when they mature.

2. The word “**distinct**” in the passage is closest in meaning to
- ☐ short
 - ☐ simple
 - ☐ separate
 - ☐ similar
3. According to paragraph 2, all of the following are characteristics of the songs of the young chaffinches in Thorpe’s experiment EXCEPT:
- ☐ They were not identical to the songs of normal adult chaffinches.
 - ☐ They lacked the complex form of the songs of wild chaffinches.
 - ☐ They were as long as the songs of normal adult chaffinches.
 - ☐ They were clearly different from each other.

4. According to paragraph 2, researchers discovered which of the following by playing recordings of songs to chaffinches?
- Chaffinches could no longer be taught to reproduce sounds after the first few weeks of life.
 - Chaffinches could not reproduce songs with exactly the same patterns of recorded songs.
 - Chaffinches at the age of eight months could recall and reproduce a song that they heard in the first few weeks of life.
 - Chaffinches that learned a song from recordings in the first few weeks of life were later unable to copy the sounds of mature chaffinches.
5. All of the following are mentioned in paragraph 2 as characteristics of wild chaffinches EXCEPT:
- They are able to copy songs very precisely.
 - Their song development requires interaction with other chaffinches.
 - Their songs are not as well-structured as the songs of other birds.
 - It is the males of the species that do the singing.

Paragraph 3

Young chaffinches normally learn only chaffinch song, though Thorpe found they could be trained to sing the song of a tree pipit (another type of bird), which is very similar to that of their own species. In general, however, the constraints on learning which birds have ensure that they only learn songs appropriate to the species to which they themselves belong. These constraints may be in their brain's circuitry, the young bird hatching with a rough idea of the sounds that it should copy. The crude song of a bird reared in isolation gives some clues as to what this rough idea may be: the length, the frequency range and the breaking up into notes are all aspects of chaffinch song shared between normal birds and those reared in isolation. In other cases the constraints are more social, young birds only being prepared to learn from individuals with whom they have social interactions. Thus, in a number of species, it has been found that they will not copy from recordings, but will do so from a live tutor. In some cases this may occur when they are young birds, but in others the main learning period is when they set up their territories and interact with neighbors for the first time, enabling them to match their neighbor's songs and so countering with them.

Whatever the nature of the learning rules in a particular species, there is no doubt that they are effective; it is very unusual to hear a wild bird singing a song which is not typical of its own species despite the many different songs which often occur in a small patch of woodland.

6. The word “enabling” in the passage is closest in meaning to
- ☐ allowing
 - ☐ challenging
 - ☐ forcing
 - ☐ preparing
7. It can be inferred from paragraph 3 that one of the functions of songs in birds is to
- ☐ bring together birds living in groups with birds living in isolation
 - ☐ help young birds distinguish other young birds from adults
 - ☐ make possible interactions between birds of different species
 - ☐ help birds to establish territories
8. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Songs produced by chaffinches reared in isolation are cruder than the songs of wild birds.
 - ☐ The song of a bird reared in isolation suggests which aspects of chaffinch song may be inborn.
 - ☐ Comparing the crude songs of chaffinches reared in isolation to the songs of wild chaffinches suggests differences as well as similarities.
 - ☐ Studying the song aspects of chaffinches reared in isolation, researchers have gained a better understanding of the songs produced by wild birds.
9. According to paragraph 3, in some species, young birds do not copy songs from recordings because
- ☐ they learn to sing only by live interactions with other birds
 - ☐ their ability to learn from recordings occurs later in life
 - ☐ they can only learn the songs of the birds living in their area of woodland

- they can only learn songs from other birds of their own species

10. Why does the author mention that it is very unusual to hear a wild bird singing a song which is not typical of its own species?

- To explain why a variety of different bird songs are often heard in a relatively small area
- To argue that social constraints have a greater impact upon learning than do genetic constraints
- To provide an example of how the process of learning rules varies from one species to another
- To illustrate how effective the different constraints upon learning are in young birds

Paragraph4

However, not all birds show the same learning pattern as do chaffinches. There are some species which produce normal sounds even if deaf, so that they cannot hear their own efforts, much less copy those of others. The cooing of doves and the crowing of cocks are examples here. In other cases, such as parrots and hill mynahs, birds can be trained to copy a huge variety of sounds, though those they learn in the wild are usually more restricted. The amazing capability of mynahs has apparently arisen simply because birds in an area learn a small number of their calls from each other, males from males and females from females, and these calls are highly varied in structure. The ability to master them has led the birds, incidentally, to be capable of saying “hello” and mimicking a wide variety of other sounds.

11. The word “restricted” in the passage is closest in meaning to

- important
- popular
- limited
- accurate

12. According to paragraph 4, why are mynahs able to learn to make a wide variety of sounds?

- They have the ability to imitate any sound that they are exposed to.

- The frequency with which mynahs travel from one small area to another exposes them to a wide variety of sounds.
- They are exposed in the wild to calls that are very different from each other.
- An acute sense of hearing allows them to listen to and copy many different sounds.

Paragraph 3

Young chaffinches normally learn only chaffinch song, though Thorpe found they could be trained to sing the song of a tree pipit (another type of bird), which is very similar to that of their own species. ■ In general, however, the constraints on learning which birds have ensure that they only learn songs appropriate to the species to which they themselves belong. ■ These constraints may be in their brain's circuitry, the young bird hatching with a rough idea of the sounds that it should copy. ■ The crude song of a bird reared in isolation gives some clues as to what this rough idea may be: the length, the frequency range and the breaking up into notes are all aspects of chaffinch song shared between normal birds and those reared in isolation. ■ In other cases the constraints are more social, young birds only being prepared to learn from individuals with whom they have social interactions. Thus, in a number of species, it has been found that they will not copy from recordings, but will do so from a live tutor. In some cases this may occur when they are young birds, but in others the main learning period is when they set up their territories and interact with neighbors for the first time, enabling them to match their neighbor's songs and so countersing with them. Whatever the nature of the learning rules in a particular species, there is no doubt that they are effective; it is very unusual to hear a wild bird singing a song which is not typical of its own species despite the many different songs which often occur in a small patch of woodland.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Are these constraints genetic, environmental, or both?

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.



The Role of Diapause

If conditions within an organism's environment occasionally or regularly become harsh, it may be advantageous for an organism to have a resistant stage built into the life cycle. In such a life history strategy, the organism suspends any growth, reproduction, or other activities for a period of time so that they may occur at a later, more hospitable time. This genetically determined resting stage, characterized by the cessation of development and protein synthesis and suppression of the metabolic rate, is called *diapause*. Many other kinds of resting stages, with different levels of suppression of physiological activities, are known. Some of these resistant stages can be extremely long-lived. In one case, seeds of the arctic lupine, a member of the pea family recovered from ancient lemming burrows in the Arctic, germinated in three days even though they were carbon-dated at more than 10,000 years old!

Unfavorable conditions that are relatively predictable probably pose a simpler problem for organisms than do unpredictable conditions. Adaptations to the regular change of seasons in the temperate and polar regions may be relatively simple. For example, many seeds require a period of *stratification*, exposure to low temperatures for some minimum period, before they will germinate. This is a simple adaptation to ensure that germination occurs following the winter conditions rather than immediately prior to their onset. In contrast, unfavorable conditions that occur unpredictably pose considerable problems for organisms. In fact, unpredictability is probably a greater problem than is the severity of the unfavorable period. How can organisms cope with the unpredictable onset of good or poor conditions?

Many adaptations to this general problem are based on a resting stage that awaits favorable conditions. We will consider two examples from the vertebrates. The first is the red kangaroo. This marsupial inhabits the deserts of central Australia where the onset of rains and the resulting sudden growth of vegetation are extremely unpredictable. Obviously, it is advantageous for a kangaroo female to produce young at a time when plant productivity is sufficient to support her offspring. For such a relatively large mammal, however, gestation (the period of development during pregnancy) is so long that if a female waited to mate and carry the young until after

the rains came, the favorable period might be past. The kangaroo's life history adaptation to this problem involves the use of embryonic diapause during gestation (development in the uterus).

After a 31-day gestation period, the female gives birth to a tiny helpless young typical of marsupials. The newborn crawls into the mother's pouch and attaches to a teat where it continues to grow and develop. After 235 days it leaves the pouch but remains with the mother and obtains milk from her. Two days after giving birth, the female mates again. The fertilized egg enters a 204-day period of embryonic diapause during which it remains in the uterus but does not attach. It then implants, and 31 days later, birth of the second young occurs. Note that the first young leaves the pouch at just this time. Again, the female mates, fertilization occurs, and another diapause follows. The eventual result is that at any one time, the female has three young at various stages of development: one in diapause, one in the pouch, and one outside the pouch. Among other benefits, this allows her to freeze the development of an embryo during times of drought and food shortage until the offspring in the pouch is able to leave.

A similar strategy—accelerated development combined with a resting stage—has also allowed amphibians to inhabit deserts. The spadefoot toads, such as Couch's spadefoot toad, inhabit some of the most severe deserts in North America. Adults of this species burrow deeply into the substrate where it is cooler and perhaps more moist. Here they enter into a resting state in which they are covered with a protective layer of dead skin. When it rains, the adults emerge and congregate to mate at temporary ponds. Development is greatly accelerated: the eggs hatch within 48 hours, and the tadpoles change into toads at 16-18 days. Consequently, they can complete the life cycle during the brief window of favorable conditions, then return to the resistant resting stage to await the next rainfall. Resting stages thus comprise a series of adaptations that allow the species to avoid the most difficult conditions for life.

Paragraph 1

If conditions within an organism's environment occasionally or regularly become harsh, it may be advantageous for an organism to have a resistant stage built into the

life cycle. In such a life history strategy, the organism suspends any growth, reproduction, or other activities for a period of time so that they may occur at a later, more hospitable time. This genetically determined resting stage, characterized by the cessation of development and protein synthesis and suppression of the metabolic rate, is called *diapause*. Many other kinds of resting stages, with different levels of suppression of physiological activities, are known. Some of these resistant stages can be extremely long-lived. In one case, seeds of the arctic lupine, a member of the pea family recovered from ancient lemming burrows in the Arctic, germinated in three days even though they were carbon-dated at more than 10,000 years old!

15. According to paragraph 1, why do some organisms have a resting stage during their life cycle?

- ☐ To recover from injuries suffered during harsh conditions
- ☐ To devote all of their energy to a period of growth and reproduction
- ☐ To wait for local conditions to become favorable for important life events
- ☐ To prepare to move to a different environment if conditions become harsh

16. Why does the author mention “seeds of the arctic lupine”?

- ☐ To argue that members of the pea family are extremely resistant to cold temperature
- ☐ To provide information about what ancient lemmings ate during their long resting periods
- ☐ To provide an example of an organism with a resting stage that has many different levels of suppression of physiological activities
- ☐ To support that some resting stages last an extremely long time.

Paragraph 2

Unfavorable conditions that are relatively predictable probably pose a simpler problem for organisms than do unpredictable conditions. Adaptations to the regular change of seasons in the temperate and polar regions may be relatively simple. For example, many seeds require a period of *stratification*, exposure to low temperatures for some minimum period, before they will germinate. This is a simple adaptation to ensure that germination occurs following the winter conditions rather than immediately prior to their onset. In contrast, unfavorable conditions that occur

unpredictably pose considerable problems for organisms. In fact, unpredictability is probably a greater problem than is the severity of the unfavorable period. How can organisms cope with the unpredictable onset of good or poor conditions?

17. According to paragraph 2, why do many seeds require a period of stratification?

- ☐ To slowly build up a tolerance for lower and lower temperatures
- ☐ To guarantee that the seeds grow after and not before unfavorable weather
- ☐ To make sure that the seeds can deal with unpredictable conditions
- ☐ To give the seeds enough time to germinate before winter /

18. The word “severity” in the passage is closest in meaning to

- ☐ length
- ☐ harshness
- ☐ unexpectedness
- ☐ completeness

Paragraph 3

Many adaptations to this general problem are based on a resting stage that awaits favorable conditions. We will consider two examples from the vertebrates. The first is the red kangaroo. This marsupial inhabits the deserts of central Australia where the onset of rains and the resulting sudden growth of vegetation are extremely unpredictable. Obviously, it is advantageous for a kangaroo female to produce young at a time when plant productivity is sufficient to support her offspring. For such a relatively large mammal, however, gestation (the period of development during pregnancy) is so long that if a female waited to mate and carry the young until after the rains came, the favorable period might be past. The kangaroo's life history adaptation to this problem involves the use of embryonic diapause during gestation (development in the uterus).

19. According to paragraph 4, all of the following statements are true about the young offspring of the red kangaroo EXCEPT:

- ☐ After birth, a newborn crawls into the mother's pouch where it grows and develops.
- ☐ After a young kangaroo leaves its mother's pouch, it still needs its mother's milk.

- A mother usually gives birth to three baby kangaroos at the same time.

Paragraph 4

After a 31-day gestation period, the female gives birth to a tiny helpless young typical of marsupials. The newborn crawls into the mother's pouch and attaches to a teat where it continues to grow and develop. After 235 days it leaves the pouch but remains with the mother and obtains milk from her. Two days after giving birth, the female mates again. The fertilized egg enters a 204-day period of embryonic diapause during which it remains in the uterus but does not attach. It then implants, and 31 days later, birth of the second young occurs. Note that the first young leaves the pouch at just this time. Again, the female mates, fertilization occurs, and another diapause follows. The eventual result is that at any one time, the female has three young at various stages of development: one in diapause, one in the pouch, and one outside the pouch. Among other benefits, this allows her to freeze the development of an embryo during times of drought and food shortage until the offspring in the pouch is able to leave.

20. Paragraph 4 supports all of the following statements about the red kangaroo of central Australia EXCEPT:

- A female kangaroo mates again shortly after her newborn enters her pouch.
- During diapause, a young kangaroo stays in the female's pouch and growth of a second fertilized egg inside the uterus is delayed.
- To put different young kangaroos at various stages of development is advantageous for the female kangaroo to handle them at the same time.
- The pause of the development of an embryo has more benefits for preparing it to avoid the harsh times than for competing with its siblings.

21. What is the main purpose of paragraph 4 in the passage?

- To give the details of an adaptation mentioned in paragraph 3
- To describe an adaptation different from the one explained in paragraph 3
- To introduce an adaptation that is described in detail in paragraph 5
- To discuss an adaptation that is set as successful as the one mentioned before.

Paragraph 5

A similar strategy—accelerated development combined with a resting stage—has also allowed amphibians to inhabit deserts. The spadefoot toads, such as Couch’s spadefoot toad, inhabit some of the most severe deserts in North America. Adults of this species burrow deeply into the substrate where it is cooler and perhaps more moist. Here they enter into a resting state in which they are covered with a protective layer of dead skin. When it rains, the adults emerge and **congregate** to mate at temporary ponds. Development is greatly accelerated: the eggs hatch within 48 hours, and the tadpoles change into toads at 16-18 days. Consequently, they can complete the life cycle during the brief window of favorable conditions, then return to the resistant resting stage to await the next rainfall. Resting stages thus **comprise** a series of adaptations that allow the species to avoid the most difficult conditions for life.

22. The word “**congregate**” in the passage is closest in meaning to

- ☐ begin
- ☐ gather
- ☐ hurry
- ☐ expect

23. The word “**comprise**” in the passage is closest in meaning to

- ☐ consist of
- ☐ bring about
- ☐ are similar to
- ☐ take the place of

24. According to paragraph 5, how do amphibians such as spadefoot toads survive the severe heat conditions in the North American deserts?

- ☐ They dig down into the ground and go into a resistant resting state.
- ☐ They remain in the ponds that develop after it has rained.
- ☐ They lose their outer layer of skin.
- ☐ Their eggs live dormant until the desert air becomes cooler and more moist.

25. According to paragraph 5, which of the following occurs during the life cycle of the spadefoot toad?

- ☐ The female’s eggs hatch under the surface of the desert.

- The adults mate during the dry period.
- The newborn grows into an adult before unfavorable conditions return.
- The newborn enters a resting stage before it becomes an adult.

Paragraph 2

Unfavorable conditions that are relatively predictable probably pose a simpler problem for organisms than do unpredictable conditions. Adaptations to the regular change of seasons in the temperate and polar regions may be relatively simple. For example, many seeds require a period of *stratification*, exposure to low temperatures for some minimum period, before they will germinate. ■ This is a simple adaptation to ensure that germination occurs following the winter conditions rather than immediately prior to their onset. ■ In contrast, unfavorable conditions that occur unpredictably pose considerable problems for organisms. ■ In fact, unpredictability is probably a greater problem than is the severity of the unfavorable period. ■ How can organisms cope with the unpredictable onset of good or poor conditions?

26. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Such adaptations to predictable conditions can also be made by animals, such as by hibernating during the coldest months.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

27. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Some organisms adapt to periodic harsh conditions by building a resistant stage, or diapause, into their life cycle.

-
-
-

Answer choices

The diapause stage evolved very early time, allowing the young to leave the and is most common in species that first pouch only when conditions are appeared more than 10,000 years ago. favorable for their growth.

Unpredictable conditions are more problematic for organisms than are fairly predictable changes such as the seasons. The female red kangaroo adapts to unfavorable conditions by delaying the development of the embryo in the uterus.

Some seeds may germinate in three days even if they have been exposed to very low temperatures for a long time. Some amphibians adapt to arid environments by completing accelerated development with resting stages deep underground.

Some marsupials can care for three newborns in their pouch at the same

The Plow and the Horse in Medieval Europe

One of the most important factors driving Europe's slow emergence from the economic stagnation of the Early Middle Ages (circa 500-1000 B.C.E.) was the improvement of agricultural technology. One innovation was a new plow, with a curved attachment (moldboard) to turn over wet, heavy soils, and a knife (or coulter) in front of the blade to allow a deeper and easier cut. This more complex plow replaced the simpler "scratch" plow that merely made a shallow, straight furrow in the ground. In the lands around the Mediterranean, with light rains and mild winters, this had been fine, but in the wetter terrain north and west of the Danube and the Alps, such a plow left much to be desired, and it is to be wondered if it was used at all. Cleared lands would more likely have been worked by hand tilling, with little direct help from animals, and the vast forests natural to Northern Europe remained either untouched, or perhaps cleared in small sections by fire, and the land probably used only so long as the ash-enriched soil yielded good crops and then abandoned for some other similarly cleared field. Such a pattern of agriculture and settlement was no basis for sustained cultural or economic life.

With the new heavy plow, however, fields could be cleared, sowed, and maintained with little more difficulty than in the long-settled lands of Southern Europe, while the richness of the new soils, the reliability of the rains, and the variety of crops now possible made for an extremely productive agriculture. The new tool, however, imposed new demands, technical, economic, and social. The heavy plow was a substantial piece of capital, unlike a simple hand hoe, and this had the same sorts of implications that capitalization always has—it favored the concentration of wealth and control. Moreover, making full use of it required more animal power, and this had a host of implications of its own. The full importance of this was even more apparent in the centuries after 1000, when oxen began to give way in certain parts of Western Europe to horses.

The powerful, rugged farm horse was itself a product of improvement during the Middle Ages, and it was part of a complex set of technical changes and capabilities. The introduction of new forms of equipment for horses transformed this animal into

the single most important assist to human labor and travel. Instead of the old harness used by the ancient Greeks and Romans, there appeared from Central Asia the rigid, padded horse collar. Now, when the horse pulled against a load, no longer did the load pull back against its neck and windpipe but rather rode on the sturdy shoulders. When this innovation was combined with the iron horseshoe, the greater speed and stamina of the horse displaced oxen wherever it could be afforded. The larger importance of this lay not only in more efficient farmwork, but in swifter and surer transportation between town and countryside. The farmer with horses could move products to market more frequently and at greater distances than with only oxen, and the urban development that was to transform the European economic and social landscape after the eleventh century was propelled in large part by these new horse-centered transport capabilities.

Another indicator of how compelling and important was the new horse agriculture was its sheer cost. Unlike oxen and other cattle, horses cannot be supported exclusively on hay and pasturage, they require, particularly in northern climates where pasturing seasons are short, cropped food, such as oats and alfalfa. Unlike grass and hay, these are grown with much of the same effort and resources applied to human nourishment, and thus their acquisition represents a sacrifice, in a real sense, of human food. The importance of this in a world that usually lived at the margins of sufficient diet is hard to overstate. The increased resources that went into making the horse central to both the medieval economy and, in a separate but related development, medieval warfare, are the surest signs of the great utility the animal now assumed.

Paragraph 1

One of the most important factors driving Europe's slow emergence from the economic **stagnation** of the Early Middle Ages (circa 500-1000 B.C.E.) was the improvement of agricultural technology. One innovation was a new plow, with a curved attachment (moldboard) to turn over wet, heavy soils, and a knife (or coulter) in front of the blade to allow a deeper and easier cut. This more complex plow replaced the simpler "scratch" plow that merely made a shallow, straight furrow in the ground. In the lands around the Mediterranean, with light rains and mild winters, this had been found, but in the wetter terrain north and west of the Danube and the Alps,

such a plow left much to be desired, and it is to be wondered if it was used at all. Cleared lands would more likely have been worked by hand tilling, with little direct help from animals, and the vast forests natural to Northern Europe remained either untouched, or perhaps cleared in small sections by fire, and the land probably used only so long as the ash-enriched soil yielded good crops and then abandoned for some others similarly cleared field. Such a pattern of agriculture and settlement was no basis for sustained cultural or economic life.

28. The word “stagnation” in the passage is closest in meaning to

- ☐ instability
- ☐ lack of growth
- ☐ dependence on others
- ☐ decline

29. According to paragraph 1, what was the main advantage of the new plow over the scratch plow?

- ☐ The new plow created straighter rows.
- ☐ The new plow was easier for animals to pull.
- ☐ The new plow could dig deeper into the soil.
- ☐ The new plow was easier to make.

30. The word “sustained” in the passage is closest in meaning to

- ☐ continued
- ☐ established
- ☐ ordinary
- ☐ active

31. According to paragraph 1, the scratch plow was particularly unsuited to

- ☐ the lands around the Mediterranean
- ☐ places where the soil was often dry
- ☐ places where land was cleared and worked by hand
- ☐ places where the soil was particularly wet and heavy

32. Paragraph 1 implies which of the following about agriculture before the

introduction of the new plow?

- ☐ Limited rainfall had prevented large-scale agricultural development.
- ☐ Most of Europe's developed agricultural communities were located in the south.
- ☐ Several other important innovations immediately preceded the development of the new plow.
- ☐ Much of Europe's forestland had been converted to agricultural use.

33. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Tilling by hand was so difficult that cleared land in Northern Europe was often abandoned and allowed to return to its natural forested state.
- ☐ Cleared land was probably tilled by hand, while the forests of Northern Europe were cleared only in small sections and used for short periods.

Paragraph 2

With the new heavy plow, however, fields could be cleared, sowed, and maintained with little more difficulty than in the long-settled lands of Southern Europe, while the richness of the new soils, the reliability of the rains, and the variety of crops now possible made for an extremely productive agriculture. The new tool, however, imposed new demands, technical, economic, and social. The heavy plow was a substantial piece of capital, unlike a simple hand hoe, and this had the same sorts of implications that capitalization always has—it favored the concentration of wealth and control. Moreover, making full use of it required more animal power, and this had a host of implications of its own. The full importance of this was even more apparent in the centuries after 1000, when oxen began to give way in certain parts of Western Europe to horses.

34. The word “implications” in the passage is closest in meaning to

- ☐ requirements
- ☐ inequalities
- ☐ consequences
- ☐ disadvantages

35. What can be inferred from the information that the new plow “favored the concentration of wealth and control”?

- Wealthy farmers in the south had a significant economic advantage over farmers in the north.
- The production and sale of the new plow became an important source of capital.
- The new plow was more popular in parts of Europe where oxen were used for farming than in parts where horses were used.
- Greater economic equality existed in northern Europe before the introduction of the new plow.

Paragraph 3

The powerful, rugged farm horse was itself a product of improvement during the Middle Ages, and it was part of a complex set of technical changes and capabilities. The introduction of new forms of equipment for horses transformed this animal into the single most important assist to human labor and travel. Instead of the old harness used by the ancient Greeks and Romans, there appeared from Central Asia the rigid, padded horse collar. Now, when the horse pulled against a load, no longer did the load pull back against its neck and windpipe but rather rode on the sturdy shoulders. When this innovation was combined with the iron horseshoe, the greater speed and stamina of the horse displaced oxen wherever it could be afforded. The larger importance of this lay not only in more efficient farmwork, but in swifter and surer transportation between town and countryside. The farmer with horses could move products to market more frequently and at greater distances than with only oxen, and the urban development that was to transform the European economic and social landscape after the eleventh century was propelled in large part by these new horse-centered transport capabilities.

36. According to paragraph 3, which of the following contributed to the dramatic rise in the agricultural use of horses in Europe?

- A powerful new breed of farm horse was brought to Europe from Central Asia.
- Farmers began using rigid, padded collars that allowed horses to pull heavy loads more easily.
- For the first time, horses became cheaper than oxen.
- Farmers began studying the farming techniques used by the horse-drawn plows.

37. According to paragraph 3, what role did horses play in the larger social changes of the eleventh century?

- ☐ The raising and selling of horses became important economic and cultural activities in Europe.
- ☐ Horses stimulated the growth of urban areas by providing quick, reliable transportation.
- ☐ Owners of successful horse-based farms became influential member of society.

Paragraph 4

Another indicator of how compelling and important was the new horse agriculture was its sheer cost. Unlike oxen and other cattle, horses cannot be supported **exclusively** on hay and pasturage, they require, particularly in northern climates where pasturing seasons are short, cropped food, such as oats and alfalfa. Unlike grass and hay, these are grown with much of the same effort and resources applied to human nourishment, and thus their acquisition represents a sacrifice, in a real sense, of human food. The importance of this in a world that usually lived at the margins of sufficient diet is hard to overstate. The increased resources that went into making the horse central to both the medieval economy and, in a separate but related development, medieval warfare, are the surest signs of the great utility the animal now assumed.

38. The word “**exclusively**” in the passage is closest in meaning to

- ☐ cheaply
- ☐ easily
- ☐ reliably
- ☐ solely

39. In paragraph 4, why does the author emphasize the amount of effort and resources needed to grow alfalfa and oats?

- ☐ To illustrate how valuable horses were by showing how much farmers were willing to sacrifice to keep them
- ☐ To provide evidence that, in medieval Europe, both horses and humans lived at the margins of a sufficient diet
- ☐ To argue that it made more sense to devote land to growing food than food for

horses

Paragraph 1

One of the most important factors driving Europe's slow emergence from the economic stagnation of the Early Middle Ages (circa 500-1000 B.C.E.) was the improvement of agricultural technology. One innovation was a new plow, with a curved attachment (moldboard) to turn over wet, heavy soils, and a knife (or coulter) in front of the blade to allow a deeper and easier cut. ■ This more complex plow replaced the simpler "scratch" plow that merely made a shallow, straight furrow in the ground. ■ In the lands around the Mediterranean, with light rains and mild winters, this had been fine, but in the wetter terrain north and west of the Danube and the Alps, such a plow left much to be desired, and it is to be wondered if it was used at all. ■ Cleared lands would more likely have been worked by hand tilling, with little direct help from animals, and the vast forests natural to Northern Europe remained either untouched, or perhaps cleared in small sections by fire, and the land probably used only so long as the ash-enriched soil yielded good crops and then abandoned for some other similarly cleared field. ■ Such a pattern of agriculture and settlement was no basis for sustained cultural or economic life.

40. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

In fact, it sliced the ground so thoroughly that fields could be planted after only one plowing rather than the two needed before.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

41. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Drag your choices to the spaces where they belong. To review the passage, click on

View Text.

Agricultural innovations with important social and economic consequences occurred in eleventh-century Europe.

-
-
-

Answer choices

Light rains and unpredictable winters had made most of the soil in Europe unsuitable for enough agriculture to sustain economic development.

Improvements in the design of plows opened up vast areas of land in Northern Europe that had previously been unusable for sustained agriculture.

Farmers switched from oxen to horses to pull their plows because inexpensive pasturage for oxen decreased significantly in the centuries after 1000 B.C.E.

With help from a new kind of harness from Asia, horses were able to pull the new heavy plow and to transport goods to market more quickly and frequently.

The introduction of iron horseshoes enabled farmers to transport goods farther than they could with the more expensive oxen.

The horse came to be valued so greatly that farmers devoted some of their land to growing crops for their horses rather than using this land to grow food for their families.

The Origins of Plant and Animal Domestication

The emergence of plant and animal domestication represented a monumental change in the ways that humans interacted with Earth's resources: the rate at which Earth's surface was modified and the rates of human population growth. The development of agriculture was accompanied by fundamental changes in the organization of human society: disparities in wealth, hierarchies of power, and urbanization.

Phrases like “plant and animal domestication” and “the invention of agriculture” create the impression that humans made the transition to cultivating plants and tending animals rather abruptly, maybe with a flash of insight. Most scholars don't think so. It seems more likely that humans used and manipulated wild plants and animals for many hundreds of thousands of years. The transition to gardens, fields, and pastures was probably gradual, the natural outgrowth of a long familiarity with the environmental requirements, growth cycles, and reproductive mechanisms of whatever plants and animals humans liked to eat, ride, or wear.

For years, scholars argued that the practices of cultivation and animal domestication were invented in one or two locations on Earth and then diffused from those centers of innovation. Genetic studies are now showing that many different groups of people in many different places around the globe learned independently to create especially useful plants and animals through selective breeding. Probably both independent invention and diffusion played a role in agricultural innovation. Sometimes the ideas of domestication and cultivation were relayed to new places. In other cases the farmers or herders themselves moved into new zones, taking agriculture or improvements such as new tools or new methods or new plants and animals with them.

Scholars used to assume that people turned to cultivating instead of gathering their food either because they had to in order to feed burgeoning populations, or because agriculture provided such obviously better nutrition. It now seems that neither of these explanations is valid. First of all, the risk attached to exploring new food sources when there were already too many mouths to feed would be too great. Second, agriculture did not necessarily improve nutrition or supplies of food. A varied diet based on gathered (and occasionally hunted) food probably provided a wider, more secure range of nutrients than an early agriculturally based diet of only one or two cultivated crops. More likely, populations expanded after agricultural successes, and not before.

Richard MacNeish, an archaeologist who studied plant domestication in Mexico and Central America, suggested that the chance to trade was at the heart of agricultural origins worldwide. Many of the known locations of agricultural innovation lie near early trade centers. People in such places would have had at least two reasons to pursue cultivation and animal raising; they would have had access to new information, plants, and animals brought in by traders, and they would have had a need for something to trade with the people passing through. Perhaps, then, agriculture was at first just a profitable hobby for hunters and gatherers that eventually, because of market demand, grew into the primary source of sustenance. Trade in agricultural products

may also have been a hobby that led to trouble.

E. N. Anderson, writing about the beginnings of agriculture in China, suggests that agricultural production for trade may have been the impetus for several global situations now regarded as problems: rapid population growth, social inequalities, environmental degradation, and famine. Briefly explained, his theory suggests that groups turned to raising animals and plants in order to reap the profits of trading them. As more labor was needed to supply the trade, humans produced more children. As populations expanded, more resources were put into producing food for subsistence and for trade. Gradually, hunting and gathering technology was abandoned as populations, with their demands for space, destroyed natural habitats. Meanwhile, a minority elite emerged when the wealth provided by trade did not accrue equally to everyone. Yet another problem was that a drought or other natural disaster could wipe out an entire harvest, thus, as ever larger populations depended solely on agriculture, famine became more common.

Paragraph 1

The emergence of plant and animal domestication represented a **monumental** change in the ways that humans interacted with Earth's resources: the rate at which Earth's surface was modified and the rates of human population growth. The development of agriculture was accompanied by fundamental changes in the organization of human society: disparities in wealth, hierarchies of power, and urbanization.

1. The word "monumental" in the passage is closest in meaning to
 - ☐ innovative
 - ☐ surprising
 - ☐ complex
 - ☐ enormous

Paragraph 2

Phrases like "plant and animal domestication" and "the invention of agriculture" create the impression that humans made the transition to cultivating plants and tending animals rather abruptly, maybe with a flash of insight. Most scholars don't think so. It seems more likely that humans used and **manipulated** wild plants and animals for many hundreds of thousands of years. The transition to gardens, fields, and pastures was probably gradual, the natural outgrowth of a long familiarity with the environmental requirements, growth cycles, and reproductive mechanisms of whatever plants and animals humans liked to eat, ride, or wear.

2. According to paragraph 2, phrases such as "plant and animal domestication" and "the invention of agriculture" encourage which of the following wrong ideas?
 - ☐ Early farmers were fairly advanced in their knowledge of plants and animals.
 - ☐ Agriculture and animal domestication arose as a result of systematic study and experimentation.
 - ☐ The change from hunting and gathering to farming and raising animals occurred quickly.
 - ☐ Early efforts to raise plants and animals were generally successful.

3. The word “manipulated” in the passage is closest in meaning to
- ☐ observed
 - ☐ consumed
 - ☐ skillfully used
 - ☐ protected
4. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ The change to land cultivation was a slow process because humans were familiar with the needs of relatively few plants and animals.
 - ☐ The cultivation of land occurred gradually as it was the product of extensive human experience observing plants and animals of value to humans.
 - ☐ Gardens, fields, and pastures were outgrowths of the desire for plants and animals that humans knew from long familiarity were good to eat, wear, or ride.
 - ☐ People learned about environmental requirements, growth cycles, and reproductive mechanisms through their long familiarity with plants and animals that they liked to eat, ride, and wear.

Paragraph 3

For years, scholars argued that the practices of cultivation and animal domestication were invented in one or two locations on Earth and then diffused from those centers of innovation. Genetic studies are now showing that many different groups of people in many different places around the globe learned independently to create especially useful plants and animals through selective breeding. Probably both independent invention and diffusion played a role in agricultural innovation. Sometimes the ideas of domestication and cultivation were relayed to new places. In other cases the farmers or herders themselves moved into new zones, taking agriculture or improvements such as new tools or new methods or new plants and animals with them.

5. What do genetic studies suggest about the theory that “cultivation and animal domestication were invented in one or two locations on Earth and then diffused from those centers of innovation”?
- ☐ The theory underestimates the speed at which cultivation and animal domestication were diffused.
 - ☐ The theory underestimates the number of locations in which cultivation and animal domestication arose independently.
 - ☐ The theory overemphasizes the importance of selective breeding in cultivation and animal domestication.
 - ☐ The theory overemphasizes the importance of cultivation and animal domestication to some groups of people.

Paragraph 4

Scholars used to assume that people turned to cultivating instead of gathering their food either because they had to in order to feed burgeoning populations, or because agriculture provided such obviously better nutrition. It now seems that neither of these explanations is valid. First of all, the risk attached to exploring new food sources when there were already too many mouths to feed would be too great. Second, agriculture did not necessarily improve nutrition or supplies of food. A varied diet based on gathered (and occasionally hunted) food probably provided a wider, more secure range of nutrients than an early agriculturally based diet of only one or two cultivated crops. More likely, populations expanded after agricultural successes, and not before.

6. According to paragraph 4, what advantages did the diet of hunter-gatherers probably have over an early agricultural diet?

- ☐ It required less effort for them to acquire food that was nutritious.
- ☐ It allowed their populations to expand across wider areas.
- ☐ It provided a greater variety of foods needed for them to stay healthy.
- ☐ It allowed them to have more children.

7. According to paragraph 4, what was the most likely relationship between population growth and food cultivation?

Population growth encouraged the development of food cultivation.

Population growth was dependent on the development of a varied diet of cultivated foods.

Successful methods of food cultivation were developed before population growth occurred.

Food cultivation was more successful in situations where population growth was limited.

Paragraph 5

Richard MacNeish, an archaeologist who studied plant domestication in Mexico and Central America, suggested that the chance to trade was at the heart of agricultural origins worldwide. Many of the known locations of agricultural innovation lie near early trade centers. People in such places would have had at least two reasons to pursue cultivation and animal raising; they would have had access to new information, plants, and animals brought in by traders, and they would have had a need for something to trade with the people passing through. Perhaps, then, agriculture was at first just a profitable hobby for hunters and gatherers that eventually, because of market demand, grew into the primary source of sustenance. Trade in agricultural products may also have been a hobby that led to trouble.

8. The word “pursue” in the passage is closest in meaning to

- ☐ practice
- ☐ encourage
- ☐ prefer
- ☐ combine

9. Which of the following most accurately reflects the statement discussed in paragraph 5, on efforts to cultivate plants and animals?

- ☐ The efforts often failed because hunter-gatherers had limited access to new information about plants and animals.

- The efforts were begun out of a desire to produce goods for trade rather than to increase their growers' food supplies.
- The efforts were sometimes abandoned so hunter-gatherers could become involved in the more profitable activity of trading.
- The efforts were not profitable until people began trading food that they had raised for other goods.

Paragraph 5

Richard MacNeish, an archaeologist who studied plant domestication in Mexico and Central America, suggested that the chance to trade was at the heart of agricultural origins worldwide. Many of the known locations of agricultural innovation lie near early trade centers. People in such places would have had at least two reasons to pursue cultivation and animal raising; they would have had access to new information, plants, and animals brought in by traders, and they would have had a need for something to trade with the people passing through. Perhaps, then, agriculture was at first just a profitable hobby for hunters and gatherers that eventually, because of market demand, grew into the primary source of sustenance. Trade in agricultural products may also have been a hobby that led to trouble.

Paragraph 6

E. N. Anderson, writing about the beginnings of agriculture in China, suggests that agricultural production for trade may have been the impetus for several global situations now regarded as problems: rapid population growth, social inequalities, environmental degradation, and famine. Briefly explained, his theory suggests that groups turned to raising animals and plants in order to reap the profits of trading them. As more labor was needed to supply the trade, humans produced more children. As populations expanded, more resources were put into producing food for subsistence and for trade. Gradually, hunting and gathering technology was abandoned as populations, with their demands for space, destroyed natural habitats. Meanwhile, a minority elite emerged when the wealth provided by trade did not accrue equally to everyone. Yet another problem was that a drought or other natural disaster could wipe out an entire harvest, thus, as ever larger populations depended solely on agriculture, famine became more common.

10. Which of the following most accurately reflect the relationship between paragraph 6 and a topic discussed in paragraph 5?
- Paragraph 6 discusses a series of events that calls into question the theory that plants and animals were raised for purposes of trade.
 - Paragraph 6 presents evidence supporting the claim that many sites of agricultural innovation were located near trade centers.
 - Paragraph 6 identifies problems that led to the raising of plants and animals as the primary source of sustenance.
 - Paragraph 6 traces negative developments that arose possibly as a result of raising plants and animals for trade.

11. The word “subsistence” in the passage is closest in meaning to

- ☐ profit
- ☐ surplus
- ☐ enjoyment
- ☐ survival

Paragraph 3

For years, scholars argued that the practices of cultivation and animal domestication were invented in one or two locations on Earth and then diffused from those centers of innovation. ■ Genetic studies are now showing that many different groups of people in many different places around the globe learned independently to create especially useful plants and animals through selective breeding. ■ Probably both independent invention and diffusion played a role in agricultural innovation. ■ Sometimes the ideas of domestication and cultivation were relayed to new places. In other cases the farmers or herders themselves moved into new zones, taking agriculture or improvements such as new tools or new methods or new plants and animals with them. ■

12. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Among the many places that are now known to be centers of independent domestication are Mesopotamia, Central China, and Mesoamerica.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

13. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Agriculture invention and animal domestication caused lasting changes to how humans live and to the physical surface of Earth.

-
-
-

Answer Choices

The transition from hunting and gathering to raising plants and animals was gradual and led to significant changes in the organization of human societies.

Scholars now believe that agriculture and animal domestication began independently in many separate locations and then spread to new areas.

As trade in agricultural products grew and social inequalities arose, new crops were developed specifically to feed the labor needed to support societies.

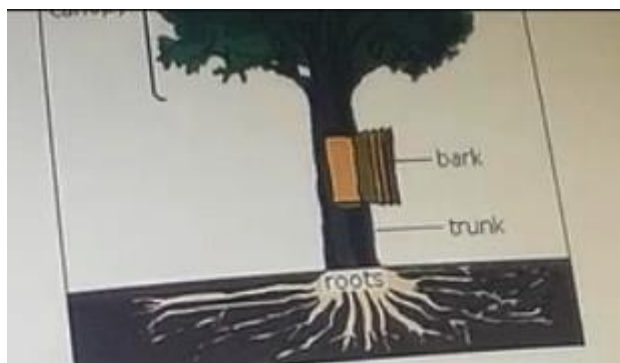
Although it is now clear that agriculture

developed independently in many places, often the most efficient techniques arose by combining practices of different cultures.

Agriculture became more widespread when human populations realized that an agricultural diet supplemented through trade could provide as much nutrition as the hunter-gatherer diet.

The earliest reason for raising plants and animals may have been to provide goods for trade, and such trade may account for the rise in social problems such as environmental destruction.





Preventing Overgrowth among Tree Branches

One way trees prevent themselves from having too many branches is simply by shedding (dropping off) branches once they have fulfilled their purpose. This happens as the tree gets bigger and grows new outer layers of foliage that shade the inner and lower branches. In most large trees, the center of the canopy contains only large branches, small branches and fine twigs are found only at the canopy's edge. In the shaded center, the small branches that would once have occupied that space are long gone. Trees like the true cypresses regularly shed small twigs complete with leaves toward the end of summer. Most other trees shed only branches that prove unproductive. If a branch is not producing enough carbohydrate to cover its own running costs—i.e., it needs to be subsidized by other branches because, for example, it is being shaded and receives little light—it will usually be got rid of. This prevents unproductive branches from being a drain on the tree and removes the wind drag (the force of air resistance) from useless branches.

Branches are shed for reasons other than lack of light. In dry parts of the world, it is common for trees and shrubs to lose smaller branches to save water. Small branches have the thinnest bark (the protective outer covering of a tree) and greatest surface area and thus are the source of most water loss once the leaves have been lost. The creosote bush of United States deserts self-prunes, or removes parts of itself, in the face of extreme heat or drought, starting from the highest and most exposed twigs and working downward to bigger and bigger branches, it's a desperate act because if the creosote bush loses too much food, it dies. Shedding branches can also be useful for self-propagation. Most poplar trees and willow trees characteristic of waterways will readily drop branches, which take root when washed up on muddy banks further downstream.

How are branches shed? In the simplest cases, dead branches rot and fall off, or healthy branches are snapped off by wind, snow, and animals. Some willows have a brittle zone at the base of small branches that encourages breaking in the wind, seemingly for propagation. Other cases of "natural pruning" are more startling: elm trees, and to a certain extent others, such as oaks, have a reputation for dropping large branches (up to half a meter in diameter) with no warning on calm, hot afternoons. Such dramatic shedding appears to be due to a combination of internal water stress coupled with heat expansion affecting cracks and decayed wood.

Many trees, however, shed branches deliberately. In this situation, branches are shed in the same way as foliage in autumn by the prior formation of a corky layer that leaves the wound sealed over with cork, which in turn is undergrown with wood the following year. In hardwoods, branches up to a meter in length and several centimeters in diameter can be shed normally after the leaves have fallen in the autumn (maples are unusual in casting branches mainly in spring and early summer). Oaks tend to shed small twigs up to the thickness of a pencil, beech may shed larger ones, and birches dump whole branches of dead twigs. Pine trees shed their clusters of needles (which really are short branches), and members of the redwood family shed their small branchlets with leaves. Typically in hardwood trees, something around 10 percent of terminal branches are lost each year through a mixture of deliberate shedding and being broken off.

Another way of reducing potential congestion is to make some branches smaller than others. Branches in the shade grow smaller than those in the sun. But trees can also regulate branch length from within. In many trees there is a clear distinction between long and short branches or shoots. The long shoots build the framework of the tree, making it bigger. The job of the short shoots (called spur shoots by horticulturalists) is to produce leaves, and commonly flowers, at more or less the same position every year. To maintain flexibility, any one shoot can switch from long to short or vice versa depending on internal factors, light levels, and damage.

Paragraph 1

One way trees prevent themselves from having too many branches is simply by shedding (dropping off) branches once they have fulfilled their purpose. This happens as the tree gets bigger and grows new outer layers of foliage that shade the inner and lower branches. In most large trees, the center of the canopy contains only large branches, small branches and fine twigs are found only at the canopy's edge. In the shaded center, the small branches that would once have occupied that space are long gone. Trees like the true cypresses regularly shed small twigs complete with leaves toward the end of summer. Most other trees shed only branches that prove unproductive. If a branch is not producing enough carbohydrate to cover its own running costs—i.e., it needs to be subsidized by other branches because, for example, it is being shaded and receives little light—it will usually be got rid of. This prevents unproductive branches from being a drain on the tree and removes the wind drag (the force of air resistance) from useless branches.

24. All of the following situations are mentioned in paragraph 1 for a tree to shed its branches EXCEPT

- ☐ endangering other branches
- ☐ building up on a tree
- ☐ wasting a tree's resources
- ☐ growing larger

25. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave

out essential information.

- ☐ A tree will usually shed branches that use more carbohydrate than they produce.
- ☐ Branches that are shaded usually do not receive enough light to produce all the carbohydrate they need.
- ☐ If a tree gets rid of a branch, it is usually because other branches lack enough carbohydrate to subsidize it.
- ☐ If a branch is shaded and cannot produce as much carbohydrate as it needs, it will usually be subsidized by other branches.

26. According to paragraph 1 most twigs of the true cypresses would be arranged in which of the following manner

- ☐ Most small twigs on true cypresses have been found located at the margins of the canopy.
- ☐ The shaded center areas of true cypresses are generally occupied by many small twigs and branches.
- ☐ True cypresses shed twigs that grow on large, unproductive branches.
- ☐ True cypresses seasonally shed small twigs without regard to whether they are productive or not.

Paragraph 2

Branches are shed for reasons other than lack of light. In dry parts of the world, it is common for trees and shrubs to lose smaller branches to save water. Small branches have the thinnest bark (the protective outer covering of a tree) and greatest surface area and thus are the source of most water loss once the leaves have been lost. The creosote bush of United States deserts self-prunes, or removes parts of itself, in the face of extreme heat or drought, starting from the highest and most exposed twigs and working downward to bigger and bigger branches; it's a desperate act because if the creosote bush loses too much wood, it dies. Shedding branches can also be useful for self-propagation. Most poplar trees and willow trees characteristic of waterways will readily drop branches, which take root when washed up on muddy banks further downstream.

27. Which of the following best describes the role of the explanation offered in paragraph 2?

- ☐ Paragraph 2 questions this explanation by providing counterexamples of some trees.
- ☐ Paragraph 2 presents additional evidence supporting this explanation.
- ☐ Paragraph 2 discusses some additional reasons why trees shed branches
- ☐ Paragraph 2 points out some additional consequences for trees besides the shedding of branches

28. The word "exposed" in the passage is closest in meaning to

- ☐ distant
- ☐ unprotected
- ☐ easily replaced
- ☐ unproductive

29. According to paragraph 2, what is true of the creosote bush of United States deserts?

- It tends to grow small branches during dry parts of the year.
- It loses more water through its bark than through its leaves.
- It loses its lower branches only after losing upper ones.
- It sheds branches for the purpose of propagating itself.

Paragraph 3

How are branches shed? In the simplest cases, dead branches rot and fall off, or healthy branches are snapped off by wind, snow, and animals. Some willows have a brittle zone at the base of small branches that encourages breaking in the wind, seemingly for propagation. Other cases of “natural pruning” are more startling: elm trees, and to a certain extent others, such as oaks, have a reputation for dropping large branches (up to half a meter in diameter) **with no warning** on calm, hot afternoons. Such dramatic shedding appears to be due to a combination of internal water stress coupled with heat expansion affecting cracks and decayed wood.

30. The phrase “**with no warning**” in the passage is closest in meaning to
- without any benefit
 - without any stress
 - without any indication beforehand
 - without any damage
31. All of the following are mentioned in paragraph 3 as a way in which branches can be lost EXCEPT:
- being broken off by the wind
 - being shed for propagation
 - becoming rotten
 - becoming too large in diameter
32. Which of the following is mentioned in paragraph 3 in the shedding of large branches by oaks on calm, hot afternoons?
- The development of a brittle zone at the base of the branches
 - The enlargement of cracks in the branches due to heat
 - The rise of sudden bursts of wind that snap off decayed wood
 - The seasonal need to propagate new trees

Paragraph 4

Many trees, however, shed branches deliberately. In this situation, branches are shed in the same way as foliage in autumn by the prior formation of a corky layer that leaves the wound sealed over with cork, which in turn is undergrown with wood the following year. In hardwoods, branches up to a meter in length and several centimeters in diameter can be shed normally after the leaves have fallen in the autumn (maples are unusual in casting branches mainly in spring and early summer). Oaks tend to shed small twigs up to the thickness of a pencil, beech may shed larger ones, and birches dump whole branches of dead twigs. Pine trees shed their clusters of needles (which really are short branches), and members of the redwood family shed their small branchlets with leaves. Typically in hardwood trees, something around 10 percent of terminal

branches are lost each year through a mixture of deliberate shedding and being broken off.

33. According to paragraph 4, what information can be learned from the deliberate shedding of branches by the trees?

- ☐ Limiting the size of branches being shed to comparatively small ones
- ☐ Forming a new layer of wood to seal the wounded area immediately after shedding
- ☐ Shedding leaves at the same time that branches are being shed
- ☐ Forming a layer of protective tissue before branch shedding begins

Paragraph 5

Another way of reducing potential **congestion** is to make some branches smaller than others. Branches in the shade grow smaller than those in the sun. But trees can also regulate branch length from within. In many trees there is a clear distinction between long and short branches or shoots. The long shoots build the framework of the tree, making it bigger. The job of the short shoots (called spur shoots by horticulturalists) is to produce leaves, and commonly flowers, at more or less the same position every year. To maintain flexibility, any one shoot can switch from long to short or vice versa depending on internal factors, light levels, and damage.

34. The word “**congestion**” in the passage is closest in meaning to

- ☐ loss
- ☐ damage
- ☐ overcrowding
- ☐ stress

35. According to paragraph 5, what is the main purpose of the long branches or shoots?

- ☐ To regulate the length of large branches
- ☐ To increase the size of the tree
- ☐ To produce leaves and flowers
- ☐ To help create shaded areas

Paragraphs 1-2

One way trees prevent themselves from having too many branches is simply by shedding (dropping off) branches once they have fulfilled their purpose. This happens as the tree gets bigger and grows new outer layers of foliage that shade the inner and lower branches. In most large trees, the center of the canopy contains only large branches, small branches and fine twigs are found only at the canopy’s edge. In the shaded center, the small branches that would once have occupied that space are long gone. Trees like the true cypresses regularly shed small twigs complete with leaves toward the end of summer. Most other trees shed only branches that prove unproductive. If a branch is not producing enough carbohydrate to cover its own running costs—i.e., it needs to be subsidized by other branches because, for example, it is being shaded and receives little light—it will usually be got rid of. This prevents unproductive branches from being a drain on the tree and removes the wind drag (the force of air resistance) from useless branches. ■

Branches are shed for reasons other than lack of light. ■ In dry parts of the world, it is common for trees and shrubs to lose smaller branches to save water. ■ Small branches have the thinnest

bark (the protective outer covering of a tree) and greatest surface area and thus are the source of most water loss once the leaves have been lost. ■ The creosote bush of United States deserts self-prunes, or removes parts of itself, in the face of extreme heat or drought, starting from the highest and most exposed twigs and working downward to bigger and bigger branches, it's a desperate act because if the creosote bush loses too much food, it dies. Shedding branches can also be useful for self-propagation. Most poplar trees and willow trees characteristic of waterways will readily drop branches, which take root when washed up on muddy banks further downstream.

36. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

A tree will also shed branches if its water supply is insufficient.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

37. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

For trees to remain healthy as they grow and as circumstances change, tree branches must change in various ways.

-
-
-

Answer Choices

Trees can benefit from shedding inefficient branches that consume more carbohydrates than they produce or that are a major source of water loss.

Branches can be lost as a result of damage from whether, animals, or disease, but they can also be shed when they are no longer useful, much as leaves are shed in autumn.

While branch shedding is common and may be necessary for a tree's survival, the corky layer that forms at sites where branches have been shed prevents additional growth in those areas for several years.

Shedding large branches is such a desperate act for survival that the creosote bush is one of the few plants to use this mechanism for removing wood.

Larger trees can self-propagate when water stress and heat expansion break off branches, as long as the shed branches fall on or can be transported to a location favorable for taking root.

Trees prevent branch overcrowding in part by varying branch length through internal mechanisms such as having shoots that can switch from long to short or from short to long, as needed.

Passage 75 2015-01-31-1

Effects of the Commercial Revolution

In the third and the second millennia B.C. long-distance trade supposedly had the character of an expedition. By the start of the last millennium B.C., however, a new approach to engaging in such trade emerged. Based on the principle of colonization, it was pioneered by the Phoenicians and Greeks, who established colonies along the Mediterranean Sea. The new approach to long-distance trade, known as the commercial revolution, led to changes in a number of political and economic patterns.

For the first time, the planting of colonies in distant lands became possible. The Phoenician settlements in the central and western Mediterranean, such as Carthage, and the slightly later establishment of Greek colonies are early examples, while the settlement of south Arabians in Eritrea around the middle of the last millennium marks the subsequent spread of this sort of commercial consequence to the Horn of Africa. In the third or second millennia B. C., a state such as Egypt might colonize areas outside its heartland, such as Nubia. But this colonization comprised military outposts and ethnic settlements that were planted to hold the contiguous territories of a land empire, not distant localities far separated from the home country.

The commercial revolution constructed the economic basis as well for a new kind of town or city, an urban center that above all serviced trade and was home to the crafts and occupational specializations that went along with commercial development. The urban locations of earlier times commonly drew trade simply because their populations had included a privileged elite of potential consumers. Such towns had arisen in the first place as political and religious centers of the society, they attracted population because power and influence resides there and access to position and wealth could be gained through service to the royal or priestly leadership.

Wherever the effects of the commercial revolution penetrated over the last millennium B. C., kings and emperors increasingly lost their ability to treat trade as a royalty sponsored activity, intended to preserve the commodities of trade as the privileges of immemorial power and position. Instead, their policies shifted toward controlling geographical accessibility to the products of commerce and to ensuring security and other conditions that attracted and enhanced the movement of goods. No longer could kings rely on agriculturally supported and religiously based claims to an ability to protect their lands and people; now they also had to overtly support the material prosperity of their people compared to other societies. And rather than exerting a monopoly over prestige commodities, as had Egyptian kings of the third and second millennia, and redistributing such commodities in ways designed to reinforce the allegiance of their subjects and enhance the awesomeness of their position, rulers turned to the taxation of trade and to the creation and control of currency, more and more relying on duties and other revenues to support the apparatus of the state. It was no historical accident that the first metal coinage in the world began to be made in eighth-century Anatolia (modern Turkey) and that the use of coins rapidly spread with the expanding commercial revolution. The material

bases and the legitimizations of state authority as we know them today had begun to take shape.

The commercial revolution tended also to spread a particular pattern of exchange. The early commercial centers of the Mediterranean most characteristically offered manufactured goods—purple dye, metal goods, wine, olive oil, and so forth—for the raw materials or the partially processed natural products of other regions. As the commercial revolution spread, this kind of exchange tended to spread with it, with the recently added areas of commerce providing new kinds of raw materials or new sources for familiar products of the natural world, and the longer established commercial centers—which might themselves have lain at the margins of this transformation—producing, or acting as the intermediaries—producing, or acting as the intermediaries in the transmission of, manufactured commodities. India, for instance, had developed by the turn of the ear into a major exporter of its own cotton textiles, as well as naturally occurring materials, such as gems of various kinds, and at the same time its merchants were the intermediaries of the silk trade.

Paragraph 1

In the third and the second millennia B.C. long-distance trade supposedly had the character of an expedition. By the start of the last millennium B.C., however, a new approach to engaging in such trade emerged. Based on the principle of colonization, it was pioneered by the Phoenicians and Greeks, who established colonies along the Mediterranean Sea. The new approach to long-distance trade, known as the commercial revolution, led to changes in a number of political and economic patterns.

1. The word “pioneered” in the passage is closest in meaning to
 - ☐ adopted
 - ☐ described
 - ☐ demonstrated
 - ☐ introduced

Paragraph 2

For the first time, the planting of colonies in distant lands became possible. The Phoenician settlements in the central and western Mediterranean, such as Carthage, and the slightly later establishment of Greek colonies are early examples, while the settlement of south Arabians in Eritrea around the middle of the last millennium marks the subsequent spread of this sort of commercial consequence to the Horn of Africa. In the third or second millennia B. C., a state such as Egypt might colonize areas outside its heartland, such as Nubia. But this colonization comprised military outposts and ethnic settlements that were planted to hold the contiguous territories of a land empire, not distant localities far separated from the home country.

2. All of the following groups are mentioned in paragraph 2 as establishing distant trading outposts in the last millennium B. C. EXCEPT
 - ☐ the Greeks
 - ☐ the Egyptians

- ☐ the Phoenicians
 - ☐ the south Arabians
3. The word “subsequent” in the passage is closest in meaning to
- ☐ initial
 - ☐ anticipated
 - ☐ later
 - ☐ increasing
4. In paragraph 2, why does the author mention the colonization of Nubia by the Egyptians?
- ☐ To prove that colonization was first carried out by the military
 - ☐ To indicate that Egypt was a major military power in the third and second millennia B. C.
 - ☐ To illustrate how large the geographic area of colonization had become over several millennia
 - ☐ To show that the purpose of colonization during the third and second millennia B. C. differed from that of the last millennium B. C.
5. The word “comprised” in the passage is closest in meaning to
- ☐ resulted in
 - ☐ focused on
 - ☐ was inspired by
 - ☐ consisted of

Paragraph 3

The commercial revolution constructed the economic basis as well for a new kind of town or city, an urban center that above all serviced trade and was home to the crafts and occupational specializations that went along with commercial development. The urban locations of earlier times commonly drew trade simply because their populations had included a privileged elite of potential consumers. Such towns had arisen in the first place as political and religious centers of the society, they attracted population because power and influence resides there and access to position and wealth could be gained through service to the royal or priestly leadership.

6. According to paragraph 3, before the emergence of the commercial revolution, trade
- ☐ enabled craftspeople and occupational specialists to gain power and influence in society
 - ☐ centered on the ruling elite and those groups closely associated with them
 - ☐ was primarily conducted by people serving the royal and religious leadership
 - ☐ was a major reason why urban centers were established

Paragraph 4

Wherever the effects of the commercial revolution penetrated over the last millennium B. C., kings and emperors increasingly lost their ability to treat trade as a royalty sponsored activity, intended to preserve the commodities of trade as the privileges of immemorial power and position. Instead, their policies shifted toward controlling geographical accessibility to the products of commerce and to ensuring security and other conditions that attracted and

enhanced the movement of goods. No longer could kings rely on agriculturally supported and religiously based claims to an ability to protect their lands and people; now they also had to overtly support the material prosperity of their people compared to other societies. And rather than exerting a monopoly over prestige commodities, as had Egyptian kings of the third and second millennia, and redistributing such commodities in ways designed to reinforce the allegiance of their subjects and enhance the awesomeness of their position, rulers turned to the taxation of trade and to the creation and control of currency, more and more relying on duties and other revenues to support the apparatus of the state. It was no historical accident that the first metal coinage in the world began to be made in eighth-century Anatolia (modern Turkey) and that the use of coins rapidly spread with the expanding commercial revolution. The material bases and the legitimizations of state authority as we know them today had begun to take shape.

7. The word “enhanced” in the passage is closest in meaning to
 - ☐ improved
 - ☐ influenced
 - ☐ protected
 - ☐ necessitated
8. The word “reinforce” in the passage is closest in meaning to
 - ☐ demand
 - ☐ strengthen
 - ☐ earn
 - ☐ repay
9. According to paragraph 4, as the commercial revolution expanded, rulers focused on
 - ☐ taxation and the development and control of money
 - ☐ monopolizing prestige commodities
 - ☐ distributing prestige commodities to ensure the allegiance of their subjects
 - ☐ protecting their land to legitimize their authority
10. What can be inferred from paragraph 4 about Anatolia?
 - ☐ Its merchants specialized in the trading of prestige commodities.
 - ☐ It was the first place to use currency for the taxation of trade.
 - ☐ It contained enormous supplies of metal compared with other states in the region.
 - ☐ Trade remained a royally sponsored activity there long after the emergence of the commercial revolution.

Paragraph 5

The commercial revolution tended also to spread a particular pattern of exchange. The early commercial centers of the Mediterranean most characteristically offered manufactured goods—purple dye, metal goods, wine, olive oil, and so forth—for the raw materials or the partially processed natural products of other regions. As the commercial revolution spread, this kind of exchange tended to spread with it, with the recently added areas of commerce providing new kinds of raw materials or new sources for familiar products of the natural world, and the

longer established commercial centers—which might themselves have lain at the margins of this transformation—producing, or acting as the intermediaries in the transmission of, manufactured commodities. India, for instance, had developed by the turn of the era into a major exporter of its own cotton textiles, as well as naturally occurring materials, such as gems of various kinds, and at the same time its merchants were the intermediaries of the silk trade.

11. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ During the commercial revolution, newer centers of trade acted as intermediaries in the exchange of different types of manufactured goods.
- ☐ Longer-established trading centers were familiar with the unprocessed products of the natural world, but depended on other areas as sources for manufactured commodities.
- ☐ Eventually, the commercial revolution led to a trading system whereby newly established commercial centers provided the resources needed for the production of goods while older trading centers produced the goods or assisted in their distribution.
- ☐ The commercial revolution depended on a system of trade where consumers valued novelty in the manufactured goods they acquired, but, at the same time, they wanted to be familiar with the natural products they received.

12. Paragraph 5 supports which of the following statements about Indian merchants at the time of the commercial revolution?

- ☐ They imported cotton, silk, and other high-quality fabrics intended for the Indian market.
- ☐ They obtained various kinds of gems from intermediaries in the silk trade.
- ☐ They were simultaneously exporters of manufactured and natural products and intermediaries for goods produced elsewhere.
- ☐ They created a highly sophisticated textile industry at the same time that they were engaged in the processing of natural products.

Paragraph 3

■ The commercial revolution constructed the economic basis as well for a new kind of town or city, an urban center that above all serviced trade and was home to the crafts and occupational specializations that went along with commercial development. ■ The urban locations of earlier times commonly drew trade simply because their populations had included a privileged elite of potential consumers. ■ Such towns had arisen in the first place as political and religious centers of the society, they attracted population because power and influence resides there and access to position and wealth could be gained through service to the royal or priestly leadership. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

It was significantly different from the typical centers that existed before the commercial revolution.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The commercial revolution of the last millennium B. C. resulted in both political and economic changes.

-
-
-

Answer Choices

New kinds of urban centers emerged that focused on commerce and encouraged craft and occupational specializations.

Rulers in the last millennium began to promote the material prosperity of their people through support and improvement of commerce.

More established commercial centers supplied final products to newer regions in exchange for raw materials.

During the first millennium B. C., new political and religious centers arose that based their power on their ability to protect their lands and people.

The focus on raw materials switched the balance of power from the manufacturing centers to the control of the exporters of the natural products.

Military occupation of neighboring lands became a major means of expanding trade into new territories.

European Context of the Scientific Revolution

The Scientific Revolution represents a turning point in world history. By 1700 European scientists had overthrown the science and worldviews of the ancient philosophers: Aristotle and Ptolemy. Europeans in 1700 lived in a vastly different intellectual world than that experienced by their predecessors in, say, 1500. The role and power of science, as a way of knowing about the world and as an agency with the potential of changing the world, likewise underwent profound restructuring as part of the Scientific Revolution.

The social context for science in Europe in the sixteenth and seventeenth centuries had changed in several dramatic ways from the Middle Ages (roughly, 500 C.E. to the 1400s C.E.). Advances in military technology, the European voyages of exploration, and contact with the New World altered the context in which the Scientific Revolution unfolded. The geographical discovery of the Americas generally undermined the closed Eurocentric cosmos of the later Middle Ages, and the science of geography provided a stimulus of its own to the Scientific Revolution. With an emphasis on observational reports and practical experience, new geographical discoveries challenged accepted knowledge. Cartography (mapmaking) thus provided exemplary new ways of learning about the world in general, ways self-evidently superior to mastering established doctrines from dusty books. Many of the scientists of the Scientific Revolution seem to have been involved in one fashion or another with geography or cartography.

In the late 1430s, Johannes Gutenberg, apparently independently of the development of woodblock printing in Asia, invented printing with movable type, and the spread of this powerful new technology after 1450 likewise altered the cultural landscape of early modern Europe. The new medium created a revolution in communications that increased the amount and accuracy of information available and made copying of books by scribes obsolete. Producing some 13,000 works by 1500, printing presses spread rapidly throughout Europe and helped to break down the monopoly of learning in universities and to create a new group of nonreligious intellectuals. Indeed, the first printshops became something of intellectual centers themselves, with authors, publishers, and workers collaborating in unprecedented ways in the production of new knowledge. Renaissance humanism, that renowned philosophical and literary movement emphasizing human values and the direct study of classical Greek and Latin texts, is hardly conceivable without the technology of printing that sustained the efforts of learned humanists. Regarding science, the advent of printing and humanist scholarship brought another wave in the recovery of ancient texts. Whereas Europeans first learned of ancient Greek science largely through translations from the Arabic in the twelfth century, in the later fifteenth century scholars brought forth new editions from Greek originals and uncovered influential new sources, notably the Greek mathematician Archimedes. Similarly, printing disseminated previously obscure handbooks of technical and magical secrets that proved influential in the developing Scientific Revolution.

Particularly in Italy, the revival of cultural life and the arts in the late fourteenth and fifteenth centuries commonly known as the Renaissance must also be considered as an urban and comparatively secular phenomenon, aligned with courts and courtly patronage but not with the universities, which were religiously base. One associates the great flourish of artistic activity of the Renaissance with such talents as Donatello, Leonardo da Vinci, Raphael, and Michelangelo. In comparison with medieval art, the use of perspective—a projection system that realistically renders the three dimensions of space onto the two dimensions of a canvas—represents a new feature typical of Renaissance painting, and through the work of Leon Battista Alberti, Albrecht Durer, and others, artists learned to practice mathematical rules governing perspective. So noteworthy was this development that historians have been inclined to place Renaissance artists at the forefront of those uncovering new knowledge about nature in the fifteenth and sixteenth centuries. Whatever one may make of that claim, early modern artists needed accurate knowledge of human muscular anatomy for lifelike renditions, and an explosion of anatomical research in the Renaissance may be attributed to this need in the artistic community.

Paragraph 1

The Scientific Revolution represents a turning point in world history. By 1700 European scientists had overthrown the science and worldviews of the ancient philosophers: Aristotle and Ptolemy. Europeans in 1700 lived in a vastly different intellectual world than that experienced by their predecessors in, say, 1500. The role and power of science, as a way of knowing about the world and as an agency with the potential of changing the world, likewise underwent profound restricting as part of the Scientific Revolution.

15. The word “profound” in the passage is closet in meaning to

- ☐ frequent
- ☐ intense
- ☐ challenging
- ☐ careful

16. According to paragraph 1, what was new about the intellectual world of 1700?

- ☐ Scientists were aware that they were participating in a turning point in world history.
- ☐ Beliefs about nature developed by ancient philosophers were no longer accepted.
- ☐ People believed that science had changed the world.
- ☐ The impact of the Scientific Revolution was being felt in all aspects of European life.

Paragraph 2

The social context for science in Europe in the sixteenth and seventeenth centuries had changed in several dramatic ways from the Middle Ages (roughly, 500 C.E. to the 1400s C.E.). Advances in military technology, the European voyages of exploration, and contact with the New World altered the context in which the Scientific Revolution unfolded. The geographical discovery of the Americas generally undermined the closed Eurocentric cosmos of the later Middle Ages, and the science of geography provided a stimulus of its own to the Scientific Revolution. With an emphasis on observational reports and practical experience, new geographical discoveries

challenged accepted knowledge. Cartography (mapmaking) thus provided exemplary new ways of learning about the world in general, ways self-evidently superior to mastering established doctrines from dusty books. Many of the scientists of the Scientific Revolution seem to have been involved in one fashion or another with geography or cartography.

17. According to paragraph 2, all of the following influenced European scientific thought during the sixteenth and seventeenth centuries EXCEPT

- ☐ progress in military technology
- ☐ explorative journeys made by Europeans
- ☐ views expressed in the scholarship of the Middle Ages
- ☐ the development of cartography

18. According to paragraph 2, how did the study of geography influence the Scientific Revolution?

- ☐ It supported established doctrines in the European-centered world.
- ☐ It created new ways of learning through recording observations and practical experiences.
- ☐ It contributed to advances in military technology.
- ☐ It allowed scientists from different regions to exchange information.

Paragraph 3

In the late 1430s, Johannes Gutenberg, apparently independently of the development of woodblock printing in Asia, invented printing with movable type, and the spread of this powerful new technology after 1450 likewise altered the cultural landscape of early modern Europe. The new medium created a revolution in communications that increased the amount and accuracy of information available and made copying of books by scribes **obsolete**. Producing some 13,000 works by 1500, printing presses spread rapidly throughout Europe and helped to break down the monopoly of learning in universities and to create a new group of nonreligious intellectuals. Indeed, the first printshops became something of intellectual centers themselves, with authors, publishers, and workers collaborating in unprecedented ways in the production of new knowledge. **Renaissance humanism**, that renowned philosophical and literary movement emphasizing human values and the direct study of classical Greek and Latin texts, is hardly conceivable without the technology of printing that sustained the efforts of learned humanists. Regarding science, the advent of printing and humanist scholarship brought another wave in the recovery of ancient texts. Whereas Europeans first learned of ancient Greek science largely through translations from the Arabic in the twelfth century, in the later fifteenth century scholars brought forth new editions from Greek originals and uncovered influential new sources, notably the Greek mathematician Archimedes. Similarly, printing disseminated previously **obscure** handbooks of technical and magical secrets that proved influential in the developing Scientific Revolution.

19. The word "**obsolete**" in the passage is closest in meaning to

- ☐ costly
- ☐ less frequent
- ☐ unappealing

☐ out of date

20. The word “obscure” in the passage is closest in meaning to

- ☐ little known
- ☐ expensive
- ☐ forbidden
- ☐ celebrated

21. Paragraph 3 suggests that before 1500 the transmission of knowledge in Europe was

- ☐ stimulated by printing developments in Asia
- ☐ dependent on collaborations between scribes and publishers
- ☐ limited to religious intellectuals in academic settings
- ☐ influenced by philosophical rather than literary sources

22. The author discusses “Renaissance humanism” in order to

- ☐ demonstrate that printing presses facilitated the spread of humanistic thought
- ☐ discuss why print shops declined as intellectual centers
- ☐ compare the beliefs of classical humanists to the Renaissance humanists
- ☐ emphasize the importance of the direct study of Greek and Latin texts

23. According to paragraph 3, what effect did the invention of printing have on science in Europe?

- ☐ Scientists were able to publish books for humanists and other non-scientific intellectuals.
- ☐ Europeans gained access to new editions of texts as well as new sources of knowledge.
- ☐ Translations of Arabic texts documenting scientific discoveries became widely available.
- ☐ Humanistic study declined as a result of the advance of scientific study.

Paragraph 4

Particularly in Italy, the revival of cultural life and the arts in the late fourteenth and fifteenth centuries commonly known as the Renaissance must also be considered as an urban and comparatively secular phenomenon, aligned with courts and courtly patronage but not with the universities, which were religiously base. One associates the great flourish of artistic activity of the Renaissance with such talents as Donatello, Leonardo da Vinci, Raphael, and Michelangelo. In comparison with medieval art, the use of perspective—a projection system that realistically renders the three dimensions of space onto the two dimensions of a canvas—represents a new feature typical of Renaissance painting, and through the work of Leon Battista Alberti, Albrecht Durer, and others, artists learned to practice mathematical rules governing perspective. So noteworthy was this development that historians have been inclined to place Renaissance artists at the forefront of those uncovering new knowledge about nature in the fifteenth and sixteenth centuries. Whatever one may make of that claim, early modern artists needed accurate knowledge of human muscular anatomy for lifelike renditions, and an explosion of anatomical research in the Renaissance may be attributed to this need in the artistic community.

24. The word “associates” in the passage is closest in meaning to

- ☐ compares
- ☐ appreciates
- ☐ connects
- ☐ presents

25. According to paragraph 4, Renaissance artistic contributed to the Scientific Revolution by
- ☐ reviving medieval mathematical and scientific sources for study
 - ☐ establishing institutions for the study of mathematics and scientific principles in art
 - ☐ creating paintings that contributed to the wealth of the courts and courtly patronage of science
 - ☐ using mathematical information to realistically represent space in art
26. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Early modern artists claim to have uncovered new knowledge about nature and human muscular anatomy before the explosion of anatomical research.
 - ☐ Artists' need for accurate knowledge in order to realistically represent the human body may have caused the sudden increase in anatomical studies in the Renaissance.
 - ☐ Whatever other claims are made about early modern art, it is accurate to state that Renaissance artists were concerned with creating lifelike representations.
 - ☐ The need for early modern artists to create lifelike renditions developed after the explosion of anatomical research made human anatomy clear.

Paragraph 2

■ The social context for science in Europe in the sixteenth and seventeenth centuries had changed in several dramatic ways from the Middle Ages (roughly, 500 C.E. to the 1400s C.E.). ■ Advances in military technology, the European voyages of exploration, and contact with the New World altered the context in which the Scientific Revolution unfolded. The geographical discovery of the Americas generally undermined the closed Eurocentric cosmos of the later Middle Ages, and the science of geography provided a stimulus of its own to the Scientific Revolution. ■ With an emphasis on observational reports and practical experience, new geographical discoveries challenged accepted knowledge. Cartography (mapmaking) thus provided exemplary new ways of learning about the world in general, ways self-evidently superior to mastering established doctrines from dusty books. ■ Many of the scientists of the Scientific Revolution seem to have been involved in one fashion or another with geography or cartography.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Given the advantages these new approaches offered, it is hardly surprising that sciences associated with exploration attracted great intellectual interest.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

European society experienced a /// period known as the Renaissance.

-
-
-

Answer Choices

The discovery of the Americas stimulated the science of geography and cartography which in turn emphasized observation and practical experience.

The invention of movable type increased the accuracy and availability of information, and a new group of nonreligious scholars emerged.

The growth of both artistic and scientific activity in Renaissance Italy was encouraged by the financial support of universities and wealthy merchants.

Johannes Gutenberg's invention of printing in the late 1430s appears to have been unrelated to earlier developments in print technology in Asia.

European intellectuals first discovered the existence of ancient Greek science and mathematic texts in the late fifteenth century, when translations from the Arabic finally became available.

The revival of culture and art in fourteenth-century Italy encouraged the development of perspective and the adherence to mathematical rules in painting.

Greek Sacred Groves and Parks

In Greek and Roman civilization, parks were associated with spirituality, public recreation, and city living. Greek philosophers pondered the meaning of nature and its innermost workings, the relationships between animals and humankind, and how matter related to spirit. The philosophy of Aristotle (384-322 B.C.) advanced the fundamental notion of nature as the embodiment of everything outside culture, an essence opposed to art and artificiality. This sense of nature and culture as distinct opposites continues to govern ideas about the environment and society today. Meanwhile, the suggestion of a state of nature, wholesome and pure, defined in opposition to civilized life, found acceptance in Aristotle's time through the concept of the Golden Age—a legendary ideal that had significance for landscape planning and artistic experiment. Described by Greek poets and playwrights, the Golden Age of perpetual spring depicted an era before the adoption of agriculture, when humans embraced nature's wonder and communicated with spirits in sacred woods. In *The Odyssey* (800 B.C.), Homer, the great Greek writer, described a garden that was a place of constant productivity, where "fruit never fails nor runs short, winter and summer alike."

Greek interest in spirituality and nature manifested itself in the tradition of the sacred grove. Usually comprised of a few trees, a spring, or a mountain crag, sacred groves became intensely mystical places by their associations with gods, spirits, or celebrated folk heroes. Twisted trees, sections of old-growth forest, and rocks or caves typically surrounded the naturalistic shrines and altars. As the Roman official and writer Pliny the Elder (A.D. 23-79) put it, "Trees were the first temples of the gods, and even now simple country people dedicate a tree of exceptional height to a god with the ritual of olden times, and we worship forests and the very silences they contain."

The Greeks were not alone in their spiritual veneration of nature. Examples of pantheism—the belief that God and the universe or nature are the same—and the worship of trees permeated many cultures. The nations of northern Europe utilized trees as places of worship. In Scandinavian mythology, the tree called Yggdrasil held up the world, its branches forming the heavens and its roots stretching into the underworld. A spring of knowledge bubbled at its base, and an eagle perched amid its sturdy branches. The Maori people of New Zealand celebrated a tree that separated the sky from the earth. For many ancient civilizations, trees signified life, permanence, and wisdom.

In some spiritual traditions, landscapes such as gardens or deserts were treated as abstract emblems of spiritual states such as innocence or despair. Rather than symbolic landscapes, as in the Judeo-Christian tradition, Greek sacred groves operated as literal homes of the gods. Instead of being confined to prehistory or celestial space, spiritual parkscapes were present within the existing cultural terrain. One could not visit a symbol of peace and severity, but one could experience these qualities in a sacred grove.

The spiritual significance of the sacred grove mandated specific preservationist measures. Civil restrictions and environmental codes of practice governed the use of such spaces. Enclosing walls prevented sheep from desecrating sacred sites, while patrolling priests issued spiritual guidance along with fines for vandalism. Laws forbade hunting, fishing, or the cutting of trees. Those not dissuaded by monetary penalties were threatened with the anger of the resident gods.

Such environmental care suggested to historian J. Donald Hughes that sacred groves represented “classical national parks.” By helping to insulate sacred groves from pressures of deforestation, erosion, and urbanization, Greek codes protected ecosystems from destruction. Sacred groves nonetheless represented imperfect parkscapes. Some encompassed relatively small areas such as a section of a hillside or a series of caves. Meanwhile, the fundamental purpose of the grove—the visitation of resident gods—sometimes promoted activities not entirely conducive to modern concepts of conservation. Animals were routinely captured to serve as sacrifices to the gods. Many groves witnessed horticultural and architectural improvements. Flowers were planted, trails cut, and statues, fountains, and caves installed for the benefit of visitors. The grove served as a recreational center for Greek society, a realm of ritual, performance, feasting, and even chariot racing.

Paragraph 1

In Greek and Roman civilization, parks were associated with spirituality, public recreation, and city living. Greek philosophers pondered the meaning of nature and its innermost workings, the relationships between animals and humankind, and how matter related to spirit. The philosophy of Aristotle (384-322 B.C.) advanced the fundamental notion of nature as the embodiment of everything outside culture, an essence opposed to art and artificiality. This sense of nature and culture as distinct opposites continues to govern ideas about the environment and society today. Meanwhile, the suggestion of a state of nature, wholesome and pure, defined in opposition to civilized life, found acceptance in Aristotle’s time through the concept of the Golden Age—a legendary ideal that had significance for landscape planning and artistic experiment. Described by Greek poets and playwrights, the Golden Age of perpetual spring depicted an era before the adoption of agriculture, when humans embraced nature’s wonder and communicated with spirits in sacred woods. In *The Odyssey* (800 B.C.), Homer, the great Greek writer, described a garden that was a place of constant productivity, where “fruit never fails nor runs short, winter and summer alike.”

1. The word “pondered” in the passage is closest in meaning to
 - ☐ explained
 - ☐ argued over
 - ☐ thought about
 - ☐ understood
2. The word “fundamental” in the passage is closest in meaning to
 - ☐ famous
 - ☐ basic

- ☐ revolutionary
 - ☐ original
3. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ Aristotle used the concept of the Golden Age to explain how the pure beauty of nature could be recreated in a new, carefully planned state.
 - ☐ During the Golden Age of Aristotle's time, human activities such as landscape planning and art reached the height of creativity.
 - ☐ In ancient Greek thought, both art and nature were characterized by purity and wholesomeness within a Golden Age.
 - ☐ During Aristotle's time, the idea that nature is pure and distinct from civilization was expressed in the idea of the perfect Golden Age, which influenced both art and landscape design.
4. In paragraph 1, why does the author include the quotation from *The Odyssey*?
- ☐ To support the idea that modern ideas about nature have not changed much since the ancient Greeks
 - ☐ To contrast Homer's ideas about nature with those of Aristotle
 - ☐ To argue that the adoption of agriculture advanced Greek culture
 - ☐ To give an example of an ancient Greek description of the Golden Age

Paragraph 2

Greek interest in spirituality and nature manifested itself in the tradition of the sacred grove. Usually comprised of a few trees, a spring, or a mountain crag, sacred groves became intensely mystical places by their associations with gods, spirits, or celebrated folk heroes. Twisted trees, sections of old-growth forest, and rocks or caves typically surrounded the naturalistic shrines and altars. As the Roman official and writer Pliny the Elder (A.D. 23-79) put it, "Trees were the first temples of the gods, and even now simple country people dedicate a tree of exceptional height to a god with the ritual of olden times, and we worship forests and the very silences they contain."

5. In paragraph 2, the author quotes Pliny the Elder in order to support the claim that
- ☐ only simple country people believed in gods who lived in forests
 - ☐ Greek beliefs about forests differed from Roman beliefs about forests
 - ☐ sacred groves had mystical meanings because of their association with gods
 - ☐ Greeks were more interested in nature than in spirituality

Paragraph 3

The Greeks were not alone in their spiritual veneration of nature. Examples of pantheism—the belief that God and the universe or nature are the same—and the worship of trees permeated many cultures. The nations of northern Europe utilized trees as places of worship. In Scandinavian mythology, the tree called Yggdrasil held up the world, its branches forming the

heavens and its roots stretching into the underworld. A spring of knowledge bubbled at its base, and an eagle perched amid its sturdy branches. The Maori people of New Zealand celebrated a tree that separated the sky from the earth. For many ancient civilizations, trees signified life, permanence, and wisdom.

6. What is the purpose of paragraph 3 in the larger discussion of ancient Greek beliefs?
- ☐ To connect the Greek view of nature to the associations between nature and religion that exist in many different cultures in the world
 - ☐ To contrast the history and development of Greek religion to the development of other religions of the time
 - ☐ To demonstrate the influences of Greek beliefs on other religions
 - ☐ To argue that ancient religions eventually rejected the spirituality of trees

Paragraph 4

In some spiritual traditions, landscapes such as gardens or deserts were treated as abstract emblems of spiritual states such as innocence or despair. Rather than symbolic landscapes, as in the Judeo-Christian tradition, Greek sacred groves operated as literal homes of the gods. Instead of being confined to prehistory or celestial space, spiritual parkscapes were present within the existing cultural terrain. One could not visit a symbol of peace and severity, but one could experience these qualities in a sacred grove.

7. It can be inferred from paragraph 4 that the ancient Greeks believed that their gods
- ☐ resided only in celestial space
 - ☐ actually lived on Earth
 - ☐ did not exist in prehistoric times
 - ☐ were only representations of ideas

Paragraph 5

The spiritual significance of the sacred grove **mandated** specific preservationist measures. Civil restrictions and environmental codes of practice governed the use of such spaces. Enclosing walls prevented sheep from desecrating sacred sites, while patrolling priests issued spiritual guidance along with fines for vandalism. Laws forbade hunting, fishing, or the cutting of trees. Those not dissuaded by monetary penalties were threatened with the anger of the resident gods.

8. The word “**mandated**” in the passage is closest in meaning to
- ☐ contributed to
 - ☐ produced
 - ☐ limited
 - ☐ required
9. All of the following are mentioned in paragraph 5 as ways the Greeks protected their sacred groves EXCEPT
- ☐ by building protective walls around the groves
 - ☐ by allowing only priests in sacred groves

- by punishing those who cut trees, hunted, or fished in the groves
- by telling people that the gods could punish vandals of the groves

Paragraph 6

Such environmental care suggested to historian J. Donald Hughes that sacred groves represented “classical national parks.” By helping to insulate sacred groves from pressures of deforestation, erosion, and urbanization, Greek codes protected ecosystems from destruction. Sacred groves nonetheless represented imperfect parkscapes. Some encompassed relatively small areas such as a section of a hillside or a series of caves. Meanwhile, the fundamental purpose of the grove—the visitation of resident gods—sometimes promoted activities not entirely conducive to modern concepts of conservation. Animals were routinely captured to serve as sacrifices to the gods. Many groves witnessed horticultural and architectural improvements. Flowers were planted, trails cut, and statues, fountains, and caves installed for the benefit of visitors. The grove served as a recreational center for Greek society, a realm of ritual, performance, feasting, and even chariot racing.

10. The word “promoted” in the passage is closest in meaning to
 - repeated
 - altered
 - encouraged
 - performed
11. Why are the sacred groves of the ancient Greeks referred to as “imperfect parkscapes” in the passage?
 - The Greeks protected their sacred groves from deforestation and erosion but allowed people to build houses on the grounds.
 - The ancient Greeks often changed the environment of the groves by adding plants, building shrines, and capturing animals for sacrifice.
 - The Greeks allowed only religious ceremonies in the groves.
 - The ancient Greeks had strict size limits on how big a sacred area could be.
12. Which of the following is NOT mentioned in paragraph 6 as a change made to the landscapes of sacred groves?
 - The introduction of new animals to the area
 - The construction of statues and fountains
 - The planting of flowers
 - The creation of trails

Paragraph 4

In some spiritual traditions, landscapes such as gardens or deserts were treated as abstract emblems of spiritual states such as innocence or despair. ■ Rather than symbolic landscapes, as in the Judeo-Christian tradition, Greek sacred groves operated as literal homes of the gods. ■ Instead of being confined to prehistory or celestial space, spiritual parkscapes were present within the existing cultural terrain. ■ One could not visit a symbol of peace and severity, but one

could experience these qualities in a sacred grove. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

This, however, was not the Greek attitude.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Ancient Greeks and Romans thought of parks and natural spaces as spiritual and recreational sites.

-
-
-

Answer Choices

Sacred groves consisting of trees, springs, or mountains were organized as shrines to Greek gods.

Because of their association with ancient Greek gods, spirits, and heroes, there are still many sections of forest in modern Greece.

The belief in a divine presence in all natural things originated with the ancient Greeks and spread to other cultures around the world.

Trees have played significant roles in the religious practices and beliefs of Greeks and numerous other cultures.

In cultures where the climate made cultivation of sacred groves impossible, trees retained only a symbolic value in mythical stories.

Because groves were spiritually important to ancient Greeks, they were often protected from damage and were the sites of various communal activities.

The Chaco Phenomenon

A truly remarkable transformation in settlement patterns occurred in the San Juan basin in northwestern New Mexico in the late tenth and early eleventh centuries, with small household farmsteads giving way to aggregated communities centered on communal masonry buildings that are now called “great houses.” These structures are found throughout the basin but are concentrated in Chaco Canyon, where several examples contained hundreds of rooms and reached four stories in height. The largest great house is Pueblo Bonito, with over 600 rooms covering two acres. The entire episode of great house construction in Chaco, the Bonito phase (A.D. 900-1140), was obviously a time of immense cooperative effort. At least 200,000 wooden beams averaging 5 meters long and 20 centimeters in diameter were brought to the canyon from distances between 40 and 100 kilometers away to build a dozen great houses, signifying a huge labor investment and a complex production process. The bulk of construction took place in the eleventh century, but by A.D. 1140 it had ceased abruptly, after which there was a rapid decline in use of the great houses and apparent abandonment of the canyon in the thirteenth century.

For more than a century archaeologists have struggled to understand the circumstances surrounding the rise and collapse of Chacoan society—dubbed the Chaco Phenomenon. In particular, research has focused on determining why such an apparently inhospitable place as Chaco, which today is extremely arid and has very short growing seasons, should have favored the concentration of labor that must have been required for such massive construction projects over brief periods of time. Until the 1970s, it was widely assumed that Chaco had been a forested oasis that attracted farmers who initially flourished but eventually fell victim to their own success and exuberance, as they denuded the canyon of trees and vegetation to build large great houses. In the 1980s this reconstruction was largely dismissed in response to evidence that there had never been a forest in Chaco, and that canyon soils had poor agricultural potential. As scientific interpretations about Chaco changed, the focus of explanatory models changed from the attractiveness of the canyon for farmers to the position of the canyon within a regional network of dispersed agricultural communities.

The adoption of a regional perspective in explaining the Chaco Phenomenon was based in part on the discovery of formal trails connecting many of the great houses in Chaco, as well as linking the canyon to smaller great houses located throughout the San Juan basin, the latter are referred to as Chaco “outliers.” These trails are densest around the concentration of great houses in the center, and the canyon itself is roughly at the center of the basin. Consequently, the canyon occupies the geographical and social center of the network formed by the connecting trails. The current consensus view is that religion provides the fundamental explanation for this centrifugal pattern.

Archaeologists now describe Chaco during the Bonito phase as a location of high devotional expression and the pilgrimage center of a sacred landscape. These descriptions emphasize aspects of the archaeological record presumed to be associated with ritual activity, including caches of turquoise beads and pendants, unusual ceramic vessels and wooden objects, several

rooms with multiple human burials, and especially the large number of kivas (multipurpose rooms used for religious, political, and social functions) found in great houses. Most of these indicators occur only at Pueblo Bonito, but archaeologists generally assume that all the great houses had a similar ritual function. In fact, some scholars have suggested that the great houses were temples rather than residences.

However, new geological field studies in Chaco have produced results that may require a significant reassessment of the assumption that the canyon was not a favorable agricultural setting. It appears that during the first half of the eleventh century, during the extraordinary boom in construction, a large volume of water and suspended sediment flowed into the canyon. A large natural lake may have existed at the western end of Chaco, near the biggest concentration of great houses. The presence of large quantities of water and, equally important, a source of sediment that replenished agricultural fields, presumably made the canyon an extremely attractive place for newly arriving people from the northern San Juan River basin.

Paragraph 1

A truly remarkable transformation in settlement patterns occurred in the San Juan basin in northwestern New Mexico in the late tenth and early eleventh centuries, with small household farmsteads giving way to aggregated communities centered on communal masonry buildings that are now called “great houses.” These structures are found throughout the basin but are concentrated in Chaco Canyon, where several examples contained hundreds of rooms and reached four stories in height. The largest great house is Pueblo Bonito, with over 600 rooms covering two acres. The entire episode of great house construction in Chaco, the Bonito phase (A.D. 900-1140), was obviously a time of immense cooperative effort. At least 200,000 wooden beams averaging 5 meters long and 20 centimeters in diameter were brought to the canyon from distances between 40 and 100 kilometers away to build a dozen great houses, signifying a huge labor investment and a complex production process. The bulk of construction took place in the eleventh century, but by A.D. 1140 it had ceased abruptly, after which there was a rapid decline in use of the great houses and apparent abandonment of the canyon in the thirteenth century.

15. The word “signifying” in the passage is closest in meaning to

- ☐ creating
- ☐ indicating
- ☐ initiating
- ☐ requiring

16. The word “ceased” in the passage is closest in meaning to

- ☐ slow down
- ☐ accelerated
- ☐ stopped
- ☐ changed in style

17. According to paragraph 1, all of the following provide evidence that the Bonito phase was a

time of immense cooperative effort EXCEPT

- ☐ the large amounts of material needed
- ☐ the size of the Pueblo Bonito complex
- ☐ the unusual materials used in construction
- ☐ the distance the materials needed to be transported

Paragraph 2

For more than a century archaeologists have struggled to understand the circumstances surrounding the rise and collapse of Chacoan society—dubbed the Chaco Phenomenon. In particular, research has focused on determining why such an apparently inhospitable place as Chaco, which today is extremely arid and has very short growing seasons, should have favored the concentration of labor that must have been required for such massive construction projects over brief periods of time. Until the 1970s, it was widely assumed that Chaco had been a forested oasis that attracted farmers who initially flourished but eventually fell victim to their own success and exuberance, as they denuded the canyon of trees and vegetation to build large great houses. In the 1980s this reconstruction was largely dismissed in response to evidence that there had never been a forest in Chaco, and that canyon soils had poor agricultural potential. As scientific interpretations about Chaco changed, the focus of explanatory models changed from the attractiveness of the canyon for farmers to the position of the canyon within a regional network of dispersed agricultural communities.

18. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Researchers have tried to establish why an area as dry as Chaco was the site of such large construction efforts.
- ☐ Researchers have tried to establish whether the concentration of massive construction projects in a brief period of time made Chaco the dry area that it is today.
- ☐ Researchers have established that Chaco's brief growing season required a concentration of labor to produce large quantities of food in a short period of time.
- ☐ Researchers have established that the hot, dry climate of Chaco forced workers to complete construction on large buildings in short periods of time.

19. According to paragraph 2, before 1970, scholars believed that Chacoan society collapsed because

- ☐ Chaco never had the forests that were needed for the development of a stable agricultural economy.
- ☐ farmers used up the natural resources in Chaco that had originally allowed the society to succeed.
- ☐ Chaco suffered a long-term drought that prevented farmers from growing enough food.
- ☐ laborers left Chaco to find other work after they finished building the great houses there.

20. It can be inferred from paragraph 2 that the pre-1970s theory about the Chaco Phenomenon

- ☐ was based on the widespread farm and tool remains found by archaeologists on the site.

- ☐ was largely reinforced by findings in the 1980s.
- ☐ was not supported by substantial evidence.
- ☐ was so strong that it went unchallenged for many decades.

21. The word “dispersed” in the passage is closest in meaning to

- ☐ connected
- ☐ scattered
- ☐ stable
- ☐ developed

22. According to paragraph 2, why did scientists change their view about the cause of the collapse of Chacoan society?

- ☐ They found evidence that Chaco had always lacked trees and good soil.
- ☐ They discovered that Chaco Canyon was much drier than they had previously believed.
- ☐ They learned that the population was not large enough to supply the laborers needed to build the great houses.
- ☐ They found evidence that the farming economy was excessively concentrated in the central canyon.

Paragraph 3

The adoption of a regional perspective in explaining the Chaco Phenomenon was based in part on the discovery of formal trails connecting many of the great houses in Chaco, as well as linking the canyon to smaller great houses located throughout the San Juan basin, the latter are referred to as Chaco “outliers.” These trails are densest around the concentration of great houses in the center, and the canyon itself is roughly at the center of the basin. Consequently, the canyon occupies the geographical and social center of the network formed by the connecting trails. The current consensus view is that religion provides the fundamental explanation for this centrifugal pattern.

Paragraph 4

Archaeologists now describe Chaco during the Bonito phase as a location of high devotional expression and the pilgrimage center of a sacred landscape. These descriptions emphasize aspects of the archaeological record presumed to be associated with ritual activity, including caches of turquoise beads and pendants, unusual ceramic vessels and wooden objects, several rooms with multiple human burials, and especially the large number of kivas (multipurpose rooms used for religious, political, and social functions) found in great houses. Most of these indicators occur only at Pueblo Bonito, but archaeologists generally assume that all the great houses had a similar ritual function. In fact, some scholars have suggested that the great houses were temples rather than residences.

23. According to paragraphs 3 and 4, which of the following best describes how archaeologists arrived at their current view of the nature of Chaco during the Bonito phase?

- ☐ They discovered a large number of kivas, which probably served as temporary houses for pilgrims on their way to the main temple.

- They found a series of paths leading to the outliers, which seem to have been centers of trade for makers of jewelry and other products.
- They found turquoise beads and pendants and other valuable objects, leading to the theory that the great houses were wealthy residences.
- They discovered many objects and rooms associated with ritual activity, leading to the theory that Chaco was a religious center.

24. The word “function” in the passage is closest in meaning to

- center
- practice
- design
- purpose

Paragraph 5

However, new geological field studies in Chaco have produced results that may require a significant reassessment of the assumption that the canyon was not a favorable agricultural setting. It appears that during the first half of the eleventh century, during the extraordinary boom in construction, a large volume of water and suspended sediment flowed into the canyon.

A large natural lake may have existed at the western end of Chaco, near the biggest concentration of great houses. The presence of large quantities of water and, equally important, a source of sediment that replenished agricultural fields, presumably made the canyon an extremely attractive place for newly arriving people from the northern San Juan River basin.

25. According to paragraph 5, what is the possible significance of new geological field studies in Chaco?

- They indicate that during the construction boom the Chaco area probably did have enough water and sediment to attract farmers to that area.
- They could undermine the theory of Chaco as a religious center.
- They show the presence of excessive amounts of water, which may have led to the departure of most of the people living there during the Bonito phase.
- They suggest that the kind of sediment present in Chaco in the eleventh century was not favorable for agriculture.

26. Why does the author state that “A large natural lake may have existed at the western end of Chaco, near the biggest concentration of great houses?”

- To suggest that geological studies are better than archaeological studies in identifying the historical uses of land
- To demonstrate that large construction projects require a large population of workers
- To support the idea that Chaco may have been favorable to agriculture during the Bonito phase
- To show that the Chacoan people preferred to build their homes near water

Paragraph 3

■ The adoption of a regional perspective in explaining the Chaco Phenomenon was based in part

on the discovery of formal trails connecting many of the great houses in Chaco, as well as linking the canyon to smaller great houses located throughout the San Juan basin, the latter are referred to as Chaco “outliers.” ■ These trails are densest around the concentration of great houses in the center, and the canyon itself is roughly at the center of the basin. ■ Consequently, the canyon occupies the geographical and social center of the network formed by the connecting trails. ■ The current consensus view is that religion provides the fundamental explanation for this centrifugal pattern.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Scholars have attempted to find a reason for this weblike arrangement of great houses around a central canyon.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The population of the Chaco Canyon in New Mexico changed significantly between the tenth and eleventh centuries, as evidenced by the remains of its great houses.

-
-
-

Answer Choices

Before the 1970s, scholars believed that the fall of Chacoan society was caused by farmers' cutting down all the trees to build their great houses.

After discovering trails connecting Chaco to surrounding communities, scholars came to believe that there were many forested oases to support those communities.

Archaeological evidence has led current scholars to believe that Chaco was a religious center during the Bonito phase.

Archaeological findings indicate that Chaco Canyon was completely abandoned by the end of the thirteenth century.

Researchers' findings in the 1980s revealed that Chaco Canyon had been a fertile agricultural area that caused the population near the center of the canyon to increase steadily during the Bonito phase.

Recent geological studies indicating the presence of water in Chaco Canyon in the eleventh century may alter scholars' belief that the area was not favorable for farming.

Dinosaurs and Parental Care

From fossil evidence alone the question of whether or not dinosaurs cared for their young is very difficult to answer. Because behaviors are not preserved in the fossil record, we can only make inferences from indirect evidence. Parental care can be divided into two types of behavior: prehatching (building nests and incubating eggs—for example, sitting on top of them so as to warm the eggs and encourage hatching) and posthatching (feeding the young and guarding the nests). Most of our evidence comes from alleged dinosaur rookeries (places where nests are built). Several have been excavated in eastern Montana, where a large concentration of dinosaur nests was found at a place now called Egg Mountain. Most of these probably belonged to the hadrosaur *Maiasaura*. Preserved in these nests are the bones of baby dinosaurs. The finds at Egg Mountain and other sites around the world document that dinosaurs laid their eggs in nests.

The nests at Egg Mountain are reported to be equally spaced, separated by a space corresponding to the length of an adult *Maiasaura*. From this arrangement scientists have inferred that the nests were separated in this way to allow incubation in a tightly packed nesting colony. Although this interpretation is open to challenge, the discovery of *Oviraptor* adults on top of *Oviraptor* egg clutches (as determined by embryos in some eggs), is relatively powerful evidence that at least these dinosaurs incubated their nests.

Evidence for parental care following hatching is much more controversial. Behavioral speculation based on indirect fossil evidence is dangerous because the data is not always as unambiguous as might appear. At Egg Mountain, many nests contain baby dinosaur bones. Not all the dinosaurs in the nest are the same size. Many of the small bones found in the nests are associated with jaws and teeth, teeth that show signs of wear. It seems reasonable to assume that the wear was caused by the chewing of the coarse plants that were the hatchlings' diet. Because the young were still in the nest, this food may have been brought to the rookery by foraging adults. This line of reasoning suggests that these animals had an advanced system of parental care. A closer look at the evidence clouds this interpretation. Analysis of dinosaur embryos indicates that worn surfaces are present on the teeth of juveniles even before hatching. Just as a human baby moves inside the mother before birth, modern-day archosaurs also grind their teeth before birth, wearing the surface in some spots. Thus, the fossil evidence for an advanced parental care system in extinct dinosaurs is suggestive but inconclusive, and it is hard even to imagine the sort of paleontologic discovery that could settle this debate for good.

The strongest evidence that extinct dinosaurs had some form of advanced parental care system is based on an understanding of the phylogenetic relationships among dinosaurs and their closest living relatives. Living dinosaurs (birds), even primitive ones such as ostriches and kiwis, exhibit parental care, so some form of parental care can be inferred to have existed in the last common ancestor of all birds. Although unappreciated, crocodiles are reptiles that are also caring parents. They build nests, guard the nests, and in some cases dig their young out of the nest when they hear the chirping young ones hatching. The young even communicate with each other while still in the egg by high-frequency squeaks (as birds do). Some evidence suggests that this squeaking is

a cue for the synchronization of the hatching. Since birds and crocodiles share a common ancestor, the simplest explanation for the characteristics they share (such as nest building and some form of parental care) is that they evolved only once—that these attributes were present in their common ancestor and passed on to its descendants. Because extinct dinosaurs also descended from that ancestor, the simplest and most general theory is that extinct dinosaurs also shared these characteristics, even though they cannot be directly observed, and we cannot be sure how elaborate their parental care was.

Paragraph 1

From fossil evidence alone the question of whether or not dinosaurs cared for their young is very difficult to answer. Because behaviors are not preserved in the fossil record, we can only make inferences from indirect evidence. Parental care can be divided into two types of behavior: prehatching (building nests and incubating eggs—for example, sitting on top of them so as to warm the eggs and encourage hatching) and posthatching (feeding the young and guarding the nests). Most of our evidence comes from **alleged** dinosaur rookeries (places where nests are built). Several have been excavated in eastern Montana, where a large concentration of dinosaur nests was found at a place now called Egg Mountain. Most of these probably belonged to the hadrosaur *Maiasaura*. Preserved in these nests are the bones of baby dinosaurs. The finds at Egg Mountain and other sites around the world document that dinosaurs laid their eggs in nests.

29. The word “**alleged**” in the passage is closest in meaning to

- ☐ scattered
- ☐ supposed
- ☐ isolated
- ☐ exposed

30. Paragraph 1 answers which of the following questions about parental care in dinosaurs?

- ☐ Which type of parental care was more important for the survival of dinosaur young, prehatching care or posthatching care?
- ☐ Why were dinosaur remains in eastern Montana preserved rather than destroyed over time?
- ☐ Did *Maiasaura* hadrosaurs provide types of parental care not provided by other dinosaurs?
- ☐ What evidence supports the view that *Maiasaura* females laid their eggs in nests?

Paragraph 1

From fossil evidence alone the question of whether or not dinosaurs cared for their young is very difficult to answer. Because behaviors are not preserved in the fossil record, we can only make inferences from indirect evidence. Parental care can be divided into two types of behavior: prehatching (building nests and incubating eggs—for example, sitting on top of them so as to warm the eggs and encourage hatching) and posthatching (feeding the young and guarding the nests). Most of our evidence comes from **alleged** dinosaur rookeries (places where nests are built). Several have been excavated in eastern Montana, where a large concentration of dinosaur nests was found at a place now called Egg Mountain. Most of these probably belonged to the hadrosaur *Maiasaura*. Preserved in these nests are the bones of baby dinosaurs. The finds at Egg Mountain and other sites around the world document that dinosaurs laid their eggs in nests.

Paragraph 2

The nests at Egg Mountain are reported to be equally spaced, separated by a space corresponding to the length of an adult *Maiasaura*. From this arrangement scientists have inferred that the nests were separated in this way to allow incubation in a tightly packed nesting colony. Although this interpretation is open to challenge, the discovery of *Oviraptor* adults on top of *Oviraptor* egg clutches (as determined by embryos in some eggs), is relatively powerful evidence that at least these dinosaurs incubated their nests.

31. According to paragraphs 1 and 2, the fossil record most clearly shows that dinosaurs engaged in which of the following behaviors?

- ☐ Laying eggs in nests
- ☐ Hiding eggs
- ☐ Feeding young
- ☐ Storing food

32. According to paragraph 2, which of the following supports the theory that the *Maiasaura* incubated their eggs?

- ☐ The examination of embryos found in some eggs
- ☐ The large concentration of nests in one location
- ☐ The amount of space between nests
- ☐ The discovery of adult *Maiasaura* bones on top of egg clutches

Paragraph 3

Evidence for parental care following hatching is much more **controversial**. Behavioral speculation based on indirect fossil evidence is dangerous because the data is not always as unambiguous as might appear. At Egg Mountain, many nests contain baby dinosaur bones. Not all the dinosaurs in the nest are the same size. Many of the small bones found in the nests are associated with jaws and teeth, teeth that show signs of wear. It seems reasonable to assume that the wear was caused by the chewing of the coarse plants that were the hatchlings' diet. Because the young were still in the nest, this food may have been brought to the rookery by foraging adults. This line of reasoning suggests that these animals had an advanced system of parental care. A closer look at the evidence clouds this interpretation. Analysis of dinosaur embryos indicates that worn surfaces are present on the teeth of juveniles even before hatching. Just as a human baby moves inside the mother before birth, modern-day archosaurs also grind their teeth before birth, wearing the surface in some spots. Thus, the fossil evidence for an advanced parental care system in extinct dinosaurs is suggestive but **inconclusive**, and it is hard even to imagine the sort of paleontologic discovery that could settle this debate for good.

33. The word "**controversial**" in the passage is closest in meaning to

- ☐ limited
- ☐ convincing
- ☐ relevant
- ☐ debatable

34. The word “inconclusive” in the passage is closest in meaning to
- ☐ not decisive
 - ☐ insignificant
 - ☐ not valid
 - ☐ misleading
35. According to paragraph 3, the patterns of wear found on the teeth of young dinosaurs may indicate which of the following?
- ☐ Baby dinosaurs were eating food brought to them by their parents.
 - ☐ Early development of jaw and teeth varied according to a dinosaur’s size.
 - ☐ Dinosaurs went foraging for food at an early age.
 - ☐ Baby dinosaurs did not begin to eat solid food until after they left the nest.
36. In paragraph 3, why does the author mention that baby archosaurs ground their teeth inside the egg?
- ☐ To support the claim that baby dinosaurs in the egg shared certain behaviors with human babies before birth
 - ☐ To contrast the behavior of baby archosaurs with that of other types of dinosaurs
 - ☐ To cast doubt on the claim that adult dinosaurs fed their hatchlings in the nest
 - ☐ To explain why the teeth of baby archosaurs were more worn than those of other juveniles

Paragraph 4

The strongest evidence that extinct dinosaurs had some form of advanced parental care system is based on an understanding of the phylogenetic relationships among dinosaurs and their closest living relatives. Living dinosaurs (birds), even primitive ones such as ostriches and kiwis, exhibit parental care, so some form of parental care can be inferred to have existed in the last common ancestor of all birds. Although unappreciated, crocodiles are reptiles that are also caring parents. They build nests, guard the nests, and in some cases dig their young out of the nest when they hear the chirping young ones hatching. The young even communicate with each other while still in the egg by high-frequency squeaks (as birds do). Some evidence suggests that this squeaking is a cue for the synchronization of the hatching. Since birds and crocodiles share a common ancestor, the simplest explanation for the characteristics they share (such as nest building and some form of parental care) is that they evolved only once—that these attributes were present in their common ancestor and passed on to its descendants. Because extinct dinosaurs also descended from that ancestor, the simplest and most general theory is that extinct dinosaurs also shared these characteristics, even though they cannot be directly observed, and we cannot be sure how elaborate their parental care was.

37. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- ☐ The simplest explanation for the similarities between birds and crocodiles is that they evolved at the same time.

- A common ancestor is probably the source of the shared traits of crocodiles and birds.
- The originally similar traits of birds and crocodiles increased after evolving through a shared ancestor.
- Only one shared pattern of behavior—that of nest building—was present in the common ancestor of birds and crocodiles.

38. The word “elaborate” in the passage is closest in meaning to

- widespread
- reliable
- well developed
- long lasting

39. Paragraph 4 answers all of the following questions about crocodiles EXCEPT:

- What is the evidence that crocodiles are caring parents?
- Why do crocodile parents communicate with the young inside their eggs?
- What is a possible reason for the high-frequency sounds that crocodiles make inside their eggs?
- How do crocodiles participate in the hatching process of their young?

40. In paragraph 4, the author discusses birds and crocodiles in order to

- contrast patterns of parenting behavior in both living and extinct animals
- provide evidence that sophisticated parental care behaviors evolved only relatively recently
- demonstrate that parental care behaviors have continued to evolve since the time of the dinosaurs
- support the theory that extinct dinosaurs probably inherited some kind of parental care system

Paragraph 1

From fossil evidence alone the question of whether or not dinosaurs cared for their young is very difficult to answer. ■ Because behaviors are not preserved in the fossil record, we can only make inferences from indirect evidence. ■ Parental care can be divided into two types of behavior: prehatching (building nests and incubating eggs—for example, sitting on top of them so as to warm the eggs and encourage hatching) and posthatching (feeding the young and guarding the nests). ■ Most of our evidence comes from alleged dinosaur rookeries (places where nests are built). ■ Several have been excavated in eastern Montana, where a large concentration of dinosaur nests was found at a place now called Egg Mountain. Most of these probably belonged to the hadrosaur *Maiasaura*. Preserved in these nests are the bones of baby dinosaurs. The finds at Egg Mountain and other sites around the world document that dinosaurs laid their eggs in nests.

41. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Evidence of the former is easier to find than that of the latter.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Scientists must use indirect evidence to determine whether extinct dinosaurs cared for their young.

-
-
-

Answer Choices

Because baby dinosaur bones and eggs were very delicate, there are relatively few preserved as fossils, so little is known about dinosaur young.

Fossils from sites like Egg Mountain indicate that dinosaurs built nests, and perhaps that they incubated their eggs and fed their hatchlings.

Fossil evidence such as the spacing of nests may indicate advanced parental care but can have different interpretations.

Tightly packed *Oviraptor* rookeries indicate that dinosaurs may have tended to nest in large colonies in order to better protect both eggs and hatchlings.

Discovery of hadrosaur bones of different sizes in the same nest may indicate that, in some species, older siblings took care of younger ones.

The strongest evidence comes from extinct dinosaurs' nearest living relatives, birds and crocodiles, who do engage in many forms of parental care.

Colonial America and the Navigation Acts

In the seventeenth and eighteenth centuries, the British parliament enacted a number of laws, called Navigation Acts, governing commerce between Britain and its overseas colonies. For example, the Navigation Acts of 1660 and 1663 barred the empire's colonial merchants from exporting such commodities as sugar and tobacco anywhere except to England and from importing goods in non-English ships. Similarly, the Molasses Act of 1733 taxed all foreign molasses (a thick liquid drained from sugarcane and used to make rum) entering the mainland American colonies at sixpence per gallon. This act was intended less to raise revenue than to serve as a protective tariff (tax) that would benefit British West Indian sugar producers at the expense of their French rivals. By 1750 a long series of Navigation Acts were in force, with several effects on the North American colonial economy.

For one thing, the laws limited all imperial trade to British ships, defined as those with British ownership and crews that were three-quarters British. For purposes of the legislation, Parliament classified all colonists as British. This restriction not only contributed to Great Britain's rise as Europe's foremost shipping nation but also laid the foundations for an American shipbuilding industry and merchant marine. By the 1750s one-third of all imperial vessels were American-owned, mostly by merchants in the northeast and in mid-Atlantic colonies. The swift growth of this merchant marine diversified the northern colonial economy and made it more self-sufficient. The expansion of colonial shipping in turn accelerated urbanization by creating a need for centralized docks, warehouses, and repair shops in the colonies. By 1770 Philadelphia and New York City had emerged as two of the British Empire's busiest ports.

The Navigation Acts also barred the export of certain "enumerated goods" to foreign nations unless those items first passed through England or Scotland. The American mainland's chief items of this sort were tobacco, rice, furs, indigo (a Carolina plant that produced a blue dye), and naval supplies (such as masts and tar). Parliament never restricted grain, livestock, fish, lumber, or rum, which altogether made up 60 percent of American colonial exports. Furthermore, Anglo-American exporters of tobacco and rice—the chief commodities affected by enumeration—had their burdens reduced by two significant concessions. First, Parliament gave tobacco growers a monopoly over the British market by excluding foreign tobacco, even though this hurt British consumers. (Rice planters enjoyed a natural monopoly because they had no competitors.) Second, Parliament tried to minimize the added cost of landing tobacco and rice in Britain (where customs officials collected duties on both) by refunding the duties on all tobacco and rice that the colonists later shipped to other countries.

The navigation system's impact on the colonies encouraged economic diversification as well. Parliament used British tax money to pay modest incentives to Americans producing such items as silk, iron, dyes, hemp, and lumber, which Britain would otherwise have had to import from other countries, and it raised the price of commercial rivals' imports by imposing protective tariffs on them. The trade laws did prohibit Anglo-Americans from competing with large-scale

British manufacturing of certain products, most notably clothing. However, colonial tailors, hatters, and other small clothes manufacturers could continue to make any item of dress in their households or small shops. Manufactured by low-paid labor, British clothing imports generally undersold whatever the colonists could have produced given their higher labor costs. The colonists were also free to produce iron and built numerous ironworks.

Finally, the Navigation Acts made the colonies a protected market for low-priced consumer goods and other exports from Britain. Steady overseas demand for colonial products created a prosperity that enabled colonists to consume ever-larger amounts not only of clothing but of dishware, home furnishings, tea, and a range of other items both produced in Britain and imported by British and colonial merchants from elsewhere. Consequently, the share of British exports sold to the colonies rapidly increased from just 5 percent in 1700 to almost 40 percent by 1760. Cheap imported goods enabled many colonists to adopt a lifestyle similar to that of middle-class Britons.

Paragraph 1

In the seventeenth and eighteenth centuries, the British parliament enacted a number of laws, called Navigation Acts, governing commerce between Britain and its overseas colonies. For example, the Navigation Acts of 1660 and 1663 barred the empire's colonial merchants from exporting such commodities as sugar and tobacco anywhere except to England and from importing goods in non-English ships. Similarly, the Molasses Act of 1733 taxed all foreign molasses (a thick liquid drained from sugarcane and used to make rum) entering the mainland American colonies at sixpence per gallon. This act was intended less to raise revenue than to serve as a protective tariff (tax) that would benefit British West Indian sugar producers at the expense of their French rivals. By 1750 a long series of Navigation Acts were in force, with several effects on the North American colonial economy.

1. According to paragraph 1, the Navigation Acts of 1660 and 1663 did not allow
 - ☐ importing goods without paying taxes on them
 - ☐ importing goods made outside the British Empire
 - ☐ using English ships to transport goods to non-British colonies
 - ☐ using non-English ships to bring in goods from outside the colonies
2. What was the purpose of "the Molasses Act of 1733"?
 - ☐ To produce the funds needed to protect British West Indian sugar producers from attack by the French
 - ☐ To give British sugar producers in the West Indies an advantage over their French rivals
 - ☐ To prevent rum from being made outside of the British West Indies
 - ☐ To discourage the mainland American colonies from importing molasses

Paragraph 2

For one thing, the laws limited all imperial trade to British ships, defined as those with British ownership and crews that were three-quarters British. For purposes of the legislation, Parliament classified all colonists as British. This restriction not only contributed to Great Britain's rise as

Europe's foremost shipping nation but also laid the foundations for an American shipbuilding industry and merchant marine. By the 1750s one-third of all imperial vessels were American-owned, mostly by merchants in the northeast and in mid-Atlantic colonies. The **swift** growth of this merchant marine diversified the northern colonial economy and made it more self-sufficient. The expansion of colonial shipping in turn accelerated urbanization by creating a need for centralized docks, warehouses, and repair shops in the colonies. By 1770 Philadelphia and New York City had emerged as two of the British Empire's busiest ports.

3. The word "**swift**" in the passage is closest in meaning to
- ☐ gradual
 - ☐ fast
 - ☐ protective
 - ☐ long-term
4. According to paragraph 2, the Navigation Acts had all of the following effects on the northern colonies in North America EXCEPT:
- ☐ The region's economic dependence on Britain declined.
 - ☐ The region's ports became increasingly busy.
 - ☐ Shipbuilding and related industries grew in strength.
 - ☐ Ownership of vessels by merchants in the northeast and mid-Atlantic colonies declined.

Paragraph 3

The Navigation Acts also barred the export of certain "enumerated goods" to foreign nations unless those items first passed through England or Scotland. The American mainland's chief items of this sort were tobacco, rice, furs, indigo (a Carolina plant that produced a blue dye), and naval supplies (such as masts and tar). Parliament never restricted **grain, livestock, fish, lumber, or rum**, which altogether made up 60 percent of American colonial exports. Furthermore, Anglo-American exporters of tobacco and rice—the chief commodities affected by enumeration—had their burdens reduced by two **significant** concessions. First, Parliament gave tobacco growers a monopoly over the British market by excluding foreign tobacco, even though this hurt British consumers. (Rice planters enjoyed a natural monopoly because they had no competitors.) Second, Parliament tried to minimize the added cost of landing tobacco and rice in Britain (where customs officials collected duties on both) by refunding the duties on all tobacco and rice that the colonists later shipped to other countries.

5. Select the TWO answer choices that, according to paragraph 3, indicate how the Navigation Acts affected rice and tobacco exporters. To receive credit, you must select TWO answer choices.
- ☐ They first had to ship their products to either England or Scotland before shipping them elsewhere.
 - ☐ Their exports were reduced by 60 percent.
 - ☐ They received special concessions by Parliament.
 - ☐ They had to use their own ships to export their products.

6. The author mentions “grain, livestock, fish, lumber, or rum” in the passage in order to
- ☐ indicate the colonial exports that were affected by enumeration
 - ☐ provide examples of important commodities that had not been allowed to be exported before the Navigation Acts were introduced
 - ☐ explain why Britain decided to introduce the Navigation Acts
 - ☐ show how North American exporters quickly increased the exports of certain products to compensate for the restrictions imposed by the Navigation Acts
7. The word “significant” in the passage is closest in meaning to
- ☐ problematic
 - ☐ considerable
 - ☐ minor
 - ☐ temporary

Paragraph 4

The navigation system's impact on the colonies encouraged economic diversification as well. Parliament used British tax money to pay modest incentives to Americans producing such items as silk, iron, dyes, hemp, and lumber, which Britain would otherwise have had to import from other countries, and it raised the price of commercial rivals' imports by imposing protective tariffs on them. The trade laws did prohibit Anglo-Americans from competing with large-scale British manufacturing of certain products, most notably clothing. However, colonial tailors, hatters, and other small clothes manufacturers could continue to make any item of dress in their households or small shops. Manufactured by low-paid labor, British clothing imports generally undersold whatever the colonists could have produced given their higher labor costs. The colonists were also free to produce iron and built numerous ironworks.

8. The word “modest” in the passage is closest in meaning to
- ☐ direct
 - ☐ relatively small
 - ☐ adequate
 - ☐ various
9. Paragraph 4 supports which of the following statements about why Britain did not pass laws restricting the small-scale manufacture of clothing in the colonies?
- ☐ Such clothing presented no serious competition to clothing imported from Britain.
 - ☐ The British realized that any attempt to enforce such laws would be ineffective.
 - ☐ Clothing manufactured in the American colonies generally did not sell well in Britain.
 - ☐ The quality of clothing produced in the colonies was lower than that of British clothing.

Paragraph 5

Finally, the Navigation Acts made the colonies a protected market for low-priced consumer goods and other exports from Britain. Steady overseas demand for colonial products created a prosperity that enabled colonists to consume ever-larger amounts not only of clothing but of dishware, home furnishings, tea, and a range of other items both produced in Britain and

imported by British and colonial merchants from elsewhere. Consequently, the share of British exports sold to the colonies rapidly increased from just 5 percent in 1700 to almost 40 percent by 1760. Cheap imported goods enabled many colonists to adopt a lifestyle similar to that of middle-class Britons.

10. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ British and colonial merchants prospered because of a demand in the colonies for increasing amounts of consumer goods made in Britain.
- ☐ Both the colonists and the British increasingly consumed clothing and other household items imported from foreign countries by colonial merchants.
- ☐ The wealth gained from exporting their products enabled the colonists to buy increasing quantities of consumer goods brought into North America by British and colonial merchants.
- ☐ As merchants increased their wealth, they began to widen the range of products they exported from North America to Britain and the British colonies elsewhere.

Paragraph 5

Finally, the Navigation Acts made the colonies a protected market for low-priced consumer goods and other exports from Britain. Steady overseas demand for colonial products created a prosperity that enabled colonists to consume ever-larger amounts not only of clothing but of dishware, home furnishings, tea, and a range of other items both produced in Britain and imported by British and colonial merchants from elsewhere. Consequently, the share of British exports sold to the colonies rapidly increased from just 5 percent in 1700 to almost 40 percent by 1760. Cheap imported goods enabled many colonists to adopt a lifestyle similar to that of middle-class Britons.

11. The word “Consequently” in the passage is closest in meaning to

- ☐ However
- ☐ Similarly
- ☐ In addition
- ☐ Therefore

12. According to paragraph 5, which of the following was one effect that the Navigation Acts had during the period from 1700 to 1760?

- ☐ They reduced the attractiveness of most locally produced commodities for the American colonists.
- ☐ They protected colonial merchants from having to compete with low-priced imports.
- ☐ They greatly increased the importance of the American colonies as purchasers of British exports.
- ☐ They raised the production costs for many commodities manufactured in the American colonies.

Paragraph 3

The Navigation Acts also barred the export of certain “enumerated goods” to foreign nations

unless those items first passed through England or Scotland. The American mainland's chief items of this sort were tobacco, rice, furs, indigo (a Carolina plant that produced a blue dye), and naval supplies (such as masts and tar). Parliament never restricted grain, livestock, fish, lumber, or rum, which altogether made up 60 percent of American colonial exports. Furthermore, Anglo-American exporters of tobacco and rice—the chief commodities affected by enumeration—had their burdens reduced by two significant concessions. ■ First, Parliament gave tobacco growers a monopoly over the British market by excluding foreign tobacco, even though this hurt British consumers. ■ (Rice planters enjoyed a natural monopoly because they had no competitors.) ■ Second, Parliament tried to minimize the added cost of landing tobacco and rice in Britain (where customs officials collected duties on both) by refunding the duties on all tobacco and rice that the colonists later shipped to other countries. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

About 85 percent of all North American tobacco and rice was eventually reexported and sold outside the British Empire.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.



14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The Navigation Acts put in place by Britain had significant effects on the economy of the American colonies.

-
-
-

Answer Choices

The acts were originally designed to force the French to gradually abandon their American colonies.

Although the acts restricted colonists from exporting certain goods directly to foreign nations, important colonial products enjoyed both reduced duties and a monopoly of the British market.

The British refusal to allow tobacco imports from foreign nations hurt both British consumers and the American colonial tobacco planters.

The acts limited trade with the Empire to British ships, but by classifying all colonists as British, the acts allowed North Americans to develop their own ships.

High tariffs on imports from the colonies served to protect British-made goods from having to compete with the less expensive goods the colonists could produce using low-cost labor.

Trade laws protected some British manufacturing from colonial competition but encouraged colonial economic prosperity while making cheap British consumer goods ready available.

Mass Production: Method and Impact

The technological and managerial innovations of Thomas Alva Edison (the inventor of electricity) and the industrial leaders Andrew Carnegie (iron and steel) and John D. Rockefeller (oil) proved readily adaptable throughout United States industry, spurring marvels of productivity. Late-nineteenth-century industrialists often discovered that their factories produced more goods than the market could absorb. This was particularly true in two kinds of businesses: those that manufactured devices for individual use, such as sewing machines and farm implements, and those that mass-produced consumer goods, such as matches, flour, soap, canned foods, and processed meats. Not surprisingly, these industries were trailblazers in developing advertising and marketing techniques. Strategies for encouraging consumer demand and for differentiating one product from another were an important component of the American post-Civil War industrial transformation.

The growth of the flour industry illustrates both the spread of mass production and the emergence of new marketing concepts. In the 1870s the nation's flour mills adopted the most advanced European manufacturing technologies and installed continuous-process machines that graded, cleaned, hulled, and packaged their product in one rapid operation. These companies, however, soon produced more flour than they could sell. To sell this excess, the mills thought up new product lines, such as cake flours and breakfast cereals, and sold them using easy-to-remember brand names.

Through brand names, trademarks, guarantees, and slogans, manufacturers built demand for their products and won remarkable consumer loyalty. Americans in large numbers bought a brand of soap first made in 1897 in Cincinnati, Ohio, because of the absurdly precise but impressive pledge that it was "99 and 44/100ths percent pure." In the photographic field, George Eastman in the 1880s developed a paper-based photographic film as an alternate to the bulky, fragile glass plates then in use. Manufacturing a cheap camera for the masses and devising a catchy slogan ("you just press the button, we do the rest"). Eastman introduced a system whereby customers returned the 100-exposure film and the camera to the Rochester, New York, factory. There the film was developed, the camera reloaded, and everything shipped back to the customer—for a charge of ten dollars. In marketing a new technology, Eastman had revolutionized an industry and democratized a visual medium previously confined to a few.

By 1900 the chaos of early industrial competition, when thousands of small companies had struggled to enter a national market, had given way to an economy dominated by a few enormous firms. An industrial transformation that originated in railroading and expanded to steel and petroleum had spread to every area of United States business, and for those who could not compete in the era's unforgiving economic environment, the cost could be measured in ruined fortunes, bankrupted companies, and shattered dreams. John D. Rockefeller, talking about businesses he wanted to acquire, said he wanted "only the big ones, one those who have already proved they can do a big business. As for the others, unfortunately they will have to die."

The cost was high, too, for millions of American workers, immigrant and native born alike. The new industrial order was built on the backs of an army of laborers who were paid subsistence wages and who could be fired on a moment's notice when hard times or new technologies made them expendable. Moreover, industrialization often devastated the environment with pollution in the relentless drive for efficiency and profit.

To be sure, this industrial revolution brought social benefits as well, in the form of labor-saving products, lower prices, and advances in transportation and communications. The benefits and liabilities were inextricably interconnected. The sewing machine, for example, created thousands of new factory jobs, made available a wider variety of clothing, and eased the lives of millions of consumers. At the same time, it encouraged greedy entrepreneurs to operate factories in which the poor worked long hours in unhealthy conditions pitifully low wages.

Whatever the final balance sheet of social gains and costs, one thing was clear: the United States had forced its way onto the world stage as an industrial nation, and the groundwork had been laid for a new social and economic order in the twentieth century.

Paragraph 1

The technological and managerial innovations of Thomas Alva Edison (the inventor of electricity) and the industrial leaders Andrew Carnegie (iron and steel) and John D. Rockefeller (oil) proved readily adaptable throughout United States industry, spurring marvels of productivity. Late-nineteenth-century industrialists often discovered that their factories produced more goods than the market could absorb. This was particular true in two kinds of businesses: those that manufactured devices for individual use, such as sewing machines and farm implements, and those that mass-produced consumer goods, such as matches, flour, soap, canned foods, and processed meats. Not surprisingly, these industries were trailblazers in developing advertising and marketing techniques. Strategies for encouraging consumer demand and for differentiating one product from another were an important **component** of the American post-Civil War industrial transformation.

15. The word "**component**" in the passage is closet in meaning to
- ☐ theme
 - ☐ development
 - ☐ part
 - ☐ tool
16. According to paragraph 1, which of the following statements is true of Edison, Carnegie, and Rockefeller?
- ☐ They were famous inventors who became rich factory owners.
 - ☐ They were the first to develop advertising and marketing techniques.
 - ☐ Their ideas and methods were used to transform United States industry.
 - ☐ Their companies produced mechanical devices and consumer goods.
17. According to paragraph 1, all of the following contributed to the industrial transformation in

the post-Civil War United States EXCEPT

- ☐ a greater variety of consumer products
- ☐ new technological and managerial methods
- ☐ mass production of goods
- ☐ development of advertising and marketing techniques

Paragraph 2

The growth of the flour industry illustrates both the spread of mass production and the emergence of new marketing concepts. In the 1870s the nation's flour mills adopted the most advanced European manufacturing technologies and installed continuous-process machines that graded, cleaned, hulled, and packaged their product in one rapid operation. These companies, however, soon produced more flour than they could see. To sell this excess, the mills thought up new product lines, such as cake flours and breakfast cereals, and sold them using easy-to-remember brand names.

18. According to paragraph 2, the new technologies used by the flour mills led the flour industry to do which of the following?

- ☐ Produce additional flour to export to Europe
- ☐ Adopt European methods of marketing flour to consumers
- ☐ Produce goods based on market research of customer needs
- ☐ Develop special new products with easily recognizable names

Paragraph 3

Through brand names, trademarks, guarantees, and slogans, manufacturers built demand for their products and won remarkable consumer loyalty. Americans in large numbers bought a brand of soap first made in 1897 in Cincinnati, Ohio, because of the absurd overly precise but impressive pledge that it was "99 and 44/100ths percent pure." In the photographic field, George Eastman in the 1880s developed a paper-based photographic film as an alternate to the bulky, fragile glass plates then in use. Manufacturing a cheap camera for the masses and devising a catchy slogan ("you just press the button, we do the rest"). Eastman introduced a system whereby customers returned the 100-exposure film and the camera to the Rochester, New York, factory. There the film was developed, the camera reloaded, and everything shipped back to the customer—for a charge of ten dollars. In marketing a new technology, Eastman had revolutionized an industry and democratized a visual medium previously confined to a few.

19. The word "remarkable" in the passage is closest in meaning to

- ☐ immediate
- ☐ extraordinary
- ☐ apparent
- ☐ steady

20. Why does the author mention a soap advertised as "99 and 44/100ths percent pure"?

- ☐ To make the point that the earliest product advertisements were very sophisticated
- ☐ To support the claim that advertising was effective in building demand and customer loyalty

- ☐ To suggest that the claims made by soap manufacturers were obviously untrue
 - ☐ To contrast the ways in which soap and photographic processing were marketed
21. The phrase “confined to” in the passage is closest in meaning to
- ☐ available to
 - ☐ known to
 - ☐ limited to
 - ☐ attractive to
22. According to paragraph 3, which of the following statements about George Eastman’s innovations in photography is true?
- ☐ He introduced a paper-based photographic film that was easy for customers to develop on their own.
 - ☐ He manufactured a camera that was more expensive but easier to operate than earlier cameras.
 - ☐ He introduced a system in which customers could exchange defective film and cameras for new ones.
 - ☐ He provided customers with the service of developing their film and reloading their cameras.

Paragraph 4

By 1900 the chaos of early industrial competition, when thousands of small companies had struggled to enter a national market, had given way to an economy dominated by a few enormous firms. An industrial transformation that originated in railroading and expanded to steel and petroleum had spread to every area of United States business, and for those who could not compete in the era’s unforgiving economic environment, the cost could be measured in ruined fortunes, bankrupted companies, and shattered dreams. John D. Rockefeller, talking about businesses he wanted to acquire, said he wanted “only the big ones, one those who have already proved they can do a big business. As for the others, unfortunately they will have to die.”

23. According to paragraph 4, which of the following is true of the economic situation of the United States in the 1900s?
- ☐ A group of national industries was controlled by a single company.
 - ☐ Many small companies failed and a few large companies dominated the economy.
 - ☐ The expansion of the steel and petroleum industries depended on the railroad companies.
 - ☐ A new national market made it possible for thousands of small companies to compete for business.

Paragraph 5

The cost was high, too, for millions of American workers, immigrant and native born alike. The new industrial order was built on the backs of an army of laborers who were paid subsistence wages and who could be fired on a moment’s notice when hard times or new technologies made them expendable. Moreover, industrialization often devastated the environment with pollution in the relentless drive for efficiency and profit.

24. The word “relentless” in the passage is closest in meaning to
- ☐ competitive
 - ☐ unceasing
 - ☐ reckless
 - ☐ passionate
25. Which of the following can be inferred about the new technologies mentioned in paragraph 5?
- ☐ They were sometimes too expensive to introduce in the workplace.
 - ☐ They forced laborers to work in more systematic ways.
 - ☐ They helped to reduce the impact of industrialization on the environment.
 - ☐ They helped make it possible for industries to reduce the workforce.

Paragraph 5

The cost was high, too, for millions of American workers, immigrant and native born alike. The new industrial order was built on the backs of an army of laborers who were paid subsistence wages and who could be fired on a moment’s notice when hard times or new technologies made them expendable. Moreover, industrialization often devastated the environment with pollution in the relentless drive for efficiency and profit.

Paragraph 6

To be sure, this industrial revolution brought social benefits as well, in the form of labor-saving products, lower prices, and advances in transportation and communications. The benefits and liabilities were inextricably interconnected. The sewing machine, for example, created thousands of new factory jobs, made available a wider variety of clothing, and eased the lives of millions of consumers. At the same time, it encouraged greedy entrepreneurs to operate factories in which the poor worked long hours in unhealthy conditions pitifully low wages.

26. According to paragraphs 5 and 6, the transformation of United States industry had all of the following negative effects EXCEPT
- ☐ higher prices for consumers
 - ☐ environmental pollution
 - ☐ unhealthy working conditions
 - ☐ low pay for hard work

Paragraph 1

The technological and managerial innovations of Thomas Alva Edison (the inventor of electricity) and the industrial leaders Andrew Carnegie (iron and steel) and John D. Rockefeller (oil) proved readily adaptable throughout United States industry, spurring marvels of productivity. ■ Late-nineteenth-century industrialists often discovered that their factories produced more goods than the market could absorb. ■ This was particular true in two kinds of businesses: those that manufactured devices for individual use, such as sewing machines and farm implements, and those that mass-produced consumer goods, such as matches, flour, soap, canned foods, and

processed meats. ■ Not surprisingly, these industries were trailblazers in developing advertising and marketing techniques. ■ Strategies for encouraging consumer demand and for differentiating one product from another were an important component of the American post-Civil War industrial transformation.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Edison's laboratory in Menlo Park, New Jersey, even became a model for the industrial research laboratories established by other corporations.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

In the late nineteenth century, technological and managerial innovations greatly raised productivity and transformed United States industry.

-
-
-

Answer Choices

Thomas Alva Edison, Andrew Carnegie, and John D. Rockefeller invented new technologies and management systems for the electrical, iron, steel, and oil industries.

Americans became loyal consumers of manufactured products like breakfast cereal and soap primarily because of the high quality associated with the brand names.

An economy characterized by competition between numerous small companies gave way to an economy in which only a few major companies were able to survive and dominate.

As manufacturers produced more goods than the market could easily absorb, they developed new advertising and marketing techniques to stimulate demand for their products.

George Eastman succeeded in creating a mass market for cameras not only by lowering manufacturing costs and selling prices but also by offering to develop film free of charge.

The industrial transformation brought a number of social benefits, but it also encouraged the exploitation of workers and polluted the environment.

Art and Culture of Pacific Northwest Communities

Pacific Northwest Culture Area

The 1,600-kilometer stretch of the northwestern Pacific coast of North America (from southern Alaska to Washington State) provided an ideal environment for the growth of stable communities. Despite the northerly latitude, the climate is temperate. Natural resources were originally so rich that the inhabitants could subsist by fishing and hunting and gathering, without the need to domesticate stock or cultivate the land. Forests yielded an abundance of wood for buildings, for boats, and for sculpture. Beyond them the Rocky Mountains were an impenetrable barrier against raids. The area appears to have been settled around 500 A.D. by tribes of diverse origins speaking mutually unintelligible languages: from north to south they include the Tlingit, the Haida, the Tsimshian, the Bella Coola, the Kwakiutl, and the Nootka. The culture to which they contributed has, nevertheless, an underlying homogeneity and a distinct visual character.

The peoples of the Northwest engaged in trade as well as warfare with one another, and this may account for the diffusion of cultural traits and artistic motifs throughout the area. Much of their art was concerned with religious ritual objects. But the rest is secular and springs from a preoccupation with the hereditary basis of their complex social structures.

The Tlingit and other nations or language groups were collections of autonomous village communities composed of one or more families, each with its own chief, who inherited his position through matrilineal descent. They had no centralized political or religious organization, but cohesion was given by extensive kinship networks established through marriage, and men and women were obliged to many outside the larger divisions of clans and moieties (tribal subdivisions) into which they were born and into which the social group was divided by matrilineal or patrilineal descent. Thus families built up riches by marriage without any one family acquiring a dominant position

Totem poles (see figure below), the most distinctive artistic product of the Northwest, were conspicuous declarations of prestige and of the genealogy (family history) by which it had been attained. These magnificent sculptures that probably originated as funerary monuments were first described by travelers in the late eighteenth century. Each one was carved from a single trunk of cedar, and the increasing availability of metal tools both permitted and encouraged more complex compositions and greater height—up to 27.4 meters. Their superimposed figures—eagles, beavers, whales, and so on—were crests (symbols of identity) that a chief inherited from his lineage, his clan, and his moiety. They were not objects of worship, though the animals carved on them might represent guardian spirits. Poles were designed according to a governing principle of bilateral symmetry, with their various elements interlocked so that they seem to grow organically out of one another, creating a unity of symbolism, form, and surface.



Masks (see figure above) are the most varied of the carvings from the Northwest, where they were an essential part of communal life. In style they range from an almost abstract symbolism to combinations of human and animal features and to a lifelike naturalism sometimes bordering on caricature (a style that strongly exaggerates features or characteristics), taken to its extreme in Tlingit war helmets. Some differences must have been due to those among the cultures in which they were created, but their place of origin cannot always be ascertained as they seem to have passed from one contiguous nation to another in the course of trade or warfare. Although carvers worked according to established conventions, no two masks are identical and those with basic similarities reveal varying degrees of skill.

The major differences between masks were determined by their purpose. Some were representations of chiefs and their ancestors and made to be displayed and treasured as heirlooms. Although they appear to record the styles of facial tattooing customary in different groups, it is difficult to say how far they were intended to be portraits rather than generalized images. Many masks, sometimes quite large, were carved to be worn in dance-dramas that re-enacted and kept alive the cohesive myths of a culture. Often, Tlingit masks were made for religious leaders and incorporated the animals that were believed to be their spirit helpers. Conjuring up forces of nature from the ocean, the forests, or the sky, they mediated between life on Earth and the inscrutable powers around and above.

Paragraph 1

The 1,600-kilometer stretch of the northwestern Pacific coast of North America (from southern Alaska to Washington State) provided an ideal environment for the growth of stable communities. Despite the northerly latitude, the climate is temperate. Natural resources were originally so rich that the inhabitants could subsist by fishing and hunting and gathering, without the need to domesticate stock or cultivate the land. Forests yielded an abundance of wood for buildings, for boats, and for sculpture. Beyond them the Rocky Mountains were an impenetrable barrier

against raids. The area appears to have been settled around 500 A.D. by tribes of diverse origins speaking **mutually unintelligible** languages: from north to south they include the Tlingit, the Haida, the Tsimshian, the Bella Coola, the Kwakiutl, and the Nootka. The culture to which they contributed has, nevertheless, an underlying homogeneity and a distinct visual character.

29. According to paragraph 1, which of the following was NOT one of the factors that made the northwestern Pacific coast perfect for the development of stable communities?

- ☐ Temperate climate
- ☐ Natural protection from raids
- ☐ Abundant natural resources
- ☐ Easily cultivated land

30. When the author states that the tribes speak "**mutually unintelligible**" languages, this means that the tribes

- ☐ speak languages of similar difficulty
- ☐ cannot understand each other's languages
- ☐ cannot understand the languages of tribes in neighboring areas
- ☐ understand the languages of tribes of similar origin

Paragraph 2

The peoples of the Northwest engaged in trade as well as warfare with one another, and this may account for the **diffusion** of cultural traits and artistic motifs throughout the area. Much of their art was concerned with religious ritual objects. But the rest is secular and springs from a preoccupation with the hereditary basis of their complex social structures.

31. The word "**diffusion**" in the passage is closest in meaning to

- ☐ development
- ☐ variety
- ☐ similarity
- ☐ spread

Paragraph 3

The Tlingit and other nations or language groups were collections of **autonomous** village communities composed of one or more families, each with its own chief, who inherited his position through matrilineal descent. They had no centralized political or religious organization, but cohesion was given by extensive kinship networks established through marriage, and men and women were **obliged** to many outside the larger divisions of clans and moieties (tribal subdivisions) into which they were born and into which the social group was divided by matrilineal or patrilineal descent. Thus families built up riches by marriage without any one family acquiring a dominant position

32. According to paragraph 3, what held together the collections of village communities?

- ☐ The control of one dominant family
- ☐ The establishment of wide kinship networks through marriage

- ☐ The participation in common religious practices
- ☐ The establishment of a central administrative authority

33. The word “autonomous” in the passage is closest in meaning to

- ☐ small
- ☐ well organized
- ☐ independent
- ☐ wealthy

34. The word “obliged” in the passage is closest in meaning to

- ☐ required
- ☐ allowed
- ☐ forbidden
- ☐ advised

Paragraph 4

Totem poles (see figure below), the most distinctive artistic product of the Northwest, were conspicuous declarations of prestige and of the genealogy (family history) by which it had been attained. These magnificent sculptures that probably originated as funerary monuments were first described by travelers in the late eighteenth century. Each one was carved from a single trunk of cedar, and the increasing availability of metal tools both permitted and encouraged more complex compositions and greater height—up to 27.4 meters. Their superimposed figures—eagles, beavers, whales, and so on—were crests (symbols of identity) that a chief inherited from his lineage, his clan, and his moiety. They were not objects of worship, though the animals carved on them might represent guardian spirits. Poles were designed according to a governing principle of bilateral symmetry, with their various elements interlocked so that they seem to grow organically out of one another, creating a unity of symbolism, form, and surface.

35. According to paragraph 4, what was the significance of totem poles in the culture of the northwestern peoples?

- ☐ They showed a family’s status and history.
- ☐ They were thought to increase success in hunting.
- ☐ They were objects of worship.
- ☐ They were battle monuments.

36. According to paragraph 4, all of the following were true of totem poles EXCEPT:

- ☐ Each was made from a single tree.
- ☐ They sometimes featured images of animals thought to provide protection against harm.
- ☐ Larger numbers began to be made after metal tools became increasingly available.
- ☐ Their decorative designs became more complex over time.

Paragraph 5

Masks (see figure above) are the most varied of the carvings from the Northwest, where they were an essential part of communal life. In style they range from an almost abstract symbolism to

combinations of human and animal features and to a lifelike naturalism sometimes bordering on caricature (a style that strongly exaggerates features or characteristics), taken to its extreme in Tlingit war helmets. Some differences must have been due to those among the cultures in which they were created, but their place of origin cannot always be ascertained as they seem to have passed from one contiguous nation to another in the course of trade or warfare. Although carvers worked according to established conventions, no two masks are identical and those with basic similarities reveal varying degrees of skill.

37. The author mentions “Tlingit war helmets” in the passage to

- ☐ explain why masks were an essential part of communal life
- ☐ provide an example of masks representing a stylistic extreme
- ☐ identify one of the uses of masks
- ☐ provide an example of masks characterized by abstract symbolism

Paragraph 5

Masks (see figure above) are the most varied of the carvings from the Northwest, where they were an essential part of communal life. In style they range from an almost abstract symbolism to combinations of human and animal features and to a lifelike naturalism sometimes bordering on caricature (a style that strongly exaggerates features or characteristics), taken to its extreme in Tlingit war helmets. Some differences must have been due to those among the cultures in which they were created, but their place of origin cannot always be ascertained as they seem to have passed from one contiguous nation to another in the course of trade or warfare. Although carvers worked according to established conventions, no two masks are identical and those with basic similarities reveal varying degrees of skill.

38. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Some characteristics of masks must be due to cultural and national factors, such as whether they were made during a period of warfare or of peace and trade.
- ☐ As one contiguous nation exchanged masks with another during trade or warfare, it became more difficult to determine the place of origin of masks.
- ☐ Some differences among masks were probably due to their different origins, but because they passed from one nation to another, their origin cannot always be determined.
- ☐ As masks of different cultural origins passed from one nation to another during trade or warfare, they acquired certain differences depending on how they were used.

Paragraph 6

The major differences between masks were determined by their purpose. Some were representations of chiefs and their ancestors and made to be displayed and treasured as heirlooms. Although they appear to record the styles of facial tattooing customary in different groups, it is difficult to say how far they were intended to be portraits rather than generalized images. Many masks, sometimes quite large, were carved to be worn in dance-dramas that re-enacted and kept alive the cohesive myths of a culture. Often, Tlingit masks were made for

religious leaders and incorporated the animals that were believed to be their spirit helpers. Conjuring up forces of nature from the ocean, the forests, or the sky, they mediated between life on Earth and the inscrutable powers around and above.

39. What can be inferred from paragraph 6 about masks representing chiefs and their ancestors?

- ☐ They were made to be exhibited and appreciated rather than used.
- ☐ They include scenes depicting the heroic achievements of chiefs and their ancestors.
- ☐ They were made from different materials than were other types of masks.
- ☐ They were sometimes worn by non-family members during religious rituals.

40. Paragraph 6 mentions all of the following as purposes of masks EXCEPT:

- ☐ They were made to be worn in dance-dramas.
- ☐ They were used as models for tattoo artists to copy.
- ☐ They were made for religious leaders and showed their animal spirit helpers.
- ☐ They were valued family possessions representing chiefs and their ancestors.

Paragraph 5

■ Masks (see figure above) are the most varied of the carvings from the Northwest, where they were an essential part of communal life. ■ In style they range from an almost abstract symbolism to combinations of human and animal features and to a lifelike naturalism sometimes bordering on caricature (a style that strongly exaggerates features or characteristics), taken to its extreme in Tlingit war helmets. Some differences must have been due to those among the cultures in which they were created, but their place of origin cannot always be ascertained as they seem to have passed from one contiguous nation to another in the course of trade or warfare. ■ Although carvers worked according to established conventions, no two masks are identical and those with basic similarities reveal varying degrees of skill. ■

41. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

If we consider 100 raven masks made by different carvers, for example, some will have been well executed, others less so, and one may have been made by an exceptional carver.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

42. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

The peoples who lived along the northwestern coast of North America had stable communities and a distinctive artistic style.

-
-
-

Answer Choices

The people lived on food from the sea, animals, and wild plants, and used wood from the forests for building construction, boat making, and sculpture.

In the absence of a central political or religious organization, wide kinship networks established through interclan marriage helped to maintain social cohesion.

Totem poles had a chief's symbols of identity imposed on them, and were thus used as objects of worship, whereas masks were used for secular purposes.

Although the peoples of the northwestern communities came from various language groups, their languages were closely related and they could generally understand one another.

The most remarkable artistic products of the northwestern peoples were totem poles and masks, which often represented families and their spirit guides.

Carvers made masks using a principle of bilateral symmetry that resulted in the creation of a unity of symbolism, form, and surface.

Architectural Change in Eighth-Century Japan

Japanese construction techniques and architectural styles changed in the eighth century C.E. from more traditional Japanese models to imported continental (especially Chinese) models. Several factors contributed to this, in particular with respect to the creation of two new capital cities. In essence, changes then occurring in Japanese political life were rendering past arrangements for the rulers' headquarters obsolete, and continental models offered an alternative.

To elaborate, before the eighth century, the elite marriage practice, which was an important instrument of political alliance making, had encouraged rulers to maintain multiple palaces: that of their own family and those of their spouses, who commonly remained at or near their native family headquarters, at least for some years after marriage. These arrangements had the effect of encouraging frequent changes in royal residence as children matured and marriage alliances changed. The customs of multiple palaces and a moveable court were feasible as long as a ruling group was modest in size and its architectural practices relatively simple.

Moreover, because buildings using the traditional construction of thatched roofs and wooden poles placed directly in the ground rotted away in two decades or so, periodic replacement of palaces, shrines, warehouses, gate towers, and fortress walls was essential. The custom of residential mobility was thus not especially wasteful of labor and material resources: when the time came, one simply erected a new building at a new site—reusing valuable timbers as appropriate—and burned the rest. The practical necessity of replacement was given religious sanction because the regular replacement of buildings was regarded as necessary to provide spiritual cleansing of the site.

As rulers of the sixth and seventh centuries expanded their realm, however, they acquired more and more underlings, administrative paraphernalia, weaponry, and tribute goods, and they needed more and more buildings to house them. As the scale of government grew, moreover, it became more important to have these people and resources close at hand where they could be more easily controlled and utilized. Under these circumstances, frequent moves by the court or replacement of buildings became more costly, even prohibitive.

A solution to the problem was advocated by experts from the continent. This was the use of continental principles of urban design and techniques of construction. These produced geometrically laid out capital cities whose major gates and buildings employed stone foundations, mortise-and-tenon framing (a technique for attaching timbers), and tile roofs that largely eliminated the problem of rot and the consequent need for replacement.

One the other hand, to construct cities and buildings of that sort required so much labor and material that their use effectively precluded periodic replacement or the transfer of a royal headquarters from site to site. Nevertheless, the notion of grand buildings and capital cities

became immensely attractive to Japanese rulers during the seventh and eighth centuries. Continental regimes, the glorious new Chinese dynasties most notably, had them: they constituted an expression of political triumph, a legitimizing symbol of the first order. Moreover, the architecture was an integral part of Buddhism, and acceptance of this religion in Japan at this time fostered adoption of its building style.

These several conflicting factors—the need to modify palace and capital arrangements but the difficulty of doing so, the wish to enjoy grandeur but the reluctance to settle for a single, immobile court—all became evident by the mid-seventh century. Change did come, but slowly, and in the end a compromise system was devised. Traditional shrines of Shinto, the native religion of Japan, and many residential buildings continued to be built in the rottable, replaceable style that accommodated religious concerns and taboos, while city gates, major government buildings, and Buddhist temples were built in the continental fashion that met the need for permanence and grandeur. Moreover, the wish of rulers to maintain multiple palaces fit with the custom of certain continental regimes that maintained summer palaces or other regional capitals where rulers could periodically reside on a temporary basis.

Paragraph1

Japanese construction techniques and architectural styles changed in the eighth century C.E. from more traditional Japanese models to imported continental (especially Chinese) models. Several factors contributed to this, in particular with respect to the creation of two new capital cities. In essence, changes then occurring in Japanese political life were rendering past arrangements for the rulers' headquarters obsolete, and continental models offered an alternative.

1. The phrase "In essence" in the passage is closest in meaning to
 - ☐ Actually
 - ☐ Basically
 - ☐ However
 - ☐ Moreover

Paragraph2

To elaborate, before the eighth century, the elite marriage practice, which was an important instrument of political alliance making, had encouraged rulers to maintain multiple palaces: that of their own family and those of their spouses, who commonly remained at or near their native family headquarters, at least for some years after marriage. These arrangements had the effect of encouraging frequent changes in royal residence as children matured and marriage alliances changed. The customs of multiple palaces and a moveable court were feasible as long as a ruling group was modest in size and its architectural practices relatively simple.

2. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - ☐ The elaborate marriage customs of the elite encouraged spouses to remain at their family palace for several years after marriage.

- Rulers maintained multiple palaces for themselves and their spouses' families.
 - Before the eighth century, it was common for the elite to form political alliances with their spouses' families at the native family headquarters for some years after marriage.
 - Before the eighth century, the practice of forming alliances through marriage encouraged rulers to maintain palaces at their spouses' family homes as well as at their own.
3. The word "feasible" in the passage is closest in meaning to
- practical
 - customary
 - acceptable
 - supported

Paragraph 3

Moreover, because buildings using the traditional construction of thatched roofs and wooden poles placed directly in the ground rotted away in two decades or so, periodic replacement of palaces, shrines, warehouses, gate towers, and fortress walls was essential. The custom of residential mobility was thus not especially wasteful of labor and material resources: when the time came, one simply erected a new building at a new site—reusing valuable timbers as appropriate—and burned the rest. The practical necessity of replacement was given religious sanction because the regular replacement of buildings was regarded as necessary to provide spiritual cleansing of the site.

4. In paragraph 3, why does the author discuss the natural decay of the wooden structures built in eighth-century Japan?
- To argue that the necessity of replacing buildings every two decades applied to all eighth-century structures, not just residences.
 - To argue that the custom of residential mobility was not unreasonable given the building practices of the eighth century
 - To explain why the elite of the eighth century had to move periodically to new residences
 - To explain why in the sixth and seventh centuries Japanese architectural practice changed to the construction of more permanent structures
5. According to paragraph 3, each of the following was true of the practice of periodic replacement of buildings EXCEPT:
- It was followed for a wide variety of structures.
 - It involved the reuse of building materials that were still good.
 - Ordinary Japanese considered it as waste of time and energy.
 - Over the years it became a religious ritual.

Paragraph 4

As rulers of the sixth and seventh centuries expanded their realm, however, they acquired more and more underlings, administrative paraphernalia, weaponry, and tribute goods, and they needed more and more buildings to house them. As the **scale** of government grew, moreover, it became more important to have these people and resources close at hand where they could be

more easily controlled and utilized. Under these circumstances, frequent moves by the court or replacement of buildings became more costly, even prohibitive.

6. The word “scale” in the passage is closest in meaning to
- ☐ importance
 - ☐ duties
 - ☐ needs
 - ☐ size
7. According to paragraph 4, what problem did traditional architectural practices create for rulers of the sixth and seventh centuries?
- ☐ It was difficult to bring the necessary people and construction materials together to replace buildings periodically.
 - ☐ It was very expensive to move and house the large number of people that were now associated with the government.
 - ☐ It was impractical to construct buildings large enough to house the growing numbers of people and resources.
 - ☐ It was too time-consuming for rulers to supervise the construction of all the necessary buildings.

Paragraph 5

A solution to the problem was advocated by experts from the continent. This was the use of continental principles of urban design and techniques of construction. These produced geometrically laid out capital cities whose major gates and buildings employed stone foundations, mortise-and-tenon framing (a technique for attaching timbers), and tile roofs that largely eliminated the problem of rot and the consequent need for replacement.

8. The word “advocated” in the passage is closest in meaning to
- ☐ discovered
 - ☐ solved
 - ☐ promoted
 - ☐ questioned

Paragraph 6

One the other hand, to construct cities and buildings of that sort required so much labor and material that their use effectively precluded periodic replacement or the transfer of a royal headquarters from site to site. Nevertheless, the notion of grand buildings and capital cities became immensely attractive to Japanese rulers during the seventh and eighth centuries. Continental regimes, the glorious new Chinese dynasties most notably, had them: they constituted an expression of political triumph, a legitimizing symbol of the first order. Moreover, the architecture was an integral part of Buddhism, and acceptance of this religion in Japan at this time fostered adoption of its building style.

9. According to paragraph 6, Japanese rulers were strongly attracted to continental architecture

because

- ☐ permanent buildings could be constructed at very low cost
 - ☐ adopting the continental architecture would not have an effect on religious practices in Japan
 - ☐ political power could be expressed by constructing grand buildings
 - ☐ important buildings could be replaced quickly by means of the latest technology
10. What can be inferred from paragraph 6 about Japanese rulers during the seventh and eighth centuries?
- ☐ They were well aware of, and strongly influenced by, developments in the royal courts of China.
 - ☐ They strongly opposed the spread of the Buddhist religion.
 - ☐ They saw the influence of continental regimes as a threat to local traditions.
 - ☐ They sought to increase their mobility by adopting changes in architecture.
11. The word “fostered” in the passage is closest in meaning to
- ☐ quickened
 - ☐ initiated
 - ☐ determined
 - ☐ encouraged
12. Which of the following is true of the compromise system mentioned in paragraph 7?
- ☐ Major government buildings combined the techniques of traditional and continental architecture.
 - ☐ The continuing desire of rulers to maintain multiple palaces was taken into account.
 - ☐ The balance of traditional and continental architecture was quickly achieved.
 - ☐ Shinto shrines and most residences were constructed using continental architecture.

Paragraph 7

These several conflicting factors—the need to modify palace and capital arrangements but the difficulty of doing so, the wish to enjoy grandeur but the reluctance to settle for a single, immobile court—all became evident by the mid-seventh century. ■ Change did come, but slowly, and in the end a compromise system was devised. ■ Traditional shrines of Shinto, the native religion of Japan, and many residential buildings continued to be built in the rottable, replaceable style that accommodated religious concerns and taboos, while city gates, major government buildings, and Buddhist temples were built in the continental fashion that met the need for permanence and grandeur. ■ Moreover, the wish of rulers to maintain multiple palaces fit with the custom of certain continental regimes that maintained summer palaces or other regional capitals where rulers could periodically reside on a temporary basis. ■

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

Such temporary residences might have enabled Japanese rulers to better control the

people living far from main capital.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

During the eighth century C.E., there was a significant change in Japanese construction techniques and architectural styles.

-
-
-

Answer Choices

Chinese architectural styles had influenced traditional Japanese architecture long before eighth-century Japanese rulers decided to create larger cities.

As religious ideas changed, it no longer was acceptable to construct buildings out of materials that required constant replacement.

Several factors complicated the architectural change, but a compromise system that considered both traditional and practical needs was eventually developed.

Before the eighth century, the palaces of the elite were relatively simple structures that could be easily built, repaired, and replaced.

Rulers' desire for grand palaces conflicted with the expense of having multiple courts, which they also wanted, but a compromise was achieved in the eighth century.

Many areas in Japan were quick to adopt the changes in architectural styles, while other areas were more reluctant.

Origin of the Solar System

The orderly nature of our solar system leads most astronomers to conclude that the planets formed at essentially the same time and from the same primordial (original) material as the Sun. This material formed a vast cloud of dust and gases called a nebula. The **nebular hypothesis** suggests that all bodies of the solar system formed from an enormous nebular cloud consisting mostly of hydrogen and helium as well as a small percent of all the other heavier elements known to exist. The heavier substances in this frigid cloud of dust and gases consisted mostly of such elements as silicon, aluminum, iron, and calcium—the substances of today's common rocky materials. Also prevalent were other familiar elements, including oxygen, carbon, and nitrogen.

Nearly five billion years ago, some external influence, such as a shock wave traveling from a catastrophic explosion (supernova), may have triggered the collapse of this huge cloud of gases and minute grains of heavier elements, causing the cloud to begin to slowly contract due to the gravitational interactions among its particles. As this slowly spiraling nebula contracted, it rotated faster and faster for the same reason ice-skaters do when they draw their arms toward their bodies. Eventually, the inward pull of gravity came into balance with the outward force caused by the rotational motion of the nebula. By this time the once vast cloud had assumed a flat disk shape with a large concentration of material at its center, called the *protosun* (pre-Sun). Astronomers are fairly confident that the nebular cloud formed a disk because similar structures have been detected around other stars.

During the collapse, gravitational energy was converted to thermal energy (heat), causing the temperature of the inner portion of the nebula to dramatically rise. At such high temperatures, the dust grains broke up into molecules and energized atomic particles. However, at distances beyond the orbit of Mars, the temperatures probably remained quite low. At -200°C , the tiny particles in the outer portion of the nebula were likely covered with a thick layer of ices made of frozen water, carbon dioxide, ammonia, and methane. Some of this material still resides in the outermost reaches of the solar system in a region called the Oort cloud.

The formation of the Sun marked the end of the period of contraction and thus the end of gravitational heating. Temperatures in the region where the inner planets now reside began to decline. The decrease in temperature caused those substances with high melting points to condense into tiny particles that began to coalesce (join together). Such materials as iron and nickel and the elements of which the rock-forming minerals are composed—silicon, calcium, sodium, and so forth—formed metallic and rocky clumps that orbited the Sun. Repeated collisions caused these masses to coalesce into larger asteroid-size bodies, called *protoplanets*, which in a few tens of millions of years accumulated into the four inner planets we call Mercury, Venus, Earth, and Mars. Not all of these clumps of matter were incorporated into the protoplanets. Rocky and metallic pieces that still remain in orbit are called *meteoroids*.

As more and more material was swept up by the inner planets, the high-velocity impact of nebular debris caused the temperatures of these bodies to rise. Because of their relatively high

temperatures and weak gravitational fields, the inner planets were unable to accumulate much of the lighter components of the nebular cloud. The lightest of these, hydrogen and helium, were eventually whisked from the inner solar system by the solar winds.

At the same time that the inner planets were forming, the larger, outer planets (Jupiter, Saturn, Uranus, and Neptune), along with their extensive satellite systems, were also developing. Because of low temperatures far from the Sun, the material from which these planets formed contained a high percentage of ices—water, carbon dioxide, ammonia, and methane—as well as rocky and metallic debris. The accumulation of ices partly accounts for the large sizes and low densities of the outer planets. The two most massive planets, Jupiter and Saturn, had surface gravities sufficient to attract and hold large quantities of even the lightest elements—hydrogen and helium.

Paragraph 1

The orderly nature of our solar system leads most astronomers to conclude that the planets formed at essentially the same time and from the same primordial (original) material as the Sun. This material formed a vast cloud of dust and gases called a nebula. The **nebular hypothesis** suggests that all bodies of the solar system formed from an enormous nebular cloud consisting mostly of hydrogen and helium as well as a small percent of all the other heavier elements known to exist. The heavier substances in this frigid cloud of dust and gases consisted mostly of such elements as silicon, aluminum, iron, and calcium—the substances of today's common rocky materials. Also prevalent were other familiar elements, including oxygen, carbon, and nitrogen.

15. According to paragraph 1, which of the following best describes the “nebular hypothesis”?

Our solar system formed from a large cloud consisting mostly of hydrogen and helium and of small amounts of other elements.

Our solar system formed from gases and heavier elements thrown off by the Sun as it rotated in the center of the nebular cloud.

The primordial matter that evolved into our solar system consisted mostly of familiar elements such as oxygen, carbon, and nitrogen.

A cloud of dust and gases gathered into a rotating nebula composed mostly of the rocky materials seen on Earth today.

16. The word “frigid” in the passage is closest in meaning to

- ☐ moving
- ☐ giant
- ☐ original
- ☐ cold

Paragraph 2

Nearly five billion years ago, some external influence, such as a shock wave traveling from a catastrophic explosion (supernova), may have triggered the collapse of this huge cloud of gases and minute grains of heavier elements, causing the cloud to begin to slowly contract due to the gravitational interactions among its particles. As this slowly spiraling nebula contracted, it rotated

faster and faster for the same reason ice-skaters do when they draw their arms toward their bodies. Eventually, the inward pull of gravity came into balance with the outward force caused by the rotational motion of the nebula. By this time the once vast cloud had assumed a flat disk shape with a large concentration of material at its center, called the *protosun* (pre-Sun). Astronomers are fairly confident that the nebular cloud formed a disk because similar structures have been detected around other stars.

17. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Possibly due to some explosion about five billion years ago, this nebular cloud began to collapse, causing gravitational contractions as its particles interacted.
- ☐ About five billion years ago, a supernova may have exploded, causing a huge cloud of gases and heavier elements to form
- ☐ Gravitational attraction among particles of gases and heavier elements caused some explosive event nearly five billion years ago.
- ☐ About five billion years ago, a shock wave from an external event caused this huge cloud of gases to collapse into small grains of heavier elements.

18. In paragraph 2, why does the author describe how ice-skaters use their arms to increase their speed of rotation?

- ☐ To help describe the armlike structures on a spiraling nebula
- ☐ To help explain why a nebula rotates faster when it contracts
- ☐ To show why spinning ice-skaters are not pulled down by gravity
- ☐ To show how the motion of a nebula differs from that of an ice-skater

19. The word “detected” in the passage is closest in meaning to

- ☐ formed
- ☐ predicted
- ☐ discovered
- ☐ recorded

20. According to paragraph 2, why do astronomers believe that the nebular cloud formed a disk around the protosun?

- ☐ They can still see some debris from the disk.
- ☐ They have observed that disks have formed around other stars.
- ☐ They know that any rotating cloud of gas tends to contract into a disk shape.
- ☐ They have conducted experiments with gravity that have confirmed their belief.

Paragraph 3

During the collapse, gravitational energy was converted to thermal energy (heat), causing the temperature of the inner portion of the nebula to dramatically rise. At such high temperatures, the dust grains broke up into molecules and energized atomic particles. However, at distances beyond the orbit of Mars, the temperatures probably remained quite low. At -200°C , the tiny

particles in the outer portion of the nebula were likely covered with a thick layer of ices made of frozen water, carbon dioxide, ammonia, and methane. Some of this material still resides in the outermost reaches of the solar system in a region called the Oort cloud.

21. According to paragraph 3, which of the following best explains why the inner part of the nebula became hotter as the nebula contracted?

- ☐ The nebula rose in temperature as its speed of rotation increased.
- ☐ Atomic particles in the nebula gave off more heat when they became energized.
- ☐ The dust grains broke up into smaller parts, allowing more light from the Sun to reach the inner portion of the nebula.
- ☐ As the nebula collapsed, its gravitational energy was changed into heat energy

Paragraph 4

The formation of the Sun marked the end of the period of contraction and thus the end of gravitational heating. Temperatures in the region where the inner planets now reside began to decline. The decrease in temperature caused those substances with high melting points to condense into tiny particles that began to coalesce (join together). Such materials as iron and nickel and the elements of which the rock-forming minerals are composed—silicon, calcium, sodium, and so forth—formed metallic and rocky clumps that orbited the Sun. Repeated collisions caused these masses to coalesce into larger asteroid-size bodies, called *protoplanets*, which in a few tens of millions of years accumulated into the four inner planets we call Mercury, Venus, Earth, and Mars. Not all of these clumps of matter were incorporated into the protoplanets. Rocky and metallic pieces that still remain in orbit are called *meteoroids*.

22. According to paragraph 4, the protoplanets formed by

- ☐ gravitational interactions among meteoroids
- ☐ increases in the gravitational pull of the Sun
- ☐ the warming of the nebular cloud
- ☐ collisions between materials orbiting the Sun

23. According to paragraph 4, all of the following occurred when the nebula stopped contracting EXCEPT:

- ☐ Gravitational interactions between heavier elements decreased.
- ☐ Temperatures declined in the area of the inner planets.
- ☐ Some elements formed rocky masses that orbited the Sun.
- ☐ Protoplanets were formed.

Paragraph 6

At the same time that the inner planets were forming, the larger, outer planets (Jupiter, Saturn, Uranus, and Neptune), along with their **extensive** satellite systems, were also developing. Because of low temperatures far from the Sun, the material from which these planets formed contained a high percentage of ices—water, carbon dioxide, ammonia, and methane—as well as rocky and metallic debris. The accumulation of ices partly **accounts for** the large sizes and low densities of the outer planets. The two most massive planets, Jupiter and Saturn, had surface

gravities sufficient to attract and hold large quantities of even the lightest elements—hydrogen and helium.

24. What can be inferred from paragraph 6 about the role that gravity on the outer planets plays in their having low densities?

- ☐ The forces of gravity from the outer planets increase the speed of the solar winds, causing the winds to carry away the heavier, denser elements.
- ☐ The outer planets' strong gravities capture low-density ices from the outer reaches of the solar system.
- ☐ Gravity on the outer planets is strong enough to prevent lighter elements from escaping.
- ☐ The strong gravity on the outer planets results in high-velocity impacts with nebular debris, causing their denser elements to be carried away into space.

25. The word “extensive” in the passage is closest in meaning to

- ☐ developing
- ☐ large
- ☐ complex
- ☐ centralized

26. The phrase “accounts for” in the passage is closest in meaning to

- ☐ explains
- ☐ creates
- ☐ encourages
- ☐ illustrates

Paragraph 2

Nearly five billion years ago, some external influence, such as a shock wave traveling from a catastrophic explosion (supernova), may have triggered the collapse of this huge cloud of gases and minute grains of heavier elements, causing the cloud to begin to slowly contract due to the gravitational interactions among its particles. ■ As this slowly spiraling nebula contracted, it rotated faster and faster for the same reason ice-skaters do when they draw their arms toward their bodies. ■ Eventually, the inward pull of gravity came into balance with the outward force caused by the rotational motion of the nebula. ■ By this time the once vast cloud had assumed a flat disk shape with a large concentration of material at its center, called the *protosun* (pre-Sun). ■ Astronomers are fairly confident that the nebular cloud formed a disk because similar structures have been detected around other stars.

27. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

In other words, as the outer parts of a rotating mass are pulled inward, the speed of rotation increases.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

28. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some answer choices do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage.

This question is worth 2 points.

Drag your choices to the spaces where they belong. To review the passage, click on **View Text**.

Our Sun and planets formed at about the same time from a collection of dust and gases called a nebular cloud.

-
-
-

Answer Choices

The early solar system was made mostly of heavier materials, such as rock-forming minerals, with a small percentage of light elements such as hydrogen, helium, oxygen, carbon, and nitrogen.

As the outer parts of the nebular cloud cooled, they became home to a region of rocky and metallic debris known as meteoroids.

The inner planets formed when certain minerals collided and coalesced into larger bodies with high temperatures and weak gravitational fields that were not able to retain the lightest elements.

Perhaps as the result of an explosive event, the nebular cloud collapsed and began contracting as gravity caused particles to interact.

A supernova sent a shock wave through the nebular cloud, causing it to expand until its heavier elements were forced to the outer solar system.

Being farther from the Sun, the outer planets were cooler than the inner planets, giving them a higher percentage of elements in the form of ice and a large quantity of the lightest gases: hydrogen and helium.

Attempts at Determining Earth's Age

Since the dawn of civilization, people have been curious about the age of Earth. In addition, we have not been satisfied in being able to state merely the relative geologic age of a rock or fossil. Human curiosity demands that we know actual age in years.

Geologists working during the nineteenth century understood rock bodies, they would have to concentrate on natural processes that continue at a constant rate and that also leave some sort of tangible record in the rocks. Evolution is one such process, and geologist Charles Lyell (1797-1875) recognized this. By comparing the amount of evolution exhibited by marine mollusks then, Lyell estimated that 80 million years had elapsed since the beginning of the Tertiary Period. He came astonishingly close to the mark, since it was actually about 65 million years. However, for older sequence of evolutionary development, estimates based on parts in the fossil record. Rates of evolution for many orders of plants and animals were not well understood.

In another attempt, geologists reasoned that if rates of deposition could be determined for sedimentary rocks, they might be able to estimate the time required for deposition of a given thickness of strata, or rock layers. Similar reasoning suggested that one could estimate total elapsed geologic time by dividing the average thickness of sediment transported annually to the oceans into the total thickness of sedimentary rock that had ever been deposited in the past. Unfortunately, such estimates did not adequately account for past difference in rates of sedimentation or losses to the total section of strata during episodes of erosion. Also, some very ancient sediments were no longer recognizable, having been converted to igneous and metamorphic rocks in the course of mountain building. Estimates of Earth's total age based on sedimentation rates ranged from as little as million to over a billion year.

Yet another scheme for approximating Earth's age had been proposed in 1715 by Sir Edmund Halley (1656-1742), whose name we associate with the famous comet. Halley surmised that the ocean formed soon after the origin of the planet and therefore would be only slightly younger than the age of the solid Earth. He reasoned that the original ocean was not salty and that subsequently salt derived from the weathering of rocks was brought to the sea by streams. Thus, if one knew the total amount of salt dissolved in the ocean

and the amount added each year, it might be possible to calculate the ocean's age in 1899, Irish geologist John Joly (1857-1933) attempted the calculation. From information provide by gauges placed at the mouths of streams. Joly was able to estimate the annual increment of salt to the oceans. Then, knowing the salinity of ocean water and the approximate volume of water, he calculated the amount of salt already held in solution in the oceans. An estimate of the age of the ocean was obtained by dividing the total salt in the ocean by the rate of salt added each year. Beginning with essentially non-saline oceans, it would have taken about 90 million years of the oceans to reach their present salinity, according to Joly. The figure, however, was off the currently accepted mark of 4.54 billion by a factor of 50, largely because there was no way to account accurately by recycled salt and salt incorporated into clay mineral deposited on the sea floors. Even though in error, Joly's calculations clearly supported those geologists who insisted on an age for Earth far in excess of a few million years. The belief in Earth's immense antiquity was also supported by Darwin, Huxley, and other evolutionary biologists, who saw the need for time in the hundreds of millions of years to accomplish the organic evolution apparent in the fossil record.

Paragraph 2

Geologists working during the nineteenth century understood rock bodies, they would have to concentrate on natural processes that continue at a constant rate and that also leave some sort of **tangible** record in the rocks. Evolution is one such process, and geologist Charles Lyell (1797-1875) recognized this. BY comparing the amount of evolution exhibited by marine mollusks then, Lyell estimated that 80 million years had elapsed since the beginning of the Tertiary Period. He came astonishingly close to the mark, since it was actually about 65 million years. However, for older **sequence** of evolutionary development, estimates based on parts in the fossil record. Rates of evolution for many orders of plants and animals were not well understood.

1. The word "**tangible**" in the passage is closest in meaning to
 - A. physical
 - B. related
 - C. significant

- D. helpful
2. It can be inferred from paragraph 2 that Charles Lyell based his study of the marine mollusk fossils on which of the following assumptions?
- A. The Tertiary Period was separated into division of time that were equal in length.
 - B. Mollusks lived under rocks in the sea during the Tertiary period.
 - C. Evolution of mollusks proceeded at a uniform rate over time
 - D. Mollusks have evolved less rapidly with the passing of time
3. The word “**sequence**” in the passage is closest in meaning to
- A. observations
 - B. senses
 - C. series
 - D. categories
4. According to paragraph 2, Lyell’s strategy for estimation geologic dates was not very accurate for periods before the Tertiary Period partly because
- A. Marine mollusks did not evolve until the Tertiary Period
 - B. fossil records of the very distant past are incomplete
 - C. there was not much agreement about how to identify or categorize earlier eras
 - D. the duration of previous geologic periods was difficult to determine

Paragraph 3

In **another attempt**, geologists reasoned that if rates of deposition could be determined for sedimentary rocks, they might be able to estimate the time required for deposition of a given thickness of strata, or rock layers. Similar reasoning suggested that one could estimate total elapsed geologic time by dividing the average thickness of sediment transported annually to the oceans into the total thickness of sedimentary rock that had ever been deposited in the past. Unfortunately, such estimates did not adequately account for past difference in rates of sedimentation or losses to the total section of strata during episodes of erosion. Also, some very ancient sediments were no longer recognizable, having been **converted** to igneous and metamorphic rocks in the course of mountain building. Estimates of Earth’s total age based on sedimentation rates ranged from as little as million to over a billion year.

5. The phrase “**another attempt**” in the passage refers to
- A. trying to understand the fossil record
 - B. trying to determine the evolutionary rate of marine mollusks
 - C. trying to understand natural processes
 - D. trying to determine Earth’s actual age
6. The word “**converted**” in the passage is closest in meaning to
- A. added
 - B. changed
 - C. restored
 - D. reduced
7. According to paragraph 3, all of the following were problems with the calculation of Earth’s age using the study of sedimentary rocks EXCEPT
- A. the inconsistency of sedimentation rates over time
 - B. the effect of geologic process on sedimentary rock
 - C. the expansion of some sedimentary rocks due to Earth’s internal heat
 - D. the loss of an unknown number of sedimentary layers due to erosion

Paragraph 4

Yet another scheme for **approximating** Earth’s age had been proposed in 1715 by Sir Edmund Halley (1656-1742), whose name we associate with the famous comet. Halley surmised that the ocean formed soon after the origin of the planet and therefore would be only slightly younger than the age of the solid Earth. He reasoned that the original ocean was not salty and that **subsequently** salt derived from the weathering of rocks was brought to the sea by streams. Thus, if one knew the total amount of salt dissolved in the ocean and the amount added each year, it might be possible to calculate the ocean’s age. In 1899, Irish geologist John Joly (1857-1933) attempted the calculation. From information provided by gauges placed at the mouths of streams, Joly was able to estimate the annual increment of salt to the oceans. Then, knowing the salinity of ocean water and the approximate volume of water, he calculated the amount of salt already held in solution in the oceans. An estimate of the age of the ocean was obtained by dividing the total salt in the ocean by the rate of salt added each year. Beginning with essentially non-saline oceans, it would have taken about 90 million years for the oceans to

reach their present salinity, according to Joly. The figure, however, was off the currently accepted mark of 4.54 billion by a factor of 50, largely because there was no way to account accurately by recycled salt and salt incorporated into clay mineral deposited on the sea floors. Even though in error, Joly's calculations clearly supported those geologists who insisted on an age for Earth far in excess of a few million years. The belief in Earth's immense antiquity was also supported by **Darwin, Huxley, and other evolutionary biologists**, who saw the need for time in the hundreds of millions of years to accomplish the organic evolution apparent in the fossil record.

8. The word "**approximating**" in the passage is closest in meaning to
- A. thinking about
 - B. researching
 - C. estimating
 - D. demonstrating
9. The word "**subsequently**" in the passage is closest in meaning to
- A. later
 - B. furthermore
 - C. evidently
 - D. accidentally
10. According to paragraph 4, John Joly's calculations were founded on all the following EXCEPT
- A. knowing how salty the ocean water is
 - B. estimating how much salt enters the ocean each year
 - C. accounting for the amount of salt that is recycled
 - D. figuring the volume of water contained in the ocean
11. According to paragraph 4, in which of the following ways could Joly's estimate of Earth's age be considered significant?
- A. It proved that Halley's idea about the age of the ocean was fairly accurate.
 - B. It indicated that Earth was much older than some scientists had claimed.
 - C. It was favored by the majority of scientists at the end of the nineteenth century
 - D. It was the basis for much modern research into the salinity of the ocean
12. The author mentions "**Darwin, Huxley, and other evolutionary**

biologists” in order to

- A. Provide evidence that Joly’s calculations inspired scientists working on other lines of scientific inquiry
- B. Support the claim that all of the leading scientists of the time believed that Earth was just over 90 million years old
- C. Argue that Joly’s calculations would have been more exact if he had collaborated with experts in other fields
- D. Provide examples of scientists who believed the age of Earth to be greater than just a few million on year, like Joly, in order to account for their findings

Paragraph 2

Geologists working during the nineteenth century understood rock bodies, they would have to concentrate on natural processes that continue at a constant rate and that also leave some sort of tangible record in the rocks. Evolution is one such process, and geologist Charles Lyell (1797-1875) recognized this. ■By comparing the amount of evolution exhibited by marine mollusks then, Lyell estimated that 80 million years had elapsed since the beginning of the Tertiary Period. He came astonishingly close to the mark, since it was actually about 65 million years. ■However, for older sequence of evolutionary development, estimates based on parts in the fossil record. ■Rates of evolution for many orders of plants and animals were not well understood. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

More fundamentally, Lyell’s evolutionary approach is intrinsically limited because Earth existed long before life and evolution began.

Where would the sentence best fit?

14. **Directions:** An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not

presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Since the dawn of civilization, people have been curious about Earth's age

Answer choices

- A. It was not until the nineteenth century that attempts were made to determine the relative geologic age of rocks and fossils.
- B. Charles Lyell made a good estimate of the Tertiary Period from the fossil record, but his method could not be extended to earlier geological periods.
- C. Attempts were made to calculate Earth's age from the thickness of surviving sedimentary rock and from the current level of the oceans' salinity.
- D. In the nineteenth century, scientists made a number of important, but unsuccessful, attempts to calculate Earth's age from the record of various natural processes.
- E. Darwin and Huxley supported the accuracy of John Joly's Calculation of Earth's age because it agree with their view of how long evolution had been in progress.
- F. Earth's true age, 4.54 billion years, was determined by combining data from the geological and fossil records.

Water Supply in Venice

The city of Venice, built on saltwater marshes and crisscrossed by canals, experienced problems with its water supply for most of its history. One fifteenth-century French traveler noted that “in a city” in which the inhabitants are in water up to their mouths, they often go thirsty “How was the community to solve this important problem?”

Water drawn from the lagoon (the large, shallow body of water between Venice and the Mediterranean Sea) and the canals within the city served many domestic uses such as washing and cooking inventories of even the most modest households list large numbers of buckets, which were emptied and rinsed, the ones used to carry the brackish (somewhat salty) canal water were kept separate from those intended for fresh water. Still, even serving such needs would have been impossible if the canals of Venice had been extremely polluted. The government was obliged to impose controls, and in the early fourteenth century, the Great Council prohibited the washing of all cloth and dyed woolens in the canals, adding that water used for dyeing could not be flushed into the canals. Henceforth dirty water of that sort was to go into the lagoon. Thanks to resistance on the part of the dyers, infractions were many, the law did not reflect common practice. A century later, however, most of the dye works that used blood or indigo (a dark blue dye) had shifted to the periphery of the city, as had all activities “that let off bad odors or smells.”, such as butchering. Blood, carcasses, and spoiled meat were to go into the lagoon. The canals of Venice began to be protected in the name of nascent ecological awareness.

Much more stringent measures were necessary to guarantee a supply of drinking water, however. In the early centuries of settlement in the lagoon basin, the populations depended on wells on the nearby coastal region. By the ninth century, however, with the increase in population density, cisterns became necessary. Basically, the cisterns were large, covered pits dug into the ground and lined with clay to hold water. The cisterns were located in the city, but unlike the wells, the cisterns were not supplied with water from the lagoon, they collected rainwater instead. Cisterns became widespread in the growing city.

Over a period of several hundred years, Venice developed an elaborate

system of cisterns and some of the gutters or pipes that carried rainwater to the cisterns and that, for a single cistern, might extend over an area of several streets. Wealthy households had their own cisterns. In less affluent areas of the city, cisterns were often owned and maintained by neighborhood groups. In crowded parts of the city where landlords offered small houses for rent, one or two cisterns were provided for each street. A network of public cisterns paralleled these private and semiprivate arrangements. Every public square in the city had a cistern to serve the poorest Venetians.

In the thirteenth century, a decision was made to create 50 additional cisterns, primarily in the recently urbanized area at the edge of the city. At the same time, a campaign was launched to repair the existing cisterns. Expansion of the cistern system stopped during much of the fourteenth century as Venice, like other cities in Europe, suffered from bubonic plague. In the fifteenth century, however, a new program of cistern construction and repair was undertaken.

In spite of the expansion of the cistern system, Venice continued to have problems with its water supply, especially during dry periods. Flotillas of boats had to be dispatched to the mouths of nearby rivers—first to the Bottenigo, then to the Brenta—to fetch fresh water. The fresh water was then sold by the bucket or poured into the cisterns. The public authorities made efforts to take bolder action to ensure the supply of fresh water from this parallel source and a number of projects were suggested during the fourteenth and fifteenth centuries to channel river water and even to construct an aqueduct. However, the high cost of such initiatives precluded their execution.

Paragraph 1

The city of Venice, built on saltwater marshes and crisscrossed by canals, experienced problems with its water supply for most of its history. One fifteenth-century French traveler noted that **“in a city” in which the inhabitants are in water up to their mouths, they often go thirsty** “How was the community to solve this important problem?

1. Why does the author include the quotation **“in a city” in which the inhabitants are in water up to their mouths, they often go thirsty**?

A. To indicate that the French traveled to Venice frequently in the fifteenth

century.

- B. To illustrate the opinion of other Europeans about the water situation in Venice
- C. To suggest that the water supply problem of Venice continued well beyond the fifteenth century.
- D. To emphasize how serious the water problem was in Venice.

Paragraph 2

Water drawn from the lagoon (the large, shallow body of water between Venice and the Mediterranean Sea) and the canals within the city served many domestic uses such as washing and cooking inventories of even the most modest households list large numbers of buckets, which were emptied and rinsed, the ones used to carry the brackish (somewhat salty) canal water were kept separate from those intended for fresh water. Still, even serving such needs would have been impossible if the canals of Venice had been extremely polluted. The government was **obliged** to impose controls, and in the early fourteenth century, the Great Council prohibited the washing of all cloth and dyed woollens in the canals, adding that water used for dyeing could not be flushed into the canals. **Henceforth**, dirty water of that sort was to go into the lagoon. Thanks to resistance on the part of the dyers, infractions were many, the law did not reflect common practice. A century later, however, most of the dye works that used blood or indigo(a dark blue dye) had shifted to the periphery of the city, as had all activities “that let off bad odors or smells.”, such as butchering. Blood, carcasses, and spoiled meat were to go into the lagoon. The canals of Venice began to be protected in the name of nascent ecological awareness.

2. The word “**Henceforth**” in the passage is closest in meaning to

- A. Instead of this
- B. In addition
- C. From this time on
- D. In effect

3. The word “**obliged**” in the passage is closest in meaning to

- A. forced
- B. allowed

- C. expected
 - D. Persuaded
4. According to paragraph 2, why did the government place restrictions on dyers?
- A. To protect the city's drinking water
 - B. To prevent the lagoon from being polluted
 - C. To keep canal water clean
 - D. To discourage the use of blood and indigo for dyeing cloth
5. According to paragraph 2, how did dyers respond to the controls imposed by the government?
- A. They switched from using dyes that let off bad odors or smells to new dyes that smelled much better
 - B. They resisted initially but eventually moved most of the dyeing operations outside the city center
 - C. They argued that the government did not consider common practice before imposing the controls
 - D. They started washing cloth and woolens dyed with blood and indigo in the lagoon

Paragraph 3

Much more stringent measures were necessary to guarantee a supply of drinking water, however. In the early centuries of settlement in the lagoon basin, the populations depended on wells on the nearby coastal region. By the ninth century, however, with the increase in population density, cisterns became necessary. Basically, the cisterns were large, covered pits dug into the ground and lined with clay to hold water. The cisterns were located in the city, but unlike the wells, the cisterns were not supplied with water from the lagoon, they collected rainwater instead. Cisterns became widespread in the growing city.

6. It can be inferred from paragraph 3 that wells on the nearby coastal region
- A. were smaller in size than the cisterns located in the city
 - B. served as a water source for the growing number of cisterns in the city
 - C. increased in number as the population density increased

D. Provided enough water for only a relatively small number of people

Paragraph 4

Over a period of several hundred years, Venice developed an elaborate system of cisterns and gome-the gutters or pipes that carried rainwater to the cisterns and that, for a single cistern, might extend over an area of several streets. Wealthy households had their own cisterns. In less affluent areas of the city, cisterns were often owned and maintained by neighborhood groups. In crowded parts of the city where landlords offered small house for rent, one or two cisterns were provided for each street. A network of public cisterns paralleled these private and semiprivate arrangements. Every public square in the city had a cistern to serve the poorest venetians.

7. According to paragraph 4, all of the following were true of Venice's system of cisterns and gome EXCEPT

- A. It was developed over several centuries
- B. It collected rainwater
- C. It was maintained with fees paid by the public
- D. It reflected the social and economic diversity of the city of Venice

Paragraph 5

In the thirteenth century, a decision was made to create 50 additional cisterns, primarily in the recently urbanized area at the edge of the city. At the same time, a campaign was **launched** to repair the existing cisterns. Expansion of the cistern system stopped during much of the fourteenth century as Venice, like other cities in Europe, suffered from bubonic plague. In the fifteenth century, however, a new program of cistern construction and repair was undertaken.

8. The word "**launched**" in the passage is closest in meaning to

- A. paid for
- B. started
- C. proposed
- D. agreed on

9. According to paragraph 5, all of the following had an effect on cisterns in Venice from the thirteenth to the fifteenth century EXCEPT

- A. the construction of cisterns in other cities in Europe
- B. the establishment of programs to construct and repair cisterns
- C. the outbreak of bubonic plague
- D. the urbanization of an area at the edge of the city

Paragraph 6

In spite of the expansion of the cistern system, Venice continued to have problems with its water supply, especially during dry periods. Flotillas of boats had to be dispatched to the mouths of nearby rivers—first to the Bottenigo, then to the Brenta—to fetch fresh water. The fresh water was then sold by the bucket or poured into the cisterns. The public authorities made efforts to take bolder action to **ensure** the supply of fresh water from **this parallel source** and a number of projects were suggested during the fourteenth and fifteenth centuries to channel river water and even to construct an aqueduct. However, the high cost of such initiatives precluded their execution.

10. The phrase “**this parallel source**” refers to
- A. flotillas of boat
 - B. nearby rivers
 - C. the cisterns
 - D. an aqueduct
11. The word “**ensure**” in the passage is closest in meaning to
- A. improve
 - B. increase
 - C. control
 - D. Guarantee
12. According to paragraph 6, how did public authorities respond to problems with the water supply during dry periods?
- A. They sent boats to fetch fresh water from nearby rivers
 - B. They channeled river water into the cisterns
 - C. They constructed an aqueduct
 - D. They sold water from the cisterns in buckets to the public

Paragraph 4

Over a period of several hundred years, Venice developed an elaborate system of cisterns and gome-the gutters or pipes that carried rainwater to the cisterns and that, for a single cistern, might extend over an area of several streets. ■Wealthy households had their own cisterns. ■In less affluent areas of the city, cisterns were often owned and maintained by neighborhood groups. ■In crowded parts of the city where landlords offered small house for rent, one or two cisterns were provided for each street. ■A network of public cisterns paralleled these private and semiprivate arrangements. Every public square in the city had a cistern to serve the poorest venetians.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

The complexity of the cistern system was social as well as physical.

Where would the sentence best fit?

14. **Directions:** An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selected THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

The city of Venice experienced problems with its water supply for most of its history.

Answer Choices

- A. The water from the lagoon between Venice and the Mediterranean Sea could not be used for drinking because it was extremely polluted.
- B. From the ninth to the fifteenth century, Venice developed a system to collect and store rainwater in cisterns for use by the population
- C. Wealthy households were able to build their own cisterns, but everyone else had to use public cisterns located in the city's many squares.
- D. By the early fourteenth century, the water in Venice's canals was becoming too polluted for household use prompting the city council to prohibit the use of the canals by dyers and butchers.
- E. By the fifteenth century, cisterns supplied by rainwater proved to be inadequate

quate, but the cost of the projects proposed for a permanent solution was too high for the projects to be undertaken.

- F. The expansion and repair of the cistern system was interrupted for much of the fourteenth century because of the bubonic plague, a situation that worsened the water supply problem.



The Early History of Motion Pictures

Motion pictures and television are possible because of two quirks of the human perceptual system: the phi phenomenon and persistence of vision. The phi phenomenon refers to what happens when a person sees one light source go out while another one close to the original is illuminated. To our eyes, it looks like the light moves from one place to another. In persistence of vision, our eyes continue to see an image for a split second after the image has disappeared from view. First observed by the ancient Greeks, persistence of vision became more widely known in 1824 when Peter Roget (who also developed the thesaurus) demonstrated that humans begin to retain an image of an object for about one-tenth of a second after the object is taken from view. Following Roget's pronouncement, a host of toys that depended on this principle sprang up in Europe. Bearing fanciful names (the Thaumatrope, the Praxinoscope), these devices made a series of hand-drawn pictures appear to move.

Before long, several people realized that a series of still photographs on celluloid film could be used instead of hand drawing. In 1878 a colorful Englishman later turned American, Edward Muybridge, attempted to settle a \$25,000 bet over whether the four feet of a galloping horse ever simultaneously left the ground. He arranged a series of 24 cameras alongside a racetrack to photograph a galloping horse. Rapidly viewing the series of pictures produced an effect much like that of a motion picture. Muybridge's technique not only settled the bet (the feet did leave the ground simultaneously at certain instances) but also photography. Instead of 24 cameras taking one picture in rapid order, it was Thomas Edison and his assistant, William Dickson, who finally developed what might have been the first practical motion-picture camera and viewing device. Edison was apparently trying to provide a visual counterpart to his recently invented phonograph. When his early efforts did not work out, he turned the project over to his assistant. Using flexible film, Dickson solved the vexing problem of how to move the film rapidly through the camera by perforating its edge with tiny holes and pulling it along by means of sprockets, projections on a wheel that fit into the holes of the film. In 1889 Dickson had perfected a machine called the Kinetoscope and even starred in a brief film demonstrating how it

worked.

These early efforts in the Edison lab were not directed at projecting movies to large crowds. Still influenced by the success of his phonograph, Edison thought a similar device could make a money by showing brief films to one person at a time for a penny a look. Edison built a special studio to produce films for his new invention, and by 1894, Kinetoscope parlors were spring up in major cities. The long-range commercial potential of his invention was lost on Edison. He reasoned that the real money would be made by selling his peep-show machine. If a large number of people were shown the film at the same time, fewer machines would be needed. Developments in Europe proved Edison wrong as inventors there devised large-screen projection devices. Faced with competition, Edison perfected the Vitascope and unveiled it in New York City in 1896.

Early monies were simple snippets of action—acrobats tumbling, horse running, jugglers juggling, and so on. Eventually, the novelty wore off and films became less of an attraction. Public interest was soon rekindled when early filmmakers discovered that movies could be used to tell story. In France, Alice Guy-Blachè produced *The Cabbage Fairy*, a one-minute film about a fairy who produces children in a Cabbage patch, and exhibited it at the Paris International Exhibition in 1896. Guy-Blachè went on to found her own studio in America. Better known is the work of a fellow French filmmaker and magician, Georges Méliès. In 1902 Méliès produced a science-fiction film that was the great-great-grandfather of *Star Wars* and *Star Trek*; it was called *A Trip to the Moon*.

Paragraph1

Motion pictures and television are possible because of two quirks of the human perceptual system: the phi phenomenon and persistence of vision. The phi phenomenon refers to what happens when a person sees one light sources go out while another one close to the original is illuminated. To our eyes, it looks like the light moves from one place to another. In persistence of vision, our eyes continue to see an image for a spit second after the image has disappeared from view. First observed by the ancient Greeks, persistence of vision became more widely known in 1824 when Peter Roget(who also

developed the thesaurus) demonstrated that human begins retain an image of an object for about one-tenth of a second after the object is taken from view. Following Roget's **pronouncement**, a host of toys that depended on this principle sprang up in Europe. Bearing fanciful manes (the Thaumatrope, the Praxinoscope), these devices made a series of hand-drawn pictures appear to move.

1. According to paragraph 1, what is the phi phenomenon?
 - A. The appearance of movement that occurs when one light is turned off while another lights up nearby
 - B. The tendency to see two lights placed close together as coming from a single light source
 - C. The fact that the human eye sees a light source for a split second after it has disappeared
 - D. The impression that there are several light sources when there is actually only one
2. According to paragraph 1, which of the following statements does NOT correctly describe persistence of vision?
 - A. It was originally noticed by the ancient Greeks
 - B. It refers to an image of an object seen by the human eye for one-tenth of a second after the object has disappeared
 - C. It is a scientific principle that was already widely accepted before Peter Roget demonstrated its validity
 - D. It provided the basis for a number of European toys, including the Thaumatrope and the Praxinoscope
3. The word "**pronouncement**" in the passage is closest in meaning to
 - A. statement
 - B. advice
 - C. theory
 - D. Experiment

Paragraph 2

Before long, several people realized that a series of still photographs on celluloid film could be used instead of hand drawing. In 1878 a colorful Englishman later turned American. Edward Muybridge, attempted to settle a

\$25,000 bet over whether **the four feet of a galloping horse ever simultaneously left the ground**. He arranged a series of 24 cameras alongside a racetrack to photograph a galloping horse. Rapidly viewing the series of pictures produced an effect much like that of a motion picture. Muybridge's technique not only settled the bet (the feet did leave the ground simultaneously at certain instances) but also photography. Instead of 24 cameras taking one picture in rapid order, it was Thomas Edison and his assistant, William Dickson, who finally developed what might have been the first practical motion-picture camera and viewing device. Edison was apparently trying to provide a visual **counterpart** to his recently invented phonograph. When his early efforts did not work out, he turned the project over to his assistant. Using **flexible** film, Dickson solved the vexing problem of how to move the film rapidly through the camera by perforating its edge with tiny holes and pulling it along by means of sprockets, projections on a wheel that fit into the holes of the film. In 1889 Dickson had perfected a machine called the Kinetoscope and even starred in a brief film demonstrating how it worked.

4. In paragraph 2, why does the author mention the bet that Edward Muybridge tried to settle about whether "**the four feet of a galloping horse ever simultaneously left the ground**"?

- A. To introduce and explain a fundamental principle of motion-picture photography
- B. To demonstrate that still photographs produced a visual effect that surpassed that of hand-drawn pictures
- C. To emphasize that photographers had to be willing to take risks in order to portray their subjects
- D. To suggest the difficulty of trying to capture animal movement in motion-picture photography

5. The word "**counterpart**" in the passage is closest in meaning to

- A. addition
- B. invention
- C. component
- D. equivalent

6. The word "**flexible**" in the passage is closest in meaning to

- A. connected
 - B. smooth
 - C. bendable
 - D. Delicate
7. According to paragraph 2, how did Muybridge contribute to the development of motion-picture technology?
- A. He invented the first motion-picture camera.
 - B. He demonstrated the technique of taking a series of photographs and viewing them in rapid succession
 - C. He asked Edison and Dickson to create a motion-picture camera that was both practical and economical
 - D. He combined hand drawings and still photographs to create movie-like effects
8. Paragraph 2 suggests that Thomas Edison's early efforts to develop a motion-picture camera failed because he could not figure out how to
- A. display the camera's pictures to an audience
 - B. move the film quickly through the camera
 - C. line the edge of the film with holes that were small enough
 - D. prevent the film from tearing

Paragraph 3

These early efforts in the Edison lab were not directed at projecting movies to large crowds. Still influenced by the success of his phonograph, Edison thought a similar device could make a money by showing brief films to one person at a time for a penny a look. Edison built a special studio to produce films for his new invention, and by 1894, Kinetoscope parlors were spring up in major cities. The long-range commercial potential of his invention was lost on Edison. He reasoned that the real money would be made by selling his peep-show machine. If a large number of people were shown the film at the same time, fewer machines would be needed. Developments in Europe proved Edison wrong as inventors there devised large-screen projection devices. Faced with competition, Edison perfected the Vitascope and unveiled it in New York City in 1896.

9. According to paragraph 3, what were Kinetoscope parlors?

- A. Places where people could pay a penny to view a short film by looking into a machine
 - B. Places where people could gather in crowds to watch short films projected onto large screens
 - C. Special studios where Edison produced films that would be shown by his newly invented machine
 - D. Places where Edison sold his phonographs, peep-show machines, and other popular inventions
10. Which of the following can be inferred from paragraph 3 about the scope?
- A. It was widely used in Europe before being adopted in the United States
 - B. It never made as much money as the equivalent European projection device
 - C. It was a larger version of the original Kinetoscope
 - D. It was designed to show motion pictures to large groups of people

Paragraph 4

Early monies were simple snippets of action—acrobats tumbling, horse running, jugglers juggling, and so on. Eventually, the novelty wore off and films became less of an attraction. Public interest was soon **rekindled** when early filmmakers discovered that movies could be used to tell story. In France, Alice Guy-Blachè produced *The Cabbage Fairy*, a one-minute film about a fairy who produces children in a Cabbage patch, and exhibited it at the Paris International Exhibition in 1896. Guy-Blachè went on to found her own studio in America. Better known is the work of a fellow French filmmaker and magician, Georges Méliès. In 1902 Méliès produced a science-fiction film that was the great-great-grandfather of *Star Wars* and *Star Trek*; it was called *A Trip to the Moon*.

11. The word “**rekindled**” in the passage is closest in meaning to
- A. reported
 - B. renewed
 - C. reinforced
 - D. Rewarded
12. In paragraph 4, the author describes the film *The Cabbage Fairy* in order to

- A. argue for the importance of continuous action to keep audiences interested
- B. suggest that early films were more popular than live performances were
- C. provide an example of one of the first films to tell a story
- D. emphasize how relatively short most early movies were

Paragraph 4

Early movies were simple snippets of action—acrobats tumbling, horse running, jugglers juggling, and so on. Eventually, the novelty wore off and films became less of an attraction. Public interest was soon rekindled when early filmmakers discovered that movies could be used to tell story. ■ In France, Alice Guy-Blaché produced *The Cabbage Fairy*, a one-minute film about a fairy who produces children in a Cabbage patch, and exhibited it at the Paris International Exhibition in 1896. ■ Guy-Blaché went on to found her own studio in America. ■ Better known is the work of a fellow French filmmaker and magician, Georges Méliès. ■ In 1902 Méliès produced a science-fiction film that was the great-great-grandfather of *Star Wars* and *Star Trek*; it was called *A Trip to the Moon*.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Although she directed hundreds of short films and produced hundreds more over the course of her career, she has largely been forgotten.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selected THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.
The phi phenomenon and persistence of vision are two characteristics of the human perceptual system that make motion pictures and television possible.

Answer Choices

- A. When the persistence of vision became widely known, it inspired the

development of toys that made hand-drawn pictures appear to move

- B. The invention of the motion-picture camera led to the discovery that a horse's feet do not leave the ground while the horse is galloping.
- C. The primary competitors in early motion-picture technology were Edison's Kinetoscope and the European-designed and manufactured large-screen projection devices
- D. The motion-picture camera develop from the experiments in sequential photography that were originally done by Edward Muybridge
- E. Later developments in film included a focus on large-screens projection rather than individual viewing machines and narrative films rather than simple action sequences.
- F. French filmmakers Alice Guy-Blachè and Georges started the first two major movie studios in America and in France, respectively.



Early Theories of Continental Drift

Paleozoic 545 to 245 million years ago

Carboniferous 360 to 286 million years ago

Pennsylvanian 320 to 286 million years ago

Permian 286 to 245 million years ago

Mesozoic 245 to 66 million years ago

Jurassic 205 to 144 million years ago

The idea that the past geography of Earth was different from today is not new. The earliest maps showing the east coast of South America and the west coast of Africa probably provided people with the first evidence that continents may have once been joined together, then broken apart and moved to their present positions.

During the late nineteenth century, Austrian geologist Eduard Suess noted the similarities between the Late Paleozoic plant fossils of India, Australia, South Africa, and South America. The plant fossils comprise a unique group of plants that occurs in coal layers just above the glacial deposits on these southern continents. In this book *The Face of the Earth* (1885), he proposed the name “Gondwanaland” (called Gondwana here) for a supercontinent composed of the aforementioned southern landmasses. Suess thought these southern continents were connected by land bridges over which plants and animals migrated. Thus, in his view, the similarities of fossils on these continents were due to the appearance and disappearance of the connecting land bridges.

The American geologist Frank Taylor published a pamphlet in 1910 presenting his own theory of continental drift. He explained the formation of mountain ranges as a result of the lateral movements of continents. He also envisioned the present-day continents as parts of larger polar continents that eventually broke apart and migrated toward equator after Earth’s rotation was supposedly slowed by gigantic tidal forces. According to Taylor, these tidal forces were generated when Earth’s gravity captured the Moon about 100

million years ago. Although we know that Taylor's explanation of continental drift is incorrect, one of his most significant contributions was his suggestion that the Mid-Atlantic Ridge—an underwater mountain chain discovered by the 1872-1876 British *HMS Challenger* expeditions—might mark the site at which an ancient continent broke apart, forming the present-day Atlantic Ocean.

However, it is Alfred Wegener, a German meteorologist, who is generally credited with developing the hypothesis of continental drift. In his monumental book, *The Origin of Continents and Oceans* (1915), Wegener proposed that all landmasses were originally united into a single supercontinent that he named "Pangaea." Wegener portrayed his grand concept of continental movement in a series of maps showing the breakup of Pangaea and the movement of various continents to their present-day locations. What evidence did Wegener use to support his hypothesis of continental drift? First, Wegener noted that the shorelines of continents fit together, forming a large supercontinent and that marine, nonmarine, and glacial rock sequences of Pennsylvanian to Jurassic ages are almost identical for all Gondwana continents, strongly indicating that they were joined together at one time. Furthermore, mountain ranges and glacial deposits seem to match up in such a way that suggests continents could have once been a single landmass. And last, many of the same extinct plant and animal groups are found today on widely separated continents, indicating that the continents must have been in proximity at one time. Wegener argued that this vast amount of evidence from a variety of sources surely indicated the continents must have been close together at one time in the past.

Alexander Du Toit, a South African geologist was one of Wegener's ardent supporters. He noted that fossils of the Permian freshwater reptile "Mesosaurus" occur in rocks of the same age in both Brazil and South Africa. Because the physiology of freshwater and marine animals is completely different, it is hard to imagine how a freshwater reptile could have swum across the Atlantic Ocean and then found a freshwater environment nearly identical to its former habitat. Furthermore, if Mesosaurus could have swum across the ocean, its fossil remains should occur in other localities besides Brazil and South Africa. It is more logical to assume that Mesosaurus lived in lakes in what are now adjacent areas of South America and Africa but were

then united in a single continent.

Despite what seemed to be overwhelming evidence presented Wegener and later Du Toit and others, most geologists at the time refused to entertain the idea that the continents might have moved in the past

Paragraph 2

During the late nineteenth century, Austrian geologist Eduard Suess noted the similarities between the Late Paleozoic plant fossils of India, Australia, South Africa, and South America. The plant fossils comprise a unique group of plants that occurs in coal layers just above the glacial deposits on these southern continents. In this book *The Face of the Earth* (1885), he proposed the name “Gondwanaland” (called Gondwana here) for a supercontinent composed of the aforementioned southern landmasses. Suess thought these southern continents were connected by land bridges over which plants and animals migrated. Thus, in his view, the similarities of fossils on these continents were due to the appearance and disappearance of the connecting land bridges.

1. According to paragraph 2, Eduard Suess believed that similarities of plant and animal fossils on the southern continents were due to
 - A. living in the southern climate
 - B. crossing the land bridges
 - C. fossilization in the coal layers
 - D. movements of the supercontinent

Paragraph 3

The American geologist Frank Taylor published a pamphlet in 1910 presenting his own theory of continental drift. He explained the formation of mountain ranges as a result of the lateral movements of continents. He also envisioned the present-day continents as parts of larger polar continents that eventually broke apart and migrated toward equator after Earth’s rotation was supposedly slowed by gigantic tidal forces. According to Taylor, these tidal forces were **generated** when Earth’s gravity captured the Moon about 100 million years ago. Although we know that Taylor’s explanation of continental drift is incorrect, one of his most significant contributions was his suggestion

that the Mid-Atlantic Ridge—an underwater mountain chain discovered by the 1872-1876 British *HMS Challenger* expeditions—might mark the site at which an ancient continent broke apart, forming the present –day Atlantic Ocean.

2. According to paragraph 3, Frank Taylor believed that
 - A. present-day continents broke off from larger continents and drifted toward the poles due to tidal forces
 - B. the lateral shifting of continents caused the formation of mountain ranges
 - C. polar continents began to join together when Earth's gravity captured the Moon 100 million years ago
 - D. Earth's gravity and speed of rotation created large polar continents
3. Which of the following can be inferred from paragraph 3 about the Mid-Atlantic Ridge?
 - A. It was once above sea level.
 - B. It formed at the same time that Earth's gravity captured the Moon.
 - C. It was much more extensive when it was first formed than it is today.
 - D. It was unknown before the *HMS Challenger* voyages.
4. The word “**generated**” in the passage is closest in meaning to
 - A. strengthened
 - B. released
 - C. produced
 - D. present

Paragraph 4

However, it is Alfred Wegener, a German meteorologist, who is generally credited with developing the hypothesis of continental drift. In his **monumental** book, *The Origin of Continents and Oceans* (1915), Wegener proposed that all landmasses were originally united into a single supercontinent that he named “Pangaea.” Wegener **portrayed** his grand concept of continental movement in a series of maps showing the breakup of Pangaea and the movement of various continents to their present-day locations. What evidence did Wegener use to support his hypothesis of continental drift? First, Wegener noted that the shorelines of continents fit together, forming a large supercontinent and that marine, nonmarine, and glacial rock sequences of Pennsylvanian to Jurassic ages are almost identical

for all Gondwana continents, strongly indicating that they were joined together at one time. Furthermore, mountain ranges and glacial deposits seem to match up in such a way that suggests continents could have once been a single landmass. And last, many of the same extinct plant and animal groups are found today on widely separated continents, indicating that the continents must have been in proximity at one time. Wegener argued that this **vast** amount of evidence from a variety of sources surely indicated the continents must have been close together at one time in the past.

5. The word “**monumental**” in the passage is closest in meaning to
 - A. final
 - B. persuasive
 - C. well-known
 - D. great and significant
6. The word “**portrayed**” in the passage is closest in meaning to
 - A. proved
 - B. formed
 - C. depicted
 - D. defended
7. The word “**vast**” in the passage is closest in meaning to
 - A. enormous
 - B. significant
 - C. convincing
 - D. additional
8. According to paragraph 4, Wegener felt confident that his theory are correct in part because
 - A. contemporary scientists were unable to successfully challenge his evidence
 - B. many different types of evidence seemed to support his theory
 - C. his theory accounted for phenomena that earlier theories could not explain
 - D. he had used the most advanced techniques available to gather his evidence
9. According to paragraph 4, Wegener pointed to all of the following in support of his theory of continental drift EXCEPT:
 - A. Plants and animals now living on some continents appear to be descended from plants and animals that originated on other continents.
 - B. Rock sequences associated with the continents are extremely similar.

- C. The coastlines of some continents seem to fit together.
- D. Mountains on some continents would be adjacent to mountains on other continents if these continents were joined.

Paragraph 5

Alexander Du Toit, a South African geologist was one of Wegener's ardent supporters. He noted that fossils of the Permian freshwater reptile "Mesosaurus" occur in rocks of the same age in both Brazil and South Africa. Because **the physiology of freshwater and marine animals is completely different**, it is hard to imagine how a freshwater reptile could have swum across the Atlantic Ocean and then found a freshwater environment nearly identical to its former habitat. Furthermore, if Mesosaurus could have swum across the ocean, its fossil remains should occur in other localities besides Brazil and South Africa. It is more **logical** to assume that Mesosaurus lived in lakes in what are now adjacent areas of South America and Africa but were then united in a single continent.

10. Why does the author mention the fact that "**the physiology of freshwater and marine animals is completely different**"?
- A. To explain why Du Toit was able to determine that Mesosaurus was a freshwater reptile
 - B. To explain why Du Toit concluded that certain fossils in rocks in Brazil and South Africa were those of the same animal
 - C. To cast doubt on the idea that Mesosaurus could have swum from one landmass to another
 - D. To show Du Toit determined which landmass Mesosaurus originated on
11. The word "**logical**" in the passage is closest in meaning to
- A. satisfactory
 - B. modern
 - C. reasonable
 - D. popular
12. Which of the following can be inferred from paragraph 5 about the Permian Mesosaurus of Brazil and South Africa?
- A. It was the dominant animal in the habitats in which it lived
 - B. It lived in similar environments in both places.

- C. It was a weak swimmer compared with other freshwater reptiles.
- D. Its physiology differed from that of modern freshwater reptiles.

Paragraph 5

Alexander Du Toit, a South African geologist was one of Wegener's ardent supporters. ■He noted that fossils of the Permian freshwater reptile "Mesosaurus" occur in rocks of the same age in both Brazil and South Africa. ■Because the physiology of freshwater and marine animals is completely different, it is hard to imagine how a freshwater reptile could have swum across the Atlantic Ocean and then found a freshwater environment nearly identical to its former habitat. ■Furthermore, if Mesosaurus could have swum across the ocean, its fossil remains should occur in other localities besides Brazil and South Africa. ■It is more logical to assume that Mesosaurus lived in lakes in what are now adjacent areas of South America and Africa but were then united in a single continent.

13. Look at the four squares [■] that indicates where the following sentence could be added to the passage.

In addition to supplying new geological evidence for continental drift, he crafted convincing arguments based on ancient life forms.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provides below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Several theories involving the movement of continents were proposed in the nineteenth and early twentieth centuries

Answer Choices

- A. Early maps showing the coastlines of South America and Africa inspired Eduard Suess to search for fossil evidence that today's southern continents had once been joined in a single landmass.

- B. To Eduard Suess, continental drift accounted for the presence of the same types of fossils on different continents that had at times been connected by land bridges.
- C. Du Toit's study of the freshwater reptile Mesosaurus added to the already considerable body of evidence that Alfred Wegener had gathered in support of the idea of continental drift.
- D. Frank Taylor expanded on Eduard Suess's theory of continental drift by arguing that tidal forces 100 million years ago had broken continents apart and caused the rise of the Mid-Atlantic Ridge.
- E. Alfred Wegener, who first developed the theory of continental drift argued that all landmasses were originally part of a supercontinent that broke up into separate continents.
- F. Early theories of continental drift were not widely accepted at the time because they failed to explain why continents moved.



Bioluminescence in Marine Creatures

At night along the sea's edge, the ocean sometimes seems to glow, as if lit from within. This glow is the result of bioluminescence, a phenomenon exhibited by many of the sea's zooplankton. Bioluminescence is the production of cold light through internal biological processes, as opposed to phosphorescence or fluorescence, both of which are re-emitted light that was initially absorbed from an external source.

Many of the sea's creatures, including squid, dinoflagellates, bacteria, worms, crustaceans, and fish, are known to produce light. The process that marine creatures use to create light is like that of the common firefly and similar to that which creates the luminous green color seen in plastic glow sticks, often used as children's toys or for illumination during nighttime events. When a glow stick is bent, two chemicals mix, react, and create a third substance that gives off light. Bioluminescent organisms do essentially the same thing; they have a substance, called luciferin, that reacts with oxygen in the presence of enzyme, luciferase. When the reaction is complete, a new molecule is formed that gives off light—glowing blue—green in the underwater world. This biologically driven chemical reaction occurs within the organism's special light-producing cells, called photocytes, or light-producing organs, called photophores. Probably one of the most complex light-producing systems is that of the squid. Some squid have both photophores and chromatophores (organs for changing color) with their skin, thus enabling them to control both the color and intensity of the light produced. Recent research has also revealed that in some squid and fish, bioluminescent light may be produced by bacteria that live in a mutually beneficial partnership inside the animal's light organs.

How and why bioluminescence occurs is not fully understood; however, in the undersea realm, it appears to be used in a variety of interesting and ingenious ways. The most commonly observed form of bioluminescence in the sea is the pinpoint sparking of light at night that can create cometlike trails behind moving objects. Almost always, this is the result of dinoflagellates reacting to water motion. The relatively short, momentary displays of light may have evolved to startle, distract, or frighten would-be predators. Collection nets brought up from the sea's depths at night frequently glow

green at great distance. Slowly fading green blobs or pulses of light can be seen coming from the organisms within, often from gelatinous creatures. This type of light display may be used to stun disorient, or lure prey. Like a wide-eyed deer caught on a road and dazed by headlights, undersea creatures living within the ocean's darkness may be momentarily disoriented by short flashes of bioluminescent light. Another of the sea's light-producing organisms is a small copepod (a type of crustacean) named *Sapphirina iris*. In the water, *Sapphirina* creates short flashes of a remarkably rich, azure blue light. But its appearance under a microscope is even more spectacular, the living copepod appears as if constructed of delicately handcrafted, multicolored pieces of stained glass. Within the deep sea, some fish also have a dangling bioluminescent lure or a patch of luminescent skin near the mouth, which may be used to entice unsuspecting prey.

Other sea creatures have both light-sensing and light-producing organs. These creatures are thought to use bioluminescence as a form of communication or as a means of identifying an appropriate mate. In the lantern fish, the pattern of photophores distinguishes one species from another. In other fish, bioluminescence may help to differentiate males from females. The squid uses light as a means of camouflage. By producing light from the photophores on its underside, the squid can match light from above and become nearly invisible to predators looking up from below. Squid, as well as some of the gelatinous zooplankton, have also been known to release luminescent clouds or strands of organic material, possibly as a decoy to facilitate escape. And finally, because what they eat is often bioluminescent, many of the transparent deep-sea creatures have red or black stomachs to hide the potentially flashing contents of ingested bioluminescent creatures. Without such a blacked-out stomach, their digestive organs would flash like a neon sign that says, "Eat me, eat me!"

Paragraph 2

Many of the sea's creatures, including squid, dinoflagellates, bacteria, worms, crustaceans, and fish, are known to produce light. The process that marine creatures use to create light is like that of **the common firefly** and similar to that which creates the luminous green color seen in plastic glow

sticks, often used as children's toys or for illumination during nighttime events. When a glow stick is bent, two chemicals mix, react, and create a third substance that gives off light. Bioluminescent organisms do essentially the same thing; they have a substance, called luciferin, that reacts with oxygen in the presence of enzyme, luciferase. When the reaction is complete, a new molecule is formed that gives off light—glowing blue—green in the underwater world. This biologically driven chemical reaction occurs within the organism's special light-producing cells, called photocytes, or light-producing organs, called photophores. Probably one of the most complex light-producing systems is that of the squid. Some squid have both photophores and chromatophores (organs for changing color) with their skin, thus enabling them to control both the color and intensity of the light produced. Recent research has also revealed that in some squid and fish, bioluminescent light may be produced by bacteria that live in a **mutually beneficial** partnership inside the animal's light organs.

1. Why does the author mention **the common firefly** in the passage?
 - A. To relate the light production of marine creature to that of a familiar light-producing species
 - B. To compare the light production of an insect with the more complex light production of marine creatures
 - C. To provide an example of a species that does not use a chemical reaction to produce light
 - D. To support the point that bioluminescence usually occurs at night
2. Which of the following statements about the chemical reaction that produces bioluminescence is NOT true, according to paragraph 2?
 - A. It occurs when luciferin reacts with oxygen.
 - B. It produces a glowing blue-green light.
 - C. It is much like the process by which children's toys are illuminated.
 - D. It requires organs called chromatophores.
3. Paragraph 2 demonstrates the complexity of certain squid by stating that
 - A. the squid employs its bioluminescent light to prevent bacteria from entering its organs
 - B. the squid controls both the color and intensity of the light it produces
 - C. the squid's light has greater intensity than that of fish

- D. the squid produces the enzyme luciferase when it produces light
4. The phrase “**mutually beneficial**” in the passage is closest in meaning to
- A. helpful to one another
 - B. generally practical
 - C. efficiently balanced
 - D. temporary by design
5. Which of the following statements about bioluminescent creatures is implied by paragraph 2?
- A. Bioluminescent creatures cannot produce light if bacteria enter their light organs.
 - B. Not all bioluminescent creatures have both photophores and chromatophores.
 - C. Most bioluminescent organisms do not need the enzyme luciferase in order to produce light.
 - D. Creatures with light-producing organs are much more common than those that have only light producing cells.

Paragraph 3

How and why bioluminescence occurs is not fully understood; however, in the undersea realm, it appears to be used in a variety of interesting and **ingenious** ways. The most commonly observed form of bioluminescence in the sea is the pinpoint sparking of light at night that can create cometlike trails behind moving objects. Almost always, this is the result of dinoflagellates reacting to water motion. The relatively short, momentary displays of light may have evolved to startle, distract, or frighten would-be predators. Collection nets brought up from the sea's depths at night frequently glow green at great distance. Slowly fading green blobs or pulses of light can be seen coming from the organisms within, often from gelatinous creatures. This type of light display may be used to stun, disorient, or lure prey. Like a wide-eyed deer caught on a road and dazed by headlights, undersea creatures living within the ocean's darkness may be momentarily disoriented by short flashes of bioluminescent light. Another of the sea's light-producing organisms is a small copepod (a type of crustacean) named *Sapphirina iris*. In

the water, *Sapphirina* creates short flashes of a remarkably rich, azure blue light. But its appearance under a microscope is even more spectacular, the living copepod appears as if constructed of **delicately** handcrafted, multicolored pieces of stained glass. Within the deep sea, some fish also have a dangling bioluminescent lure or a patch of luminescent skin near the mouth, which may be used to entice unsuspecting prey.

6. The word “**ingenious**” in the passage is closet in meaning to
- A. inventive
 - B. important
 - C. unusual
 - D. specialized
7. According to paragraph 3, when do dinoflagellates produce pinpoint sparkling displays of light?
- A. When they are caught in collection nets
 - B. When they are hunting food
 - C. When there is a sudden movement of the water around them
 - D. When they are stunned or disoriented
8. According to paragraph 3, what is notable when looking at *Sapphirina iris* under a microscope?
- A. It produces brief flashes of light.
 - B. Its body is a rich azure blue color.
 - C. It has luminescent skin on its mouth.
 - D. It looks as if it is made of glass of many colors.
9. Which of the following is NOT mentioned in paragraph 3 as a possible reason for the use of bioluminescence ?
- A. To provide a means of lighting the dark marine waters
 - B. To momentarily disorient other creatures
 - C. To frighten away potential predators
 - D. To attract prey
10. The word “**delicately**” in the passage is closet in meaning to
- A. individually
 - B. brilliantly
 - C. unusually
 - D. finely

Paragraph 4

Other sea creatures have both light-sensing and light-producing organs. These creatures are thought to use bioluminescence as a form of communication or as a means of identifying an appropriate mate. In the lantern fish, the pattern of photophores distinguishes one species from another. In other fish, bioluminescence may help to differentiate males from females. The squid uses light as a means of camouflage. By producing light from the photophores on its underside, the squid can match light from above and become nearly invisible to predators looking up from below. Squid, as well as some of the gelatinous zooplankton, have also been known to release luminescent clouds or strands of organic material, possibly as a decoy to facilitate escape. And finally, because what they eat is often bioluminescent, many of the transparent deep-sea creatures have red or black stomachs to hide the potentially flashing contents of **ingested** bioluminescent creatures. Without such a blacked-out stomach, their digestive organs would flash like a neon sign that says, "Eat me, eat me!"

11. According to paragraph 4, squid use bioluminescence to
 - A. communicate with other squid
 - B. locate mates
 - C. tell males and females apart
 - D. hide from predators
12. The word "**ingested**" in the passage is closest in meaning to
 - A. remaining
 - B. eaten
 - C. living
 - D. nutritious

Paragraph 4

Other sea creatures have both light-sensing and light-producing organs. These creatures are thought to use bioluminescence as a form of communication or as a means of identifying an appropriate mate. ■In the lantern fish, the pattern of photophores distinguishes one species from another. ■In other fish, bioluminescence may help to differentiate males from females. ■The squid uses light as a means of camouflage. ■By producing light from the photophores on its underside, the squid can match light from above

and become nearly invisible to predators looking up from below. Squid, as well as some of the gelatinous zooplankton, have also been known to release luminescent clouds or strands of organic material, possibly as a decoy to facilitate escape. And finally, because what they eat is often bioluminescent, many of the transparent deep-sea creatures have red or black stomachs to hide the potentially flashing contents of ingested bioluminescent creatures. Without such a blacked-out stomach, their digestive organs would flash like a neon sign that says, “Eat me, eat me!”

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage

Yet, certain species use bioluminescence for exactly the opposite purpose—to blend in with surroundings and become less identifiable.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Many of the sea’s creatures produce light through bioluminescence.

Answer Choices

- A. Bioluminescent creature use a substance called luciferin and an enzyme to produce light in a biologically driven chemical reaction.
- B. Bioluminescence can be used to achieve various and sometimes seemingly opposite goals such as frightening away predators or luring prey.
- C. Bioluminescence can be used by some sea creatures as a form of communication or as a means of identifying an appropriate mate.
- D. The majority of bioluminescent creatures are gelatinous in nature, allowing them to produce light.
- E. The same bioluminescent creature cannot have both light-sensing and light-producing organs.
- F. Sea creatures that eat bioluminescent material possess special chemicals that keep their stomach contents from flashing.

Egypt Circa 3100 B.C.

The city of Memphis, located on the Nile near the modern city of Cairo, was founded around 3100 B.C., as the first capital of a recently united Egypt. The choice of Memphis by Egypt's first kings reflects the site's strategic importance. First, and most obvious, the apex of the Nile River delta was a politically opportune location for the state's administrative center, standing between the united lands of Upper and Lower Egypt and offering ready access to both parts of the country. The older predynastic (pre- 3100 B.C.) centers of power. This and Hierakonpolis, were too remote from the vast expanse of the delta, which had been incorporated into the unified state. Only a city within easy reach of both the Nile valley to the south and the more spread out, difficult terrain to the north could provide the necessary political control that the rulers of early dynastic Egypt (roughly 3000-2600 B.C.) required.

The region of Memphis must have also served as an important node for transport and communications, even before the unification of Egypt. The region probably acted as a conduit for much, if not all, of the river-based trade between northern and southern Egypt. Moreover, commodities (such as wine, precious oils, and metals) imported from the Near East by the royal courts of predynastic Upper Egypt would have been channeled through the Memphis region on their way south. In short, therefore, the site Memphis offered the rulers of the Early Dynastic Period an ideal location for controlling internal trade within their realm, an essential requirement for a state-directed economy that depended on the movement of goods.

Equally important for the national administration was the ability to control communications within Egypt. The Nile provided the easiest and quickest artery of communication, and the national capital was, again, ideally located in this respect. Recent geological surveys of the Memphis region have revealed much about its topography in ancient times. It appears that the location of Memphis may have been even more advantageous for controlling trade, transport, and communications than was previously appreciated. Surveys and

drill cores have shown that the level of the Nile floodplain has steadily risen over the last five millennia. When the floodplain was much lower, as it would have been in predynastic and early dynastic times, the outwash fans (fan-shaped deposits of sediments) of various wadis (stream-beds or channels) that carry water only during rainy periods) would have been much more prominent features on the east bank. The fan associated with the Wadi Hof extended a significant way into the Nile floodplain, forming a constriction in the vicinity of Memphis. The valley may have narrowed at this point to a mere three kilometers, making it the ideal place for controlling river traffic.

Furthermore, the Memphis region seems to have been favorably located for the control not only of river-based trade but also of desert trade routes. The two outwash fans in the area gave access to the extensive wadi systems of the eastern desert. In predynastic times, the Wadi Digla may have served as a trade route between the Memphis region and the Near East, to judge from the unusual concentration of foreign artifacts found in the predynastic settlement of Maadi. Access to, and control of, trade routes between Egypt and the Near East seems to have been a preoccupation of Egypt's rulers during the period of state formation. The desire to monopolize foreign trade may have been one of the primary factors behind the political unification of Egypt. The foundation of the national capital at the junction of an important trade route with the Nile valley is not likely to have been accidental. Moreover, the Wadis Hof and Digla provided the Memphis region with accessible desert pasturage. As was the case with the cities of Hierakonpolis and Elkab, the combination within the same area of both desert pasturage and alluvial arable land (land suitable for growing crops) was a particularly attractive one for early settlement, this combination no doubt contributed to the prosperity of the Memphis region from early predynastic times.

Paragraph 1

The city of Memphis, located on the Nile near the modern city of Cairo, was founded around 3100 B.C., as the first capital of a recently united Egypt. The choice of Memphis by Egypt's first kings reflects the site's strategic importance. First, and most obvious, the apex of the Nile River delta was a politically opportune location for the state's administrative center, standing between the united lands of Upper and Lower Egypt and offering ready access to both parts of the country. The older predynastic (pre- 3100 B.C.) centers of power, This and Hierakonpolis, were too remote from the **vast** expanse of the delta, which had been incorporated into the unified state. Only a city within easy reach of both the Nile valley to the south and the more spread out, difficult terrain to the north could provide the necessary political control that the rulers of early dynastic Egypt (roughly 3000-2600 B.C.) required.

1. The word "**vast**" in the passage is closest in meaning to
 - A. fertile
 - B. huge
 - C. unique
 - D. irregular
2. According to paragraph 1, why was Memphis a better choice for the capital of a united Egypt than either This or Hierakonpolis?
 - A. Memphis was in a better location for maintaining administrative control.
 - B. Memphis had long been a regional administrative center by the time Egypt was united.
 - C. This and Hierakonpolis had never actually been incorporated into the unified state.
 - D. Egyptian rulers had failed to keep political control over This and Hierakonpolis in predynastic times.
3. It can be inferred from paragraph 1 that one consequence of the unification of Egypt was
 - A. the reduction of the strategic importance of older centers of power

- B. the opportunity for the recently united Egypt to become economically self-sufficient
 - C. the increase in political tensions between the rulers of Upper and Lower Egypt
 - D. the reduction of Egypt's dependence upon the Nile for trade and communications
4. Which of the following best describes how paragraph 1 is organized?
- A. Two simultaneous developments are described, as well as the reasons why neither one would have occurred without the other.
 - B. A hypothesis is presented, and then points in favor of that hypothesis as well as points against it are discussed.
 - C. A major event is described, and then the most obvious effects of that event are presented.
 - D. A decision is described, and then one likely motivation for that decision is presented.

Paragraph 2

The region of Memphis must have also served as an important node for transport and communications, even before the unification of Egypt. The region probably acted as a conduit for much, if not all, of the river-based trade between northern and southern Egypt. Moreover, commodities (such as wine, precious oils, and metals) imported from the Near East by the royal courts of predynastic Upper Egypt would have been channeled through the Memphis region on their way south. In short, therefore, the site Memphis offered the rulers of the Early Dynastic Period an ideal location for controlling internal trade within their realm, an essential requirement for a state-directed economy that depended on the movement of goods.

5. According to paragraph 2, when did Egypt import goods from the Near East?
- A. Once internal trade was fully controlled from Memphis

- B. Not until early dynastic Egypt established its state-directed economy
 - C. As early as predynastic times
 - D. Only when local supplies of those goods had been completely used up
6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- A. Thus in Memphis, the rulers of the Early Dynastic Period were ideally placed to control internal trade, which they had to do in order to run their economy.
 - B. Therefore the rulers of the Early Dynastic Period thought Memphis was the ideal location for trade with nearby countries.
 - C. In short, a state-directed economy like that of the Early Dynastic Period requires choosing a single location to which goods can be moved-Memphis, in this case
 - D. In sum, then, a state-directed economy first developed during Egypt's Early Dynastic Period because Memphis was an ideal location for controlling trade.

Paragraph 3

Equally important for the national administration was the ability to control communications within Egypt. The Nile provided the easiest and quickest artery of communication, and the national capital was, again, ideally located in this respect. Recent geological surveys of the Memphis region have revealed much about its topography in ancient times. It appears that the location of Memphis may have been even more advantageous for controlling trade, transport, and communications than was previously **appreciated**. Surveys and drill cores have shown that the level of the Nile floodplain has steadily risen over the last five millennia. When the floodplain was much lower, as it would have been in predynastic and early dynastic times, the outwash fans (fan-shaped deposits of sediments) of various wadis (stream-beds or channels)

that carry water only during rainy periods) would have been much more prominent features on the east bank. The fan associated with the Wadi Hof extended a significant way into the Nile floodplain, forming a constriction in the **vicinity** of Memphis. The valley may have narrowed at this point to a mere three kilometers, making it the ideal place for controlling river traffic.

7. The word “**appreciated**” in the passage is closest in meaning to
- A. proposed
 - B. understood
 - C. approved
 - D. expected
8. According to paragraph 3, recent research into the topography of the Memphis region in ancient times suggests which of the following?
- A. The level of the Nile floodplains was much higher in predynastic and dynastic times than in later times.
 - B. The sediment deposits of wadis were not as noticeable in predynastic and dynastic times than in later times.
 - C. The Nile valley at the point of Memphis was narrower in predynastic and dynastic times than it was in later times
 - D. Frequent rainy periods may have caused a significant reduction of trade traffic during the predynastic and dynastic times.
9. The word “**vicinity**” in the passage is closest in meaning to
- A. center
 - B. fields
 - C. city
 - D. surrounding area

Paragraph 4

Furthermore, the Memphis region seems to have been favorably located for the control not only of river-based trade but also of desert trade routes. The two outwash fans in the area gave access to the extensive wadi systems of the

eastern desert. In predynastic times, the Wadi Digla may have served as a trade route between the Memphis region and the Near East, to judge from the unusual concentration of foreign artifacts found in the predynastic settlement of Maadi. Access to, and control of , trade routes between Egypt and the Near East seems to have been a preoccupation of Egypt's rulers during the period of state formation. The desire to monopolize foreign trade may have been one of the primary factors behind the political unification of Egypt. The foundation of the national capital at the junction of an important trade route with the Nile valley is not likely **to have been accidental**. Moreover, the Wadis Hof and Digla provided the Memphis region with accessible desert pasturage. As was the case with the cities of Hierakonpolis and Elkab, the combination within the same area of both desert pasturage and alluvial arable land (land suitable for growing crops) was a particularly attractive one for early settlement, this combination no doubt contributed to the prosperity of the Memphis region from early predynastic times.

10. According to paragraph 4, which of the following is NOT a reason Memphis was chosen as the capital of a united Egypt?

- A. It was at the junction of a major trade route with the Nile valley.
- B. It was near land that could be used for animal grazing and for growing crops.
- C. The nearby outwash fans led into wadis that could be used as desert trade routes.
- D. Since foreign traders had settled in nearby Maadi, trade between the two cities could be established.

11. The phrase "**to have been accidental**" in the passage is closest in

- A. to have gone wrong
- B. to have been helpful
- C. to have occurred by chance
- D. to have made a difference

12. In paragraph 4, why does the author mention the cities of Hierakonpolis

and Elkab?

- A. To give an indication of the level of prosperity that Memphis is thought to have enjoyed from its earliest days
- B. To compare the Memphis region to them in terms of their similar combinations of characteristics providing advantages for early settlement
- C. To identify the models that the founders of Memphis followed in laying out the national capital
- D. To suggest that the combination of desert pasturage and alluvial arable land in the same area was very common

■ The region of Memphis must have also served as an important node for transport and communications, even before the unification of Egypt. ■ The region probably acted as a conduit for much, if not all, of the river-based trade between northern and southern Egypt. ■ Moreover, commodities (such as wine, precious oils, and metals) imported from the Near East by the royal courts of predynastic Upper Egypt would have been channeled through the Memphis region on their way south. ■ In short, therefore, the site Memphis offered the rulers of the Early Dynastic Period an ideal location for controlling internal trade within their realm, an essential requirement for a state-directed economy that depended on the movement of goods.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

While considerations of political power and ease of administration were decisive in choosing the location of the new capital, the site clearly had other advantages.

Where would the sentence best fit?

14. **Directions:** An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some

sentences do not belong in the summary because the express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Around 3100 B.C. Memphis was chosen for its strategic importance to be the first capital of a recently united Egypt.

Answer Choices

- A. River-based trade from northern Egypt and imported goods going south all passed through the Memphis region, making Memphis an ideal location for controlling trade.
- B. Recent geological surveys suggest that the topographical features of the Memphis region made it particularly well-suited for controlling communications and trade.
- C. The rulers of unified Egypt enjoyed a monopoly over foreign trade because all such trade was required to go through the Wadi Digla, to which the rulers controlled all access.
- D. After Memphis became the capital city, river-based trade along the Nile gained in importance, while land-based desert trade declined in importance.
- E. The Nile, despite a constriction of its valley near Memphis, was the most advantageous route for communication and travel once the floodplain had begun to rise.
- F. While the location of Memphis was agriculturally favorable, it was particularly attractive because it enabled Egypt's rulers to control trade moving through the desert from the Near East.

Population Growth in Nineteenth-Century Europe

Because of industrialization, but also because of a vast increase in agricultural output without which industrialization would have been impossible, Western Europeans by the latter half of the nineteenth century enjoyed higher standards of living and longer, healthier lives than most of the world's peoples. In Europe as a whole, the population rose from 188 million in 1800 to 400 million in 1900. By 1900, virtually every area of Europe had contributed to the tremendous surge of population, but each major region was at a different stage of demographic change.

Improvements in the food supply continued trends that had started in the late seventeenth century. New lands were put under cultivation, while the use of crops of American origin, particularly the potato, continued to expand. Setbacks did occur. Regional agricultural failures were the most common cause of economic recessions until 1850, and they could lead to localized famine as well. A major potato blight (disease) in 1846-1847 led to the deaths of at least one million persons in Ireland and the emigration of another million, and Ireland never recovered the population levels the potato had sustained to that point. Bad grain harvests at the same time led to increased hardship throughout much of Europe.

After 1850, however, the expansion of foods more regularly kept pace with population growth, though the poorer classes remained malnourished. Two developments were crucial. First, the application of science and new technology to agriculture increased. Led by German universities, increasing research was devoted to improving seeds, developing chemical fertilizers, and advancing livestock. After 1861, with the development of land-grant universities in the United States that had huge agricultural programs, American crop-production research added to this mix. Mechanization included the use of horse-drawn harvesters and seed drills, many developed initially in the United States. It also included mechanical cream separators and other food-processing devices that improved supply.

The second development involved industrially based transportation. With trains and steam shipping, it became possible to move foods to needy regions within Western Europe quickly. Famine (as opposed to malnutrition) became a thing of the past. Many Western European countries, headed by Britain, began also to import increasing amounts of food, not only from Eastern Europe, a traditional source, but also from the Americas, Australia, and New Zealand. Steam shipping, which improved speed and capacity, as well as new procedures for canning and refrigerating foods (particularly after 1870), was fundamental to these developments.

Europe's population growth included an additional innovation by the nineteenth century: it combined with rapid urbanization. More and more Western Europeans moved from countryside to city, and big cities grew most rapidly of all. By 1850, over half of all the people in England lived in cities, a first in human history. In one sense, this pattern seems inevitable. Growing numbers of people pressed available resources on the land, even when farmwork was combined with a bit of manufacturing, so people crowded into cities seeking work or other resources. Traditionally, however, death rates in cities surpassed those in the countryside by a large margin, cities had maintained population only through steady in-migration. Thus rapid urbanization should have reduced overall population growth, but by the middle of the nineteenth century this was no longer the case. Urban death rates remained high, particularly in the lower-class slums, but they began to decline rapidly.

The greater reliability of food supplies was a factor in the decline of urban death rates. Even more important were the gains in urban sanitation, as well as measures such as inspection of housing. Reformers, including enlightened doctors, began to study the causes of high death rates and to urge remediation. Even before the discovery of germs, beliefs that disease spread by "miasmas" (noxious forms of bad air) prompted attention to sewers and open garbage. Edwin Chadwick led an exemplary urban crusade for underground sewers in

England in the 1830s. Gradually, public health provisions began to cut into customary urban mortality rates. By 1900, in some parts of Western Europe life expectancy in the cities began to surpass that of the rural areas. Industrial societies had figured out ways to combine large and growing cities with population growth, a development that would soon spread to other parts of the world.

Paragraph 1

Because of industrialization, but also because of a vast increase in agricultural output without which industrialization would have been impossible, Western Europeans by the latter half of the nineteenth century enjoyed higher standards of living and longer, healthier lives than most of the world's peoples. In Europe as a whole, the population rose from 188 million in 1800 to 400 million in 1900. By 1900, virtually every area of Europe had contributed to the tremendous surge of population, but each major region was at a different stage of demographic change.

1. According to paragraph 1, which of the following is true about Europe in the nineteenth century?
 - A. A large increase in food production led to industrialization.
 - B. Population changes occurred at the same pace in the major regions.
 - C. The standard of living rose to the level of that in most parts of the world.
 - D. The tremendous rise in population led to greater agricultural output in every region.

Paragraph 2

Improvements in the food supply continued trends that had started in the late seventeenth century. New lands were put under cultivation, while the use of crops of American origin, particularly the potato, continued to expand. Setbacks did occur. Regional agricultural failures were the most common cause of economic recessions until 1850, and they could lead to localized

famine as well. A major potato blight (disease) in 1846-1847 led to the deaths of at least one million persons in Ireland and the emigration of another million, and Ireland never recovered the population levels the potato had sustained to that point. Bad grain harvests at the same time led to increased hardship throughout much of Europe.

2. According to paragraph 2, which of the following caused the food supply to increase in most of Western Europe during the nineteenth century?
 - A. Replacement of seventeenth-century farming techniques with more modern ones
 - B. Improved grain harvests in most European countries
 - C. Reduced demand for food as a result of a decreased population
 - D. Use of new land to grow crops
3. In paragraph 2, why does the author mention the potato blight that occurred in Ireland?
 - A. To identify a crop that was more successful in the United State than it was in Western Europe
 - B. To support a claim about regional agricultural failures
 - C. To give an example of a problematic trend that had started in the late seventeenth century
 - D. To provide evidence that many countries in Europe experienced a loss of population in the nineteenth century

Paragraph 3

After 1850, however, the expansion of foods more regularly **kept pace with** population growth, though the poorer classes remained malnourished. Two developments were crucial. First, the application of science and new technology to agriculture increased. Led by German universities, increasing research was devoted to improving seeds, developing chemical fertilizers, and advancing livestock. After 1861, with the development of land-grant universities in the United States that had huge agricultural programs,

American crop-production research added to this mix. Mechanization included the use of horse-drawn harvesters and seed drills, many developed initially in the United States. It also included mechanical cream separators and other food-processing devices that improved supply.

4. The phrase “**kept pace with**” in the passage is closest in meaning to
- A. exceeded
 - B. matched the increase in
 - C. increased the rate of
 - D. caused
5. According to paragraph 3, all of the following factors helped the supply of food meet the needs of a growing population EXCEPT
- A. increased agricultural research in Germany
 - B. introduction of new crops
 - C. development of food-processing devices
 - D. agricultural programs in universities in the United States

Paragraph 4

The second development involved industrially based transportation. With trains and steam shipping, it became possible to move foods to needy regions within Western Europe quickly. Famine (as opposed to malnutrition) became a thing of the past. Many Western European countries, headed by Britain, began also to import increasing amounts of food, not only from Eastern Europe, a traditional source, but also from the Americas, Australia, and New Zealand. Steam shipping, which improved speed and **capacity**, as well as new procedures for canning and refrigerating foods (particularly after 1870), was fundamental to these developments.

6. The word “**capacity**” in the passage is closest in meaning to
- A. variety of goods
 - B. distance
 - C. reliability

- D. available storage space
7. According to paragraph 4, famine became less of a problem in Western Europe during the nineteenth century because of
- A. the decline of malnutrition
 - B. the construction of more food-storage facilities
 - C. faster means of transportation
 - D. improved agricultural methods in Eastern Europe

Paragraph 5

Europe's population growth included an additional innovation by the nineteenth century: it combined with rapid urbanization. More and more Western Europeans moved from countryside to city, and big cities grew most rapidly of all. By 1850, over half of all the people in England lived in cities, a first in human history. In one sense, this pattern seems **inevitable** growing numbers of people pressed available resources on the land, even when farmwork was combined with a bit of manufacturing, so people crowded into cities seeking work or other resources. Traditionally, however, death rates in cities surpassed those in the countryside by a large margin, cities had maintained population only through steady in-migration. Thus rapid urbanization should have reduced overall population growth, but by the middle of the nineteenth century this was no longer the case. Urban death rates remained high, particularly in the lower-class slums, but they began to decline rapidly.

8. The word "**inevitable**" in the passage is closest in meaning to
- A. unexplainable
 - B. undesirable
 - C. unavoidable
 - D. unpredictable
9. According to paragraph 5, which of the following factors led to rapid urbanization in the first half of the nineteenth century?

- A. The destruction of many farms due to bad harvests
- B. The reduction in the amount of good-quality farmland
- C. The rise in death rates in the countryside
- D. The lack of jobs in the countryside

Paragraph 6

The greater reliability of food supplies was a factor in the decline of urban death rates. Even more important were the gains in urban sanitation, as well as measures such as inspection of housing. Reformers, including enlightened doctors, began to study the causes of high death rates and to urge remediation. Even before the discovery of germs, beliefs that disease spread by “miasmas” (noxious forms of bad air) prompted attention to sewers and open garbage. Edwin Chadwick led an exemplary urban crusade for underground sewers in England in the 1830s. Gradually, public health provisions began to cut into customary urban mortality rates. By 1900, in some parts of Western Europe life expectancy in the cities began to **surpass** that of the rural areas. Industrial societies had figured out ways to combine large and growing cities with population growth, a development that would soon spread to other parts of the world.

10. The word “**surpass**” in the passage is closest in meaning to
- A. exceed
 - B. influence
 - C. equal
 - D. differ from
11. Which of the following can be inferred from paragraph 6 about underground sewers?
- A. They became common in most of Western Europe in the 1830s.
 - B. They helped reduce deaths caused by disease in cities.
 - C. They led to the discovery that disease could be caused by germs.
 - D. They encouraged people to leave rural areas and move to the cities.

12. Paragraph 6 mentions all of the following as factors that contributed the rapid decline of urban death rates EXCEPT
- A. the greater reliability of food supplies
 - B. improvements in sanitation
 - C. advances in the treatment of disease
 - D. provisions for inspecting houses

The greater reliability of food supplies was a factor in the decline of urban death rates. ■ Even more important were the gains in urban sanitation, as well as measures such as inspection of housing. ■ Reformers, including enlightened doctors, began to study the causes of high death rates and to urge remediation. Even before the discovery of germs, beliefs that disease spread by “miasmas” (noxious forms of bad air) prompted attention to sewers and open garbage. Edwin Chadwick led an exemplary urban crusade for underground sewers in England in the 1830s. ■ Gradually, public health provisions began to cut into customary urban mortality rates. ■ By 1900, in some parts of Western Europe life expectancy in the cities began to surpass that of the rural areas. Industrial societies had figured out ways to combine large and growing cities with population growth, a development that would soon spread to other parts of the world.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Such individual efforts had substantial, concrete effects on society.

Where would the sentence best fit?

14. **Directions:** An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are

not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Western Europe experienced a tremendous growth in population in the nineteenth century.

Answer Choices

- A. Agricultural failures became less damaging after 1850 because of advances in science and technology as well as improvements in the transportation and preservation of foods.
- B. The development of better food-processing technologies allowed many Western European countries to grow their own food without having to import it from other countries.
- C. High death rates in the cities began to decline as food supplies became more reliable and as reformers prompted improvements in sanitation and housing.
- D. Although agricultural failures led to deaths and emigration, population levels were restored within a short time.
- E. As the population in the countryside began increasing faster than the supply of food and living space, people began moving to the cities in search of jobs and other resources.
- F. The improvements in crop-growing methods created new jobs on the farms, causing people from the overcrowded cities to move to the countryside to find those jobs.

Costs of Quitting a Job

Economic theory predicts that when the costs of quitting one's job are relatively low, mobility is more likely. This observation underlines the analysis of the rise in quit rates during periods of prosperity, and the effects of mobility costs can be seen when looking at residential location and job turnover. Industries with high concentrations of employment in urban areas, where a worker's change of employer does not necessarily require investing in a change of residence, appear to have higher rates of job turnover than industries concentrated in nonmetropolitan areas do.

Beyond the costs that can be associated with such measurable characteristics as age and residential location are those that are psychic in nature. These latter costs, though unobservable to the researcher, are very likely to differ widely across individuals. Some people adapt more quickly to new surroundings than others do, for example. Recent studies have found considerable heterogeneity among workers in their propensity to change jobs, with one study reporting that almost half of all permanent separations that took place over a three-year period involved a small number (13 percent) of workers who had three or more separations during the period (in contrast, 31 percent of workers had no separations at all during the period).

It is also possible that the costs of job changing by employees vary internationally. Data suggest that workers in the United States may well be more likely to change employers than workers elsewhere may be. Indeed, data confirm that, on average, American workers have been with their current employers fewer years than workers in most other developed countries, particularly workers in Europe and Japan, have been with theirs. It is not known why Americans are more mobile than most others are, but one possibility relates to the lower levels of company training received by American workers. Another possibility, however, is that the costs of mobility are lower in the United States (despite the fact that Japan and Europe are more densely populated and hence more urban). What would create these lower costs?

One hypothesis that has received at least some investigation is that housing policies in Europe and Japan increase the costs of residential, and therefore job, mobility. Germany, the United Kingdom, and Japan, for

example, have controls on the rent increases that proprietors can charge to existing renters while tending to allow proprietors the freedom to negotiate any mutually agreeable rent on their initial lease with the renter. Thus, it is argued that renters who move typically face very large rent increases in these countries. Similarly, subsidized housing is much more common in these countries than in the United States, but since it is limited relative to the demand for it, those British, German, or Japanese workers fortunate enough to live in subsidized units are reluctant (it is argued) to give them up. The empirical evidence on the implications of housing policy for job mobility, however, is both limited and mixed.

It could also be hypothesized that the United States, Australia, and Canada, all of which exhibit shorter job tenures than do most European countries or Japan, are large, sparsely populated countries that historically have attracted people willing to emigrate from abroad or resettle internally over long distances. In a country of “movers,” moving may not be seen by either worker or employer as an unusual or especially traumatic event.

While questions remain about the causes of different job mobility rates across countries, the social desirability of job mobility can also be debated. On one hand, mobility can be seen as socially useful because it promotes both individual well-being and the quality of job matches. Moreover, the greater the number of workers and employers “in the market” at any given time, the more flexibility an economy has in making job matches that best adapt to a changing environment. Indeed, when focusing on this aspect of job mobility, economists have long worried whether economies have enough mobility. On the other hand, lower mobility costs (and therefore greater mobility) among workers may well serve to reduce the incentives of their employers to provide job training. Whether the presence of job changing costs is a social boon or bane, these costs and the mobility associated with them are factors with which all employers must contend.

Paragraph 1

Economic theory predicts that when the costs of quitting one's job are relatively low, mobility is more likely. This observation underlines the analysis of the rise in quit rates during periods of prosperity, and the effects of mobility

costs can be seen when looking at residential location and job turnover. Industries with high concentrations of employment in urban areas, where a worker's change of employer does not necessarily require investing in a change of residence, appear to have higher rates of job turnover than industries concentrated in nonmetropolitan areas do.

1. According to paragraph 1, people are more likely to quit their jobs in which **TWO** of the following situations? To receive credit, you must select **TWO** answers.
 - A. They are living in good economic times.
 - B. They are moving from urban areas to non-urban areas.
 - C. They are living in urban areas.
 - D. They do not like their employers.
2. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
 - A. Workers in urban areas change jobs less frequently than do those in nonmetropolitan areas because the costs associated with a change of residence are higher in urban areas.
 - B. Industries located in urban areas experience higher rates of job turnover than do those in nonmetropolitan areas because workers can often change employers without having to change where they live.
 - C. Industries located in urban areas tend to have lower rates of job turnover than do those in nonmetropolitan areas because they are more likely to invest in residences for workers.
 - D. Workers in urban industries are likely to change jobs more frequently than are those in nonmetropolitan industries because it is less costly for workers to change residences in urban than in nonmetropolitan areas.

Paragraph 2

Beyond the costs that can be associated with such measurable characteristics as age and residential location are those that are psychic in nature. These latter costs, though unobservable to the researcher, are very likely to differ widely across individuals. **Some people adapt more quickly to new surroundings than others do**, for example. Recent

studies have found considerable heterogeneity among workers in their propensity to change jobs, with one study reporting that almost half of all permanent separations that took place over a three-year period involved a small number (13 percent) of workers who had three or more separations during the period (in contrast, 31 percent of workers had no separations at all during the period).

3. Why does the author note that “**Some people adapt more quickly to new surroundings than others do**”?
 - A. To argue that some people experience little psychic cost when they change jobs
 - B. To support the claim that the psychic cost of changing jobs is likely to differ widely between individuals
 - C. To illustrate why some psychic costs are unobservable to researchers
 - D. To argue that psychic costs alone cannot fully explain changes in mobility rates
4. According to paragraph 2, what do students reveal about the tendency for workers to change jobs?
 - A. It relates more to economic costs than to psychic costs.
 - B. It decreases with age.
 - C. It vanes greatly.
 - D. It increases over time.

Paragraph 3

It is also possible that the costs of job changing by employees vary internationally. Data suggest that workers in the United States may well be more likely to change employers than workers elsewhere may be. Indeed, data confirm that, on average, American workers have been with their current employers fewer years than workers in most other developed countries, particularly workers in Europe and Japan, have been with theirs. It is not known why Americans are more mobile than most others are, but one possibility relates to the lower levels of company training received by American workers. Another possibility, however, is that the costs of mobility are lower in the United States (despite the fact that Japan and Europe are more densely populated and hence more urban). What would create these

lower costs?

5. According to paragraph 3, what is one possible explanation for why American workers change jobs more frequently than workers elsewhere do?
- A. The relatively small percentage of American workers who live in urban areas
 - B. The tendency of American employers to provide less training for workers
 - C. The recent decrease in mobility costs in the United States
 - D. The lower average population density in the United States

Paragraph 4

One hypothesis that has received at least some investigation is that housing policies in Europe and Japan increase the costs of residential, and therefore job, mobility. Germany, the United Kingdom, and Japan, for example, have controls on the rent increases that proprietors can charge to existing renters while tending to allow proprietors the freedom to negotiate any mutually agreeable rent on their initial lease with the renter. Thus, it is argued that renters who move typically face very large rent increases in these countries. Similarly, subsidized housing is much more common in these countries than in the United States, but since it is limited relative to the demand for it, those British, German, or Japanese workers fortunate enough to live in subsidized units are reluctant (**it is argued**) to give them up. The empirical evidence on the implications of housing policy for job mobility, however, is both limited and mixed.

6. According to paragraph 4, what may contribute to high mobility costs in Germany, the United Kingdom, and Japan?
- A. Rent control does not apply to a renter's first lease on a property.
 - B. Governments have significantly reduced housing subsidies.
 - C. There is little control on the rent increases that can be charged after the initial agreement.
 - D. When a renter leaves a housing unit that has been subsidized, the unit will not be subsidized for the next renter.
7. What is the author's purpose in using the parenthetical phrases "**it is argued**"?

- A. To indicate that the view that workers are reluctant to leave subsidized housing is widely accepted
- B. To suggest that the view that workers are reluctant to leave subsidized housing has not been proved true
- C. To emphasize that the view that workers are reluctant to leave subsidized housing was based on careful reasoning
- D. To oppose the view that workers are reluctant to leave subsidized housing

Paragraph 5

It could also be hypothesized that the United States, Australia, and Canada, all of which exhibit shorter job tenures than do most European countries or Japan, are large, **sparsely** populated countries that historically have attracted people willing to emigrate from abroad or resettle internally over long distances. In a country of “movers,” moving may not be seen by either worker or employer as an unusual or especially **traumatic** event.

- 8. By indicating that the United States, Australia, and Canada are “**sparsely**” populated, the author means that the population of these countries
 - A. is widely scattered
 - B. has increased rapidly
 - C. is very large
 - D. has remained stable over the years
- 9. Paragraph 5 supports which of the following ideas about mobility costs for American workers compared with mobility costs for workers in most European countries?
 - A. Mobility costs are higher for American workers because they have shorter job tenures.
 - B. American workers get more help from employers in covering the costs of moving to a new job.
 - C. The psychic costs of resulting internally to take a job are lower for American workers
 - D. The economic costs of emigrating to take a job are higher for American workers

10. The word “**traumatic**” in the passage is closest in meaning to

- A. Important
- B. Unreasonable
- C. Expensive
- D. Upsetting

Paragraph 6

While questions remain about the causes of different job mobility rates across countries, the social desirability of job mobility can also be debated. On one hand, mobility can be seen as socially useful because it promotes both individual well-being and the quality of job matches. Moreover, the greater the number of workers and employers “in the market” at any given time, the more flexibility an economy has in making job matches that best adapt to a changing environment. Indeed, when focusing on this aspect of job mobility, economists have long worried whether economies have enough mobility. On the other hand, lower mobility costs (and therefore greater mobility) among workers may well serve to reduce the **incentives** of their employers to provide job training. Whether the presence of job changing costs is a social boon or bane, these costs and the mobility associated with them are factors with which all employers must contend.

11. According to paragraph 6, high job mobility rates may benefit an economy by

- A. making it more likely that the economic environment will change significantly
- B. encouraging employers to increase the sizes of their workforces
- C. making it more likely that jobs will be filled by people who are suited to them
- D. encouraging workers to improve their skills

12. The word “**incentives**” in the passage is closest in meaning to

- A. authority
- B. obligation
- C. ability
- D. motivation

Paragraph 6

While questions remain about the causes of different job mobility rates across countries, the social desirability of job mobility can also be debated. On one hand, mobility can be seen as socially useful because it promotes both individual well-being and the quality of job matches. ■ Moreover, the greater the number of workers and employers “in the market” at any given time, the more flexibility an economy has in making job matches that best adapt to a changing environment. ■ Indeed, when focusing on this aspect of job mobility, economists have long worried whether economies have enough mobility. ■ On the other hand, lower mobility costs (and therefore greater mobility) among workers may well serve to reduce the incentives of their employers to provide job training. ■ Whether the presence of job changing costs is a social boon or bane, these costs and the mobility associated with them are factors with which all employers must contend.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage

According to this view, greater mobility could result in a less skilled workforce.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selected THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

According to economic theory, workers are more likely to change jobs when the associated costs are low.

Answer Choices

- A. Since job changes by a minority of workers can radically alter overall job mobility rates, job mobility rates tell us little about the mobility costs facing the average worker

- B. Presidential location and individual psychic factors could account for some differences in job mobility among workers
- C. Economists argue strongly for job mobility despite the social and economic costs associated with it.
- D. Outside of the United States, subsidized housing and controls on rent increases explain why there is less job mobility in urban areas than in nonmetropolitan areas.
- E. Possibly due to housing and other differences, mobility costs may be lower in the United States than in most other developing nations which could help account for differences in job mobility.
- F. High job mobility arguably has both benefits, in terms of individual well-being and economic flexibility, and economic costs.



Stone Tools and Pottery Fragments

Aside from ancient buildings, in sheer bulk the largest part of the archaeological record is made up of stone tools and pottery fragments (shards). Stone tools are the earliest known artifacts, having been first used more than two million years ago, and they have remained in use to the present day. When a chunk of fine-grain stone is struck with sufficient force at the proper angle with another rock or with a wood or bone baton, a shock wave will pass through the stone and detach a flake of the desired size and shape. In analyzing ancient stone tools, many archaeologists have mastered the skills needed to make stone tools themselves. Few things are sharper than a fragment struck from fine-grain flint or from obsidian (volcanic glass). Obsidian is so fine grained that flakes of it can have edges only about twenty molecules thick-hundreds of times thinner than steel tools.

Through experimentation, some archaeologists are able to produce copies of almost every stone tool type used in antiquity. A common research strategy is to make flint tools, use them to cut up meat, saw wood, clean hides, bore holes, etc, and then compare the resulting wear traces with the marks found on ancient artifacts. Sometimes electron-scanning microscopes are used to study minute variations in these use marks. Some rough correspondence can be found between the types of uses and the characteristics of wear marks, but there are many ambiguities.

Ethnographic data from people who still use these tools, like one study of how the IKung hunter-gatherers use different styles of stone spear points to identify their different social groupings, indicate that even crude-looking stone tools may reflect a great deal of the social and economic structure.

Ceramics were in use much later than the first stone tools (appearing in quantity in many places about 10,000 years ago), but they were used in such massive quantities in antiquity that, for many archaeologists, work life consists mainly of the slow sorting and analyzing of pottery fragments. Ceramic pots were first made by hand and dried in the sun or in low temperature kilns, a process that did not produce a very durable material. But in many areas of Africa, Asia and Europe high-temperature kilns produced pottery that is nearly a form of glass, and fragments of these pots survive even when the pottery is broken.

Ceramics form such a large part of archaeologists' lives because ceramics express so much about the people who made them. Pots are direct indicators of function in that they show how diets and economies changed over time. Archaeologists have documented how pottery in the American Southwest changed in prehistoric times as a diet developed that included boiled seeds of various native plants, and pottery was developed to withstand the heat and mechanical stresses of the boiling process.

Ceramics are almost always analyzed on the basis of their style. This idea of style is hard to define, but changing styles are the basis on which archaeologists date much of the archaeological record. But for many archaeologists, ceramic styles are more than just convenient devices of dating. For many archaeologists, stylistic decoration of artifacts is the primary means by which one can enter the cognitive world of the ancients. Societies throughout history have invested their objects with styles that have profound and complex meanings and effects. In the case of the Maya and every other early civilization, rulers used particular symbols and styles as mechanisms through which they portrayed, communicated, and implemented their power. In all societies, styles fix social meaning and are powerful ways in which social groups define and construct their culture. Styles of objects, language, and personal behavior identify people in terms of gender, age group, ethnic group, socioeconomic class, and in many other important ways. Some researchers, for example, have argued that a particular kind of pottery, called Ramey incised (which is incised with figures of eyes, fish, arrows, and abstract objects and was used by the people in the area of present-day Missouri and Illinois at about A.D. 900), was primarily used to distribute food but was also used to communicate the idea that the society's elite, for whom the pots were made, were mediators of cosmic forces.

Paragraph 1

Aside from ancient buildings, in sheer bulk the largest part of the archaeological record is made up of stone tools and pottery fragments (shards). Stone tools are the earliest known artifacts, having been first used more than two million years ago, and they have remained in use to the present day. When a chunk of fine-grain stone is struck with sufficient force at the proper

angle with another rock or with a wood or bone baton, a shock wave will pass through the stone and **detach** a flake of the desired size and shape. In analyzing ancient stone tools, many archaeologists have mastered the skills needed to make stone tools themselves. Few things are sharper than a fragment struck from fine-grain flint or from obsidian (volcanic glass). Obsidian is so fine grained that flakes of it can have edges only about twenty molecules thick-hundreds of times thinner than steel tools.

1. The word “**detach**” in the passage is closest in meaning to
 - A. separate
 - B. sharpen
 - C. loosen
 - D. produce
2. According to paragraph 1, each of the following is true of stone tools EXCEPT
 - A. They were first produced more than two million years ago.
 - B. They are still being used today.
 - C. They are made of fine-grained stones such as flint or obsidian.
 - D. Their edges are never as thin or as sharp as those of steel tools.

Paragraph 2

Through experimentation, some archaeologists are able to produce copies of almost every stone tool type used in antiquity. A common research strategy is to make flint tools, use them to cut up meat, saw wood, clean hides, bore holes, etc, and then compare the resulting wear traces with the marks found on ancient artifacts. Sometimes electron-scanning microscopes are used to study **minute** variations in these use marks. Some rough correspondence can be found between the types of uses and the characteristics of wear marks, but there are many **ambiguities**.

3. The word “**minute**” in the passage is closest in meaning to
 - A. unusual
 - B. important
 - C. hidden
 - D. tiny

4. The word “**ambiguities**” in the passage describes things that
 - A. do not make any sense
 - B. are not well documented
 - C. can be interpreted in more than one way
 - D. do not fit with the available data
5. According to paragraph 2, archaeologists make and use their own stone tools in order to
 - A. find out how strong different types of stone tools are
 - B. find out what kinds of tasks such tools were used for in ancient times
 - C. study the copies under electron microscopes and to avoid damaging the originals
 - D. show that ancient multipurpose tools were practical and easy to use

Paragraph 3

Ethnographic data from people who still use these tools, like one study of how the IKung hunter-gatherers use different styles of stone spear points to identify their different social groupings, indicate that even crude-looking stone tools may reflect a great deal of the social and economic structure.

6. Which of the following questions about the IKung is answered in paragraph 3?
 - A. Are the IKung rare among today's hunter-gatherers in using stone tools?
 - B. Is the social structure of the IKung more complex than that of most hunter-gatherer societies?
 - C. Does the IKung's use of several styles of stone tools have a social function?
 - D. Do the IKung use stone tools other than spear points?

Paragraph 4

Ceramics were in use much later than the first stone tools (appearing in quantity in many places about 10,000 years ago), but they were used in such massive quantities in antiquity that, for many archaeologists, work life consists mainly of the slow sorting and analyzing of pottery fragments. Ceramic pots were first made by hand and dried in the sun or in low temperature kilns, a process that did not produce a very durable material. But in many areas of Africa, Asia and Europe high-temperature kilns produced

pottery that is nearly a form of glass, and fragments of these pots survive even when the pottery is broken.

7. According to paragraph 4, which of the following is true of the earliest ceramics?

- A. They were in use before the earliest stone tools.
- B. They were used in only a few places 10,000 years ago.
- C. They appeared in many places about 10,000 years ago.
- D. They were all baked in low-temperature kilns.

8. Paragraph 4 suggests sun-dried pots

- A. interest archaeologists less than stone tools because they are not as old.
- B. account for comparatively few of the pottery fragments that archaeologists study.
- C. are more common in Africa, Asia, or Europe than other parts of the world.
- D. are easier to sort and analyze than pottery made in high-temperature kilns.

Paragraph 5

Ceramics form such a large part of archaeologists' lives because ceramics express so much about the people who made them. Pots are direct indicators of function in that they show how diets and economies changed over time. Archaeologists have documented how pottery in the American Southwest changed in prehistoric times as a diet developed that included boiled seeds of various native plants, and pottery was developed to withstand the heat and mechanical stresses of the boiling process.

9. In paragraph 5, the author discusses pottery in the American Southwest in order to

- A. emphasize that ceramics are usually the only means of reconstructing the diet and economic activity of prehistoric peoples.
- B. argue that changes in the style of pottery are usually a result of changes in food preparation.
- C. explain why certain types of prehistoric pottery have been able to survive better than others.
- D. support the claim that ceramics provide important dining and economical information about their users.

Paragraph 6

Ceramics are almost always analyzed on the basis of their style. This idea of style is hard to define, but changing styles are the basis on which archaeologists date much of the archaeological record. But for many archaeologists, ceramic styles are more than just convenient devices of dating. For many archaeologists, stylistic decoration of artifacts is the primary means by which one can enter the cognitive world of the ancients. Societies throughout history have invested their objects with styles that have profound and complex meanings and effects. In the case of the Maya and every other early civilization, rulers used particular symbols and styles as mechanisms through which they **portrayed**, communicated, and implemented their power. In all societies, styles fix social meaning and are powerful ways in which social groups define and construct their culture. Styles of objects, language, and personal behavior identify people in terms of gender, age group, ethnic group, socioeconomic class, and in many other important ways. Some researchers, for example, have argued that a particular kind of pottery, called Ramey incised (which is incised with figures of eyes, fish, arrows, and abstract objects and was used by the people in the area of present-day Missouri and Illinois at about A.D. 900), was primarily used to distribute food but was also used to communicate the idea that the society's elite, for whom the pots were made, were mediators of cosmic forces.

10. The word "**portrayed**", in the passage is closest in meaning to
- A. claimed
 - B. represented
 - C. defended
 - D. established
11. According to paragraph 6, decorating artifacts in particular styles served each of the following functions in ancient societies EXCEPT
- A. to maintain the power of the societies' rulers
 - B. to mark socially relevant distinctions between groups
 - C. to establish the superiority of a society's artistic values
 - D. to define important aspects of the society's culture

12. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- A. Some researchers argue that pottery of the Ramey Incised style was used not only to distribute food but also to communicate that the society's elite were mediators of cosmic forces.
 - B. Some researchers argue that the figures of eyes, fish, arrows and abstract objects were incised on Ramey Incised pottery to communicate that those who provided the pots were members of the elite.
 - C. Some researchers argue that the primary function of Ramey Incised pottery was food distribution rather than a way of communicating the status of the society's elite.
 - D. Some researchers argue that, based on the kinds of symbols incised on pottery, that pottery was sometimes used to mediate cosmic forces.

Paragraph 6

Ceramics are almost always analyzed on the basis of their style. This idea of style is hard to define, but changing styles are the basis on which archaeologists date much of the archaeological record. But for many archaeologists, ceramic styles are more than just convenient devices of dating. For many archaeologists, stylistic decoration of artifacts is the primary means by which one can enter the cognitive world of the ancients. Societies throughout history have invested their objects with styles that have profound and complex meanings and effects. ■ In the case of the Maya and every other early civilization, rulers used particular symbols and styles as mechanisms through which they portrayed, communicated, and implemented their power. ■ In all societies, styles fix social meaning and are powerful ways in which social groups define and construct their culture. ■ Styles of objects, language, and personal behavior identify people in terms of gender, age group, ethnic group, socioeconomic class, and in many other important ways. ■ Some researchers, for example, have argued that a particular kind of pottery, called Ramey incised (which is incised with figures of eyes, fish, arrows, and abstract objects and was used by the people in the area of present-day Missouri and Illinois at about A.D 900), was primarily used to distribute food but was also

used to communicate the idea that the society's elite, for whom the pots were made, were mediators of cosmic forces. Where would the sentence best fit?

13. Look at the four squares [■] that indicates where the following sentence could be added to the passage.

Maya rulers, for example, used symbols on their war banners to communicate that they had great power because of the support of patron gods.

14. Direction: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.**

Stone tools and pottery fragments are two of the most common kinds of archaeological finds.

Answer Choices

- A. Stone tools are the oldest known artifacts, predating the development of ceramics by about two million years.
- B. The styles of stone tools produced by prehistoric peoples are more reliable indicators of their society's social and economic structure than the styles of their ceramics.
- C. Some researchers believe that the figures and symbols found on pottery may have multiple meanings, but this has not yet been fully established.
- D. Archaeologists know how stone tools were made and can produce copies themselves, but it is often difficult to determine how any particular ancient tool was used.
- E. The earliest evidence of ceramic production comes from Africa, Asia, and Europe, where the development of specialized technologies made pots nearly indestructible.
- F. Because pottery had both practical and symbolic uses, it can tell researchers a lot about the diet, economy, and social structure of ancient societies.

The Role of the Ocean in Controlling Climate

To predict what the climate will be like in the future, scientists must rely on sophisticated computer models. These models use mathematical equations to represent physical processes and interactions in the atmosphere, ocean, and on land. A starting point is usually based on current measurements or estimates of past conditions. Then, using a spherical grid laid out over the entire globe, thousands of calculations are performed at grid intersections to represent and assess how conditions in the air, in the sea, and on land will change over time. Because of their complexity and size, supercomputers are used to run full-scale climate models. Much of the uncertainty in their outputs comes from the way that various aspects of the climate are represented by different models, and even more so, because there are aspects of climate that are not well understood—one of which is how the ocean impacts climate.

The ocean's role in global warming stems principally from its huge capacity to absorb carbon dioxide and to store and transport heat. In the sea, photosynthesis by marine plants and algae, especially phytoplankton, removes great quantities of carbon dioxide from the atmosphere. Hence, the greater the growth (productivity) of phytoplankton in the sea, the greater the removal of carbon dioxide. But what controls the ocean's productivity? There are several limiting factors, but results from a recent experiment suggest that in areas of the ocean where other nutrients are plentiful, iron may be one of the most important and, until recently, unrecognized variables controlling phytoplankton production. Some have proposed a radical, highly controversial and uncertain means to counteract global warming—adding iron to the oceans to induce phytoplankton blooms. Perhaps increased phytoplankton growth would use up a significant amount of carbon dioxide in the atmosphere, but perhaps not, and there might well be side effects that could be detrimental to the ocean ecosystem.

Within the ocean, the production of limestone, in the form of calcium carbonate skeletons or shells, also reduces atmospheric carbon dioxide. However, when deposits of limestone become exposed and weathered on land or are recycled in the sea, carbon dioxide is released back into the atmosphere. What is not well understood is how much carbon dioxide resides in the sea and at what rate it is taken up and recycled. Relatively new research has also

discovered beneath the sea a new and potentially significant threat to skyrocketing Earth temperature: gas hydrates. Gas hydrates are a solid, crystalline form of water, like ice, except that they contain additional gas, typically methane, and are often found stored in ocean sediments. Increased ocean temperatures could cause gas hydrates to dissociate, releasing massive amounts of methane gas into the atmosphere and cause undersea landslides in the process. Consequently, hydrates may, if released, significantly increase global warming as well as create a geologic hazard to offshore drilling operations.

The ocean is also a great reservoir and transporter of heat. Heat from the ocean warms the atmosphere and fuels tropical storms. Heat is transported by currents from the equator to the poles. Ocean circulation is strongly controlled by wind and by the sea's balance of salt and heat. Scientists think that climate warming may slow down circulation, while cooling may speed it up, but these responses are not well understood. Evaporation from the ocean also supplies the precipitation that creates fields of snow and ice at high latitudes. Snow and ice coverage change the reflectivity Earth's surface and are an important influence on how much incoming radiation is either absorbed or reflected. Furthermore, clouds and water vapor in the atmosphere come mainly from the sea and strongly influence climate. Surprisingly, clouds are one of the least understood and most poorly modeled parts of the climate change equation. Most climate modeling grids fail to take into account common-sized cloud formations. Aerosols, tiny particles of soot, dust, and other materials, are thought to seed cloud formation scatter incoming radiation and promote cooling, but this effect, which would counteract warming, is also only superficially understood. Computer models of climate change must take into account all of the processes within the ocean, over land, and in the sky that potentially influence warming. No wonder there is such uncertainty.

Paragraph 1

To predict what the climate will be like in the future, scientists must rely on sophisticated computer models. These models use mathematical equations to represent physical processes and interactions in the atmosphere, ocean, and on land. A starting point is usually based on current measurements or

estimates of past conditions. Then, using a spherical grid laid out over the entire globe, thousands of calculations are performed at grid intersections to represent and assess how conditions in the air, in the sea, and on land will change over time. Because of their complexity and size, supercomputers are used to run full-scale climate models. Much of the uncertainty in their outputs comes from the way that various aspects of the climate are represented by different models, and even more so, because there are aspects of climate that are not well understood—one of which is how the ocean impacts climate.

1. According to paragraph 1, the results of full-scale climate models are questionable in part because
 - A. the supercomputers used for such modeling are large and complex
 - B. thousands of calculations have to be performed to assess conditions
 - C. past conditions cannot always be estimated accurately
 - D. there are multiple ways to represent the same aspect of climate

Paragraph 2

The ocean's role in global warming stems **principally** from its huge capacity to absorb carbon dioxide and to store and transport heat. In the sea, photosynthesis by marine plants and algae, especially phytoplankton, removes great quantities of carbon dioxide from the atmosphere. Hence, the greater the growth (productivity) of phytoplankton in the sea, the greater the removal of carbon dioxide. But what controls the ocean's productivity? There are several limiting factors, but results from a recent experiment suggest that in areas of the ocean where other nutrients are plentiful, iron may be one of the most important and, until recently, unrecognized variables controlling phytoplankton production. Some have proposed a radical, highly **controversial** and uncertain means to counteract global warming—adding iron to the oceans to **induce** phytoplankton blooms. Perhaps increased phytoplankton growth would use up a significant amount of carbon dioxide in the atmosphere, but perhaps not, and there might well be side effects that could be detrimental to the ocean ecosystem.

2. The word “**principally**” in the passage is closest in meaning to
 - A. obviously
 - B. apparently

- C. mainly
- D. originally
3. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
- A. Iron may be one of the most important factors in controlling phytoplankton production in ocean waters that are rich in other nutrients.
- B. Results from a recent experiment suggest that several factors limiting phytoplankton production in ocean waters have gone unrecognized.
- C. Although it was not recognized until recently, nutrients are plentiful in areas of the ocean where iron controls phytoplankton production.
- D. Until recently, the importance of iron was not taken into account in experiments concerning phytoplankton production.
4. The word “**controversial**” in the passage is closest in meaning to
- A. experimental
- B. fascinating
- C. producing disagreement
- D. demonstrating poor judgment
5. The word “**induce**” in the passage is closest in meaning to
- A. supply nutrients to
- B. cause the formation of
- C. expand
- D. strengthen
6. According to paragraph 2, how might increasing phytoplankton growth help lower global temperatures?
- A. By cooling the oceans
- B. By decreasing carbon dioxide levels in the ocean
- C. By reducing the amount of carbon dioxide in the atmosphere
- D. By transporting heat from the ocean’s surface to deeper levels

Paragraph 3

Within the ocean, the production of limestone, in the form of calcium carbonate skeletons or shells, also reduces atmospheric carbon dioxide. However, when deposits of limestone become exposed and weathered on land

or are recycled in the sea, carbon dioxide is released back into the atmosphere. What is not well understood is how much carbon dioxide resides in the sea and at what rate it is taken up and recycled. Relatively new research has also discovered beneath the sea a new and potentially significant threat to skyrocketing Earth temperature: gas hydrates. Gas hydrates are a solid, crystalline form of water, like ice, except that they contain additional gas, typically methane, and are often found stored in ocean sediments. Increased ocean temperatures could cause gas hydrates to dissociate, releasing massive amounts of methane gas into the atmosphere and cause undersea landslides in the process. Consequently, hydrates may, if released, significantly increase global warming as well as create a geologic hazard to offshore drilling operations.

7. According to paragraph 3, which of the following reduces atmospheric carbon dioxide?
- A. The weathering of limestone
 - B. The production of limestone
 - C. The recycling of carbon dioxide
 - D. The presence of methane in gas hydrates
8. According to paragraph 3, why are gas hydrates a possible threat to the global climate?
- A. If disturbed by offshore drilling, they can destroy limestone deposits.
 - B. They can replace regular ice at certain locations.
 - C. If melted, they may release a lot of carbon dioxide into the atmosphere.
 - D. They contain a lot of methane, which may be released as the ocean warms

Paragraph 4

The ocean is also a great reservoir and transporter of heat. Heat from the ocean warms the atmosphere and **fuels** tropical storms. Heat is transported by currents from the equator to the poles. Ocean circulation is strongly controlled by wind and by the sea's balance of salt and heat. Scientists think that climate warming may slow down circulation, while cooling may speed it up, but these responses are not well understood. Evaporation from the ocean also supplies the precipitation that creates fields of snow and ice at high latitudes. Snow and ice coverage change the reflectivity Earth's surface and

are an important influence on how much incoming radiation is either absorbed or reflected. Furthermore, clouds and water vapor in the atmosphere come mainly from the sea and strongly influence climate. Surprisingly, clouds are one of the least understood and most poorly modeled parts of the climate change equation. **Most climate modeling grids fail to take into account common-sized cloud formations.** Aerosols, tiny particles of soot, dust, and other materials, are thought to seed cloud formation scatter incoming radiation and promote cooling, but this effect, which would counteract warming, is also only superficially understood. Computer models of climate change must take into account all of the processes within the ocean, over land, and in the sky that potentially influence warming. No wonder there is such uncertainty.

9. The word “**fuels**” in the passage is closest in meaning to
 1. provides energy for
 2. determines the route of
 3. carries
 4. breaks up
10. Which of the following is NOT mentioned in paragraph 4 as a way in which the ocean affects the climate?
 - A. It stores heat
 - B. It moves heat from the equator toward the poles.
 - C. It speeds up wind circulation.
 - D. It warms up the atmosphere.
11. Paragraph 4 suggests that a significant decrease in snow and ice fields at high latitudes would have what effect?
 - A. More clouds and water vapor would be produced in the atmosphere.
 - B. More of the Sun’s radiation would be absorbed by Earth.
 - C. The oceans would cool more quickly.
 - D. More precipitation would occur at low latitudes.
12. Why does the author mention that “**Most climate modeling grids fail to take into account common-sized cloud formations**”?
 - A. To suggest why the influence of clouds on climate change is still undetermined
 - B. To explain why research on climate change does not focus on clouds

- C. To help explain why it is unclear whether aerosols have the effect of counteracting warming
- D. To explain in part why scientists are uncertain how much incoming radiation is absorbed or reflected

Paragraph 3

Within the ocean, the production of limestone, in the form of calcium carbonate skeletons or shells, also reduces atmospheric carbon dioxide. ■ However, when deposits of limestone become exposed and weathered on land or are recycled in the sea, carbon dioxide is released back into the atmosphere. ■ What is not well understood is how much carbon dioxide resides in the sea and at what rate it is taken up and recycled. ■ Relatively new research has also discovered beneath the sea a new and potentially significant threat to skyrocketing Earth temperature: gas hydrates. ■ Gas hydrates are a solid, crystalline form of water, like ice, except that they contain additional gas, typically methane, and are often found stored in ocean sediments. Increased ocean temperatures could cause gas hydrates to dissociate, releasing massive amounts of methane gas into the atmosphere and cause undersea landslides in the process. Consequently, hydrates may, if released, significantly increase global warming as well as create a geologic hazard to offshore drilling operations.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage

Nor is carbon dioxide the only gaseous substance in the ocean that may affect climate.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selected THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

The oceans affect the climate in numerous ways, some of which are poorly understood and therefore cannot be accurately modeled in computer climate programs.

Answer Choices

- A. Estimates of future conditions are entered into supercomputers to calculate climate possibilities at various places on earth.
- B. Oceans absorb a great deal of carbon dioxide from the air through limestone production and photosynthesis or phytoplankton.
- C. Gases are stored in the sea in the form of shells and hydrates, but gases stored in these ways can be recycled to the atmosphere where they may cause warming.
- D. The ocean's capacity to absorb carbon dioxide remains great despite recent reduction of marine plant nutrients such as iron.
- E. Ocean circulation is strongly controlled by wind and by the sea's balance of salt and heat.
- F. The ocean holds and moves a great deal of heat, and as water evaporates, it produces clouds, snow, and ice, which all affect global temperatures.

Animal Behavior

By the early 1900s the field of animal behavior had split into two major branches. One branch, ethology, developed primarily in Europe. To ethologists, what is striking about animal behaviors is that they are fixed and seemingly unchangeable? For example, kittens and puppies play in characteristic but different ways. Present a kitten with a ball of yarn and invariably it draws back its head and bats the yarn with claws extended. Kittens are generally silent as they play, and their tails twitch. Puppies, by contrast, are most likely to pounce flat-footed on a ball of yarn. They bit and bark and their tails wag. Ethologists came to believe that ultimately even the most complex animal behaviors could be broken down into a series of unchangeable stimulus/response reactions. They became convinced that the details of these patterns were as distinctive of a particular group of animals as were anatomical characteristics. For well over half a century, their search for and description of innate patterns of animal behavior continued.

Meanwhile, mainly in North America, the study of animal behavior took a different tack, developing into comparative behavior. Of interest to comparative behaviorists was where a particular came from, that is, its evolutionary history, how the nervous system controlled it, and the extent to which it could be modified. In 1894, C. Lloyd Morgan, an early comparative behaviorist, insisted that animal behavior be explained as simply as possible without reference to emotions or motivations since these could not be observed or measured. In Morgan's research, animals were put in simple situations, presented with an easily described stimulus, and their resultant behavior described.

The extension to animals of behaviorism—the idea that the study of behavior should be restricted to only those elements that can be directly observed—was an important development in comparative behavior. Studies of stimulus/response and the importance of simple rewards to enforce and modify animal behavior were stressed. Not surprisingly, comparative behaviorists worked most comfortably in the laboratory. Comparative behaviorists stressed the idea that animal behavior could be modified, while their ethologist colleagues thought it was innate and unchangeable. Inevitably, the two approaches led to major disagreements.

To early ethologists, the major driving force in behavior was instinct, behaviors that are inherited and unchangeable. Moths move towards light because they inherit the mechanism to so respond to light. Although dogs have more options available to them, they bark at strangers for much the same reasons. The comparative behaviorists disagreed: learning and rewards are more important factors than instinct in animal behavior. Geese are not born with the ability to retrieve lost eggs when they roll out the nest, they learn to do so. If their behavior seems sometimes silly to humans because it fails to take new conditions into account, that is because the animal's ability to learn is limited. There were too many examples of behaviors modified by experience for comparative behaviorists to put their faith in instincts.

The arguments came to a peak in the 1950s and became known as the nature or nurture controversy. Consider how differently an ethologist and a comparative behaviorist would interpret the begging behavior of a hatchling bird. The first time a hatchling bird is approached by its parent, it begs for food. All baby birds of a particular species beg in exactly the same way. Obviously, said the ethologists, they inherited the ability and the tendency to beg. Baby birds did not have to learn the behavior, they were born with it—a clear example of innate, unchanging behavior. Not so, countered the comparative behaviorists. Parent birds teach their young to beg by stuffing food in their open mouths. Later experiments showed that before hatching, birds make and respond to noises of their nest mates and adults. Is it not possible that young birds could learn to beg prenatally?

It was hard for ethologists to accept that innate behaviors could be modified by learning. It was equally difficult for comparative behaviorists to accept that genetic factors could dominate learning experiences. The controversy raged for over a decade. Eventually, however, the distinctions between the two fields narrowed. The current view is that both natural endowments and environmental factors work together to shape behavior.

Paragraph 1

By the early 1900s the field of animal behavior had split into two major branches. One branch, ethology, developed primarily in Europe. To ethologists, what is striking about animal behaviors is that they are fixed and

seemingly unchangeable? For example, kittens and puppies play in characteristic but different ways. Present a kitten with a ball of yarn and invariably it draws back its head and bats the yarn with claws extended. Kittens are generally silent as they play, and their tails twitch. Puppies, by contrast, are most likely to pounce flat-footed on a ball of yarn. They bit and bark and their tails wag. Ethologists came to believe that **ultimately** even the most complex animal behaviors could be broken down into a series of unchangeable stimulus/response reactions. They became convinced that the details of these patterns were as distinctive of a particular group of animals as were anatomical characteristics. For well over half a century, their search for and description of innate patterns of animal behavior continued.

1. The word “**ultimately**” in the passage is closest in the meaning to
 - A. noticeably
 - B. importantly
 - C. some of the time
 - D. in the end
2. According to paragraph 1, what do ethologists think is the most notable characteristic of animal behavior?
 - A. Animal responses in most situations are predictable and do not vary
 - B. In similar situations, different animal species often behave in similar ways.
 - C. Even in ordinary situations, animal behavior can be unusually complex.
 - D. Animal behavior may sometimes include stimulus/response reactions.

Paragraph 2

Meanwhile, mainly in North America, the study of animal behavior took a different tack, developing into comparative behavior. Of interest to comparative behaviorists was where a particular came from, that is, its evolutionary history, how the nervous system controlled it, and the extent to which it could be modified. In 1894, C. Lloyd Morgan, an early comparative behaviorist, insisted that animal behavior be explained as simply as possible without reference to emotions or motivations since these could not be observed or measured. In Morgan’s research, animals were put in simple situations, presented with an easily described stimulus, and their resultant behavior described.

3. According to paragraph 2, C. Lloyd Morgan agreed with which of the following statements about animal behavior?
- A. Only those elements of animal behavior that could be observed and measured should be used to explain it.
 - B. Any study of animal behavior should include an explanation of emotions and motivations.
 - C. Emotions and motivations can be measured indirectly using simple experimental situations.
 - D. Experimental situations are less than ideal if researchers want to develop a comprehensive explanation of animal behavior.
4. According to paragraph 2, comparative behaviorists were interested in finding answers to all of the following questions EXCEPT
- A. How has animal behavior changed over time?
 - B. How can emotions causing a specific behavior in one animal species help explain behavior in other animal species?
 - C. To what degree can animal behavior be changed?
 - D. How does the nervous system regulate animal behavior?

Paragraph 3

The extension to animals of behaviorism—the idea that the study of behavior should be restricted to only those elements that can be directly observed—was an important development in comparative behavior. Studies of stimulus/response and the importance of simple rewards to enforce and modify animal behavior were stressed. Not surprisingly, comparative behaviorists worked most comfortably in the laboratory. Comparative behaviorists stressed the idea that animal behavior could be modified, while their ethologist colleagues thought it was innate and unchangeable. Inevitably, the two approaches led to major disagreements.

5. Paragraph 3 suggests that comparative behaviorists' conclusions concerning animal behavior were based
- A. on the observation that rewards do not affect inherited animal behavior
 - B. on the application of stress to modify animal behavior
 - C. most often on the results of laboratory experiments
 - D. more on stimulus/response reactions than on simple rewards

Paragraph 4

To early ethologists, the major driving force in behavior was instinct, behaviors that are inherited and unchangeable. Moths move towards light because they inherit the mechanism to so respond to light. Although dogs have more options available to them, they bark at strangers for much the same reasons. The comparative behaviorists disagreed: learning and rewards are more important factors than instinct in animal behavior. Geese are not born with the ability to **retrieve** lost eggs when they roll out the nest, they learn to do so. If their behavior seems sometimes silly to humans because it fails to take new conditions into account, that is because the animal's ability to learn is limited. There were too many examples of behaviors modified by experience for comparative behaviorists to put their faith in instincts.

6. The word "**retrieve**" in the passage is closest in meaning to
- A. find
 - B. recover
 - C. remember
 - D. hatch
7. According to paragraph 4, why did comparative behaviorists believe that their view of instinct in animal behavior was correct?
- A. They had observed that animals can respond to the same stimulus in different ways.
 - B. They had demonstrated that animals could use learned behaviors in new conditions.
 - C. They had acquired sufficient evidence that instincts vary from one animal to another.
 - D. They had shown that the behavior of many different animals had been changed by learning.

Paragraph 5

The arguments came to a peak in the 1950s and became known as the nature or nurture controversy. Consider how differently an ethologist and a comparative behaviorist would interpret the begging behavior of a hatchling bird. The first time a hatchling bird is approached by its parent, it begs for food. All baby birds of a particular species beg in exactly the same way.

Obviously, said the ethologists, they inherited the ability and the tendency to beg. Baby birds did not have to learn the behavior, they were born with it—a clear example of innate, unchanging behavior. Not so, **countered** the comparative behaviorists. Parent birds teach their young to beg by stuffing food in their open mouths. Later experiments showed that before hatching, birds make and respond to noises of their nest mates and adults. Is it not possible that young birds could learn to beg prenatally?

8. The word “**Obviously**” in the passage is closest in meaning to
- A. Originally
 - B. Clearly
 - C. Similarly
 - D. Consequently
9. The word “**countered**” in the passage is closest in meaning to
- A. learned
 - B. argued back
 - C. assumed
 - D. predicted
10. In paragraph 5, why does the author discuss the begging behavior of a hatchling bird?
- A. To support the view that instinct explains animal behavior better than learning does
 - B. To demonstrate that ethologists are correct about the limited ability of animals to learn
 - C. To contrast an ethologist’s explanation of a particular animal behavior with that of a comparative behaviorist
 - D. To question whether the discussion about the roles of nature and nurture was a valid one

Paragraph 6

It was hard for ethologists to accept that innate behaviors could be modified by learning. It was equally difficult for comparative behaviorists to accept that genetic factors could dominate learning experiences. The controversy raged for over a decade. Eventually, however, the distinctions between the two fields narrowed. The **current** view is that both natural

endowments and environmental factors work together to shape behavior.

11. The word “**current**” in the passage is closest in meaning to

- A. ideal
- B. basic
- C. alternative
- D. present

Paragraph 4

To early ethologists, the major driving force in behavior was instinct, behaviors that are inherited and unchangeable. ■Moths move towards light because they inherit the mechanism to so respond to light. ■Although dogs have more options available to them, they bark at strangers for much the same reasons. ■The comparative behaviorists disagreed: learning and rewards are more important factors than instinct in animal behavior. ■Geese are not born with the ability to retrieve lost eggs when they roll out the nest, they learn to do so. If their behavior seems sometimes silly to humans because it fails to take new conditions into account, that is because the animal’s ability to learn is limited. There were too many examples of behaviors modified by experience for comparative behaviorists to put their faith in instincts.

Look at the four squares [■] that indicate where the following sentence could be added to the passage. Where would the sentence best fit?

This view is supported by the behavior of insects as well as animals.

12. Directions: Select from the seven phrases below the phrases that correctly characterize ethologists and the phrases that correctly characterize comparative behaviorists. Drag each phrase you select into the phrases will NOT be used. This question is worth 3 points.

Ethologists

-
-

Comparative Behaviorists

-
-
-

Answer Choices

- A. Worked primarily in North America
- B. Argued that animal behavior is passed on from one generation to another without change over time
- C. Maintained from the start that behaviors that are inherited could be influenced by learning
- D. Believed that stimulus-response reactions serve to distinguish one animal from another just as their physical features do
- E. Studied stimulus-response reactions and emphasized the importance of rewards for enforcing and changing behavior
- F. Conducted more experiments with birds than with any other species
- G. Studied primarily how physical characteristics often determine behavior.



Grain in Colonial North America

Although the colonists of seventeenth- and early-eighteenth-century British North America consumed most of the grain produced in the colonial economy, few households were self-sufficient. Instead, they traded with their neighbors for what they did not produce themselves. In any given year, farmers who produced more grain than they needed would exchange their surpluses locally with other farmers who had different surpluses, with local laborers who supported themselves by selling their labor, or with the local storekeeper, who might also be the miller (trade person who ground grain into flour). Satisfying the domestic demand for breadstuff, then, depended on trade between neighbors. The colonists recorded these myriad transactions as credits and debts in their individual account books. Debts and credits could remain outstanding for years before being settled. Trading based on book credit gave more value to maintaining equilibrium between local supply and demand and to preserving a cooperative spirit among neighbors than to expanding production beyond the immediate needs of the locality.

Colonists also traded grain surpluses long-distance, responding to impersonal demand beyond the community. Some of the long-distance trade catered to regional and urban domestic demand. As the urban areas matured, they increasingly relied on producers in distant areas for grain and other agricultural supplies. In the early 1750s, the most densely populated towns of eastern and southern New England had begun importing substantial quantities of flour and rice from the middle and southern colonies to compensate for grain deficits that developed in their region. Other urban areas followed their example, though their greater proximity to grain regions enabled them to tap supplies closer to home. Assuming that in the early 1770s at least half of the demand for grain from farmers with surpluses was satisfied through long-distance channels, the proportion of grain produced for consumption beyond the local market probably accounted for about a quarter of total grain production consumed by humans.

The colonists organized the long-distance grain economy differently from their local economy. New mechanisms enabled the long-distance economy to respond sensitively to variations in demand, and these in turn gave it greater dynamism than the community-centered, local economy possessed. The

contrast between the local and long-distance grain trade is best illustrated by looking at the flour-milling industry.

Nearly every area of colonial settlement had a local gristmill to which farmers brought grain to be custom ground. The limited capital value of most custom mills, the need to process rye and corn as well as wheat, together with seasonal factors affecting the water supply, restricted the volume of wheat flour that could be produced. The production of flour for long-distance exchange and particularly for export usually took place in merchant mills that were larger, had more capital, and were increasingly specialized.

The difference between a merchant mill and a custom mill was one of degree as much as kind. Most merchant mills had started as custom mills, and the colonial and state governments often compelled merchant mills to set aside certain days for custom work. Mills that acquired the designation “merchant” did so because they catered to the demand of merchants in the principal ports. These merchants enabled certain millers to specialize in wheat flour by placing orders for large parcels of it and paying in cash. That in turn allowed the millers to offer cash to the primary producers and grain brokers who delivered wheat to their mills. Cash was the economic motivator of this export-oriented economy for the simple reason that farmers would prepare and haul their grain to landings and mills and even increase their wheat acreage to obtain this commodity.

Cash gave farmers choices they did not enjoy when they traded with neighbors alone. Beyond opening up access to a range of products that could not be produced locally, it freed them from the web of mutual indebtedness and allowed more choices in the selection of trading partners. In other words, the cash economy allowed producers to seek the best bargains in that wider, impersonal market of which the export of agricultural surpluses formed the principal part. Of course, few in this age would have welcomed total release from the support and obligations that local trade conferred.

Paragraph 1

Although the colonists of seventeenth- and early-eighteenth-century British North America consumed most of the grain produced in the colonial economy, few households were self-sufficient. Instead, they traded with their

neighbors for what they did not produce themselves. In any given year, farmers who produced more grain than they needed would exchange their surpluses locally with other farmers who had different surpluses, with local laborers who supported themselves by selling their labor, or with the local storekeeper, who might also be the miller (trade person who ground grain into flour). Satisfying the domestic demand for breadstuff, then, depended on trade between neighbors. The colonists recorded these myriad transactions as credits and debts in their individual account books. Debts and credits could remain outstanding for years before being settled. Trading based on book credit gave more value to maintaining equilibrium between local supply and demand and to **preserving** a cooperative spirit among neighbors than to expanding production beyond the immediate needs of the locality.

1. According to paragraph 1, why did North American colonists trade with their neighbors?
 - A. They preferred not to accept cash for goods they had to trade.
 - B. They were not able to produce everything that they needed.
 - C. They had difficulty establishing book credit with the local storekeepers.
 - D. They wanted to balance the credits and debts in their account books.
2. The word “**preserving**” in the passage is closest in meaning to
 - A. limiting
 - B. revealing
 - C. maintaining
 - D. creating

Paragraph 2

Colonists also traded grain surpluses long-distance, responding to impersonal demand beyond the community. Some of the long-distance trade catered to regional and urban domestic demand. As the urban areas matured, they increasingly relied on producers in distant areas for grain and other agricultural supplies. In the early 1750s, the most densely populated towns of eastern and southern New England had begun importing **substantial** quantities of flour and rice from the middle and southern colonies to compensate for grain deficits that developed in their region. Other urban areas followed their example, though their greater proximity to grain regions

enabled them to tap supplies closer to home. Assuming that in the early 1770s at least half of the demand for grain from farmers with surpluses was satisfied through long-distance channels, the proportion of grain produced for consumption beyond the local market probably accounted for about a quarter of total grain production consumed by humans.

3. According to paragraph 2, what happened as urban areas matured?
- A. Urban areas became increasingly dependent on grain produced in distant areas.
 - B. Urban areas began to rely on their own stored surpluses of wheat flour.
 - C. Urban consumers were forced to reduce their consumption of grain by one fourth.
 - D. Urban areas relied more on rice and less on wheat.
4. The word “**substantial**” in the passage is closest in meaning to
- A. modest
 - B. sufficient
 - C. necessary
 - D. large

Paragraph 3

The colonists organized the long-distance grain economy differently from their local economy. New mechanisms enabled the long-distance economy to respond sensitively to variations in demand, and these in turn gave it greater dynamism than the community-centered, local economy possessed. The contrast between the local and long-distance grain trade is best illustrated by looking at the **flour-milling industry**.

5. What is the purpose of the discussion of the “**flour-milling industry**”?
- A. To contrast the organization of the grain trade with the organization of the flour-milling industry.
 - B. To explain how the economic organization of the long-distance trade in grain differed from that of the local economy.
 - C. To illustrate how the local flour-milling industry differed from the long-distance flour-milling industry.
 - D. To argue that the flour-milling industry made the long-distance grain trade more dynamic than the local grain trade.

Paragraph 4

Nearly every area of colonial settlement had a local gristmill to which farmers brought grain to be custom ground. The limited capital value of most custom mills, the need to process rye and corn as well as wheat, together with seasonal factors affecting the water supply, restricted the volume of wheat flour that could be produced. The production of flour for long-distance exchange and particularly for export usually took place in merchant mills that were larger, had more capital, and were increasingly specialized.

6. According to paragraph 4, all of the following limited the milling capacity of local gristmills to produce wheat flour EXCEPT
- A. limitations on their capital value
 - B. seasonal shortages of water
 - C. the lack of trained mill workers
 - D. the need to process several types of grain

Paragraph 5

The difference between a merchant mill and a custom mill was one of degree as much as kind. Most merchant mills had started as custom mills, and the colonial and state governments often **compelled** merchant mills to set aside certain days for custom work. Mills that acquired the designation “merchant” did so because they catered to the demand of merchants in the principal ports. These merchants enabled certain millers to specialize in wheat flour by placing orders for large parcels of it and paying in cash. That in turn allowed the millers to offer cash to the primary producers and grain brokers who delivered wheat to their mills. Cash was the economic motivator of this export-oriented economy for the simple reason that farmers would prepare and haul their grain to landings and mills and even increase their wheat acreage to obtain this commodity.

7. The word “**compelled**” in the passage is closest in meaning to
- A. required
 - B. advised
 - C. selected
 - D. helped

8. According to paragraph 5, a custom mill became a merchant mill by
- A. agreeing to sell grain to the state government at a reduced price
 - B. providing goods to fulfill the demand of traders in the main port cities
 - C. limiting the amount of work a mill would do for individual farmers to only a few days a week
 - D. hiring specialized millers to produce each type of flour
9. According to paragraph 5, why could merchant mills specialize in producing wheat flour?
- A. Grain farmers would accept cash from merchant mills for wheat, but not for other grains.
 - B. Merchant mills were not required by colonial and state government to mill grains other than wheat.
 - C. Millers who specialized in producing wheat flour for merchants were paid in advance.
 - D. Merchants ordered large amounts of wheat flour from the mills they dealt with
10. Paragraph 5 implies which of the following about the export trade in grain?
- A. The growth of the export trade caused most custom mills to go out of business.
 - B. The export trade did not have a significant economic impact beyond the large port cities.
 - C. The export trade's use of cash for payments led farmers to produce surpluses for this market.
 - D. In the export trade, most of the profits were made by the primary products and grain brokers

Paragraph 6

Cash gave farmers choices they did not enjoy when they traded with neighbors alone. Beyond opening up access to a range of products that could not be produced locally, it freed them from the web of mutual indebtedness and allowed more choices in the selection of trading partners. In other words, the cash economy allowed producers to seek the best bargains in that wider,

impersonal market of which the export of agricultural surpluses formed the principal part. Of course, few in this age would have welcomed total release from the support and **obligations** that local trade conferred.

11. The word “**obligations**” in the passage is closest in meaning to
 - A. trust
 - B. safety
 - C. responsibilities
 - D. advantages
12. Which of the following can be inferred from paragraph 6 about the result of the expanding export economy?
 - A. Some commercial relationships became less local and more impersonal.
 - B. Farmers had a greater choice in the variety of crops to grow.
 - C. The demand for agricultural surpluses was greater than the supply.
 - D. The prices of local products decreased.

Paragraph 6

Cash gave farmers choices they did not enjoy when they traded with neighbors alone. ■ Beyond opening up access to a range of products that could not be produced locally, it freed them from the web of mutual indebtedness and allowed more choices in the selection of trading partners. ■ In other words, the cash economy allowed producers to seek the best bargains in that wider, impersonal market of which the export of agricultural surpluses formed the principal part. ■ Of course, few in this age would have welcomed total release from the support and obligations that local trade conferred. ■

13. Look at the four squares [■] that indicates where the following sentence could be added to the passage.

Consequently, most farmers who participated in the long distance cash economy continued to deal with local merchants through the system of trade credit.

Where would the sentence best fit?

14. Direction: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the important ideas in the passage. Some sentences do

not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Grain played an important role in the economy of colonial North America.

Answer Choices

- A. Trade between neighbors was necessary to satisfy the local demand for grain and other products.
- B. Large amounts of grain from the countryside were brought to cities and ports, both for the residents' use and for the export trade.
- C. Merchant mills required large economic investments that were not available locally.
- D. Long-distance trade eventually replaced the need for local trade.
- E. Wheat farmers were attracted to the export trade because it allowed them to find the best price and because the cash it provided could be used for goods not available locally.
- F. After some time, cash replaced the trade system and individual account books disappeared.

Sea Turtle Hatchling Strategies for Navigation

Sea turtles' eggs are laid at night to minimize the likelihood of their discovery by predators, and the offspring, when ready to emerge from their eggshells and dig their way out of the sand, hatch at night for the same reason. Since the offspring are especially vulnerable immediately after hatching, it is vital for them to get to the sea as soon as possible. Turtle hatchlings use a number of cues to tell them where the sea is.

The most important cue seems to be light. The night sky is usually brightest over the sea. Cover a turtle hatchling's eyes, and it cannot find the sea even if there is other information available, such as a downward slope of the sand toward the water's edge. The hatchlings respond to light cues covering a vertical range of only about 30° above the horizon or, depending on the species, even less. Responding only to lights that are close to the horizon decreases the risk that hatchlings will become confused. They seem less attracted to yellow light than to other colors—loggerhead turtles show an aversion to yellow light—and this preference may keep them from becoming disoriented by the rising Sun.

It is usually safest to have more than one internal compass, and hatchlings seem to be guided by more than light alone. They steer away from sand dunes and vegetation. Possibly these objects merely block light behind them that might mislead turtle hatchlings about where the sea is, but it is also possible that turtles are sensitive to the shape of such objects and process these shapes as signals that the sea is located in some other direction. Such reinforcing cues, however, are not enough to guide hatchlings away from the artificial lights that now burn on many a beach environment. Artificial lighting is often strong enough to completely overcome the signals a hatchling sea turtle is programmed to recognize. Artificial light, if it is bright enough, becomes a stimulus so powerful that the hatchlings respond to nothing else, crawling toward it from hundreds of meters away.

If all goes well and the hatchlings scramble over the sand in the right direction, avoid their enemies, and reach the surf, a new set of orienting mechanisms takes over. As soon as they are afloat, the hatchlings begin to swim at something over 1.5 kilometers per hour. They dive into the path of the wave undertow, where the receding waters sweep them outward, away from

the beach. When they surface again, they head for open sea. This time, they are guided not by sight but apparently exclusively by the direction of the incoming waves. Experiments with loggerheads, greens, and leatherbacks have shown that hatchlings swim toward approaching waves; but if the sea is calm, they swim randomly or in circles. Under experimental conditions, hatchlings will swim into the waves even if doing so sends them back to the beach again.

The farther a hatchling gets from shore, the less reliable wave direction becomes as a pointer to the open sea. Researchers have shown that hatchling green sea turtles released from a hatchery in Borneo, East Malaysia, are able to navigate around small islands and keep swimming offshore, even when there are few waves to guide them. They may be relying on yet another internal compass this time oriented to Earth's magnetic field. Recent experiments suggest that leatherback and olive ridley hatchlings "switch on" their geomagnetic compass almost as soon as they are out of the nest. Though the hatchlings position themselves geomagnetically as soon as they leave the nest and appear to be able to use that position as a reference point, they will not follow it automatically if other cues, such as light and sound, are available. Hatchlings find their geomagnetic compass useful only after they have already been able to determine the direction they should swim. A simple directional compass—one that always sent the turtles westward, for instance—would be useless if the open sea lay in some other direction. Therefore, a magnetic compass does not so much tell a hatchling turtle which way to go as keep it on course once it has determined the direction it should swim from some other cue.

Paragraph 1

Sea turtles' eggs are laid at night to minimize the likelihood of their discovery by predators, and the offspring, when ready to emerge from their eggshells and dig their way out of the sand, hatch at night for the same reason. Since the offspring are especially vulnerable immediately after hatching, it is vital for them to get to the sea as soon as possible. Turtle hatchlings use a number of cues to tell them where the sea is.

1. Which of the sentences below best express the essential information in the highlighted sentence in the passage? Incorrect choices change the

meaning in important ways or leave out essential information.

- A. Sea turtle eggs are laid at night and hatch at night for the same reason.
- B. To minimize the likelihood of their discovery by predators, sea turtle hatchlings dig their way out of the sand at night.
- C. To minimize the likelihood of discovery by predators, sea turtle eggs are laid and hatch at night.
- D. Eggs that laid at night and offspring that emerge from the eggs at night are less likely to be discovered by predators

Paragraph 2

The most important cue seems to be light. The night sky is usually brightest over the sea. Cover a turtle hatchling's eyes, and it cannot find the sea even if there is other information available, such as a downward slope of the sand toward the water's edge. The hatchlings respond to light cues covering a vertical range of only about 30° above the horizon or, depending on the species, even less. Responding only to lights that are close to the horizon decreases the risk that hatchlings will become confused. They seem less attracted to yellow light than to other colors—loggerhead turtles show an aversion to yellow light—and this preference may keep them from becoming disoriented by the rising Sun.

- 2. In paragraph 2, why does the author provide the information that sea turtle hatchlings cannot find the sea when their eyes are covered?
 - A. To identify a major reason that such turtles might not reach the sea
 - B. To support the claim that light is an important directional cue for such turtles
 - C. To point out the effect of yellow light on the attempts of such turtles to reach the sea
 - D. To provide evidence showing that such turtles easily lose their sense of direction
- 3. According to paragraph 2, which of the following is true of turtle hatchlings that are trying to make their way to the sea?
 - A. They are unable to distinguish the light of the rising Sun from other types of light.
 - B. They require the presence of yellow light in order to reach the water's

edge.

- C. They have trouble crossing downward slopes near the water's edge.
- D. They respond to light only when it is close to the horizon.

Paragraph 3

It is usually safest to have more than one internal compass, and hatchlings seem to be guided by more than light alone. They steer away from sand dunes and vegetation. Possibly these objects merely block light behind them that might mislead turtle hatchlings about where the sea is, but it is also possible that turtles are sensitive to the shape of such objects and process these shapes as signals that the sea is located in some other direction. Such **reinforcing** cues, however, are not enough to guide hatchlings away from the artificial lights that now burn on many a beach environment. Artificial lighting is often strong enough to completely overcome the signals a hatchling sea turtle is programmed to recognize. Artificial light, if it is bright enough, becomes a stimulus so powerful that the hatchlings respond to nothing else, crawling toward it from hundreds of meters away.

- 4. The word "**reinforcing**" in the passage is closest in meaning to
 - A. guiding
 - B. supporting
 - C. indirect
 - D. valuable
- 5. Which of the following is offered in paragraph 3 as a possible explanation for sea turtle hatchlings' behavior of avoiding sand dunes and vegetation?
 - A. Sand dunes and vegetation may provide hiding places for animals that attack hatchlings.
 - B. Sand dunes and vegetation may block the pathway between hatchlings and the water's edge.
 - C. The shapes of sand dunes and vegetation may indicate the wrong direction to travel in.
 - D. The shapes of sand dunes and vegetation may resemble the shapes of some hatchling predators
- 6. Paragraph 3 supports which of the following statements about the artificial lights now found on many beaches?

- A. The lights have sometimes helped sea turtle hatchlings find their way to the sea.
- B. The lights can be bright enough to affect sea turtle hatchlings within hundreds of meters of them.
- C. The lights are commonly turned off during periods when sea turtle offspring are hatching.
- D. The lights tend to affect the behavior of sea turtle hatchlings less than other programming signals do.

Paragraph 4

If all goes well and the hatchlings scramble over the sand in the right direction, avoid their enemies, and reach the surf, a new set of orienting mechanisms takes over. As soon as they are afloat, the hatchlings begin to swim at something over 1.5 kilometers per hour. They dive into the path of the wave undertow, where the receding waters sweep them outward, away from the beach. When they surface again, they head for open sea. This time, they are guided not by sight but apparently **exclusively** by the direction of the incoming waves. Experiments with loggerheads, greens, and leatherbacks have shown that hatchlings swim toward approaching waves; but if the sea is calm, they swim randomly or in circles. Under **experimental** conditions, hatchlings will swim into the waves even if doing so sends them back to the beach again.

- 7. The word “**exclusively**” in the passage is closest in meaning to
 - A. with great accuracy
 - B. strongly
 - C. in part
 - D. only
- 8. Based on paragraph 4, sea turtle hatchlings are most likely to lose their sense of where the open sea is located when they
 - A. swim in calm seas
 - B. face incoming waves
 - C. collide with other objects floating in the water
 - D. swim in fast-moving, outgoing waters
- 9. The word “**experimental**” in the passage is closest in meaning to

- A. favorable
- B. trial
- C. Certain
- D. difficult

Paragraph 5

The farther a hatchling gets from shore, the less reliable wave direction becomes as a pointer to the open sea. Researchers have shown that hatchling green sea turtles released from a hatchery in Borneo, East Malaysia, are able to navigate around small islands and keep swimming offshore, even when there are few waves to guide them. They may be relying on yet another internal compass this time oriented to Earth's magnetic field. Recent experiments suggest that leatherback and olive ridley hatchlings "**switch on**" their geomagnetic compass almost as soon as they are out of the nest. Though the hatchlings position themselves geomagnetically as soon as they leave the nest and appear to be able to use that position as a reference point, they will not follow it automatically if other cues, such as light and sound, are available. Hatchlings find their geomagnetic compass useful only after they have already been able to determine the direction they should swim. A simple directional compass—one that always sent the turtles westward, for instance—would be useless if the open sea lay in some other direction. Therefore, a magnetic compass does not so much tell a hatchling turtle which way to go as keep it on course once it has determined the direction it should swim from some other cue.

10. In paragraph 5, why does the author provide the information that hatchling green sea turtles can navigate around small islands and keep swimming offshore even when few waves are present?
- A. To point out a benefit of the fact that hatchlings use their geomagnetic compasses almost as soon as they leave the nest
 - B. To provide evidence that green sea turtles use different navigational techniques than leatherbacks and olive ridley turtles do
 - C. To provide a reason for considering the possibility that sea turtles navigate by detecting Earth's magnetic field
 - D. To provide evidence showing that hatchling do not automatically follow

their geomagnetic compass if other cues are available

11. The word “**switch on**” in the passage is closest in meaning to
- A. follow
 - B. change
 - C. control
 - D. activate
12. According to paragraph 5, all the following claims about the geomagnetic compass of sea turtle hatchlings are true EXCEPT
- A. In some species it is switched on almost as soon as a turtle hatches.
 - B. It helps keep the hatchlings heading in a direction that it is initially determined by some cue such as light or sound.
 - C. Its direction is followed automatically to swim toward open water
 - D. It can be used to navigate around islands when the waves stop indicating the direction of the open sea.

Paragraph 5

The farther a hatchling gets from shore, the less reliable wave direction becomes as a pointer to the open sea. Researchers have shown that hatchling green sea turtles released from a hatchery in Borneo, East Malaysia, are able to navigate around small islands and keep swimming offshore, even when there are few waves to guide them. They may be relying on yet another internal compass this time oriented to Earth’s magnetic field. Recent experiments suggest that leatherback and olive ridley hatchlings “switch on” their geomagnetic compass almost as soon as they are out of the nest. Though the hatchlings position themselves geomagnetically as soon as they leave the nest and appear to be able to use that position as a reference point, they will not follow it automatically if other cues, such as light and sound, are available. ■ Hatchlings find their geomagnetic compass useful only after they have already been able to determine the direction they should swim. ■ A simple directional compass—one that always sent the turtles westward, for instance—would be useless if the open sea lay in some other direction. ■ Therefore, a magnetic compass does not so much tell a hatchling turtle which way to go as keep it on course once it has determined the direction it should swim from some other cue. ■

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage

This preference for other cues points to the special role of the geomagnetic compass in hatchling navigation.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

To minimize attacks from predators, turtle hatchlings must quickly find their way to the water's edge.

Answer Choices

- A. When light cues are absent, hatchlings use other navigational strategies, such as following the downward slope of the beach to find the water's edge.
- B. If the incoming waves are too strong to allow hatchlings to reach the open sea, they are carried back to the beach where they wait for calmer waters
- C. Once hatchlings reach the open sea, they begin to use their geomagnetic compasses as simple directional compasses.
- D. Hatchlings use light and cues such as shapes to help them locate the sea, though strong artificial light on land may lead them in the wrong direction.
- E. After they reach the water, hatchlings rely on incoming waves to indicate the direction in which the open sea is located.
- F. Hatchlings' geomagnetic compasses can help keep hatchlings on course toward the open sea, once other cues have been used to get them headed in the right direction.

1. Controversy about Causing Emotion

The fact that we react to certain experiences with "Emotion" is obvious. For example, the feeling of embarrassment, which triggers a physiological response that may cause blushing, is caused by a foolish act committed in the company of friends. Although this description of an embarrassed reaction seems logical, the American psychologist William James, in 1884, believed that the course of an emotional experience follows another sequence of events.

Following the argument of James, what subjective experience tells us is completely opposite that the sequence of events in an emotional experience. First, he insisted that both physiological excitement and physical reaction are generated by an incident. Only then does the individual perceive or interpret the physical response as an emotion. That is, we associate blushing that caused by physical reaction with embarrassment, such as saying something silly may cause us to blush. In 1890, James went on to claim that "people feel sorry because they cry, furious because they strike, afraid because they shudder." Simultaneously with James' proposition, Carl Lange, a Danish physiologist and psychologist, independently formulated virtually similar theory. The James-Lange theory of emotion (Lange and James, 1922) suggests that different patterns of arousal in the autonomic nervous system create the different emotions people feel, and that physiological arousal occurs prior to the emotion is perceived.

In 1927, Another early theory of emotion that challenged the James-Lange theory was proposed by Walter Cannon. He claimed that physical changes caused by the diverse emotions are not sufficiently distinct to allow people to distinguish one emotion from another. After Cannon stated his original theory, in 1934, it was further developed by physiologist Philip Bard. ■ The Cannon-Bard theory suggests that the following chain of events takes place when an emotion is felt. ■ Stimuli which trigger emotion are received by the senses and then are relayed simultaneously to the cerebral cortex, which imparts the conscious mental experience of the emotion, and to the sympathetic nervous system, which generates the physiological state of arousal. ■ In other words, the feeling of emotion occurs roughly the same time when the physiological arousal is experienced. One does not cause the other.

In 1962, Schachter and Singer proposed a two-factor theory. Stanley Schachter thought that the early theories of emotion excluded a critical component that the subjective cognitive interpretation of why a state of arousal has occurred. According to this theory, two things must happen in order for a person to feel an emotion. At first, the person must experience physiological arousal. Then, for the person can label it as specific emotion, there must be a cognitive interpretation or explanation. Thus, Schachter delivered the conclusion that a true emotion can appear only if a person is physically aroused and can find the reason for it. When people are in a state of physiological arousal but do not know why they are aroused, they tend to label the state as an emotion that is appropriate to their situation at the time. There were several attempts to replicate the findings of this theory, but they have not been successful.

Richard Lazarus, in the 1990, proposed the emotion theory that most heavily emphasizes the cognitive aspect. According to his theory, the first step in an emotional response is cognitive appraisal, and all other aspects of emotion, including physiological arousal, rely on the cognitive appraisal. This theory is most compatible with the subjective experience of an emotion's sequence of events-the sequence that William James reversed long ago. People first appraise a stimulus, or an event, when they encounter it. This cognitive appraisal determines whether the person will have an emotional response, and, if so, what type of response. From this appraisal, the physiological arousal and all other aspects of the emotion arise. In brief, Lazarus contends that emotions are roused when cognitive appraisals of events or circumstances are positive or negative-but not neutral.

Some critics criticize the Lazarus theory by saying that some emotional reactions are instantaneous, which means they occur too rapidly to pass through a cognitive appraisal. In respond to the criticisms, Lazarus remarks that some mental processing occurs without conscious awareness, meaning that a person should not know what he or she is responding to or what emotion to feel, or else, some form of cognitive realization must manifest but brief.

1. The author mentions a foolish act committed in the company of friends in order to

- (A) give an idea that most people easily get embarrassed in a social environment
- (B) show that most experiences activate a physiological reaction
- (C) insist that an embarrassed reaction is usually unwarranted
- (D) provide an example of an experience that generates emotion

2. According to paragraph 2 , which of the following is true of what James believed about an emotional experience?
- (A) Emotions can be caused only when there is a specific response occurs in the body.
 - (B) The sequence of events is not always the same for every emotional experience.
 - (C) We are able to figure out the right sequence of events in an emotional experience through subjective experience.
 - (D) Both a physical reaction and physiological arousal are triggered by the feeling of an emotion.
3. The word formulated in paragraph 2 is closest in meaning to
- (A) published
 - (B) developed
 - (C) revealed
 - (D) duplicated
4. According to paragraph 3, Walter Cannon claimed that the James-Lange theory of emotion is not convincing for which of the following reasons?
- (A) The body can be affected by different emotions in similar ways.
 - (B) It is often not easy for people to distinguish one emotion from another.
 - (C) It is not significant enough for bodily changes to be perceived.
 - (D) It is often not obvious that bodily changes are caused by emotions or other factors.
5. According to the Canon-Bard theory arguments in paragraph 3 , which of the following is the direct cause of mental experience of emotion?
- (A) the cerebral cortex
 - (B) emotion-rousing stimuli
 - (C) aroused physiological state
 - (D) the sympathetic nervous system
6. The word component in paragraph 4 is closest in meaning to
- (A) procedure
 - (B) explanation
 - (C) argument
 - (D) element
7. According to paragraph 4, the Schachter-Singer theory is different from earlier theories of emotion in which of the following ways?
- (A) It suggests that there are several steps in an emotional experience.
 - (B) It shows the reason that people do not understand why they are physiologically aroused.
 - (C) It says that the first step of an emotional experience is physiological arousal.
 - (D) It explains that people must decide why they are aroused to experience emotion.
8. According to paragraph 4 , what can be inferred about an emotion suggested by Schachter-Singer theory?
- (A) It should be occurred just before a person physiologically aroused.
 - (B) It happens regardless of the fact that people know the reason why they are aroused.
 - (C) It has to be passed through a stimulation and clarification process, or people can't experience it.
 - (D) It People can experience several emotions at once.
9. The word replicate in paragraph 4 is closest in meaning to
- (A) interpret
 - (B) reproduce
 - (C) revise
 - (D) defend
10. According to paragraph 5, which of the following is NOT true of the Lazarus' theory
- (A) interpretation of experience is more emphasized than any other theory.

- (B) It indicates that a person first evaluate an event in order to have an emotion.
- (C) It explains almost identical sequence of events to that of the subjective experience of an emotion.
- (D) The theory of emotion proposed by William James is generally similar to this theory.

11 • The word contends in paragraph 5 is closest in meaning to

- (A) doubts
- (B) argues
- (C) proves
- (D) reports

12. Which of the following best describes the organization of this passage?

- (A) The author draws a comparison among different theories of emotion and claims that the last one is correct.
- (B) The author chronologically introduces theories of emotion and identifies criticisms of each theory.
- (C) The author verifies several theories of emotion and argues that each theory is not accurate enough.
- (D) The author develops his own theory summarizing different theories of an emotional experience.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

The emotions of rage and fear, for instance, both include a rapid heartbeat, perspiration, and increased energy supplied to the muscles.

Where would the sentence best fit?

14. Summary

There has been a controversy among researchers about the sequence of events in an emotional experience. Answer

Choices

- (A) Cognitive appraisal probably does not play a large role in an emotional experience, evidenced by the very brief span of time between an event and an emotional reaction.
- (B) Two early theory of emotion both assert that that different emotions result from distinct changes in the body that correspond to particular emotions.
- (C) There is a disagreement among researchers who emphasize the cognitive aspect of an emotion about whether cognitive interpretation comes before or after physiological arousal.
- (D) Theories in disagreement with each other about the sequence of events in emotional experience concentrate on the extent to which cognition and interpretation are involved in an emotional experience.
- (E) Some theories claim that physiological arousal comes either before or at the same time as the feeling of an emotion, and contradict the subjective experience of emotion.
- (F) The Lazarus theory emphasized that an emotion comes from a positive or negative appraisal of a physiological arousal.

2.Three Theories about Origin of Life

Oxygen and nitrogen are major components of our current atmosphere. But the kinds of hydrogen reactions with other gases that are required to transform simple organic molecules into complex ones are interrupted by oxygen, which combines with hydrogen atoms from other compounds. Therefore, life on Earth must have originated when there was very little oxygen in Earth's atmosphere. The modern scientific theory of life's origin was first formulated in the 1920s by Russian scientist Aleksandr Oparin and independently by British scientist J. B. S. Haldane. The assumption that life sprang up from chemical reactions that were initiated in the early atmosphere(oxygen-poor/hydrogen-rich) and came to completion in the early oceans was posited by the Oparin-Haldane hypothesis, as it came to be called. Oparin and Haldane suggested that the hydrogen-containing gases caused to react with each other to form organic compounds by energy sources such as sunlight and lightning.

With regard to the view that these complex organic compounds could have begun to shape in Earth's oceans, some researchers remain skeptical. The probability that the fundamental building blocks of life, formaldehyde (H_2CO) and hydrogen cyanide(HCN), even though they were probably available, would have been concentrated sufficiently to allow further reactions to occur was likely small. And the more complex organic compounds that might have formed in this way would not have lasted long in the surface-ocean environment, because photochemical and thermal reactions would have destroyed them. Therefore, researchers have sought alternative explanations for how complex organic compounds formed.

There is one possibility that the relevant organic compounds were created in space, and asteroids or comets brought them to Earth, probably as tiny dust particles. Recovered from the stratosphere (an upper region of Earth's atmosphere), interplanetary dust particles (IDPs) is tiny particles that are known to be extraterrestrial origin. From various researches, we know that organic compounds, including amino acids, exist in IDPs as well as in some meteorite. Now, we actually have identified the fact that amino acids and many other complex organic compounds in interstellar dust clouds. It is believed that they form from reactions between charged particles and neutral molecules. Those atoms appear in interstellar dust clouds at very low temperatures-on the order of 200 or more degrees below zero on the Celsius scale. It may seem surprising that organic chemistry could occur in the interstellar environment, but it is precisely the extremely low temperatures involved that allow complex organic molecules to exist because temperatures are too cold to allow them to decompose.

■ It is thought that some of the molecules created in the interstellar environment have survived the collapse of the gas and dust cloud that formed our solar nebula and Sun. ■ They would have been incorporated into solid materials that condensed out of the nebula and formed asteroids and comets. ■ Such materials might have been delivered to Earth in great quantities during the heavy bombardment period of solar system history, between 4.5 and 3.8 billion years ago. ■

The hypothesis that life took place in or around hydrothermal vents (hot springs), where new seafloor is being created along mid-ocean ridges (underwater mountain chains) at the ocean's bottom is the third theory of life's origin. By seawater that flows a kilometer or more down through crevices in the rock, is heated, and then rises rapidly back to the surface, the ridges are cooled. During the process, the water gathers substances such as hydrogen, hydrogen sulfide, and dissolved ferrous iron. When it meets the cold water, the extremely hot (350°C) vent water generates a dark plume comprised mostly of iron sulfide, a compound produced by the reaction between ferrous iron and hydrogen sulfide.

Still, the explanation that submarine hydrothermal vents a likely place for life to have originated is controversial. In vent systems, there are various types of materials from which organic molecules can be synthesized. However, complex organic molecules are not stable at the high temperatures observed in vents positioned directly on the axis of a ridge. If life did originate at the mid-ocean ridges, it probably did so in cooler, off-axis vents. Some researchers claim that perfect place for life to have begun would be in some near-freezing surface environment because even the off-axis vents are too warm. The dispute as to whether life originated in a hot or cold environment is unlikely to stop.

1. By suggesting that the theory of life's origin was first formulated by Oparin and independently by Haldane, the author means that
(A) haldane planed the theory in greater detail tan Oparin

- (B) Oparin devised the theory in cooperation with Haldane
- (C) haldane formulated the theory separately from Oparin
- (D) Oparin developed the theory much earlier than Haldane

2. The word skeptical in the paragraph 2 is closest in meaning to

- (A) doubtful
- (B) confident
- (C) hopeful
- (D) puzzled

3. According to paragraph 2, what is a problem for the explanation that life began in the oceans?

- (A) Some life-forms would have been killed by high concentrations of H_2CO and HCN in the early Earth's oceans.
- (B) If complex organic compounds formed in Earth's oceans, they have been broken down by photochemical and thermal reaction.
- (C) The essential elements for complex organic compounds did not exist on the early Earth.
- (D) The creation of complex organic compounds on Earth's oceans was influenced by the photochemical and thermal reactions.

4. The word sufficiently in the paragraph 2 is closest in meaning to

- (A) quickly
- (B) repeatedly
- (C) initially
- (D) adequately

5. The word sought in the paragraph 2 is closest in meaning to

- (A) accepted
- (B) proposed
- (C) challenged
- (D) looked for

6. Which of the following most accurately describes the relationship of paragraph 2 to paragraph 1?

- (A) Paragraph 2 accounts for the reason why some scientists doubt the theory presented in paragraph 1.
- (B) Paragraph 2 give information about the way that some scientists strengthened the theory introduced in paragraph 1.
- (C) Paragraph 2 exhibits another theory to the one mentioned in paragraph 1.
- (D) Paragraph 2 provides information that reinforces the theory introduced in paragraph 1.

7. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- (A) In consideration of the extreme cold, decomposition of complex organic molecules in the interstellar environment is astonishing.
- (B) Extremely cold temperatures prevent organic molecules from decomposing, making organic chemistry possible in the interstellar environment.
- (C) Although the extreme cold makes complex organic molecules exist, organic chemistry rarely occurs in the interstellar environment.
- (D) Since one would expect the extreme cold to prevent the formation of complex organic molecules, organic chemistry in the interstellar environment is not understandable.

8. In paragraph 3, all of the following are evidences that the organic compounds relevant to the formation of life may have formed in space EXCEPT

- (A) Organic compounds are present in interplanetary dust particles and some meteorites.
- (B) The severe cold conditions of interstellar dust clouds keep organic molecules preserved.
- (C) Interplanetary dust particles have been discovered in the stratosphere of Earth.

(D) Organic compounds created by molecules on Earth react with charged particles found in comets.

9. According to paragraphs 3 and 4, complex organic compounds probably have first appeared on Earth by the way that

- (A) they might have been transformed from comets and asteroids.
- (B) they might have arrived to Earth as our solar system passed through an interstellar nebula.
- (C) they might have created when the newly appeared Sun led solid materials of the solar nebula to decompose.
- (D) they might have come from reactions inside the solid materials that made up the early Earth.

10. The phrase been incorporated into in the paragraph 4 is closest in meaning to

- (A) been attracted by
- (B) been changed into
- (C) formed around
- (D) become part of

11 , According to paragraph 5, all of the following are true of the water emitted by undersea hydrothermal vents EXCEPT

- (A) It has very high temperature.
- (B) It includes a dissolved form of iron.
- (C) It releases hydrogen as it rises to the surface at low rate.
- (D) It creates iron sulfide when it hits the surrounding ocean.

12. According to paragraph 6 , what evidence supports the theory that undersea hydrothermal vents are a possible location for the origin of life?

- (A) Undersea hydrothermal vents probably were much colder during the period of the early Earth than they are observed to be today.
- (B) The fact that materials with the kind of chemical composition from which organic molecules can be produced are in undersea hydrothermal vents.
- (C) Located on the ridge axis, complex organic materials are more stable.
- (D) They are located directly on the ridge axis, near-freezing surface environments.

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

As the modern study of a meteorite that crashed in western Canada has revealed, even today, organic compounds continuously arrive on Earth in this way.

Where would the sentence best fit?

14. Summary

The origin of life on Earth has been the subject of scientific study, and several theories developed to clarify it.

Answer Choices

- (A) The Oparin-Haldane hypothesis about the origin of life explained the relationship between early atmosphere and chemical reactions but it has been abandoned by modern scientists.
- (B) All of the hypotheses about the origin of life have been disproved, and it seems impossible to devise better hypotheses.
- (C) Organic compounds can occur and remain stable in interstellar space and these chemicals may have led to the origin of life after they reached Earth.
- (D) One theory assumed that organic molecules formed at hydrothermal vents are responsible for the origin of life, and the high temperatures in some vents make such molecules unstable.
- (E) One theory argues that molecules in the cold interstellar environment reacted in the heat of hydrothermal vents, creating the stable organic molecules which are essential for life.
- (F) Modern hypothesis suggested life may come from chemical reactions in Earth's ocean, but some scientists still doubt that the chemical products would have concentrated or long lasting enough.

3.The Distribution of Gliding Animals

Generally, gliding is used for some animal species as a mean of fleeing from the predators since it enables them to move between trees without the need to descend to the ground and it also is an energy-efficient way to travel long distances between scattered food resources. For scientist, gliding animals (flying squirrels, flying frogs, and flying lizards with wings of skin that allow them to glide through the tropical forest) have long been the intriguing subject of study. Recently, researchers have found that Southeast Asia has a unique abundance and diversity of these animals. This observation leads them to the following questions. What could be an explanation about biological diversity of these animals found in the forest of Southeast Asia and what could explain the scarcity of gliding animals in other regions? Most of all, what makes Southeast Asian rain forests unusual?

Several theories have been proposed by many scientists to explain the diversity of gliding animals in Southeast Asia. The first theory might be called the tall-trees hypothesis. According to this theory, taller trees in Southeast Asia could offer longer glides as well as the opportunity to boost in a dive before gliding because the forests of this region are taller than any other forests in the world, which comes from the domination of dipterocarp family, a family of tall, tropical hardwood trees. And by providing a more advantageous situation for gliding between tall-trees, the lower wind speeds might also contribute to the great number of appearance of gliding animals. This speculation, however, has several flaws. First, gliding animals are found throughout the Southeast Asian region, even in relatively short-stature forests located in the northern area of the rain forest in China, Thailand and Vietnam. Also, some gliders thrive in low secondary forests, plantations, and even city parks. It is obvious that gliding animals do not need tall trees for their activities. In addition, many gliding animals initiate their glides from the middle of tree trunks, not necessarily ascending to the tops of trees to take off.

Another theory, known as broken-forest hypothesis, speculates that animals in Southeast Asia must risk descending to the ground or glide to move between trees because the top layer of the forest-the tree canopy-has fewer woody vines connecting tree crowns in Southeast Asian forests than in New World and African forests. It also presumes that the tree canopy in Asian forests is more uneven in height, due to the existence of tall dipterocarp trees with lower trees between them, and this imbalance is favored by gliding animals. But it is observed by ecologist working in different regions of world that, depending on the site conditions of soil, climate, slope elevation, and local disturbance, there is a tremendous local variation in tree height, canopy structure, and abundance of vines. Indeed, we can find many locations with abundant woody vines and numerous connections between trees in Southeast Asia and similarly many Amazonian forests with few woody vines.

A last theory differs from the others in suggesting that it is the presence of dipterocarp trees themselves that is promoting the evolution of gliding species. ■ According to this theory, dipterocarp forests can be "food-deserts" for the animals that live in them. _ The animals living in dipterocarp forests that have developed gliding divide into two main groups: leaf eaters and carnivores that eat small prey such as insects and small vertebrates. _ For leaf-eating gliders the problems is not the absence of any leaves but the desert-like absence of edible leaves. Dipterocarp trees often account for 50 percent or more of the total number of canopy trees in a forest and over 95 percent of the large trees, yet dipterocarp leaves are unavailable to most vertebrate plant eaters because of the high concentration of toxic chemicals in their leaves. ■ Many species of gliding animals avoid eating dipterocarp leaves and so must travel widely through the forest, bypassing the dipterocarp tree, to find the leaves they need to eat. And gliding is a more efficient way of traveling between trees than descending to the ground and walking or else jumping between trees.

Since there is the lower abundance of prey and other insects, many carnivorous animals also may need to forage more widely for food. This scarcity of food source is caused by dipterocarps' irregular flowering and fruiting cycles of two- to seven-year intervals, resulting in a shortage of the flowers, fruits, seeds, and seedlings that are the starting point of so many food chains. The lower abundance of prey in dipterocarp forests forces animals such as lizards and geckos to move between tree crowns in search of food, with gliding being the most efficient means.

1 • The word scattered in paragraph 1 is closest in meaning to

- (A) seasonally available
- (B) hard to find
- (C) highly varied

(D) widely separated

2. According to paragraph 1, why is the ability to glide considered useful to forest-dwelling species?

- (A) Because gliding removes the need to travel long distances in search of food.
- (B) Because gliding serves as a rapid, energy efficient way of descending from trees.
- (C) Because gliding gives them an advantage of moving through the forest without being exposed to predators in the ground.
- (D) Because gliding makes them possible to adapt to various forest conditions.

3. According to paragraph 1, researchers are trying to answer which of the following questions about gliding species?

- (A) Why gliding animals, which are uncommon in most tropical forests, have developed in so many different families in Southeast Asia.
- (B) Why gliding animals developed in many tropical forests in Southeast Asia before they evolved in other tropical forests in the world.
- (C) Why gliding animals limitedly evolved in tropical rain forests.
- (D) Why gliding animals evolved so slowly in the tropical forests of Southeast Asia.

4. According to paragraph 2, which of the following does NOT support the tall-trees hypothesis?

- (A) tall trees let longer glides possible.
- (B) gliding from the middle of tree trunks can be possible because trees are tall.
- (C) tall trees make increased speed in a dive possible.
- (D) there are lower wind speeds between tall trees.

5. Select the TWO answer choices that point to flaws in the tall-trees hypothesis, according to paragraph 2. To receive credit, you must select TWO answers.

- (A) Many gliding animals begin their glides from midway positions on tree trunks.
- (B) Gliding animals are not evenly spread throughout the forests of the Southeast Asian region.
- (C) Most gliding animals are incapable of ascending to the tops of trees.
- (D) Many gliding animals are found in locations where trees tend to be relatively short.

6. The word speculates in paragraph 3 is closest in meaning to

- (A) claims
- (B) puts forward as a possibility
- (C) recognizes
- (D) concludes from evidence

7. Paragraph 3 suggests which of the following ideas about forests in which there are abundant woody vines connecting tree crowns?

- (A) There are likely fewer predators on the ground in such forests than in other forests.
- (B) In such forest, the tree canopy is more even than that in other forests.
- (C) In such forest, there is a wider diversity of animals than in other forests.
- (D) Animals in such forests can move between trees by traveling on vines.

8. The word tremendous in paragraph 3 is closest in meaning to

- (A) seasonal
- (B) unexpected
- (C) enormous
- (D) endless

9. Paragraph 3 supports the idea that one problem with the broken-forest hypothesis is that

- (A) in broken forests with an uneven canopy structure, gliding is difficult, and it is easy in forests where the trees are all about the same height
- (B) ecologists in different regions have found that gliding animals are as abundant and varied in some forests of Africa and

the New World as they are in Southeast Asian forests

- (C) ecologists have found gliding animals in areas of Southeast Asia where trees are connected by vines and not found them in Amazonian forests where trees are not connected by vines
- (D) with the fewest woody vines connecting the tops of trees, the forest in Southeast Asia turn out to have the most gliding animals

10. According to paragraph 4, what difficulty do leaf-eating animals have in a dipterocarp forest?

- (A) There is no efficient method of obtaining from one tree to another.
- (B) There is a large gap between trees that have edible leaves.
- (C) Leaves of most trees locates very high tree trunk so it makes animals difficult to reach.
- (D) Dipterocarp trees have less leaves than those of other canopy trees.

11. How does paragraph 5 related to paragraph 4?

- (A) Paragraph 5 finishes the explanation of the food-desert theory initiated in paragraph 4.
- (B) Paragraph 5 explains why the author calls the theory set out in paragraph 4 the food-desert theory.
- (C) Paragraph 5 suggests an alternative to the food-desert theory described in paragraph 4.
- (D) Paragraph 5 mentions the limitations of the food-desert theory introduced in paragraph 4.

12. According to paragraph 5, what is responsible for the relative scarcity insects and other prey in dipterocarp forests?

- (A) The proficient hunting method of lizards and geckos
- (B) A plethora of carnivorous animals in dipterocarp forests
- (C) The unusual behavior of insects and other prey to eat the toxic seeds, flowers, and fruits of dipterocarp trees
- (D) Irregular flowering and fruiting cycles of Dipterocarps trees

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

For each group, a dipterocarp forest is like a desert in that food resources are few and far apart.

Where would the sentence best fit?

14. Summary

Various theories have been suggested to explain the unique abundance and diversity of gliding animals in the rain forests of Southeast Asia.

Answer Choices

- (A) The fact that the abundance of gliding animals in different regions of the world corresponds to difference in tree height, canopy structure and amount of vines has been explained by ecologists.
- (B) One theory suggests that so many gliding species evolved in Southeast Asia because the forests are exceptionally tall, but there is evidence that the theory is not enough.
- (C) Jumping from tree to tree or descending to the ground and walking in forests that are dominated by tall trees may be a more energy-efficient way of traveling through the forest than gliding.
- (D) The fact that gliding animals are most abundant in the short-structure forests of Southeast region indicates that gliding did not evolve as an adaptation to an environment of tall trees.
- (E) Dipterocarp trees create an environment in which many species must travel widely to find food, and gliding may have evolved as a rapid and efficient way of moving between tree crowns.
- (F) The hypothesis that gliding evolved to compensate for a scarcity of vines linking tree canopies overlooks problematic evidence from both Southeast Asian and Amazonian forests.



參考答案：（第一篇第四题疑为 A，答案可能有误）

Controversy about Causing Emotion	DABCA DDCBD BBAfCDE]
Three Theories about Origin of Life	CABDD ABDDBD CBD[ACF]
The Distribution of Gliding Animals	DCAB[AD] BDCCB ADC[BEF]

4.The Sentinel Behavior of Meerkats

A species of small mongooses in Africa called meerkats share sentinel (guard) duties to warn other group members by repeating alarm calls if a predator is seen. This is an important job, because when meerkats are foraging, their heads are in the ground seeking prey, and they cannot see a predator coming.

The question is, why do group members take turns acting as sentinels? Kin selection, that is, being able to save the lives of family members can be one hypothesis for this type of sentinel behavior. Family members share copies of a meerkat's genes. Kin selection is achieved by helping a meerkat's own offspring as well as non descendant kin, including sibling, nieces, nephews, aunts, and uncles. Therefore, if members of a certain group are closely related, a sentinel ensures that copies of its genes can be passed on to future generations by saving the majority of family members by alerting others, even at the expense of its own life.

Assuming this hypothesis is true, we can predict that group members have close genetic ties. Otherwise, kin selection would not work. But this prediction does hold true. A dominant, breeding female is mother to 75 percent of all the litters in a group, and one dominant male fathers 75 percent of all the pups born. Even though a typical meerkat group includes a few immigrants, most subordinate adults are siblings or half siblings. Therefore, it is likely that subordinate adults share 25 or 50 percent of their genes.

On account of most meerkat group members being family, it is possible that kin selection has favored sentinel behavior. Nonetheless, by itself, a close inherent relationship is not enough evidence to conclude that kin selection has played a role. Thus, we need further evidence, and must improve the prediction.

Based on the same hypothesis, a more specific prediction is that each mongoose should increase the frequency of sentinel behavior when they are guarding family members. This new prediction needed testing, so the group was observed to determine which members stand guard and when. The immigrants without any kin relations to other group members acted as sentinels just as much as the individuals with many relatives nearby. Therefore, the result of this test does not support the kin-selection hypothesis.

Another hypothesis that is often suggested to explain such cooperative behavior is that it results from reciprocal altruism—each individual takes turns standing guard to benefit the rest of the group, rather than itself. The reciprocal altruism theory can work only when those who cheat by avoiding guard duty can be identified and punished by the rest of the group. This hypothesis produces the prediction that there should be a regular rotation of sentry duty within the group and that the ones who neglect this duty should be chastised. However, this is not observed. In fact, the group members do take turns on sentry duty, but there is no predetermined order for this. In addition, when some members shorten their shift, other group members increase their contributions to compensate. The predictions and observations of the reciprocal-altruism hypothesis do not coincide with each other.

Yet another hypothesis for the evolution of meerkat sentinel behavior is that it results from selfish antipredator behavior. This idea stems from the fact that the meerkat watching for predators increases its personal safety, and warning others does not harbor any disadvantage. So, when a meerkat has had enough to eat, it should watch for predators. The sentinel on duty can then return to foraging. This hypothesis produces a prediction that sentinel duty is not dangerous or risky in any way. This does seem to be true. Over the course of 2,000 hours of observation, no sentinels were attacked or killed by predators. They may actually be safer because they are the first to sense the predator. Moreover, they generally stand guard within 5 meters of a burrow, and are the first underground when a predator comes close. ■

If a meerkat's personal safety is increased with serving as a sentinel, it would be possible to predict that an individual would spend a proportion of its time guarding, whether it was solitary or part of a group. ■ As predicted, individual meerkats spend about the same time on guard duty as members of large groups. ■ Groups with more members suffer less predation because there is a sentinel for a longer portion of foraging time than in small groups. ■

1. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- (A) A sentinel attempts to save as many closely related family members as possible, even if it dies while doing so.
- (B) The group will not be able to pass their genes to future generations if a sentinel dies before alerting other family members.
- (C) The genes of all members of the group will be passed to future generations even if a sentinel alerts only its close family members.
- (D) Although a sentinel dies while alerting family members, it ensures that its genes will survive in future generations.

2. According to paragraph 3, the theory that meerkats' sentinel behavior is the result of kin selection is supported by the fact that

- (A) sentinels warn the dominant males and females of the meerkat group first
- (B) most members of a meerkat group are hereditarily related to each other
- (C) immigrant members of the meerkat group will benefit from the sentinels' warnings
- (D) only a small portion of a meerkat group serves as sentinels

3. The word includes in the passage is closest in meaning to

- (A) contains
- (B) attracts
- (C) allows
- (D) brings

4. What role does paragraph 4 play in the passage?

- (A) It explains the difficulties in collecting evidence about kinship relationships.
- (B) It introduces a new hypothesis to replace the one presented in paragraph 2.
- (C) It explains the relationship between hypotheses, prediction, and evidence.
- (D) It introduces another observation test of the hypothesis presented in paragraph 2.

5. The word improve in the passage is closest in meaning to

- (A) reject
- (B) clarify
- (C) refine
- (D) discard

6. Paragraph 5 suggests that before the more specific prediction based on the kin-selection hypothesis is tested, researchers involved in the testing should know

- (A) the minimum number of members needed as guards
- (B) which group members were related to each other
- (C) which group members were more skilled at standing guard
- (D) how well group members were protected by the sentinels

7. The phrase In addition in the passage is closest in meaning to

- (A) Over time
- (B) Astonishingly
- (C) Furthermore
- (D) Ultimately

8. According to paragraph 6, what observation was made that weakened the reciprocal-altruism hypothesis?

- (A) Some group members never spent any time guarding.

- (B) There were some moments when no sentinel was present.
- (C) No established order of guard duty existed.
- (D) Some group members were less severely punished than others.

9. According to the hypothesis discussed in paragraph 7, why would a meerkat choose to stand guard once it has eaten?

- (A) To look for new foraging grounds as well as predators
- (B) To protect itself from being attacked by a predator
- (C) To improve its position within the group
- (D) To warn others in its group of any predator attacks, even at great risk to itself

10. According to paragraph 7, the fact that during the 2,000 hours of observation no sentinels were attacked by predators may show that

- (A) sentinels are extremely effective in warning members of their group about dangers
- (B) it is safer for an individual to stand guard than to be guarded from predation
- (C) it is hard to determine how hard guard duty is even from 2,000 hours of observation
- (D) sentinels can ensure that foraging group members stay within 5 meters of a burrow

11. The phrase comes close in the passage is closest in meaning to

- (A) approaches
- (B) move about
- (C) assaults
- (D) assails

12. According to paragraph 8, which of the following statements is supported by studies of individual meerkats' guarding behavior?

- (A) Solitary meerkats spend more time on foraging than ones belonging to a group, but they also watch for predators some of the time.
- (B) Individual animals in large groups tend to spend more time on guard duty than ones in small groups.
- (C) The time spent watching for predators increases a meerkat's personal safety even when the animal is solitary.
- (D) Sentinels prefer to spend their guarding time as solitary individuals rather than as members of a large group.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

This results in meerkats gaining an advantage in survival by gathering in large groups rather than small ones.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Many hypotheses have been suggested to explain why meerkats take turns acting as sentinels, warning other group members of any threats from predators.

- (A) Kin selection is thought to explain sentinel behavior, but this hypothesis does not fully explain how unrelated immigrants can be part of a group.
- (B) Even though sentinel behavior can be explained with reciprocal altruism, there is lack of evidence that group members are

ever held responsible for not taking their turn to guard.

- (C) Guard duty increases the personal safety of every group member, but the increase is at its least for the member who actually stands guard.
- (D) The proposed theory that sentinel behavior increases the safety of only family members is contradicted by the sentinel behavior of immigrant group members.
- (E) A plausible reason for sentinel behavior may be to allow other meerkats a chance to forage for food in groups rather than as individuals.
- (F) Sentinel behavior is most likely the result of self-protection by individuals and only secondarily a means to protect the group.



5. Portraits as Art

According to the Oxford English Dictionary, portraiture is, "a representation or delineation of a person, especially of the face, made by life, by drawing, painting, photography, engraving... a likeness." However, this simplistic definition disregards the complexities of portraiture. Portraits are works of art that engage with ideas of identity as they are perceived, represented, and understood in different times and places, rather than simply aim to represent a likeness. These concepts of identity can encompass social hierarchy, gender, age, profession, and the character of the subject, among other things. Rather than being fixed, these features are expressive of the expectations and circumstances of the time when the portrait was made. It is impossible to reproduce the aspects of identity; it is only possible to evoke or suggest them. Consequently, even though portraits represent individuals, it is generally conventional or typical - rather than unique - qualities of subject that are stressed by the artist. Portrait art has also undergone significant shifts in artistic convention and practice. Despite the fact that the majority of portraits portray the subject matter in some amount of verisimilitude, (an appearance of being true or real), they are still the outcome of prevailing artistic fashions and favored styles, techniques, and media. Therefore, portrait art is a vast art category which provides a wide range of engagements with social, psychological, and artistic practices and expectations.

Since portraits are distinct from other genres or art categories in the ways they are produced, the nature of what they represent, and how they function as objects of use and display, they are worthy of separate study. _ First, during their production, portraits require the presence of a specific person, or an image of the individual to be represented, in almost all cases. ■ In the majority of instances, the production of portraiture has necessitated sittings, which result in interaction between the subject(s) and artist throughout the creation of the work. ■ If the sitter is of high social standing or is occupied and unavailable to sit in the studio regularly, portraitists could use photographs or sketches of their subject. ■ In Europe, during the seventeenth and eighteenth century, the sitting time was sometimes decreased by focusing solely on the head and using professional drapery painters to finish the painting. For instance, Sir Peter Lily, the English artist, had a collection of poses in a pattern book that enabled him to focus on the head and require fewer sittings from his aristocratic patrons. Portrait painters could be asked to present the likeness of individuals who were deceased. In this sort of instance, photographs or prints of the subject could be reproduced. Theoretically, portraitists could work from impressions or memories when creating a painting, but this is a rare occurrence according to documented records. Nonetheless, whether the work is based on model sittings, copying a photograph or sketch, or using memory, the process of painting a portrait is closely linked with the implicit or explicit attendance of the model.

Furthermore, portrait painting can be differentiated from other artistic genres like landscape, still life, and history by its connection with appearance, or likeness. As such, the art of portrait painting got a reputation for imitation, or copying, instead of for artistic innovation or creativity; consequently it is sometimes viewed as being of a lower status than the other genres. According to Renaissance art theory, (which prevailed until the start of the nineteenth century) fine art was supposed to represent idealized images, as well as to be original and creative instead of to copy other works. Portraiture, in comparison, became linked with the level of a mechanical exercise as opposed to a fine art. Michelangelo's well known protest that he would not paint portraits because there were not enough ideally beautiful models is only one example of the dismissive attitude to portraiture that persisted among professional artist - even those who, ironically, made their living from portraiture. In the time of modernism, during the nineteenth and twentieth centuries, the attitude towards portraiture was critical. Even so, artists from around the globe persisted painting portraits in spite of their theoretical objections. Picasso, for instance, became renowned for cubist still-life painting early in his career, but some of his most effective early experiments in this new style were his portraits of art dealers.

1. The word engage in the passage is closest in meaning to

- (A) construct
- (B) are pleased
- (C) are altered
- (D) are involved in

2. According to paragraph 1, which of the following gives support of portrait painting's complexity?

- (A) Portraits representing faces are more true to life than portraits that portray a whole figure
- (B) Portrait art comes in many varieties, which include painting, photography, and drawing
- (C) Portraiture tries to portray the most uncommon attributes of a given subject
- (D) Portraiture is an interpretation of a subject rather than a copy of it

3. The word prevailing in the passage is closest in meaning to

- (A) apparent
- (B) distinct
- (C) steady
- (D) current

4. According to paragraph 1, which of the following best illustrates the development of portrait painting as an art form?

- (A) It started as a minor art form, but became progressively more popular
- (B) It is a comparatively stable art form due to its fixed subject matter
- (C) It has been subjected to consistent and major style shifts
- (D) It was the first art form concerned with the identity of the subject

5. Paragraph 2 suggests which of the following differences between portraiture and other types of art?

- (A) Portraits portray the subject matter in a more accurate manner than other forms of art
- (B) Portraits typically take less time to produce than other art forms
- (C) Portraits typically necessitate an increased level of personal interaction between the subject matter and artist than other art forms
- (D) In opposition to other art forms, portraiture usually necessitates collaboration among several artists

6. In the passage, the word solely is closest in meaning to

- (A) only
- (B) primarily
- (C) particularly
- (D) directly

7. The author discusses the "Sir Peter Lily, the English artist," to provide an example of an artist who

- (A) invented a method to reduce the necessary number of sittings for his rich patrons
- (B) employed professional drapery painters to help him finish his portraits
- (C) concentrated on painting different parts of the subject body at each sitting
- (D) had an uncommon range of patrons as subjects

8. All of the following are mentioned in paragraph 2 as techniques employed by artists to create portraits EXCEPT

- (A) combining facial traits from different subjects
- (B) observing the subjects directly during painting
- (C) copying a photograph
- (D) recalling what the subject looked like from memory

9. According to paragraph 3, portraiture grew to be regarded as a mechanical practice due to its association with which of the following?

- (A) innovation
- (B) imitation
- (C) perfectionism
- (D) creativity

10. What can be inferred from paragraph 3 regarding Michelangelo's view of portraiture?

- (A) He felt that imitating and copying were prerequisites of achieving creative portraits.
- (B) He thought that portrait artists ought to select subjects from long ago rather than present day.
- (C) He felt that portrait art should be viewed as a form of fine art.
- (D) He felt that portraits should only portray idealized beauty.

11. The word critical in the passage is closest in meaning to

- (A) misunderstanding
- (B) fault-finding
- (C) ignorant

(D) insignificant

12. In paragraph 3, the author talks about Picasso as an example of an artist who

- (A) altered the way other artists felt about portrait art
- (B) relied on portrait art to establish a high reputation
- (C) had fewer theoretical objections to portraiture than most modern artists
- (D) created portraits in spite of his doubts about portraiture as a fine art form

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage

In certain instances, portrait artists depended on a combination of direct and indirect involvement with their subjects.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Portraiture as an art form is more complex than is suggested by its definition.

- (A) The definitions of portrait art in the dictionary have regularly transformed throughout the years to reflect shifting attitudes regarding the genre.
- (B) Portrait art should be considered as a distinct artistic genre due to its intense occupation with the subject and the way in which it was produced.
- (C) Beginning in the Renaissance and continuing into the start of the nineteenth century, portrait art was idealized to a greater degree than it is in today.
- (D) Portraits generally mirror the conventions of the time rather than the unique qualities of the individual.
- (E) Throughout history, the majority of professional artists avoided portrait art since they regarded it as a mechanical art form, and not as fine art.
- (F) Portrait art was at times viewed in a negative light since it was considered as simple copying void of artistic innovation.

7.Agriculture in Medieval Japan

A rapid population of Japan occurred during its medieval times. Japan's population was around 7 million but it rose keenly to 12 million from year 1200 to 1600. In this period, numerous hamlets formed throughout the country. They were mostly formed in the lands listed as "unsettled" or as a "wasteland" before 1300. There were many facets in increase in number of new hamlets, but by far the most significant characteristic of newly formed hamlets were that they were much bigger in terms of size compared to that of the hamlets built before 1300. There are many factors for forming of such large towns that contributed to increase in population mass. Some factors that can be considered are people's demand for local authority, voluntarily, to defend themselves against outside threats or to form religious communities. Whatever the impetus, such formation of large villages was due to improvements in the agricultural technologies. Some improvements in technologies involved turning over of fields, irrigation methods, and usage of waterwheels, iron tools and diversification of crop output.

Among many improvements in agriculture, field leveling was the most basic practice used to optimize the land for farming. The farmers would create flat land for farming by leveling a field. They then would use the surfeit soil from the field to level the slightly slanted field. As a result, two fields of difference in altitude would be formed. Such difference in elevation allowed farmers to use the lower for rice paddy and the higher for dry crops. Practice of field leveling allowed a paddy culture to settle and allowed vast variety in dry crops to be produced due to the formation of drop crop field. Though the labor involved in formation of fields was enormous, the field preparation enabled marshlands alongside the rivers to be used for husbandry even if the rivers were uncontrolled.

Rice crops require ample water for growth and it takes much time until they are ready to be harvested. So farmers naturally worked by places where they had access to ample supply of water, such as riverbanks, streams and ponds. However, natural water supply was inefficient for the growth of rice, especially in sweltering summers. This led to the usage and development of ditches and dikes. Development in drawing the water from the distant locations led number of dams to increase and directed them to wherever they needed them. This was most evident in Yamato Basin where farmers built permanent dams. The water detained in the pools was kept for times when they needed water for farming in droughty seasons. Such development led to keen proliferation of crop output as the heated water metabolized the germination of crops and caused crops to mature even faster.

By mid 1500s, one quarter of all paddy land were used to double crop. The farmers not only used fields to grow two crops in a year, but they even grew three in a single annual cycle. ■ An envoy from Korea stated that Japanese farmers from Hyogo region would grow barley and sow in winter and harvest them in summer. ■ Followed by rice cropping in summer and fall, buckwheat was harvested in winter. ■ As time went on agriculture advanced, such technique progressed from generation to generation. ■ Farming became more consistent and the crop output became even greater. A greater sense of discipline in land tilling and wide range of crops being planted in the same piece of land broadened the understanding of agriculture in farmers.

Crops harvested were used for farmers themselves. In many cases, one hectare of decent land was enough to sustain their entire family. They would only plow enough land for food to cater families for several reasons. Much of the medieval Japanese reclamation of land was due to the search for enough arable land to meet the food needed for just a single household. In case a farmer having enough fields for crop output required for his family, he would expand his fields so that one had to put into farming for his crops to grow. This was further compounded by the scarcity of land for farming as well as limiting capacity for water and fertilizer supplies, not to mention the likelihood of antagonizing neighbors. Taking these variable factors into consideration,

farmers of this period persisted with single hectare or less of arable land, just enough to sustain their families.

1. What can be inferred about the reason for the consolidation of population in medieval Japan?

- (A) Whether from the orders of local government or due to a voluntary nature, the goal was to avoid danger.
- (B) A list of possible catalysts is known but the exact combination of reasons is unclear.
- (C) The desire to settle previously undeveloped land allured people, whatever their particular situation.
- (D) Population boomed because there were more lands given out to people by the authority.

2. What was the significance of villages that appeared after 1300?

- (A) They exemplified advancing capacity of food output that could support the increasing masses.
- (B) They had been settled in areas that had been previously described as “unsettled” or as “wastelands.”
- (C) They tended to gain greater area per person than the cities built before 1300.
- (D) The population had grown by 5 million in a very short amount of time which forced villages to appear.

3. Why does the author mention various agronomical techniques?

- (A) To explain the types of prowess needed to be learned in order for villages to prosper
- (B) To begin a discussion on the relevance of these techniques in increasing productivity of fields
- (C) To isolate the practices that facilitated the increased population size of the villages
- (D) To point out the different jobs that had to be performed on a successful field

4. The word impetus in the passage is closest in meaning to

- (A) situation
- (B) purpose
- (C) motivation
- (D) outcome

5. Which of the following is NOT true about field leveling?

- (A) Extra soil was used to level the neighboring field.
- (B) It was not a very arduous process.
- (C) It resulted in two different growing platforms.
- (D) It didn't involve a high degree of apparatuses.

6. The word optimize in the passage is closest in meaning to

- (A) harness
- (B) take advantage of
- (C) enhance
- (D) make the best use of

7. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage?
Incorrect answer choices change the meaning in important ways or leave out essential information.

- (A) The innovation of this technique facilitated the utilized water from uncontrolled rivers.
- (B) This allowed for the swampy marshlands near rivers to be uncontrolled by the rivers themselves.
- (C) Farmers were now able to invest labor resources into the development of marshlands from rivers.
- (D) Field leveling was a technique that allowed farmers to exploit the natural layout of their terrain.

8. What TWO benefits did the shallow holding pools in the Yamato Basin provide?

- (A) A faster harvesting process for the annual yield
- (B) An additional source of water
- (C) The reduced need of relying on ditches and dikes
- (D) A source of heat for the water

9. The word ample in the passage is closest in meaning to

- (A) constant
- (B) reliable
- (C) plenty
- (D) nearby

10. Why does the author mention farmers in the Hyogo region?

- (A) To point out the immense amount of work required in triple cropping
- (B) To express veneration for work carried out by farmers in Japan's medieval period
- (C) To introduce a topic and explain its reasoning in the next paragraph
- (D) To illustrate an example of year long utilization of a field

11. What can be inferred about the desire for land during the medieval reclamation?

- (A) People tried to secure only enough land to look after their rations.
- (B) Land use was limited by the fact that Japan was not a very large country.
- (C) The existence of several physical barriers prevented the desire for any superfluity of lands.
- (D) Most of the land was undeveloped and owned by landlords.

12. The word antagonizing in the passage is closest in meaning to

- (A) altering
- (B) belittling
- (C) alienating
- (D) angering

13. Look at the four squares [■] that indicate where the following sentence can be added to the passage. **The potential benefit of this was not the amounts of crop output.**

What would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

The drastic increase in Japan's population during its medieval period signified the ability of the country to create enough fields close to a population to be able to support a denser population. Answer Choice

- (A) The reason for the consolidation of citizens lied in a number of different factors, such as the formation of religious communities or simply voluntarily.

- (B) Field leveling was a practice that allowed the cultivation of two entirely different types of crops with very efficient use of land.
- (C) Farmers now had access to the marshlands of Japan close to rivers that were, as yet, not controlled by humans.
- (D) Numerous advancements in land utilization, such as dams and dikes, as well as the practice of double cropping allowed farmers to maximize the possible output from a single patch of land.
- (E) Koreans noted that farmers in the Hyogo region of Japan had accomplished a smooth triple cropping annual cycle that used the land during the entire year.
- (F) The aspiration for land above that required to achieve fulfillment of food needs was curbed by the existence of several factors, such as limited land resources.



8.Grassland Environment

As opposed to forests, grasslands receive markedly less precipitation. ■ This is one of the reasons why trees, which usually require a significant amount of moisture, are spaced relatively sparsely in grasslands. ■ Due to the facts that most grasslands are situated in the center of continental landmasses, and that coastal mountain ranges can block west-to-east winds as is the case with the Great Plains in North America, air masses from the north and south are highly influential on grasslands. ■ The Great Plains receive cold Arctic air in winter and hot tropical air in summer. ■ A typical grassland climate includes hot summers with long periods of desiccation and cold winters with erratic snow cover. Consequently, plants in these grassland areas experience low soil moisture and, additionally, are exposed to **fixll** sun, strong dry winds, and extreme summer heat and winter cold. Grassland plants must adapt to these conditions.

To support research on various types of ecosystems, during the 1970s, The National Science Foundation started what is known as the Long-Term Ecological Research (LTER) program. With the LTER program, scientists working at far-away sites can share data and collaborate to identify and understand large scale ecological patterns. At one LTER site in the Flint Hills of Northeastern Kansas, scientists made an important discovery about the influence of precipitation on plant productivity, the fuel on which ecosystems run. It was found that grasslands respond more strongly to pulses in rainfall than any other ecosystem. Dramatic bursts of plant growth and significant increases in primary productivity result from intermittent rainfall in grasslands. This pattern indicates that grasslands have a high underlying growth potential that surfaces when enough water is suddenly available. Because of these findings, LTER scientists have proposed that grassland annual primary productivity can be a useful indicator of global climate change. Similar to the canaries that miners used to carry into mines to warn of poisonous gases, grasslands may be able to serve as a warning of climate change and how it is affecting plants and humans.

When maintaining grasslands, fire plays an essential role. In the summer, dry grasses and their dead remains which have accumulated over previous years on the upper layer of soil (known as mulch) make a highly combustible fuel for fires started by people or lightning. When there is an absence of rain, dry lightning is a common occurrence in grassland areas, as is the accidental escape of campfires or other human-related fires. These fires in grasslands can burn for many kilometers before being stopped by rain or wet areas. Studying charcoal buried in soil layers have suggested that under natural (pre-European settlement) conditions, at least a hectare of North American prairie may have burned once every 5 to 30 years.

Plants in the grasslands are adapted to survive fire in many ways, but, just as importantly, fire is an essential component of grassland ecosystems. The invasion of both fire-sensitive plants and most trees are prevented by fire, and fire clears away dead plant material, thereby releasing necessary nutrients that facilitate new growth. Since the development of new grass was favored by Native Americans' horses and was also attractive to the buffalo they hunted, they often set grassland fires to stimulate it. These fires on grasslands was also helpful in increasing productivity of the Native Americans' wild food plants, improving visibility (which brings security), and helping control pests such as ticks. Nowadays, people use fire as an important tool in restoring and preserving grasslands.

eatrs create a disturbance to which grassland plants are adapted in many ways. In grasslands, grazing is more significant than in other types of ecological communities. Up to 60 percent of energy and grassland materials flows through primary and other

consumers. In other terrestrial ecosystems, on the other hand, less than 5 percent of the ecosystem's material and energy usually flows through consumer food webs. This difference coordinates with the relatively large herbivore populations that grasslands support.

Benefits brought about from grazing include helping lower-growing plants by preventing shading by tall species, and providing a rich source of nutrients from grazer excrement. Grazing and fire both prevent just a few plants from dominating grasslands and help maintain the high natural plant diversity. Researchers at the University of Minnesota have found that high plant diversity has been pivotal in establishing and maintaining the high fertility in natural grassland soils.

1. According to paragraph 1, grassland regions have extreme summer heat and winter cold as a result of those areas being

- (A) affected primarily by air masses that move from east to west
- (B) surrounded by mountains on all sides
- (C) in the middle area from both the tropics and the Arctic
- (D) isolated from west-to-east air movements

2. The word collaborate in the passage is closest in meaning to

- (A) work together
- (B) make an attempt
- (C) compare outcomes
- (D) acquire evidence

3. Why does the author use the phrase the fuel on which ecosystems run?

- (A) To explain why grasslands are a difficult ecological pattern to study
- (B) To illustrate what discovery scientists made at the LTER site in the Flint Hills of northeastern Kansas
- (C) To point out that research run by the LTER program can lead to practical results
- (D) To emphasize the fundamental importance of precipitation on ecosystems

4. The word intermittent in the passage is closest in meaning to

- (A) strong
- (B) regular
- (C) occasional
- (D) unusual

5. According to paragraph 2, what indicates the strong growth potential underlying in grasslands?

- (A) The frequency with which they occur geographically
- (B) The average rate in which they reproduce
- (C) The way they respond to rainfall
- (D) The speed with which they respond to global climate changes

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage?

Incorrect choices change the meaning in important ways or leave out essential information.

- (A) Dead and dry grasses that form on the soil surface in grasslands are called mulch.
- (B) In summer, people use dry grasses and their dead remains to fuel fires in grasslands.
- (C) In grasslands, summer fires started either by lightning or by people are fueled by dry grasses and their dead remains at the soil surface.
- (D) It is always easy to find fuel to use for fires in grasslands because dry grasses and mulch accumulate at the soil surface during summers

7. According to paragraph 3, which of the following is true about lightning in grasslands?

- (A) Only in the dry months of summer do storms with lightnings occur in grasslands.
- (B) Only during the most dry months of summer do storms with lightning occur.
- (C) Almost all the severe storms that have happened in grasslands involve a great deal of lightning.
- (D) Lightning at the absence of rain is a common phenomenon in grasslands.

8. The word essential in the passage is closest in meaning to

- (A) integral
- (B) busy
- (C) likely
- (D) known

9. According to paragraphs 3 and 4 , what is one way in which fire helps maintain the character of grasslands?

- (A) It prevents trees from becoming permanently established.
- (B) It helps new plants grow.
- (C) It causes charcoal to deposit in layers on grassland soils.
- (D) It kicks out the wild grazing animals that can quickly consume too many of the grassland plants.

10. According to paragraph 4 , each of the following is a benefit that Native Americans gained from grassland fires EXCEPT

- (A) increased height of new plants in which to hide
- (B) decrease of insect pest populations
- (C) enhanced productivity of wild food plants
- (D) raised availability of the food their horses preferred

11. According to paragraph 5 , what indicates that grazing is more essential in grasslands than in other ecological communities?

- (A) The animals that graze on grasslands include more natural herbivores than livestock, which is usually not true of other ecological communities.
- (B) Adaptation to the disturbance of grazing causes grassland plants to grow more slowly than plants in other ecological communities.
- (C) In grasslands herbivores consume a higher percentage of the plant material than they do in other ecological communities.

(D) Grazing is a harmful disturbance in grasslands but not in other ecological communities.

12. According to paragraph 6 , one way grazing contributes to maintaining high natural plant diversity of grasslands is by

- (A) controlling how fast nutrients leave the soil
- (B) guaranteeing that enough sunlight is given to low-growing plants
- (C) allowing tall plant species to compete for space
- (D) improving the distribution of the plants with the highest productivity throughout grassland areas

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

A typical grassland area averages only 25 to 100 centimeters per year.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the, most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Grassland plants must be able to cope with low moisture levels and extreme climates. Answer Choice

- (A) Before being stopped by wet areas or rain, fires started by humans have resulted in grasslands , being burned over many square kilometers.
- (B) Grassland plants are not only well adapted to survive grazing, but grazing actually helps maintain high plant diversity.
- (C) Recurring grassland fire serves an essential part in maintaining grasslands , characteristic vegetation.
- (D) Grassland ecosystems have been preserved with the help of Native Americans hunting herbivores such as buffalo and horses.
- (E) Despite grasslands having hot dry summers, plant growth dramatically increases in response to some rainfall.
- (F) Grassland summers are dry, so grassland plant productivity relies largely on the amount of snowfall that accumulates throughout winter.

9.The British Economy Under the Roman Empire

Following the Roman Empire conquering the area in the first century A.D., there is a great deal of archaeological evidence for the economic growth of the British Isles. Prior to this event, the economy of the British Isles, which was based on manufacturing, was centered mainly on the household and on craft skills, and where the best quality and greatest range of goods were largely a monopoly of the tribal aristocracies. This was the nature of the economy which lasted in regions of Britain that were unconquered by the Roman Empire, even though some Roman products were utilized in such areas. The majority of these Roman artifacts were glass vessels, pots, as well as small metal objects that were dispersed over a vast region. They perhaps held a symbolic value and were not necessarily used for their originally designed purposes. The spread of Roman objects beyond Roman Britain does not seem to have happened on an enormous scale. In areas where artifacts are more numerous, it is likely due to gift giving during close interactions between the Roman government and the tribes.

In regions that experienced direct economic control under the Romans, however, economic growth is clearly notable. There was an enormous increase in the number and variety of goods in circulation and the range of settlements in which they were found. This is clearly true in the overwhelming majority of excavated sites in Roman Britain, with the only exceptions being some rural regions that continued the pre-Roman, Iron Age pattern. The majority of sites resulted in the discovery of an abundance of iron, glass, and pottery, and good quantities of copper alloys, lead, tin, silver, and occasionally gold. For example, the humble iron nail is found in numbers not repeated until the Industrial Revolution.

The technology levels and range of the manufacturing of these objects also developed alongside the sheer increase in their quantity. During the Iron Age, the typical household objects were usually manufactured using a low technology of craft manufacture. Later, this changed to more specialized and larger-scale production methods. During this time, specialized workers could utilize equipment manufactured through time and resource investments. In these regions, small-scale workshops used by specialized craftsmen betoken full-time employment in this work. Regardless of the large increase in the scale of manufacturing, there is little evidence of major growth in the size of productive units. We are left with the impression of an economy still based on small-scale craft production.

Where we do see an important change is in the removal of any exclusive association between the best traditional craftsmen and the governing elite. The powerful could show off their status in new ways, particularly by using Roman architecture and domestic decoration, but the traditional classes of decorative metalwork manufacture no longer seem to have been under the control of the tribal leaders. Rich objects from a wide range of archaeological sites imply the deterioration of this monopoly. There are a number of contributing factors. The control of precious metals moved to the imperial government immediately after the conquest, and gold and silver were also removed from circulation when captured as booty during the invasion. Similarly, changes in taste and the fashions of wealth and status display were stimulated by the arrival of new things like Roman dress, architecture, and sculpture.

These changes in manufacture were accompanied by increased distances over which many goods were transported to their consumers. ■ The bulk of pottery and other items originated locally, during the Iron Age; but after the Roman invasion, these objects had been produced over a far greater range of distances. ■ In this way, vast regions of the Roman province were incorporated into a society where there was wide access to material wealth. _ New changes in manufacturing production were coupled with huge increase in the importation of goods from elsewhere in the empire. ■ These commodities, which included Mediterranean foodstuffs such as olive oil as well as comparatively low-value objects such as decorated pottery, also achieved a wide distribution and are found in many different types of site.

- 1 • Which of the sentences below best expresses the essential information in the highlighted sentence in the passage?

Incorrect choices change the meaning in important ways or leave out essential information.

- (A) Prior to the Roman conquest, the majority of manufacturing was focused on supplying the aristocratic tribal households.
- (B) Before the first century A.D., the manufacturing economy was not as developed as it would become under the Roman Empire, but there was a vast range of articles available to the aristocracy.
- (C) Before the Roman conquest, the majority of manufacturing in the British Isles was mostly devoted to household product! with the highest quality products monopolized by the tribal aristocracies.
- (D) The pre-Roman British society was ruled by tribal aristocracies and evolved with a focus on high-quality craftsmanship of household goods.

2. The word lasted in the passage is closest in meaning to

- (A) endured
- (B) spread
- (C) was copied
- (D) was presented

3. The word dispersed in the passage is closest in meaning to

- (A) found
- (B) spread
- (C) amassed
- (D) manufactured

4. According to paragraph 1, the presence of a multitude of Roman products in the regions of Britain that were not under Roman was caused by

- (A) the trade of goods between traveling Roman craftspeople through Britain and local inhabitants
- (B) gifts Roman government officials provided to local tribes due to close relationships with the tribes and their officials
- (C) trade between centers of manufacturing in Roman Britain and in centers in other regions of the Roman Empire
- (D) the settlement of a high number of Roman people in regions around the Roman Britain province

5. Paragraph 2 mentions all of the following as evidence of Roman Britain's economic development EXCEPT

- (A) a rise in the variety of products available
- (B) a growth in the quantity of settlements where products became available
- (C) an increase in the diversity of materials uncovered at archaeological sites
- (D) a rise in the rural population

6. In paragraph 2, the author talks about the iron nail in order to

- (A) imply that most products manufactured were considered to be practical or useful
- (B) illustrate that iron was a key player in Britain's economy before the Roman invasion
- (C) give support for the statement that there was much economic activity in Britain following the Roman conquest
- (D) prove that archaeologists have underestimated the degree of the Roman conquest on the British economy

7. According to paragraph 3, manufacturing in the period of Roman economic control over the British was characterized by

- (A) specialized production techniques

- (B) a growth in the productive units' size
- (C) the creation of workshops that hired a large number of staff
- (D) the utilization of simple equipment for the making of household items

8. Paragraph 3 suggests which of the following statements about craft manufacture in Roman Britain?

- (A) Workers did not have to invest much resources and time in the production of crafts.
- (B) A number of domestic products was increasingly being made using iron.
- (C) Workshops remained small despite the rise in production.
- (D) Production thrived although craftspeople continued to use old-fashion production techniques.

9. The word Similarly in the passage is closest in meaning to

- (A) Although
- (B) Unavoidably
- (C) Shockingly
- (D) Likewise

10. Paragraph 4 discusses all of the following as reasons for the breakdown of the tribal leaders' monopoly in decorative metalwork manufacture EXCEPT

- (A) the tribal leaders' reduced influence within their communities
- (B) the effect of Roman culture on the elite's display of wealth and status
- (C) the removal of gold and silver from circulation
- (D) the Roman government's control of precious metals

11. The word huge in the passage is closest in meaning to

- (A) instant
- (B) vast
- (C) steady
- (D) quick

12. The author mentions olive oil and decorated pottery to support the claim that

- (A) the inhabitants of Roman Britain had access to daily and specialty products
- (B) the rise in product manufacture caused a notable increase in waste material
- (C) a number of commodities were manufactured in Britain prior to the Roman conquest
- (D) imported goods had become a common feature of Roman Britain's economy

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

Furthermore, this provincial society was itself economically integrated with markets beyond its borders.

Where would the sentence best fit?

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question

is worth 2 points.

The British Isles' economy experienced significant changes after the Roman conquest. Answer Choice

- (A) British craftspeople slowly stopped producing items that generally were of little practical value in favor of the manufacturing of household products.
- (B) Manufacturers invested an increased amount of resources and time towards developing specialty equipment.
- (C) The production of goods still remained small scale , but the exclusive relationship between the governing elite and the best craftsmen broke down.
- (D) Trade goods manufactured in Roman Britain were spread throughout the Roman Empire.
- (E) The number and variety of goods found at sites in Roman Britain as well as the development of sophisticated manufacturing processes are evidence of economic development under the Romans.
- (F) Roman Britain experienced a rise in product imports from elsewhere in the empire.



參考答案：

Agriculture in Medieval Japan	1. BBBCB 6. DC[BD]CD 11. ADC[BDE]
Grassland Environment	1. DADCC 6. CDABA 11. CBA[BCE]
The British Economy Under the Roman Empire	1. CABBD 6. CACDA 11. BDC[CEF]

講了 Kelp 和 urchin 數量關係(說講這兩種數量關係的目的是?)Kelp 越來越少了,大家就懷疑是 urchin 的錯。然後就提出 Assumption 啊,說 Urchin 的 increase 會導致 kelp 的 decrease。可是又說,這種 assumption 不對,因為 kelp 可能是受了 XX 和厄爾尼諾的影響,跟 Urchin 不關係。

第二和第三段分別舉了兩個例子,大概的意思是證明 kelp 的 decrease 不是因為 Urchin 的錯(有題,問二三段的關係),有可能是因為什麼病的影響而導致 kelp 的減少。

不過,因為 Urchin 的 increase 而導致 kelp decrease 的例子是有的,然後就舉了兩個 support 的例子。就說了 otter、lobster(會吃 Urchin 的東東)多的時候,kelp 也會多,因為 otter 和 lobster 會吃了 Urchin, U 減少, K 就自然增加。

單字: contention: claim ; whereas

Foundational and Keystone Species

The harmony present in any ecological system relies on the inclusion of different types of species, each with a specific role or set of roles. Although to some extent all species contribute in important ways, scientists are now learning about the pivotal parts played by, what are being called foundational and keystone species. Interestingly, the latter represent only a small fraction of the overall animal population within an ecosystem while the former can make up a significant portion of it. In both cases, though, their removal often results in massive changes to, if not the complete destruction of, their ecosystems. Efforts to conserve these species are currently based on the new idea that they prevent the start of a domino effect that would, once started, be unstoppable.

Foundational species are the backbones of the systems they inhabit. These species are referred to as primary producers, which means that they generate a large amount of the abundance other species require to survive. In most cases, they are a type of vegetation or stationary animal. Coral, which tends to grow in large colonies, is often pointed to as an example. Eventually a group of colonies will form a coral reef, which can sustain a complete ecosystem around it. The reef contains the skeletal remains of coral at its bottom and living coral on the top. Numerous animals, including zooplankton and sponges, live in the small crevices located in the rock-like bottom. At the top, the coral interacts with seaweed and other forms of vegetation to regulate the levels of nutrients and gases in the water. There are also more than four thousand types of fish that live on or around most coral reefs. Fish rely on the holes in the reef for protection. The more colorful species of coral provide excellent camouflage for fish, as well. Obviously, without its foundational species, an ecosystem would collapse.

Keystone species play less apparent but equally important roles. The name comes from the keystones used in the building of stone archways. The keystone receives the smallest amount of pressure of all the stones in the arch, but, if it is removed, the arch will collapse. In the same way, research is now showing, certain species within ecosystems, although they are smaller in numbers and biomass than most of the other species present, act like keystones.

These species are being categorized as either predators or engineers, depending on their relationships with the various species around them. These categories are not absolute, though, and animals are moved from one to the other as new facts come to light.

In the case of predator keystone species, there are four carnivores that prey on herbivores and other animals. Also, the herbivores usually have no other natural predator in the ecosystem. The sea otter is now considered by many scientists to be a keystone species, because it controls the number of sea urchins, which have few other predators. Both species can be found around kelp forests, which are in the warm parts of the world's oceans. Because kelp is an underwater plant, its roots are not used for the collection of nutrients. Instead, they are there to anchor the plant. As soon as enough sea urchins chewed on the roots, which are fragile, the kelp would be removed from the ecosystem. If the sea otter were to disappear, the urchins would quickly grow in number and destroy all the kelp.

■ Engineer keystone species maintain the balance in ecosystems in different ways. ■ Although bears tend to live in forests, they bring in important sources of nutrients from ocean and sea-based ecosystems. ■ The bears capture large salmon from the water and take them into the forest to eat them. ■ This distributes large amounts of protein in the form of bear waste matter as well as uneaten portions of salmon. The protein eventually supports life in the forest, either as a food source for smaller animals or the vegetation. Another species that plays a part similar to that of bears is the beaver. Through the construction of dams, beavers convert small rivers into ponds and marshes. The new landscape in turn supports a variety of fish, which in turn provides a dependable source of food for the beavers. The removal of beavers from an ecosystem would also remove the landscape it depends on.

1. The word **extent** in the passage is closest in meaning to
 - (A) amount
 - (B) explanation
 - (C) purpose
 - (D) average
2. According to paragraph 1, foundational species are usually
 - (A) responsible for environmental damage
 - (B) present only in certain ecosystems
 - (C) close to complete extinction
 - (D) found in the largest number
3. The word **sustain** in the passage could best be replaced by
 - (A) endure
 - (B) create

- (C) receive
- (D) support

4. The word **it** in the passage refers to

- (A) coral
- (B) coral reef
- (C) ecosystem
- (D) colony

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- (A) The foundational species become obvious once the system has collapsed.
- (B) Ecosystems will not collapse as the foundational species do not exist.
- (C) It is clear that the system would fall apart if it did not have its foundational species.
- (D) Without the collapse, the foundational species are not obvious in the ecosystem.

6. The author mentions arches in paragraph 3 in order to

- (A) help understand what keystone species means
- (B) provide an example of an ecological imbalance
- (C) show the problems with a foundational species
- (D) explain the role of a type of animal species

7. Which of the following can be inferred from paragraph 3 about keystone species?

- (A) Their roles have been understood for a long time.
- (B) They are related to a large number of ecosystems.
- (C) They are now seen as less important in nature.
- (D) They are the subjects of ongoing research.

8. The phrase **In the same way** in the passage could best be replaced by

- (A) similarly
- (B) usually
- (C) conversely
- (D) strongly

9. It is NOT true according to paragraph 4 that sea otters

- (A) hunt sea urchins
- (B) damage kelp roots
- (C) eat keystone species

(D) live near water

10. According to paragraph 5, which of the following is true about beavers?

- (A) They are predator keystone species.
- (B) They live in the same areas as bears.
- (C) They usually eat fish from ponds.
- (D) They can create large, strong dams.



11. Look at the four squares [■] that indicate where the following sentence can be added to the passage.

The bears that live on the West Coast of North America are often cited as an engineer species.

Where would the sentence best fit?

12. Directions: Ecological systems require the contributions of various species types. Complete the table by matching the appropriate species with their characteristics. This question is worth 3 points.

Foundational Species	• • •
Keystone Species	• •

- A. They are large in number.
- B. Ecosystems form around them.
- C. They are more beautiful.
- D. They are smaller in number.
- E. Four carnivorous animals belong to them.
- F. They are more dangerous than anything else.
- G. Without them, ecosystems would collapse.

参考答案：

1. A 2. D 3. D 4. B 5. C
6. A 7. D 8. A 9. B 10. C
11. B 12. Foundational Species (ABG) Keystone Species(DE)

1. Putting a date on the first appearance of eyes depends on what one means by eye. If the term refers to a multicellular organ, even if it has just a few cells, then by definition, eyes could not form before there were multicellular animals. But many protists (animal-like, plantlike, or fungus-like unicellular organisms that require a water-based environment) can detect light by using aggregations of pigment molecules, and they use this information to modify their metabolic activity or motility (the ability to move spontaneously and independently). One of the familiar living examples, probably known to anyone who has taken a biology class, is the aquatic protozoan *Euglena*, which has an eyespot near its motile flagellum (hairlike structure). Some living protists are very like their ancestral forms embedded in ancient sedimentary rocks, and this similarity suggests that the ability to detect light and modify behavior in response to light has been around for a very long time. Animals arose from one of such unicellular creatures, perhaps from one already specialized for a primitive kind of vision.
2. An eye is a collection of cells that are specialized for light detection through the presence of photosensitive pigment as well as a means of restricting the direction of incoming light that will strike the photosensitive cells. This definition says nothing about image formation, lenses, eye movements, or any of the other features we associate with our own eyes, but it does recognize the simplest form of functional and anatomical specialisation namely, detection of light. Everything else can be built up from this simple beginning, and some animals appear to have had eyes almost from the beginning of the animal kingdom.
3. Animals were scarce 600 million years ago in the geological era called the Precambrian. There are very few fossil remains from that time (though more keep turning up), and most evidence of the presence of animals is indirect, such as small tunnels in rock that could be ancient worm burrowings. But just 50 million years or so later, fossilized bits and pieces of animals abound, suggesting that a great burst of evolutionary creativity occurred in the 50-million-year interval. This surge of new life, marked by an abundance of animals, is called the Cambrian explosion.
4. The first direct evidence for the early origin of eyes comes from fossils that are about 530 million years old, a time shortly after the Cambrian explosion; they were found on a mountainside in British Columbia in a deposit known as the Burgess Shale. The Burgess Shale fossils are extraordinarily important because among them are remains of soft-bodied creatures, many of them lacking shells and other hard parts that fossilize easily. Consequently, their preservation is little short of miraculous (as are the delicate methods used to reconstruct three-dimensional structure from these flattened fossils), and they are one of the few known repositories of early soft-bodied animals.
5. Not all of the Burgess animals had eyes. However, some did. (Gross features location, size, and hemispheric shape are responsible for the designation of some structures as eyes). The reconstructed eyes of these Burgess animals look superficially like eyes of some living crustaceans, particularly those of shrimp and crabs whose eyes are mounted on stalks that improve the range of vision by raising the eyes above the surface of the head. The eyes of some Burgess organisms sat on stalks; those of others were on or a part of the body surface. One animal, *Opabinia*, had five eyes: two lateral pairs and a single medial eye; at least one of the lateral pairs had stalks that could have been movable. And some trilobite-like animals in

the Burgess Shale had faceted eyes much like those of later fossil trilobites.

6. Although the presence of eyes on some of the Burgess animals indicates that eyes have been around for a very long time, it is unlikely that these were the first eyes; they seem much too large and (potentially) well developed to be brand new inventions. The best we can do is put the origin of eyes somewhere between the beginning of the Cambrian explosion, about 600 million years ago, and the death of the Burgess animals, some 530 million years ago.

1..The word aggregations in the passage is closest in meaning to

- A. parts
- B. reactions
- C. groups
- D. types

2..Paragraph 1 supports all of the following statements about protists EXCEPT:

- A. Some are multicellular.
- B. Some are able to move.
- C. Some have pigment molecules.
- D. They live in environments that contain moisture.

3..According to paragraph 1, what have scientists concluded from the fact that some living protists are very like their ancestral forms

- A. The eye did not evolve until multicellular organisms arose.
- B. The ability to detect light and change behavior in response to light has existed for a long time.
- C. The ancestral forms of these living protists likely had an eyespot near the motile flagellum.
- D. The ancestral forms of these living protists depended primarily on light as the mechanism for modifying their metabolic activity or motility.

4..The word lateral in the passage indicates a location at the

- A. front
- B. back
- C. top
- D. side

5..Paragraph 2 implies which of the following about the early eyes

- A. They were able to detect simple movements almost from the beginning of their evolution.
- B. They were not as sensitive to light as once thought.
- C. They could not form images.
- D. Their cells had more photosensitive pigment than do human eyes.

6..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. There are few fossils from the Precambrian, though more keep turning up.
- B. Most evidence of animals in the fossil record is indirect and little of it is from the Precambrian.

- C. Tunnels in Precambrian rocks that may have been made by worms provide indirect evidence of these animals existing at that time.
- D. There are very few fossils of animals from the Precambrian and most evidence of animal life from that period is indirect.

7..According to paragraph 3, the Cambrian period was characterized by

- A. a great abundance of animals
- B. a slow rate of animal extinction
- C. the rapid fossilization of animals
- D. an increase in the life span of some animals

8..The phrase little short of miraculous is closest in meaning to

- A. very highly valued
- B. amazing because almost impossible
- C. causing controversy
- D. almost but not quite complete

9..According to paragraph 4, all of the following are true of the Burgess Shale EXCEPT:

- A. Its fossils were in a flattened condition when discovered.
- B. Its fossils provide direct evidence about the origin of eyes.
- C. It contains fossils of both Precambrian and Cambrian animals.
- D. It contains fossilized remains of soft-bodied organisms.

10..The word designation in the passage is closest in meaning to

- A. evolution
- B. identification
- C. reconstruction
- D. confusion

11..Why does the author point out that The eyes of some Burgess organisms sat on stalks

- A. To suggest that some Burgess organisms had a greater range of vision than do living shrimp and crabs
- B. To explain why it is thought that one of the lateral pairs of eyes in Opabinia may have been movable
- C. To explain why the eyes of some Burgess animals were not recognizable as such before they were reconstructed
- D. To support the statement that the reconstructed eyes of Burgess animals look superficially like the eyes of some living crustaceans

12..Paragraph 6 suggests that the first eyes probably

- A. came into existence long before 600 million years ago
- B. came into existence at a late point in the Cambrian period
- C. existed before the animals of the Burgess Shale existed

D. were larger than those of animals found in the Burgess Shale

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Molaria spinifera and H. optata, both of which lived in water levels beyond the reach of light, fit into this category.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. The ability of some unicellular organisms to detect light and change their behavior accordingly suggests that eyes did not originate with multicellular animals.
- B. The earliest eyes apparently contained molecules that were capable of forming and focusing images.
- C. Too few fossils from the Precambrian have been found to determine which if any Precambrian organisms had eyes.
- D. Evidence from the Burgess Shale suggests that eyes of some early animals were similar to the eyes of living crustaceans.
- E. Fossil evidence suggests that organisms in the Burgess Shale with faceted eyes developed later than organisms in the Burgess Shale with nonfaceted eyes.
- F. The large size and possible complexity of the eyes of some organisms in the Burgess Shale suggest that their eyes were not the first eyes.

1. In order to understand the origin of Earth's atmosphere, we must go back to the earliest days of the solar system, before the planets themselves were formed from a disk of rocky material spinning around the young Sun. This material gradually coalesced into lumps called planetesimals as gravity and chance smashed smaller pieces together, a chaotic and violent process that became more so as planetesimals grew in size and gravitational pull. Within each orbit, collisions between planetesimals generated immense heat and energy. How violent these processes were is suggested by the odd tilt and spin of many of the planets, which indicate that each of the planets was, like a billiard ball, struck at some stage by another large body of some kind. Visual evidence of these processes can be seen by looking at the Moon. Because the Moon has no atmosphere, its surface is not subject to erosion, so it retains the marks of its early history. Its face is deeply scarred by millions of meteoric impacts, as you can see on a clear night with a pair of binoculars. The early Earth did not have much of an atmosphere. Before it grew to full size, its gravitational pull was insufficient to prevent gases from drifting off into space, while the solar wind (the great stream of atomic particles emitted from the Sun) had already driven away much of the gaseous material from the inner orbits of the solar system. So we must imagine the early Earth as a mixture of rocky materials, metals, and trapped gases, subject to constant bombardment by smaller planetesimals and without much of an atmosphere.
2. As it began to reach full size, Earth heated up, partly because of collisions with other planetesimals and partly because of increasing internal pressures as it grew in size. In addition, the early Earth contained abundant radioactive materials, also a source of heat. As Earth heated up, its interior melted. Within the molten interior, under the influence of gravity, different elements were sorted out by density. By about 40 million years after the formation of the solar system, most of the heavier metallic elements in the early Earth, such as iron and nickel, had sunk through the hot sludge to the center, giving Earth a core dominated by iron. This metallic core gives Earth its characteristic magnetic field, which has played an extremely important role in the history of our planet.
3. As heavy materials headed for the center of Earth, lighter silicates (such as the mineral quartz) drifted upward. The denser silicates formed Earth's mantle, a region almost 3,000 kilometers thick between the core and the crust. With the help of bombardment by comets, whose many impacts scarred and heated Earth's surface, the lightest silicates rose to Earth's surface, where they cooled more rapidly than the better-insulated materials in Earth's interior.
4. These lighter materials, such as the rocks we call granites, formed a layer of continental crust about 35 kilometers thick. Relative to Earth as a whole, this is as thin as an eggshell. Seafloor crust is even thinner, at about 7 kilometers; thus, even continental crust reaches only about 1/200th of the way to Earth's core. Much of the early continental crust has remained on Earth's surface to the present day.
5. The lightest materials of all, including gases such as hydrogen and helium, bubbled through Earth's interior to the surface. So we can imagine the surface of the early Earth as a massive volcanic field. And we can judge pretty well what gases bubbled up to that surface by analyzing the mixture of gases emitted by volcanoes. These include hydrogen, helium, methane, water vapor, nitrogen, ammonia, and hydrogen sulfide. Other materials, including

large amounts of water vapor, were brought in by cometary bombardments. Much of the hydrogen and helium escaped; but once Earth was fully formed, it was large enough for its gravitational field to hold most of the remaining gases, and these formed Earth's first stable atmosphere.

1..The word chaotic in the passage is closest in meaning to

- A. rapid
- B. disorganized
- C. intense
- D. long-lasting

2..All of the following are true of the planetesimals mentioned in paragraph 1 EXCEPT:

- A. They were formed of rocky material spinning around the early Sun.
- B. They collided violently with each other.
- C. They gradually grew in size.
- D. They lost their atmospheres as they were hit by larger bodies.

3..The word retains in the passage is closest in meaning to

- A. reveals
- B. acquires
- C. hides
- D. preserves

4..The author discusses the Moon in paragraph 1 in order to

- A. help explain why Earth had fewer meteoric impacts than other planets in the solar system
- B. show why it is difficult to understand how the first planetary atmospheres developed
- C. help explain the processes that took place in the formation of large planetary bodies in the solar system
- D. illustrate why the Moon's spin and tilt are unique among other planetary bodies in the solar system

5..The word constant in the passage is closest in meaning to

- A. considerable
- B. unpredictable
- C. continual
- D. violent

6..Paragraph 3 answers which of the following questions about early Earth

- A. What caused materials on Earth to become radioactive
- B. What percentage of Earth's core was nickel
- C. What internal pressures caused Earth to heat up as it grew in size
- D. What caused Earth's magnetic field

7..According to paragraph 3, Earth's core is mostly iron because, compared to most other elements on early Earth, iron

- A. was denser
- B. melted more easily
- C. was more radioactive
- D. was more plentiful

8..Select the TWO answer choices that, according to paragraph 4, indicate true statements about Earth's formation. To obtain credit, you must select TWO answer choices.

- A. Comets hitting Earth helped the lightest silicates to reach Earth's surface.
- B. Silicates such as mineral quartz drifted downward and mixed with denser materials as they reached Earth's core.
- C. When Earth's mantle became approximately 3,000 kilometers thick, the heaviest materials in it began to cool.
- D. Lighter materials reaching Earth's surface formed Earth's continental crust.

9..According to paragraph 4, Earth's continental crust

- A. has changed significantly in composition over time
- B. was as thick as Earth's mantle in its early stages
- C. is very thin relative to Earth's size
- D. caused the temperatures of Earth's early core and mantle to gradually increase

10..The word coalesced in the passage is closest in meaning to

- A. collided
- B. joined
- C. changed
- D. shrank

11..The word emitted in the passage is closest in meaning to

- A. released
- B. consumed
- C. contained
- D. heated

12..What can be inferred from paragraph 5 about Earth's first stable atmosphere

- A. It existed before Earth was yet fully formed.
- B. It contained very little hydrogen and helium.
- C. It contained only materials that had bubbled up through Earth's surface.
- D. It lacked water vapor.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Even some of its oldest portions as old as 3.8 billion years can still be found in parts of Canada,

Australia, South Africa, and Greenland.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. Early Earth's lack of an atmosphere explains why it was bombarded with much more frequency and violence than other planetesimals.
- B. Continued bombardments and internal pressures made the growing Earth hotter, causing its interior to melt and the heavier elements to sink and form Earth's core.
- C. Lighter elements from Earth's interior rose and formed the mantle, a denser layer of silicates around the core, and the crust, a thinner layer of silicates at Earth's surface.
- D. The formation of Earth's crust protected the inner layers of Earth from the high-energy particles in space, reducing the temperatures of the mantle and the core.
- E. Once Earth had gone through the final stages of its formation, gases bubbled to the surface and were held by Earth's gravitational field to form the atmosphere.
- F. Volcanoes today are the result of gases that were trapped in Earth's interior during the planet's early stages of formation.

Passage 110 2015-04-18-1 Historical Trends in European Urban Design

1. European city planning and design have a long history. Most Greek and Roman settlements were deliberately laid out on the grid system, within which the siting of key buildings was carefully thought out. The roots of modern Western urban planning and design can be traced to the Renaissance and Baroque periods (between the fifteenth and seventeenth centuries) in Europe, when artists and intellectuals dreamed of ideal cities, and rich and powerful regimes used urban design to produce extravagant symbolizations of wealth, power, and destiny. Inspired by the classical art forms of ancient Greece and Rome, Renaissance urban design sought to recast cities in a deliberate attempt to show off the power and the glory of the state and church.
2. Spreading slowly from its origins in Italy at the beginning of the fifteenth century, Renaissance design successfully diffused to most of the larger cities of Europe. Dramatic advances in weaponry brought a surge of planned redevelopment that featured impressive geometric-shaped fortifications and an extensive sloping, clear zone of fire. Inside new walls, cities were recast according to a new aesthetic of grand design: fancy palaces, geometrical plans, streetscapes, and gardens that emphasized views of dramatic perspectives. These developments were often so extensive and so interconnected with each other that they effectively fixed the layout of cities well into the eighteenth, and even into the nineteenth, century, when walls and/or open spaces eventually made way for urban redevelopment in the form of parks, railway lines, or beltways.
3. As societies and economies became more complex with the transition to industrial capitalism, national rulers and city leaders looked to urban design to impose order, safety, and efficiency, as well as to symbolize the new seats of power and authority. The most important early precedent was set in Paris by Napoleon III, who presided over a comprehensive program of urban redevelopment and monumental urban design. The work was carried out by Baron Georges-Eugene Haussmann between 1853 and 1870. Haussmann demolished large sections of old Paris to make way for broad, new, tree-lined avenues, with numerous public open spaces and monuments. In doing so, he made the city not only more efficient (wide boulevards meant better flows of traffic) and a better place to live (parks and gardens allowed more fresh air and sunlight in a crowded city and were held to be a civilizing influence) but also safer from revolutionary politics (wide boulevards were hard to barricade; monuments and statues helped to instill a sense of pride and identity).
4. The preferred architectural style for these new designs was the Beaux Arts style. In this school, architects were trained to draw on Classical, Renaissance, and Baroque styles, synthesizing them in designs for new buildings for the Industrial Age. The idea was that the new buildings would blend artfully with the older palaces, cathedrals, and civic buildings that dominated European city centers. Haussmann's ideas were widely influential and extensively copied.
5. Early in the twentieth century there emerged a different intellectual and artistic reaction to the pressures of industrialization and urbanization. This was the Modern movement, which was based on the idea that buildings and cities should be designed and run like machines. Equally important to the Modernists was that urban design should not simply reflect dominant social and cultural values but, rather, help to create a new moral and social order. The movement's best-known advocate was Le Corbusier, a Paris-based Swiss who provided the inspiration for

technocratic urban design. Modernist buildings sought to dramatize technology, exploit industrial production techniques, and use modern materials and unembellished, functional design. Le Corbusier's ideal city featured linear clusters of high-density, medium-rise apartment blocks, elevated on stilts and segregated from industrial districts; high-rise tower office blocks; and transportation routes all separated by broad expanses of public open space.

6. After 1945 this concept of urban design became pervasive, part of what became known as the International Style: boxlike steel-frame buildings with concrete-and-glass facades. The International Style was avant-garde yet respectable and, above all, comparatively inexpensive to build. This tradition of urban design, more than anything else, has imposed a measure of uniformity on cities around the world.

1..In paragraph 1, why does the author mention that most Greek and Roman settlements were laid out on the grid system

- A. To show how they resembled one another in terms of their layout
- B. To support the claim that city planning had a long history in Europe
- C. To help explain why cities of Renaissance and Baroque design were typically laid out in the form of a grid
- D. To contrast the sophistication of Greek and Roman urban design with the simplicity of the urban design of the Renaissance and Baroque periods

2..The word regimes in the passage is closest in meaning to

- A. cities
- B. builders
- C. governments
- D. planners

3..According to paragraph 1, an important goal of Renaissance urban design was to

- A. serve as an expression of the wealth and power of the ruling class
- B. improve the classical forms of ancient Greek and Roman cities
- C. show that the state rather than the church was the most powerful institution in a city
- D. restore the religious and civic buildings of a city to their previous glory

4..Paragraph 2 supports the idea that important features typical of Renaissance urban design resulted from

- A. Renaissance designers' improved understanding of geometry
- B. the characteristics of new weaponry
- C. an increased interest in highly productive gardens
- D. the need to reduce the likelihood of fires

5..The phrase a surge of is closest in meaning to

- A. a combination of
- B. an altered approach to
- C. a sudden increase in
- D. a return to

6..Paragraph 3 mentions each of the following as an accomplishment of Haussmann's redevelopment of Paris EXCEPT

- A. improving the flow of traffic
- B. making it harder for revolutionaries to be effective
- C. improving housing in large sections of old Paris
- D. bringing more fresh air and sunlight into the city

7..The word synthesizing in the passage is closest in meaning to

- A. combining
- B. simplifying
- C. using
- D. copying

8..According to paragraph 4, what was an advantage of the Beaux Arts style

- A. It was especially well suited for industrial buildings.
- B. It fit in well with important older buildings in European cities.
- C. It could be easily copied by builders everywhere.
- D. It allowed new buildings to be constructed much more efficiently.

9..According to paragraph 5, Modernist urban design differed from previous urban design styles in that it

- A. meant to contribute to a new moral and social order
- B. was heavily influenced by the work of one urban planner
- C. was a reaction to social and economic changes
- D. was intended to make cities more beautiful

10..The word elevated in the passage is closest in meaning to

- A. raised
- B. imagined
- C. separated
- D. designed

11..Paragraph 5 supports the idea that Le Corbusier held which of the following views

- A. Industrial production techniques should be used only for buildings in industrial districts.
- B. Different types of activities that go on in a city should be kept physically separated from each other.
- C. All the buildings in a city should be about the same height and of similar design.
- D. Major transportation routes should be kept at a significant distance from cities.

12..According to paragraph 6, the International Style became widespread partly because

- A. it was a style that combined many traditional national styles
- B. it was a style that emerged independently in city after city
- C. building in that style tended to produce uniform results
- D. building in that style tended to cost less

13..Look at the four squares that indicate where the following sentence could be added to the passage.

This mechanical analogy was a significant departure from earlier attitudes that emphasized the civilizing influence of cities and their buildings.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. Beginning in fifteenth-century Italy, advances in weaponry led to the redesign of cities, and a new aesthetic of grand design inspired by classical art forms took hold.
- B. The walls and open spaces typical of Renaissance urban design were re-discovered in the early twentieth century and became a major component of the Modern movement.
- C. The social changes brought about by the transition to industrial capitalism placed new demands on urban planners that could only be met by adopting new design styles.
- D. The redevelopment of Paris in the mid-1800s displayed a new idea of urban design to make cities orderly, efficient, and healthier and to positively reflect the new social and economic order.
- E. The goal of Beaux Arts style architects was to replace the older palaces, cathedrals, and civic buildings that dominated European cities with modern buildings.
- F. The twentieth-century International Style's boxlike steel, glass, and concrete buildings arose from the Modernist view that buildings and cities should be designed and run like machines.

Passage 111 2015-04-18-2 Origins of the Megaliths

1. Since the days of the earliest antiquarians, scholars have been puzzled by the many Neolithic (~ 4000 B.C. - 2200 B.C.) communal tombs known as megaliths along Europe's Atlantic seaboard. Although considerable variations are found in the architectural form of these impressive monuments, there is a general overriding similarity in design and, particularly, in the use of massive stones.
2. The construction of such large and architecturally complex tombs by European barbarians struck early prehistorians as unlikely. The Bronze Age seafaring civilizations that lived in the region of the Aegean Sea (~ 3000 B.C. - 1000 B.C.), among whom collective burial and a diversity of stone-built tombs were known, seemed a probable source of inspiration. It was suggested that Aegean people had visited Iberia in southwestern Europe in search of metal ores and had introduced the idea of collective burial in massive tombs, which then spread northward to Brittany, Britain, North Germany, and Scandinavia.
3. Radiocarbon dates for a fortified settlement of megalith builders at Los Millares in Spain appeared to confirm this picture, though dates for megaliths in Brittany seemed too early. When calibrated, however, it became clear that radiocarbon dates were universally too early to support a Bronze Age Aegean origin. It is now clear that the megaliths are a western and northern European invention, not an introduced idea. Even so, they are still a subject of speculation and inquiry. What induced their builders to invest massive efforts in erecting such monumental tombs? How was the necessary labor force assembled? What underlies their striking similarities?
4. One answer to the last question was proposed by Professor Grahame Clark, one of Britain's greatest prehistorians. Investigating the megaliths of southern Sweden, he noted that one group was concentrated in coastal locations from which deep-sea fish such as cod, haddock, and ling could have been caught in winter. Historically, much of the Atlantic was linked by the travels of people who fished, and this could well have provided a mechanism by which the megalith idea and fashions in the style of tomb architecture spread between coastal Iberia, Brittany, Ireland, western England and Scotland, and Scandinavia. The high concentrations of megaliths on coasts and the surprising number of megaliths found on small islands may support a connection with fishing.
5. Professor Colin Renfrew of the University of Cambridge, England, however, views the similarities as similar responses to similar needs. At the structural level, the passage that forms a major element of many graves could have been devised independently in different areas to meet the need for repeated access to the interior of these communal tombs. Other structural resemblances could be due to similarities in the raw materials available. In answer to the question of why the idea of building monumental tombs should arise independently in a number of areas, he cites the similarities in their backgrounds.
6. Most megaliths occur in areas inhabited in the postglacial period by Mesolithic hunter-gatherers (~8500 B.C. - 4000 B.C.). Their adoption of agriculture through contact with Neolithic farmers, Renfrew argues, led to a population explosion in the region and consequent competition for farmland between neighboring groups. In the face of potential conflict, the groups may have found it desirable to define their territories and emphasize their boundaries. The construction of megaliths could have arisen in response to this need.
7. Renfrew has studied two circumscribed areas, the Scottish islands of Arran and Rousay, to

examine this hypothesis more closely. He found that a division of the arable land into territories, each containing one megalith, results in units that correspond in size to the individual farming communities of recent times in the same area. Each unit supported between 10 and 50 people. The labor needed to put up a megalith would probably be beyond the capabilities of a community this size. But Renfrew argues that the cooperation of other communities could be secured by some form of recognized social incentive perhaps a period of feasting at which communal building was one of several activities.

8. Most megaliths contain collective burials. Different tombs used different arrangements, but there seems to have been an underlying theme: people placed in these tombs were representative of their society, but their identity as individuals was not important. The tombs belonged to the ancestors, through whom the living society laid claim to their land. This interpretation reinforces Renfrew's view of the megaliths as territorial markers.

1..The word collective in the passage is closest in meaning to

- A. aboveground
- B. public
- C. elaborate
- D. group

2..According to paragraph 2, early prehistorians thought the Aegean people of the Bronze Age might have influenced megalith building along the Atlantic seaboard because they

- A. had established commercial routes along the Atlantic seaboard
- B. had been in Iberia, where they introduced the idea of burial in very large tombs
- C. were thought to have found megaliths in Iberia when searching for metals
- D. were thought to have passed along the concept of burial in monumental tombs as they explored Brittany, Britain, North Germany, and Scandinavia

3..In paragraph 3, why does the author discuss the results of radiocarbon dates

- A. To support the idea that megaliths spread rapidly during the Bronze Age
- B. To question the idea that megaliths have a religious origin
- C. To provide evidence against the theory that Bronze Age Aegeans inspired the megaliths
- D. To argue that the megaliths in Brittany are older than the megaliths in Los Millares

4..The word erecting in the passage is closest in meaning to

- A. designing
- B. constructing
- C. protecting
- D. decorating

5..According to paragraph 4, what did Professor Clark propose as a result of studying the megaliths of southern Sweden

- A. Swedish megaliths are nearly identical to megaliths elsewhere.
- B. People who traveled for fishing may have been responsible for the spread of megaliths in Europe.
- C. Swedish megaliths were probably built after other European megaliths were built.
- D. Megaliths in Europe were usually located near sites for deep-sea fishing in winter.

6..What is the purpose of discussing the passage that forms a major element of many graves

- A. To provide an example of a commonly occurring feature of megaliths that might be related to a similar need
- B. To argue that similarities in raw materials were responsible for the similarity of passages
- C. To explain how repeated access to the interior of the communal tombs was possible
- D. To provide evidence that the builders of the megaliths had similar backgrounds

7..The word consequent in the passage is closest in meaning to

- A. intense
- B. increasing
- C. resulting
- D. continual

8..According to paragraph 6, Professor Renfrew has argued that one factor leading to the rise of megaliths in Europe was

- A. the adoption of farming by Mesolithic hunter-gatherers
- B. the transition from a glacial to a postglacial climate
- C. the relocation of Mesolithic populations from one region to another
- D. the conflict over whether areas inhabited by Mesolithic peoples would be used for farming

9..The phrase this hypothesis in the passage refers to the idea that

- A. there was competition for territory between Mesolithic hunter-gathers and invading Neolithic farmers
- B. a population explosion brought about a division of the region's Neolithic farmers into neighboring groups
- C. the need of neighboring groups to define their territories led to the construction of megaliths
- D. the construction of megaliths was a way of competing for farmland

10..According to paragraph 7, what did Renfrew conclude about the megaliths of Arran and Rousay

- A. Each megalith was associated with a specific agricultural community.
- B. Each megalith was built by between 10 and 50 people.
- C. Some megaliths were built using stones quarried at other places.
- D. Some megaliths were built gradually over time rather than all at once.

11..The word reinforces in the passage is closest in meaning to

- A. represents
- B. differs from
- C. expands on
- D. supports

12..According to the passage, all of the following were true of the megaliths along the Atlantic seaboard EXCEPT:

- A. They often had a main passageway.
- B. They identified the individuals buried within them.
- C. They were built before the Aegean Bronze Age.
- D. They differed somewhat in style from region to region.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

So it might seem that megaliths could not have been used by an individual community to mark its land.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. The Bronze Age Aegeans most probably built some of the tombs in Iberia.
- B. Scientific evidence supports the idea that the megaliths were a western and northern European invention.
- C. Most megaliths are found in coastal regions, suggesting that people fishing may have used them to guide their travels.
- D. Archeologists have found enough indicators to believe that the layout of tombs in the landscape reflects each community's social organization.
- E. The high proportion of coastal megaliths has given rise to the idea that megalith building was related to fishing.
- F. It has been suggested that megaliths arose in response to the spread of agriculture and competition for farmland.

1. A stage that is imperative in any archaeological process is the reconstruction of the physical environments in which a particular segment of the archaeological record was formed. Climates and the world's geomorphologythe shape and constituents of land surfaceshave changed greatly over the past several million years of human history, and each archaeological analysis begins with an effort to reconstruct the physical world of the culture being analyzed.
2. Ancient climates can often be reconstructed from floral and faunal remains. The study of animal remains, or faunal analysis, is a complex field in which, in most cases, the archaeologist is trying to reconstruct human diet and local environments. Faunal analysts usually count the numbers and kinds of animals represented by the remains they find, and then use statistical methods to estimate the food values, ages, and sexes of the animals being exploited. The prehistoric record of the meat-eating habits of early humans is far from clear about the prevalence of scavenging. One faction of prehistorians argues there is evidence that early humans were primarily scavengers who found the remains of animals killed by lions and other carnivores, and butchered them. Another faction disagrees and proposes that early humans hunted for their own meat. Marks left by humans cutting up animals with stone tools are now being analyzed to help distinguish between cases in which people butchered animals they had killed themselves and those in which they butchered animals they scavenged from kills of other animals.
3. Throughout human history, plants have been our main source of food, and so floral analysesstudies of the remains of plantsare an extremely important part of archaeology, particularly in studies of how domesticated plants and animals and agricultural economies evolved. Carbon is chemically quite stable, so charred plants (plants converted to charcoal or carbon) and seeds preserve well. Carbonized plant remains can be retrieved by flotation: excavated sediments are mixed with water or some other fluid and the charred plant fragments rise to the surface, where they can be skimmed off and identified. The importance of such analyses lies in the fact that these plants indicate much about the climates and vegetation of the periods in which the animals lived. For example, there are debates about when and where various animals were domesticated. If phytoliths (tiny mineral particles formed inside plants) of domesticated grains are found on the teeth of these animals, the probability is high that they were part of an agricultural economy.
4. Human bodies are also valuable sources of information for archaeologists, particularly if the bodies are well preserved. For example, eleven naturally mummified bodies were found in beach sand in northern Chile and date to about 1000 B.C. When they were analyzed, it was found that one of them was a coca leaf chewer (the earliest known), while other bodies showed the changes of the bones of the inner ear that are characteristic of people who spend a lot of time diving in cold water. In addition, they had the kinds of dental problems and missing teeth associated with the sticky starches of an agricultural dietalthough about 40 percent of their diet came from marine resources.
5. A rapidly growing technical specialty within archaeology is geoarchaeology, which combines archaeological and geological analyses. Geology and archaeology form a natural marriage in many obvious ways because both disciplines are concerned with the alteration of natural landscapes. Glaciers, changing rainfall patterns, and many other natural forces cause changes to landscapes, and of course, so do people. Geologists are broadly concerned with ancient

physical environments, and archaeologists require knowledge of these environments to interpret their finds.

6. Geoarchaeological analyses involve many different kinds of questions and techniques. In the Egyptian Delta region, for example, many of the earliest communities were built on large sand-and-gravel mounds created by the Nile River as it deposited the sediments it carried. But many of these communities have been buried under many meters of sediments from numerous ancient floods since that time and by other factors as well. Moreover, the streams feeding into the Nile River in the delta have changed course many times, leaving a maze of crisscrossed buried river channels. Finding these buried sand-and-gravel mounds and the archaeological sites on them often requires complex geological analyses involving special digging, satellite image analysis, and many other techniques.

1..The word constituents in the passage is closest in meaning to

- A. components
- B. locations
- C. size
- D. temperatures

2..According to paragraph 2, prehistorians disagree about which of the following

- A. Whether humans hunted animals themselves or ate meat from animals killed by other carnivores
- B. The number and kinds of animals humans ate
- C. Whether lions and other carnivores were hunted by humans
- D. Whether or not humans butchered animals to eat

3..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Humans using stone tools to butcher animals left marks on the bones of the animals they killed and the animals they scavenged.
- B. Humans scavenged animals killed by other animals and butchered them using stone tools.
- C. Tool marks on butchered animal bones are analyzed to help determine whether humans killed or merely scavenged the animals.
- D. Analysis of the stone tools used by humans to butcher scavenged animals is now being carried out.

4..The word excavated in the passage is closest in meaning to

- A. dug from the ground
- B. concentrated in solid form
- C. deposited
- D. dried

5..According to paragraph 3, what is one reason that scientists use flotation

- A. To estimate the fluid content of ancient plants and seeds
- B. To identify the sediments in which ancient plants grew
- C. To learn about the climate of places where ancient plants grew
- D. To determine whether the remains of ancient plants and seeds have carbonized

6..According to paragraph 3, which of the following is true about carbonized plant remains

- A. Each fragment of carbonized plant remains contains sediments of a wide variety of plants.
- B. Carbonized plant remains are usually found near the remains of domesticated animals.
- C. Carbonized plant remains can be recovered by mixing sediments with fluid.
- D. Carbonized plant remains from an area are used to settle debates about the age of animal remains located in the same area.

7..According to paragraph 3, which of the following do archaeologists consider to be a strong indicator that an agricultural economy existed in a particular site

- A. An unusually high number of animal remains near the surface of the site
- B. Remains of domesticated plants in the teeth of animals
- C. High numbers of mineral particles inside carbonized plants
- D. Evidence of little vegetation change for long periods of time

8..According to paragraph 4, all of the following were true of some people in northern Chile around 1000 B.C. EXCEPT:

- A. They were divers.
- B. They chewed coca leaves.
- C. They had dental problems caused by eating food from marine resources.
- D. They ate plants grown on land.

9..The word characteristic in the passage is closest in meaning to

- A. expected
- B. typical
- C. suggestive
- D. destructive

10..The word alteration in the passage is closest in meaning to

- A. destruction
- B. preservation
- C. authenticity

D. modification

11..The purpose of paragraph 6 is to

- A. answer questions about the equipment and techniques used in the Egyptian Delta area
- B. describe the earliest communities built on large, sand-and-gravel mounds
- C. explain how streams feeding the Nile River in the delta change course over time
- D. provide an example of a situation where complex geoarchaeological analyses and techniques are necessary

12..According to paragraph 6, what is one reason that complex geological analyses are needed for locating many of the earliest Nile River communities

- A. Flood waters can carry evidence of human habitation far from its place of origin.
- B. The streams associated with early communities have changed course over time.
- C. More recent communities have been built on top of ancient communities.
- D. The types of sediments carried by Nile floods have changed since ancient times.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Such a variety of methods are especially important at sites that are no longer visible on the surface.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. Studying plant and animal remains can reveal details of ancient climates, food sources, and agricultural activities.
- B. Faunal analysis deals mainly with analyzing the tools ancient humans used for hunting and killing animals.
- C. Once identified, carbonized plant remains can help modern agriculturalists determine the most stable varieties of plants to grow.
- D. An examination of human remains can tell archaeologists about such things as the individual's diet and habits.
- E. To help understand the physical world of ancient communities, archaeologists may work with geologists and employ complex new techniques.
- F. Geoarchaeology, a growing field within archaeology, has proved to be more effective in explaining the alterations made to natural landscapes than in explaining how members of ancient communities lived.

Passage 113 2015-05-09-3 Vocalization in Frogs

1. The t[†]hgara frog is a small terrestrial vertebrate that is found in Central America. T[†]hgara frogs breed in small pools, and breeding groups range from a single male to choruses of several hundred males. The advertisement call of a male t[†]hgara frog is a strange noise, a whine that starts at a frequency of 900 hertz and sweeps downward to 400 hertz in about 400 milliseconds. The whine may be produced by itself, or it may be followed by one or several chucks or clucking sounds. When a male t[†]hgara frog is calling alone in a pond, it usually gives only the whine portion of the call, but as additional males join a chorus, more and more of the frogs produce calls that include chucks. Scientists noted that male t[†]hgara frogs calling in a breeding pond added chucks to their calls when they heard the recorded calls of other males played back. That observation suggested that it was the presence of other calling males that incited frogs to make their calls more complex by adding chucks to the end of the whine.
2. What advantage would a male frog in a chorus gain from using a whine-chuck call instead of a whine? Perhaps the complex call is more attractive to female frogs than the simple call. Michael Ryan and Stanley Rand tested that hypothesis by placing female t[†]hgara frogs in a test arena with a speaker at each side. One speaker broadcast a pre-recorded whine call, and the second speaker broadcast a whine-chuck. When female frogs were released individually in the center of the arena, fourteen of the fifteen frogs tested moved toward the speaker broadcasting the whine-chuck call.
3. If female frogs are attracted to whine-chuck calls in preference to whine calls, why do male frogs give whine-chuck calls only when other males are present? Why not always give the most attractive call possible? One possibility is that whine-chuck calls require more energy than whines, and males save energy by only using whine-chucks when competition with other males makes the energy expenditure necessary. However, measurements of the energy expenditure of calling male t[†]hgara frogs showed that energy cost was not related to the number of chucks. Another possibility is that male frogs giving whine-chuck calls are more vulnerable to predators than frogs giving only whine calls. T[†]hgara frogs in breeding choruses are preyed upon by a species of frog-eating bats, *Trachops cirrhosus*, and it was demonstrated that the bats locate the frogs by homing on their vocalizations.
4. In a series of playback experiments, Michael Ryan and Merlin Tuttle placed pairs of speakers in the forest and broadcast vocalizations of t[†]hgara frogs. One speaker played a recording of a whine and the other a recording of a whine-chuck. The bats responded as if the speakers were frogs: they flew toward the speakers and even landed on them. In five experiments at different sites, the bats approached speakers broadcasting whine-chuck calls twice as frequently as those playing simple whines (168 approaches versus 81). Thus, female frogs are not alone in finding whine-chuck calls more attractive than simple whines; an important predator of frogs also responds more strongly to the complex calls.
5. Ryan and his colleagues measured the rates of predation in t[†]hgara frog choruses of different sizes. Large choruses of frogs did not attract more bats than small choruses, and consequently the risk of predation for an individual frog was less in a large chorus than in a small one. Predation was an astonishing 19 percent of the frogs per night in the smallest chorus and a substantial 1.5 percent per night even in the largest chorus. When a male frog shifts from a simple whine to a whine-chuck call, it increases its chances of attracting a female, but it simultaneously increases its risk of attracting a predator. In small choruses, the competition

from other males for females is relatively small, and the risk of predation is relatively large. Under these conditions it is apparently advantageous for a male t̃hgara frog to give simple whines. However, as chorus size increases, competition with other males also increases while the risk of predation falls. In that situation, the advantage of giving a complex call apparently outweighs the risks.

1..The word incited in the passage is closest in meaning to

- A. allowed
- B. stimulated
- C. forced
- D. helped

2..According to paragraph 1, male t̃hgara frogs add chucks to the whine they produce when

- A. potential mates are unable to hear the frequency of their whine sounds
- B. other males produce louder whine sounds than they do
- C. the frogs breed in large pools rather than small ones
- D. other males are present in their breeding pool

3..What is the relationship of paragraph 2 in the passage to paragraph 1

- A. Paragraph 2 provides additional support for a scientific hypothesis discussed in paragraph 1.
- B. Paragraph 2 questions the accuracy of a scientific observation discussed in paragraph 1.
- C. Paragraph 2 provides a possible explanation for a scientific observation discussed in paragraph 1.
- D. Paragraph 2 identifies some strengths and weaknesses of a scientific hypothesis discussed in paragraph 1.

4..According to paragraph 2, Ryan and Rand conducted an experiment to find out whether which of the following theories was true

- A. Male frogs in a chorus use a whine-chuck call in place of a whine call.
- B. Female frogs prefer a whine-chuck call to a simple whine call.
- C. Male frogs tend to produce more whine-chuck calls than whine calls.
- D. Female frogs respond differently to live calls from males than they do to recorded calls.

5..To be attracted to whine-chuck calls in preference to whine calls means

- A. to like whine-chuck calls instead of whine calls
- B. to like whine-chuck calls in addition to whine calls
- C. to like whine-chuck calls followed by whine calls
- D. to like whine-chuck calls more than whine calls

6..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Males may save energy when competing for mates by using only whine-chuck calls rather than both whines and whine-chucks.
- B. Males expend as much of their energy on whine-chuck calls as on whine calls when competing with other males.
- C. Males save energy by using whine-chuck calls only when competing with other males.
- D. Males that save energy by using only whines are less able to compete with other males.

7..According to paragraph 4, all of the following are true of the playback experiments EXCEPT:

- A. Female frogs and predator bats approached the broadcasting speakers.
- B. The bats responded more strongly to the whine-chuck calls than they responded to the whine calls.
- C. Each speaker played a different kind of male frog call.
- D. The same experiment was repeated at different locations.

8..According to paragraph 4, the playback experiments of Ryan and Tuttle demonstrated which of the following

- A. Túngara frogs use both whines and whine-chucks in their vocalizations.
- B. Female túngara frogs are attracted to both whine and whine-chuck vocalizations.
- C. Female túngara frogs and predators of túngara frogs are attracted to different types of vocalizations.
- D. Frog-eating bats are attracted to whine-chuck calls more than to whines alone.

9..The word substantial in the passage is closest in meaning to

- A. average
- B. smaller
- C. considerable
- D. surprising

10..The word outweighs in the passage is closest in meaning to

- A. exceeds
- B. ignores
- C. minimizes
- D. disguises

11..According to paragraph 5, all of the following are true about túngara frog vocalizations EXCEPT:

- A. The larger the frog chorus, the smaller the chance there is of a particular frog being eaten by a predatory bat.
- B. The larger the frog chorus, the louder each individual frog calls.
- C. The smaller the frog chorus, the easier it is for a frog to attract a female.
- D. The smaller the frog chorus, the more likely it becomes that a frog using the whine-chuck vocalization will be attacked by a bat.

12..Which of the following can be inferred from paragraph 5 about the behavior of male t̃h̃gara frogs

- A. When in small choruses they use less effective mating calls to decrease their risk of predation.
- B. They avoid joining a large chorus in a breeding pool because it increases the risk of predation.
- C. They avoid the use of the whine-chuck call whenever there is the risk of predators.
- D. They attempt to avoid predation by making their calls at night.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Predation, therefore, is a risk in choruses of all sizes, but the risk varies depending on the type of call used.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. T̃h̃gara frogs generally use simple calls when they wish to attract a mate, and complex calls when they wish to avoid predation.
- B. Two hypotheses have been put forward about why females and frog-eating bats are more attracted to males using whine-chuck calls.
- C. The hypothesis that whine calls are used to save energy when males are not in immediate competition with each other has been disproved by showing that chuck calls do not require more energy.
- D. T̃h̃gara females overwhelmingly favor the whine-chuck call used by the males, but so do certain bats that prey upon t̃h̃gara frogs.
- E. Most males gather in groups of several hundreds when calling because the rate of predation from bats is so high in small groups.
- F. Male t̃h̃gara frogs use the whine-chuck call in large groups, where their risk of predation is lower, and the whine call in small groups, where their risk is higher.

1. Southeast Asia has a unique abundance and diversity of gliding animals flying squirrels, flying frogs, and flying lizards with wings of skin that enable them to glide through the tropical forest. What could be the explanation for the great diversity in this region and the scarcity of such animals in other tropical forests? Gliding has generally been viewed as either a means of escaping predators, by allowing animals to move between trees without descending to the ground, or as an energetically efficient way of traveling long distances between scattered resources. But what is special about Southeast Asian rain forests?
2. Scientists have proposed various theories to explain the diversity of gliding animals in Southeast Asia. The first theory might be called the tall-trees hypothesis. The forests of Southeast Asia are taller than forests elsewhere due to the domination of the dipterocarp family: a family of tall, tropical hardwood trees. Taller trees could allow for longer glides and the opportunity to build up speed in a dive before gliding. The lower wind speeds in tall-tree forests might also contribute by providing a more advantageous situation for gliding between trees. This argument has several flaws, however. First, gliding animals are found throughout the Southeast Asian region, even in relatively short-stature forests found in the northern range of the rain forest in China, Vietnam, and Thailand. Some gliders also thrive in low secondary forests, plantations, and even city parks. Clearly, gliding animals do not require tall trees for their activities. In addition, many gliding animals begin their glides from the middle of tree trunks, not even ascending to the tops of trees to take off.
3. A second theory, which we might call the broken-forest hypothesis, speculates that the top layer of the forest the tree canopy has fewer woody vines connecting tree crowns in Southeast Asian forests than in New World and African forests. As a result, animals must risk descending to the ground or glide to move between trees. In addition, the tree canopy is presumed to be more uneven in height in Asian forests, due to the presence of the tall dipterocarp trees with lower trees between them, again favoring gliding animals. Yet ecologists who work in different regions of the world observe tremendous local variation in tree height, canopy structure, and abundance of vines, depending on the site conditions of soil, climate, slope elevation, and local disturbance. One can find many locations in Southeast Asia where there are abundant woody vines and numerous connections between trees and similarly many Amazonian forests with few woody vines.
4. A final theory differs from the others in suggesting that it is the presence of dipterocarp trees themselves that is driving the evolution of gliding species. According to this view, dipterocarp forests can be food-deserts for the animals that live in them. The animals living in dipterocarp forests that have evolved gliding consist of two main feeding groups: leaf eaters and carnivores that eat small prey such as insects and small vertebrates. For leaf-eating gliders the problem is not the absence of any leaves but the desert-like absence of edible leaves. Dipterocarp trees often account for 50 percent or more of the total number of canopy trees in a forest and over 95 percent of the large trees, yet dipterocarp leaves are unavailable to most vertebrate plant eaters because of the high concentration of toxic chemicals in their leaves. Many species of gliding animals avoid eating dipterocarp leaves and so must travel widely through the forest, bypassing the dipterocarp trees, to find the leaves they need to eat. And gliding is a more efficient manner of traveling between trees than descending to the ground and walking or else jumping between trees.

5. Many carnivorous animals also may need to search more widely for food due to the lower abundance of insects and other prey. This is caused by dipterocarps' irregular flowering and fruiting cycles of two- to seven-year intervals, causing a scarcity of the flowers, fruits, seeds, and seedlings that are the starting point of so many food chains. The lower abundance of prey in dipterocarp forests forces animals such as lizards and geckos to move between tree crowns in search of food, with gliding being the most efficient means.

1..According to paragraph 1, what question about gliding species are researchers trying to answer

- A. Why it took millions of years for gliding animals to evolve in the tropical forests of Southeast Asia
- B. Why gliding animals, though rare in most tropical forests, have evolved in so many different families in Southeast Asia
- C. Why gliding animals evolved in many tropical forests in Southeast Asia before they evolved in any of the tropical forests elsewhere in the world
- D. Why gliding animals evolved only in tropical rain forests

2..According to paragraph 1, it is generally thought that the ability to glide is useful to forest-dwelling species because gliding

- A. allows them to adapt to a wide variety of forest conditions
- B. eliminates the need to travel long distances in search of food
- C. provides a rapid, energy-efficient way of descending from the top of a tree to the ground
- D. enables them to move through the forest without being exposed to predators on the ground

3..The word scattered in the passage is closest in meaning to

- A. hard to find
- B. seasonally available
- C. widely separated
- D. highly varied

4..All of the following are mentioned in paragraph 2 in support of the tall-trees hypothesis EXCEPT:

- A. Tall trees make longer glides possible.
- B. Tall trees make building up speed in a dive possible.
- C. Tall trees make gliding from the middle of tree trunks possible.
- D. Tall-tree forests have lower wind speeds.

5..Select the TWO answer choices that point to flaws in the tall-trees hypothesis, according to paragraph 2. To receive credit, you must select TWO answers.

- A. Many gliding animals are unable to ascend to the tops of tall trees.

- B. Gliding animals are not evenly distributed throughout the forests of the Southeast Asian region.
- C. In Southeast Asia, many gliding animals are found in places where trees tend to be relatively short.
- D. Many gliding animals begin their glides from positions midway up the trunks of trees.

6..The word speculates in the passage is closest in meaning to

- A. concludes from evidence
- B. recognizes
- C. puts forward as a possibility
- D. claims

7..Paragraph 3 implies which of the following ideas about forests in which there are abundant woody vines connecting tree crowns

- A. The tree canopy is more even than it is in other forests.
- B. In such forests, animals can move between trees by traveling on vines.
- C. Such forests generally contain a wider diversity of animals than other forests do.
- D. There are likely to be fewer predators on the ground in such forests than in other forests.

8..The word tremendous in the passage is closest in meaning to

- A. seasonal
- B. endless
- C. unexpected
- D. enormous

9..Paragraph 3 supports the idea that one problem with the broken-forest hypothesis is that

- A. ecologists have found gliding animals in areas of Southeast Asia where trees are connected by vines and not found them in Amazonian forests where trees are not connected by vines
- B. in Southeast Asia, the forests with the fewest woody vines connecting the tops of trees turn out to have the most gliding animals
- C. according to ecologists in different regions of the world, gliding animals are as abundant and varied in some forests of Africa and the New World as they are in Southeast Asian forests
- D. gliding is no easier in broken forests with an uneven canopy structure than it is in forests where the trees are all about the same height

10..According to paragraph 4, what special difficulty do leaf-eating animals face in a dipterocarp forest

- A. Dipterocarp trees are less leafy than other canopy trees.
- B. There is no efficient method of getting from one tree to another.

- C. Most trees are very tall with leaves that are difficult to reach.
- D. There is a large distance between trees that have edible leaves.

11..How does paragraph 5 relate to paragraph 4

- A. Paragraph 5 shows that the food-desert theory introduced in paragraph 4 can account for only part of what needs to be explained.
- B. Paragraph 5 explains why the author calls the theory set out in paragraph 4 the food-desert theory.
- C. Paragraph 5 completes the account of the food-desert theory begun in paragraph 4.
- D. Paragraph 5 outlines an alternative to the food-desert theory described in paragraph 4.

12..According to paragraph 5, what is responsible for the relative scarcity of insects and other prey in dipterocarp forests

- A. The inability of insects and other prey to eat the toxic seeds, flowers, and fruits of dipterocarp trees
- B. The efficiency with which lizards and geckos hunt their prey
- C. The abundance of carnivorous animals in dipterocarp forests
- D. Dipterocarps' irregular flowering and fruiting cycles

13..Look at the four squares that indicate where the following sentence could be added to the passage.

For each group, a dipterocarp forest is like a desert in that food resources are few and far apart. Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text. Answer Choices

- A. One theory is that so many gliding species evolved in Southeast Asia because the forests are exceptionally tall, but there is evidence that calls that theory into question.
- B. The fact that gliding animals are most abundant in the short-stature forests of China, Vietnam, and Thailand shows that gliding did not evolve as an adaptation to an environment of tall trees.
- C. Ecologists have shown that the abundance of gliding animals in different regions of the world corresponds to variations in tree height, canopy structure, and abundance of vines.
- D. The hypothesis that gliding evolved to compensate for a scarcity of vines linking tree canopies overlooks problematic evidence from both Southeast Asian and Amazonian forests.
- E. In forests that are dominated by tall trees, jumping from tree to tree or descending to the ground may be a more efficient way of traveling through the forest than gliding.
- F. Dipterocarp trees create an environment in which many species must travel widely to find food, and gliding may have evolved as a rapid and efficient way of moving between tree crowns.

Passage 115 2015-05-24-2 Saving Soil and Cropland

1. The world's farmers are literally losing ground on two fronts the loss of soil from erosion and the conversion of cropland to nonfarm uses. Both are well-established trends that reduce agricultural output, but since both are gradual processes, they are often not given the attention they deserve.
2. The 1930s that threatened to turn the United States Great Plains into a vast desert was a traumatic experience that led to revolutionary changes in American agricultural practices, such as the planting of tree shelterbelts rows of trees planted beside fields to slow wind and thus reduce wind erosion. Perhaps the most lasting change is strip cropping, the planting of crops on alternate strips with fallowed (not planted) land each year. This permits soil moisture to accumulate on the fallowed strips, while the planted strips reduce wind speed and hence the wind erosion on the idled strips. The key to controlling wind erosion is to keep the land covered with vegetation as much as possible and to slow wind speed at ground level.
3. One of the time-tested methods of dealing with water erosion is terracing creating hill-side ridges to reduce runoff. Another newer, highly effective tool in the soil conservation tool kit is conservation , which includes both no tillage and minimum tillage. In conventional farming, land is plowed, disked, or harrowed to prepare the seedbed, seed is drilled into the soil with a planter, and row crops are cultivated with a mechanical cultivator two or three times to control weeds. With minimum tillage, farmers simply drill seeds directly into the soil. The only tillage is a one-time disturbance in a narrow band of soil where the seeds are inserted, leaving the remainder of the soil undisturbed, covered by crop residues and thus resistant to both water and wind erosion.
4. In the United States, where farmers during the 1990s were required to implement a soil-conservation plan on erodible cropland to be eligible for crop price supports, the no-till area went from 7 million hectares in 1990 to nearly 21 million hectares (51 million acres) in 2000, tripling within a decade. An additional 23 million hectares were minimum-tilled, for a total of 44 million hectares of conservation tillage. This total included 37 percent of the corn crop, 57 percent of soybeans, and 30 percent of the wheat. Outside the United States, data for crop year 1998-1999 show Brazil using conservation tillage on 11 million hectares and Argentina on 7 million hectares. Canada, using conservation tillage on 4 million hectares, rounds out the big four. And now no-till farming is catching on in Europe, Africa, and Asia. In addition to reducing soil losses, minimum-till and no-till practices also help retain water and reduce energy use.
5. Another example of an effort to control soil erosion is the Conservation Reserve Program (CRP). Created in the United States in 1985, the CRP aimed to convert 45 million acres of highly erodible land into permanent vegetative cover under ten-year contracts. Under this program, farmers were paid to plant grass or trees on fragile cropland. The retirement of 35 million acres under the CRP, together with the adoption of conservation practices on 37 percent of all cropland, reduced soil erosion in the United States from 3.1 billion tons in 1982 to 1.9 billion tons in 1997.
6. Saving cropland is sometimes more difficult than saving the topsoil on the cropland. This is particularly the case when dealing with urban sprawl, where strong commercial forces have influence. With cropland becoming scarce, efforts to protect prime farmland from urban spread are needed everywhere. Japan provides a good example of such efforts. It has

successfully protected rice paddies even within the boundaries of Tokyo, thus enabling it to remain self-sufficient in rice, its staple food.

7. In the United States, Portland, Oregon, provides another example. The state adopted boundaries to urban growth twenty years ago, requiring each community to project its growth needs for the next two decades and then, based on the results, draw an outer boundary that would accommodate that growth. This has worked in Oregon because it has forced development back to the city.

1..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. The 1930s Dust Bowl was a revolutionary event that threatened to destroy United States agriculture by turning the Great Plains into a vast desert.
- B. The 1930s Dust Bowl in the United States resulted in radical changes in agricultural practices aimed at reducing wind erosion, such as the planting of tree shelterbelts.
- C. Tree shelterbelts, which are often used in the Great Plains area, are made up of trees that are planted in long rows beside agricultural fields.
- D. Of all the innovative techniques used to control wind erosion after the 1930s Dust Bowl, only tree shelterbelts proved effective.

2..The word hence in the passage is closest in meaning to

- A. therefore
- B. most importantly
- C. to a large extent
- D. indirectly

3..According to paragraph 2, which of the following is true about strip cropping

- A. It increases crop yields annually.
- B. It forces farmers to plant crops that absorb less water and fewer nutrients from the soil.
- C. It requires the use of shelterbelts.
- D. It prevents wind erosion and allows moisture to collect on sections of land left unplanted.

4..According to paragraph 3, all of the following are practices involved in minimum tillage EXCEPT

- A. using mechanical devices to control weeds
- B. leaving unseeded soil undisturbed
- C. disturbing the soil only once where the seeds are inserted
- D. protecting against water and wind erosion by leaving parts of the soil covered with crop residues

5..The word implement in the passage is closest in meaning to

- A. reduce the cost of
- B. put into effect
- C. give consideration to
- D. improve

6..According to paragraph 4, why did the amount of no-till area increase between 1990 and 2000 in the United States

- A. More land area had become available for farming during this period.
- B. Fewer crops were needed since no-till farming had increased the soybean, wheat, and corn crop yields.
- C. Because conventional farming practices were too expensive, farmers decided to use the cheaper no-till conservation plan.
- D. The government provided financial support to farmers who practiced soil conservation.

7..Paragraph 4 suggests that all of the following were among the largest users of conservation tillage during the late 1990s EXCEPT

- A. Argentina
- B. Europe
- C. Canada
- D. Brazil

8..What can be inferred from paragraphs 4 and 5 about soil conservation efforts in the United States

- A. Encouraging minimum tillage practices resulted in much more efficient soil conservation than converting erodible land into vegetative cover.
- B. Complete retirement of land combined with soil-conservation practices significantly reduced soil erosion.
- C. Measuring the success of government-supported conservation programs over extended periods of time was sometimes as difficult as getting the programs started.
- D. The reduction of energy use due to practices such as conservation tillage and land retirement was much larger in the United States than in any other country.

9..The word particularly in the passage is closest in meaning to

- A. usually
- B. obviously
- C. especially
- D. currently

10..The word scarce in the passage is closest in meaning to

- A. short in supply
- B. more threatened
- C. more expensive
- D. less productive

11..In paragraph 6, the author refers to Tokyo, Japan, in order to

- A. explain why Japan is not likely to experience problems with soil erosion in the future
- B. provide evidence of the importance of maintaining cropland close to big cities
- C. point to an approach for reducing urban spread into croplands that has had positive results
- D. argue for the use of Japanese techniques to prevent erosion in the United States

12..Select the TWO answer choices that, according to paragraph 7, indicate true statements about Oregon. To obtain credit, you must select TWO answer choices.

- A. It planned and set long-term limits to urban growth.
- B. Its urban development within the city limits increased.
- C. Its surrounding farmland provided what the city needed to make it self-sufficient.
- D. It allowed each of its communities to deal with the commercial forces behind urban spread independently.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

These methods differ from traditional farming practices.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Terracing, probably one of the oldest methods for controlling soil erosion, led to the creation of the more sophisticated and more efficient conservation practices used in contemporary farming.
- B. After the 1930s Dust Bowl, the United States made significant changes to its agricultural practices, including the planting of tree shelterbelts and strip cropping.
- C. The United States has successfully instituted programs that encourage conservation tillage and the retirement of highly erodible land.
- D. The Conservation Reserve Program created in the United States in 1985 is gradually being adopted in other areas of the world such as Europe, Africa, and Asia.
- E. City governments in the United States and Japan developed conservation programs that encourage farmers near big cities to cultivate crops that minimize soil erosion.
- F. Saving cropland is sometimes difficult because it involves dealing with commercial forces, but some efforts like those in Tokyo and Oregon have been successful.

Passage 116 2015-05-24-3 Poetry Writing Course to be Canceled

1. To predict what the climate will be like in the future, scientists must rely on sophisticated computer models. These models use mathematical equations to represent physical processes and interactions in the atmosphere, ocean, and on land. A starting point is usually based on current measurements or estimates of past conditions. Then, using a spherical grid laid out over the entire globe, thousands of calculations are performed at grid intersections to represent and assess how conditions in the air, in the sea, and on land will change over time. Because of their complexity and size, supercomputers are used to run full-scale climate models. Much of the uncertainty in their outputs comes from the way that various aspects of the climate are represented by different models, and even more so, because there are aspects of climate that are not well understood one of which is how the ocean impacts climate.
2. The ocean's role in global warming stems principally from its huge capacity to absorb carbon dioxide and to store and transport heat. In the sea, photosynthesis by marine plants and algae, especially phytoplankton, removes great quantities of carbon dioxide from the atmosphere. Hence, the greater the growth (productivity) of phytoplankton in the sea, the greater the removal of carbon dioxide. But what controls the ocean's productivity. **There are several limiting factors, but results from a recent experiment suggest that in areas of the ocean where other nutrients are plentiful, iron may be one of the most important and, until recently, unrecognized variables controlling phytoplankton production.** Some have proposed a radical, highly controversial, and uncertain means to counteract global warming adding iron to the oceans to induce phytoplankton blooms. Perhaps increased phytoplankton growth would use up a significant amount of carbon dioxide in the atmosphere, but perhaps not, and there might well be side effects that could be detrimental to the ocean ecosystem.
3. Within the ocean, the production of limestone, in the form of calcium carbonate skeletons or shells, also reduces atmospheric carbon dioxide. However, when deposits of limestone become exposed and weathered on land or are recycled in the sea, carbon dioxide is released back into the atmosphere. What is not well understood is how much carbon dioxide resides in the sea and at what rate it is taken up and recycled. Relatively new research has also discovered beneath the sea a new and potentially significant threat to skyrocketing Earth temperatures: gas hydrates. Gas hydrates are a solid, crystalline form of water, like ice, except that they contain additional gas, typically methane, and are often found stored in ocean sediments. Increased ocean temperatures could cause gas hydrates to dissociate, releasing massive amounts of methane gas into the atmosphere and cause undersea landslides in the process. Consequently, hydrates may, if released, significantly increase global warming as well as create a geologic hazard to offshore drilling operations.
4. The ocean is also a great reservoir and transporter of heat. Heat from the ocean warms the atmosphere and fuels tropical storms. Heat is transported by currents from the equator to the poles. Ocean circulation is strongly controlled by wind and by the sea's balance of salt and heat. Scientists think that climate warming may slow down circulation, while cooling may speed it up, but these responses are not well understood. Evaporation from the ocean also supplies the precipitation that creates fields of snow and ice at high latitudes. Snow and ice coverage change the reflectivity of Earth's surface and are an important influence on how much incoming radiation is either absorbed or reflected. Furthermore, clouds and water vapor

in the atmosphere come mainly from the sea and strongly influence climate. Surprisingly, clouds are one of the least understood and most poorly modeled parts of the climate change equation. Most climate modeling grids fail to take into account common-sized cloud formations. Aerosols, tiny particles of soot, dust, and other materials, are thought to seed cloud formation, scatter incoming radiation and promote cooling, but this effect, which would counteract warming, is also only superficially understood. Computer models of climate change must take into account all of the processes within the ocean, over land, and in the sky that potentially influence warming. No wonder there is such uncertainty.

1..According to paragraph 1, the results of full-scale climate models are questionable in part because

- A. the supercomputers used for such modeling are large and complex
- B. thousands of calculations have to be performed to assess conditions
- C. past conditions cannot always be estimated accurately
- D. there are multiple ways to represent the same aspect of climate

2..The word principally in the passage is closest in meaning to

- A. obviously
- B. apparently
- C. mainly
- D. originally

3..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Iron may be one of the most important factors in controlling phytoplankton production in ocean waters that are rich in other nutrients.
- B. Results from a recent experiment suggest that several factors limiting phytoplankton production in ocean waters have gone unrecognized.
- C. Although it was not recognized until recently, nutrients are plentiful in areas of the ocean where iron controls phytoplankton production.
- D. Until recently, the importance of iron was not taken into account in experiments concerning phytoplankton production.

4..The word controversial in the passage is closest in meaning to

- A. experimental
- B. fascinating
- C. producing disagreement
- D. demonstrating poor judgment

5..The word induce in the passage is closest in meaning to

- A. supply nutrients to
- B. cause the formation of
- C. expand
- D. strengthen

6..According to paragraph 2, how might increasing phytoplankton growth help lower global temperatures

- A. By cooling the oceans
- B. By decreasing carbon dioxide levels in the ocean
- C. By reducing the amount of carbon dioxide in the atmosphere
- D. By transporting heat from the ocean's surface to deeper levels

7..According to paragraph 3, which of the following reduces atmospheric carbon dioxide

- A. The weathering of limestone
- B. The production of limestone
- C. The recycling of carbon dioxide
- D. The presence of methane in gas hydrates

8..According to paragraph 3, why are gas hydrates a possible threat to the global climate

- A. If disturbed by offshore drilling, they can destroy limestone deposits.
- B. They can replace regular ice at certain locations.
- C. If melted, they may release a lot of carbon dioxide into the atmosphere.
- D. They contain a lot of methane, which may be released as the ocean warms.

9..The word fuels in the passage is closest in meaning to

- A. provides energy for
- B. determines the route of
- C. carries
- D. breaks up

10..Which of the following is NOT mentioned in paragraph 4 as a way in which the ocean affects the climate

- A. It stores heat.
- B. It moves heat from the equator toward the poles.
- C. It speeds up wind circulation.
- D. It warms up the atmosphere.

11..Paragraph 4 suggests that a significant decrease in snow and ice fields at high latitudes would have what effect

- A. More clouds and water vapor would be produced in the atmosphere.
- B. More of the Sun's radiation would be absorbed by Earth.
- C. The oceans would cool more quickly.
- D. More precipitation would occur at low latitudes.

12..Why does the author mention that Most climate modeling grids fail to take into account common-sized cloud formations

- A. To suggest why the influence of clouds on climate change is still undetermined
- B. To explain why research on climate change does not focus on clouds
- C. To help explain why it is unclear whether aerosols have the effect of counteracting warming
- D. To explain in part why scientists are uncertain how much incoming radiation is absorbed or reflected

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Nor is carbon dioxide the only gaseous substance in the ocean that may affect climate.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Estimates of future conditions are entered into supercomputers to calculate climate possibilities at various places on Earth.
- B. The ocean's capacity to absorb carbon dioxide remains great despite recent reductions of marine plant nutrients such as iron.
- C. Oceans absorb a great deal of carbon dioxide from the air through limestone production and the photosynthesis of phytoplankton.
- D. Ocean circulation is strongly controlled by wind and by the sea's balance of salt and heat.
- E. Gases are stored in the sea in the form of shells and hydrates, but gases stored in these ways can be recycled to the atmosphere where they may cause warming.
- F. The ocean holds and moves a great deal of heat, and as ocean water evaporates, it produces clouds, snow, and ice, which all affect global temperatures.

Passage 117 2015-05-30-1 Thermal Stratification

1. Physical characteristics of aquatic environments at different depths such as salt level, light, inorganic nutrients, degree of acidity, and pressure all play key roles in the distribution of organisms. One of the most important physical features is thermal stratification.
2. When solar radiation strikes water, some is reflected, but most penetrates the surface and is ultimately absorbed. Although water may appear transparent, it is much denser than air and absorbs radiation rapidly. In clear water, 99 percent of the solar radiation is absorbed in the upper 50 to 100 meters. Longer wavelengths of light are absorbed first; the shorter wavelengths (which have more energy) penetrate farther, giving the depths their characteristic blue color.
3. This rapid absorption of sunlight by water has two important consequences. First, it means that photosynthesis—the process by which plants use the energy of sunlight to produce the organic carbon compounds necessary for life—can only occur in surface waters where the light intensity is sufficiently high. Species that produce their own organic carbon compounds are called primary producers, and they are the base of the marine food web. Virtually all of the photosynthesis that supports the rich life of oceans and lakes comes from plants living in the upper 10 to 30 meters of water. Along shores and in very shallow bodies of water, some species such as kelp are rooted to the bottom. These plants may attain considerable size and structural complexity, and may support diverse communities of organisms. In the open waters that cover much of the globe, however, the primary producers of organic carbon are tiny, often one-celled algae (called phytoplankton), which are suspended in the water. Zooplankton, tiny invertebrates that feed on phytoplankton, migrate vertically on a daily cycle: up into the surface waters at night to feed and down into the dark, deeper waters during the day to escape predatory fish that rely on light to detect prey.
4. Second, the rapid absorption of sunlight by water means that only surface water is heated. The density of pure water is greatest at 4 °C and declines as the water's temperature rises above or falls below this point. When solar radiation heats the water surface above 4 °C, the warm surface water becomes lighter than the cool, deeper water, and so tends to remain on the surface, where it may be heated further and become even less dense. In tropical areas and in temperate climates during the summer, the surfaces of oceans and lakes are usually covered by a thin layer of warm water. Unless these bodies of water are shallow, the deep water below this layer is much colder (sometimes near 4 °C). The change in temperature between the warm surface water and the cold, deep water is called the thermocline. Mixing of the surface water by wave action determines the depth of the thermocline and maintains relatively constant temperatures in the water above it.
5. Tropical lakes and oceans show pronounced permanent stratification of their physical properties, with warm, well-oxygenated, and lighted surface water giving way to frigid, dark, deep water almost devoid of oxygen. Oxygen cannot be replenished at great depths where there are no photosynthetic organisms to produce it, and the stable thermal stratification prevents mixing and reoxygenation by surface water. Only relatively few organisms can live in such extreme conditions. The waste products and dead bodies of organisms living in the surface waters sink to the depths, taking their mineral nutrients with them. The lack of vertical circulation thus limits the supply of nutrients to the phytoplankton above. Consequently, deep tropical lakes are often relatively unproductive and depend on continued

input from streams for the nutrients required to support life.

6. The situation is somewhat different in temperate and polar waters. Deep lakes, in particular, undergo dramatic seasonal changes: they develop warm surface temperatures and a pronounced thermocline in summer, but freeze over in winter. Twice each year, in spring and fall, the entire water column attains equal temperature and equal density; moderate winds may then generate waves that mix deep and shallow water, producing what is called overturn. This semiannual mixing carries oxygen downward and returns inorganic nutrients to the surface. Phosphorus and other nutrients may be depleted during the summer; overturn replenishes these nutrients by stimulating the growth of phytoplankton.

1..The word ultimately in the passage is closest in meaning to

- A. probably
- B. quickly
- C. eventually
- D. frequently

2..According to paragraph 2, which of the following is true about solar radiation when it strikes water

- A. Longer wavelengths travel farther.
- B. Most of it is absorbed near the surface.
- C. It is not absorbed by cloudy water.
- D. More long wavelengths than short wavelengths are absorbed.

3..The word Virtually in the passage is closest in meaning to

- A. Apparently
- B. Usually
- C. Nearly
- D. Fortunately

4..According to paragraph 3, all of the following are true of kelp EXCEPT:

- A. They are found at the bottom of shallow waters.
- B. They are the primary producers of organic carbon.
- C. They can grow very large.
- D. They are a source of food for a variety of organisms.

5..The word suspended in the passage is closest in meaning to

- A. dissolved
- B. floating
- C. invisible

D. released

6..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Zooplankton move up into surface waters at night to feed on phytoplankton, and down into deep, dark waters in daytime to avoid detection by predatory fish.
- B. Zooplankton migrate vertically on a daily basis in order to feed on phytoplankton, following them up into surface waters by day and down into dark, deeper waters at night.
- C. Unlike zooplankton, which migrate between surface waters and deeper waters, predatory fish must rely on the daytime light of surface waters to detect prey.
- D. Zooplankton are so tiny that they can only be detected by predatory fish in the light conditions of surface water, where they feed on phytoplankton.

7..According to paragraph 4, what is a thermocline

- A. The deep, cold layer of water below the surface of oceans and lakes
- B. The thin layer of warm water on the surface of oceans and lakes
- C. The change in density of surface water as it is continually heated
- D. The point at which the temperature of water in oceans and lakes changes from warm to cold

8..The word replenished in the passage is closest in meaning to

- A. absorbed
- B. restored
- C. contained
- D. obtained

9..According to paragraph 5, all of the following are true of the water at great depths in tropical lakes EXCEPT:

- A. It has less oxygen than the surface water does.
- B. It is very dark.
- C. It contains relatively few living organisms.
- D. It has low levels of mineral nutrients.

10..The word dramatic in the passage is closest in meaning to

- A. partial
- B. complex
- C. frequent
- D. striking

11..According to paragraph 6, after overturn, deep water levels of lakes in temperate climates will contain increased amounts of

- A. warm water
- B. phosphorus
- C. oxygen
- D. phytoplankton

12..According to paragraphs 5 and 6, temperate lakes are in general more productive than tropical lakes because

- A. temperate lakes receive more nutrients from incoming streams
- B. the summer thermocline is relatively deep in temperate lakes
- C. solar radiation penetrates more of the water in temperate lakes
- D. temperate lakes experience vertical circulation of water

13..Look at the four squares that indicate where the following sentence could be added to the passage.

If the water is clouded by many microscopic organisms, absorption occurs even closer to the surface.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Sunlight is rapidly absorbed by the upper layer of water, which makes it warmer and consequently less dense than the water below.
- B. Photosynthesizing plants, which provide the food and oxygen for most other forms of aquatic life, are limited to the levels of oceans and lakes where sunlight is available.
- C. The diversity of plant life in shallow water makes it attractive to zooplankton.
- D. As water becomes less dense and moves to the surface of the ocean or lake, it cools down and moderates the temperature of the surface.
- E. Because more solar energy is absorbed by bodies of water in the tropics, the topmost layer of the thermocline is widest there.
- F. In the tropics, permanent thermal stratification limits the amount of life supported in deep bodies of water; however, in temperate climates, seasonal thermal stratification allows vertical circulation and mixing of water, allowing the support of more life.

Passage 118 2015-05-30-2 Live Performance

1. Unlike video and cinema (although sometimes employing elements of both), the theater is a living, real-time event, with both performers and audience mutually interacting, each aware of the other's immediate presence. This turns out to be an extremely important distinction. Distinguished film stars, particularly those with theater backgrounds (as most have), routinely return to the live dramatic stage despite the substantially greater financial rewards of film work and invariably prefer stage acting because of the immediate audience response theater provides, with its corresponding sensations of excitement and presence.
2. The first of these is the rapport existing between actor and audience. Both are breathing the same air; both are involved at the same time and in the same space with the stage life depicted by the play. Sometimes their mutual fascination is almost palpable; every actor's performance is affected by the way the audience yields or withholds its responses: its laughter, sighs, applause, gasps, silences. Live theatrical performance is always a two-way communication between stage and house.
3. Second, theater creates a relationship among the audience members. Having arrived at the theaters as individuals or in groups of two or three, the audience members quickly find themselves fused into a common experience with total strangers: laughing at the same jokes, empathizing with the same characters, experiencing the same revelations. This broad communal response is never developed by television drama, which is played chiefly to solitary or clustered viewers who (because of frequent commercial advertisements) are only intermittently engaged, nor is it likely to happen in movie houses, where audience members essentially assume a one-on-one relationship with the screen and rarely (except in private or group screenings) break out in a powerful collective response, much less applause. By contrast, live theatrical presentations generate audience activity that is broadly social in nature: the crowd arrives at the theater at about the same time, people mingle and chat during intermissions, and all depart together, often in spirited conversation about the play. Moreover, they communicate during the play: laughter and applause build upon themselves and gain strength from the recognition that others are laughing and applauding. The final ovation unique to live performance inevitably involves the audience applauding itself, as well as the performers, for understanding and appreciating the theatrical excellence they have all seen together. And plays with political themes can even generate collective political response. In a celebrated example, 1935's Waiting for Lefty was staged as if the audience were a group of union members; by the play's end the audience was yelling Strike! Strike! in response to the play's issues. Obviously, only a live performance could evoke such a response.
4. Finally, live performance inevitably has the quality of immediacy. The action of the play is taking place right now, as it is being watched, and anything can happen. **Although in most professional productions the changes that occur in performance from one night to another are so subtle that only an expert would notice, the fact is that each night's presentation is unique, and everyone present the audience, the cast, and those behind the scenes knows it.** This awareness lends an excitement that cannot be achieved by theatrical events that are wholly in the can. One reason for the excitement, of course, is that in live performance, mistakes can happen; this possibility occasions a certain abiding tension, perhaps even an edge of stage fright, which some people say creates the ultimate thrill of the theater. But just as disaster can come without warning, so too can splendor. On any given

night, each actor is trying to better his or her previous performance, and no one knows when this collective effort will coalesce into something sublime. The actors' constant striving toward self-transcendence gives the theater a vitality that is missing from performances fixed unalterably on videotape or celluloid. But perhaps most appropriately, the immediacy of live performance embodies the fundamental uncertainty of life. One prime function of theater is to address the uncertainties of human existence, and the very format of live performance presents a moment-to-moment uncertainty right before our eyes. Ultimately, this immediate theater helps us define the questions and confusions of our lives and lets us grapple, in the present, with their implications.

1..The word distinction in the passage is closest in meaning to

- A. idea
- B. blend
- C. definition
- D. difference

2..Paragraph 1 makes which of the following points about theater and film

- A. Theater audiences tend to be more critical than film audiences.
- B. Actors in the theater are usually not as well-known as film actors.
- C. Theater companies tend to pay more than film companies do for the most distinguished actors.
- D. Audiences respond to actors differently in theater than in film.

3..Paragraph 1 suggests that the reason distinguished film stars return to live theater is that they

- A. are able to command higher fees as well-known actors
- B. enjoy the excitement of performing before a live audience
- C. have great respect for theatrical drama as an art form
- D. are dissatisfied with the roles they are offered in films and television

4..The word rapport in the passage is closest in meaning to

- A. excitement
- B. balance
- C. bond
- D. fascination

5..In paragraph 3, which of the following is mentioned as support for the statement that This broad communal response is never developed by television drama

- A. Television drama is rarely about serious social issues.
- B. People do not usually talk to each other while watching television.

- C. Television audiences vary greatly in their interest in television dramas.
- D. People do not typically watch television in large groups.

6..According to paragraph 3, movie house audiences are different from audiences at live theatrical performances because movie house audiences do not

- A. enjoy humor and jokes as much as theater audiences do
- B. develop broad communal responses
- C. sympathize with the characters they see dramatized
- D. generally applaud unless everyone else is applauding

7..Why does the author mention the play Waiting for Lefty

- A. To illustrate the power of the communal response to plays
- B. To argue that plays about political subjects have more power to evoke deep feelings in an audience than nonpolitical plays do
- C. To provide an example of a play that was a popular success because it dealt with important political issues
- D. To compare the political importance of plays in recent times with the political importance of earlier plays

8..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Although experts can detect the changes that occur in different performances, the changes are too subtle to be noticed by anyone else.
- B. Although their performances vary only subtly from one night to another, actors in most professional productions want audiences to believe that every performance is unique.
- C. Everyone involved in the professional production of a play knows that very small, almost unnoticeable changes make each performance unique.
- D. In most professional productions, changes are included from one performance to another that are intended to make every performance a unique one.

9..The word thrill in the passage is closest in meaning to

- A. goal
- B. weakness
- C. meaning
- D. excitement

10..The word vitality in the passage is closest in meaning to

- A. style

- B. energy
- C. purpose
- D. quality

11..According to paragraph 4, on any given night the result of actors' efforts to better their previous performances is that the actors

- A. form long-lasting relationships with the audience
- B. are better able to overcome their stage fright
- C. create a quality that is not present in film or television
- D. are more likely to be admired by audiences

12..It can be inferred from paragraph 4 that one of the reasons filmed performances are less exciting than live theatrical performances is because

- A. there is little chance that a mistake will occur in a filmed performance
- B. most movies portray situations that audiences have seen before
- C. audiences are interested in seeing famous actors live rather than on a screen
- D. most people are accustomed to going to the movies but view the theater as a special event

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Such signs of an audience's engagement thus become part of every performance.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Although live theater is unlike either video or cinema, the different genres sometimes overlap.
- B. While live theater is regarded by most film actors as the most exciting place to perform, many are too troubled by stage fright to perform live.
- C. In the theater, there is a two-way communication between actor and audience that has an effect on the actor's performance.
- D. Audiences at live theatrical performances form a special connection, sharing the common experience of reacting to a live theatrical performance.
- E. Many people go to the theater excited to see in a live performance an actor that they have never seen before except on television or in films.
- F. The excitement of live theater cannot be achieved in any other medium, and its immediacy and uncertainty help us deal with life's fundamental uncertainty.

Passage 119 2015-05-30-3 Earthquake Prediction

1. Accurate prediction of earthquakes is not currently possible, although intensive research is proceeding in many areas.
2. Two types of earthquake prediction are theoretically possible. The first type is long-term forecasting, in which the probability of an earthquake along a particular segment of a within a certain time interval is calculated by studying seismic gaps and historical records of earthquakes that have occurred along that fault segment. By plotting the number of earthquakes within specific time intervals against their magnitudes, diagrams can be constructed for a local area. From this plot it is possible to determine the recurrence interval, or the average time interval between earthquakes of a specific magnitude. Predictions can then be made that an earthquake of that magnitude has a high probability of occurrence within a specified time interval, if the date of the last earthquake is known.
3. Research leading to short-term forecasting, which involves a shorter time interval, has been focused on precursors observed prior to previous earthquakes. Precursors are physical or chemical phenomena that occur in a typical pattern before an earthquake. These phenomena include changes in the velocity of seismic waves, the electrical resistance of rocks, the frequency of the usually minor preliminary earthquakes (foreshocks), the deformation of the land surface, and the water level or water chemistry of wells in the area. Many of these precursors can be explained by a theory called the dilatancy model. Under this hypothesis, rocks in the process of strain along a fault show significant dilation or swelling before rupture. This volume increase is caused by the opening of microcracks, which are minute failure zones in weaker mineral grains in the rock and along grain boundaries. Groundwater flows into the highly stressed areas during the formation of microcracks. These changes in density and water content affect the ability of the rock to transmit seismic waves and conduct electricity. Therefore, seismic-wave velocity and electrical resistance progressively change as the overall rupture along the fault draws near. Localized changes in land-surface elevation are also related to volume changes at depth. An area of recent uplift along the San Andreas Fault near Los Angeles, which has been named the Palmdale Bulge, is being monitored in great detail as a possible indicator of a future earthquake.
4. Volume changes and groundwater movement may be reflected by changes in water levels in wells and also by changes in the chemical composition of groundwater. Radon gas has been observed to increase in wells prior to earthquakes. These increases are perhaps related to the release of radon gas from rocks during the formation of microcracks. The pattern of seismic activity is also significant in the vicinity of a fault area where rupture is imminent. This pattern consists of an initial rise in the number of small events, followed by a decline in foreshocks just prior to the major earthquake. The decline may represent a temporary increase in rock strength before the newly formed microcracks are filled with water.
5. The precursor phenomena can be grouped into stages according to the dilatancy model. Stage I consists of a gradual stress buildup along the fault. Stages II and III are correlated with dilatancy and water influx. Stage IV is the major earthquake, and stage V is the aftermath of the event. If every earthquake followed the sequence with uniform stage duration, earthquake prediction would be a simple matter. Instead of following the same patterns, each earthquake is unique in terms of specific precursor behavior patterns and length of precursor stages. A magnitude 6.9 North American earthquake in 1989 was preceded by a

substantially smaller magnitude 5 earthquake fifteen months before the event. Another foreshock of similar size occurred two months before the event. In each case, a public advisory was issued stating that those smaller earthquakes could be foreshocks to a stronger earthquake within five days. However, the fault did not cooperate, and those predictions were not successful. Continued research and study of future earthquakes will certainly lead to refinement of the dilatancy model or to a replacement model with more accurate predictive capabilities.

1..The word specified in the passage is closest in meaning to

- A. probable
- B. stated
- C. short
- D. typical

2..According to paragraph 2, all of the following information is used in the process of long-term earthquake prediction EXCEPT the

- A. analysis of seismic gaps
- B. record of past earthquakes in the fault area
- C. date of the last recorded earthquake in the area
- D. pattern of earthquake activity in other nearby fault segments

3..According to paragraph 2, long-term forecasting can be used to predict which of the following

- A. The influence of earthquake activity in one segment of the fault area on other segments
- B. The frequency with which earthquakes of a certain size will occur
- C. The possible date of the next earthquake
- D. The magnitude of the next earthquake

4..The word focused in the passage is closest in meaning to

- A. dependent
- B. funded
- C. concentrated
- D. published

5..Paragraph 3 mentions all of the following as examples of precursors EXCEPT

- A. changes in the speed of seismic waves
- B. changes in the availability of electricity
- C. changes in the frequency of foreshocks
- D. changes in land surfaces

6..According to the dilatancy model, what happens to rocks shortly before an earthquake

- A. They lose significant amounts of moisture.
- B. They show signs of expanding.
- C. They move downward at great speed.
- D. They increase in temperature.

7..According to paragraph 3, the groundwater that flows into microcracks before an earthquake causes

- A. changes in seismic waves and electrical activity
- B. increases in the mineral content of rocks
- C. the disappearance of grain boundaries in rocks
- D. a release in the tension of highly stressed areas of rocks

8..The author discusses the San Andreas Fault near Los Angeles in order to

- A. contrast past and future patterns of earthquake activity in the area
- B. give an example of an area where underground earthquake activity is apparent from land changes above the ground
- C. explain why recent earthquake predictions have increased accuracy
- D. suggest that some areas of earthquake activity are easier to monitor than others

9..According to paragraph 4, which of the following occurs just before an earthquake

- A. The chemical content of groundwater drops.
- B. The rocks weaken as they fill with water.
- C. Seismic activity decreases.
- D. Radon gas causes microcracks to form.

10..The word imminent in the passage is closest in meaning to

- A. frequent
- B. well understood
- C. known to occur
- D. about to happen

11..How is paragraph 5 organized

- A. The sequence of earthquake stages is given, and the effect of variable stage length on earthquake prediction is explained.
- B. The earthquake stages are named, and the most important stage is illustrated with a specific earthquake event.
- C. The sequence of earthquake stages is given, and evidence is presented that the intervals between stages are roughly equal in length.

D. The earthquake stages are first named, and each is then described in greater detail.

12..The word refinement in the passage is closest in meaning to

- A. reconsideration
- B. acceptance
- C. improvement
- D. extension

13..Look at the four squares that indicate where the following sentence could be added to the passage.

But the reality of earthquake forecasting is considerably more complex.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Short-term forecasting has been used more widely than long-term forecasting in the prediction of earthquakes.
- B. Long-term forecasting of earthquakes uses data on past seismic activity to determine the likelihood that an earthquake will occur in a certain area within a certain time period.
- C. Short-term forecasting research has studied earthquake precursors such as volume increases in rocks and unusual movements in underground water that occur shortly before an earthquake takes place.
- D. The dilatancy model has been used to successfully forecast some recent earthquakes.
- E. Attempts to improve forecasting by using five stages of earthquake predictors have been unsuccessful because each earthquake has unique precursor patterns and durations.
- F. The magnitude 6.9 North American earthquake in 1989 was not successfully predicted because the many foreshocks before the event were too small to measure.

Passage 120 2015-06-13-2 Gondwana

1. Among the enduring legacies of the famous European voyages of discovery in the eighteenth and nineteenth centuries are a collection and scientific description of plants and animals from around the world. These form the nucleus of the great collections in modern museums and have been responsible for a radical revision in the way that we perceive the structure of Earth and the forces that have shaped its surface over time. As the fauna and flora from far-flung lands came to be described and incorporated into the body of knowledge about the world, it was noted that there were some striking similarities among living and extinct organisms of the Southern Hemisphere continents. In the 1840s, the English botanist Sir Joseph Dalton Hooker commented on the remarkable fact that the flora of South America and Oceania (mainly Australia, New Zealand, New Guinea, and the Malay Archipelago) shared seven families of flowering plants and 48 genera that were not to be found elsewhere. Later, similar patterns were observed in other groups of plants and animals, such as liverworts, lichens, mayflies, midges, and various types of vertebrates. How could these similarities be explained in view of the enormous stretches of ocean that separate the Southern Hemisphere continents today **One idea developed during the late nineteenth century was that there existed in the remote geological past a vast Southern Hemisphere continent in other words, that the modern continents of the Southern Hemisphere were somehow connected long ago, thus explaining the similarities in fauna and flora. The name given to this hypothetical continent was Gondwana.**
2. One of the most distinctive fossil plants of this hypothetical continent is called Glossopteris. When first described by the French paleobotanist Adolphe Brongniart in 1828, Glossopteris was thought to be a type of fern. Now, however, it is known to be a woody seed-bearing shrub or tree. The trunks of Glossopteris could reach 4 meters in height. Seeds and pollen-containing organs were borne in clusters at the tips of slender stalks attached to the leaves, but some species may have borne seeds in cones. It is thought that Glossopteris lived in a seasonal environment, and this is consistent with the occurrence of growth rings in the wood. Also, there is evidence that the plant was deciduous (that is, that it shed its leaves annually at the end of the growing season) and that it grew under very wet soil conditions, like the modern swamp cypress. The large leaves of Glossopteris which exceeded 30 centimeters in length are common fossils in rocks of the Permian period (299 'C251 million years ago) in India, Africa, South America, Australia, and Antarctica.
3. At the time the Gondwana hypothesis was conceived, the prevailing theory of Earth saw continents as fixed in their relative positions. The problem of linking up the various elements of Gondwana was solved by hypothesizing the existence of ancient land bridges. This changed in 1912 with the proposal of the theory of continental drift by the German meteorologist and geophysicist Alfred Wegener, an idea that was later developed and championed by the famous South African geologist Alex Logan du Toit. Wegener and du Toit argued that the continents are not fixed; rather, they have moved apart or drifted to their present-day positions. In the past, Gondwana was a single contiguous landmass comprising the present-day Southern Hemisphere continents.
4. These ideas seemed incredible at the time, but in support of their theory Wegener and du Toit pointed to similarities in fauna and flora, and the distributions of fossils such as Glossopteris provided an important piece of evidence in the assembly of the Gondwana jigsaw puzzle.

Wegener and du Toit also drew together other different sources of evidence, such as the remarkable geometric fit of South America and Africa, and similarities between the ages and types of rock found in areas of Southern Hemisphere continents that are now thousands of miles apart. The notion of drifting continents only became widely accepted in the 1960s following the discovery of paleomagnetism (the study of changes in the polarity of Earth's magnetic field through time) and the development of the theory of plate tectonics, which explained the growth and movement of continents and other geological phenomena.

1..The word enduring in the passage is closest in meaning to

- A. exciting
- B. lasting
- C. unexpected
- D. well-known

2..Why does the author mention the enormous stretches of ocean that separate the Southern Hemisphere continents today

- A. To emphasize the importance of the famous European voyages of discovery in the eighteenth and nineteenth centuries
- B. To suggest a reason why so many genera of flowering plants are found only in the Southern Hemisphere
- C. To question the accuracy of Sir Joseph Dalton Hooker's observations about the similarities between the flora of Australia and that of South America
- D. To explain why the similarities between flora pointed out by Sir Joseph Dalton Hooker seemed so remarkable

3..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. The similarities in fauna and flora across Southern Hemisphere continents were explained in the late nineteenth century.
- B. In the nineteenth century it was discovered that the Southern Hemisphere continents contain fauna and flora that are highly similar.
- C. In the nineteenth century, it was suggested that the modern continents of the Southern Hemisphere were once connected.
- D. The fauna and flora of the modern continents of the Southern Hemisphere were found to be very similar in the late nineteenth century.

4..In paragraph 2, the author discusses *Glossopteris* in order to

- A. provide an example of a fossil species that was attributed to Gondwana on the basis of its wide distribution
- B. explain why the descriptions of ancient plants made by Adolphe Brongniart were not

completely accurate

- C. establish the importance of nineteenth-century paleobotanist Adolphe Brongniart
- D. show that similarities among flora in the Southern Hemisphere continents were not limited to flowering plants

5..According to paragraph 2, current evidence indicates that *Glossopteris* had all of the following characteristics EXCEPT:

- A. It was deciduous.
- B. It was a type of fern.
- C. It had large leaves.
- D. It grew in wet soil conditions.

6..According to paragraph 2, what suggests that *Glossopteris* may have lived in a seasonal environment

- A. It grew in what is now Antarctica.
- B. It was a seed-bearing plant.
- C. There is evidence that it lost its leaves annually.
- D. There were stalks attached to its leaves.

7..The word prevailing in the passage is closest in meaning to

- A. best understood
- B. most controversial
- C. generally accepted
- D. recently proposed

8..According to paragraph 3, why did geologists initially hypothesize the existence of ancient land bridges across continents

- A. To explain how the continents may have drifted to their present-day positions
- B. To explain how organisms could spread across distant continents
- C. To argue against Wegener's theory of continental drift
- D. To help explain the present-day positions of the Southern Hemisphere continents

9..According to paragraph 3, what did Alfred Wegener and Alex Logan du Toit have in common

- A. Both were originally trained as meteorologists.
- B. Both had doubts about the hypothesis that Gondwana was once a single contiguous landmass.
- C. Both believed that continents change their position over time.
- D. Both believed that the present-day Southern Hemisphere continents were too far apart to have been linked by a land bridge.

10..The word incredible in the passage is closest in meaning to

- A. unfamiliar
- B. unsupported
- C. hard to understand

D. impossible to believe

11..According to paragraph 4, Wegener and du Toit offered all of the following evidence in support of their hypothesis EXCEPT

- A. distributions of fossils across the Southern Hemisphere continents
- B. the geometric fit of South America and Africa
- C. similarities in rocks across the Southern Hemisphere continents
- D. differences in geological phenomena across the Southern Hemisphere continents

12..Paragraph 4 strongly suggests that the theory of continental drift was not widely accepted before the 1960s in part because

- A. the distributions of fossils such as *Glossopteris* were not yet generally known
- B. other explanations for the geometric fit of South America and Africa were available
- C. there was no satisfactory explanation for the movement of continents
- D. few scientists accepted the claims made by Wegener and du Toit about the similarities in the ages of groups of rocks

13..Look at the four squares that indicate where the following sentence could be added to the passage.

This led to its name, which means tongue fern in Greek and is a reference to its tongue-shaped leaves.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. The plants and animals collected during the eighteenth- and nineteenth-century voyages of discovery made it possible to advance important scientific theories.
- B. By 1828, fossils of *Glossopteris* had been discovered to have inhabited both the Northern and Southern Hemisphere continents in the remote geological past.
- C. To explain the observed distribution patterns of flora and fauna, scientists hypothesized that land bridges had linked the widely separated Southern Hemisphere continents in the geological past.
- D. The theory of continental drift, proposed in 1912, held that the Southern Hemisphere continents once formed an unbroken landmass and subsequently moved apart to their present locations.
- E. To support their hypothesis that continents are not fixed in their relative positions, Alfred Wegener and Alex Logan du Toit used paleomagnetic evidence to develop the theory of plate tectonics.
- F. Continental drift explained the distribution of organisms as well as the similarities in rocks and the geometric fit of Africa and South America, but the theory was not generally accepted until the 1960s.

Passage 121 2015-06-13-3 Olmec Art

1. The earliest Mesoamerican art and architecture to combine ideological complexity, craft, and permanence was that of the Olmecs, whose civilization flourished between about 1500 B.C. and 100 B.C. The early Olmecs established major ceremonial centers along the rich lowlands of the modern Mexican states of Veracruz and Tabasco. At distant Teopantecuanitlan, the Olmecs established a sacred precinct, the first monumental evidence of the Olmecs in the highlands. But the Olmecs had an advanced social and economic system, with networks for commerce extending far to the west and south. The fertile gulf plain probably allowed for an agricultural surplus, controlled by only a handful of individuals. From the art and architecture of their ceremonial centers (we know too little about Olmec domestic life to call their sites cities), it is clear that for the Olmecs, social stratification was sufficiently advanced for their society to place great importance on the records of specific individuals, particularly in the form of colossal heads (enormous stone sculptures of human heads and faces).
2. Long before modern radiocarbon dating testified to the antiquity of this culture, archaeologists and art historians had become aware of the power of Olmec art through individual objects. Some even identified the Olmec culture as the oldest of Mesoamerican civilizations, perhaps a mother culture from which all others derived, as the art historian Miguel Covarrubias once thought. Eventually the antiquity of Olmec culture was confirmed, and today many important elements of Mesoamerican art and architecture can be seen to have had a probable Olmec origin: ball courts, pyramids, portraiture, and mirrors. Some later Mesoamerican deities probably derive from Olmec gods, and even the famous Maya calendar was already in use by peoples in the Olmec area at the dawn of Maya civilization.
3. One of the first important Olmec objects to come to modern attention was the Kunz axe, acquired in the 1860s in Oaxaca, Mexico. The ceremonial axe puzzled and intrigued investigators for years because on the one hand, it was clearly neither Aztec nor Maya, the best-known ancient Mesoamerican cultures, and in fact it had no features that could be linked with any known civilization, while on the other hand, it had surely been made in Mesoamerica in antiquity.
4. The axe exhibits many qualities of the style we now call Olmec: precious blue-green translucent jade, worked to reveal a figure in both two and three dimensions. More than half the axe is devoted to the creature's face: an open, toothless mouth, and closely set, slanting eyes which has often been likened to the face of a howling human infant. The creature's hands are worked in lower relief, and in them he grasps a miniature version of himself. Feet and toes are indicated only by incision (carved lines), and incision also marks the face, ears, and upper body, perhaps to suggest tattooing, ear ornaments, and a tunic. For over two millennia this large, precious axe was presumably kept as a treasure or heirloom. **It was not until 1955, after several seasons of excavation at La Venta had produced many fine jade objects and a convincing series of radiocarbon dates in the first millennium B.C., that objects such as the Kunz axe were at last understood by scholars to embody the principles of the first great art style of Mesoamerica.**
5. Early scholars of the Olmec style noticed a pattern of imagery repeated on many of the carved stone objects. Many howling baby faces were found, and other faces seemed to combine the features of humans and jaguars (large cats). Today, while the presence of jaguar imagery is still acknowledged, scholars have discovered that aspects of many other tropical

rainforest fauna can be identified in the carvings. The caiman (a kind of alligator), eagle, toad, jaguar, and snake all appear in the Olmec supernatural repertoire. Anthropologist Peter David Joralemon has suggested that most of the motifs and images can be allocated to a few Olmec deities. The paw-wing motif, for example, can be shown to be an element of the winged dragon, itself perhaps derived from the eagle and caiman. This whole intricate symbolic code appears to have been in use from the first appearance of the Olmecs, and to have been employed consistently for a thousand years.

1..The word that in the passage refers to

- A. Mesoamerican art and architecture
- B. the ideological complexity, craft, and permanence
- C. the earliest civilization
- D. the permanent art and architecture

2..The word surplus in the passage is closest in meaning to

- A. advantage
- B. excess quantity
- C. system
- D. supply

3..According to paragraph 1, which of the following is true about the ceremonial centers established by the Olmecs

- A. The centers served as the burial places of ancient Olmec rulers.
- B. The inhabitants of each center had little or no contact with those in other centers.
- C. The centers featured major works of art and architecture that were made to last.
- D. The first and most important centers were built in the highlands.

4..Paragraph 1 supports which of the following ideas about Olmec society

- A. Major artists and successful traders had roughly equal status.
- B. The most important members of Olmec society resided in the highlands.
- C. More people were engaged in producing monumental works of art than were engaged in agriculture.
- D. There was a well developed social structure in which some individuals held more power than others.

5..The author put the word Maya in quotation marks in order to indicate that

- A. few Mesoamericans were familiar with the Maya calendar
- B. the calendar commonly attributed to the Maya was not actually developed by them
- C. the names of Mesoamerican gods were included in the Maya calendar

D. it is doubtful that the Olmec and the Maya used the same calendars

6..According to paragraph 2, how was the antiquity of Olmec culture confirmed

- A. Through close study of the Maya calendar
- B. By archaeologists' success at tracing later Mesoamerican gods back to those of the Olmec
- C. By radiocarbon dating of Olmec objects
- D. By comparing different references to an ancient mother culture

7..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Because the Kunz axe could not be linked with known Mesoamerican cultures of antiquity, investigators concluded that it was neither Aztec nor Maya.
- B. The ceremonial axe puzzled and intrigued investigators because it was neither Aztec nor Maya, nor was it from any other ancient Mesoamerican civilization.
- C. On the one hand the ceremonial axe was puzzling because it was not Aztec or Maya, and on the other hand it was intriguing because no other Mesoamerican culture made ceremonial axes.
- D. The Kunz axe puzzled investigators for years because, although it was clearly made in ancient Mesoamerica, it could not be attributed to any known Mesoamerican culture.

8..The word exhibits in the passage is closest in meaning to

- A. displays
- B. includes
- C. combines
- D. introduces

9..The word embody in the passage is closest in meaning to

- A. utilize
- B. reveal
- C. incorporate
- D. clarify

10..It can be inferred from paragraph 4 that the author provides a very detailed description of the Kunz axe because

- A. the Kunz axe is more like later Mesoamerican art than it is like Olmec art
- B. the Kunz axe is a characteristic example of Olmec artistic style and principles
- C. the Kunz axe is the single most important and valuable piece of Olmec art so far discovered
- D. the face of the creature represented on the Kunz axe resembles a human infant

11..In paragraph 5, the author uses the example of the paw-wing motif in order to illustrate

- A. how Olmec images may be related to a few Olmec deities
- B. why jaguar imagery is the most important of Olmec animal imagery
- C. the importance of the paw-wing motif in cultures before the Olmec
- D. how images of animals from beyond the rainforest were represented in Olmec art

12..According to paragraph 5, which of the following is true about the Olmec symbolic code

- A. It included only animals that have paws or wings.
- B. It did not change significantly from one century to the next.
- C. It was not strongly connected to Olmec religion.
- D. It developed gradually over a thousand-year period.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

But these opinions lacked proof.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. Between 1500 B.C. and 100 B.C., the Olmecs developed complex ceremonial centers, an extensive agricultural and trading economy, and a highly distinctive art.
- B. Early in their history, the Olmec left the fertile gulf plain and moved to Teopantecuanitlan.
- C. The frequent reappearance of Olmec images in the art of the Maya and Aztecs suggests that the Olmecs gave rise to these later civilizations.
- D. The Kunz axe, once linked with Maya culture after being found at an ancient Maya site, was eventually attributed to Olmec artists.
- E. Many cultural innovations are now attributed to the Olmecs that were once attributed to other Mesoamerican cultures, including the calendar used by the Maya.
- F. Olmec art involved a complex symbolic code, including various animal images and the howling baby seen on the Kunz axe and elsewhere, that was used consistently for a thousand years.

1. On the basis of available evidence, there existed in ancient state-level societies a variety of urban types. These have been classified under a number of different headings, ranging from city-states to territorial- or village-states. Mesopotamia and Egypt, for example, traditionally represent the two opposing extremes along a spectrum of possible settlement distributions and types.
2. **Mesopotamian city-state systems were made up of densely populated urban areas that shared a common language, status symbols, and economic systems, but their elites tended to compete with each other, often militarily, to control territory, trade routes, and other resources.** Each city-state controlled a relatively small territory, often only a few hundred square kilometers, and had its own capital city, which in many cases was enclosed by a wall. In addition to its capital, a city-state might govern a number of smaller centers, as well as numerous farming villages and hamlets. Ancient Sumer is a classic example of such a system.
3. In ancient Mesopotamia, urban centers tended to be relatively large, with populations ranging from less than 1,000 to more than 100,000 inhabitants, depending on the ability of a particular city-state to control and collect payments from its neighbors. Often, a considerable number of farmers lived in these centers to secure greater protection for themselves and their possessions. It is estimated that in southern Mesopotamia (circa 2900 'C2350 BC) more than 80 percent of the total population lived in cities.
4. These cities also supported craft production, which sought to satisfy the demands of the urban elite and society as a whole. The development of craft specialization and commercial exchanges between town and countryside as well as between neighboring urban centers encouraged the growth of public markets. Although the evidence for actual marketplaces is less than clear for southern Mesopotamia, the remnants of shop-lined streets indicate vigorous commercial activity involving large numbers of people. This activity in turn promoted competition among city-states to obtain supplies of exotic raw materials. As a result of widespread access to goods produced by full-time specialists and the development of more intensive agriculture close to urban centers, Mesopotamian city-states were able to support numerous nonfood producers, possibly as high a proportion as 20 percent of the total population.
5. In contrast to Mesopotamia, ancient Egypt's population has traditionally been perceived as more evenly dispersed across the landscape, a characteristic of village-states. Topography and the formation of the early state were the major factors contributing to this dispersal. Unlike Mesopotamia, Egypt had relatively secure and defined borders, allowing a single state to dominate the area. Additionally, the villages and towns of Egypt, all of which were situated near the Nile on the river's narrow flood plain, had approximately equal access to the river and did not have to compete among themselves for water as their contemporaries in Mesopotamia were forced to do. As the main highway through Egypt, the Nile offered innumerable harbors for shipping and trading, so there was no strong locational advantage to be gained in one area as opposed to another; hence the Egyptian population generally remained dispersed throughout the valley and delta in low densities. Trade specialists apparently were evenly spread throughout Egypt, supported by both independent workshops in small towns and royal patronage in the territorial capitals. In contrast to the defensive walls

of Mesopotamian city-states, the walls of Egyptian towns primarily defined and delineated sections of the town (for example, a temple precinct from a residential area).

6. Egypt, however, was not without urban centers. At points where goods entered the Nile valley via maritime routes or overland routes from the Red Sea via wadis (stream beds that remain dry except during the rainy season), the right circumstances existed for the growth of larger cities. Egyptian cities and towns shared certain characteristics with other contemporary societies but also displayed unique traits influenced by the culture and environment of the Nile valley. Thus, the geopolitical system that evolved in ancient Egypt was different from that of Mesopotamia; Egypt developed a village or territorial state characterized by dispersed settlements of varying size, a form of urbanism that gave Egypt its distinctive identity.

1..According to paragraph 1, which of the following best describes how ancient societies were organized

- A. Ancient societies were classified as either city-states or village-states.
- B. Most ancient societies started out as city-states and then became territorial- or village-states.
- C. With the exception of Mesopotamia and Egypt, ancient societies were generally not urbanized.
- D. Ancient societies likely followed a number of different urban settlement patterns.

2..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Although composed of very similar societies, Mesopotamian city-states were also characterized by conflicts among elites over trade, territory, and resources.
- B. City-states that shared a common language, status symbols, and economic systems were more likely to compete militarily than were other city-states.
- C. Most military conflicts among Mesopotamian city-states were about economic issues, such as territory or trade routes, but some were over the status symbols of elites.
- D. Despite the military control of elites, Mesopotamian city-states tended to compete with each other.

3..The author mentions Ancient Sumer as an example of

- A. an unusual settlement that differed from the classic city-state
- B. a small farming village under the control of a large city
- C. a city-state consisting of a capital and outlying settlements
- D. a city-state that was particularly small in size for Mesopotamia

4..According to paragraph 3, what determined the size of an urban center in ancient Mesopotamia

- A. The number of people defending it

- B. The amount of available space between the city and its nearest neighbor
- C. The extent of its political and economic enforcement power over its neighbors
- D. The number of farmers and the amount of food they produced

5..The word remnants in the passage is closest in meaning to

- A. remains
- B. locations
- C. number
- D. existence

6..According to paragraph 4, which of the following is NOT true of commercial activity in ancient Mesopotamia

- A. Perhaps 20 percent of the population was involved in commercial activity rather than food production.
- B. Commercial exchanges took place not only between urban and rural areas, but also between cities.
- C. Although most urban centers had marketplaces, the largest ones were located in southern Mesopotamia.
- D. Goods were plentiful and widely available to inhabitants of Mesopotamian cities.

7..The word exotic in the passage is closest in meaning to

- A. high-quality
- B. popular
- C. diverse
- D. foreign

8..The word dominate in the passage is closest in meaning to

- A. enclose
- B. control
- C. protect
- D. acquire

9..In paragraph 5, why does the author provide the information that all Egyptian villages and towns were located near the Nile and had equal access to the river

- A. To explain why flooding was a frequent problem for the Egyptian people
- B. To identify a contributing cause of the dispersal of Egypt's population
- C. To support the claim that Egypt had well-defined borders
- D. To demonstrate the similarity between Egyptian and Mesopotamian settlement patterns

10..According to paragraph 5, the primary purpose of city walls in ancient Egypt was to

- A. distinguish territorial capitals from other urban areas
- B. prevent the city's population from becoming too spread out
- C. protect the city from outside attack
- D. separate parts of the city designated for different uses

11..Paragraph 6 suggests that Egypt's urban centers were located near stream beds called wadis because these areas

- A. had the most fertile soil
- B. provided opportunities for trade
- C. had increased their water supplies
- D. could easily be protected from invaders

12..The phrase contemporary societies in the passage means societies that

- A. existed at the same time
- B. were located in the same region
- C. were the same size
- D. had the same resources

13..Look at the four squares that indicate where the following sentence could be added to the passage.

For example, Egypt's capital, Memphis, was located at a strategic point near the mouth of the Nile and grew to be one of the largest settlements of its time.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Mesopotamia was characterized by large, densely-populated urban centers, while the population of Egypt was more evenly distributed across the Nile valley.
- B. Unlike Mesopotamian city-states, which were culturally and economically distinct, Egyptian villages and towns shared a common language and economy.
- C. While defense was crucial in Mesopotamian cities due to competition for territory, trade routes, and raw materials, it was less important in Egypt.
- D. Once they realized that craft production was more profitable than crop production, many Mesopotamians moved from rural to urban areas.
- E. Differences in settlement patterns help to explain why the Egyptian central government survived and the Mesopotamian city-states did not.
- F. Trade specialists were evenly spread throughout Egypt, while Mesopotamia's vibrant commercial trade was concentrated in its large urban centers.

Passage

123

2015-06-14-3

Extinctions at the End of the Cretaceous

1. It has long been recognized that the dinosaurs disappeared from the fossil record at the end of the Cretaceous period (65 million years ago), and as more knowledge has been gained, we have learned that many other organisms disappeared at about the same time. The microscopic plankton (free-floating plants and animals) with calcareous shells suffered massively. The foundation of the major marine food chain that led from the minute plankton to shelled animals to large marine reptiles had collapsed.
2. On land it was not only the large animals that became extinct. The mammals, most of which were small, lost some 35 percent of their species worldwide. Plants were also affected. For example, in North America 79 percent did not survive, and it has been noted that the survivors were often deciduous—they could lose their leaves and shut down while others could survive as seeds. As in the sea, it seems that on the land one key food chain collapsed: the one with leaves as its basic raw material. These leaves were the food of some of the mammals and of the herbivorous dinosaurs, which in turn were fed on by the carnivorous dinosaurs. Furthermore, it is most likely that these large dinosaurs had slow rates of reproduction, which always increases the risk of extinction. Crocodiles, tortoises, birds, and insects seem to have been little affected. The two first named are known to be able to survive for long periods without food, and both can be scavengers (feed on dead material). Indeed, with the deaths of so many other animals and with much dead plant material, the food chain based on detritus would have been well-supplied. Many insects feed on dead material; furthermore, most have at least one resting stage in which they are very resistant to damage. In unfavorable conditions some may take a long time to develop: there is a record of a beetle larva living in dead wood for over 40 years before becoming an adult. Some birds were scavengers, but the survival of many lineages is a puzzle.
3. What happened in the biological story just after these extinctions? What is found in and just above the boundary layer between the deposits of the Cretaceous and those of the Tertiary (65 ± 2.6 million years ago), termed the K/T boundary. For a very short period the dominant microorganisms in marine deposits were usually diatoms and dinoflagellates (both single-celled types of plankton). The important feature for the survival of both these groups was the ability to form protective cysts (sacs around organisms) that rested on the sea floor. Above these, in the later deposits, are the remains of other minute plankton, but the types are quite different from those of the Late Cretaceous. In terrestrial deposits a sudden and dramatic increase in fern plant spores marks the boundary in many parts of the world; ferns are early colonizers of barren landscapes. **The fern spike (sudden increase), as it is termed, has been found also in some marine deposits (such as the abundance of fern spores blown around the world), and it occurs in exactly the same layer of deposit where the plankton disappear.** We can conclude that the major marine and terrestrial events occurred simultaneously.
4. Many theories have been put forward for the extinction of the dinosaurs, but most of them can be dismissed. Since 1980 there have been more focused, but still controversy-ridden, investigations. In that year Louis and Walter Alvarez and colleagues from the University of California published their research on the amounts of various metals in the boundary between Cretaceous and Tertiary rocks (K/T boundary) in Italy, Denmark, and New Zealand. They had found, accidentally, that a rare metal, iridium, suddenly became very abundant exactly at the

boundary and then slowly fell away. This phenomenon, known as the iridium spike, has now been identified in K/T boundary deposits in over a hundred other sites in the world. Iridium occurs in meteorites and volcanic material, but in the latter case it is accompanied by elevated levels of nickel and chromium. These other metals are not especially abundant at the K/T boundary. The Alvarezes concluded that the iridium spike was due to a large asteroid that struck Earth 65 million years ago.

1..The word recognized in the passage is closest in meaning to

- A. suspected
- B. argued
- C. assumed
- D. accepted

2..According to paragraph 1, which of the following was true of small planktonic organisms at the end of the Cretaceous

- A. They lost their main sources of food.
- B. They developed calcareous shells.
- C. They decreased greatly in number as did many other types of organisms.
- D. They replaced other minute organisms as a food source in the major marine food chain.

3..Which of the following statements is NOT supported by the information provided in paragraph 2 about extinctions at the end of the Cretaceous

- A. About 35 percent of mammal species were lost.
- B. 79 percent of North American plants disappeared.
- C. Most birds, tortoises, and crocodiles escaped extinction.
- D. Deciduous trees were especially likely to go extinct.

4..According to paragraph 2, which of the following factors probably contributed to the extinction of the dinosaurs

- A. The length of time it took dinosaurs to reproduce
- B. Large quantities of dead material disturbing their habitats
- C. Increased competition for food from scavengers
- D. An increase in carnivore populations

5..In paragraph 2, why does the author provide the information that there is a record of a beetle larva living in dead wood for over 40 years before becoming an adult

- A. To help explain why insects were less likely to go extinct than other species
- B. To show that not all species that relied on trees disappeared during the late Cretaceous

- C. To suggest that insects that lived long ago had much longer life spans than those living today
- D. To support the claim that conditions at the end of the Cretaceous were highly unfavorable

6..The word simultaneously in the passage is closest in meaning to

- A. rapidly
- B. repeatedly
- C. at the same time
- D. for different reasons

7..According to paragraph 3, which of the following is true of plankton after the extinctions at the end of the Cretaceous

- A. Diatoms and dinoflagellates suddenly became very rare.
- B. Single-celled types of plankton were replaced by more complicated microorganisms.
- C. The plankton found in later deposits are closely related to single-celled Late Cretaceous microorganisms.
- D. The only types of Late Cretaceous plankton to survive extinction were those protected by cysts.

8..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. The fern spike is the term given to this sudden increase in the abundance of fern spores blown into the oceans around the world.
- B. The sudden increase in fern spores occurred in exactly the same parts of the world where the plankton was disappearing.
- C. Fern spores have even been found in some marine deposits from exactly the same layer as that showing the disappearance of the plankton.
- D. Most marine deposits from around the world contain either plankton or a lot of fern spores but not both together in the same layer.

9..The word dismissed in the passage is closest in meaning to

- A. further explored
- B. rejected
- C. completely revised
- D. replaced

10..The word elevated in the passage is closest in meaning to

- A. high
- B. varying

- C. unusual
- D. adequate

11..According to paragraph 4, what evidence is there that an asteroid hit Earth

- A. The fact that the metals nickel and chromium were found at sites around the world
- B. The presence in K/T boundary deposits of large amounts of a metal that is found in meteorites
- C. The fact that iridium amounts decreased at the same time that the Cretaceous ended and the Tertiary began
- D. The differences in the types of metals found in deposits in Italy, Denmark, and New Zealand

12..Paragraph 4 suggests which of the following about the Alvarezes' theory

- A. Their theory was accepted until 1980, when more focused investigations showed it to be incorrect.
- B. It rules out the possibility that the iridium spike was the result of volcanic activity.
- C. Before it can be accepted, more evidence must be gathered from locations outside Europe and New Zealand.
- D. Experts believe the research done by the Alvarezes was too broad.

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

This change in plankton found in marine deposits is what marks the boundary between the Cretaceous and the Tertiary.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Aside from the dinosaurs, most of the organisms affected were very small, such as single-celled plankton and insects.
- B. Herbivores and carnivores were both affected, but the hardest-hit group was the scavengers, including many birds and some mammals.
- C. Two major food chains were eliminated, one in the oceans based on plankton, and one on land based on leaves.
- D. Examinations of marine and terrestrial deposits show clearly that ocean extinctions occurred well before those on land.
- E. In geological samples from around the world, the K/T boundary is marked by a sudden change in plankton and by a spike in fern spores.
- F. Sudden, high levels of iridium found at the K/T boundary suggest that a meteorite might have caused the extinctions.

1. Termites are insects that collect vegetation, chew it up, and leave the chemical breakdown to other organisms. There are two strategies. The most primitive termites swallow the chewed vegetation and pass it to a fermentation chamber in their bodies. There, anaerobic bacteria break down the cellulose, an organic compound that forms about 33 percent of all plant matter. The termites are nourished by the ever-growing population of microorganisms in their guts that turn the grass, leaves, and twigs the insects ingest into glucose. Cattle do much the same thing: they allow bacteria to ferment the cellulose in an airtight rumen (digestive chamber), and then digest the bacteria.
2. Termite evolution has several obvious trends, from primitive species, which live in small hidden colonies, to groups millions strong, the builders of enormous mounds that allow for heat and gas exchange. The less advanced groups digest microorganisms, which do the real work of breaking food down. The culture (colony) of cellulose digesters is passed along through a special exchange. Young termites feed on a special liquid secretion provided by adults, rich in the group's digestive heritage. When reproductive termites those destined to produce offspring leave the nest, they carry in their stomachs the microorganisms essential for the digestive success of their offspring. Treat a colony of these termites with an antibiotic solution, and they will slowly starve to death.
3. More advanced species have a different feeding strategy. The energy source is still cellulose, but it is digested outside the termite's body. Not having to carry around large chambers of slowly fermenting cellulose solution makes these species more nimble and efficient. Foragers bring twigs and leaves back to special areas and chew them. They then transplant bits of fungus growing on other pieces of nearby vegetation onto the gnawed edges, where the fungi break down the cellulose. Fungi is the only kingdom of organisms able to digest cellulose in air, though they need warmth and humidity to do the job efficiently. This is just what the termites provide. Moreover, these social insects carefully tend the fungus-covered vegetation by treating it with antibiotics they secrete to keep bacterial growth to a minimum. When it is time for the fungus to reproduce, pieces are carried into the open to complete the life cycle. Some species of fungi are found only in termite mounds of a particular species; without their caretakers, these fungi would die. Needless to say, the termites eat the fungi; neither can live without the other. Reproductive termites even carry a chunk of fungi when they leave on mating flights.
4. The evolutionary trend in termites is to forsake excavated nests in soil or wood, like those of most ants, for carton nests constructed inside excavations or on trees. (When referring to termites, carton means, broadly, nesting material consisting of a mix of adhesive saliva or feces with earth or pulp, and even sand, to create cells, floors, walls, graceful arches, tiered roofs, chimney stacks, and buttressed towers up to twenty feet high.) Primitive termites do not store food; they live from hand to mouth, inside a rotting tree, for instance. **Advanced termites have special carton areas for food they hold in reserve; these supplies consist of nonperishable material such as grass clippings, analogous to the hay and straw fed to cattle in the winter, and are kept in a dry carton loft. Primitive species need wet cellulose, such as damp wood; more advanced species can also process dry material.**
5. To expand their niche in this way, dry-diet termites require a source of water. In arid habitats, they excavate vertical tunnels down to the water table, as much as 150 feet below, which fan

out at the base to increase the area of contact and thus maximize the rate of subsurface water accumulation. Finally, less advanced termites remain their entire lives in tunnels and cells excavated in or near wood. More complex species, on the other hand, search for food away from a central nest. To ensure that they can work in safety, they burrow shallowly through the earth or build mud-covered tunnels on the surface of the ground or trees and around the food they wish to harvest.

1..According to paragraph 1, in what way are the most primitive termites like cattle

- A. They have two ways of breaking down the cellulose in vegetation.
- B. They get no nutritional benefit from eating vegetation.
- C. Their digestive chambers are not airtight.
- D. They rely on bacteria to process the vegetation they ingest.

2..The phrase special exchange in the passage refers to

- A. the transfer of cellulose digesters from one colony of termites to another
- B. the development of small colonies into much larger ones
- C. the transfer of liquid secretions from adult termites to young termites
- D. the departure of certain reproductive termites and the arrival of others

3..Why does the author include the statement Treat a colony of these termites with an antibiotic solution, and they will slowly starve to death

- A. To suggest one reason why termites sometimes leave a colony
- B. To emphasize how dependent the termites are on bacteria
- C. To account for the elimination of certain termite lineages
- D. To help explain why termites protect essential microorganisms by carrying them in their stomachs

4..The phrase Needless to say in the passage is closest in meaning to

- A. On the other hand
- B. Most importantly
- C. Obviously
- D. Besides

5..According to paragraph 3, each of the following is true about the fungi found in termite nests EXCEPT:

- A. The fungi can digest cellulose in the presence of air.
- B. The fungi secrete antibiotics to keep bacterial growth to a minimum.
- C. The fungi reproduce in the open with the help of termites.
- D. The fungi need a warm, humid environment to function effectively.

6..The word adhesive in the passage is closest in meaning to

- A. sticky
- B. wet
- C. ordinary
- D. released

7..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. In winter, when fresh grass is not available, termites must rely on nonperishable food sources such as hay and straw.
- B. Advanced termites store different foods in different chambers, reserving the least perishable material for winter.
- C. Advanced termites have special chambers where they store nonperishable food supplies.
- D. Advanced termites' supplies of grass clippings are like hay and straw fed to cattle in winter.

8..Paragraph 4 suggests that for primitive termites, a major advantage of nesting inside a rotting tree is that a rotting tree

- A. is a very easy site to excavate
- B. provides a ready source of wet cellulose
- C. provides a lot of raw material for making carton
- D. helps preserve stored food

9..The word ensure in the passage is closest in meaning to

- A. show
- B. prove
- C. suggest
- D. guarantee

10..According to paragraph 5, which of the following is true about the vertical tunnels constructed by dry-diet termites

- A. They allow water from the surface to accumulate underground.
- B. They are generally wider at the top than at the bottom.
- C. They provide extra space so that dry-diet termite populations can expand.
- D. They are constructed to provide direct access to water.

11..According to paragraph 5, why do termites build mud-covered tunnels on the surface of the ground

- A. To make harvesting food less dangerous
- B. To create a safe place to live
- C. To avoid contact with surface water
- D. To avoid competition with species that create deep underground tunnels

12..Look at the four squares that indicate where the following sentence could be added to the passage.

This remarkable adaptation, however, is not unique to termites.

Where would the sentence best fit Click on a square to add the sentence to the passage.

13..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

Grow fungi on chewed vegetation

Do not chew vegetation

Must remain close to or inside wood

Are able to digest cellulose in air

Can consume dry vegetation

Take care of fungi that break down cellulose

Have a body chamber specifically for breaking down cellulose

Passage 125 2015-07-05-2 Elements of Life

1. The creation of life requires a set of chemical elements for making the components of cells. Life on Earth uses about 25 of the 92 naturally occurring chemical elements, although just 4 of these elements—oxygen, carbon, hydrogen, and nitrogen—make up about 96 percent of the mass of living organisms. Thus, a first requirement for life might be the presence of most or all of the elements used by life.
2. Interestingly, this requirement can probably be met by almost any world. Scientists have determined that all chemical elements in the universe besides hydrogen and helium (and a trace amount of lithium) were produced by stars. These are known as heavy elements because they are heavier than hydrogen and helium. Although all of these heavy elements are quite rare compared to hydrogen and helium, they are found just about everywhere.
3. Heavy elements are continually being manufactured by stars and released into space by stellar deaths, so their amount compared to hydrogen and helium gradually rises with time. **Heavy elements make up about 2 percent of the chemical content (by mass) of our solar system; the other 98 percent is hydrogen and helium. In some very old star systems, which formed before many heavy elements were produced, the heavy-element share may be less than 0.1 percent.** Nevertheless, every star system studied has at least some amount of all the elements used by life. Moreover, when planetesimals—small, solid objects formed in the early solar system that may accumulate to become planets—condense within a forming star system, they are inevitably made from heavy elements because the more common hydrogen and helium remain gaseous. Thus, planetesimals everywhere should contain the elements needed for life, which means that objects built from planetesimals—planets, moons, asteroids, and comets—also contain these elements. The nature of solar-system formation explains why Earth contains all the elements needed for life, and it is why we expect these elements to be present on other worlds throughout our solar system, galaxy, and universe.
4. Note that this argument does not change, even if we allow for life very different from life on Earth. Life on Earth is carbon based, and most biologists believe that life elsewhere is likely to be carbon based as well. However, we cannot absolutely rule out the possibility of life with another chemical basis, such as silicon or nitrogen. The set of elements (or their relative proportions) used by life based on some other element might be somewhat different from that used by carbon-based life on Earth. But the elements are still products of stars and would still be present in planetesimals everywhere. No matter what kinds of life we are looking for, we are likely to find the necessary elements on almost every planet, moon, asteroid, and comet in the universe.
5. A somewhat stricter requirement is the presence of these elements in molecules that can be used as ready-made building blocks for life, just as early Earth probably had an organic soup of amino acids and other complex molecules. Earth's organic molecules likely came from some combination of three sources: chemical reactions in the atmosphere, chemical reactions near deep-sea vents in the oceans, and molecules carried to Earth by asteroids and comets. The first two sources can occur only on worlds with atmospheres or oceans, respectively. But the third source should have brought similar molecules to nearly all worlds in our solar system.
6. Studies of meteorites and comets suggest that organic molecules are widespread among both

asteroids and comets. Because each body in the solar system was repeatedly struck by asteroids and comets during the period known as the heavy bombardment (about 4 billion years ago), each body should have received at least some organic molecules. However, these molecules tend to be destroyed by solar radiation on surfaces unprotected by atmospheres. Moreover, while these molecules might stay intact beneath the surface (as they evidently do on asteroids and comets), they probably cannot react with each other unless some kind of liquid or gas is available to move them about. Thus, if we limit our search to worlds on which organic molecules are likely to be involved in chemical reactions, we can probably rule out any world that lacks both an atmosphere and a surface or subsurface liquid medium, such as water.

1..The word components in the passage is closest in meaning to

- A. parts
- B. bodies
- C. combinations
- D. characteristics

2..According to paragraphs 1 and 2, living cells contain which of the following

- A. All chemical elements in the universe except lithium
- B. About 25 different elements
- C. About 96 percent of all known elements
- D. Ninety-two naturally occurring elements

3..Why does the author provide the information that "Heavy elements are continually being manufactured by stars and released into space by stellar deaths"

- A. To explain how it is that the elements required for life can be found everywhere
- B. To provide evidence that our solar system is relatively young
- C. To argue that some solar systems are more likely to support life than others
- D. To explain why heavy elements have greater mass than hydrogen and helium

4..Paragraph 3 suggests that which of the following may be a difference between very old star systems and newer star systems

- A. Older star systems are likely to have fewer planets, moons, asteroids, and comets than newer star systems.
- B. Newer star systems probably contain more hydrogen and helium than older star systems.
- C. Newer star systems probably contain more heavy elements than older star systems.
- D. The process of solar-system formation may have been fundamentally different in older star systems than in newer star systems.

5..The word Nevertheless in the passage is closest in meaning to

- A. additionally
- B. however
- C. in particular
- D. on the contrary

6..The word inevitably in the passage is closest in meaning to

- A. typically
- B. unsurprisingly
- C. necessarily
- D. naturally

7..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Planetesimals may remain in star systems when hydrogen and helium combine with less common heavier elements.
- B. Planetesimals are composed of heavy elements because hydrogen and helium stay in the form of gases.
- C. Planetesimals are small, solid objects that condense within a forming star system and may become planets.
- D. When planetesimals accumulate to form planets, they inevitably contain gaseous as well as heavy elements.

8..According to paragraph 4, which of the following is true about possible life on other planets

- A. It cannot be based on silicon or nitrogen.
- B. It could not survive on Earth.
- C. It probably would not be made of elements produced by stars.
- D. It is likely to have carbon as its chemical basis.

9..According to paragraph 5, all of the following are true of the organic molecules on early Earth EXCEPT:

- A. Some of them were probably brought to Earth by asteroids or comets.
- B. Some of them probably formed in the atmosphere and oceans.
- C. They were probably significantly different from the organic molecules present on other planets in the solar system.
- D. They included complex molecules.

10..The phrase widespread among is closest in meaning to

- A. frequently present in

- B. widely separated in
- C. a significant part of
- D. found throughout the bodies of

11..The word intact in the passage is closest in meaning to

- A. buried
- B. whole
- C. confined
- D. active

12..According to paragraph 6, why is life unlikely to be found on any planet that lacks both an atmosphere and a surface or subsurface liquid medium

- A. Organic molecules must be protected from solar radiation by a surface layer of liquid.
- B. Planets that lack both of these features are probably too small to have been hit by many asteroids or comets carrying organic matter.
- C. Organic molecules need a liquid or gaseous environment to bring them together so they can interact.
- D. An atmosphere is needed to protect organic molecules from being destroyed by asteroids and comets.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

To answer the question Could life exist on other planets we must first look at the necessary preconditions for life.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Somewhere between 4 and 25 naturally occurring elements are necessary for life, depending on the complexity of the organism.
- B. The most common elements used by lifeoxygen, carbon, hydrogen, and nitrogenare also some of the most widely distributed elements in the universe.
- C. Life is most likely to be found in the oldest star systems, where heavy elements have been continually produced since those systems were formed.
- D. Planets, moons, asteroids, and comets are all composed of heavy elements, which means they contain the basis for any life form, carbon based or otherwise.
- E. Life is most likely to exist on those bodies that were not heavily bombarded with asteroids and comets during the formation of the solar system.
- F. Organic molecules are widely available, but chemical reactions among these molecules probably require either an atmosphere or a liquid medium.

1. Throughout its history, agriculture was the economic mainstay of the Ottoman Empire, which dominated North Africa, the Middle East, Turkey, and southeastern Europe for over 600 years until the early twentieth century. Most cultivators possessed small landholdings, engaging in a host of tasks, with their crops and animal products mainly dedicated to self-consumption. But enormous changes over time prevailed in the agrarian sector. Beginning in the late eighteenth century, agriculture became more and more commercialized, with increasing amounts of produce going to sale to domestic and international consumers.
2. At least three major engines increased this agricultural production devoted to the market, the first being rising demand, both international and domestic. Abroad, especially after 1840, the living standards and buying power of many Europeans improved substantially, permitting them to buy a wider choice and quantity of goods. Rising domestic markets within the empire were also important, thanks to increased urbanization as well as mounting personal consumption. In the late nineteenth century, newly opened railroad districts brought a flow of domestic wheat and other cereals to major coastal cities. Railroads also attracted market gardeners who now could grow and ship fruits and vegetables to the expanding and newly accessible markets of these cities.
3. The second engine driving agricultural output concerns cultivators' increasing payment of their taxes in cash rather than in kind (that is, in agricultural or other products). **Some historians have asserted that the increasing commitment to market agriculture was a product both of a mounting per capita tax burden and the state's growing preference for tax payments in cash rather than in kind.** In this view, such government decisions forced cultivators to grow crops for sale in order to pay their taxes. Thus, state policy is seen as the most important factor influencing the cultivators' shift from subsistence farming to market agriculture.
4. However, cultivators' rising involvement in the market was not simply a reactive response to the state's demands for cash taxes; other factors were at work. There was a third engine driving increased agricultural production: cultivators' own desires for consumer goods. Among Ottoman consumers, increasingly frequent changes in taste, along with the rising availability of cheap imported goods, stimulated a rising consumption of goods. This pattern of rising consumption began in the eighteenth century, as seen by the urban phenomenon of the Tulip Period (1718-1730) a time of urban revival and orientation toward the West and accelerated subsequently. Wanting more consumer goods, cultivators needed more cash. Thus, rural families worked harder than they had previously, not merely because of cash taxes. In such circumstances, leisure time diminished, cash incomes rose, and the flow of consumer goods into the countryside accelerated.
5. Increases in agricultural production both promoted and accompanied a vast expansion of the area of land under cultivation. At the beginning of the eighteenth century and indeed until the end of the empire, there remained vast stretches of uncultivated, sometimes nearly empty, land on every side. These spaces began to fill in, a process finally completed only in the 1950s in most areas of the former empire. Many factors were involved. In many cases, families increased the amount of time at work, bringing into cultivation uncultivated land already under their control. They also engaged in sharecropping, agreeing to work another's land and paying that person a share of the output. Often such acreage had been pastureland

for animals but now was given over to crop production. The extraordinarily fertile lands of Moldavia and Wallachia (modern Romania), for example, had been among the least populated lands of the Ottoman empire in the eighteenth century, but now saw large amounts of land brought under the plow. Significant concentrations of commercial agriculture first formed in areas easily accessible by water, such as the Danube River basin. During the nineteenth century, expansion in such areas continued, and interior regions joined the list as well. There were also some increases in productivity. Irrigation projects, one form of intensive agriculture, developed in some areas, and the use of modern agricultural tools increased. But more intensive exploitation of existing resources remained comparatively unusual, and most increases in production derived from placing additional land under cultivation.

1..According to paragraph 1, in which of the following ways did agricultural production in the Ottoman empire begin to change at end of the eighteenth century

- A. Agricultural products no longer contributed as much to the Ottoman economy.
- B. Agricultural workers left their farms to work in commercial industries in the cities.
- C. Farmers with small landholdings began to focus on a single task rather than on many tasks.
- D. Farm products were sold commercially instead of being kept for personal use.

2..The word substantially in the passage is closest in meaning to

- A. without interruption
- B. significantly
- C. rapidly
- D. unexpectedly

3..The word accessible in the passage is closest in meaning to

- A. profitable
- B. competitive
- C. created
- D. reachable

4..According to paragraph 2, all of the following contributed to a rising demand for the agricultural products of the Ottoman empire during the nineteenth century EXCEPT

- A. the sale of domestic wheat in place of other cereals in coastal cities
- B. the development of railroad systems leading to coastal cities
- C. the rise in living standards and buying power among Europeans
- D. the emergence of new domestic markets in the Ottoman Empire

5..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Some historians have asserted that the increasing commitment to market agriculture allowed the government to raise taxes and have them paid in cash.
- B. The government raised taxes and required they be paid in cash, so farmers had to use the cash they obtained from selling their farm products in the market to pay their taxes
- C. Some historians have asserted that the increasing commitment to market agriculture was a product of the state's growing preference for tax payments in cash rather than kind.
- D. According to some historians, the growth of market agriculture was the result of mounting taxes which the government wanted individuals to pay in cash.

6..Which of the following best represents the explanation for the change in agricultural production mentioned in paragraph 3

- A. The state allowed cultivators to pay their taxes in crops as well as in cash.
- B. Cultivators needed crops they could sell to pay taxes in cash.
- C. State policies began favoring farmers who shifted to market agriculture by lowering those farmers' taxes.
- D. Cultivators had more money to invest in crops because of a lower tax burden.

7..The word stimulated in the passage is closest in meaning to

- A. encouraged
- B. followed
- C. suggested
- D. demonstrated

8..According to paragraph 4, the increasing desire of Ottoman cultivators to purchase consumer goods led to all of the following changes EXCEPT:

- A. Cultivators worked harder and for longer hours.
- B. Cultivators had less cash available to use for tax payments.
- C. Cultivators succeeded in increasing the amount of cash income they earned.
- D. More consumer goods became available in rural areas.

9..The word engaged in the passage is closest in meaning to

- A. succeeded
- B. believed
- C. invested
- D. participated

10..The word exploitation in the passage is closest in meaning to

- A. improvement
- B. use
- C. management

D. investigation

11..According to paragraph 5, which of the following was true of the process of bringing new land under cultivation

- A. It began in interior areas and quickly spread to areas near water.
- B. It was completed near the end of the eighteenth century.
- C. It occurred slowly because most uncultivated land was not very fertile.
- D. It often occurred as a result of farming families working longer hours.

12..What can be inferred from paragraph 5 about agricultural production during the nineteenth century

- A. Irrigation and the use of modern tools contributed little to increased production in comparison with other factors.
- B. Interior regions increased their agricultural production much more than regions near river basins did.
- C. Agricultural production was aided by using less-productive land for animal pasture instead of for growing crops.
- D. Agricultural production increased in some areas but decreased in others during the nineteenth century.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

These increases resulted from using technology to improve yields on existing farmland, a system known as **intensive agriculture**.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Until the late eighteenth century, farmers were limited to selling their crops and animal products for cash in nearby towns.
- B. Starting in the late eighteenth century, farmers increasingly produced crops for sale in domestic and international markets.
- C. Many farmers sold their products in order to be able to pay their taxes in cash and buy consumer goods for themselves.
- D. Railroads changed the way farming was done, since they brought modern tools from towns to the agricultural areas.
- E. Increased productivity resulted more from the cultivation of additional land than from the use of new tools and intensive agricultural processes.
- F. Farming was concentrated in river basins throughout the nineteenth century, while interior areas were devoted to pastureland for animals.

1. A person's performance on tasks can be enhanced or impaired by the mere presence of others, and a person's behavior as part of a group can be quite different from the person's behavior when acting alone.
2. In certain cases, individual performance can be either helped or hindered by the physical presence of others. The term social facilitation refers to any effect on performance, whether positive or negative, that can be attributed to the presence of others. Research on this phenomenon has focused on two types of effects: audience effects (the impact of passive spectators on performance) and coaction effects (the effect on performance caused by the presence of other people engaged in the same task).
3. In one of the first studies in social psychology, psychologist Norman Triplett looked at coaction effects. He had observed in official bicycle records that bicycle racers pedaled faster when they were pedaling against other racers than when they were racing against the clock. Was this pattern of performance peculiar to competitive bicycling Or was it part of a more general phenomenon whereby people work faster and harder in the presence of others than when performing alone Triplett set up a study in which he told 40 children to wind fishing reels as quickly as possible under two conditions: alone or in the presence of other children performing the same task. He found that the children worked faster when other reel turners were present than when they performed alone.
4. Later studies on social facilitation found just the opposite effect that the presence of others, whether co-acting or just watching, could hurt or diminish individual performance. Social psychologist Robert Zajonc proposed an explanation for these seemingly contradictory effects. He reasoned that we become aroused by the presence of others and that arousal facilitates the dominant response the one most natural to us. On simple tasks and on tasks at which we are skilled, the dominant response is to perform effectively. However, on tasks that are difficult or tasks we are just learning, the incorrect response (making a mistake or not performing effectively) is dominant. This reasoning accounts for the repeated findings that, in the presence of others, performance improves on tasks that people do easily but suffers on difficult tasks. Other researchers have suggested that concern over the observers' evaluation is what most affects people's performance, particularly if they expect a negative evaluation.
5. What happens in cooperative tasks when two or more people are working together instead of competing Do they increase their effort or slack off Researcher Bibb Latan "used the term social loafing to refer to people's tendency to exert less effort when working with others on a common task than when they work alone. **Social loafing occurs in situations where no one person's contribution to the group can be identified and individuals are neither praised for a good performance nor blamed for a poor one. In one experiment, Latan " and others asked male students to shout and clap as loudly as possible, first alone and then in groups.** In groups of two, individuals made only 71 percent of the noise they had made alone; in groups of four, each student put forth 51 percent of his solo effort; and with six students, each made only a 40 percent effort.
6. Harkins and Jackson found that social loafing disappeared when participants in a group believed that each person's performance could be monitored and evaluated; indeed, even the idea that the group performance may be evaluated against some standard can be sufficient to eliminate the loafing effect. When a group is relatively small and group evaluation is

important, some members will even expend extra effort if they know that some of their coworkers are unwilling, unreliable, or incompetent to perform well. Moreover, social loafing is unlikely when participants can evaluate their own individual contribution or when they have a personal stake in the outcome. It is also unlikely when participants feel that the task is challenging or when they are working with close friends or teammates. Some 80 experimental studies have been conducted on social loafing in diverse cultures. Based on evidence these studies have produced, social loafing probably occurs in almost all cultures.

1..The word enhanced in the passage is closest in meaning to

- A. interrupted
- B. improved
- C. influenced
- D. hurt

2..What role does paragraph 1 play in the passage

- A. It introduces a common opinion that the rest of the passage challenges on the basis of scientific evidence.
- B. It explains why the passage focuses on actions people take individually rather than on actions people take as part of a group.
- C. It offers a specific example of a general principle discussed in the rest of the passage.
- D. It describes two phenomena, possible explanations of which are considered in the rest of the passage.

3..According to paragraph 2, the term "social facilitation" refers to the phenomenon that a person's performance

- A. is intended to help someone
- B. is influenced by the presence of other people
- C. is evaluated by other people
- D. has an effect on others working on the same task

4..The phrase peculiar to in the passage is closest in meaning to

- A. damaging to
- B. unique to
- C. rare in
- D. new in

5..The study conducted by Norman Triplett described in paragraph 3 supported the hypothesis that

- A. coercion effects are stronger on the performance of children than they are on the performance of adults
- B. coercion effects are limited to situations in which the time taken for a task matters

- C. people perform better when they know that their performance is being measured by someone
- D. people perform better in the presence of others who are doing the same thing they are

6..According to paragraph 4, Robert Zajonc proposed that whether the presence of others hurts or helps a person's performance is determined by

- A. how skilled the observers are in the task that they are observing
- B. how closely the person is being watched
- C. whether or not the person finds the task difficult
- D. whether or not the person likes the people who are watching

7..According to paragraph 4, if other people are present, a person's performance on a task that he or she is just learning would most likely be

- A. somewhat worse than if no one else is there
- B. somewhat better than if no one else is there
- C. completely unaffected by the presence of those other people
- D. dependent on the number of people who are present

8..The word exert in the passage is closest in meaning to

- A. put forth
- B. waste
- C. demand
- D. accept

9..According to paragraph 5, people tend to engage in social loafing when they

- A. prefer to work alone
- B. fear being blamed for a poor performance
- C. believe that their individual performance will not be identified
- D. desire to be loyal to their group

10..Which of the following can be inferred from paragraph 5 about Bibb Latan 's research on social loafing

- A. The less a person likes to work alone, the harder that person is likely to work as a member of a group.
- B. The less a person contributes to a group, the more likely it is that person will be blamed if the group performs poorly.
- C. The more people there are in a group, the more likely they are to compete with each other.
- D. The fewer people there are in a group, the less likely it is that social loafing will occur.

11..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out

essential information.

- A. Social loafing does not occur when participants in the group believe that evaluating individual performance will improve the group's performance.
- B. The loafing effect is eliminated if the members of the group accept the standard by which their performance and that of the group is evaluated.
- C. Social loafing tends to disappear if the members of the group believe that their or the group's performance may be evaluated.
- D. Researchers have found that when group members receive monitoring and evaluation of their performance, their performance improves.

12..According to paragraph 6, which of the following has NOT been shown to decrease social loafing

- A. Participants know that fellow group members are willing, reliable, and competent.
- B. The group's task is seen as a challenge.
- C. Group members know and like each other.
- D. Participants know that their group is being judged on its performance.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

However, it appears to be most common in individualistic Western cultures such as that of the United States.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. When people are present, the performance of individuals generally improves on tasks they already do well but worsens on tasks they generally do poorly.
- B. Studies show that bicycle racers pedal faster when they are competing against other racers, but children wind fishing reels slower when in the presence of others than when alone.
- C. People's performance on a task is more affected by the presence of others when those others are engaged in the same task than when the others are passive spectators.
- D. When people work together on a common task but no one's contribution is measured, there is a tendency for individuals to work less hard than if they were working alone.
- E. Social loafing decreases under certain conditions, such as when the performance of the group or its members is evaluated or when a positive outcome matters to the participants.
- F. While social loafing occurs in almost all groups across cultures, the extent to which it occurs in any particular group depends on the individual personalities of the group's members.

Passage 128 2015-07-11-2 The Identification of the Genetic Material

1. The history of biology is filled with incidents in which research on one specific topic has contributed richly to another, apparently unrelated area. Such a case is the work of Frederick Griffith, an English physician whose attempts to prevent the disease pneumonia led to the identification of the material in cells that contains genetic informationthe information that determines an organism's characteristic structure. In the 1920s, Griffith was studying the bacterium *Streptococcus pneumoniae*, or pneumococcus, one of the organisms that cause pneumonia in humans. He was trying to develop a vaccine against this devastating illness. He was working with two strains of the bacteria pneumococcus. A bacterial strain is a population of cells descended from a single parent cell; strains differ in one or more inherited characteristics. Griffith's strains were designated S and R because, when grown in the laboratory, one produced shiny, smooth (S) colonies or groups of bacteria, and the other produced colonies that look rough (R).
2. When the S strain was injected into mice, the mice became diseased. When the R strain was injected, the mice did not become diseased. Bacteria of the S strain are virulent (able to cause disease) because they are surrounded by a protective jelly-like coating that prevents the mouse's immune defense mechanisms from destroying the bacteria before they can multiply. The R strain lacks this coating.
3. With the hope of developing a vaccine against pneumonia, Griffith injected some mice with heat-killed S pneumococci. These heat-killed bacteria did not produce infection. Griffith assumed the mice would produce antibodies to the bacteria that would allow them to fight the virulent form if they were exposed to it. However, when Griffith inoculated other mice with a mixture of living R bacteria and heat-killed S bacteria, to his astonishment, the mice became ill with pneumonia. When he examined blood from these mice, he found it full of living bacteriamany with characteristics of the virulent S strain. Griffith concluded that, in the presence of the dead S pneumococci, some of the living R pneumococci had been transformed into virulent S-strain organisms.
4. Did this transformation of the bacteria depend on something the mouse did to the bacteria No. It was shown that simply putting living R and heat-killed S bacteria together in a test tube yielded the same transformation. Next it was discovered that a cell-free extract of heat-killed S cells also transformed R cells. (A cell-free extract contains all the contents of cells, but no intact cells.) **This result demonstrated that some substancecalled at the time a chemical transforming principlefrom the extract of S pneumococci could cause a heritable change (a change that could be passed on to future generations) in the affected R cells.** From these observations, some scientists concluded that this transforming material carried heritable information, and thus was the genetic material that scientists had been searching for.
5. The identification of the transforming material was a crucial step in the history of biology, accomplished over a period of several years by Oswald Avery and his colleagues at what is now Rockefeller University. They treated samples of the transforming extract in a variety of ways to destroy different types of substancesproteins, nucleic acids, carbohydrates, and lipidsand tested the treated samples to see if they had retained transforming activity. The answer was always the same: If the DNA (deoxyribonucleic acid) in the extract was destroyed, transforming activity was lost; everything else could be eliminated without

removing the transforming ability of the extract. As a final step, Avery, with Colin MacLeod and Maclyn McCarty, isolated virtually pure DNA from a sample of pneumococcal transforming extract and showed that it caused bacterial transformation.

6. In retrospect, the work of Avery, MacLeod, and McCarty, published in 1944, was a milestone in establishing that DNA is the genetic material. However, at the time, it had little impact on scientists' view about the physical basis of inheritance. The genetic material had to encode all the information needed to specify an organism, and the chemical complexity and diversity of proteins were known to be impressive. So during the first half of the twentieth century, the hereditary material was generally assumed to be a protein. Nucleic acids, by contrast, were known to have only a few components and seemed too simple to carry such complex information.

1..The word apparently in the passage is closest in meaning to

- A. seemingly
- B. surprisingly
- C. relatively
- D. previously

2..According to paragraph 1, Griffith experimented with strains of the pneumococcus bacteria because he wanted to discover which of the following

- A. A strain of bacteria that could be used to develop a vaccine
- B. How bacterial strains developed under laboratory conditions
- C. Why the strains of bacteria differed in appearance
- D. Which bacterial strains were most infectious in humans

3..Why does the author provide the information that The R strain lacks this coating

- A. To provide an example of variations within strains of pneumococcus bacteria
- B. To explain why the R strain is not able to cause disease
- C. To suggest that the R strain has other ways to defend itself from immune defense mechanisms
- D. To explain why mice became diseased when injected with the R strain

4..The word astonishment in the passage is closest in meaning to

- A. alarm
- B. surprise
- C. disappointment
- D. interest

5..According to paragraph 3, why did Griffith conclude from his experiment injecting both R and S strains pneumococci into mice that some of the R strain bacteria transformed into disease-causing S strain pneumococci

- A. All the living bacteria he found in the blood of the injected mice were S strain bacteria.
- B. He already knew from earlier experiments that R strain pneumococci sometimes transform into S strain pneumococci.
- C. He could tell from examining the bacteria under a microscope that some individual pneumococci cells had characteristics of both the S and R strains.
- D. He observed living cells in the mice's blood with S strain characteristics, but the only living cells injected were R strain pneumococci.

6..According to paragraph 4, why was Griffith's experiment repeated in a test tube

- A. To provide additional support for the transformation of R-strain into S-strain pneumococci
- B. To establish whether or not the transformation of R cells was caused by something the mouse's body did
- C. To determine why the S-strain pneumococci somehow survived if they were in the presence of the R-strain
- D. To test the results of adding a cell-free extract to the mixture

7..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. This result showed that the chemical transforming principle in S pneumococci was passed on to future generations of S pneumococci.
- B. After exposure to the cell-free extract from the S pneumococci, R pneumococci strain cells acquired the ability to transform themselves into S pneumococci.
- C. The transformation of R cells by a cell-free extract of S pneumococci demonstrated the existence of a chemical transforming principle that brought about heritable change.
- D. This transformation showed that the characteristics that the S pneumococci possess are superior to the characteristics of R pneumococci.

8..According to paragraph 5, why did Oswald Avery and his colleagues treat the transforming extract in a variety of destructive ways

- A. They hoped to destroy the virulent part of the transforming extract.
- B. They wanted to identify the substance responsible for the transforming activity.
- C. They wanted to identify which methods would destroy particular substances in the transforming extract.
- D. They needed to determine which treatments were most successful in destroying DNA.

9..The word virtually in the passage is closest in meaning to

- A. perfectly
- B. nearly
- C. partially
- D. relatively

10..The phrase In retrospect in the passage is closest in meaning to

- A. By general agreement
- B. In reality
- C. Looking back
- D. Practically speaking

11..According to paragraph 6, why did scientists continue to believe that the hereditary material was a protein

- A. Scientists thought that the research of Avery and his colleagues provided insufficient information about the nature of DNA.
- B. Scientists believed that only proteins were complex enough to carry genetic information.
- C. Scientists thought Avery and his colleagues had little understanding of the physical basis of inheritance.
- D. Scientists ignored important milestones that indicated the chemical complexity of DNA.

12..Which of the following can be inferred from the passage about the transformed R-strain pneumococci

- A. They had acquired the genetic information for producing a protective coating.
- B. They were unable to cause transformation in other strains of pneumococci.
- C. In the presence of heat-killed R-strain bacteria, they lost their virulence.
- D. They did not multiply as quickly as nontransformed cells did.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Why

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. From the 1920s through 1944, researchers used pneumococcus bacteria to discover the properties of DNA because the bacteria was relatively simple, having only two strains.
- B. Frederick Griffith discovered that a nonvirulent strain of bacteria could be transformed into a virulent strain by being exposed to dead cells from the virulent strain.
- C. By selectively destroying various substances in the cells of pneumococci bacteria, Oswald Avery and his colleagues identified DNA as the substance that caused bacterial transformation.
- D. Oswald Avery injected the combination of heat-killed, virulent cells and nonvirulent cells into mice because he hoped this would lead to a vaccine for pneumonia.
- E. Avery and his colleagues were able to isolate Griffith's transforming principle by injecting mice with the extract that contained the transforming principle.
- F. Scientists did not initially recognize the importance of the discovery that DNA could cause genetic transformation because the hereditary material was assumed to be a protein.

1. Most songbirds hatch in the spring and then merely listen to the songs of adult male birds until sometime in late summer or autumn, when the adults stop singing, not to resume until the end of winter the following year. It is usually male birds that are doing the singing in northern latitudes, though female singing is common in the tropics. Many young songbirds do no singing of their own until nearly a year after their birth. With the coming of their second spring, their testosterone levels rise and this in turn prompts them to begin singing, with their song development following a predictable pattern over a period of weeks. At first, their songs may be a quiet, jumbled series of chirps and whistles. Over time, young birds begin to use the syllables of their species' songs, though the order in which these syllables appear will vary. Finally, their songs crystallize (take form) into the clear, orderly song of their species.
2. There is a songbird, called the white-crowned sparrow, whose song development follows this general script while providing some variations that are instructive about the interplay of internal influences and learning in birdsong. White-crowned sparrows raised in captivity will follow the pattern of song acquisition just described: they listen to songs in their first spring and summer but do not themselves begin singing until they are perhaps six months old. In nature, however, things are different. For example, the white-crown found year-round in the San Francisco area sings a particular regional variant or dialect of the basic white-crown song and begins singing within six weeks or so of birth and may progress to fully crystallized song as early as three months after birth, meaning about September.
3. Why would there be a difference between singing in nature and singing in the laboratory? The pressures of nature. As year-round residents, the San Francisco white-crowns do not fly into an area in spring and then establish territories. Rather, they establish territories as early as their first autumn. **One function of birdsong is to announce, I have a territory here. Young white-crowns, like many species, will extend this practice by countersinging, meaning a male, upon hearing the song of a nearby male of its species, will repeat the exact song he has heard, thus setting off a back-and-forth duel, like two children in an argument, each of them saying, I'm still here.**
4. Internal influences and learning are also on display in white-crowns in the way they acquire their songs. We know that there is often a so-called sensitive period for animal learning a kind of window in which an animal is able to acquire certain skills or information. In laboratory-raised white-crowns, the sensitive period starts at about ten days after birth and extends until about fifty days after birth. A white-crown that became deaf prior to the opening of the sensitive period eventually will sing individual notes, but it will never learn to sing its species' song. Meanwhile, white-crowns that are raised in nature through part of their sensitive period and then taken to the laboratory will begin singing the following winter in the dialect of the area in which they were hatched. Two points are worth observing about this. First, note that these birds are learning the white-crown song months before they ever start practicing it themselves. Indeed, the learning window will be closed completely (in their first summer) before these lab-reared birds ever sing a note (the following winter). Second, learning is important enough in song acquisition that white-crowns learn not just their species' song but local or regional variants of it, which they are able to recall months after last hearing them.

5. But what about internal influences Interestingly, all white-crowns that are reared in isolation from birth eventually sing nearly identical versions of a kind of standard white-crown song. In other words, there seems to be a built-in version of the white-crown song that becomes modified with local dialects only when birds are raised in the wild. Beyond this, isolated white-crowns that are exposed to tapes of other species' songs will ignore the other birds' songs entirely and go on to sing the basic white-crown song. White-crowns are thus genetically disposed to learn their own song while ignoring the songs of others.

1..The word prompts in the passage is closest in meaning to

- A. stimulates
- B. strengthens
- C. prepares
- D. forces

2..According to paragraph 1, which of the following is true of male songbirds in the first year of life

- A. They do not begin singing until sometime in late summer or autumn.
- B. They begin singing earlier in the tropics than in northern latitudes.
- C. They listen to songs of adults for an extended period of time before they themselves sing.
- D. Their earliest songs contain the characteristic order of syllables for their species.

3..The word particular in the passage is closest in meaning to

- A. popular
- B. specific
- C. well-known
- D. complex

4..According to paragraphs 2 and 3, all of the following are true about San Francisco white-crowns EXCEPT:

- A. They do not migrate to another area in spring to establish territories.
- B. They completely acquire their song as early as three months after birth.
- C. They establish territories in their first autumn.
- D. They begin singing much earlier in captivity than they do in nature.

5..In paragraph 3, the author points out that San Francisco white-crowns establish their territories in the area in which they are born in order to explain which of the following

- A. Why they practice counter-singing

- B. Why they get better territories than white-crowns that establish territories in areas in which they are not born
- C. Why they are more competitive than white-crowns raised in captivity
- D. Why in their natural habitat they start singing earlier than white-crowns raised in captivity

6..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Many species, including white-crowns, use a dueling technique to spread the practice of countersinging to other males of the species.
- B. A young white-crown male uses countersinging to learn the songs of nearby males by repeatedly practicing them.
- C. A young white-crown male engages in countersinging with a nearby male in order to assert its continuing presence in its territory.
- D. Young white-crown males, much like young children, are competitive.

7..The word eventually in the passage is closest in meaning to

- A. generally
- B. probably
- C. in the end
- D. at the least

8..The word recall in the passage is closest in meaning to

- A. repeat
- B. remember
- C. recognize
- D. complete

9..According to paragraph 4, white-crowns with which of the following life histories demonstrate the importance of memory in song acquisition

- A. White-crowns that learn a dialect before they learn their standard song
- B. White-crowns that first heard a dialect of the white-crown song before they were ten days old
- C. White-crowns that were moved from where they were born to a different region during their sensitive period
- D. White-crowns that were raised in nature through part of their sensitive period and then transferred to the laboratory

10..What can be inferred from paragraph 4 about the local dialect of the species song that a white-crown sings after the sensitive period has closed

- A. Those dialects must be learned during the sensitive period and are retained thereafter even in new environments.
- B. Those dialects can be learned after the sensitive period if they are common in the local area.
- C. Those dialects can be learned after the sensitive period if the birds are raised in the laboratory.
- D. Those dialects are learned during the sensitive period and afterward used only when they hear others sing them.

11..According to paragraph 5, which of the following statements is true about white-crowns reared in isolation that are exposed to tapes of other species' songs

- A. The tapes cause the white-crowns to begin singing earlier than those not exposed to the tapes.
- B. The tapes do not affect the white-crowns' singing development or change their song.
- C. The tapes help the white-crowns learn the standard song of their species.
- D. The tapes aid the white-crowns in learning local dialects of other species.

12..What can be inferred from paragraph 5 about the song of white-crowns raised in the wild

- A. It is less complex than the song of birds raised in isolation.
- B. It is the standard song of the white-crown species.
- C. It is a mixture of the basic white-crown song and the dialects of other white-crowns that inhabit the local area.
- D. It is identical to dialects learned from exposure to tapes.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

So San Francisco white-crowns need to start quickly making their presence known.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Male birds have a standard process for acquiring their songs that involves a sensitive period when learning takes place.
- B. Birds raised in the wild can differ from those raised in captivity in when they begin to sing their song, which is used to establish their territory.
- C. Testosterone levels in male birds affect the quality of the young bird's song.
- D. Birds that do not need to establish their territory, such as those raised in isolation, are unlikely to learn to sing.
- E. The learning of local dialects demonstrates that song acquisition involves both internal and environmental influences.
- F. Birds raised in the wild may acquire the songs of other species in their local area.

Passage 130 2015-07-11-4 Cereals and Legumes: A Partnership

1. Cereals are flowering grasses that sprout, flower, seed, and die in the space of a year, which is why gardeners refer to them as annuals. Grown for their seeds or kernels, cereals are excellent sources of energy: although they lack some amino acids, as well as calcium, vitamin A, and vitamin C, they provide starch and oil, and in some cases, considerable amounts of protein. Once ripe, the kernels are relatively easy to store, and they retain their nutrients for a long time. Even the stalks of cereals are useful as animal food, as bedding in stables and barns, and as a building material. A major drawback with cereals is that they depend on the soil for nitrogen. Without fertilization they eventually exhaust the fields they are growing in, but despite this, two cereals (wheat and barley) were the very first plants to be domesticated (grown for human use); and a third (rye) may have been cultivated, or even domesticated, at about the same time. Today, cereal crops including wheat, rice, maize, sorghum, millet, and oats provide most of the calories in the human diet.
2. Like cereals, legumes are annuals. Some legumes are grown for animal fodder. Many other legumes, however, are cultivated for their seeds, which ripen in pods. The seeds are rich in B vitamins and iron, contain on average two times the protein but less starch than cereals, and can be eaten, sometimes pods and all, while they're still green. (Snow peas and green beans are familiar examples.) Legumes are characterized by a long period of sequential ripening, during which a single plant may have ripe pods, green pods, and flowers, all at the same time, which means that a stand of legumes can be harvested again and again over several weeks. Like cereals, legumes can be dried and stored for later use (the pods open easily when dry), and again like cereals, legumes provide food for both people and animals. However, legume plants add nitrogen to the soil, so when they are grown in the same fields as cereals, they can replace much of the nitrogen the cereals have depleted.
3. Growing cereals and legumes together is good for the fields, and eating them together is good for the farmers. In order to build and maintain body tissue, people need protein or more specifically, the amino acids in protein. Some amino acids are synthesized in the adult human body, but eight essential ones cannot be and have to come from food. Although all eight are present in animal protein, plant proteins are usually missing one or two. When cereals and legumes are eaten together, they provide all eight of the essential amino acids, a fact that the ancestors of early agriculturalists undoubtedly understood at least on a practical level and their descendants took advantage of that knowledge. In Asia, rice, wheat, and barley were grown along with soybeans; in India rice was paired with hyacinth bean, black gram, and green gram; in the African savanna, pear millet and sorghum were domesticated along with cow pea and Bambara groundnut; and in the New World, maize and Phaseolus beans in Central America and maize and groundnuts in South America were the bases for agriculture.
4. Cereals and legumes are technically dry fruits (they have a hard dry layer around their seeds). Early agriculturalists also experimented with growing succulent fruits like apples, olives, grapes, and melons, but most of these were brought into domestication much later than cereals and legumes, and in most cultures they've always been supplementary foods rather than staples. **Many of them are propagated vegetatively asexually by using a plant part such as a bulb or cutting rather than sexually through seeds, so they are more complicated to grow than cereals and legumes, and this may account for their typically late addition to agricultural assemblages.** It should be noted, however, that recent research

in Israel suggests that figs may have been domesticated at a site near Jericho in the Jordan Valley at about the same time as the first experiments with cereals and legumes, and some archaeologists believe that in New Guinea, tubers may have been domesticated long before other crops were imported.

1..According to paragraph 1, all of the following are advantages of cereals EXCEPT:

- A. They provide large amounts of energy when consumed.
- B. They store easily and retain nutrients for a long time.
- C. They provide considerable amounts of calcium, vitamin A, and vitamin C.
- D. They have multiple uses, including as bedding or building material.

2..According to paragraph 1, a major disadvantage of cereals is that they

- A. cannot be used as animal food
- B. must be planted in a different field every year
- C. take a long time to ripen before they can be used or stored
- D. use up all of the nitrogen in a field unless fertilizer is used

3..According to paragraph 2, one way in which legumes differ from cereals is that legumes

- A. are a better source of starch
- B. contain far more protein
- C. take much longer to ripen
- D. must be dried before being stored

4..According to paragraph 2, all of the following statements about legumes are true EXCEPT:

- A. Legumes have pods that help seeds ripen quickly.
- B. Legumes contain a lot of iron and B vitamins.
- C. Legumes return nitrogen to the soil.
- D. Legume plants can be harvested many times during a growing season.

5..The word specifically in the passage is closest in meaning to

- A. precisely
- B. importantly
- C. frequently
- D. likely

6..The word undoubtedly in the passage is closest in meaning to

- A. possibly

- B. typically
- C. certainly
- D. initially

7..Paragraph 3 supports which of the following ideas about amino acids

- A. Amino acids are not produced by the human body and must be obtained from food.
- B. Certain amino acids that people need for building and maintaining body tissue cannot be acquired from plant proteins.
- C. When legumes or cereals are consumed alone, they do not provide all of the essential amino acids.
- D. Legumes are missing many more of the eight essential amino acids than cereals are.

8..In paragraph 3, why does the author discuss crops grown in Asia, India, the African savanna, and the New World

- A. To show how widely the understanding of the benefits of combining legumes and cereals was applied
- B. To suggest that it was most effective for the same crops to be grown year after year in many parts of the world
- C. To emphasize that proteins that come from plants were recognized as valuable in many parts of the world
- D. To demonstrate that a wide variety of very different cereals and legumes could be grown together

9..The word technically in the passage is closest in meaning to

- A. more complex than
- B. generally understood to be
- C. often confused with
- D. scientifically classified as

10..The word supplementary in the passage is closest in meaning to

- A. valued
- B. rare
- C. seasonal
- D. extra

11..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Many of the cereals and legumes can be propagated asexually through offshoots, cuttings,

- tubers, bulbs and corns or sexually through seeds, which are less complicated to grow.
- B. Fruits were typically domesticated later than cereals and legumes, possibly because they tend not to be propagated through seeds and are thus more complicated to grow.
 - C. Plants that are propagated sexually through seeds are generally much less complicated to grow than asexually propagated plants are.
 - D. In addition to being propagated asexually, rather than sexually through seeds, many fruits can be added and grown later in the season than cereals and legumes.

12..Paragraph 4 supports which of the following ideas about the figs that may have been domesticated in the Jordan Valley near Jericho

- A. Their early domestication casts doubt on the idea that succulent fruits were grown much later than cereals and legumes.
- B. They were a more important crop to inhabitants of the Jordan Valley than cereals and legumes were.
- C. They are closely related to the plants domesticated in New Guinea before other crops were imported.
- D. They are much easier to grow than any other succulent fruit.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Although they must be replanted each year, they can be grown for a variety of uses.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Cereals, annual flowering grasses with usable stalks and nutritious seeds that dry and store well, were the first domesticated plants, but they must be grown with legumes to preserve soil nitrogen.
- B. Legumes contain a large amount of protein and, when they are eaten with cereals, provide all the amino acids essential to humans.
- C. Both the seeds and stalks of wheat and barley were eaten by early agriculturalists, who needed to take in enough nutrients and calories to work the fields.
- D. Planting cereals and legumes together sped up their growing period, which meant that they could both be harvested many times over a period of several weeks to several months.
- E. Cereals and legumes were typically domesticated well before succulent fruits, which are harder to cultivate, and became staples in early agricultural societies.
- F. Because soil conditions in New Guinea and the Jordan Valley made it difficult to grow cereals such as wheat and barley, these regions began to cultivate figs and tubers instead.

Passage 131 2015-07-11-5 Alaska and Bark Beetles

1. Over the twentieth century, global temperatures increased by an average of about 0.7 degrees Celsius, but some places have warmed a lot more than this, and other places have warmed less. These temperature increases have been enough to trigger changes in ecosystems all over the world, especially in places where the warming has been the greatest. In some places, the changes have been subtle, perhaps a slight shift in vegetation that only a careful observer would notice. In other cases, small changes in climate have sparked a chain of larger effects, leading to massive changes.
2. The biggest climate-caused ecosystem shifts today are happening at the world's most northern latitudes, where the temperature over the last century has been rising about two times faster than the global average. In the northernmost state of the United States, Alaska, for example, warming has paved the way for a spike in the numbers of spruce bark beetles. Bark beetles have been a pest to Alaskan white spruce trees for thousands of years, but their numbers were held in check by the cold climate, which forced the insects to hide in the bark of individual trees for most of the year. As the length of the warm season increased over the 1980s and 1990s, however, bark beetles had more time to fly from one tree to the next, burrow, and lay their eggs between the bark and the wood. The beetles had another thing going for them, too: a multi-year drought had weakened many of the spruce trees, leaving them vulnerable to attack. In the mid-1990s, the bark beetle population exploded, and over the next few years the pests wiped out white spruce forests over an area the size of the U.S. state of Connecticut. In the years since, the combined forces of a longer insect-breeding season and forest management practices that left forests overcrowded gave way to similar epidemics farther south. Large swaths of pine and spruce have been destroyed by insects in several other parts of the United States.
3. In the late 1990s, the effects of the bark beetle epidemic rippled throughout Alaska's white spruce ecosystem and affected virtually every population of living organisms, but not all of the impacts were negative. Fewer spruce trees meant a sunnier area in the forest below the treetops, which allowed grasses to move in and take hold. The grasses, in turn, changed the soil temperature, making the environment more friendly for some other types of vegetation. Animals that feed on grasses, including moose, elk, and some birds, also benefited. But the beetle infestation was bad news for organisms that rely on white spruce for their habitat, like hawks, owls, red squirrels, and voles. Voles are a type of small, mouse-like rodent, an especially vital part of the ecosystem because they help spread mycorrhizal fungi, which attach to the roots of plants and help them take in water and nutrients. Voles are also an important food for a number of predators.
4. Ecosystem changes always hurt some living creatures and help others. It's hard to say, therefore, whether a change is good or bad overall. Instead, ecologists (people who study ecosystems) often focus on the impacts on a single species: for instance, us. In the short term, the Alaskan spruce beetle epidemic supplied a lot of people with firewood, but only by destroying tons of otherwise valuable timber and threatening the livelihoods of loggers. And no one knows for sure what the long-term impacts on the forest will be. **Ecosystems tend to return to their previous states after disturbances like pest outbreaks, fires, or major storm events, but if the Alaskan spruce ecosystem is disturbed too often or too much, it might shift to a different type of forest, a woodland, or a grassland instead.**

5. In extreme cases, major assaults on ecosystems can lead to a total collapse in which the ecosystem doesn't bounce back to the way it was or transition to a new, healthy state. The result is an area with very little life; in the oceans, biologists refer to these areas as dead zones. One such example is the coral reef die-off that happened in the Indian Ocean in the late 1990s.

1..The word subtle in the passage is closest in meaning to

- A. limited
- B. unimportant
- C. not obvious
- D. gradual

2..The phrase paved the way for in the passage is closest in meaning to

- A. come together with
- B. made possible
- C. increased the intensity of
- D. made absolutely certain

3..The phrase wiped out in the passage is closest in meaning to

- A. damaged
- B. threatened
- C. spread through
- D. killed off

4..Paragraph 2 suggests that the warming of the Alaskan climate affected bark beetles in which of the following ways

- A. By making it possible for a beetle to deposit its eggs in a greater number of trees
- B. By making it possible for beetles to survive in the bark of trees for longer lengths of time
- C. By making it unnecessary for a beetle to protect its eggs by laying them between the bark and the wood
- D. By increasing the number of spruce trees, thereby providing the beetles with far more places to live

5..According to paragraph 2, all of the following contributed to the destruction of forests in different parts of the United States EXCEPT

- A. a drought that had lasted for several years
- B. a lack of forest management practices
- C. overcrowding in forests
- D. a huge increase in spruce tree pest populations

6..Which of the following statements most accurately describes the relationship of paragraph 3 to paragraph 2

- A. Paragraph 2 explains the causes of the spruce bark beetle epidemic in Alaska, and paragraph 3 discusses the chain of events that occurred as a result of that epidemic.
- B. Paragraph 2 shows that warming air temperatures can affect a large number of species, and paragraph 3 shows that warming soil temperatures can have even greater effects.
- C. Paragraph 2 discusses one explanation for the disappearance of spruce trees from a part of Alaska, but paragraph 3 shows that an alternative explanation is more likely to be correct.
- D. Paragraph 2 describes the negative consequences of climate warming for some species, but paragraph 3 shows that there are also some positive consequences for these same species.

7..According to paragraph 3, which of the following effects did the bark beetle epidemic have on moose, elk, and some birds

- A. The epidemic increased the availability of water for these animals.
- B. The epidemic increased the availability of food for these animals.
- C. The epidemic destroyed the habitat of these animals.
- D. The epidemic meant that these animals experienced more competition from hawks, owls, red squirrels, and voles.

8..According to paragraph 3, a decline in the vole population in Alaska may have which TWO of the following consequences To receive credit, you must select TWO answer choices.

- A. Some predators may have less to eat.
- B. Hawk and red squirrel populations may be more successful.
- C. Plants may find it more difficult to absorb water and nutrients.
- D. Mycorrhizal fungi numbers may increase.

9..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Ecosystems like the spruce ecosystem in Alaska tend to return to their previous states after disturbances such as pest outbreaks, fires, or major storm events.
- B. While ecosystems tend to return to their previous states after disturbances, the Alaskan spruce ecosystem might not if it is disturbed too often or too much.
- C. Ecosystems tend to return to their previous states after disturbances, so Alaska might again become covered with woodlands or grasslands.
- D. After certain types of disturbances such as pest outbreaks, ecosystems do not always return to their previous states but shift to being woodlands or grasslands instead.

10..The phrase assaults on in the passage is closest in meaning to

- A. imbalances in
- B. changes in
- C. problems for
- D. attacks on

11..In paragraph 5, coral reefs in the Indian Ocean are presented as an example of which of the following

- A. Ecosystems that totally collapsed
- B. Ecosystems that transitioned to a new, healthy state
- C. Ecosystems that bounced back to the way they were
- D. Ecosystems that were affected by a nearby dead zone

12..The passage provides an answer to which of the following questions

- A. Why has the temperature at northern latitudes been rising faster than the global average
- B. Why did corals in the Indian Ocean die off in the late 1990s
- C. What types of vegetation benefited from the change in soil temperatures in Alaska
- D. What were some of the effects of the bark beetle epidemic for humans

13..Look at the four squares that indicate where the following sentence could be added to the passage.

But even from this limited perspective, the answer is not completely straightforward.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Global warming has led to changes in ecosystems all over the world, with ecosystems at northern latitudes being affected the most.
- B. A longer warm season in Alaska caused a sharp increase in the number of bark beetles, leading to the destruction of spruce forests, which in turn seriously affected many other species.
- C. Sometimes ecosystems are able to recover from disturbances or to develop into different, but healthy, systems, but in extreme cases, they may collapse completely.
- D. The loss of spruce forests caused an epidemic in mycorrhizal fungi, and these fungi damaged the roots of many plants, making them unable to take in water and nutrients.
- E. Whereas some types of changes are good for the majority of species in an ecosystem, ecologists believe that most disturbances to ecosystems are bad overall.
- F. Coral reefs may die off as a result of the global increase in temperatures, but after a transition period as a dead zone, they are able to return to their original state.

Passage 132 2015-07-11-6 Motor Development in Children

1. Control over one's motor behavior ranks among the infant's greatest achievements. Psychologists who study the acquisition of motor skills in children find it useful to distinguish between gross motor development, that is, motor skills which help children to get around in their environment such as crawling and walking, and fine motor development, which refers to smaller movement sequences like reaching and grasping.
2. The development of motor skills has implications beyond simply learning how to perform new actions: motor skills can have profound effects on other areas of development. For example, researchers have shown that infants with locomotor experience (experience moving around their environment) were less likely to make errors while searching for hidden objects. The ability to initiate movement around one's environment stimulates the development of , making hidden object tasks easier to solve. Psychology professor Carolyn Rovee-Collier argues that the onset of independent locomotion at around nine months old marks an important transition in memory development. Children who can move about the environment develop an understanding of locations such as here and there. Because infant memory is initially highly dependent on context that is, the similarity between the situation where information is encoded (stored in memory) and where it is recalled infants who have experience moving about the environment and who learn to spatially encode information become less dependent on context for successful recall. These examples show that gross motor development has implications beyond the immediately apparent benefits of crawling and walking.
3. Renowned psychologist Jean Piaget argued that the development of reaching and grasping was a key aspect of development because it formed an important link between biological adaptation and intellectual adaptation. Reaching and grasping are voluntary actions under the infant's control, and as such, they open up exciting new possibilities in their ability to explore the environment. An infant who reaches for and grasps an object so as to explore it pushes his development forward as he engages in processes such as adapting his grip to the size and shape of the object. Piaget argued that these early processes drive cognitive development in the first two years of an infant's life.
4. The development of reaching begins early on in life. Newborn infants seated in an upright position will swipe and reach towards an object placed in front of them, a behavior labeled "prereaching." These poorly coordinated behaviors start to decline around two months of age and are replaced by "directed reaching" which begins at about three months of age. At this time reaching becomes more coordinated and efficient, and improves in accuracy. According to research conducted by Clifton et al., the infant's reaching does not depend simply on the guidance of the hand and arm by the visual system but is controlled by proprioception, the sensation of movement and location based on the stimulation arising from bodily sources such as muscle contractions. By about nine months old, infants can adjust their reaching to take into account a moving object. However, nine month olds are far from expert reachers. A good deal of skill must still develop.
5. Once infants begin reaching they also begin to grasp the objects that are the target of their reaches. The ulnar grasp is seen when infants first engage in directed reaching. The ulnar grasp is a primitive form of grasping in which the infant's fingers close against its palm. The fingers seem to act as a whole, requiring the use of the palm in order to hold an object.

Shortly after this accomplishment, when infants can sit upright on their own, they can acquire the ability to transfer objects from hand to hand. **Around the end of the first year, infants will have graduated to using the pincer grasp where they use their index finger and thumb in an opposable manner (placing them opposite each other), resulting in a more coordinated and finely tuned grip which allows for the exploration of very small objects or those objects which demand specific actions for their operation, such as the knobs on a stereo system which require turning to the left or right to adjust volume.**

1..According to paragraph 1, the distinction between gross motor development and fine motor development is based primarily on

- A. how much control the infant has over the motor skills
- B. when the motor skills are developed
- C. the size of the movement sequences involved
- D. the usefulness of the movement sequences involved

2..The word onset in the passage is closest in meaning to

- A. achievement
- B. beginning
- C. improvement
- D. practice

3..The phrase immediately apparent in the passage is closest in meaning to

- A. available
- B. obvious
- C. desirable
- D. useful

4..According to paragraph 2, why do infants with locomotor experience have less trouble locating hidden objects

- A. Moving around their environment helps infants to develop a better memory for spatial locations.
- B. Moving around their environment increases infants' ability to make use of context to identify objects.
- C. Moving around their environment gives infants more opportunity to correct their errors when searching for objects.
- D. Moving around their environment reduces the time infants have to spend spatially encoding information.

5..According to paragraph 2, as a result of developing an understanding of here and there, infants are better able to

- A. describe the locations of objects in space
- B. feel comfortable in new and unfamiliar situations
- C. use context as an aid to recalling previously encoded information
- D. recall information in situations unlike the one in which it was originally encoded

6..The word Renowned in the passage is closest in meaning to

- A. Educational
- B. Controversial
- C. Famous
- D. Theoretical

7..The phrase engages in in the passage is closest in meaning to

- A. repeats
- B. learns about
- C. performs
- D. imitates

8..What can be inferred from paragraph 3 about the cognitive development of an infant in its first two years of life as described by Piaget

- A. It is a sign of advanced development when an infant is able to control its urges to reach and grasp.
- B. Repeated practice reaching for and grasping objects results in important biological adaptations.
- C. Infants who spend large amounts of time in exciting environments have more difficulty in their intellectual development.
- D. An infant's development will be slower if it is not given the opportunity to reach for and hold objects.

9..In paragraph 4, why does the author emphasize the point that nine month olds are far from expert reachers

- A. To support the idea that nine-month-olds vary a great deal in their reaching abilities
- B. To distinguish between the directed reaching and the non directed reaching of infants
- C. To stress that an infant's reaching skills continue to improve even after the first nine months
- D. To call into question the accuracy of the results presented by Clifton et al

10..According to paragraph 4, which of the following statements about directed reaching is true

- A. Directed reaching behaviors have typically developed by the time an infant is about two months old.

- B. Directed reaching behaviors do not appear until an infant is able to account for the movement of an object.
- C. Directed reaching is the earliest form of reaching behavior that infants develop.
- D. Directed reaching is controlled both by the visual system and by proprioception.

11..All of the following statements about the ulnar grasp are true EXCEPT:

- A. It is a relatively uncoordinated form of grasping.
- B. It is used by infants when they first engage in directed reaching.
- C. It develops only after infants become able to sit upright on their own.
- D. It makes use of the palm as well as the fingers to hold an object.

12..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Infants of about a year old begin using their index finger and thumb to make more coordinated and finely tuned movements, allowing them to explore and manipulate small objects.
- B. Around the end of the first year, infants begin to use a type of grip on small objects that is more coordinated and finely tuned than was the pincer grasp.
- C. Infants begin handling very small or difficult-to-operate objects at about one year of age, resulting in improvements in their ability to grip objects with their thumb and fingers.
- D. When one-year-old infants begin using the pincer grasp, they become much more interested in very small objects (such as knobs on a stereo system).

13..Look at the four squares that indicate where the following sentence could be added to the passage.

What accounts for this greater accuracy

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. The ability to move around in an environment gives infants an understanding of location and thus reduces the extent to which their memory is dependent on context.
- B. Infants become better at finding hidden objects once they have developed the type of grasp that allows them to handle and explore very small objects in their environment.
- C. Piaget argues that an infant's cognitive development is related to the development of fine motor skills that make it possible for infants to interact with and adapt their actions to their environment.
- D. Fine motor skills tend to develop later than do gross motor skills because fine motor skills require smaller, more finely tuned movements and a great deal of coordination.
- E. When infants first begin to grasp objects, they transfer the objects from hand to hand and firmly close their fingers against their palms.
- F. Reaching and grasping begin early; by about three months, reaching is more coordinated and efficient, and by one year, infants begin to develop a coordinated and finely tuned grip.

Passage 133 2015-07-11-7 Bison and Humans

1. When human beings first migrated from Asia into North America at the end of the last ice age, they found an enormous, now extinct creature known as the giant long-horned bison (*Bison priscus*). We know that early Americans hunted these beasts because excavated skeletons of the bison bear stone spear tips. The style of the points dates them to twelve to thirteen thousand years ago, not long after the first wave of human immigrants washed south and east across the continent. These early Americans ate a variety of plants and animals, but judging from the campsite remains, they had a special taste for long-horned bison. It was their favorite prey, perhaps because one animal filled so many stomachs.
2. The giant horns that gave *Bison priscus* its common name tell us some important things about its lifestyle. Animals with gigantic weapons on their heads usually live alone or in small groups. Animals that live in herds usually have small horns. Horns and antlers help males in several ways. Animals use these horns and antlers to fight with other members of the same species, to increase their appeal to potential mates, and to protect themselves from predators. Fossil bones suggest that giant bison used their long, outward-facing horns to injure their opponents. **An individual with longer horns had a better chance of circumventing its opponents' horns and fatally wounding them than one with shorter horns, and females probably preferred to mate with winners of these contests rather than with losers, either because they liked what they saw in the male or because they liked the territory that the male could defend from competitors.**
3. The giant bison's architecture served it well for thousands of years, but its body shrank and changed shape starting about twelve thousand years ago. The timing gives us an important clue about the cause. Only two major predators, wolves and lions, had hunted giant bison for tens of thousands of years. If they caused the change, it would have happened much earlier. The big change in the bison's environment twelve to thirteen thousand years ago was the arrival of a new predator. This one walked on two feet, hunted in cooperative bands, and carried spears with well-designed stone points. Its remarkable efficiency at hunting seems to have caused a reduction in the body size of other large mammals, too. Over the past ten thousand years, North American sheep, elk, moose, musk ox, bears, antelope, and wolves have all shrunk.
4. Scholars have offered various explanations for these changes, but it seems likely that these new hunters converted the giant bison's shape and habits from virtues into liabilities. Hunters who needed to get close to their prey, such as wolves and human beings armed with spears, preferred to attack lone individuals rather than many victims at once. Hunting punished solitary, territorial giant bison and rewarded those that stayed close together. Clumps of bison became more common and grew into herds.
5. Herding is a classic response to heavy predation. It brings a statistical advantage to herd members because the odds that a predator will hone in on any one individual will decrease with the size of the herd. Herds further improved odds for members through cooperative behavior. Members warned each other of danger, and they fought off predators by joining forces (e.g. by forming a circle with vulnerable backsides to the center and dangerous horns facing the periphery).
6. But bison paid a price for herding. In a given area, the supply of food per individual declined along with the chances of being attacked. Smaller bodies probably resulted from a decline in

food availability as bison crowded together. Herding changed the bison's shape as well as size. Now survival depended on the ability to crop grass, bison's main food, quickly. Shifting the head closer to the ground, reducing horn size, and growing a hump to cantilever, or support, the head's weight enabled bison to graze for long periods without strain. Giant horns, which enabled males to defend territory, may also have become a liability as being able to stay close together became more valuable.

1..According to paragraph 1, which of the following best describes the relationship between humans twelve to thirteen thousand years ago and the giant long-horned bison

- A. Humans first came to the Americas as a result of following long horned bison that were migrating from Asia to the Americas.
- B. Humans in the Americas preferred hunting long-horned bison to hunting other animals.
- C. Humans in the Americas were forced to migrate south and east across the continent as a result of the presence of long-horned bison.
- D. Humans in the Americas generally ate plants and small animals because long-horned bison were difficult to hunt.

2..The word gigantic in the passage is closest in meaning to

- A. very big
- B. very dangerous
- C. powerful
- D. sharp

3..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Females may have chosen their mates based on the male's ability to win contests with other males, or perhaps they decided based on territory.
- B. The longer-horned opponent was more likely to win in a fight, which probably made him more attractive to females because of his physical characteristics or his ability to protect territory.
- C. Males engaged in contests in which they tried to wound one another with their horns while avoiding being hit by the horns of their opponent.
- D. An individual that was able to avoid its opponents' horns had a better chance of fatally wounding them and winning the contest for territory or females.

4..According to paragraph 2, the fact that Bison priscus had giant horns suggests which of the following about its lifestyle

- A. The bison were probably more concerned with protecting themselves from predators than with fighting each other.
- B. The horns were probably more for display to attract mates than for use as actual weapons.

- C. Those individuals with smaller horns probably banded together to fight the males with larger horns.
- D. Individuals probably lived by themselves or with only a few other bison.

5..In paragraph 3, why does the author include the information that wolves and lions "had hunted giant bison for tens of thousands of years"

- A. To emphasize how good the bison's defenses were
- B. To support the claim that only wolves and lions hunted the bison for long enough to cause changes in its evolution
- C. To help explain why predation by wolves and lions cannot account for the changes that occurred in bison twelve thousand years ago
- D. To introduce a discussion of how changes in bison directly affected other large animals in North America

6..Which of the following can be inferred from paragraph 3 about North American sheep, elk, moose, musk ox, bears, antelope, and wolves

- A. They have all changed as a result of hunting by humans.
- B. They all originated in North America at about the same time.
- C. They all contributed to the change in the bison's environment.
- D. They all contributed to the change in the bison's size and shape.

7..According to paragraph 4, which of the following is true about humans and wolves when hunting large animals

- A. Both needed to attack their prey when their prey were outside their territory.
- B. Both attacked the animals in the closest part of a herd.
- C. Both preferred to attack animals that were far from others.
- D. Both preferred to attack multiple animals at the same time.

8..The word vulnerable in the passage is closest in meaning to

- A. hidden
- B. defenseless
- C. harmless
- D. useless

9..Which of the following is NOT one of the reasons given in paragraph 5 for why herding is a classic response to heavy predation

- A. Herd members can teach one another strategies for avoiding predators.
- B. Herd members can alert one another to approaching predators.
- C. Herd members can fight off predators together.
- D. Being in a herd reduces the chance that any particular individual will be a target.

10..The word strain in the passage is closest in meaning to

- A. stopping
- B. rest
- C. movement
- D. stress

11..The word valuable in the passage is closest in meaning to

- A. common
- B. possible
- C. important
- D. natural

12..According to paragraph 6, bison in herds needed to be able to crop grass quickly because

- A. the amount of food available for an individual had decreased
- B. the bison body had become smaller
- C. the chances of being attacked by a predator had increased
- D. bison had grown a hump to support the head's weight

13..Look at the four squares that indicate where the following sentence could be added to the passage.

In sum, by forcing bison to live in herds, humans encouraged the development of a new kind of bison, the short-horn, humpbacked bison that we know today.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. Judging from campsite remains, the first wave of immigrants to North America carried with them hunting tools they had successfully used to hunt large animals in Asia.
- B. Fossil bones suggest that several species of the long-horned bison once existed in North America, but by the time humans arrived, most of them had become extinct.
- C. The size of the bison's horns suggest that individuals lived alone or in small groups, a habit that made them attractive prey for human hunters.
- D. The bison began living in herds, which would have provided protection from hunters but also decreased the amount of food available for each individual.
- E. Early Americans preferred hunting long-horned bison for their horns which could be used to create sharp spear tips for hunting other large animals such as moose and musk ox.
- F. When bison began to live close together, this led to other adaptations in size and shape necessary for survival.

Passage 134 2015-07-12-1 The Western Roman Empire in the Fifth Century

1. Shortly after the death of emperor Theodosius in 395 A.D., the Roman Empire was permanently divided into Eastern and Western empires. By the fifth century A.D., the power of the Western Roman Empire had declined considerably, though the Eastern Roman Empire centered in Byzantium continued to flourish. Various problems contributed to this undermining of the West.
2. The accessions of Arcadius and Honorius, sons of Theodosius, as emperors in the East and West, respectively, illustrate the unfortunate pattern of child heirs that had unfavorable effects for both empires. When Arcadius died in 408, he was succeeded by his seven-year-old son, Theodosius II. Reigning until 423, Honorius was succeeded by his nephew Valentinian III, who was only five. **Because of their young ages, Theodosius' sons and grandsons could not rule without older advisors and supervising regents upon whom they naturally became dependent and from whom they were unable to break away after reaching maturity.** As powerful individuals vied for influence and dominance at court, the general welfare was often sacrificed to private rivalries and ambitions. Moreover, it was the women of the dynasty who were the more capable and interesting characters. Holding the keys to succession through birth and inheritance, they became active players in the political arena.
3. Compared with the East, however, the West faced a greater number of external threats along more permeable frontiers. Whereas the East could pursue war and diplomacy more effectively with their enemies on the long eastern frontier, the West was exposed to the more volatile tribal Germanic peoples on a frontier that stretched along the Rhine and Danube rivers for 1,000 miles. The East, however, only had to guard the last 500 miles of the Danube. In addition, the East had many more human and material resources with which to pursue its military and diplomatic objectives. The East also had a more deeply rooted unity in the Greek culture of the numerous Greek and Near Eastern cities that Rome had inherited from earlier Grecian empires. Latin culture had not achieved comparable penetration of the less urbanized West outside of Italy. The penetration of Germanic culture from the north had been so extensive along the permeable Rhine-Danube frontier that it was often difficult to distinguish between barbarians (speakers of German and other languages unrelated to Latin) and Romans in those regions by the fifth century anyway.
4. One of the most outstanding features at the beginning of this period was the prominence of Germanic generals in the high command of the Roman Imperial army. The trend became significant, and several practical reasons can explain it. The foremost probably was the sheer need for military manpower that made it attractive to recruit bands of Germanic peoples for the armies, which, in turn, gave able chieftains and warlords the opportunity to gain Imperial favor and advance in rank. Second, one way to turn Germanic chieftains from potential enemies into loyal supporters was to offer them a good position in the Roman military. Third, although Theodosius had risen to power as a military leader, he was also a cultured aristocrat and preferred to emphasize the civilian role of the emperor and to rely for protection on Germanic generals whose loyalties were primarily to him, their patron.
5. Unfortunately, the high positions achieved by Germanic officers often aroused the jealousy and hostility of high-ranking Roman military and civilian officials. Such positions also gave their Germanic holders a chance to act on both personal and tribal animosities in the arena of Imperial politics. Internal Roman rivalries and power struggles aggravated the situation.

Rival factional leaders often granted Imperial titles and conceded territory to one Germanic leader or another in return for help against fellow Romans. While the Romans were thus distracted by internal conflict, other tribes seized the opportunity to cross into Roman territory unopposed. When the Romans could not dislodge them, peace was bought with further titles and territorial concessions as allies. In the midst of it all, alliances and coalitions between Roman emperors or powerful commanders and various tribes or tribal kings were made, unmade, and remade so often that it is nearly impossible to follow their course. Accordingly, all of these situations proved dangerous to the peace and safety of the West.

1..The word unfavorable in the passage is closest in meaning to

- A. negative
- B. uncontrollable
- C. unexpected
- D. long lasting

2..According to paragraph 2, which of the following was one result of the pattern of rule by child emperors

- A. The common people lost respect for the position of emperor.
- B. Regents and advisors attempted to put an end to traditional rivalries for dominance within the court.
- C. Women within the dynasty gained increased influence and power.
- D. Traditional rules of succession by inheritance were changed.

3..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. As young rulers, the sons and grandsons of Theodosius necessarily depended on older advisors, and as adults, they were unable to rule independently of these advisors.
- B. The sons and grandsons of Theodosius were too young when they came to power to rule without the assistance of older advisors.
- C. On reaching maturity, the sons and grandsons of Theodosius attempted to break away from the older officials who had advised them since childhood.
- D. Because the sons and grandsons of Theodosius were young when they became rulers, older advisors were able to prevent them from breaking away.

4..In describing the frontiers of the Western Empire as more permeable the author means that these frontiers

- A. had more places where crossings could occur
- B. were more distant from the center
- C. were more likely to be changed

D. were more poorly equipped

5..The word pursue in the passage is closest in meaning to

- A. expand
- B. engage in
- C. control
- D. avoid

6..The word comparable in the passage is closest in meaning to

- A. similar
- B. desirable
- C. necessary
- D. noticeable

7..Which of the following is NOT identified in paragraph 3 as a factor contributing to the greater stability and success of the Eastern empire

- A. A shorter border subject to invasion by Germanic tribes
- B. Greater cultural unity among the inhabitants
- C. More resources available for achieving political goals
- D. Lower population densities outside of urban areas

8..In paragraph 3, why does the author discuss the Germanic culture

- A. To compare the less urbanized West outside of Italy to the more urbanized East
- B. To explain why Roman military and political objectives necessarily changed in the fifth century
- C. To emphasize that the Romans had more of a cultural disadvantage in the West than in the East
- D. To explain why there were as many speakers of German as there were Romans on the western frontier

9..Which of the following is NOT identified in paragraph 4 as a reason the practice arose of making Germanic chieftains generals in the Roman high command

- A. It helped reduce the number of possible enemies against the empire.
- B. It helped make it possible to maintain an imperial military force of sufficient size.
- C. It was cheaper than recruiting and training Roman generals.
- D. It gave Theodosius confidence that his generals would remain loyal while he focused on other matters.

10..According to paragraph 4, by becoming generals in the Roman army, Germanic chieftains were given a chance to

- A. obtain benefits from the emperor
- B. influence Roman civilian life
- C. help shape military policy

D. attract Germanic recruits into the Roman army

11..Which of the following is identified in paragraph 5 as a negative consequence of making Germanic chieftains high-ranking officers in the Roman army

- A. Romans no longer sought achievement through the military.
- B. Germanic generals sometimes used their military power to advance their own and their tribes' interests.
- C. Germanic soldiers focused on achieving imperial titles rather than military success.
- D. Greater divisions developed between the Western Empire and the Eastern Empire, which lacked military leadership.

12..According to paragraph 5, what is one way that internal conflict in Rome endangered the peace and safety of the West

- A. The conflict made it more difficult to make peace through the process of granting imperial titles and territorial concessions.
- B. The conflict made it easier for invaders to cross the frontier and enter Roman territory.
- C. The conflict discouraged Roman leaders from creating alliances and coalitions with Germanic tribes.
- D. The conflict made it nearly impossible to track the activities of enemy tribes outside Roman territory.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Once within Roman borders, they proved difficult to remove.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. The division of the Roman Empire into two parts was particularly damaging for the Western Empire because it relied on the Eastern Empire for economic support.
- B. The heirs of Theodosius came to the throne as young children, allowing them to be dominated by advisors who competed for influence at the expense of the empire's welfare.
- C. Western emperors after Theodosius were unable to emphasize their civilian role because of their need to rely on the protection of Germanic generals loyal to them.
- D. Compared to the Eastern Empire, the Western Empire had many disadvantages, including more foreign enemies and fewer material and human resources.
- E. Resentment against Germanic chieftains achieving high rank in the Roman military and factionalism among Roman leaders were among the causes of the period's considerable instability.
- F. As the resources needed to secure Rome's borders increased, serious conflicts developed among Roman leaders over how best to protect Roman territory against invading tribes.

Passage 135 2015-07-12-2 Features of Tropical Mammals

1. There are several important features of tropical mammals and their habitats that differentiate them from temperate-zone mammals. First, tropical mammals face different environmental stresses than do temperate-zone mammals, and they respond to stresses in different ways. Many temperate-zone mammals, of course, must endure extreme variation within a year; from cold winters with snow and low food supplies to hot summers with dry weather and abundant food. Many mammals respond with hibernation, staying more or less dormant for several months until conditions improve. **Tropical mammals, except in the high-altitude mountains, do not encounter such extreme annual changes, but they do face dry seasons, up to five months long, that sometimes severely reduce food supplies.** For some surprising reasons, they cannot alleviate this stress by hibernating, waiting for the rainy season to arrive with its increased food supplies. When a mammal in Canada or Alaska hibernates, many of its predators leave the area. This is not the case in the tropics. A mammal sleeping away the dry season in a burrow would be easy prey to snakes and other predators. Moreover, a big danger to sleeping mammals would be army ants. These voracious insects are very common in the tropics and would quickly eat a sleeping mouse or squirrel. Also, external parasites, such as ticks and mites, which are inactive in extreme cold, would continue to be very active on sleeping tropical mammals, sucking blood and doing considerable damage. Last, the great energy reserves needed to be able to sleep for an extended period through warm weather may be more than any mammal can physically accumulate. Therefore, tropical mammals need to stay active throughout the year. One way they counter the dry season's reduction in their normal foods is to switch food types seasonally. For instance, some rodents that eat mostly insects during the rainy season switch to seeds during the dry season; some bats that feed on insects switch to dry-season fruits.
2. The abundance of tropical fruit brings up another interesting difference between temperate and tropical mammals: a surprising number of tropical mammals eat a lot of fruit, even among the carnivore group, which, as its name implies, should be eating meat. All the carnivores in Brazil, save pumas, jaguars, and otters, are known to eat fruit on occasion. Upon reflection, however, it makes sense that these mammals consume fruit. Fruit is very abundant in the tropics, available throughout much of the year, and, at least when it is ripe, easily digested by mammalian digestive systems. A consequence of such frugivory (fruit eating) is that many mammals have become, together with frugivorous birds, major dispersal agents of fruit seeds, which they spit out or which travel unharmed through their digestive tracts to be deposited in feces far from the mother tree. Some biologists believe that, even though the carnivores plainly are specialized for hunting down, killing, and eating animal prey, it is likely that fruit has always been a part of their diet.
3. Finally, there are some differences in the kinds of animals inhabiting tropical and temperate regions. For instance, in tropical regions there are few social rodents like beavers and prairie dogs and very few rabbit species. On the other hand, some groups occur solely in the tropics and do extremely well there. There are about 75 to 100 species of New World monkeys (depending on which primate specialist you consult), all of which occur in tropical areas. Arboreal (tree-living) mammals such as monkeys and sloths are plentiful in tropical forests, probably because there is a rich, resource-filled, dense canopy to occupy and feed in. The closed canopy blocks light to the ground, which allows only an undergrowth that is

sparse and poor in resources, and consequently permits few opportunities for mammals to live and feed there. Bats thrive in the tropics, being very successful both in terms of number of species and in their abundances. Nine families of bats occur in Brazil, including more than 140 species; only four families and 40 species occur in the entire United States, an area similar in size to Brazil. While most North American bats feed on insects, the diets of Brazilian bats are more varied and include fruit, nectar, and fish.

1..The word extreme in the passage is closest in meaning to

- A. great
- B. repeated
- C. unusual
- D. constant

2..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Most tropical mammals do not experience extreme temperature changes but can suffer severe food losses from long dry seasons.
- B. Tropical mammals in high-altitude mountains encounter more severe food shortages than other tropical mammals.
- C. Dry seasons up to five months long cause as much damage to food resources of tropical mammals as extreme annual changes.
- D. Dry seasons do not severely reduce the food supplies of tropical mammals every year but they do occur sometimes.

3..According to paragraph 1, mammals in the tropics encounter extreme temperature variations in which of the following situations

- A. During hot summers
- B. During the dry season
- C. In high-altitude mountains
- D. During the rainy season

4..According to paragraph 1, a tropical mammal hibernating throughout the dry season would encounter all of the following difficulties EXCEPT:

- A. It would be easy prey for predators such as snakes.
- B. It would not be able to maintain a steady body temperature.
- C. It might be attacked by army ants.
- D. It could not accumulate sufficient energy reserves.

5..The phrase Upon reflection in the passage is closest in meaning to

- A. For this reason
- B. After consideration
- C. Most importantly
- D. In general

6..Paragraph 2 indicates which of the following about fruit trees in the tropics

- A. They have been the plants most studied by biologists in the tropics.
- B. They don't need to grow much before they begin to produce fruit that tropical animals can eat.
- C. They can spread to areas far from the mother tree, in part because of the eating habits of some carnivores.
- D. They have become the main source of food for pumas, jaguars, and otters.

7..The word consult in the passage is closest in meaning to

- A. believe
- B. admire
- C. find
- D. ask

8..Why does the author provide the information that The closed canopy blocks light to the ground

- A. To illustrate why rodents living in tropical forests live in social groups
- B. To illustrate why tropical forests have so many species of bats
- C. To explain why there are few mammals that live on the ground
- D. To explain why some species of monkeys do very well in the tropics

9..It can be inferred from paragraph 3 that the animals most likely to inhabit tropical regions are animals that

- A. live in trees
- B. outnumber social rodents
- C. do not require a variety of food resources
- D. prefer the dark

10..Select the TWO answer choices that indicate ways in which Brazilian and North American bats discussed in paragraph 3 differ. To obtain credit, you must select TWO answers.

- A. Their diets
- B. Their ability to co-exist with arboreal mammals
- C. The size of their habitats
- D. The number of species in each location

11..What can be inferred from paragraph 3 about the greater variety of bats in Brazil compared to bats in North America

- A. North America is too cold for many varieties of bats.
- B. The more varied diets of Brazilian bats allow more diversity in the bat population.
- C. An insect diet is less nutritious than a diet of fruit, nectar, and fish.
- D. Brazilian bats are arboreal and North American bats are not.

12..Look at the four squares that indicate where the following sentence could be added to the passage.

A possible explanation for this difference is related to what they eat.

Where would the sentence best fit Click on a square to add the sentence to the passage.

13..Directions: Select from the seven phrases below the phrases that correctly characterize tropical mammals and the phrases that correctly characterize temperate-zone mammals. Drag each phrase you select into the appropriate column of the table. Two of the phrases will NOT be used. This question is worth 3 points.

Answer Choices

Endure extreme temperature fluctuations

May reproduce several times a year

Endure a long dry season

Have thick fur coats

Some amass large energy reserves

Some have predators that are not present all year

Remain active all year

Passage 136 2015-07-12-3 Honeybee Society

1. Honeybee colonies are essentially societies of females. In a hive of perhaps 20,000 bees, only a few hundred will be male bees, called drones. They are around only in the spring or summerlong enough to rise to treetop level in a comet-like swarm, chasing after one of the queen bees that have assembled from various hives at a mating site. Of the many drones assembled, only 10 to 15 will actually mate with a queen during one of her mating flights. Each drone that is successful dies in the process, however, and a similar fate awaits drones that aren't successful; once mating is done, they will be expelled from their hives or killed.
2. The week of mating flights prepares the queen for a lifetime of prodigious egg laying; she will produce up to 2,000 fertile eggs a day for years. Nearly all of the offspring that hatch from these eggs are female; they are the hive's worker bees; and they are well named, for it is they who will maintain the hive, forage for food, store the food away, care for newly laid eggs, and more. It is they who will do everything for the colony, in other words, except lay eggs and mate with the queen.
3. Over their brief adult lives of perhaps six weeks, every worker bee takes on, in a predictable order, nearly all the worker tasks that the hive has to offer. For the first three days of her life, a worker is primarily a cleaner of the cells that the bee larvae (immature, wormlike bees) are stored in. As the days pass, she becomes primarily a larvae feeder, then a hive construction worker, then an entrance guard and food storer, and finally a forager, going out to secure nectar, pollen, and water for the colony. Within this structure, however, a worker's life is one of surprising flexibility. After becoming a construction worker, for example, she still engages in some cell cleaning; and throughout her life, she spends a good deal of time resting and patrolling the hive.
4. Importantly, there is no chain of command in a colony no group of workers communicating the message more food needed now or cell cleaning needed over here. How, then, does all this work get organized among tens of thousands of bees? Bees are prompted to act either because of environmental conditions (the temperature of the hive, for example) or because of signals or cues they receive from other bees. The signals are explicit acts of communication, as with the famous waggle dance that bees perform to inform their fellow workers of the location of food sites.
5. Quite often, however, bees are reacting to cues they get from other bees that simply imply a given condition. Take, as an example, a cue that researcher Thomas Seeley confirmed that has to do with unloading time at the hive. In a well-fed hive, forager bees gather food only from flower patches that have lots of nectar. When a hive is near starvation, however, the foragers aren't so choosy; then low-yield flower patches will do. So, how does a forager know whether to be choosy or not? How is she informed of the nutritional status of the colony, in other words? Her informational source is the length of time it takes her to unload her food. Providing the cues are the food-storer bees, which receive the food the foragers bring back and then process it into honey and pack it away in the hive. It takes a returning forager a relatively long time to make contact with a food-storer bee in a well-fed hive, but a relatively short time in a starving hive. Why? Because in a well-fed hive, the food storers have plenty to keep them busy there is plenty of food to store away. If, however, a forager can make contact with a food storer within 15 seconds of entering the hive, the forager knows the colony is low on food and will start paying visits to low-yield sites. This is but one example

of how life in the colony is self-organizing; each bee's behavior is shaped by the behavior of other bees.

1..The phrase expelled from in the passage is closest in meaning to

- A. forced from
- B. carried from
- C. left by
- D. guided from

2..It can be inferred from paragraph 1 that drones are around only in spring or summer because

- A. they are born in these seasons
- B. mating flights occur in these seasons
- C. they are expelled from the hive in these seasons
- D. they are unable to survive in colder seasons

3..According to paragraph 1, all of the following are true about honeybee drones EXCEPT:

- A. They are a small percentage of the bees in a hive.
- B. They die in the process if they mate with a queen bee.
- C. They are accepted back in the hive if they fail to mate.
- D. They swarm at mating sites to chase after a queen bee.

4..Which of the following can be inferred from the discussion of bee society in paragraphs 1 and 2

- A. Male bees have no function other than to mate with the queen.
- B. Male bees have higher status than female bees in the hive.
- C. Female bees produce numerous offspring.
- D. Female bees play a relatively unimportant role in the hive.

5..According to paragraph 3, the main task for the most mature worker bees is

- A. storing nectar and pollen
- B. cleaning bee larvae cells
- C. guarding the entrance to the hive
- D. obtaining food and water for the hive

6..The word prompted in the passage is closest in meaning to

- A. able
- B. required
- C. programmed
- D. stimulated

7..The word explicit in the passage is closest in meaning to

- A. unusually creative
- B. clearly expressed
- C. ordinary
- D. necessary

8..Why does the author mention the famous waggle dance that bees perform in the passage

- A. To give an example of a signal shared between worker bees
- B. To indicate how worker bees are able to find good food sites
- C. To show how environmental conditions affect bee behavior
- D. To illustrate how the presence of food stimulates bee activity

9..The word imply in the passage is closest in meaning to

- A. introduce
- B. suggest
- C. state
- D. reveal

10..According to paragraph 5, which of the following describes forager bee activity when a hive has not been receiving sufficient food

- A. The foragers spend time looking for high-yield flower patches.
- B. The foragers return to the hive only infrequently.
- C. The foragers bring nectar from low-yield as well as high-yield flower patches.
- D. The foragers travel long distances looking for nectar.

11..According to paragraph 5, how does a returning forager bee know that the hive is well fed

- A. The food-storer bees signal the foragers to remain in the hive.
- B. The food-storer bees are producing honey, not storing food.
- C. The food-storer bees ignore forager bees that are bringing low-quality food.
- D. The food-storer bees are busy and not readily available to unload food.

12..In which of the following ways does paragraph 5 relate to paragraph 4

- A. Paragraph 5 continues the discussion of the location of food sites begun in paragraph 4.
- B. Paragraph 5 elaborates on the topic introduced in paragraph 4 about types of communication between bees.
- C. Paragraph 5 discusses the research that made the waggle dance in paragraph 4 famous.
- D. Paragraph 5 explains in further detail the foraging activities of worker bees mentioned in

paragraph 4.

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

Lacking the body parts to collect nectar and pollen, they have no function in the community once the opportunity to mate has passed.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Although a hive may contain as many as 20,000 bees, only a few of its females become queens and mate with drones.
- B. Once a queen has completed mating, she lays thousands of eggs each day, thereby supplying the colony with offspring for many years.
- C. Worker bees accomplish all the work necessary to maintain the hive by following an orderly pattern of taking on new tasks as they mature.
- D. Mature workers are typically required to play a variety of roles daily to ensure that the needs of the hive are being met.
- E. Rather than following orders from leaders, workers determine what needs to be done from environmental cues or from signals from other workers.
- F. Bees communicate most efficiently when they provide explicit signals to their fellow workers.

1. The transition from the historical period known as the Middle Paleolithic to the Upper Paleolithic around 40 to 35 thousand years ago (kya) represents one of the major developments in the prehistory of humankind. The basic features of this transition include more versatile stone implements and the use of antler, bone, and ivory for tools, figurative art, music, and personal decoration. So striking were the strides in human achievement during this period that it is sometimes referred to as the Upper Paleolithic Revolution.
2. Until recently it had been argued that the Upper Paleolithic Revolution was an archaeological phenomenon found only in Eurasia. The apparent lack of equivalent evidence in other regions suggested that a fundamental change had occurred in human intellectual development around 40 kya in Europe. The recent discovery in the Blombos Cave in South Africa of a block of decorated ochre and then sets of shell beads, dated to around 77 kya, opened up the debate. This supports other evidence of more versatile stone implements and bone tools found in Africa from the same period. Now the Upper Paleolithic Revolution is being seen as simply the most visible example of the evolving process of modern human behavior that had been developing over a much longer timescale.
3. This raises two further questions. First, what was happening to the human cognitive process during the 40,000 years or so between the creations in the Blombos Cave and the flourishing of human creativity in Europe around 35 kya, and second, was climate change a component. Climate change is associated with the sudden occurrence of creative activity in Europe at the beginning of the Upper Paleolithic.
4. The question of whether the sudden transition seen in Europe was built on earlier developments in Africa has been addressed at length by anthropologists Sally McBrearty and Alison Brooks. They argue that the whole issue of the Upper Paleolithic Revolution stems from a profound Eurocentric bias and a failure to appreciate the depth and breadth of the African archaeological record. In fact, many of the components of this revolution are found earlier in the African Middle Paleolithic tens of thousands of years before they appeared in Europe. These features include blade and microlithic technology, bone tools, increased geographic range, specialized hunting, exploitation of aquatic resources, long-distance exchange networks, systematic processing and use of pigment, and art and decoration. These items do not occur suddenly together as predicted by the revolutionary model, but at sites that are widely separated in space and time. This suggests a gradual assembling of the package of modern human behaviors in Africa and its later export to other regions of the Old World.
5. **The extraordinary range of rock art in Australia adds great weight to the idea that artistic creativity was part and parcel of the intellectual capacity of modern humans that migrated out of Africa around 70 kya.** The fact that these people almost certainly arrived in Australia before 60 kya and were, in any case, completely isolated from any evolutionary events that may have occurred in Europe around 40 kya makes this argument compelling.
6. The consequence of this analysis is that the question of the sudden emergence of creative activity that appears to constitute the Upper Paleolithic Revolution falls to the ground. The obvious explanation is that the gap between African developments and the subsequent better-known European events is a matter of the limitations of the archaeological record. This does not altogether cover the question of why there was the sudden flowering of creativity at

the beginning of the Upper Paleolithic in Europe. It may be that earlier creative efforts have either been lost in or have yet to emerge from the mists of time. Recent finds of decorative pierced shells dating from 43 kya or even earlier in caves in parts of western Asia near Europe may be examples of a process extending the evidence back in time. The creative flowering may also be a result of the climatic conditions at the time that governed the movement of modern humans into Europe. Following a period of extreme cold around 39 kya, a period of warming around 35 kya rendered the region more hospitable. As the ancestors of today's Europeans moved into a largely depopulated region, their presence in the archaeological record appeared revolutionary.

1..Why does the author mention a a block of decorated ochre and sets of shell beads

- A. To help make the point that archaeologists regard artistic creations as the highest kind of human achievement
- B. To illustrate how the discovery of certain objects makes the discovery of certain other objects more likely
- C. To give some of the evidence that has changed archaeologists' thinking about human intellectual development
- D. To help explain why archaeologists have been slow to recognize the importance of certain evidence available to them

2..How far back in time do the origins of the more versatile stone implements and bone tools found in Africa go

- A. To around 40 kya
- B. To around 77 kya
- C. To the time of the Upper Paleolithic Revolution
- D. To a time before modern human behavior had begun to evolve

3..The word equivalent in the passage is closest in meaning to

- A. comprehensible
- B. concrete
- C. comparable
- D. widely debated

4..According to paragraph 3, what do archaeologists want to know about the climate during the 40,000 years between around 75 kya and 35 kya

- A. Whether it was a reason that humans lived in caves
- B. Whether it was stable throughout that period
- C. Whether it changed in similar ways in Africa and in Europe
- D. Whether it was a factor in how the human cognitive process developed

5..According to paragraph 4, anthropologists McBrearty and Brooks argue that archaeologists' traditional understanding of the Upper Paleolithic Revolution was flawed because

- A. they underestimated available African archaeological evidence
- B. the archaeological evidence available to them contained errors
- C. they could not distinguish artistic creations from objects meant for practical use
- D. they based their judgments on the limited archaeological record available at that time

6..Anthropologists McBrearty and Brooks consider such components of the Upper Paleolithic Revolution as blade and microlithic technology to have

- A. first emerged in the Middle Paleolithic Period, in Africa rather than in Europe
- B. emerged in Europe independently in many different places at different times
- C. first emerged in Europe, then to have been further developed in Africa
- D. been part of a similarly sudden, but earlier, cultural revolution in Africa

7..The word exploitation in the passage is closest in meaning to

- A. maintenance
- B. discovery
- C. transference
- D. use

8..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- A. The idea that in modern humans creativity and intellect developed together is supported by ancient rock art ranging from Australia to Africa.
- B. Australian rock art strongly suggests that modern humans that migrated out of Africa around 70 kya already possessed artistic creativity.
- C. Since Australian rock art dated to around 70 kya shows that Australians possessed artistic creativity, Africans of that period probably possessed it, too.
- D. The wide range of Australian rock art strongly suggests that intellectual capacity and artistic creativity were connected as early as 70 kya.

9..What role does paragraph 5 play in relation to paragraph 4

- A. It accepts the arguments made in paragraph 4 but asserts that the supporting examples given are not the best ones available.
- B. It shows that the conclusion reached in paragraph 4 is almost certainly false.
- C. It presents additional reasons for accepting the position argued for in paragraph 4.
- D. It describes a possible alternative to the position developed in paragraph 4.

10..The word altogether in the passage is closest in meaning to

- A. even
- B. necessarily
- C. merely
- D. fully

11..The fact that decorative pierced shells were recently found in caves in parts of western Asia near Europe suggests that

- A. there may be evidence of earlier creative activity in Europe that has not yet been discovered
- B. the creative flowering may have originated in Asia, not in Africa
- C. there was really an Upper Paleolithic Revolution but it did not happen in Europe
- D. the people who used the decorative shells were unrelated to the people who inhabited Europe during the Upper Paleolithic

12..According to paragraph 6, the apparent sudden flowering of creativity in Europe might be related to a change of climate in which of the following ways

- A. People became very creative in order to survive climatic extremes.
- B. As the climate became warmer, ancient European populations moved around much less.
- C. Improving climatic conditions drew substantial human migration into Europe.
- D. Climatic conditions became more favorable to preserving the archaeological record.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

The adoption of these new materials and practices moved early human culture forward at a rapid pace.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Recent evidence that in Africa the transition from the Middle Paleolithic to the Upper Paleolithic was also marked by enormous progress strengthens the European evidence.
- B. The idea that events in Europe around 40 kya mark an advance in basic human capabilities has been discredited by African and Australian evidence that points to a much earlier date.
- C. Evidence suggests that the striking set of human accomplishments that emerged in Europe around 40 kya was built on earlier development in Africa.
- D. Europe was the center of the Upper Paleolithic Revolution not so much because individual innovations originated there, but because these innovations were creatively fitted together there.
- E. The people who settled Australia were, without a doubt, modern humans whose origins were in Africa, but whether they reached Australia before 60 kya has not been definitively established.
- F. It is still unclear why the entire package of modern human behaviors seems to show up in the European archaeological record so suddenly, although various answers have been proposed.

Passage 138 2015-08-22-3 Two Kinds of Lizards

1. Lizards can be divided into two types according to the way they look for food: sit-and-wait foragers and active foragers. Sit-and-wait lizards normally remain in one spot from which they can survey a broad area. These motionless lizards detect the movement of an insect visually and capture it with a quick run from their observation site. Sit-and-wait lizards may be most successful in detecting and capturing relatively large insects like beetles and grasshoppers. Active foragers, on the other hand, spend most of their time on the ground surface, moving steadily and poking their heads under fallen leaves and into crevices in the ground. These lizards apparently rely largely on chemical cues to detect insects, and they probably seek out local concentrations of prey such as termites. Active foragers appear to eat more insects than do lizards that are sit-and-wait predators. Thus, the different foraging behaviors of lizards lead to differences in their diets, even when the two kinds of lizards occur in the same habitat.
2. The different foraging modes also have different consequences for lizards regarding their exposure to predators. A lizard that spends 99 percent of its time resting motionless is relatively inconspicuous, whereas a lizard that spends most of its time moving is easily seen. Sit-and-wait lizards are probably most likely to be discovered and captured by predators that are active searchers, whereas widely foraging lizards are likely to be caught by sit-and-wait predators. Because of this difference, foraging modes may alternate at successive levels in the food chain: insects that move about may be captured by lizards that are sit-and-wait foragers, and those lizards may be eaten by active predators, whereas insects that are sedentary are more likely to be discovered by lizards that are active foragers, and those lizards may be caught by sit-and-wait predators.
3. The body forms of sit-and-wait foragers may reflect selective pressures different from those that act on active foragers. Sit-and-wait lizards are often stout bodied, short tailed, and colored to match their background. Many of these species have patterns of different-colored blotches that probably obscure the outlines of the lizard's body as it rests motionless on a rock or a tree trunk. Active foragers are usually slim and elongated with long tails, and they often have patterns of stripes that may produce optical illusions as they move. However, one predator-avoidance mechanism, the ability to break off their tails when they are seized by predators, does not differ among lizards with different foraging modes.
4. What physiological characteristics are necessary to support different foraging modes . The energy requirements of a quick motion that lasts for only a second or two are quite different from those of locomotion that is sustained nearly continuously for several hours. Sit-and-wait lizards and active foragers differ in their relative emphasis on the two ways that most animals use adenosine triphosphate (ATP) a molecule that transports energy within cells for activity and in how long that activity can be sustained. Sit-and-wait lizards move in brief spurts, and they rely largely on anaerobic metabolism to sustain their movements, namely the kind of metabolism that does not use oxygen. Anaerobic metabolism uses glycogen stored in the muscles and produces lactic acid as its end product. It is a way to synthesize ATP quickly (because the glycogen is already in the muscles), but it is not good for sustained activity because the glycogen is quickly exhausted and lactic acid inhibits cellular metabolism. Lizards that rely on anaerobic metabolism can make brief sprints but become exhausted when they are forced to run continuously. In contrast, aerobic metabolism uses glucose that is

carried to the muscles by the circulatory system, and it produces carbon dioxide and water as end products. Aerobic exercise can continue for long periods because the circulatory system brings more glucose and carries carbon dioxide away. As a result, active foragers can sustain activity for long periods without exhaustion. Active species of lizards have larger hearts and more red blood cells in their blood than do sit-and-wait species. As a result, each beat of the heart pumps more blood, and that blood carries more oxygen to the tissues of an active species than a sit-and-wait species.

1..The word steadily in the passage is closest in meaning to

- A. slowly
- B. at an unvarying rate
- C. cautiously
- D. without making noise

2..The word concentrations in the passage is closest in meaning to

- A. varieties
- B. dense groupings
- C. hidden traces
- D. shelters

3..According to paragraph 1, sit-and-wait foragers and active foragers are different in all of the following EXCEPT

- A. the methods that they primarily use to detect insects
- B. the number of insects they typically eat
- C. the habitats in which they can be found
- D. the amount of time they spend moving

4..According to paragraph 1, compared to the insects typically consumed by sit-and-wait lizards, insects typically consumed by active lizards

- A. move more quickly
- B. are generally found in smaller groups
- C. are often hidden from view
- D. spend more time on the surface of the ground

5..In paragraph 2, why does the author contrast the visibility of lizards that remain motionless most of the time with the visibility of lizards that move most of the time

- A. To show that it is possible for lizards to alternate their foraging modes at successive levels in the food chain
- B. To suggest that sit-and-wait lizards are more likely than active lizards to be attacked by

predators

- C. To explain why sit-and-wait lizards are more successful than active lizards at preying on insects
- D. To explain how a lizard's foraging strategy affects the type of predator likely to attack it

6..The word reflect in the passage is closest in meaning to

- A. indicate
- B. create
- C. resist
- D. require

7..The word obscure in the passage is closest in meaning to

- A. change
- B. hide
- C. fill
- D. expand

8..Paragraph 3 supports which of the following ideas about active-forager lizards

- A. They are less likely to break off their tails when seized by predators than sit-and-wait lizards are.
- B. They tend to vary widely in their individual colors.
- C. They lack well-developed mechanisms for avoiding predators.
- D. They may be easier to identify when they are not moving than when they are moving.

9..According to paragraph 4, compared with active lizards, the movements of sit-and-wait lizards are

- A. more sudden
- B. more sustained
- C. more predictable
- D. more frequent

10..According to paragraph 4, all of the following are true about anaerobic metabolism in lizards EXCEPT:

- A. It uses the glycogen in the muscles of lizards.
- B. It produces lactic acid that interferes with metabolism within the cells of lizards.
- C. It allows lizards to maintain their foraging activity over a long period of time.
- D. It allows lizards to run fast for short periods of time.

11..Which of the following can be inferred from paragraph 4 about sit-and-wait lizards

- A. They have less glycogen than active-forager lizards.
- B. They have highly variable heart rates.
- C. They have metabolic systems that do not depend on the efficient removal of carbon dioxide.
- D. They synthesize adenosine triphosphate continuously.

12..According to paragraph 4, how are active-foraging lizards and sit-and-wait lizards different from each other

- A. Sit-and-wait lizards tend to have more red blood cells in their blood than active-foraging lizards do.
- B. Active-foraging lizards' blood carries less oxygen to the tissues than sit-and-wait foraging lizards' blood does.
- C. Sit-and-wait lizards carry out cellular metabolism more efficiently than active-foraging lizards do.
- D. Active-foraging lizards tend to have larger hearts than sit-and-wait lizards do.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Such visual effects make it harder for predators to catch these moving targets.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.
Answer Choices

- A. Sit-and-wait lizards use brief spurts of energy to forage, while the movements of active lizards are more sustained.
- B. Lizards that are active foragers are able to detect insects even when those insects are stationary, but sit-and-wait lizards are able to detect only moving insects.
- C. Sit-and-wait lizards tend to have predators that are active, while active lizards tend to have sit-and-wait predators.
- D. Sit-and-wait lizards have less need for bodies that match their environment than active lizards do because of the ease with which they can break off their tails to escape from predators.
- E. The body forms and metabolic functions of the two types of lizards differ in ways that support the requirements of their different foraging behaviors.
- F. Sit-and-wait lizards derive energy for movement from anaerobic metabolism alone, while active lizards rely almost equally on aerobic and anaerobic metabolisms.

Passage 139 2015-09-05-2 The Cambrian Explosion

1. The earliest fossil evidence for eukaryotes complex organisms whose cells contain a distinct nucleus dates to only about 1.2 billion years ago. The fossil record suggests that animal evolution progressed slowly, with relatively little change seen between fossils from 1.2 billion years ago and those from a half-billion years later. But then something quite dramatic happened as can be judged by the many different animal groups that suddenly appear in the fossil record.
2. Biologists classify animals according to their basic body plans. For example, the basic body plan shared by mammals and reptiles is fundamentally different from that of insects. Animals are grouped by body plan into what biologists call phyla. Mammals and reptiles both belong to the single phylum Chordata, which includes animals with internal skeletons. Insects, crabs, and spiders belong to the phylum Arthropoda, which contains animals with body features such as jointed legs, an external skeleton, and segmented bodies. Classifying animals into phyla is an ongoing project for biologists, but modern animals appear to comprise about 30 different phyla, each representing a different body plan.
3. Remarkably, nearly all of these different body plans, plus a few others that have gone extinct, make their first known appearance in the geological record during a period spanning only about 40 million years less than about 1 percent of Earth's history. This remarkable flowering of animal diversity appears to have begun about 545 million years ago, which corresponds to the start of the Cambrian period. Hence it is called the Cambrian explosion.
4. The fact that the Cambrian explosion marks the only major diversification of body plans in the geological record presents us with two important and related questions: Why, so long after the origin of eukaryotes, did the pace of evolution suddenly accelerate dramatically at the beginning of the Cambrian, and why hasn't there been another period of similarly explosive diversification since then?
5. We can identify at least four factors that might have contributed to the Cambrian explosion. First, the oxygen level in our atmosphere may have remained well below its present level until about the time of the Cambrian explosion. Thus, the rapid diversification in animal life may have occurred at least in part because oxygen reached a critical level for the survival of larger and more energy-intensive life forms.
6. A second factor that may have been important was the evolution of genetic complexity. As eukaryotes evolved, they developed more and more genetic variation in their DNA. Some scientists believe that the Cambrian explosion marks the point at which organisms developed certain kinds of genes (homeobox genes) that control body form and that could be combined in different ways, allowing the evolution of a great diversity of forms over time.
7. A third factor may have been climate change. Geological evidence points to a series of episodes in which Earth froze over before the Cambrian began. The extreme climate conditions of these episodes eliminated many species, leaving a wide array of ecological niches available into which new species could rapidly evolve when climate conditions eased at the beginning of the Cambrian.
8. A fourth factor may have been the absence of efficient predators. **Early predatory animals were probably not very sophisticated, so some evolving animals that later might have been eliminated by predation were given a chance to survive, making the beginning of the Cambrian period a window of opportunity for many different adaptations to**

establish themselves in the environment.

9. This last idea may partly explain why no similar explosion of diversity has taken place since the Cambrian: once predators were efficient and widespread, it may have been virtually impossible for animals with entirely new body forms to find an environmental niche in which they could escape predation. Or it may be that while more body plans may have been possible at some early point in evolution, it was not possible to evolve into those other body plans from the body plans that evolved in the Cambrian. Or perhaps the various body forms that arose during the Cambrian explosion represent the full range of forms possible given the basic genetic resources that characterize all Earth's organisms. In any case, no fundamentally new body forms have emerged since the Cambrian explosion.

1..Paragraph 1 implies which of the following about evolutionary change

- A. Eukaryotes have a very slow rate of evolution.
- B. The fossil record of evolutionary change is incomplete for the first half-billion years of animal evolution.
- C. Evolution has not always proceeded at the same rate.
- D. Evolutionary rates of change in animals were slowing down considerably before a dramatic reversal happened 1.2 billion years ago.

2..The word ongoing in the passage is closest in meaning to

- A. uncertain
- B. full-time
- C. important
- D. continuing

3..In paragraph 2, why does the author provide the information that Arthropoda represents animals with features like jointed legs, an external skeleton, and segmented body parts

- A. To indicate basic physical differences among insects, crabs, and spiders
- B. To illustrate the types of physical characteristics considered when classifying animals
- C. To show the complexity of features that have evolved in organisms
- D. To demonstrate that some phyla include a wider range of body plans than others do

4..The phrase corresponds to in the passage is closest in meaning to

- A. causes
- B. occurs before
- C. differs from
- D. matches

5..According to paragraphs 3 and 4, what was remarkable about the Cambrian explosion

- A. That the evolution of species occurred so soon after the first eukaryotes appeared
- B. That most of the known animal types appeared in a relatively short period in history
- C. That many of the animal types that appeared in the period have survived until today
- D. That the pace of evolution slowed before it accelerated

6..The function of the two questions in paragraph 4 is to

- A. recognize two common questions that cannot be addressed within the passage
- B. present the two different points of view contrasted in the passage
- C. provide important objections to the central idea of the passage
- D. indicate two important questions that will be explored in the passage

7..Paragraph 5 implies which of the following about oxygen

- A. It was not essential for the life forms that appeared before the Cambrian period.
- B. It has remained at relatively the same level since the beginning of the fossil record.
- C. Its changes in levels are associated with animal extinctions.
- D. Its levels before the Cambrian period were too low for large animals to survive.

8..The word complexity in the passage is closest in meaning to

- A. sophistication
- B. adaptation
- C. improvement
- D. variation

9..The word allowing in the passage is closest in meaning to

- A. resulting in
- B. making possible
- C. preceding
- D. spreading

10..According to paragraph 7, all of the following occurred before the Cambrian began EXCEPT:

- A. Almost all of Earth froze over.
- B. New ecological niches were filled by new species.
- C. A series of extreme climate episodes occurred.
- D. Many species became extinct.

11..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Predators at the beginning of the Cambrian period had to become more sophisticated in order to survive in environments with newly adapted animals.
- B. New adaptations had an opportunity to survive at the beginning of the Cambrian period because early predators were not yet sophisticated enough to eliminate the animals with these adaptations.
- C. Early predatory animals lacked sophisticated adaptations because they did not develop them during a window of opportunity in the early Cambrian period.
- D. Early predators had an opportunity at the beginning of the Cambrian period to remove new adaptations before they established themselves in the environment.

12..Paragraph 9 suggests all of the following possible explanations for the uniqueness of the Cambrian explosion EXCEPT

- A. the inability of later animals to evolve body plans different from those that appeared during the Cambrian period
- B. the post-Cambrian appearance of efficient predators occupying nearly every environmental niche
- C. the decline in the number of habitats having sufficient resources to support the rapid evolution of new species
- D. the limited range of genetically possible body types

13..Look at the four squares that indicate where the following sentence could be added to the passage.

After all, evolution of body structure can act only on the structure that already exists.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text.

Answer Choices

- A. Biologists want to find out why the rapid diversification of body forms did not happen soon after the appearance of eukaryotes around 1.2 billion years ago.
- B. Biologists classify animals according to their body plans into phyla such as Chordata, animals with a backbone, and Arthropoda, animals with segmented exoskeletons.
- C. The Cambrian explosion was a unique episode in the history of evolution that produced nearly all of the 30 or so animal body plans that have ever been seen.
- D. The Cambrian explosion may have been aided by genes that could yield a variety of body forms and the inability of early predators to eliminate the new forms.
- E. Once predators became efficient after the Cambrian explosion, they were able to eliminate any animals that began to evolve a new body plan.
- F. At the beginning of the Cambrian, an increase in oxygen needed for animal growth and the return of a hospitable climate may have contributed to the Cambrian explosion.

1. Some blind elephantnose fish produce weak electric signals that are used for detecting objects in their surroundings a phenomenon called active electrolocation. These fish have specialized electric organs that discharge either in pulses or in a wave-like fashion, depending on the species. **Although discharges follow one another almost continuously throughout the life of the fish, their power level is much too low to be detected by human handlers but potent enough to create a stable electric field around the body of the fish.** When an object enters into this electric field, it causes distortions in the current that are detected by electroreceptor organs distributed over the fish's skin.
2. A weak electric system may have several uses, including the exploration of novel environments. For example, blind elephantnose fish can easily find the only opening that allows them to cross through a newly installed partition within their aquarium, even though they cannot see it with their eyes. Their electric sense must be implicated because when these individuals become electrically silent (unable to use their electric system through denervation of their electric organs), they can no longer find the opening.
3. During the 1970s, biologists became interested in the role of the weak electric system not only as a means of electrolocation but also as a means of electrical communication between individual fish. Communication is possible because the rate and waveform of the electric discharges can vary between species, between sexes, between individuals, or even between situations in the same individual. Moreover, some fish can temporarily interrupt their normally continuous train of discharges, and these pauses can be full of meaning. The effective range of communication by electric signals can reach a little over 1 meter depending on water resistance.
4. In terms of functions, electric communication is strikingly similar to acoustical vocalization (vocal sounds). Some of these functions are concerned with reproductive activity. In some species, males switch to new electric calls during courtship, resuming their regular programming only after the mating season is over. In species in which each sex has its own distinctive pattern of discharges, females are attracted to the pattern of males, and males to the pattern of females. Females can even be induced to release their eggs in the vicinity of electrodes that imitate a male signal 'Dthe spark of love. As expected, through natural selection, both males and females prefer the electric pattern of their own species to that of other species.
5. Other functions relate to aggression. Aggressive individuals often precede their attacks with an increase in discharge rate, whereas submissive fish may stop emitting altogether. This submissive behavior seems to work. Researchers have found that individuals rendered electrically silent through denervation of their electric organs are seldom attacked by dominant fish. Finally, individual recognition can also be based on electric signatures. In banded knifefish, territory neighbors recognize each other through individually distinctive discharge waveforms.
6. The fact that weak electric fish can use their electric sense to communicate with one another leads to an interesting question: How can a fish distinguish between its own electric bursts and those from another fish In blind elephantnose fish, the problem is solved by the presence of two types of electroreceptors. One of these two types is automatically and briefly shut down each time the fish discharges. Therefore, any signal picked up by these

electroreceptors has to come from another animal. Elephantnose fish also have the habit of echoing the discharges of other individuals. They discharge their own electric organ a fixed time after sensing the electric signal of another fish. This response time is extremely short—approximately 12 milliseconds—probably the most rapid form of communication in the animal kingdom.

7. Knifefish also display a peculiar behavior called the jamming avoidance response. This response allows knifefish to prevent interference with their electric system when they meet other knifefish. In order to avoid confusion, an electric fish must somehow keep track of the discharge rate of another knifefish while remaining aware of its own. If the two rates are too close, each fish alters its frequency of discharge so as to widen the gap between the two. In a sense, they do not want to get their wires crossed. In the laboratory, it is possible, using artificial signals, to force a knifefish to decrease its frequency of firing just by exposing it to a high but slowly decreasing signal rate or to increase its frequency of firing by switching to a low but slowly rising signal rate.

1..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- A. Throughout the life of the fish, its electric discharges vary in power from undetectable to fairly potent.
- B. Nearly continuous discharges undetectable by humans create a stable, low-level electric field around the body of the fish.
- C. Human handlers cannot feel the electric discharges because the discharges are at such low power levels.
- D. The discharges are so weak that they would have to be fired almost continuously to create a detectable electric field.

2..The word distortions in the passage is closest in meaning to

- A. decreases
- B. signals
- C. concentrations
- D. irregularities

3..Paragraph 2 describes an experiment with elephantnose fish that were electrically silent and therefore unable to

- A. distinguish between new and familiar objects in their aquarium
- B. swim around the partition in their aquarium
- C. find their way through a hole in a barrier
- D. see with their eyes

4..All of the following are mentioned in paragraph 3 as components of communication in electric fish EXCEPT

- A. the rate of electric discharges
- B. the direction in which discharges travel
- C. the waveform of electric discharges
- D. pauses between electric discharges

5..The word temporarily in the passage is closest in meaning to

- A. deliberately
- B. suddenly
- C. partially
- D. briefly

6..The author provides the information that Females can even be induced to release their eggs in the vicinity of electrodes that imitate a male signal in order to

- A. emphasize the importance of electric signals in some fish mating behavior
- B. argue that many fish are not very particular when choosing a mate
- C. show that electric signals alone do not provide adequate information for mate selection
- D. imply that identifying a potential mate by its electric signal is more difficult than it might seem

7..All of the following are mentioned in paragraphs 4 and 5 as ways fish communicate using discharge patterns EXCEPT:

- A. Some fish increase their rate of discharge right before they attack.
- B. Some fish reproduce the electric pattern of another species to hide from a predator.
- C. Some male fish switch to a different electric call during mating season.
- D. Some fish temporarily stop discharging as a sign of submission.

8..In saying that Aggressive individuals often precede their attacks with an increase in discharge rate the author means that

- A. aggressive individuals often prepare for their attacks by increasing the discharge rate
- B. aggressive individuals often intensify their attacks by increasing the discharge rate
- C. attacks by aggressive individuals often increase when there is an increase in the discharge rate
- D. an increase in the discharge rate often occurs before aggressive individuals attack

9..According to paragraph 5, becoming electrically silent can affect a fish by

- A. causing it to become more aggressive
- B. making it a more frequent target of attacks
- C. damaging its organs for sending electric signals
- D. making it less likely to be attacked by dominant fish

10..According to paragraph 6, one of the types of electroreceptors of the elephantnose fish shuts down at times in order to

- A. avoid confusing its own signals with those of other fish
- B. prevent the receptor from becoming damaged
- C. distinguish between signals of short and long duration
- D. enable the fish to produce echoes of the discharges of other fish

11..The word peculiar in the passage is closest in meaning to

- A. clever
- B. frequent
- C. unusual
- D. helpful

12..According to paragraph 7, what happens when a knifefish comes into contact with another knifefish that has a similar discharge rate

- A. Both fish speed up their discharge rate, perhaps out of a sense of competition.
- B. Both fish slow down their discharge rate so that they are better able to hear each other.
- C. One fish increases its discharge rate while the other decreases it, making the signals less similar.
- D. Each fish slowly changes its discharge rate until they match up perfectly.

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

Each fish in an encounter is discharging electric signals at a specific rate.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Some fish have special organs that produce a weak electric field around their body and have receptor organs that allow them to detect disruptions to this field.
- B. Electrolocation is used primarily by fish who are blind, whereas electric communication is used by fish who are not equipped for acoustical vocalization.
- C. Research suggests that the effective range of communication by electric signals can reach a little over 1 meter, depending on water resistance.
- D. Distinctive discharge patterns allow fish with a weak electric system to identify each other by sex or species, or even recognize particular individuals.
- E. Most fish with a weak electric system have two forms of electroreceptors so that they can detect both very fast and very slow discharge rates.
- F. Fish can deliberately slow, speed up, or pause their rate of discharge in order to communicate and to avoid confusing their own signals with those of another fish.

1. For 500 years after the fall of the Western Roman Empire in 476 A.D., a period known as the early Middle Ages, Europe endured an age of political instability, economic decline, and reduced population. But as the millennium approached, the situation began to improve. Toward the end of the tenth century, an increase in the amount of crop-producing land was accompanied by an increase in population, with the potential for that number to rise even higher. The increase in agricultural production came about as a result of a combination of factors, the most prominent of which were changing methods of field management and improvements in agricultural technology.
2. For much of the early Middle Ages, peasants continued the Roman practice of dividing their fields in two — leaving one fallow, or uncultivated, for a year, and planting their crops in the other half. Fallow land restored its nutrients, but the practice meant that half the land produced nothing every year. In southern Europe with its drier climate this system of two-field crop rotation continued, but in northern Europe, peasants improved on this system by dividing their land into three parts. One they left fallow, another they planted in the spring, and the third they planted with winter crops. This three-field crop rotation, dependent on more rainfall than southern Europe received, meant that two-thirds instead of one-half of a peasant's land was under production in one year.
3. Related to the changes in crop rotation were improvements in plows and animal harnessing. More land under cultivation spurred experimentation in the construction of plows. **Peasants attached wheels to their plows, which made it easier for oxen to pull them through the heavier, wetter soil of northern Europe, and made it possible for a plow to move more quickly down a row provided it had a speedy animal pulling it.**
4. Oxen are slow and unintelligent compared to horses, but peasants could not use horses to pull plows until they devised a different kind of harnessing than the strap that circled an ox's neck. With a harness resting on its shoulders instead of its neck, a horse could be used to plow, and horses could walk more quickly and work longer hours than oxen. They also required less guidance, since they understood verbal signals to turn or to stop. Heavier, wheeled plows pulled by suitably harnessed horses meant that peasants could work more land in a day than ever before. Whether an increase in population across western Europe, but particularly in the north, stimulated innovations or whether such innovations contributed to a rise in population, the cumulative effect of these changes in agriculture was apparent in the tenth century. Conditions in Europe were ripe for an economic and cultural upswing.
5. Even before trade with the eastern Mediterranean increased starting in the twelfth century, trade and towns were on the rise. Travel was still dangerous, but merchants were willing to risk transporting goods over long distances. By the late thirteenth century, a few merchants from Italy had even reached China. Greater surpluses in crops meant people had more to sell at market. More people and goods led to regularly held markets in the most populated location in a region. It would be impossible to say whether trade gave rise to towns or vice versa. What is clear is that each fostered the other in conditions of greater social stability.
6. Travel on trade routes increased, and some towns sprang up to provide rest and refreshment to traders. The distance between towns often corresponded to the distance that traders could cover in a day. Merchants kept their eyes open for customers with money to spend. The residences of kings, nobles, and powerful officials became sites of markets for local and

long-distance traders. In Champagne, in northeastern France, six large annual markets attracted merchants from all over Europe in the twelfth century. Their different currencies prompted the first development of banking techniques. With the use of coins now the norm, money changers daily posted changing exchange rates so that merchants would know the worth of their coins in relation to the worth of other merchants' coins. By 1300, trade had transformed life for the better throughout western Europe.

1..Paragraph 2 suggests that the land-management practices developed in the north could not have been adopted in southern Europe because

- A. the southern climate was too dry
- B. southern farmers were too strongly tied to traditional Roman farming practices
- C. the new practice would have required much additional farmland, which was lacking in the south
- D. southern farmers had already developed a new crop-rotation system

2..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. Peasants had to attach wheels to their plows to make it possible for their oxen to plow the heavy, wet soil of northern Europe.
- B. With speedy oxen pulling the plow, peasants were able to plow the heavy soil of northern Europe.
- C. Adding wheels to a plow made plowing the heavy soil of northern Europe much easier and potentially faster.
- D. Attaching wheels to a plow made it necessary to use speedier animals.

3..According to paragraph 4, what initially kept farmers from using horses to pull their plows

- A. Horses moved too quickly for traditional farmwork.
- B. Farmers did not have access to many horses.
- C. Horses were thought to be too weak to pull plows.
- D. The traditional harness was unsuitable for horses.

4..According to paragraph 4, farmers found horses to be better than oxen for pulling plows for all of the following reasons EXCEPT:

- A. Horses are smarter than oxen.
- B. Horses can pull plows faster.
- C. Horses are able to plow for longer periods of time.
- D. Horses do not need verbal signals to turn and stop.

5..The word cumulative in the passage is closest in meaning to

- A. eventual
- B. positive
- C. combined
- D. practical

6..The word apparent in the passage is closest in meaning to

- A. welcome
- B. noticeable
- C. predictable
- D. widespread

7..The word each in the passage refers to which of the following

- A. people and goods
- B. crop surpluses and markets
- C. trade and towns
- D. merchants from Italy and merchants from China

8..In paragraph 5, why does the author state that a few Italian merchants had reached China

- A. To emphasize how much farther European merchants traveled during this time period than in previous centuries
- B. To support the idea that Chinese goods were important for European merchants
- C. To show how European merchants influenced Chinese markets
- D. To indicate the wide-ranging tastes of European consumers during the thirteenth century

9..The phrase corresponded to in the passage is closest in meaning to

- A. approached
- B. equaled
- C. exceeded
- D. determined

10..Paragraph 6 suggests that markets developed near the residences of kings, nobles, and powerful officials for which of the following reasons

- A. Nobles and officials made it legal for merchants to sell their goods.
- B. Merchants could sell their goods in relative safety at these sites.
- C. Nobles and officials established their residences along traditional trade routes.
- D. Merchants could sell their goods more easily to wealthy nobles and officials.

11..The phrase the norm in the passage is closest in meaning to

- A. acceptable
- B. desirable
- C. common
- D. available

12..According to paragraph 6, what led to the development of the first banking techniques

- A. Merchants from different European regions needed to compare the value of their currency at large markets.
- B. Merchants participating in large markets wanted a place to safely store the money they made from their sales.
- C. Organizers of large markets developed methods for determining the value of goods from long-distance traders.
- D. Merchants needed to borrow money to buy the goods of other merchants at large markets.

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

A stronger part of the body had to bear the strain of pulling for the horse to become a useful farm animal.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Peasants in southern Europe developed a three-field system of crop rotation that was based on agricultural practices used by Roman farmers.
- B. By rotating their land, using three fields rather than the traditional two, many northern farmers were able to harvest twice during the year, significantly increasing their annual production.
- C. Agricultural innovations led to population growth everywhere in Europe, and the resulting towns were the reason that trade increased.
- D. By using wheeled plows and replacing horses with oxen, farmers could plow the soil of Northern Europe more quickly, contributing to economic improvements.
- E. As European town markets expanded to become trade centers selling goods from distant locations, the social and economic power of merchants increased.
- F. A growing system of trade that included long-distance merchants and large annual markets contributed to innovations in economic practice and the flourishing of town culture.

Passage 142 2015-09-12-2 Gliding and Soaring

1. Gliding is gravity-powered flight where the movement of the glider has a downward tilt. But many birds are capable of ascending without flapping their wings, and this is called soaring. Birds usually soar by finding air that is rising as fast as or faster than the gliding bird's sinking speed. For example, a turkey vulture might glide with a sinking speed of about 0.8 meters per second. If the vulture can find a place where the air is rising at 0.8 meters per second, it will be able to maintain a constant altitude. If it finds air rising faster than that, it will be able to climb.
2. Two common processes produce updrafts, or rising air. When heated air rises, it is called a thermal, and when wind blows up a hill or over a large obstacle, it is called ridge lift or slope lift. Thermals occur when the Sun heats some parts of the ground more than others. For example, a freshly plowed field may heat up faster than an adjacent meadow. The warm ground heats the air above it, and the air starts to rise. As the warm air rises, it is replaced by cool air from the surrounding terrain, and this new air is heated until it rises. Thermals may be continuous chimneys of rising air, or a series of discrete, doughnut-shaped bubbles (ring thermals) formed at intervals by the warmed ground.
3. If they could be made visible, ring thermals would look like giant, rising smoke rings. Some airplane pilots and biologists disagree about the exact form of continuous thermal chimneys. Pilots have traditionally interpreted thermals as large, tall columns of rising air, usually with a cumulus (white, fluffy) cloud marking the top of the column. In contrast, observers of animal flight find only small, localized thermal chimneys, which usually take the form of dust devils, which are small columnar thermals with intense rotation. Colin Pennycuik, a prolific researcher on bird flight, discounts thermal chimneys and recognizes only ring thermals as sources of large-scale, long-lasting updrafts. In any case, thermals can rise 2 or 3 kilometers above the ground. Also, they tend to increase in size and intensity as they rise, sometimes reaching over 1,000 meters in diameter. **Thermals are usually capped by a cloud, because the upper limit of a thermal is set by the altitude where the temperature is low enough to condense water vapor in the thermal, which cools the air and forms a cloud.**
4. As long as the upward speed of the thermal is greater than the sinking speed of a glider, the glider will ascend in the thermal. Of course, the glider will quickly fly out of the thermal if it flies in a straight line, so it must circle to stay in the rising air. (A glider should stay on the inside of the ring, because the air on the outer edge of the ring is actually rolling downward.) Imagine a vulture ascending to 1,500 meters above the ground by circling in a ring thermal. From this height, it will be able to fly out of the thermal and glide for about 30 minutes (traveling over 23 kilometers) before it runs out of altitude and needs to either start flapping, find another thermal, or land. Many soaring birds use just this pattern: climbing up in a thermal, gliding a long distance, then finding another thermal in which to soar. This type of flight is an efficient way to cover long distances at a low energy cost, making it a handy way to migrate or search for food.
5. Slope soaring is useful when wind blows upward along a slope. The speed of the wind's upward motion can be calculated in the same manner that the sinking speed of a glider is calculated. If the upward speed of this wind is greater than or equal to the sinking speed of a glider, the glider will be able to maintain altitude. Such ridge lift has a characteristic that is both an advantage and a disadvantage: ridge lift is usually predictably tied to a particular

slope, so it is easy to find. But it is usable only in that fixed, local area.

1..Which of the following plays a role in helping a bird to soar

- A. Gravity
- B. Wing flapping
- C. Gliding
- D. Rising air

2..In paragraph 1, why does the author discuss the example of a turkey vulture in flight

- A. To explain why birds sink unless they find currents of rising air
- B. To provide information about sinking and soaring speeds typical for large birds
- C. To illustrate why gliding birds seek out rapidly rising air currents
- D. To clarify the relationship between sinking speed and the speed of rising air needed to soar

3..The word adjacent in the passage is closest in meaning to

- A. open
- B. nearby
- C. densely covered
- D. sunny

4..The phrase at intervals in the passage is closest in meaning to

- A. at low elevations
- B. in areas without vegetation
- C. periodically
- D. unevenly

5..The word discrete in the passage is closest in meaning to

- A. separate
- B. rising
- C. rotating
- D. flexible

6..Paragraph 2 implies that which of the following is a factor that contributes to the formation of thermals

- A. In some areas, characteristics of the ground cause it to heat more than surrounding areas.
- B. The sun heats the air, causing it to start to rise.
- C. Warm air continues to rise until the ground beneath it is cooled by colder air.
- D. Uneven heating of the air causes it to rotate as it rises.

7..The word prolific in the passage is closest in meaning to

- A. skillful
- B. experienced
- C. highly productive
- D. highly respected

8..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- A. Thermals capped by clouds have upper limits when the temperature gets low enough to condense water vapor.
- B. Thermals usually stop rising when they reach a cloud, because thermals are limited in the amount of water vapor they can hold before the vapor condenses.
- C. Most thermals have clouds at their upper end, because thermals stop rising in places cool enough to condense the thermal's water vapor.
- D. Thermals usually rise to the same altitude as clouds, because cool air in clouds allows the water vapor to condense.

9..Which TWO statements below about thermal chimneys would the biologists discussed in paragraph 3 regard as likely true? To receive credit, you must select TWO answers.

- A. They cover only a small area.
- B. They sometimes contain dust devils.
- C. They usually spin very quickly.
- D. They usually end in a fluffy cloud.

10..Paragraph 3 indicates that all of the following are true of thermals EXCEPT:

- A. Air movement is more intense at their higher reaches.
- B. They can rise above cloud level.
- C. They usually become wider as they rise.
- D. They sometimes rise more than a kilometer from Earth's surface.

11..According to paragraph 4, a bird that flies by alternating between soaring and gliding gains which of the following advantages

- A. It can often cover short distances faster than it could by flapping its wings to travel.
- B. It is able to travel a long way without using a lot of energy.
- C. It can find thermals more quickly than it could by flapping its wings to travel.
- D. It can stay inside the thermal regardless of how thermal speeds change.

12..Which of the following is discussed in paragraph 5 as a disadvantage of slope soaring as compared to thermal soaring

- A. Slope soaring does not allow a bird to remain at a constant altitude.
- B. Gliders tend to travel at slower speeds during slope soaring than during thermal soaring.
- C. Slope soaring is limited to the area where the particular slope is located.
- D. It is difficult to predict the location of an area suitable for slope soaring.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

Soaring uses energy such as rising air currents to allow birds to increase time aloft.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Birds can soar by finding an updraft, which is created either by air rising from ground that is warmer than the surrounding ground or by wind blowing up a hill or over a large obstacle.
- B. Soaring is possible in updrafts in which the air's upward speed is greater than the bird's sinking speed.
- C. Birds flying over a flat meadow often need to flap their wings in order to stay at the same altitude because thermal chimneys and ring thermals typically do not form in such areas.
- D. Birds can travel in an energy-efficient way by soaring upward in thermals and then leaving the thermal and gliding to lower altitudes.
- E. Birds that need to fly over long distances generally do not use thermals because they limit the distance the birds can soar.
- F. Thermals that form as a result of ridges have predictable speeds that depend on the incline of the particular slope and the climate of the local area.

Passage 143 2015-09-12-3 Pleistocene Extinctions

1. At the end of the Pleistocene (roughly 11,500 years ago), many large mammals became extinct. Large mammals in the Americas and Australia were particularly hard-hit. In Australia, 15 of the continent's 16 of large mammals died out; North America lost 33 of 45 genera of large mammals, and in South America 46 of 58 such genera went extinct. In contrast, Europe lost only 7 of 23 such genera, and in Africa south of the Sahara only 2 of 44 died out. What caused these extinctions? Why did these extinctions eliminate mostly large mammals? Why were the extinctions most severe in Australia and the Americas? No completely satisfactory explanation exists, but two competing hypotheses are currently being debated. One holds that rapid climatic changes at the end of the Pleistocene caused extinctions, whereas another, called prehistoric overkill, holds that human hunters were responsible.
2. Rapid changes in climate and vegetation occurred over much of Earth's surface during the late Pleistocene, as glaciers began retreating. The North American and northern Eurasian open steppe tundras (treeless and permanently frozen land areas) were replaced by conifer and broadleaf forests as warmer and wetter conditions prevailed. The Arctic region changed from a productive herbaceous one that supported a variety of large mammals, to a relatively barren waterlogged tundra that supported a far sparser fauna. The southwestern United States region also changed from a moist area with numerous lakes, where saber-tooth cats, giant ground sloths, and mammoths roamed, to a semiarid environment unable to support a diverse fauna of large mammals.
3. Rapid changes in climate and vegetation can certainly affect animal populations, but the climate hypothesis presents several problems. First, why did the large mammals not migrate to more suitable habitats as the climate and vegetation changed? After all, many other animal species did. For example, reindeer and the arctic fox lived in southern France during the last glaciation and migrated to the Arctic when the climate became warmer.
4. The second argument against the climatic hypothesis is the apparent lack of correlation between extinctions and the earlier glacial advances and retreats throughout the Pleistocene Epoch. Previous changes in climate were not marked by episodes of mass extinctions.
5. Proponents of the prehistoric overkill hypothesis argue that the mass extinctions in North and South America and Australia coincided closely with the arrival of humans. Perhaps hunters had a tremendous impact on the faunas of North and South America about 11,000 years ago because the animals had no previous experience with humans. The same thing happened much earlier in Australia soon after people arrived about 40,000 years ago. No large-scale extinctions occurred in Africa and most of Europe because animals in those regions had long been familiar with humans.
6. One problem with the prehistoric overkill hypothesis is that archaeological evidence indicates the early human inhabitants of North and South America, as well as Australia, probably lived in small, scattered communities, gathering food and hunting. How could a few hunters destroy so many species of large mammals? However, it is true that humans have caused major extinctions on oceanic islands. For example, in a period of about 600 years after arriving in New Zealand, humans exterminated several species of the large, flightless birds called moas. A second problem is that present-day hunters concentrate on smaller, abundant, and less dangerous animals. The remains of horses, reindeer, and other small animals are found in many prehistoric sites in Europe, whereas mammoth and woolly rhinoceros remains

are scarce. Finally, few human artifacts are found among the remains of extinct animals in North and South America, and there is usually little evidence that the animals were hunted. Countering this argument is the assertion that the impact on the previously unhunted fauna was so swift as to leave little evidence.

7. The reason for the extinctions of large Pleistocene mammals is still unresolved and probably will be for some time. It may turn out that the extinctions resulted from a combination of different circumstances. Populations that were already under stress from climate changes were perhaps more vulnerable to hunting, especially if smaller females and young animals were the preferred targets.

1..According to paragraph 1, which of the following groups of mammals experienced a high extinction rate at the end of the Pleistocene

- A. Large mammals living in North America
- B. Small mammals living in South America
- C. Large mammals living in South Africa
- D. Large mammals living in Europe

2..According to paragraph 1, researchers have been able to answer which of the following questions about late Pleistocene extinctions

- A. Why did some parts of the world experience more extinctions than others
- B. Which parts of the world experienced the greatest number of extinctions
- C. Did the large mammals of the Americas or Australia become extinct first
- D. How rapidly did the climate change during the extinctions

3..What can be inferred from paragraph 1 about the extinctions that occurred at the end of the Pleistocene

- A. They were caused by a single factor.
- B. They had relatively little impact on small mammals.
- C. They wiped out nearly all of the world's large mammal species.
- D. They occurred slowly over a period of thousands of years.

4..The word sparser in the passage is closest in meaning to

- A. more thinly distributed
- B. more threatened
- C. less adapted
- D. less mobile

5..All of the following are mentioned in paragraph 2 as changes that occurred during the late Pleistocene EXCEPT:

- A. The Eurasian tundras became more forested as both temperature and rainfall increased.
- B. The Arctic region became less productive, and much of its fauna and flora began to disappear.
- C. The southwestern United States became much drier, resulting in a decline in species diversity.
- D. The North American open steppe tundras became cooler, resulting in a decrease in vegetation.

6..In paragraph 3, why does the author point out that some animals living in southern France migrated to the Arctic when the climate became warmer

- A. To show that more suitable habitats existed at the time that the megafauna became extinct
- B. To question the idea that the megafauna were able to migrate to more suitable habitats
- C. To provide evidence that weakens the climate hypothesis for the megafauna extinctions
- D. To argue that smaller animals are more successful at adapting to rapid changes in climate

7..The word episodes in the passage is closest in meaning to

- A. threats
- B. patterns
- C. reports
- D. occurrences

8..The word Proponents in the passage is closest in meaning to

- A. creators
- B. opponents
- C. reviewers
- D. supporters

9..In paragraph 5, why does the author discuss what happened in Australia 40,000 years ago

- A. To suggest that humans most likely arrived in North and South America much earlier than 11,000 years ago
- B. To make a comparison that supports the prehistoric overkill hypothesis
- C. To argue that most extinctions can be traced to the impact of humans on the environment
- D. To emphasize the similarities between the extinctions that occurred in Australia and those that occurred in Africa and Europe

10..The word swift in the passage is closest in meaning to

- A. effective
- B. unexpected
- C. quick
- D. complete

11..According to paragraph 6, archaeological evidence of settlement patterns in North and South

America indicates which of the following

- A. Human populations may have been too small and too far apart to have caused the extinction of large mammals.
- B. Humans may have lived too far away from the habitats of large mammals to have been responsible for their extinctions.
- C. Humans probably did not cause the extinction of large mammals, because they spent far more time gathering food than hunting.
- D. Humans probably did not remain in their settlements long enough to have a significant impact on populations of large mammals.

12..In paragraph 6, the author identifies all of the following as being problems with the prehistoric overkill hypothesis EXCEPT:

- A. There were not enough people to kill so many species of large animals.
- B. There is little evidence to show that extinct animals were hunted.
- C. Prehistoric Europeans apparently preferred hunting smaller animals.
- D. It took 600 years for humans in New Zealand to exterminate just a few species of moa birds.

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

It should be noted, however, that island conditions that lead to extinction, such as limited space to escape predators, do not apply to landmasses such as continents.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14..Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Rapid climate change has been proposed as a cause of the extinctions, though there is strong evidence that mammals were able to survive similar climate changes in the past.
- B. The climatic hypothesis has gained more support than the prehistoric overkill hypothesis because climate change can explain why very few extinctions occurred in Europe and Africa.
- C. Some researchers have begun to doubt the idea that mass extinctions occurred at the end of the Pleistocene, because large animal remains are rarely found at settlements dating from this time.
- D. Late Pleistocene hunters may have killed off large mammals when they first arrived in certain areas, but evidence from settlements and animal remains does not often support this hypothesis.
- E. According to the prehistoric overkill hypothesis, the extinctions were concentrated in North and South America because these areas had more hunters and smaller populations of large mammals.
- F. Although neither the climatic hypothesis nor the prehistoric overkill hypothesis alone explains late Pleistocene extinctions, elements of both and other factors may have contributed to the extinctions.

Passage 144 2015-09-19-2 Models of Egg Development

- Several different theories have been put forward to explain how the hard-shelled eggs of land-dwelling reptiles (e.g. lizards) evolved from the soft eggs that amphibians (e.g. frogs and toads) lay in water. The Romer model of egg development is named after the late Alfred Romer, a paleontologist who also became director of the Harvard University Museum of Comparative Zoology. His specialty was early reptiles because, he felt, they were the key to understanding the great reptile diversification seen in the Late Paleozoic and Mesozoic Eras (around 230 million years ago). Romer's hypothesis was that some aquatic amphibians 'that is, amphibians living in water' called anthracosaurs began to lay their eggs on land at about the time that they were evolving reptile-like skeletal features. **Indeed, some of these early amphibians and earliest reptiles are so similar in their skeletons that the exact transition point from one to the other is still difficult to determine.** Eventually, though, the transition was made, but these early reptiles remained aquatic. The advantage for laying eggs on land was primarily to avoid the aquatic larval (pre-adult) stage during which immature amphibians live exclusively in water with its inherent risk of predators and drying of ponds. However, the land has its own set of dangers, not least of which is the drying effect of the atmosphere. To cope with these problems, a series of protective membranes developed around the egg, including a hard shell. Only later did the reptiles completely abandon an aquatic lifestyle.
- Another hypothesis was proposed by German paleontologist Rolf Kohring, whose specialty is fossil eggs. In Kohring's model, amphibians during the Mississippian epoch (360 'C320 million years ago) spread into nutrient-poor or cooler water. Because of the harsher conditions, eggs were produced with larger yolks, that is, more nutrients for the embryo. With larger yolks, the eggs were bigger, and fewer of them could be produced by the female 'Hundreds rather than thousands. To keep the larger egg intact, one or more membranes were developed, including one that surrounded and protected the egg. This outer membrane provided a place to safely store calcium ions, which are poisonous. Accumulating the calcium in a hard shell then made it possible for the egg to be laid on land (it was pre-adapted to be laid there.)
- One other model we should consider is the anti-predator hypothesis proposed by Gary and Mary Packard to explain the evolution of the hard-shelled egg. Their model was not concerned with the development of membranes surrounding the egg but continues the story after these membranes appeared. The Packards assume that the earliest reptiles laid leathery shelled eggs on very wet ground where they could absorb water during the embryos' growth. But life on the ground is not without hazards, based on studies of modern reptiles with leathery shelled eggs. Predatory insects and microbes can be a major cause of egg mortality. To counter this loss of eggs, some of the early reptiles began secreting a thin calcareous (containing calcium carbonate) layer. This hard layer gave the embryos a better chance of surviving until hatching. And these survivors in turn would probably leave more progeny once a few of them reached reproductive age. In time, a thicker, more resistant shell developed. However, a thicker eggshell meant that less water could be absorbed for the needs of the embryo. To compensate, larger eggs were produced, containing a great deal more albumen (egg white, a water-soluble protein). At this point, the rigid eggshell had reached the bird egg level of complexity.

4. Mary Packard presented yet another model with her colleague Roger Seymour. They note that amphibian eggs can never get very large because the gelatin coat surrounding the developing larva is not very good at transmitting oxygen. Because of this restriction, we will never see frog eggs the size of a chicken's. For Packard and Seymour, the major evolutionary breakthrough in reptile eggs was the elimination of the thick gelatin coat and replacing part of it with a fibrous membrane. This change allowed larger eggs to be developed.

1..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information.

- A. During the period that early amphibians changed into reptiles, their skeletons were exactly the same.
- B. It is difficult to identify the time that early amphibians changed to reptiles because their skeletons look so much alike.
- C. The skeletons of some early amphibians and reptiles are so similar that it is difficult to say which are amphibians and which are reptiles.
- D. Early amphibians and reptiles had the same kind of skeleton at the same point in time.

2..The word exclusively in the passage is closest in meaning to

- A. only
- B. initially
- C. primarily
- D. temporarily

3..According to paragraph 1, aquatic amphibians laid their eggs on land in order to

- A. enable young amphibians to benefit from a dry atmosphere
- B. ensure the rapid development of a hard shell
- C. enable young amphibians to evolve features necessary for living on land
- D. protect young amphibians from the dangers associated with life in the water

4..The word Accumulating in the passage is closest in meaning to

- A. Forcing
- B. Collecting
- C. Distributing
- D. Isolating

5..In paragraph 2, why does the author mention the information about calcium ions

- A. To explain Koherig's theory that amphibian eggs developed at least two protective outer membranes

- B. To explain that the calcium in a hard shell is not poisonous when the egg is laid on land
- C. To explain why, according to Kohring, a hard shell evolved
- D. To explain why only the outer membrane stored calcium

6..Which of the following can be inferred from paragraph 2 about the relationship between eggs and water temperature

- A. Eggs needed adaptations to survive in cold water.
- B. Eggs needed warm water to survive.
- C. Smaller eggs were produced in cold water.
- D. Fewer eggs were produced in warm water.

7..According to paragraph 2, all of the following are true of the eggs of amphibians during the Mississippian epoch EXCEPT:

- A. They had hard shells made from calcium ions.
- B. They had larger yolks than previously.
- C. They had protective membranes.
- D. They were produced in larger quantities than previously.

8..The word counter in the passage is closest in meaning to

- A. escape
- B. stop
- C. combat
- D. delay

9..According to paragraph 3, early reptiles began to develop a thin calcareous layer around the egg so that

- A. the embryo could survive attacks from predatory insects and microbes
- B. the embryo could absorb sufficient water during its growth
- C. the surviving embryo could reach reproductive age
- D. the egg could be laid on land

10..Which of the following is mentioned in paragraph 3 as a disadvantage of the hard eggshell

- A. It increased the hatching period.
- B. It prevented the development of large-size eggs.
- C. It made it more difficult for the embryo to obtain water.
- D. It made it harder for the embryo to survive until hatching.

11..The word breakthrough in the passage is closest in meaning to

- A. effect
- B. development
- C. requirement
- D. goal

12..Which of the following can be inferred from paragraph 4 about the fibrous membrane

- A. It served the same function as the gelatin coat.
- B. It was larger than the gelatin coat.
- C. It allowed amphibians to produce eggs as large as those of reptiles.
- D. It allowed for better transmission of oxygen.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

And the relatively few that were produced had to be properly protected.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Alfred Romer's hypothesis was that early amphibians started evolving reptilian skeletal features long before they started laying eggs on land.
- B. Alfred Romer theorized that early reptiles developed protective membranes around the egg in response to the dangers to young amphibians in water and threats to the egg on land.
- C. In Rolf Kohring's view, early reptiles developed protective membranes around the large eggs that were produced in harsh water conditions, making it possible for the egg to be laid on land.
- D. Gary and Mary Packard claimed that reptiles developed the hard-shelled egg in order to reduce the rate at which eggs were destroyed by predatory insects and microbes.
- E. Rolf Kohring argued that egg development was poor during the Mississippian epoch due to nutrient-poor waters.
- F. According to Mary Packard and Roger Seymour, reptiles could not successively develop very large eggs because of the elimination of the gelatin coat.

1. In late seventeenth-century Europe, what had been evolution in population followed by stabilization changed to population revolution. Increasing contacts with the Americas brought more sophisticated knowledge of the advantages of new foods, particularly the potato. Originally a cool-weather mountain crop in the Americas, potatoes did well in the Pyrenees, Alps, and Scottish Highlands. They also grew well in the long, damp springtime of the northwest European plain. Whatever hesitancy peasants may have felt about eating potatoes quickly passed when famine threatened; after all, people who in famines desperately consumed grass, weeds, and the bark of trees hardly would have hesitated to eat a potato. By the later eighteenth and the nineteenth century, American foods had become the principal foodstuffs of many rural folk. Various agricultural publicists promoted adoption of these foods, and peasants found that potatoes could allow subsistence on smaller plots of land. Fried potatoes soon began to be sold on the streets of Paris in the 1680s the original French fries. Governments, eager to promote population growth as a source of military and economic strength, also backed the potato.
2. Along with new foods, some landowners began to introduce other innovations. The nutritional base for a population revolution combined regional changes with the use of American foods. Dutch and English farmers drained more swamps and so increased cultivable land. Agricultural reformers further promoted the use of crops such as the turnip that return valuable nitrogen to the soil. Improvements in available tools, such as growing use of the scythe instead of the sickle for harvesting, and better methods of raising livestock also spread. All this took shape from the late seventeenth century onward, building on earlier agricultural changes. At the same time, rates of epidemic disease declined, in part because of more effective government controls over the passage of people and animals along traditional plague routes from the Middle East. It was the change in foods that really counted, however.
3. These developments provided a framework for an unprecedented surge. In virtually every area of Europe, the population increased by 50 to 100 percent in the eighteenth century, with the greatest growth coming after 1750. The Hapsburg Empire grew from 20 million to 27 million people; Spain rose from 5 million to 10 million, and Prussia rose from 3 million to 6 million. Growth would continue throughout the nineteenth century. In Europe as a whole, population rose from 188 million in 1800 to 401 million in 1900. This was an upheaval of truly impressive proportions.
4. The population explosion resulted from a break in the traditional, if approximate, balance between births and deaths in European society. In England between 1700 and 1750, approximately 32.8 people were born annually for every 1,000 inhabitants, and 31.5 people died. Similarly, in Lombardy in the eighteenth century, 39 people were born and 37 people died for every 1,000 inhabitants. Clearly, a major alteration had to occur in either the birth or the mortality rate before the expansion of population could begin. In fact, both rates changed: families began to have more children, and a lower percentage of the population died each year. Lower infant death rates meant more people living to produce children of their own, though falling adult death rates also increased the number of older Europeans.
5. While historians continue to debate the precise balance of causes involved in these dramatic changes, basic outlines are clear. Better food and a reduction in the epidemic-disease cycle allowed more children to live to adulthood, which increased the population directly and also

provided more parents for the next generation a double impact. Rapidly increasing populations provided a new labor force for manufacturing. In the eighteenth century, this mainly involved hundreds of thousands of people, mostly rural, producing thread, cloth, and other products for market sale. This manufacturing expansion helped sustain the growing population, but it could also encourage a higher birth rate. Some people, able to earn money by their late teens, began to produce children earlier; the rate of illegitimate births went up. Others realized that having an extra child or two might help the family economy by providing additional worker-assistants. While death-rate decline was the most important source of Europe's population explosion, various changes on the birth rate side, though quite short-lived, pushed the population up as well.

1..Paragraph 1 suggests that the European population before the late seventeenth century had been

- A. growing slowly and then not at all
- B. changing in distribution but not in the overall number of people
- C. decreasing at a small but stable rate
- D. alternating between periods of slow and fast growth

2..The word sophisticated in the passage is closest in meaning to

- A. quickly obtained
- B. highly developed
- C. widely distributed
- D. easily understood

3..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- A. The constant threat of famine caused peasants to become desperate and eat unusual foods like grass, weeds, and the bark of trees.
- B. Because famine forces people to eat foods they normally would not want to eat, peasants were willing to eat potatoes.
- C. Although some people ate foods like potatoes during famine, others preferred to eat easily accessible foods like grass, weeds, and the bark of trees.
- D. Famine had the greatest impact on peasants, whose regular diet of potatoes expanded to include other vegetation.

4..According to paragraph 1, all of the following contributed to the widespread adoption of the potato in Europe EXCEPT:

- A. Peasants growing potatoes for their own use could support themselves on smaller plots of land.
- B. Potatoes grew well in a variety of locations in Europe.

- C. Potatoes were the preferred food of European military forces.
- D. Agricultural publicists encouraged the public to eat potatoes.

5..According to paragraph 2, regional farmers did all of the following to improve food production in Europe EXCEPT:

- A. They improved the way they raised farm animals.
- B. They used both the sickle and the scythe to harvest crops.
- C. They grew special crops that nourished the soil.
- D. They created more farmland by draining swamps.

6..The word surge in the passage is closest in meaning to

- A. event
- B. benefit
- C. sudden increase
- D. important change

7..In paragraph 3, the author mentions the Hapsburg Empire, Spain, and Prussia in order to

- A. support the claim that the population explosion covered most of the European continent
- B. give examples of population growth during the nineteenth century
- C. suggest that the population of Prussia grew more slowly than the populations of other countries
- D. demonstrate that the fastest population growth took place in Spain

8..The word proportions in the passage is closest in meaning to

- A. origins
- B. consequences
- C. growth
- D. size

9..According to paragraph 4, the expansion of Europe's population was made possible by

- A. a major improvement in the care of older Europeans
- B. increased variation in the ages at which people gave birth to children
- C. a change in traditional beliefs about family size
- D. increased birth rates accompanied by a decline in mortality

10..The word sustain in the passage is closest in meaning to

- A. support
- B. stimulate

- C. explain
- D. unite

11..According to paragraph 5, what effect did the epidemic-disease cycle have on population during the eighteenth century

- A. Childhood diseases kept population growth rates from rising even higher.
- B. Periodic epidemics caused population growth rates to rise and fall in cycles.
- C. The effect varied by area, with urban populations more affected by disease than rural areas.
- D. Fewer childhood deaths from disease led to an increased number of children in the current and future generations.

12..According to paragraph 5, how did the manufacturing expansion affect population growth

- A. It caused a small decline, because families working in manufacturing needed fewer children as worker-assistants than did farming families.
- B. It made teenage workers delay childbearing, which caused a decline in population growth.
- C. It caused an increase in population by allowing workers to support a family at an earlier age.
- D. It caused the growth rate to rise in the cities and to decline in rural areas.

13..Look at the four squares that indicate where the following sentence could be added to the passage.

What were the factors that led to this population revolution

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text .

Answer Choices

- A. Climate conditions in Europe allowed for the introduction of new crops that competed with American foods for popular consumption.
- B. An important cause of population growth was improved nutrition, due in large part to the addition of the potato and other American foods to the standard European diet.
- C. Regional landowners developed improved agricultural techniques, and mortality rates declined as governments gained control over traditional plague routes.
- D. Growth rates varied widely across the continent but were highest in France, Spain, and Britain and lowest in Prussia and the Hapsburg Empire.
- E. Birth rates went up as more people lived long enough to have their own children, had children earlier, and had larger families.
- F. Government policies promoting population growth helped to create a large labor force for the manufacturing industry.

1. Identifying tree species in tropical rain forests may be harder than you think .
2. Plant species identification can be difficult for all kinds of reasons even identification of trees , which are big and conspicuous . For example, for some willow trees, both leaves and flowers may be needed for identification, but the two may not be present at the same time. Yet whatever problems may confront us in temperate climates, we can be sure that the tropics will pose far worse difficulties.
3. In tropical rain forests, the flowers of a given tree species are typically not in bloom and so cannot be observed. In seasonal rain forests (with a distinct wet season and a distinct dry season) , many trees adjust their flowering to the rains, so flowering is to some extent predictable. But much rain forest (as in much of the Amazon region) is nonseasonal , and trees may flower at any time. To be sure, different trees of the same species generally flower simultaneously , for if they did not, they could not pollinate each other. 80 they must be responding to signals from the environment at large, or else (or in addition) they must be communicating with one another. But what those signals are is unknown, at least to us. To the human observer, the flowering seems random. In any case , in a tropical forest (at least in a secondary forest , which is forest that is regrowing after previous harvesting or clearance) , the trees grow very close together, and most are remarkably thin, like poles, and grow straight up and disappear into the gloom, twenty meters overhead . Even if there are flowers, you would not necessarily see them .
4. The leaves may not be accommodating either, at least when viewed from the ground. Rain-forest trees all face the same kinds of conditions and have adapted in the same general kinds of way. Rain forests are wet by definition. But in some there is a dry season , and even when there is not, it doesn't rain all the time. Thus the forest floor may be moist, but the topmost leaves of the canopy are far above it and are exposed to the fiercest sun. 80 the uppermost leaves must resist desiccation (drying out). Yet from time to time, and in due season every day, they must also endure tremendous downpours. Leaves that can cope with such contrasts tend to be thick and leathery (to resist drought) , oval in shape, and have a prection at the end known as a drip tip to let surplus rain run off the leaf. Many hundreds of trees from dozens of only distantly related families have leaves of this general type. But even if you can distinguish individual leaves, it is hard to be certain if they belong to the tree you are interested in or to the one next to it or to some epiphyte (a plant that grows on other plants) or liana (vine) slung over its branches. Often, in short, researchers must base their identification of a tree on the bark of its trunk. The trunks of tropical trees are sometimes highly characteristic , being deeply furrowed or twisted , but in most species the bark is simply smooth and gray, dappled with lichen and moss.
5. In a temperate forest you can be fairly sure that any one tree is the same species as the one next to it or, at least, it will be one of a list that is unlikely to exceed more than half a dozen (oak with ash in much of Britain; lodgepole pine with aspen in the northernmost reaches of North America ; alder, 8 scotch pine, and spruce in the Baltic ; and so on). But in the Amazon in particular, you can be fairly sure that any one tree is not the same species as the one next to it. Often there is a third of a mile between any two trees of the same species , and there can be up to 120 different species of trees in any one acre . 80 the task , often, is to identify an

individual tree that may be not much thicker than your arm from the appearance of its bark, out of a totallist of several hundred (or thousand) possibilitieswhich may well include some that have not been described before, so that ther e is nothing to refer back to.

1..The word conspicuous in the passage is closest in meaning to

- A. close to each other
- B. common
- C. easily noticed
- D. solitary

2..In paragraph 1, why does the author discuss willow trees

- A. To provide an example of a tree that is unusually large
- B. To explain the process of tree species identification
- C. To support the idea that tropical trees can be more difficult to identify than temperate trees
- D. To demonstrate one reason why it can be problematic to determine what species a tree belongs to

3..Paragraph 2 implies which of the following about tree flowering in seasonal rainforests

- A. It is less predictable than tree flowering in tropical rain forests
- B. It results from an unknown means of communication between the individual trees of each species.
- C. It is more useful to researchers trying to identify tree species than is the tree flowering in tropical rain forests.
- D. It follows the same pattern that tree flowering follows in a tropical secondary forest.

4..In paragraph 2, why does the author discuss pollination

- A. To provide evidence that in many tropical tree species flowering is at least somewhat predictable
- B. To help explain why flowers on trees in tropical rainforests can be absent most of the time
- C. To help explain why different trees of the same species bloom at the same time
- D. To suggest one reason why trees might benefit from adjusting their flowering to the rains

5..According to paragraph 2, what is true about much of the Amazon region

- A. It has tree species that adjust their pollination to rainfall.
- B. It has not yet been harvested or cleared.
- C. It has trees that typically flower during periods of rain.
- D. It does not have separate wet and dry seasons.

6..The word endure in the passage is closest in meaning to

- A. withstand
- B. benefit from
- C. avoid

D. respond to

7..The word surplus in the passage is closest in meaning to

- A. heavy
- B. steady
- C. sudden
- D. extra

8..Paragraph 3 offers an explanation for each of the following characteristics of rain-forest tree leaves EXCEPT

- A. their having a drip-tip at the end
- B. their thickness
- C. their leathery texture
- D. their oval shape

9..According to paragraph 3, what is one reason that looking at leaves may not be very useful when trying to determine the species of a rain-forest tree

- A. Leaves of rainforest trees often look very different when they are wet than when they are dry.
- B. Leaves that are exposed to the hot tropical sun dry out and lose their distinguishing characteristics.
- C. It can be hard to determine whether a leaf has a drip tip or not.
- D. It can be hard to determine whether a leaf belongs to a tree or to another plant growing on the tree.

10..According to paragraph 3, one reason that researchers in tropical forests must often rely on a tree's bark for species identification is that

- A. distinct species of tree often have similar kinds of lichens and moss growing on their trunks
- B. the leaves of many different tropical tree species are very similar to each other in appearance
- C. many rainforest trees have neither epiphytes nor lianas
- D. the bark of the trees is less affected by changes in light and moisture than leaves are

11..Which of the sentences below best expresses the essential information in the highlighted sentence in the passage Incorrect choices change the meaning in important ways or leave out essential information

- A. Different temperate forests contain different combinations of tree species but in all such forests , all the trees of any one species are likely to be found close together.
- B. Temperate forests rarely contain more than about six species of tree, and trees growing next to each other tend to be of the same species.
- C. You can easily make a list of all the tree species that are likely to be found in temperate

- forests, and if one tree is not on that list, you can be confident that the tree next to it will be.
- D. Forests in Britain, in North America, and in the Baltic generally contain no more than half a dozen tree species, and all these species can be found on lists.

12..The word task in the passage is closest in meaning to

- A. difficulty
- B. job
- C. strategy
- D. requirement

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

As a result, the topmost level of a rainforest can be desert-like.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text Answer Choices

- A. In tropical forests different trees of the same species flower at different times so they are not able to pollinate one another.
- B. Flowering in tropical trees is often unpredictable and when it occurs, the flowers themselves tend to be too high up to be clearly visible.
- C. Many tropical tree species have leaves that look very similar from the ground, and because the trees often grow close together, correctly matching a leaf with a particular tree can be difficult.
- D. The leaves and bark of a single rainforest tree are often very different at different heights due to the need to adapt to different conditions, which adds to the difficulty of species identification.
- E. Because trees in rain forests are spaced closely together, it is easy to tell whether neighboring plants are the same species, but that is not much help in determining which species they are.
- F. Often, a rain-forest tree has to be identified by its bark, but trees of like species are widely scattered, the list of possible species is very long, and many have similar-looking bark.

Passage 147 2015-11-15-4 The Extinction of the Dinosaurs

1. Geologists define the boundary between sediment layers of the Cretaceous period (144..C65 million years ago) and the Paleocene period (65..C55 million years ago) in part by the types and amounts of rocks and fossils they contain or lack. Before the limit of 65 million years ago, marine
2. strata are rich in calcium carbonate due to accumulations of fossils of microscopic algae deposited on the sea floor. Above the 65-million-year limit, sea-floor sediments contain much less calcium carbonate, and fossils of several families of mollusks are no longer found. In continental sediments, dinosaur fossils, though frequent before 65 million years ago, are totally absent. By
3. contrast, new families of mammals appear, including large mammals for the first time. Scientists wondered for many years about what could have caused the dinosaurs' rapid disappearance at the end of the Cretaceous period, coming up with a great variety of theories and scenarios. For some, it could have been due to unfavorable genetic changes triggered by a dramatic increase by a factor of 10, 100, 1, 000 in cosmic-ray particles reaching the Earth after a supernova explosion somewhere in the neighborhood of the solar system. For these high-energy particles to affect life, they would have to get through the protective barrier of the Earth's magnetosphere, the region of the upper atmosphere controlled by Earth's magnetic field. That could have happened if the cloud of particles from the supernova explosion reached the Earth during a period when the magnetosphere was weakened, something that may happen when the Earth's magnetic field changes direction. And we know that the magnetic north and south poles of the Earth switch on the average twice every million years. However, this is not the only possible explanation for dinosaur destruction.
4. Other theories have raised the possibility of strong climate changes in the tropics (but they then must be explained). Certainly, if climate changes, the changed distributions of temperature and rainfall modify the conditions that favor one ecosystem over another. The extinction of a particular family, genus, or species may result from a complicated chain of indirect causes and effects. Over thirty years ago, scientist Carl Sagan quoted one suggestion that the demise of the dinosaurs resulted from the disappearance of a species of fern plant that was important for dinosaur digestion. Other theories involved a worldwide cold wave following the spread of a layer of cold but not very salty water in the world's oceans, which floated on the surface because, with its low salinity, the water was less dense.
5. Proponents of another theory that remains under consideration today postulate that the extinction of the dinosaurs corresponds to a period of intense volcanic activity. It's not a question of just one or even of a thousand eruptions comparable to the explosion of Krakatoa in 1883, one of the largest volcanic events in modern times, but rather of a prolonged period of activity. On the Deccan plateau in India, basalt (volcanic) rocks cover more than 500, 000 square kilometers (nearly 200, 000 square miles), and correspond to massive lava outflows occurring precisely at the end of the Cretaceous. This sort of outflow could correspond to volcanic activity similar to the activity that drives sea-floor spreading, with lava emerging from elongated fractures in the crust rather than from craters.
6. The volcanic convulsion that buried the Deccan plateau in lava must also have changed the composition of the atmosphere and severely affected climate. Initially, there must have been strong sudden cooling resulting from the blocking of sunlight by sulfate aerosol veils in the

stratosphere (part of the Earth's atmosphere). If strong cooling lasted a year after the formation of the aerosols, it would have been the death of tropical species unable to adapt to such a volcanic winter.

7. However, a long period of strong volcanic activity (again, remember thousands of Krakatoas) would at the same time have added a substantial amount of carbon dioxide to the atmosphere, reinforcing the greenhouse effect. This would gradually warm things up, ending the extended cold snap and producing global warming together with geographic shifts of humid and arid (dry) zones. Certainly things would change to upset living conditions, leading to the extinction of some species while others would profit, if only from the disappearance of predators.

1..According to paragraph 1, which of the following is true of Paleocene sediments

- A. They lack fossils from some families of mammals found in Cretaceous sediments.
- B. They contain fossils of dinosaurs.
- C. They contain fossils of some animals that did not exist during the Cretaceous.
- D. They contain fossils of more kinds of mollusks than are found in Cretaceous sediments.

2..The word unfavorable in the passage is closest in meaning to

- A. Unusual
- B. Disadvantageous
- C. Sudden
- D. major

3..The word triggered in the passage is closest in meaning to

- A. initiated
- B. intensifie
- C. followed
- D. aided

4..Why does the author include the information that the magnetic north and south poles of the Earth switch on the average twice every million years

- A. To provide evidence showing that Earth's protective barrier is difficult to get through
- B. To show that it is reasonable to think that particles from a supernova explosion could have reached Earth
- C. To explain why some scientists believe a supernova explosion may have occurred near our solar system
- D. To help explain why some scientists have rejected the theory presented in paragraph 2

5..According to the theory discussed in paragraph 2, a weakening of Earth's magnetosphere may have

- A. caused a supernova to explode near our solar system

- B. allowed gene-altering particles to reach Earth's surface
- C. forced Earth's magnetic field to change direction
- D. allowed clouds of protective particles to escape from Earth's upper atmosphere

6..All of the following are mentioned in paragraph 3 as possible causes for the extinction of the dinosaurs EXCEPT

- A. a change in the diet of dinosaurs
- B. a change in the climate of the tropics
- C. a decrease in global temperatures
- D. a decrease in deep ocean salinity

7..In paragraph 3, why does the author include the quotation by Carl Sagan

- A. To explain the connection between dinosaur extinction and the extinction of other animal species
- B. To support the claim about species extinction being due to indirect causes and effects
- C. To show that scientists have revised their ideas greatly in the last thirty years
- D. To identify the differences between the various theories for the extinction of dinosaurs

8..According to paragraph 4, what was one unusual aspect of the volcanic activity at the end of the Cretaceous

- A. Some explosions were much larger than Krakatoa.
- B. Eruptions occurred over a long period of time .
- C. Active volcanoes were sometimes separated by many kilometers.
- D. There were active volcanoes in the sea as well as on land.

9..Which of the following is presented in paragraph 4 as evidence that intense volcanic activity occurred at about the time that the dinosaurs became extinct

- A. The size of the volcanic craters on the Deccan plateau
- B. An increase in sea-floor spreading
- C. The formation in India of large amounts of a type of rock associated with volcanoes
- D. The occurrence of a thousand or more volcanic explosions the size of Krakatoa

10..The word severely in the passage is closest in meaning to

- A. certainly
- B. consequently
- C. greatly
- D. permanently

11..The word reinforcing in the passage is closest in meaning to

- A. making possible

- B. spreading
- C. introducing
- D. strengthening

12..According to paragraph 5, all of the following are theorized to have occurred as a result of volcanic activity EXCEPT

- A. a decrease in the amount of sunlight reaching Earth's surface
- B. a reduction in the number of sulfate aerosol veils in the stratosphere
- C. increased dryness in some areas that were once more humid
- D. changes in the atmosphere's composition resulting in an increase in temperature

13.. Look at the four squares that indicate where the following sentence could be added to the passage.

Thus , ecosystems following periods of extensive volcanic activity necessarily had different combinations of species than earlier ecosystems did.

Where would the sentence best fit Click on a square to add the sentence to the passage.

14.. Drag your choices to the spaces where they belong. To review the passage, click on View Text Answer Choices

- A. Cosmic rays from a nearby supernova explosion may have penetrated Earth's atmosphere , causing genetic changes that dinosaur populations could not survive.
- B. Climate changes in the tropics may have set off a chain of indirect effects that negatively affected the ecosystems in which dinosaurs lived.
- C. The spread of lava over large portions of previously favored habitats such as the Deccan Plateau may have made these favored areas uninhabitable for many years.
- D. Although the volcanic winter resulting from the formation of sulfate aerosols eventually ended, temperatures may have remained below levels required by dinosaurs to survive.
- E. Temperature changes and geographic shifts in climate zones due to the atmospheric effects of volcanic activity may have been significant enough to cause extinction.
- F. To be convincing , theories about what caused dinosaurs to become extinct must be able to explain the disappearance of other predators in the food chain at the same time

Passage 1 2011-01-08-1 Crown of Thorns Starfish and
Coral Reefs

CBBAC, BDABB, ADA, 236

Passage 2 2011-01-08-2 The Rise of Moscow

41334,42313,122 (134)

Passage 3 2011-01-08-3 Forest Succession

22312,24312,414 (145)

Passage 4 2013-01-12-1 England's Economy in the
Sixteenth Century

ABDAB, CBCDC, (CD)DC, 234

Passage 5 2013-01-12-2 Documenting the Incas

DDBBB, DDBAC, BAB, 345

Passage 6 2013-01-12-3 What Controls Flowering

DCABD, AACCB, BDC, 123

Passage 7 2014-03-02-1 Cell Theory

(只有文章没有题目)

Passage 8 2014-03-02-2 The Great Depression

(只有文章没有题目)

Passage 9 2014-03-02-3 Poikilotherms

(只有文章没有题目)

Passage 10 2014-03-16-1 Costs and Benefits of Social
Life

CABDD, BACBD, AD, 345

Passage 11 2014-03-16-2 Comets

DDABC, AC(BD)BC, BCB, 345

Passage 12 2014-03-16-3 Agricultural Society in
Eighteenth-Century British America

CDBBC, DDABB, ABA, 356

Passage 13 2014-03-22-1 The Beringia Landscape

AABBD, CBBDC, BDB, 156

Passage 14 2014-03-22-2 Wind pollination

DAACC, (AC)BBBB, ADB, 235

Passage 15 2014-03-22-3 Feeding Strategies in the
Ocean

ABCDC, DCCCC, BAB, 345

Passage 16 2014-03-23-1 From Fish to Terrestrial
Vertebrates

BACAD, CAA, 156

Passage 17 2014-03-23-2 Early Modern Industrialization

CBACB, BCDCC, B, 146

Passage 18 2014-03-23-3 Environmental Impact of the
Anasazi

CDBDC, ACBAC, DAB, 235

Passage 19 2014-04-12-1 Surface Fluids on Venus and

Earth

BCBAB, BDAAA, BCB, 345

Passage 20 2014-04-12-2 Water Management in Early
Agriculture

BAADC, ABCCB, CDD, 126

Passage 21 2014-04-12-3 Plant and Animal Life of the
Pacific Islands

ACBBC, CBBAD, ADA, 134

Passage 22 2014-04-19-1 The Cosmological Principle

ADDCC, DBCAB, BAA, 345

Passage 23 2014-04-19-2 Distribution of Seaweeds

BBDCA, CDABD, CDB, 135

Passage 24 2014-04-19-3 Bird Colonies

BBADD, CDBAB, ADB, 456

Passage 25 2014-04-27-1 Urban Development in the
United States in the Nineteenth Century

BB(AB)AC, CDDDC,

Passage 26 2014-04-27-2 The Climate of Japan

CCADB, BDACB, BBB, 123

Passage 27 2014-04-27-3 Economic Decline in Europe
During the Fourteenth Century

BDCAB, ABACD, BDB, 125



Passage 28 2014-05-11-1 Determining the Ages of the Planets and the Universe

BCDAA, ABCBD, ACA, 123

Passage 29 2014-05-11-2 Costs and Benefits of Dispersal

ADDCB, DDCCD, BBB, 356

Passage 30 2014-05-11-3 The Multiplier Effect

CBBDA, CBDCA, ADC, 156

Passage 31 2014-05-17-1 Cognitive Maps in Animals

ACABD, CBDBA, 234

Passage 32 2014-05-17-2 The Use of the Camera Obscura

BCDAC, DCCCB, ADD, 126

Passage 33 2014-05-17-3 Habitat Selection

DCACD, AACDD, DB, 156

Passage 34 2014-05-24-1 Sumerian Contributions

BCADB, BDDCB, CAA, 346

Passage 35 2014-05-24-2 Impacts and Mass Extinctions

BBBCA, CCDBB, DCC, 456

Passage 36 2014-05-24-3 Milankovitch Cycles and Glaciation

DCDDB, ACBBA, ACC, 123

Passage 37 2014-05-25-1 The Long History of
Overexploitation

ADDBC, DACC, 134

Passage 38 2014-05-25-2 How Animals in Rain Forests
Make Themselves Heard

ABDCD, DCDCB, CBC, 345

Passage 39 2014-05-25-3 Pest Control

BCAAB, CDBDC, BDD, 246

Passage 40 2014-06-15-1 Early life-forms and Earth's
Atmosphere

ADCBC, ACBDA, BCA, 234

Passage 41 2014-06-15-2 Constraints on Natural
Selection

BABCD, AADAC, BBC, 256(a single, a single, both)

Passage 42 2014-06-15-3 Newspapers in Western
Europe

CDDCC, BCABC, ABA, 145

Passage 43 2014-06-21-1 Climate Change and the
Natufian People

CABAC, ADDBD, BAB, 156

Passage 44 2014-06-21-2 Metamorphosis

ABBCC, DBDCC, ABB, 345

Passage 45 2014-06-21-3 The Origin of Coral Reefs

CCDAC, BABCD, ABD, 245

Passage 46 2014-06-28-1 Southwest Agriculture

CCCAA,DBBCA, DD, 346

Passage 47 2014-06-28-2 The Development of Printing

CCBCD, CCDBA, ACD, 235

Passage 48 2014-06-28-3 Earth's Energy Cycle

BBBDC, ADAAC, BDC, 123(Incoming short-wavelength, Heat energy, Some of)

Passage 49 2014-07-06-1 Removing Dams

DCCBB, ADCAD, BBC, 134

Passage 50 2014-07-06-2 Coral Reef Communities

AADBB, B(C/D)BCB, CDB, 135

Passage 51 2014-07-06-3 Sociality in Animals

BBDCC, ABACD, DDC, 234

Passage 52 2014-07-12-1 Understanding Ancient Mesoamerican Art

BDBCBC, DBADC, BBB, 456

Passage 53 2014-07-12-2 How Soil is Formed

CBBCC, BDBBC, CCA, 234

Passage 54 2014-07-12-3 The Emergence of Civilization

ABABA, DBCAB, CCB, 234



Passage 55 2014-08-16-1 Hunting and the Setting of
Inner Eurasia

CAADB, DCBCB, BAD, 356

Passage 56 2014-08-16-2 Written Records

BCDCA, BAADD, CBA, 126

Passage 57 2014-08-16-3 Temperature Regulation in
Marine Organisms

CBBDC, CDABD, ABC, 135

Passage 58 2014-08-30-1 Mesolithic Complexity in
Scandinavia

BBDCA, ABB(AD)C, DDB, 346

Passage 59 2014-08-30-2 Evolution of the Flowering
Plants

CBCDB, AAD(题干不全)BA, CAA, 246

Passage 60 2014-08-30-3 How Herding Can Provide
Safety

DBBDA, CDABA, BBA, 234

Passage 61 2014-10-18-1 The Origins of Writing

BCABC, CDCDA, BCB, 234

Passage 62 2014-10-18-2 Soil Fertilization

BCBBC, BCBDA, BBA, 156

Passage 63 2014-10-18-3 The Collapse of the Mays

BCADB, BDDCB, CAA, 346

Passage 64 2014-12--1 Forms of Locomotion

CDA(AD)D, CBDBD, CAD, 245

Passage 65 2014-12--2 How Plants and Animals Arrived
in the Hawaiian Islands

CADDC, ABBDB, BCA, 这篇最后一题残

Passage 66 2014-12--3 Mexican Mural Art

CCBDC, DBDAB, ADB, 346

Passage 67 2015-01-10-1 Artisans in Sixteenth-Century
Europe

CACBB, DCDAABCD, 124

Passage 68 2015-01-10-2 Disease and History

BDCDC, BB(与上一题重复)DCA, CC B, 123

Passage 69 2015-01-10-3 Earth's Atmosphere

ACDBD, AABCD, ABD, 135

Passage 70 2015-01-11-1 Birdsong

CCDCC, ADBAD, CCB

Passage 71 2015-01-11-2 The Role of Diapause

CDBBC, DABAA, CB, 256

Passage 72 2015-01-11-3 The Plow and the Horse in
Medieval Europe

BCADB, BCDBB,DAA, 456

Passage 73 2015-01-25-1 The Origins of Plant and Animal Domestication

DCCBB, CCABD, DB, 126

Passage 74 2015-01-25-3 Preventing Overgrowth among Tree Branches

BADCB, CCDBD, CBB, 126

Passage 75 2015-01-31-1 Effects of the Commercial Revolution

DBCDD, BABAB, CCB, 123(New kinds of, Rulers in, More established)

Passage 76 2015-01-31-2 European Context of the Scientific Revolution

BBCBD, ACABC, DBD, 126(The discovery of, The invention of, The revival of)

Passage 77 2015-02-01-1 Greek Sacred Groves and Parks

CBDDC, ABDBC, BAA, 156(Sacred groves, Trees have, Because groves)

Passage 78 2015-02-01-2 The Chaco Phenomenon

CBDDC, ABDBC, BAA, 156(Sacred groves, Trees have, Because groves)

Passage 79 2015-02-01-3 Dinosaurs and Parental Care

BDACD, AACBC, BDC, 236

Passage 80 2015-03-07-1 Colonial America and the
Navigation Acts

BBBD(AC), DBBAC, DCD, 246

Passage 81 2015-03-07-2 Mass Production: Method and
Impact

CCADB, BCDBB, DAA, 346



Passage 82 2015-03-07-3 Art and Culture of Pacific Northwest Communities

DBDBC, AACBC, ABD, 125

Passage 83 2015-04-12-1 Architectural Change in Eighth-Century Japan

BDABC, DBCCA, DBD, 134

Passage 84 2015-04-12-2 Origin of the Solar System

ADABC, BDDAC, BAB, 346

Passage 85 2015-tfyd-30-1 Attempts at Determining Earth's Age

1. A 2. C 3. C 4. B 5. D 6. B 7. C 8. C 9. A 10. C 11. B 12. D 13. D 14. BCD

Passage 86 2015-tfyd-30-2 Water Supply in Venice

1. D 2. C 3. A 4. C 5. B 6. D 7. C 8. B 9. A 10. B 11. D 12. A 13. A 14. BEF

Passage 87 2015-tfyd-30-3 The Early History of Motion Pictures

1. A 2. C 3. A 4. B 5. D 6. C 7. B 8. B 9. A 10. D 11. B 12. C 13. C 14. CDE

Passage 89 2015-tfyd-30-10 Early Theories of Continental Drift

1. B 2. B 3. D 4. C 5. D 6. C 7. A 8. B 9. A 10. C 11. C 12. B

13. A 14. BCE

Passage 90 2015–tfyd–30–11 Bioluminescence in Marine Creatures

1.A 2. D 3. B 4. A 5. B 6. A 7. C 8. D 9. A 10. D 11. D 12. B 13.

C 14. ABC

Passage 91 2015–tfyd–30–22 Egypt Circa 3100 B.C

B 2. A 3. A 4. D 5. C 6. A 7. B 8. C 9. D 10. D 11. C 12. B 13.

A 14. ABF

Passage 92 2015–tfyd–30–24 Population Growth in Nineteenth-Century Europe

1. A 2. D 3. B 4. B 5. B 6. D 7. C 8. C 9. B 10. A 11. B 12. D

13. C 14. ACE

Passage 93 2015–tfyd–30–25 Costs of Quitting a Job

1.AC 2. B 3. B 4. C 5. B 6. A 7. C 8. A 9. C 10. D 11. C 12. D

13. D 14. BCF

Passage 94 2015–tfyd–30–26 Stone Tools and Pottery Fragments

1. A 2. D 3. D 4. C 5. B 6. C 7. C 8. B 9. D 10. B 11. C 12. A

13. B 14. ADF

Passage 95 2015–tfyd–30–27 The Role of the Ocean in Controlling Climate

1. D 2. C 3. A 4. C 5. B 6. C 7. B 8. D 9. A 10. C 11. B 12. A 13.

C 14. BCF

Passage 96 2015–tfyd–30–28 Animal Behavior

1. D 2. A 3. A 4. B 5. C 6. B 7. D 8. B 9. B 10. C 11. D 12. A

13 Ethologists: BD; Comparative Behaviorists: ACE

Passage 97 2015–tfyd–30–29 Grain in Colonial North America

1. B 2. C 3. A 4. D 5. B 6. C 7. A 8. B 9. D 10. C 11. C 12. A

13. D 14. ABE

Passage 98 2015–tfyd–30–30 Sea Turtle Hatchling Strategies for Navigation

1. C 2. B 3. D 4. B 5. C 6. B 7. D 8. A 9. B 10. C 11. D 12. C 13.

A 14. DEF

Passage 99 2015–kls–12–1 Controversy about Causing Emotion

DABCA DDCBD BBAfCDE]

Passage 100 2015–kls–12–2 Oxygen and nitrogen are major components

CABDD ABDBD CBD[ACF]

Passage 101 2015–kls–12–3 The Distribution of Gliding Animals

DCAB[AD] BDCCB ADC[BEF]

Passage 102 2015-cls-12-4 The Sentinel Behavior
of Meerkats

1. DBADC 6. BCCBB 11. ACDfBDF]

Passage 103 2015-cls-12-5 Portraits as Art

1. DDDCC 6. AAABD 11. BDC[BDF]

Passage 104 2015-cls-12-7 Agriculture in Medieval
Japan

1. BBBCB 6. DC[BD]CD 11. ADC[BDE]

Passage 105 2015-cls-12-8 Grassland Environment

1. DADCC 6. CDABA 11. CBA[BCE]

Passage 106 2015-cls-12-9 The British Economy Under
the Roman Empire

1. CABBD 6. CACDA 11. BDC[CEF]

Passage 107 2015-cls-12-10 Foundational and Keystone
Species

1. A 2. D 3. D 4. B 5. C 6. A 7. D 8. A 9. B 10. C 11. B 12.

Foundational Species (ABG) Keystone Species(DE)